FUNCTORIAL ALGEBRAIC GEOMETRY

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Abstract. These are unofficial lecture notes typed by Timon Tausendpfund and Yiming Wang for the course *Algebraic Geometry* taught by Marc Hoyois during 2025–2026 at the University of Regensburg. The sole contribution of the note-takers is to record some proofs for some stated results, any mistakes and misconceptions are due to the note-taker.

Contents

1. Some examples 3
1.1. Punctured elliptic curves 3
1.2. Singular cubic curves 6
1.3. Compactifying affine curves 7
1.4. Fermat curves 9
1.5. Affine, projective, and general schemes 10
2. Affine geometry 12
2.1. Affine spaces 12
2.2. Presheaves 12
2.3. Polynomial equations 16
2.4. Example of affine scheme 19
2.5. Base change 20
2.6. Functions 26
2.7. Closed and open subfunctor 27
2.8. Zariski descent 34
2.9. Finiteness properties 44
2.10. The Nullstellensatz 46
3. Projective geometry 48
3.1. Projective spaces over a field 48
3.2. Projective spaces 53
3.3. Graded rings 54
3.4. Homogeneous polynomial equations 56
3.5. Loci associated with linear maps 58
3.6. The functor Proj 63
3.7. Saturation 73
3.8. Projective closure 77
3.9. Example of projective schemes 78
3.10. The projective Nullstellensatz 80
4. Quasi-coherent modules 82
4.1. Limits of categories 82
4.2. Quasi-coherence 84
4.3. Classification of closed and open subfunctors 86
4.4. Relative Spec and Proj 88
4.5. Modules over quasi-projective schemes 90

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4.6. Serre twists 94
5. Locales and topological spaces 96
5.1. Pointless topology 96
5.2. Locales of radical ideals 98
5.3. The topological space of an algebraic functor 99
5.4. Open coverings 104
6. Sheaves 107
6.1. Sieves and descent 107
6.2. Grothendieck topologies 109
6.3. Sheafification 115
7. Schemes 120
7.1. Separatedness 123
7.2. Quasi-compactness and quasi-separatedness 124
7.3. Locally ringed spaces 126
7.4. Schemes as locally ringed spaces 128
7.5. Quasi-coherent modules on schemes 129
8. Properties of schemes 131
8.1. Local properties 131
8.2. Reduced schemes 135
8.3. Noetherian schemes 136
8.4. Irreducible components 138
8.5. Dimension theory 140
8.6. Regular schemes 143
8.7. Normal schemes 145
9. Smoothness 148
9.1. Smooth and étale morphisms 148
9.2. Cotangent and conormal modules 150
9.3. The cotangent module of an algebraic functor 158
9.4. Smoothness over a field 163
9.5. Formal smoothness 164
9.6. Local structure of étale maps 170
9.7. Bertini's theorem 171
10. Divisors and blowups 174
10.1. Effective divisors 175
10.2. Blowing up 177
10.3. Regular closed immersions 182
10.4. Cartier divisors 185
10.5. Weil divisors 187
10.6. The Picard group 189
11. Properness 191
11.1. Prim and proper morphisms 191
11.2. Valuative criteria 192
11.3. Relation to projectivity and finiteness 194
Appendix A. Facts about commutative algebra 196
References 197

1. Some examples

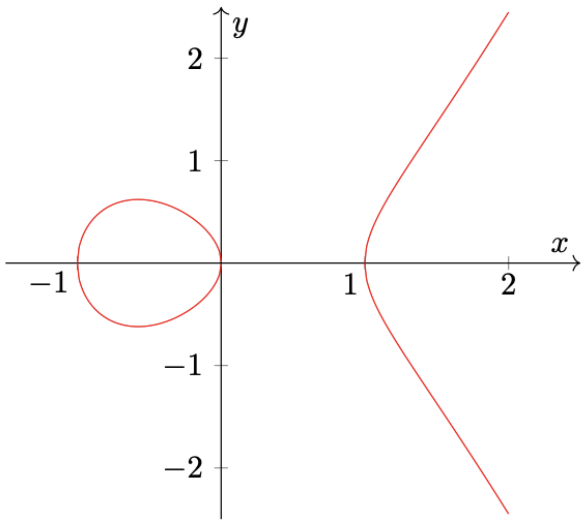
1.1. **Punctured elliptic curves.** Consider the polynomial equation in two variables $y^2 = x^3 - x$. We can consider its set of solutions in any ring R, namely

$$X(R) = \{(a, b) \in R^2 \mid b^2 = a^3 - a\}.$$

This defines a functor $X\colon \mathrm{CAlg} \to \mathrm{Set}$ from the category of rings to the category of sets, which is an example of an *affine scheme*. Being defined by a single equation in two variables, X is called a *family of algebraic curves* of simply a *curve*. It is also called an *arithmetic surface*: "arithmetic" because the equation has integral coefficients, and "surface" because it turns out to be a 2-dimensional object from the perspective of dimension theory in commutative algebra.

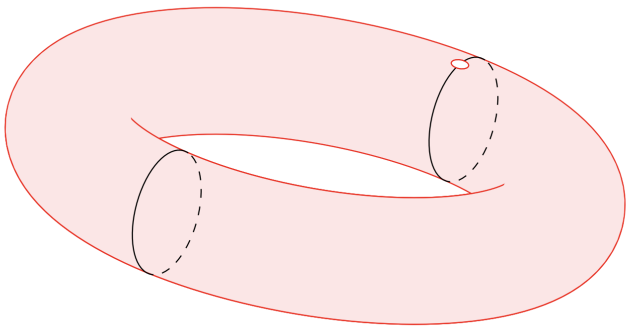
Let us consider explicitly the solution sets $X(R)$ for various rings R:

(a) $X(\mathbf{R})$ is a 1-dimensional real submanifold of $\mathbf{R}^2$, which is diffeomorphic to $S^1 \sqcup \mathbf{R}$:



(b) For any subring $R \subseteq \mathbf{R}$, $X(R)$ is the subset of $X(\mathbf{R})$ consisting of all points whose coordinates lie in R. One can show for example that the only points of $X(\mathbf{R})$ with rational coordinates are those on the x-axis, so that $X(\mathbf{Q}) = X(\mathbf{Z}) = \{(0, 0), (\pm 1, 0)\}$.

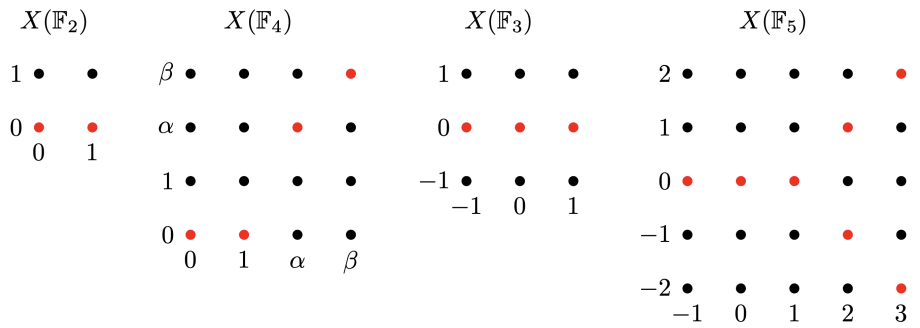
(c) $X(\mathbf{C})$ is a 1-dimensional complex submanifold of $\mathbf{C}^2$, which can be shown to be diffeomorphic to a punctured torus:



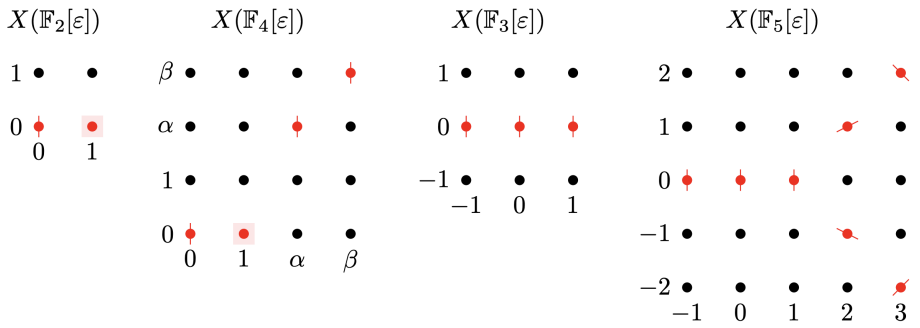
The two black circles indicates the intersection of $X(\mathbf{C})$ with $\mathbf{R}^2 \subseteq \mathbf{C}^2$, which is the set $X(\mathbf{R})$ from (a). Furthermore, $X(\mathbf{C})$ is biholomorphic to $(\mathbf{C}/\Lambda) \setminus \{0\}$, where Λ is the lattice $\mathbf{Z} \oplus \mathbf{Z}i$ (and $\mathbf{C} \setminus \Lambda$ means the quotient of abelian groups). This is an example of a noncompact *Riemann surface*.

(d) Let $\mathbf{R}[\varepsilon] = \mathbf{R}[x]/(x^2)$ be the ring of dual numbers over $\mathbf{R}$. A pair of udal numbers $(a + u\varepsilon, b + v\varepsilon)$ belongs to $X(\mathbf{R}[\varepsilon])$ if and only if $(a, b) \in X(\mathbf{R})$ and (u, v) is a tangent vector to $X(\mathbf{R})$ at the point (a, b) (see Remark 1.1 below). Thus, $X(\mathbf{R}[\varepsilon])$ is naturally identified with the *tangent bundle* of the 1-dimensional manifold $X(\mathbf{R})$. As a subspace of $\mathbf{R}[\varepsilon]^2 \simeq \mathbf{R}^4$, $X(\mathbf{R}[\varepsilon])$ is diffeomorphic to $(S^1 \times \mathbf{R}) \sqcup (\mathbf{R} \times \mathbf{R})$.

(e) We can consider solutions in finite fields:



(f) In analogy with (d), we can interpret $X(R[\varepsilon])$ as the set of tangent vectors to $X(R)$ for any ring R. Given $(a, b) \in X(R)$, the set of all (u, v) such that $(a + u\varepsilon, b + v\varepsilon)$ belongs to $X(R[\varepsilon])$ is an R-submodule of R^2, called the *tangent space* of X at (a, b). The tangent spaces over the first few finite fields are as follows:



For any field k, the tangent spaces at all points of $X(k)$ are 1-dimensional k-vector spaces, except for the point $(1, 0)$ in characteristic 2, where the tangent space is 2-dimensional. This reflects the fact that the point $(1, 0)$ is *singular* in characteristic 2, as it is a meeting point of the two branches $(1, 0)$ and $(-1, 0)$. The equation $y^2 = x^3 - x$ is said to have *bad reduction* at the prime 2, and *good reduction* at the other primes.

As the picture (c) over the complex numbers strong suggests, one should be able to *compactify* X by filling in the puncture. In the picture (a) over the real numbers, this corresponds to adding a point "at infinity" in the vertical direction that closes up the right-hand component. We will explain how to make sense of this compactified object in Section 1.3.

Remark 1.1 (Dual numbers and tangent vectors). Let us give some details on the claim in (d). Given a ring R, a polynomial $f \in R[x_1, \ldots, x_n]$, and an n-tuple of dual numbers $a + u\varepsilon \in R[\varepsilon]^n$, we have

$$f(a + u\varepsilon) = f(a) + \left(\frac{\partial f}{\partial x_1}(a) u_1 + \ldots + \frac{\partial f}{\partial x_n}(a) u_n \right) \varepsilon$$

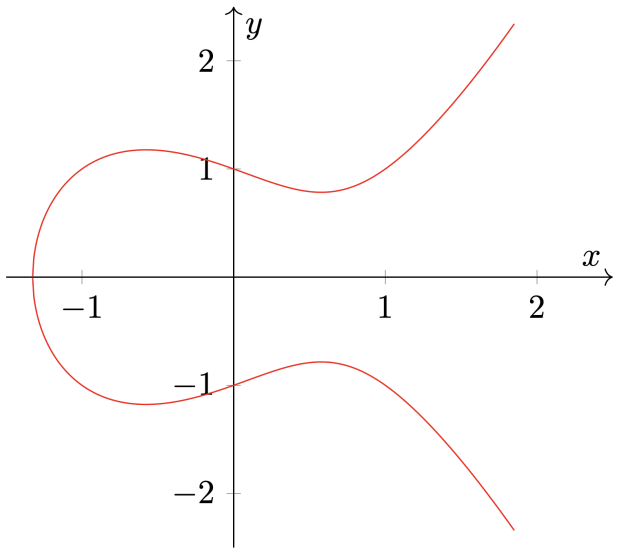
in $R[\varepsilon]$. This expression vanishes if and only if a is a zero of f and u is orthogonal to the gradient $\nabla f(a)$. If $R = \mathbf{R}$ and if this gradient is not zero, so that $f^{-1}(0)$ is a submanifold of $\mathbf{R}^n$ in a neighborhood of a, this precisely means that u is tangent to $f^{-1}(0)$ at a. On the other hand, if this gradient is zero, then every vector is orthogonal to it. Analogous statements hold for $R = \mathbf{C}$.

Consider now the slightly different equation $y^2 = x^3 - x + 1$ and associated solution sets

$$Y(R) = \{(a, b) \in R^2 \mid b^2 = a^3 - a + 1\}.$$

Despite the similarity to the previous equation $y^2 = x^3 - x$ defining X, Y turns out to be qualitatively quite different from X:

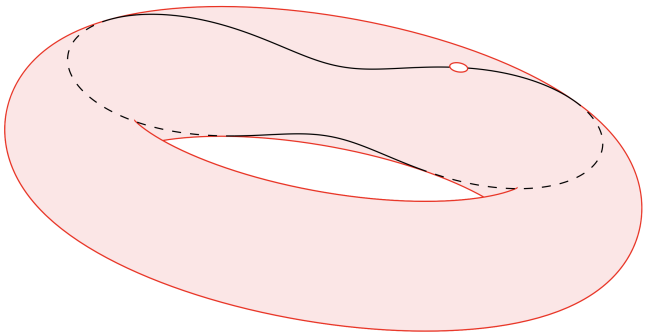
(g) The set $Y(\mathbf{R}) \subseteq \mathbf{R}^2$ now has a single component and is diffeomorphic to $\mathbf{R}$:



(h) The set $Y(\mathbf{Q})$ of rational solutions is infinite, and there are exactly 12 solutions in $\mathbf{Z}$:

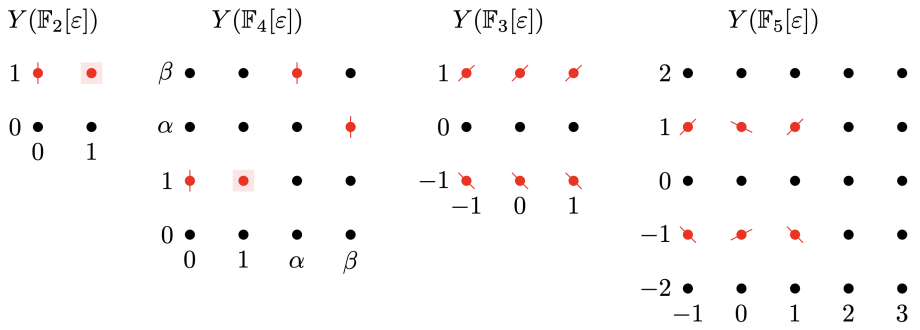
$$Y(\mathbf{Z}) = \{(-1, \pm 1), (0, \pm 1), (1, \pm 1), (3, \pm 5), (5, \pm 11), (56, \pm 419)\}.$$

(i) The set $Y(\mathbf{C}) \subseteq \mathbf{C}^2$ is again a punctured torus, but its intersection with $\mathbf{R}^2$ now has a single component:



It is biholomorphic to $(\mathbf{C}/\Lambda) \setminus \{0\}$ for some lattice $\Lambda = \mathbf{Z} \oplus \mathbf{Z}\tau$, where $\tau \approx \frac{1}{2} + 0.233i$. An exact expression for τ can be written using certain integrals called *elliptic integrals*, which also appear in the formula for the arc length of an ellipse; this is the origin of the term "elliptic curve". It turns out that $X(\mathbf{C})$ and $Y(\mathbf{C})$ are *not* biholomorphic, even though they are diffeomorphic.

(*j*) Here are the solutions over some finite fields together with their tangent spaces:

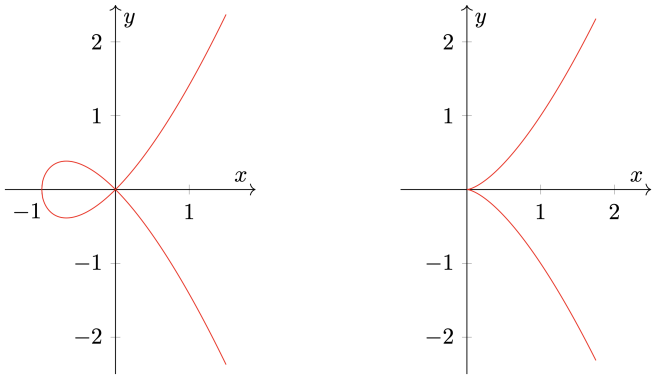


In addition to the prime 2 shown above, the equation $y^2 = x^3 - x + 1$ also has bad reduction at the prime 23, as $(0, 13)$ is a singular point of $Y(\mathbf{F}_{23})$. It has good reduction at all other primes.

1.2. **Singular cubic curves.** Consider the polynomial equations $y^2 = x^3 + x^2$ and $y^2 = x^3$. As in Section 1.1, they define functors $N, C\colon \mathrm{CAlg} \to \mathrm{Set}$ given by

$$N(R) = \{(a, b) \in R^2 \mid b^2 = a^3 + a^2\},$$
$$C(R) = \{(a, b) \in R^2 \mid b^2 = a^3\}.$$

We call N the *nodal cubic* and C the *cuspidal cubic*. Here is what $N(\mathbf{R})$ and $C(\mathbf{R})$ look like:



In contrast to the curves in Section 1.1, these curves have singularities and hence do not define submanifolds of $\mathbf{R}^2$. We can "confirm" the singular nature of the node and of the cusp by computing tangent spaces as explained in Remark 1.1. A vector (u, v) belongs to the tangent space at $(a, b) \in N(R)$ if and only if

$$2bv = (3a^2 + 2a)u.$$

For the point $(a, b) = 0$, this holds for all (u, v), so that the tangent space at $(0, 0)$ is 2-dimensional. Similarly, the equation for the tangent space at $(a, b) \in C(R)$ is

$$2bv = 3a^2u,$$

which always holds if $(a, b) = (0, 0)$. These equations also show that, for any field k, $(0, 0)$ is the only singular point in both $N(k)$ and $C(k)$.

Over the complex numbers, the nodal cubic $N(\mathbf{C}) \subseteq \mathbf{C}^2$ looks like a punctured *pinched torus*, i.e., a torus in which one of the circles bounding a hold is collapsed to a point; this is equivalently a punctured sphere in which two points have been identified. The cuspidal cubic $C(\mathbf{C}) \subseteq \mathbf{C}^2$ looks like a punctured sphere with a single thorn.

The two ways of visualizing $N(\mathbf{C})$ corresponds to two ways to algebraically "resolve" the nodal singularity of N at the origin:

(a) The "pinched torus" picture suggests viewing N as a degenerate member of a *family of curves* N_λ given (for example) by the equations $y^2 = x(x+\lambda)(x+1)$. We have $N_0 = N$ and, for nonzero values of λ, $N_\lambda(\mathbf{C})$ is a punctured torus with no singular points, similar to both examples in Section 1.1. Thus, we can think of N as the limit of the nonsingular curves N_λ as $\lambda \to 0$. In this situation, we say that N_λ with $\lambda \neq 0$ is a *deformation* of N.

(b) The "sphere with two points identified" picutre suggests viewing N as a *quotient* of a nonsingular curve $\tilde{N}$ such that $\tilde{N}(\mathbf{C})$ is a punctured sphere. One can achieve this algebraically by a change of variable: if we replace the coordinate y by the "slope" coordinate $s = y/x$, the equation $y^2 = x^3 + x^2$ becomes $s^2 = x + 1$. If $\tilde{N}\colon \mathrm{CAlg} \to \mathrm{Set}$ is the functor defined by the latter equation, i.e.,

$$\tilde{N}(R) = \{(a, c) \in R^2 \mid c^2 = a + 1\},$$

then there is map $\tilde{N} \to N$ sending (a, c) to (a, ac), and one can check that $\tilde{N}$ has no singular points. The curve $\tilde{N}$ is called the *blowup* of N at the origin. Over the real numbers, $\tilde{N}(\mathbf{R}) \to N(\mathbf{R})$ looks like the quotient map $\mathbf{R} \twoheadrightarrow \mathbf{R}/((-1) \sim 1)$.

While an arbitrary system of polynomial equations does not in general have a nonsingular deformation as in (a), it is a deep theorem that, if w work over a field of characteristic zero, it is always possible to resolve singularities by blowing up as in (b). Whether this is always possible in positive characteristic is a major open question in algebraic geometry.

1.3. **Compactifying affine curves.** The curves considered in Section 1.1 and Section 1.2 can be compactified by replacing the ambient 2-dimensional affine space $\mathbf{A}^2$ by the 2-dimensional projective space $\mathbf{P}^2$.

The *affine n-space* $\mathbf{A}^n$ is the functor $\mathrm{CAlg} \to \mathrm{Set}$ given by $\mathbf{A}^n(R) = R^n$. The *projective n-space* $\mathbf{P}^n$ is a functor $\mathrm{CAlg} \to \mathrm{Set}$ containing $\mathbf{A}^n$; for simplicity, we shall only define it here on certain subcategory of CAlg (the general definition will be given in Section 3). Let $(\mathbf{A}^n \setminus \{0\})(R) \subseteq R^n$ be the set of n-tuples that generate the unit ideal of R. The group of units $R^\times$ acts on the set R^n by scalar multiplication, and this action preserves the subset $(\mathbf{A}^n \setminus \{0\})(R)$. If R is a local ring or a principal ideal domain, we define

$$\mathbf{P}^n(R) = (\mathbf{A}^{n+1} \setminus \{0\})(R)/R^\times$$

(for arbitrary rings R, the right-hand side is only a subset of the left hand side). We write

$$[a_0 : \ldots : a_n] \in \mathbf{P}^n(R)$$

for the equivalence class of $(a_0, \ldots, a_n) \in (\mathbf{A}^{n+1} \setminus \{0\})(R)$. We can identify $\mathbf{A}^n(R) = R^n$ with the subset of $\mathbf{P}^n(R)$ where a_0 is a unit:

$$\mathbf{A}^n(R) \hookrightarrow \mathbf{P}^n(R), \quad (a_1, \ldots, a_n) \mapsto [1 : a_1 : \ldots : a_n].$$

If k is a field, then $\mathbf{P}^n(k)$ is obtained from $\mathbf{A}^n(k)$ by adding a "point at infinity" $[0 : a_1 : \ldots : a_n]$ in the direction of every nonzero vector $(a_1, \ldots, a_n)$; these points at infinity form a copy of $\mathbf{P}^{n-1}(k)$. Inductively, we therefore obtain a decomposition

$$\mathbf{P}^n(k) = k^n \sqcup k^{n-1} \sqcup \ldots \sqcup k^0.$$

A polynomial is called *homogeneous of degree d* if it is a linear combination of monomials of degree exactly d. For example, $x^3 + x^2y - 2xz^2 \in \mathbf{Z}[x, y, z]$ is homogeneous of degree 3. If $f \in R[x_0, \ldots, x_n]$ is homogeneous of degree d and $(a_0, \ldots, a_n) \in R^{n+1}$, then

$$f(\lambda a_0, \ldots, \lambda a_n) = \lambda^d f(a_0, \ldots, a_n)$$

for any $\lambda \in R^\times$. Because of this, the statement "f vanishes at $(a_0, \ldots, a_n)$" depends only on the point $[a_0 : \ldots : a_n]$ in $\mathbf{P}^n(R)$. In other words, a *homogeneous* polynomial equation in $n + 1$ variables has a well-defined solution set in projective n-space.

Let us now return to the equation $y^2 = x^3 - x$ from Section 1.1 defining the subfunctor $X \subseteq \mathbf{A}^2$. We can *homogenize* this equation by introducing a new variable w and multiplying each term by the minimal power of

w so that the equation becomes homogeneous: this yields the degree 3 homogeneous equation $wy^2 = x^3 - w^2x$. Let $\bar{X} \subseteq \mathbf{P}^2$ be the subfunctor of solutions to this equation, given by

$$\bar{X}(R) = \{[s : a : b] \in \mathbf{P}^2(R) \mid sb^2 = a^3 - sa\}$$

(this makes sense for an arbitrary ring R, although we have only defined it so far when R is a local ring or a principal ideal domain). If we set $s = 1$, we recover precisely the subset $X(R)$ of $\mathbf{A}^2(R) \subseteq \mathbf{P}^2(R)$:

$$X = \bar{X} \cap \mathbf{A}^2.$$

On the other hand, we can find the solutions "at infinity" by setting $s = 0$, which yields the equation $a^3 = 0$. If k is a field, we see that the only point of $\bar{X}(k)$ with $s = 0$ is $[0 : 1 : 1]$, which is the point at infinity in the vertical direction $(0, 1)$:

$$\bar{X}(k) = X(k) \sqcup \{[0 : 0 : 1]\} \subseteq \mathbf{P}^2(k).$$

As we surmisses in Section 1.1, $\bar{X}(\mathbf{R})$ is a submanifold of the real projective plane $\mathbf{P}^2(\mathbf{R})$ diffeomorphic to $S^1 \sqcup S^1$, while $\bar{X}(\mathbf{C})$ is a complex submanifold of the complex projective plane $\mathbf{P}^2(\mathbf{C})$, which is diffeomorphic to a torus $S^1 \times S^1$ and biholomorphic to $\mathbf{C}/(\mathbf{Z} \oplus \mathbf{Z}i)$.

One can still define tangent spaces via dual numbers. Let us work out the tangent space at the point at infinity $[0 : 0 : 1] \in \bar{X}(R)$ for a local ring R. Note that $R[\varepsilon]$ is again local and $R[\varepsilon]^\times = R^\times + R\varepsilon$. Given $[s : a : b] \in \bar{X}(R)$, a point $[s + t\varepsilon : a + u\varepsilon : b + v\varepsilon]$ belongs to $\bar{X}(R[\varepsilon])$ if and only if

$$(b^2 + 2sa)t + (-3a^2 + s^2)u + 2sbv = 0.$$

If $[s : a : b] = [0 : 0 : 1]$, this reduces to $t = 0$, so that tangent vectors at infinity have the form $[0 : u\varepsilon : 1 + v\varepsilon] = [0 : u\varepsilon : 1]$. Thus, we see that $u \mapsto [0 : 0 : 1]$ is a bijection from R to the tangent space at $[0 : 0 : 1]$. Since this is a free R-module of rank 1, the point $[0 : 0 : 1]$ is nonsingular.

Similarly, the affine curve $Y \subseteq \mathbf{A}^2$ defined by the equation $y^2 = x^3 - x + 1$ is compactified to the projective curve $\bar{Y} \subseteq \mathbf{P}^2$ given by

$$\bar{Y}(R) = \{[s : a : b] \in \mathbf{P}^2(R) \mid sb^2 = a^3 - s^2a + s^3\}.$$

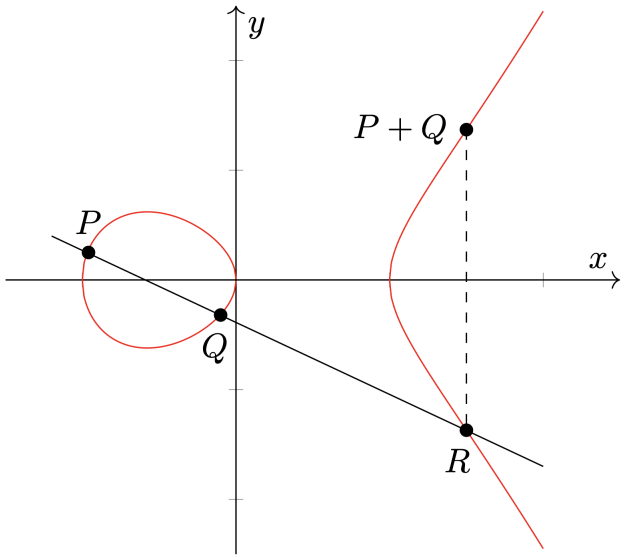
For a field k, we have $\bar{Y}(k) = Y(k) \sqcup \{[0 : 0 : 1]\}$ as before. The real solution set $\bar{Y}(\mathbf{R})$ is now diffeomorphic to a circle, while $\bar{Y}(\mathbf{C})$ is diffeomorphic to a torus but not biholomorphic. Finally, the nodal cubic N and cuspical cubic C from Section 1.2 also have compactifications $\bar{N}, \bar{C} \subseteq \mathbf{P}^2$, which over a field add the single point at infinity $[0 : 0 : 1]$. Over the complex numbers, this has the effect of filling in the picture.

Away from the primes of bad reduction (which means: when restricted to rings in which these primes are invertible), $\bar{X}$ and $\bar{Y}$ are examples of *elliptic curves*. A remarkable fact is that elliptic curves have an essentially unique group structure. More precisely, if we choose a point e in $\bar{X}(\mathbf{Z}[\frac{1}{2}])$ to serve as the unit element, there is a unique lift

$$\begin{array}{ccc} \mathrm{CAlg}_{\mathbf{Z}[\frac{1}{2}]} & \xrightarrow{(\bar{X},e)} & \mathrm{Set}_* \\ & \overset{\exists!}{\searrow} & \uparrow \mathrm{fgt} \\ & & \mathrm{Grp}\,. \end{array}$$

Moreover, this lift lands in the subcategory $\mathrm{Ab} \subseteq \mathrm{Grp}$ of abelian groups. Over the complex numbers, the group structure on $\bar{X}(\mathbf{C})$ is that of the quotient $\mathbf{C}/\Lambda$ (with unit 0). In general, if we tak the unit to be the point at infinity, then the group law is determined by the requirement that $P + Q + R = 0$ whenever P, Q and R lie on a line (if $P = Q$, this means that R lies on the tangent line at P, and if P is the point at infinity, this means that Q

and R lie on the same vertical line). Here is an illustration over the real numbers:



The same discussion applies to the elliptic curve $\bar{Y}\colon \mathrm{CAlg}_{\mathbf{Z}[\frac{1}{46}]} \to \mathrm{Set}$ (recall that Y has bad reduction at the prime 2 and 23, and we have to invert both to get an elliptic curve).

1.4. **Fermat curves.** Let $n \in \mathbf{N}$ and consider the homogeneous equation $x^n = y^n + z^n$. Let $X_n(R)$ be its set of solutions in the projective plane $\mathbf{P}^2(R)$ (which we have only defined so far when R is a local ring or principal ideal domain):

$$X_n(R) = \{[a : b : c] \in \mathbf{P}^2(R) \mid a^n = b^n + c^n\}.$$

(a) For $n = 0$, the equation is $1 = 1 + 1$, which holds in a ring R if and only if R is the zero ring. In other words, $X_0(R) = \varnothing$ if $R \neq 0$ and $X_0(0) = \{0\}$. The functor $X_0(R)$ is the so called *empty scheme*.

(b) For $n = 1$, the equation $x = y + z$ cuts out a line in $\mathbf{P}^2$, which is isomorphic to $\mathbf{P}^1$. Indeed, there is a natural bijection

$$\mathbf{P}^1 \xrightarrow{\sim} X_1, \quad [a : b] \mapsto [a + b : a : b].$$

The real projective line $\mathbf{P}^1(\mathbf{R})$ is diffeomorphic to a circle, while the complex projective line $\mathbf{P}^1(\mathbf{C})$, also called the *Riemann sphere*, is diffeomorphic to a sphere.

(c) For $n = 2$, we have the equation $x^2 = y^2 + z^2$. The solutions to this equation in $\mathbf{Z}^3$ are the so-called *Pythagoren triples*. It turns out that these can be explicitly determined, as we now explain. Call a Pythagorean truple (a, b, c) *primitive*, if $a > 0$ and $(a, b, c) = \mathbf{Z}$. Every Pythagorean triple has the form $n(a, b, c)$ for an integer n and a primitive Pythagorean triple (a, b, c) (which are uniquely determined if $n \neq 0$). Sending (a, b, c) to $[a : b : c]$ defines a bijection between the set of primitive Pythagorean triples and $X_2(\mathbf{Z})$. Furthermore, the inclusion $\mathbf{Z} \to \mathbf{Q}$ induces a bijection $X_2(\mathbf{Z}) \xrightarrow{\sim} X_2(\mathbf{Q})$, with inverse given by clearing denominators (this is true for X_n for all n and reflects the fact that the scheme X_n is *proper*).

Now, for any local $\mathbf{Z}[\frac{1}{2}]$-algebra R (in fact any $\mathbf{Z}[\frac{1}{2}]$-algebra), the map

$$\sigma\colon \mathbf{P}^1(R) \to X_2(R), \quad [a : b] \mapsto [a^2 + b^2 : a^2 - b^2 : 2ab]$$

is well-defined and bijective. The inverse is given by

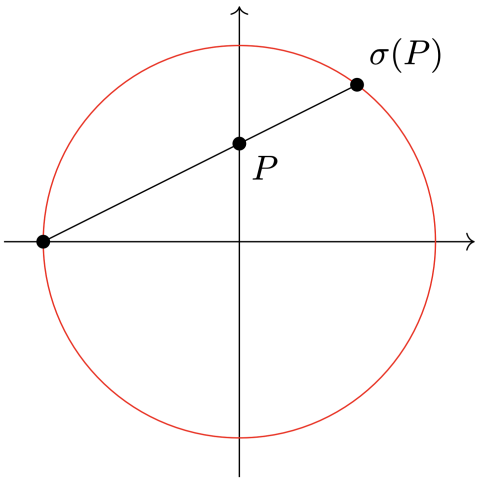
$$[u : v : w] \mapsto \begin{cases} [u + v : w] & u + v \in R^\times, \\ [w : u - v] & u - v \in R^\times. \end{cases}$$

In particular, for $R = \mathbf{Q}$, σ gives a complete enumeration of all primitive Pythagorean triples, which is know as *Euclid's formula*.

The map $\sigma\colon \mathbf{P}^1(k) \to X_2(k)$ has a concrete geometric interpretation when k is a subfield of $\mathbf{R}$. If $[a : b : c] \in X_2(k)$, then a cannot be zero as then b and c would also be zero (note that this is not true for $k = \mathbf{C}$). Hence

$$X_2(k) = \{[1 : b : c] \in \mathbf{P}^2(k) \mid b^2 + c^2 = 1\} = |(b, c) \in \mathbf{A}^2(k) \mid b^2 + c^2 = 1.$$

In other words, $X_2(k)$ is the set of points on the unit circle in $\mathbf{R}^2$ with coordinates in k. The map σ is then the inverse stereographic projection from the point $(-1, 0)$ on the unit circle:



(d) For $n \geq 3$, Fermat's Last Theorem states that

$$X_n(\mathbf{Q}) = X_n(\mathbf{Z}) = \begin{cases} \{[0 : 1 : -1], [1 : 0 : 1], [1 : 1 : 0]\} & \text{if } n \text{ is odd,} \\ \{[1 : 0 : \pm 1], [1 : \pm 1 : 0]\} & \text{if } n \text{ is even.} \end{cases}$$

Away from the prime 3, $X_3\colon \mathrm{CAlg}_{\mathbf{Z}[\frac{1}{2}]} \to \mathrm{Set}$ is another example of an elliptic curve, which has a unique group structure with unit element $[0 : 1 : -1]$.

1.5. **Affine, projective, and general schemes.** In Section 1.1 and Section 1.2, we say examples of *affine schemes*, and in Section 1.3 and Section 1.4, we saw examples of *projective schemes*. The former are the solutions in affine space of systems of polynomial equations, while the latter are the solutions in projective space of systems of homogeneous polynomial equations. If we allow *inequations* in addition to equations, we obtain the notions of *quasi-affine* and *quasi-projective* schemes. We will study affine schemes in Section 2 and projective schemes in Section 3. All of these objects are examples of *schemes*, which we will define in Section 7. The following diagram summarizes the situation:

$$\begin{array}{ccccc} \{\text{affine schemes}\} & \subseteq & \{\text{quasi-affine schemes}\} & & \\ & & \cap\!\!| & & \\ \{\text{projective schemes}\} & \subseteq & \{\text{quasi-projective schemes}\} & \subseteq & \{\text{schemes}\}. \end{array}$$

(Strictly speaking, quasi-projective schemes are defined using polynomials in finitely many variables; for the vertical inclusion to hold, on should either remove this finiteness condition on the quasi-projective side or add it on the quasi-affine side.)

(a) An example of a quasi-affine scheme that is not affine is the *punctured affine n-space* $\mathbf{A}^n \setminus \{0\}$, for $n \geq 2$, which is defined as

$$\mathbf{A}^n \setminus \{0\}\colon \mathrm{CAlg} \to \mathrm{Set}, \quad R \mapsto \{a \in R^n \mid (a) = R\}.$$

As the notation suggests, this is in a precise sense the complement of 0 in $\mathbf{A}^n$, where 0 is the joint vanishing locus of the coordinate functions $x_1, \ldots, x_n$ on $\mathbf{A}^n$.

(b) An example of a quasi-projective scheme that is neither projective nor quasi-affine is the *punctured projective n-space* $\mathbf{P}^n \setminus \{0\}$ for $n \geq 2$. This is again the complement of 0 in $\mathbf{P}^n$, where 0 is the vanishing locus of the projective coordinates $x_1, \ldots, x_n$ on $\mathbf{P}^n$. For a local ring R, we have

$$(\mathbf{P}^n \setminus \{0\})(R) = \{[a_0 : \ldots : a_n] \in \mathbf{P}^n(R) \mid (a_1, \ldots, a_n) = R\}.$$

(c) Schemes that are not quasi-projective are more difficult to come by, but they can be constructed by explicitly gluing affine schemes. The simplest example is the *affine line with doubled origin*, which is the functor

$$\mathrm{CAlg} \to \mathrm{Set}, \quad R \mapsto \{(f, e) \mid f \in R, e \in R/(f), \text{ and } e^2 = e\}.$$

4

2. Affine geometry

2.1. **Affine spaces.** Affine geometry studies the solutions in affine spaces to systems of polynomial equations, while projective geometry studies the solutions in projective spaces to systems of homogeneous polynomial equations. In both cases, the solutions form a functor $\mathrm{CAlg}_k \to \mathrm{Set}$ from the category of k-algebras to the category of sets. Such functors are the basic objects of algebraic geometry:

Definition 2.1 (Algebraic functor)**.** Let k be a ring. An *algebraic k-functor* is a functor $\mathrm{CAlg}_k \to \mathrm{Set}$. An algebraic $\mathbf{Z}$-functor is simply called an *algebraic functor*. Given an algebraic k-functor X and a k-algebra R, the elements of $X(R)$ are called the *R-valued* points or *R-points* of X.

A basic example is given by affine spaces:

Definition 2.2 (Affine space)**.** Let I be a set. The *affine I-space* over k is the algebraic k-functor

$$\mathbf{A}_k^I \colon \mathrm{CAlg}_k \to \mathrm{Set}, \quad R \mapsto R^I$$

We simply write $\mathbf{A}^I$ when $k = \mathbf{Z}$. For $n \geq 0$, the *affine n-space* over k is $\mathbf{A}_k^n = \mathbf{A}_k^{\{1,\dots,n\}}$. It is also called the *affine line*, if $n = 1$ and the *affine plane*, if $n = 2$.

Remark 2.3. The following are some basics properties of affine spaces:

(a) $\mathbf{A}_k^0$ is a final object $*$ of $\mathrm{Fun}(\mathrm{CAlg}_k, \mathrm{Set})$.
(b) $\mathbf{A}_k^1$ is isomorphic to the forgetful functor $\mathrm{CAlg}_k \to \mathrm{Set}$.
(c) $\mathbf{A}_k^I$ is contravariantly functorial in the set I: a map $f \colon J \to I$ induces a natural transformation $\mathbf{A}_k^I \to \mathbf{A}_k^J$ given by precomposition with f.
(d) By the universal property of polynomial rings, the functor $\mathbf{A}_k^I$ is represented by the polynomial k-algebra $k[x_i \mid i \in I]$, i.e., there is an natural isomorphism of functors

$$\mathbf{A}_k^I \simeq \mathrm{Map}(k[x_i \mid i \in I], -) \colon \mathrm{CAlg}_k \to \mathrm{Set}\,.$$

Indeed, given an I-tuple $(r_i)_{i \in I} \in R^I$, there is a unique k-algebra map $k[x_i \mid i \in I] \to R$ sending x_i to r_i.

2.2. **Presheaves.** We start with some categorical preliminaries on set-valued functors, also known as *presheaves*.

Definition 2.4 (Presheaves)**.** Let $\mathcal{C}$ be a category. A *presheaf* on $\mathcal{C}$ is a functor $\mathcal{C}^{\mathrm{op}} \to \mathrm{Set}$. We denote by

$$\mathrm{P}(\mathcal{C}) = \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathrm{Set})$$

the category of presheaves on $\mathcal{C}$. More generally, given an arbitrary category $\mathcal{E}$, an *$\mathcal{E}$-valued presheaf* on $\mathcal{C}$ is a functor $\mathcal{C}^{\mathrm{op}} \to \mathcal{E}$.

For example, an algebraic k-functor is exactly a presheaf on $\mathrm{CAlg}_k^{\mathrm{op}}$.

Remark 2.5. The category $\mathrm{P}(\mathcal{C})$ always admits limits and colimits, which are computed *pointwise* in the category of sets. Many properties of the category of sets are thereby inherited by the category of presheaves $\mathrm{P}(\mathcal{C})$, such as the fact that filtered colimits commutes with finite limits, the fact the monomorphism and epimorphisms are effective, etc.

Definition 2.6 (Yoneda embedding)**.** Let $\mathcal{C}$ be a category. The *Yoneda embedding* of $\mathcal{C}$ is the functor

$$\text{ｙ} \colon \mathcal{C} \to \mathrm{P}(\mathcal{C}), \quad \text{ｙ}(X) = \mathrm{Map}(-, X) \colon \mathcal{C}^{\mathrm{op}} \to \mathrm{Set}\,.$$

A presheaf on $\mathcal{C}$ is called *representable*, if it lies in the essential image of ｙ.

Notation 2.7 (The functor Spec)**.** When $\mathcal{C} = \mathrm{CAlg}^{\mathrm{op}}$, the Yoneda embedding is denoted by

$$\mathrm{Spec} \colon \mathrm{CAlg}_k^{\mathrm{op}} \to \mathrm{P}(\mathrm{CAlg}_k^{\mathrm{op}}) = \mathrm{Fun}(\mathrm{CAlg}_k, \mathrm{Set}).$$

Thus $\mathrm{Spec}(A)$ is the algebraic k-functor represented by the k-algebra A:

$$\mathrm{Spec}(A)(R) = \mathrm{Map}(A, R).$$

For example, $\mathbf{A}^I_k \simeq \mathrm{Spec}(k[x_i \mid i \in I])$.

Definition 2.8 (Category of elements)**.** Let $\mathcal{C}$ be a category and let $F \in \mathrm{P}(\mathcal{C})$ be a presheaf on $\mathcal{C}$. The *category of elements* $\mathrm{El}(F)$ of F is defined by the cartesian square of categories

$$\begin{array}{ccc} \mathrm{El}(F) & \longrightarrow & (\mathrm{Set}_*)^{\mathrm{op}} \\ \downarrow & & \downarrow{\scriptstyle \text{forget}} \\ \mathcal{C} & \xrightarrow{F} & \mathrm{Set}^{\mathrm{op}} \end{array}$$

It is also denoted by $\int F$. Explicitly, objects of $\mathrm{El}(F)$ are pairs (X, x) with $X \in \mathcal{C}$ and $x \in F(X)$, and morphisms $(X, x) \to (Y, y)$ are morphisms $f\colon X \to Y$ in $\mathcal{C}$ such that $f^*(y) = x$.

Theorem 2.9 (Properties of Yoneda embedding)**.** *Let $\mathcal{C}$ be a small category.*

(a) (The Yoneda lemma) Let $X \in \mathcal{C}$ and $F \in \mathrm{P}(\mathcal{C})$. Then the map

$$\mathrm{Map}(\text{ɤ}(X), F) \to F(X), \quad f \mapsto f(\mathrm{id}_X),$$

is a bijection with inverse $x \mapsto ((f\colon Y \to X) \mapsto f^*(x))$.

(b) The Yoneda embedding ɤ$\colon \mathcal{C} \to \mathrm{P}(\mathcal{C})$ is fully faithful.

(c) The Yoneda embedding ɤ$\colon \mathcal{C} \to \mathrm{P}(\mathcal{C})$ preserves all limits that exists in $\mathcal{C}$.

(d) Every presheaf $F \in \mathrm{P}(\mathcal{C})$ is canonically a colimit of representable presheaves:

$$\mathrm{colim}(\mathrm{El}(F) \xrightarrow{\mathrm{fgt}} \mathcal{C} \xrightarrow{\text{ɤ}} \mathrm{P}(\mathcal{C})) \xrightarrow{\sim} F.$$

(e) (Universal property of ɤ) Let $\mathcal{E}$ be a cocomplete category. Then the functor

$$\text{ɤ}^*\colon \mathrm{Fun}^{\mathrm{colim}}(\mathrm{P}(\mathcal{C}), \mathcal{E}) \to \mathrm{Fun}(\mathcal{C}, \mathcal{E}),$$

is an equivalence of categories, where $\mathrm{Fun}^{\mathrm{colim}}$ denotes the category of colimit-preserving functors; the inverse is given by left Kan extension along ɤ. In particular, any functor $\mathcal{C} \to \mathcal{E}$ extends uniquely (up to unique isomorphism) to a colimit-preserving functor $\mathrm{P}(\mathcal{C}) \to \mathcal{E}$.

(f) If $\mathcal{E}$ is any category, then any colimit-preserving functor $K\colon \mathrm{P}(\mathcal{C}) \to \mathcal{E}$ has a right adjoint $\mathcal{E} \to \mathrm{P}(\mathcal{C})$ given by $e \mapsto \mathrm{Map}(K(\text{ɤ}(-)), e)$.

Proof. We will only give a proof sketch. For more details, we refer to [Rie16].

(a) We will construct an inverse. Let $x \in F(X)$. For any $Y \in \mathcal{C}$ we define a morphism by the rule

$$\text{ɤ}(X)(Y) \to F(Y), \quad g \mapsto F(g)(x) =: g^*(x).$$

These maps assemble into a natural transformation. Indeed, naturality amounts to that, for each map $\alpha\colon Y' \to Y$, the following diagram commutes:

$$\begin{array}{ccc} \text{ɤ}(X)(Y) & \longrightarrow & F(Y) \\ {\scriptstyle \alpha^*}\downarrow & & \downarrow{\scriptstyle \alpha^*} \\ \text{ɤ}(X)(Y') & \longrightarrow & F(Y'). \end{array}$$

It is left to the reader to check that this assertion holds true, and that the map, sending an element $x \in F(x)$ to the natural transformation defined in this way really is an inverse to the evaluation at the identity.

(b) One can check that the following diagram commutes

$$\begin{array}{ccc} \mathrm{Map}_{\mathcal{C}}(X, Y) & \xrightarrow{\text{ɤ}} & \mathrm{Map}(\text{ɤ}(X), \text{ɤ}(Y)) \\ & {\scriptstyle \mathrm{id}}\searrow & \downarrow{\scriptstyle \simeq} \\ & & \mathrm{Map}_{\mathcal{C}}(X, Y), \end{array}$$

where the right isomorphism follows from Yoneda lemma, see part a).

(c) This follows from the fact that $\mathrm{Map}(-,-)$ preserves limits in both variables.

(d) By the Yoneda lemma, an element $(X,x) \in \mathrm{El}(F)$ corresponds to a map $x\colon \text{よ}(X) \to F$, and a map $(X,x) \to (Y,y)$ correspond to a commutative diagram

$$\begin{array}{ccc} \text{よ}(X) & \xrightarrow{f_*} & \text{よ}(Y). \\ & \searrow^{x} & \downarrow{\scriptstyle y} \\ & & F \end{array}$$

This induces a map $\mathrm{colim}_{(X,x)\in\mathrm{El}(F)} \text{よ}(X) \to F$. We want to show that for each $Y \in \mathcal{C}$, the map

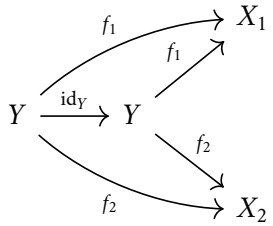
$$\beta\colon (\operatorname*{colim}_{(X,x)} \mathrm{Map}(Y,X)) \to F(Y)$$

is bijective. The left hand side consists of equivalence classes of elements of the form (x,f), where $f\colon Y \to X$ and $x \in F(X)$, where $(x',f') \sim (x,f)$ if and only there exists a commutative diagram of the form

$$\begin{array}{ccc} Y & \xrightarrow{f} & X, \\ & \searrow^{f'} & \downarrow{\scriptstyle \alpha} \\ & & X' \end{array}$$

such that $\alpha^*(x) = x'$. The map β is then explicitly given by sending an equivalence class $[(x,f)]$ to the element $f^*x \in F(Y)$

Since $y \in F(y)$ is the image of (y, id_Y), the map β is surjective. For injectivity, assume that we have (x_1,f_1) and (x_2,f_2), where $f_1\colon Y \to X_1$ and $f_2\colon Y \to X_2$ and $x_1 \in F(X_1)$, $x_2 \in F(X_2)$ such that $F(f_1)(x_1) = F(f_2)(x_2) = y \in F(Y)$. Then the commutative diagram



indicates that $(x_1,f_1) \sim (y,\mathrm{id}_Y) \sim (x_2,f_2)$.

(e) For essentially surjectivity, let $F \in \mathrm{Fun}(\mathcal{C},\mathcal{E})$ be a functor, we define $\hat{F}\colon \mathrm{P}(\mathcal{C}) \to \mathcal{E}$ by

$$\hat{F}(H) = \operatorname*{colim}_{(X,x)\in\mathrm{El}(H)} F(X).$$

We need to show that $\hat{F}(\text{よ}(Y)) \simeq F(Y)$. However, by the above

$$\hat{F}(\text{よ}(Y)) \simeq \operatorname*{colim}_{(X,x)\in\mathrm{El}(\text{よ}(Y))} F(X).$$

We have $\mathrm{El}(\text{よ}(Y)) \simeq \mathcal{C}_{/Y}$, given by sending a tuple $(X, f \in \text{よ}(Y)(X))$ to the morphism $f : X \to Y$, and a morphism $\varphi : (X,f) \to (X',f')$ to the commutative triangle $f = f' \circ \alpha$. Essential surjectivity is clear, while fully faithfulness follows directly from the fully faithfulness of the Yoneda embedding, see part b). Now, $\mathcal{C}_{/Y}$ admits a final object, given by (Y,id_Y). Therefore, the colimit in the right hand side is isomorphic to $F(Y)$.

For fully faithfulness, let $\hat{F},\hat{G}\colon \mathrm{P}(\mathcal{C}) \to \mathcal{E}$ be colimit preserving functors. Furthermore, let $F = \hat{F} \circ \text{よ}$ and $G = \hat{G} \circ \text{よ}$. We need to show that there is an isomorphism

$$\mathrm{Map}(\hat{F},\hat{G}) \xrightarrow{\sim} \mathrm{Map}(F,G).$$

Let $\varphi : F \to G$ be a natural transformation, and let $H \in \mathrm{P}(\mathcal{C})$ be a presheaf, then $H \simeq \mathrm{colim}_{(X,x)\in \mathrm{El}(H)} \text{よ}(X)$. Since $\hat{F}$ preserves colimits, we have an isomorphism $\hat{F}(H) \simeq \mathrm{colim}_{(X,x)\in\mathrm{El}(H)} F(X)$. Therefore, we have the following commutative diagram induced by the naturality of φ

$$\begin{array}{ccc} \hat{F}(H) & \xrightarrow{\exists! \hat{\varphi}_H} & \hat{G}(H) \\ \downarrow{\scriptstyle \simeq} & & \downarrow{\scriptstyle \simeq} \\ \mathrm{colim}_{(X,x)\in\mathrm{El}(H)} F(X) & \xrightarrow{\varphi} & \mathrm{colim}_{(X,x)\in\mathrm{El}(H)} F(X), \end{array}$$

There exists a unique map $\hat{\varphi}_H$ filling the diagram. Varying H, we obtain that the $\hat{\varphi}$ assemble into a natural transformation $\hat{F} \to \hat{G}$, where we implicitly use that the colimit is natural in the diagram, and the constriction $\mathrm{El}(F)$ is natural in F. This proves the claim.

(f) It suffices to show that, for each $e \in \mathcal{E}$ and $H \in \mathrm{P}(\mathcal{C})$, there exists a natural isomorphism

$$\mathrm{Map}_{\mathcal{E}}(K(H), e) \simeq \mathrm{Map}_{\mathrm{P}(\mathcal{C})}(H, \mathrm{Map}(K(\text{よ}(-)), e)).$$

Since K preserves colimits and H can be written as a colimit of representable presheaves, we may assume that H is representable. In this case, the right hand side is isomorphic to the left hand side, by Yoneda lemma.

□

Remark 2.10. (a) By the Yoneda lemma, the category of elements of a presheaf $F \in \mathrm{P}(\mathcal{C})$ can be equivalently be described as the pullback

$$\begin{array}{ccc} \mathrm{El}(F) & \longrightarrow & \mathrm{P}(\mathcal{C})_{/F} \\ \downarrow & & \downarrow{\scriptstyle \text{forgetful}} \\ \mathcal{C} & \xrightarrow{\text{よ}} & \mathrm{P}(\mathcal{C}), \end{array}$$

whose objects are pairs (X, x) with $X \in \mathcal{C}$ and $x \colon \text{よ}(X) \to F$. Indeed, the equivalence between the two pullbacks is explicitly given by sending tuple $(X, x \in F(X))$ to the tuple $(X, x : \text{よ}(X) \to F)$, which clearly admits an inverse by the Yoneda lemma.

Theorem 2.9 then says that every presheaf is the colimit of all representable presheaves mapping to it.

(b) As seen in the proof of Theorem 2.9, we have $\mathrm{El}(\text{よ}(X)) \simeq \mathcal{C}_{/X}$. Together with the statement of Theorem 2.9, this shows that a presheaf is representable if and only if its category of elements has a final object (which is then the representing object).

Corollary 2.11. *Let $F, G \in \mathrm{P}(\mathcal{C})$ be presheaves. Then*

$$\mathrm{Map}(F, G) = \lim_{(X,x)\in\mathrm{El}(F)} G(X).$$

Proof. By Theorem 2.9, we may write F as a colimit of representable presheaves. Then the statement follows from that $\mathrm{Map}(-,-)$ preserves limits in both variables. □

Example 2.12. By Theorem 2.9, the functor

$$\mathrm{Spec} \colon \mathrm{CAlg}_k^{\mathrm{op}} \to \mathrm{Fun}(\mathrm{CAlg}_k, \mathrm{Set})$$

preserves all limits. Moreover, limits in $\mathrm{CAlg}_k^{\mathrm{op}}$ are colimits in CAlg_k. For example k is the initial object of CAlg_k, so that $\mathrm{Spec}(k)$ is the final object of $\mathrm{Fun}(\mathrm{CAlg}_k, \mathrm{Set})$. The coproduct of two k-algebras A and B is the tensor product $A \otimes_k B$, so that

$$\mathrm{Spec}(A \otimes_k B) \simeq \mathrm{Spec}(A) \times \mathrm{Spec}(B)$$

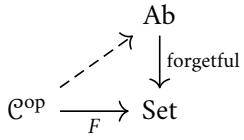
in $\mathrm{Fun}(\mathrm{CAlg}_k, \mathrm{Set})$. More generally, the pushout of a diagram $A \leftarrow C \to B$ in CAlg_k is the relative tensor product $A \otimes_C B$, so that

$$\mathrm{Spec}(A \otimes_k B) \simeq \mathrm{Spec}(A) \times_{\mathrm{Spec}(C)} \mathrm{Spec}(B).$$

Remark 2.13 (Algebraic structures on presheaves). Algebraic objects like monoids, groups, abelian groups, rings, modules over a ring, etc. make sense in any category with finite products. In categories of presheaves, since finite products are computed objectwise, algebraic objects are the same as presheaves valued in the category of algebraic object of the same type in Set. For example, abelian group objects in presheaves are the same as presheaves valued in abelian groups:

$$\mathrm{Ab}(\mathrm{P}(\mathcal{C})) \simeq \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathrm{Ab}).$$

Thus, given a presheaf $F \in \mathrm{P}(\mathcal{C})$, equipping F with an abelian group structure is equivalent to lifting F along the forgetful functor $\mathrm{Ab} \to \mathrm{Set}$:



2.3. **Polynomial equations.**

Definition 2.14 (System of polynomial equations)**.** Let k be a ring and let I and J be sets. A *system of J polynomial equations in I variables* over k is a J-tuple $\Sigma = (f_j)_{j\in J}$ in the polynomial ring $k[x_i \mid i \in I]$. We denote by (Σ) the ideal in $k[x_i \mid i \in I]$ generated by $(f_j)_{j\in J}$ and by $k[\Sigma]$ the k-algebra $k[x_i \mid i \in I]/(\Sigma)$.

Remark 2.15. Every k-algebra R is isomorphic to $k[\Sigma]$ for some system of polynomial equations Σ. A choice of isomorphism $R \simeq k[\Sigma]$ is exactly a *presentation* of R by generators and relations.

Given a system of polynomial equations over k, we can consider its solutions in any k-algebra. To that end, recall that there is, for any k-algebra R, an *evaluation map*

$$k[x_i \mid i \in I] \times R^I \to R, \quad (f, a) \mapsto f(a),$$

which is defined as follows: for each $a \in R^I, f \mapsto f(a)$ is the unique k-algebra map $k[x_i \mid i \in I] \to R$ sending x_i to a_i.

Definition 2.16 (Vanishing locus)**.** Let $F \subseteq k[x_i \mid i \in I]$ be a subset. The *vanishing locus* of F in $\mathbf{A}^I_k$ is the subfunctor $\mathrm{V}(F) \subseteq \mathbf{A}^I_k$ given by

$$\mathrm{V}(F)(R) = \{a \in R^I \mid f(a) = 0 \text{ for all } f \in F\} \subseteq R^I.$$

This is indeed a subfunctor: for any k-algebra map $R \to S$, the induced map $R^I \to S^I$. sends $\mathrm{V}(F)(R)$ to $\mathrm{V}(F)(S)$.

Remark 2.17. It is clear that the vanishing locus of F depends only on the ideal generated by F: if $(F) = (F')$, then $\mathrm{V}(F) = \mathrm{V}(F')$. We will see below that the converse also holds (Corollary 2.24).

Definition 2.18 (Solution functor)**.** Let $\Sigma = (f_j)_{j\in J}$ be system of J polynomial equations in I variables over k. Its *solution functor* $\mathrm{Sol}_\Sigma \colon \mathrm{CAlg}_k \to \mathrm{Set}$ is the vanishing locus of $\{f_j \mid j \in J\}$ in $\mathbf{A}^I_k$:

$$\mathrm{Sol}_\Sigma = \mathrm{V}(\{f_j \mid j \in J\}) \subseteq \mathbf{A}^I_k.$$

By the universal property of polynomial rings, there is a one-to-one correspondence between systems of J polynomial equations in I variables and k-algebra maps

$$k[x_j \mid j \in J] \to k[x_i \mid i \in I].$$

By the Yoneda lemma, these are in turn equivalent to natural transformations

$$\mathbf{A}^I_k \to \mathbf{A}^J_k \colon \mathrm{CAlg}_k \to \mathrm{Set}\,.$$

Unraveling these equivalences, the map $\mathbf{A}^I_k \to \mathbf{A}^J_k$ corresponding to a system $\Sigma = (f_j)_{j\in J}$ is given on a k-algebra R by

$$R^I \to R^J, \quad a \mapsto (f_j(a))_{j\in J}.$$

By definition, the solution functor Sol_Σ is the kernel of this map, i.e., there is a pullback square

$$\begin{array}{ccc} \mathrm{Sol}_\Sigma & \hookrightarrow & \mathbf{A}_k^I \\ {\scriptstyle 0}\downarrow & & \downarrow \\ 0 & \hookrightarrow & \mathbf{A}_k^J, \end{array}$$

where 0 is the subfunctor of $\mathbf{A}_k^J$ given by $0(R) = \{0\} \subseteq R^J$.

Definition 2.19 (Affine scheme)**.** A functor $\mathrm{CAlg}_k \to \mathrm{Set}$ is called an *affine k-scheme*, if it is isomorphic to Sol_Σ for some system of polynomial equations Σ over k. We denote $\mathrm{Aff}_k \subset \mathrm{Fun}(\mathrm{CAlg}_k, \mathrm{Set})$ the full subcategory spanned by the affine k-schemes. An *affine scheme* is an affine $\mathbf{Z}$-scheme.

Example 2.20. The affine I-space $\mathbf{A}_k^I$ is an affine k-scheme, as it is the solution functor of the empty system of equations in I variables.

Lemma 2.21. *Let Σ be a system of polynomial equations over k. Then the solution functor Sol_Σ is represented by the k-algebra $k[\Sigma]$, i.e., there is an isomorphism*

$$\mathrm{Sol}_\Sigma \simeq \mathrm{Spec}(k[\Sigma]) \colon \mathrm{CAlg}_k \to \mathrm{Set}\,.$$

Proof. This follows from the terminal object in affine schemes being represented by the initial object in CAlg_k, that is, by k itself. Thus the pullback defining Sol_Σ as a functor is equivalent to a pushout square in CAlg_k. But the pushout of commutative algebras is given by the tensor-product, and hence Sol_Σ is represented by the k algebra $k \otimes_{k[x_i]} k[x_j] \cong k[\Sigma]$. □

Theorem 2.22 (Characterization of affine schemes)**.** *Let k be a ring. The following conditions are equivalent for an algebraic k-functor $X \colon \mathrm{CAlg}_k \to \mathrm{Set}$:*

(a) X is an affine k-scheme.
(b) X is representable, i.e., isomorphic to $\mathrm{Spec}(A)$ for some k-algebra A.
(c) X preserves limits and is accessible[1]*.*

Proof. $(a) \Rightarrow (b)$ follows from Lemma 2.21. $(b) \Rightarrow (a)$ follows from the fact every k-algebra A is isomorphic to some $k[\Sigma]$ (take the generator set to be A). $(b) \Leftrightarrow (c)$ follows from the more general fact that if a category $\mathcal{C}$ is *presentable*, that is $\mathcal{C}$ is locally small, admits all small colimits and there exists a generating set of κ-compact generators, which are closed under κ-filtered colimits for some regular cardinal κ, a functor $F \colon \mathcal{C} \to \mathrm{Set}$ is representable if and only if F preserves limits and is accessible. This is a corollary of the adjoint functor theorem, see [AR94, Theorem 1.66]. Now we use that if $\mathcal{C}$ is a small, then category of presheaves is always presentable. Indeed, it is an easy exercise that the representabel functors are compact[2], and so is any finite colimits of representables. As any colimit can be build from finite coproducts, equalizers, and filtered colimits, and any presheaf is given as a colimit of representables, this shows that the collection of compact objects $\mathrm{P}(\mathcal{C})^\omega$ contains a generating set closed under filtered filtered under filtered colimits, and hence, $\mathrm{P}(\mathcal{C})$ is presentable. □

Corollary 2.23. *The Yoneda embedding of $\mathrm{CAlg}_k^{\mathrm{op}}$ induces an equivalence of categories*

$$\mathrm{Spec} \colon \mathrm{CAlg}_k^{\mathrm{op}} \xrightarrow{\sim} \mathrm{Aff}_k \subseteq \mathrm{Fun}(\mathrm{CAlg}_k, \mathrm{Set}).$$

Under this equivalence, the affine k-scheme Sol_Σ corresponds to the k-algebra $k[\Sigma]$.

Proof. This follows immediately from Theorem 2.9 and Theorem 2.22, since the Yoneda embedding is fully faithful and the essential image consists of the representable presheaves. The second claim is Lemma 2.21 □

[1] Accessibility of X is a technical condition saying that X is a small colimit of representables. It is equivalent to the condition that X preserves κ-filtered colimits for some infinite cardinal κ, which is usually easy to check in practice.

[2] Here, compact means ω-compact.

Under the equivalence of Corollary 2.23, the embedding $\mathrm{Sol}_\Sigma \hookrightarrow \mathbf{A}_k^I$ of affine k-schemes corresponds to the quotient map $k[x_i \mid i \in I] \twoheadrightarrow k[\Sigma]$. This implies the following result:

Corollary 2.24 (Functorial Nullstellensatz)**.** *Sending a subset $F \subseteq k[x_i \mid i \in I]$ to its vanishing locus $\mathrm{V}(F) \subseteq \mathbf{A}_k^I$ induces an order reversing bijection*

$$\mathrm{V}\colon \{\textit{ideal in } k[x_i \mid i \in I]\} \xrightarrow{\sim} \{\textit{vanishing loci in } \mathbf{A}_k^I\}.$$

Proof. Surjectivity follows from Remark 2.17. For injectivity, consider the following commutative diagram:

$$\begin{array}{ccc} \mathrm{V}(F) & \xrightarrow{\sim} & \mathrm{Spec}(k[x_i \mid i \in I]/F) \\ \downarrow & & \downarrow \\ \mathbf{A}_k^I & \xrightarrow{\sim} & \mathrm{Spec}(k[x_i \mid i \in I]). \end{array}$$

By Yoneda lemma, the right hand side correpsonds to a map $f\colon k[x_i \mid i \in I] \twoheadrightarrow k[x_i \mid i \in I]/F$. The ideal F can be recovered as $\ker(f)$. This implies injectivity. □

Example 2.25. Consider the following systems of polynomial equations over $\mathbf{R}$ in one variable:

$$\Sigma_1 = (x^2 + 1), \quad \Sigma_2 = ((x^2 + 1)^2), \quad \Sigma_3 = (x^2 + x + 1), \quad \Sigma_4 = (x^4 + 1).$$

Then

$$\mathrm{Sol}_{\Sigma_1}(\mathbf{R}) = \varnothing, \quad \mathrm{Sol}_{\Sigma_2}(\mathbf{R}) = \varnothing, \quad \mathrm{Sol}_{\Sigma_3}(\mathbf{R}) = \varnothing, \quad \mathrm{Sol}_{\Sigma_4}(\mathbf{R}) = \varnothing,$$

and

$$\mathrm{Sol}_{\Sigma_1}(\mathbf{C}) = \{\pm i\}, \quad \mathrm{Sol}_{\Sigma_2}(\mathbf{C}) = \{\pm i\}, \quad \mathrm{Sol}_{\Sigma_3}(\mathbf{C}) = \{\zeta_3, \bar{\zeta}_3\}, \quad \mathrm{Sol}_{\Sigma_4}(\mathbf{C}) = \{\pm\zeta_8, \pm\bar{\zeta}_8\},$$

where $\zeta_n = \exp(\frac{2\pi i}{n}) \in \mathbf{C}$. All four equations have the same solutions in $\mathbf{R}$. However, as the four ideals (Σ_i) in $\mathbf{R}[x]$ are pairwise distinct, they define four different subfunctors of $\mathbf{A}^1_{\mathbf{R}}$ by Corollary 2.24. The solutions in $\mathbf{C}$ distinguish them, except for Sol_{Σ_1} and Sol_{Σ_2}. To see that $\mathrm{Sol}_{\Sigma_1} \neq \mathrm{Sol}_{\Sigma_2}$ as subfunctors of $\mathbf{A}^1_{\mathbf{R}}$, we can compute the solutions in the $\mathbf{R}$-algebra $\mathbf{C}[\varepsilon]$ of dual complex numbers (where $\varepsilon^2 = 0$):

$$\mathrm{Sol}_{\Sigma_1}(\mathbf{C}[\varepsilon]) = \{\pm i\}, \quad \mathrm{Sol}_{\Sigma_2}(\mathbf{C}[\varepsilon]) = \{\pm i + a\varepsilon \mid a \in \mathbf{C}\}.$$

On the other hand, the associated $\mathbf{R}$-algebras are

$$\mathbf{R}[\Sigma_1] \simeq \mathbf{C}, \quad \mathbf{R}[\Sigma_2] \simeq \mathbf{C}[\varepsilon], \quad \mathbf{R}[\Sigma] \simeq \mathbf{C}, \quad \mathbf{R}[\Sigma_4] \simeq \mathbf{C} \times \mathbf{C}.$$

By Lemma 2.21, Sol_{Σ_1} and Sol_{Σ_3} are both isomorphic to $\mathrm{Spec}(\mathbf{C})$. The different ideals (Σ_1) and (Σ_3) correspond to two different embeddings of the affine $\mathbf{R}$-scheme $\mathrm{Spec}(\mathbf{C})$ into $\mathbf{A}^1_{\mathbf{R}}$, and the systems Σ_1 and Σ_3 themselves are two different presentations of the $\mathbf{R}$-algebra $\mathbf{C}$.

Remark 2.26. In summary, given a system of polynomial equations Σ over k, we have the following relations between Σ and Sol_Σ:

(a) The data of the pullback square below Definition 2.18 is equivalent to the data of Σ itself.
(b) The data of the embedding $\mathrm{Sol}_\Sigma \hookrightarrow \mathbf{A}_k^I$ is equivalent to the data of the ideal (Σ) in the polynomial ring $k[x_i \mid i \in I]$.
(c) The data of the affine k-scheme Sol_Σ alone is equivalent the k-algebra $k[\Sigma]$.

This can be compared with the following types of data in differential geometry:

(a) A smooth manifold M given as the vanishing locus of a smooth functioon $\mathbf{R}^n \to \mathbf{R}^m$.
(b) A smooth manifold M given as a closed submanifold of $\mathbf{R}^n$.
(c) A smooth manifold M.

Smooth manifolds are the basic objects of interest in differential geometry. Embedding a manifold M into a Euclidean space or realizing it as the vanishing locus of a function are often useful ways to understand M, but we do not consider this additional data to be part of the manifold M itself. The situation in algebraic geometry is entirely similar: the basic objects of interest are affine schemes. Any affine scheme X can be embedded into

an affine space ($X \hookrightarrow \mathbf{A}^I$) or realized as the solution functor of a system of polynomial equations ($X \simeq \mathrm{Sol}_\Sigma$), but this data is not part of the affine scheme X itself.

A key different between differential geometry and algebraic geometry is that it is much easier to embed smooth manifolds into $\mathbf{R}^n$ than it is to embed schemes into $\mathbf{A}^n$. In fact, the former is always possible under mild technical assumptions[3] (which are usually taken as a part of the definition of smooth manifolds). but many interesting schemes are not affine. For example, the real projective space $\mathbf{P}^n(\mathbf{R})$ can be embedded in the Euclidean space $\mathbf{R}^{2n}$, but we will see that the algebraic projective space $\mathbf{P}^n$ with $n \geq 1$ cannot be embedded in $\mathbf{A}^N$ for any N.

Remark 2.27 (System of linear equations). Let us spell out the analogy with linear algebra. A system Λ of J-linear equations in I-variables over a ring k is J-indexed family in the free k-module $k^{(I)}$, or equivalently a k-linear map $k^{(J)} \to k^{(I)}$. If $(a_{ij})_{i\in I, j\in J}$ is the corresponding $I \times J$-matrix, a solution to Λ in a k-module M is a family $(m_i)_{i\in I}$ in M such that $\sum_{i\in I} a_{ij} m_i = 0$ for all $j \in J$. This defines a *solution functor*

$$\mathrm{Sol}_\Lambda \colon \mathrm{Mod}_k \to \mathrm{Set}\,.$$

Unraveling the definitions, $\mathrm{Sol}_\Lambda(M)$ is exactly the kernel of the map $M^I \to M^J$, obtained by appyling $\mathrm{Map}(-, M)$ to the given map $k^{(J)} \to k^{(I)}$. It follows that $\mathrm{Sol}_\Lambda \simeq \mathrm{Map}(C, -)$, where C is the cokernel of $k^{(J)} \to k^{(I)}$. Thus, we can think of a system of linear equations over k as a k-module C equipped with a presentation, and its solution functor as the k-module C itself.

2.4. **Example of affine scheme.**

Example 2.28 (The final scheme). The constant functor $\mathrm{CAlg} \to \mathrm{Set}$ sending every ring to a one-point set is isomorphic to $\mathrm{Spec}(\mathbf{Z})$ and hence is an affine scheme. This is the final object of $\mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})$.

Example 2.29 (The empty scheme). The functor

$$\mathrm{CAlg} \to \mathrm{Set}, \quad R \mapsto \begin{cases} \varnothing & R \neq 0, \\ * & R = 0, \end{cases}$$

is an affine scheme, isomorphic to $\mathrm{Spec}(0)$. It is called the *empty scheme* and denoted by $\varnothing$. Note that $\varnothing$ is the initial object of Aff, but it is *not* the initial object of $\mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})$, which is the constant functor with value $\varnothing$.

Example 2.30 (The idempotent classifier). Let $\mathrm{Idem} \colon \mathrm{CAlg} \to \mathrm{Set}$ be the functor sending R to the set of idempotent elements of R. Then Idem is an affine scheme, isomorphic to $\mathrm{Spec}(\mathbf{Z} \times \mathbf{Z})$. Indeed, there is a bijection

$$\mathrm{Map}_{\mathrm{CAlg}}(\mathbf{Z} \times \mathbf{Z}, R) \xrightarrow{\sim} \mathrm{Idem}(R), \quad \varphi \mapsto \varphi(1, 0),$$

which is natural in $R \in \mathrm{CAlg}$.

Example 2.31 (The multiplicative group). The functor $\mathbf{G}_m \colon \mathrm{CAlg} \to \mathrm{Ab}$ sending R to the group of units $R^\times$ is called the *multiplicative group*. It is an *affine group scheme*, meaning that the composition

$$\mathrm{CAlg} \xrightarrow{\mathbf{G}_m} \mathrm{Ab} \xrightarrow{\mathrm{fgt}} \mathrm{Set}$$

is an affine scheme. Indeed, it is isomorphic to $\mathrm{Spec}(\mathbf{Z}[u^{\pm 1}])$: for every ring R, there is a bijection

$$\mathrm{Map}_{\mathrm{CAlg}}(\mathbf{Z}[u^{\pm 1}], R) \xrightarrow{\sim} R^\times, \quad \varphi \mapsto \varphi(u).$$

Example 2.32 (The additive group). The functor $\mathbf{G}_a \colon \mathrm{CAlg} \to \mathrm{Ab}$ sending R to the underlying group $(R, +)$ is called the *additive group*. The composition

$$\mathrm{CAlg} \xrightarrow{\mathbf{G}_a} \mathrm{Ab} \xrightarrow{\mathrm{fgt}} \mathrm{Set}$$

is simply the forgetful functor, also known as the affine line $\mathbf{A}^1$. Hence $\mathbf{G}_a$ is an affine group scheme.

[3] namely: Hausdorff, second countable, and of bounded dimension.

Example 2.33 (The matrix ring). Let $n \geq 0$ and let $\mathrm{Mat}_n \colon \mathrm{CAlg} \to \mathrm{Alg}$ be the functor sending R to the associated ring of $n \times n$ matrices over R. This is an associative ring object in affine schemes. Indeed, since a matrix over R is simply a family of n^2 elements of R, the composition

$$\mathrm{CAlg} \xrightarrow{\mathrm{Mat}_n} \mathrm{Alg} \xrightarrow{\mathrm{fgt}} \mathrm{Set}$$

is isomorphic to $\mathbf{A}^{n^2}$ and hence is an affine scheme.

Example 2.34 (The general linear group). Let $n \geq 0$ and let $\mathrm{GL}_n \colon \mathrm{CAlg} \to \mathrm{Grp}$ be the functor sending R to the group $\mathrm{GL}_n(R)$ of invertible $n \times n$ matrices. Then GL_n is an affine group scheme. Indeed, let $A = \mathbf{Z}[x_{ij} \mid (i,j) \in \{1,\dots,n\}^2]$ be the ring representing Mat_n, which contains the universal $n \times n$-matrix $X = (x_{ij})_{i,j}$. A matrix $M \in \mathrm{Mat}_{n\times n}(R)$ is invertible if and only if its determinant $\det(M) \in R$ is a unit. Hence, for any ring R, there is an isomorphism

$$\mathrm{Map}_{\mathrm{CAlg}}(A_{\det(X)}, R) \xrightarrow{\sim} \mathrm{GL}_n(R), \quad \varphi \mapsto (\varphi(x_{ij}))_{i,j},$$

so that $\mathrm{GL}_n \simeq \mathrm{Spec}(A_{\det(X)})$.

Example 2.35 (The special linear group). Let $n \geq 0$ and let $\mathrm{SL}_n \colon \mathrm{CAlg} \to \mathrm{Grp}$ be the functor sending R to the special linear group $\mathrm{SL}_n(R)$ of $n \times n$ matrices with determinant 1. Let $X \in \mathrm{Mat}_n(A)$ be the universal $n \times n$ matrix as in Example 2.34. We then have

$$\mathrm{Map}_{\mathrm{CAlg}}(A/(\det(X)-1), R) \xrightarrow{\sim} \mathrm{SL}_n(R), \quad \varphi \mapsto (\varphi(x_{ij}))_{i,j},$$

so that $\mathrm{SL}_n \simeq \mathrm{Spec}(A/\det(X)-1)$. The subfunctor inclusions $\mathrm{SL}_n \subseteq \mathrm{GL}_n \subseteq \mathrm{Mat}_n$ corresponding to the ring maps

$$A \hookrightarrow A_{\det(X)} \twoheadrightarrow A/(\det(X)-1).$$

Example 2.36 (The affine space of a module). Let k be a ring and let M be a k-module. Consider the functor $\mathbf{A}(M) \colon \mathrm{CAlg}_k \to \mathrm{Mod}_k$ defined by

$$\mathbf{A}(M)(R) = \{k\text{-linear maps } M \to R\} = (M \otimes_k R)^\vee.$$

Then $\mathbf{A}(M)$ is a k-module valued affine k-schemes. Indeed, if $\mathrm{Sym}_k(M)$ is the free k-algebra on M, there is a bijection

$$\mathrm{Map}_{\mathrm{CAlg}_k}(\mathrm{Sym}_k(M), R) \xrightarrow{\sim} \mathbf{A}(M)(R), \quad \varphi \mapsto \varphi|_M,$$

so that $\mathbf{A}(M) \simeq \mathrm{Spec}(\mathrm{Sym}_k(M))$. The affine I-space is a special case of this construction: $\mathbf{A}^I_k \simeq \mathbf{A}(k^{(I)})$.

Remark 2.37. Let k be a ring and let M be a k-module. Given Example 2.36, it is tempting to consider the following *predual* of $\mathbf{A}(M)$: define $\mathbf{A}^\vee \colon \mathrm{CAlg}_k \to \mathrm{Mod}_k$ by

$$\mathbf{A}^\vee(M)(R) = M \otimes_k R.$$

There is a canonical map $\mathbf{A}^\vee(M^\vee) \to \mathbf{A}(M)$, which is an isomorphism if and only if M is a vector space (see Definition 3.3). Otherwise, $\mathbf{A}^\vee(M)$ does not preserve limits and hence is not an affine k-scheme (in fact, it is not even a scheme). For that reason, the functor $\mathbf{A}^\vee(M)$ is rarely used.

2.5. **Base change.** Given a ring map $\varphi \colon k \to k'$, we can transform any system of polynomial equations Σ over k into a system $\varphi^*(\Sigma)$ over k' by applying φ to all the coefficients. More generally, many types of data over k can be transformed into data over k' using φ, a process known as *base change*, *change of coefficients*, or *extension of scalars*. Other examples are the functor $\varphi^* \colon \mathrm{CAlg}_k \to \mathrm{CAlg}_{k'}$ sending a k-algebra A to the k'-algebra $A \otimes_k k'$. Unraveling these constructions, we see that there is a canonical isomorphism of k'-algebras

$$k'[\varphi^*(\Sigma)] \simeq \varphi^*(k[\Sigma]).$$

In this section, we investigate the related process of transforming an algebraic k-functor into an algebraic k'-functor.

Theorem 2.38 (Functoriality of presheaves)**.** *Let $u\colon \mathcal{C} \to \mathcal{D}$ be a functor, and let*

$$u^*\colon \mathrm{P}(\mathcal{D}) \to \mathrm{P}(\mathcal{C}), \quad F \mapsto F \circ u,$$

be the restriction along *u functor.*

(a) The functor u^* admits a left adjoint $u_\sharp$ and a right adjoint u_* given by

$$u_\sharp(F)(d) = \mathrm{colim}((\mathcal{C} \times_{\mathcal{D}} \mathcal{D}_{d/})^{\mathrm{op}} \to \mathcal{C}^{\mathrm{op}} \xrightarrow{F} \mathrm{Set}) = \operatorname*{colim}_{d \to u(c)} F(c),$$

$$u_*(F)(d) = \lim((\mathcal{C} \times_{\mathcal{D}} \mathcal{D}_{/d})^{\mathrm{op}} \to \mathcal{C}^{\mathrm{op}} \xrightarrow{F} \mathrm{Set}) = \lim_{u(c) \to d} F(c),$$

provided these colimits and limits exists (e.g., if $\mathcal{C}$ is small).

(b) The functor $u_\sharp$ extends u: there is a canonical isomorphism

$$u_\sharp \circ \text{よ}_{\mathcal{C}} \simeq \text{よ}_{\mathcal{D}} \circ u$$

(c) If u is fully faithful, then $u_\sharp$ and u_* are fully faithful.

(d) If u is a localization, then u^* is fully faithful.

(e) If the functor u has a left adjoint u_L (resp. a right adjoint u_R), then there is a canonical isomorphism $u_\sharp \simeq u_L^*$ (resp. $u_* \simeq u_R^*$).

Proof. We will not always develop the full machinery of category theory in these notes; instead, the reader is referred to standard texts on category theory for the necessary background, see for example [Rie16] or [Mac98].

(a) The left (resp. right) adjoint of u^* is given by the left (resp. right) Kan extension. The explicit formula is simply the explicite pointwise formulas for Kan extensions, see [Rie16, ??].

explicit reference

(b) Let $c \in \mathcal{C}$. By Yoneda lemma, it suffices to show that for each $F \in \mathrm{P}(\mathcal{D})$, there is an isomorphism

$$\mathrm{Map}_{\mathrm{P}(\mathcal{D})}(u_\sharp \circ \text{よ}_{\mathcal{C}}(c), F) \simeq \mathrm{Map}_{\mathrm{P}(\mathcal{D})}(\text{よ}_{\mathcal{D}} \circ u(c), F).$$

By adjunction and Yoneda lemma, both sides are isomorphic to $F(u(c))$.

(c) We record a general fact: let $L \vdash R$ be an adjunction, then L is fully faithful if and only if the unit is an isomorphism, and R is fully faithful if and only if the counit is an isomorphism. In our case, if u is fully faithful, then u^* is again fully faithful and we may deduce the result from pointwise formula for Kan extensions.

(d) This follows immediately from the universal property of localization, and that if $\mathcal{C} \to \mathcal{D}$ is a localization at the classes of morphism W, then $\mathcal{C}^{\mathrm{op}} \to \mathcal{D}^{\mathrm{op}}$ is the localization at the classes of morphism W^{op}.

(e) If u has a left adjoint u_L (resp. a right adjoint u_R), then u_L^* (resp. u_R^*) is a left (resp. right) adjoint of u^*. The result then follows from the uniqueness of left (resp. right) adjoint.

□

Remark 2.39. (a) The above shows that $F\colon \mathcal{C}^{\mathrm{op}} \to \mathrm{Set}$, the presheaves $u_\sharp(F)$ and $u_*(F)$ are special cases of *Kan extensions*: $u_\sharp(F)$ is the left Kan extension of F along u^{op}, and $u_*(F)$ is the right Kan extension of F along u^{op}.

(b) By the universal property of $\text{よ}_{\mathcal{C}}$, the functor $u_\sharp$ is the *unique* colimit preserving extension of u (up to unique isomorphism). However, there is no analogous characterization of u_*.

Corollary 2.40. *Let $\mathcal{C}$ be a category and $Y \to X$ be a morphism in $\mathcal{C}$, and let $u\colon \mathcal{C}_{/Y} \to \mathcal{C}_{/X}$ be the forgetful functor.*

(a) The functor $u^*\colon \mathrm{P}(\mathcal{C}_{/X}) \to \mathrm{P}(\mathcal{C}_{/Y})$ has a left adjoint $u_\sharp$ given by

$$u_\sharp(F)(U \to X) = \coprod_{(U \to Y) \in \mathcal{C}_{/X}} F(U \to Y).$$

(b) If pullback along $Y \to X$ exist in $\mathcal{C}$, the functor $u^*\colon \mathrm{P}(\mathcal{C}_{/X}) \to \mathrm{P}(\mathcal{C}_{/Y})$ has a right adjoint u_* given by

$$u_*(F)(U \to X) = F(U \times_X Y \to Y).$$

Proof. (a) From Theorem 2.38 (a), we know that for a fixed presheaf $F \in \mathrm{P}(\mathcal{C})$, the left presheaf $u_\sharp F \in \mathrm{P}(\mathcal{C}_{/X})$ is given as the left-Kan extension in the following diagram

$$\begin{array}{ccc} (\mathcal{C}_{/Y})^{\mathrm{op}} & \xrightarrow{F} & \mathrm{Set} \\ {\scriptstyle u^{\mathrm{op}}}\downarrow & \nearrow_{u_\sharp F} & \\ (\mathcal{C}_{/X})^{\mathrm{op}} & & \end{array}$$

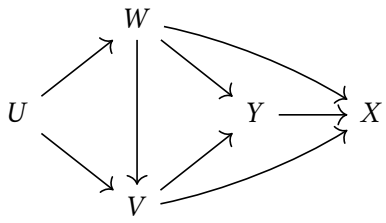
From the explicit formula for pointwise-left-Kan-extensions, we see that

$$u_\sharp(F)(U \xrightarrow{f} X) = \operatorname*{colim}_{(V \to Y) \in \mathcal{J}^{\mathrm{op}}} F(V \to Y).$$

Here, the (opposite of the) indexing category $\mathcal{J}$ has as objects all commutative squares

$$\begin{array}{ccc} U & \xrightarrow{f} & X \\ \downarrow & & \uparrow \\ V & \longrightarrow & Y \end{array}$$

where $Y \to X$ is the fixed morphism defining the forgetful functor u. A morphism in this category is a map $W \to V$ such that the following diagram commutes



such that the composition $U \to X$ is given by f.

Consider now the discrete category $\mathcal{J}$ consisting of the different possible maps $U \to Y$, which factor the map $U \to X$. Consider this as a subcategory $\mathcal{J} \subseteq \mathcal{J}$ by sending $U \to Y$ to the factorization $U \xrightarrow{\mathrm{id}} U \to Y \to X$. This discrete category is clearly initial, and hence

$$u_\sharp(F)(U \to X) = \operatorname*{colim}_{(V \to Y) \in \mathcal{J}^{\mathrm{op}}} F(V \to Y) \cong \operatorname*{colim}_{(U \to Y) \in \mathcal{J}^{\mathrm{op}}} F(U \to Y) \cong \coprod_{U \to Y} F(U \to Y).$$

This shows the claim.

(b) From Theorem 2.38 (a) we deduce that the right adjoint u_* always exists for presheaves. However, in the case that $\mathcal{C}$ admits pullback, the forgetful functor $u : \mathcal{C}_{/Y} \to \mathcal{C}_{/X}$ admits a right adjoint $- \times_X Y : \mathcal{C}_{/X} \to \mathcal{C}_{/Y}$. That this is really the right adjoint can be checked quickly from the universal property of the fiber product. By Theorem 2.38 (e) we conclude that the right adjoint $u_* \mathrm{P}(\mathcal{C}_{/Y}) \to \mathrm{P}(\mathcal{C}_{/X})$ is given by the induced functor $(- \times_X Y)^*$ implying the claim.

□

If k is a ring, then $\mathrm{CAlg}_k \simeq \mathrm{CAlg}_{k/}$ and hence $\mathrm{CAlg}_k^{\mathrm{op}} \simeq (\mathrm{CAlg}^{\mathrm{op}})_{/k}$. Using this identification, we obtain the following special case of Corollary 2.40 with $\mathcal{C} = \mathrm{CAlg}^{\mathrm{op}}$:

Corollary 2.41. *Let $\varphi\colon k \to k'$ be a ring map. Then there is a triple of adjoint functors*

$$\mathrm{Fun}(\mathrm{CAlg}_{k'}, \mathrm{Set}) \underset{\varphi_*}{\overset{\varphi_\sharp}{\overset{\varphi^*}{\leftarrow}}} \mathrm{Fun}(\mathrm{CAlg}_k, \mathrm{Set})$$

where:

- *φ^* is precomposition with the forgetful functor* $\mathrm{CAlg}_{k'} \to \mathrm{CAlg}_k$*, and it is the unique colimit preserving extension of* $\varphi^*\colon \mathrm{CAlg}_k^{\mathrm{op}} \to \mathrm{CAlg}_{k'}^{\mathrm{op}}$*;*
- *φ_* is precomposition with* $\varphi^*\colon \mathrm{CAlg}_k \to \mathrm{CAlg}_{k'}$*;*
- *$\varphi_\sharp$ is given by*

$$\varphi_\sharp(X)(A) = \coprod_{(k' \to A) \in \mathrm{CAlg}_k} X(A),$$

and it is the unique colimit-preserving extension of the forgetful functor $\mathrm{CAlg}_{k'}^{\mathrm{op}} \to \mathrm{CAlg}_k^{\mathrm{op}}$.

Proof. This is a direct consequence of Corollary 2.40, with the uniqueness of φ^* being the universal property of the Yoneda embedding, while the uniqueness of $\varphi_\sharp$ being Remark 2.39. □

Definition 2.42. Let $\varphi\colon k \to k'$ be a ring map, giving rise to the adjoint triple of Corollary 2.41.

(a) The functor φ^* is called the *base change* or *extension of scalars* along φ and is also denoted by $X \mapsto X_{k'}$.
(b) The functor φ_* is also called *Weil restriction* or *restriction of scalars* along φ and is also denoted by R_φ or $\mathrm{R}_{k'/k}$.

Remark 2.43. Corollary 2.41 says in particular that the functors φ^* and $\varphi_\sharp$ preserve affine schemes:

(a) For a k-algebra A, $\varphi^*(\mathrm{Spec}(A)) \simeq \mathrm{Spec}(A \otimes_k k')$ in $\mathrm{Fun}(\mathrm{CAlg}_{k'}, \mathrm{Set})$. Indeed, for any k'-algebra B, we have

$$\varphi^*(\mathrm{Spec}(A))(B) = \mathrm{Spec}(A)(\varphi^*(B)) = \mathrm{Map}_{\mathrm{CAlg}_k}(A, \varphi^*B) \cong \mathrm{Map}_{\mathrm{CAlg}_{k'}}(A \otimes_k k', B) = \mathrm{Spec}(A \otimes_k k')(B).$$

Here, the middle isomorphism is the base change adjunction. Thus $\varphi^*(\mathrm{Spec}(A))$ and $\mathrm{Spec}(A \otimes_k k')$ are isomorphic for all k' algebras, and these isomorphisms are clearly natural. Hence the functors are isomorphic. In particular, for any system of polynomial equations Σ over k, $\varphi^*(\mathrm{Sol}_\Sigma) \simeq \mathrm{Sol}_{\varphi^*(\Sigma)} = \mathrm{Spec}(k'[\Sigma])$.

(b) For a k'-algebra B we have $\varphi_\sharp \mathrm{Spec}(B) \cong \mathrm{Spec}(\varphi^*B) \in \mathrm{Fun}(\mathrm{CAlg}_k, \mathrm{Set})$, that is, $\varphi_\sharp \mathrm{Spec}(B)$ is the affine scheme associated to B considered as a k-algebra. Indeed, for any algebraic k-functor F, we obtain from the Yoneda lemma

$$\mathrm{Nat}(\varphi_\sharp \mathrm{Spec}(B), F) \cong \mathrm{Nat}(\mathrm{Spec}(B), \varphi^*F) \cong \varphi^*F(B) = F(\varphi^*B) = \mathrm{Nat}(\mathrm{Spec}(\varphi^*B), F).$$

Fully faithfulness of the Yoneda embedding implies the claim. Alternatively, this follows from the explicit formula

$$\varphi_\sharp \mathrm{Spec}(B)(A) \cong \coprod_{k' \to A \in \mathrm{CAlg}_k} \mathrm{Spec}(B)(A) \cong \coprod_{k' \to A \in \mathrm{CAlg}_k} \mathrm{MapM}_{\mathrm{CAlg}_{k'}}(B, A) = \mathrm{Map}_{\mathrm{CAlg}_k}(B, A).$$

Here we use that any morphism $B \to A$ defines a k'-algebra structure on A given by the composition $k' \to B \to A$. In particular, any k-algebra homomorphism $B \to A$ is also a k'-algebra homomorphism.

On the other hand, the functor φ_* does not always preserve affine schemes, as the next example shows.

Example 2.44. Consider the affine scheme $\mathbf{G}_m\colon \mathrm{CAlg} \to \mathrm{Set}$, $R \mapsto R^\times$. Let $\mathbf{G}_{m,\mathbf{C}}\colon \mathrm{CAlg}_{\mathbf{C}} \to \mathrm{Set}$ be its base change to $\mathbf{C}$, i.e., its restriction along the forgetful functor $\mathrm{CAlg}_{\mathbf{C}} \to \mathrm{CAlg}$. The Weil restriction $\mathrm{R}_{\mathbf{C}/\mathbf{R}}(\mathbf{G}_{m,\mathbf{C}})\colon \mathrm{CAlg}_{\mathbf{R}} \to \mathrm{Set}$ is given by

$$\mathrm{R}_{\mathbf{C}/\mathbf{R}}(\mathbf{G}_{m,\mathbf{C}})(A) = \mathbf{G}_{m,\mathbf{C}}(A \otimes_{\mathbf{R}} \mathbf{C}) = (A \otimes_{\mathbf{R}} \mathbf{C})^\times.$$

One can check that $\mathrm{R}_{\mathbf{C}/\mathbf{R}}(\mathbf{G}_{m,\mathbf{C}})$ is an affine $\mathbf{R}$-scheme: for any $\mathbf{R}$-algebra A, there is a bijection

$$\mathrm{Map}(\mathbf{R}[x,y,z,w]/(xz - yw - 1, yz + xw), A) \xrightarrow{\sim} (A \otimes_{\mathbf{R}} \mathbf{C})^\times, \quad \varphi \mapsto \varphi(x) + i\varphi(y).$$

These equations ensure that $\varphi(x) + i\varphi(y)$ is inverse to $\varphi(z) + i\varphi(w)$. More generally, if $k \subseteq k'$ is finite field extension, one can show that Weil restriction $\mathrm{R}_{k'/k}$ sends affine k'-schemes to affine k-schemes.

A fundamental property of sets is that maps of sets are equivalent to family of sets: there is an equivalence of categories

$$\mathrm{Ar}(\mathrm{Set}) \simeq \mathrm{Fam}(\mathrm{Set}),$$

where $\mathrm{Ar}(\mathrm{Set}) = \mathrm{Fun}([1], \mathrm{Set})$ is the arrow category of Set and $\mathrm{Fam}(\mathrm{Set})$ is the category whose objects are (set-indexed) families of sets $(X_i)_{i\in I}$, where a map $(X_i)_{i\in I} \to (Y_j)_{j\in J}$ consists of a map $u\colon I \to J$ and maps $X_i \to Y_{u(i)}$ for all $i \in I$. In one direction, a map $f\colon X \to I$ corresponds to the family of its fibers $(f^{-1}\{i\})_{i\in I}$. In the other direction, a family $(X_i)_{i\in I}$ corresponds to the map $\coprod_{i\in I} X_i \to I$. If we fix the target/indexing set I, we obtain an equivalence categories

$$\mathrm{Set}_{/I} \simeq \mathrm{Set}^I.$$

The following proposition generalizes this fact to presheaves of sets (we recover the last equivalence by taking $\mathcal{C} = *$):

Proposition 2.45 (Slices of presheaf categories)**.** *Let $\mathcal{C}$ be a category and $F \in \mathrm{P}(\mathcal{C})$ be a presheaf on $\mathcal{C}$. Then there is an equivalence of categories*

$$\mathrm{fib}_F : \mathrm{P}(\mathcal{C})_{/F} \simeq \mathrm{P}(\mathrm{El}(F)) : \coprod_F,$$

described as follows.

If $u\colon \mathrm{El}(F) \to \mathcal{C}$ is the forgetful functor, there is a tautological map $* \to u^*(F)$, whose adjoint $u_\sharp(*) \to F$ is an isomorphism. We can thus define

- *Given $H \in \mathrm{P}(\mathrm{El}(F))$, the presheaf $\coprod_F H$ over F is $u_\sharp(H) \to u_\sharp(*) \simeq F$. Explicitly,*

$$\left(\coprod_F H\right)(X) = \coprod_{x\in F(X)} (H(X,x) \to *).$$

- *Given $G \in \mathrm{P}(\mathcal{C})_{/F}$, the presheaf $\mathrm{fib}_F(G)$ on $\mathrm{El}(F)$ is the pullback $u^*(G) \times_{u^*(F)} *$. Explicitly,*

$$\mathrm{fib}_F(G)(X,x) = G(X) \times_{F(X)} \{x\} \simeq \left\{ \begin{array}{ccc} & & G \\ & \nearrow & \downarrow \\ \text{ɟ}(X) & \xrightarrow{x} & F \end{array} \right\}.$$

Proof. Fix $(X,x) \in \mathrm{El}(F)$. The value of the tautological natural transformation $\mathrm{can} : * \to u^*(F)$ on (X,x) is the morphism $* \to u^*(F)(X,x) = F(X)$ picking out the element $x \in F(X)$. By the Yoneda density, every presheaf is the colimit of representable presheaves mapping into the given presheaf. As the presheaf $* \in \mathrm{P}(\mathrm{El}(F))$ is terminal, this implies

$$* \cong \operatorname*{colim}_{\mathrm{El}(F)} \text{ɟ}_{\mathrm{El}(F)}((X,x)).$$

Since $u_\sharp$ is the unique colimit preserving extension of u, so that $u_\sharp \text{ɟ}_{\mathrm{El}(F)} = \text{ɟ}_{\mathcal{C}} u$, we further have

$$u_\sharp(*) \cong u_\sharp(\operatorname*{colim}_{\mathrm{El}(F)} \text{ɟ}_{\mathrm{El}(F)}((X,x))) \cong \operatorname*{colim}_{\mathrm{El}(F)} u_\sharp \text{ɟ}_{\mathrm{El}(F)}((X,x)) \cong \operatorname*{colim}_{\mathrm{El}(F)} \text{ɟ}_{\mathcal{C}}(X) \cong F. \tag{2.46}$$

We claim that this composition of equivalences is exactly the adjoint of the canonical natural transformation $\mathrm{can} : * \to u^*F$. Indeed, the adjoint of the canonical natural transformation is the composition $u_\sharp * \xrightarrow{u_\sharp \mathrm{can}} u_\sharp u^* F \xrightarrow{\varepsilon_F} F$ where we denote by ε the counit of the $(\sharp, *)$-adjunction.

We now notice that, using the identification $F \cong \operatorname*{colim}_{\mathrm{El}(F)} \text{ɟ}_{\mathcal{C}}(X)$, the morphism $u_\sharp \mathrm{can}$ can be identified with the morphism

$$u_\sharp * \to u_\sharp u^* F \cong u_\sharp u^* \operatorname*{colim}_{\mathrm{El}(F)} \text{ɟ}_{\mathcal{C}}(X) \cong u_\sharp u^* u_\sharp \operatorname*{colim}_{\mathrm{El}(F)} \text{ɟ}_{\mathrm{El}(F)}((X,x))$$

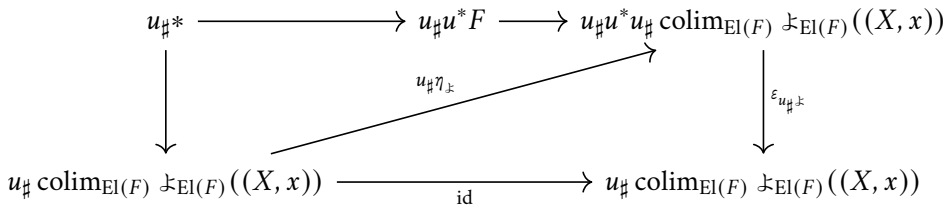
Here we used again that $u_\sharp$ is the unique colimit preserving extension of u to presheaves. Using the identification $F \cong \operatorname*{colim}_{\mathrm{El}(F)} \text{よ}_{\mathcal{C}}(X)$ again, we can see that the counit map can be identified with the map

$$u_\sharp u^* u_\sharp \operatorname*{colim}_{\mathrm{El}(F)} \text{よ}_{\mathrm{El}(F)}((X,x)) \xrightarrow{\varepsilon_{u_\sharp \text{よ}}} u_\sharp \operatorname*{colim}_{\mathrm{El}(F)} \text{よ}_{\mathrm{El}(F)}((X,x)) \cong \operatorname*{colim}_{\mathrm{El}(F)} u_\sharp \text{よ}_{\mathrm{El}(F)}((X,x)) \cong \operatorname*{colim}_{\mathrm{El}(F)} \text{よ}_{\mathcal{C}}(X) \cong F$$

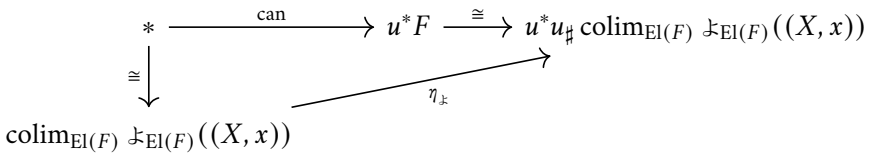
Combining both compositions and comparing with (2.46), we conclude that it suffices to show that the composition

$$u_\sharp * \to u_\sharp u^* F \cong u_\sharp u^* u_\sharp \operatorname*{colim}_{\mathrm{El}(F)} \text{よ}_{\mathrm{El}(F)}((X,x)) \xrightarrow{\varepsilon_{u_\sharp \text{よ}}} u_\sharp \operatorname*{colim}_{\mathrm{El}(F)} \text{よ}_{\mathrm{El}(F)}((X,x))$$

is equal to $u_\sharp(* \xrightarrow{\cong} \operatorname{colim}_{\mathrm{El}(F)} \text{よ}_{\mathrm{El}(F)}((X,x)))$. Consider the diagram



Here the right triangle commutes by the triangle identity. The commutativity of the left triangle can be checked before applying $u_\sharp$, that is, we have to check that the following diagram commutes



Using the two isomorphisms, this is easily checked on sections. Indeed, given $(Y,y) \in \mathrm{El}(F)$, it suffices to show that the two compositions from $\operatorname{colim}_{\mathrm{El}(F)} \text{よ}_{\mathrm{El}(F)}((X,x))(Y,y) = \operatorname{colim}_{\mathrm{El}(F)} \mathrm{Map}_{\mathrm{El}(F)}((Y,y),(X,x))$ to $u^*F(Y,y) = F(Y)$ are the same. The clockwise composition factors over $* \xrightarrow{y} F(Y)$, so hits only y, while the counter-clockwise composition is given by $[f,(X,x)] \mapsto [f,X] \mapsto f^*(x) = y$. Thus, the outer diagram commutes, implying the claim.

For $H \in \mathrm{P}(\mathrm{El}(F))$, by the pointwise Kan extension formula, we have

$$u_\sharp(H)(X) \simeq \operatorname{colim}((\mathrm{El}(F) \times_{\mathcal{C}} \mathcal{C}_{X/})^{\mathrm{op}} \to \mathrm{El}(F)^{\mathrm{op}} \xrightarrow{H} \mathrm{Set}).$$

For each object (α, y) in $\mathrm{El}(F) \times_{\mathcal{C}} \mathcal{C}_{X/}$, where $\alpha \colon X \to Y$, there exists a unique map from $(\mathrm{id}_X, F(\alpha)(y)) \to (\alpha, y)$. By a similar cofinality argument as in Corollary 2.40, we obtain the desired description. The explicit description of $\mathrm{fib}_F(G)$ follows from that pullback of functors are computed pointwise. Finally, the proof that the two functors are inverses to each other is analogous to that of the case of sets, using universality of sets. □

Corollary 2.47. *Let $\mathcal{C}$ be a category, let $X \in \mathcal{C}$, and let $u \colon \mathcal{C}_{/X} \to \mathcal{C}$ be the forgetful functor. Then the functor $u_\sharp$ induces an equivalence of categories*

$$\mathrm{P}(\mathcal{C}_{/X}) \xrightarrow{\sim} \mathrm{P}(\mathcal{C})_{/\text{よ}(X)}.$$

Proof. This follows by specializing Proposition 2.45 to the case of a representable presheaf, using that $\mathrm{El}(\text{よ}(X)) \simeq \mathcal{C}_{/X}$. □

Corollary 2.48. *Let k be a ring and let $\varphi \colon \mathbf{Z} \to k$ be the unique map. Then the functor $\varphi_\sharp$ induces an equivalence of categories*

$$\mathrm{Fun}(\mathrm{CAlg}_k, \mathrm{Set}) \xrightarrow{\sim} \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})_{/\mathrm{Spec}(k)}.$$

Proof. This follows by specializing Corollary 2.47 to the case $\mathcal{C} = \mathrm{CAlg}^{\mathrm{op}}$. □

Remark 2.49. Because of Corollary 2.48, algebraic geometry over a base ring k is subsumed by algebraic geometry over $\mathbf{Z}$. In other words, working in the category $\mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})$ of algebraic functors does not restrict the generality, and we will often do so from now on. Note that the functor Spec of Notation 2.7 is independent of k, in the sense that the following square commutes:

$$\begin{array}{ccc} \mathrm{CAlg}_k^{\mathrm{op}} & \xrightarrow{\sim} & (\mathrm{CAlg}^{\mathrm{op}})_{/k} \\ {\scriptstyle \mathrm{Spec}}\downarrow & & \downarrow{\scriptstyle \mathrm{Spec}} \\ \mathrm{Fun}(\mathrm{CAlg}_k, \mathrm{Set}) & \xrightarrow{\sim} & \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})_{/\mathrm{Spec}(k)}. \end{array}$$

Remark 2.50. If $f\colon X' \to X$ is a morphism or algebraic functors, there is a triple of adjoint functors

$$\mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})_{/X'} \underset{f_*}{\overset{f_\sharp}{\underset{\longleftarrow}{\overset{\longrightarrow}{f^*}}}} \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})_{/X},$$

where $f^*(Y) = Y \times_X X'$ and $f_\sharp(Y') = Y'$. Given Proposition 2.45, this can be seen by applying Theorem 2.38 to the functor $u\colon \mathrm{El}(X') \to \mathrm{El}(X)$. This recovers Corollary 2.41 when f is $\mathrm{Spec}(\varphi)\colon \mathrm{Spec}(k') \to \mathrm{Spec}(k)$.

2.6. **Functions.** Recall that the affine line $\mathbf{A}^1$ is the forgetful functor

$$\mathbf{A}^1\colon \mathrm{CAlg} \to \mathrm{Set}, \quad R \mapsto R.$$

Tautologically, $\mathbf{A}^1$ has a structure of a ring object in $\mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})$, given by the factorization

$$\begin{array}{ccc} & & \mathrm{CAlg} \\ & {\scriptstyle \mathrm{id}}\nearrow & \downarrow{\scriptstyle \mathrm{forget}} \\ \mathrm{CAlg} & \xrightarrow[\mathbf{A}^1]{} & \mathrm{Set} \end{array}$$

Recall also that $\mathbf{G}_m$ is the subfunctor of $\mathbf{A}^1$ given by $R \mapsto R^\times$, which has the structure of abelian group (see Example 2.32).

Definition 2.51 (Function and nonvanishing function)**.** Let X be an algebraic functor.

(a) A *function* on X is a map $X \to \mathbf{A}^1$. We denote by
$$\mathcal{O}(X) = \mathrm{Map}(X, \mathbf{A}^1)$$
the set of functions on X. The ring structure on $\mathbf{A}^1$ induces a ring structure on $\mathcal{O}(X)$.

(b) An *nonvanishing function* on X is a map $X \to \mathbf{G}_m$. We denote by
$$\mathcal{O}^\times(X) = \mathrm{Map}(X, \mathbf{G}_m)$$
the set of nonvanishing functions on X. The abelian group structure on $\mathbf{G}_m$ induces an abelian group structure on $\mathcal{O}^\times(X)$.

Remark 2.52. (a) Since $\mathbf{G}_m \subseteq \mathbf{A}^1$, we have $\mathcal{O}^\times(X) \subseteq \mathcal{O}(X)$. In fact, monomorphism can be defined via pullbacks, and $\mathrm{Map}(-,-)$ preserves limits, hence pullbacks in both variables.

(b) By the Yoneda lemma, there is a canonical isomorphism $\mathcal{O}(\mathrm{Spec}(R)) \simeq R$, i.e., the ring of functions on $\mathrm{Spec}(R)$ is R itself. Hence, when restricted to affine schemes, the functor $\mathcal{O}\colon \mathrm{Aff}^{\mathrm{op}} \to \mathrm{CAlg}$ is an equivalence of categories, which is inverse to Spec. Similarly, $\mathcal{O}^\times(\mathrm{Spec}(R)) \simeq R^\times$.

(c) For a general algebraic functor X, we have
$$\mathcal{O}(X) \simeq \lim_{x\colon \mathrm{Spec}(R) \to X} R, \quad f \mapsto (f \circ x)_x,$$
where the limit is indexed by the category of elements $\mathrm{El}(X)^{\mathrm{op}}$ (see Corollary 2.11). Since the unit group functor $R \mapsto R^\times$ preserves limits (it is the right adjoint of the group ring functor $\mathbf{Z}[-]$), it follows that $\mathcal{O}^\times$ is precisely the unit group $\mathcal{O}(X)^\times$.

(d) By definition, the functor

$$\mathcal{O}\colon \operatorname{Fun}(\mathrm{CAlg}, \mathrm{Set})^{\mathrm{op}} \to \mathrm{CAlg}, \quad X \mapsto \mathcal{O}(X),$$

is limit-preserving. By (b) and the universal property of presheaves, it is the unique-limit preserving extension of the identity $\mathrm{CAlg} \to \mathrm{CAlg}$. Similarly, $\mathcal{O}^\times$ is the unique limit-preserving extension of the functor $\mathrm{CAlg} \to \mathrm{Ab}$, $R \mapsto R^\times$.

Remark 2.53 (Size issues). The statement (d) of Remark 2.52 is actually nonsensical due to *size issues*. Since the category CAlg is large, the limit in the formula for $\mathcal{O}(X)$ is not an object of CAlg, which is the category of small rings. There are two standard ways to rectify this issue:

(a) One can simply replace the target of $\mathcal{O}$ by the category $\widehat{\mathrm{CAlg}}$ of large rings. One may then also replace the category of sets in the source by the category $\widehat{\mathrm{Set}}$ of large sets. We obtain the functor

$$\mathcal{O}\colon \operatorname{Fun}(\mathrm{CAlg}, \widehat{\mathrm{Set}})^{\mathrm{op}} \to \widehat{\mathrm{CAlg}},$$

which is the unique extension of the embedding $\mathrm{CAlg} \hookrightarrow \widehat{\mathrm{CAlg}}$ that preserves *large* limits.

(b) One can replace the source of $\mathcal{O}$ by the subcategory $\operatorname{Fun}^{\mathrm{acc}}(\mathrm{CAlg}, \mathrm{Set})$ of *accessible* functors, which are the functors that are *small* colimits of representables. For an accessible functor X, $\mathcal{O}(X)$ is a small limit of small rings and hence is small. We therefore have a functor

$$\mathcal{O}\colon \operatorname{Fun}^{\mathrm{acc}}(\mathrm{CAlg}, \mathrm{Set})^{\mathrm{op}} \to \mathrm{CAlg}$$

which is the unique extension of the identity $\mathrm{CAlg} \to \mathrm{CAlg}$ that preserves *small* limits. All algebraic functors that arise in practice (including all schemes) are accessible, and all relevant constructions preserve accessibility, so that this restriction does not have any undesirable consequences.

Remark 2.54 (The adjunction between Spec and $\mathcal{O}$). Let X be an algebraic functor and A a ring. By Corollary 2.11 and the Yoneda lemma, we have a sequence of natural isomorphisms

$$\begin{aligned}
\operatorname{Map}(X, \operatorname{Spec}(A)) &\simeq \lim_{(R,x)\in \mathrm{El}(X)^{\mathrm{op}}} \operatorname{Map}(\operatorname{Spec}(R), \operatorname{Spec}(A)) \\
&\simeq \lim_{(R,x)\in \mathrm{El}(X)^{\mathrm{op}}} \operatorname{Map}(A, R) \\
&\simeq \operatorname{Map}(A, \lim_{(R,x)\in \mathrm{El}(X)^{\mathrm{op}}} R) \\
&\simeq \operatorname{Map}(A, \mathcal{O}(X)).
\end{aligned}$$

Thus, modulo the size issues discussed in Remark 2.53, we have an adjunction

$$\operatorname{Fun}(\mathrm{CAlg}, \mathrm{Set}) \underset{\operatorname{Spec}}{\overset{\mathcal{O}}{\rightleftarrows}} \mathrm{CAlg}^{\mathrm{op}}.$$

2.7. **Closed and open subfunctor.** Roughly speaking, a *closed* subfunctor of an algebraic functor is a subfunctor defined by the vanishing of functions, while an *open* subfunctor is one defined by the nonvanishing of functions. This terminology is borrowed from topology, where the vanishing locus of a continuous function is closed and its nonvanishing locus is open. We will see later in Section 5.4 that there is in fact a topological interpretation of open subfunctors, though not of closed subfunctors.

Definition 2.55 (Vanishing and nonvanishing loci)**.** Let X be an algebraic functor and $F \subseteq \mathcal{O}(X)$ a set of functions on X.

(a) The *vanishing locus* of F is the subfunctor $\mathrm{V}(F) \subseteq X$ given by

$$\mathrm{V}(F)(R) = \{x \in X(R) \mid f(x) = 0 \text{ for all } f \in F\}.$$

(b) The *nonvanishing locus* of F is the subfunctor of $\mathrm{D}(F) \subseteq X$ given by

$$\mathrm{D}(F)(R) = \{x \in X(R) \mid (f(x))_{f\in F} \text{ generates the unit ideal in } R\}.$$

Remark 2.56. (a) It is clear that $\mathrm{V}(F)$ depends on the ideal (F) generated by F, and $\mathrm{D}(F)$ only on the *radical ideal* $\sqrt{(F)}$ generated by F, since an ideal is the unit ideal if and only if its radical is.

(b) We have the following implications:

$$(F) = \mathcal{O}(X) \quad \Longrightarrow \quad \mathrm{V}(F) = \varnothing_X \quad \Leftrightarrow \quad \mathrm{D}(F) = X.$$

Here $\varnothing_X$ is the functor with $\varnothing_X(0) = X(0)$ and $\varnothing_X(R) = \varnothing$ for $R \neq 0$. The reverse implication holds if X is affine, by Proposition 2.66 below.

Example 2.57 (Punctured affine spaces)*.* Let I be a set. The *punctured affine I-space* $\mathbf{A}^I \setminus \{0\}$ is the nonvanishing locus of the coordinate functions $\{x_i \mid i \in I\}$ on $\mathbf{A}^I = \mathrm{Spec}(\mathbf{Z}[x_i \mid i \in I])$. Explicitly:

$$(\mathbf{A}^I \setminus \{0\})(R) = \{(a_i)_{i\in I} \mid a_i \in R, (a_i)_{i\in I} = R\}.$$

Note that $\mathbf{A}^1 \setminus \{0\}$ is another name for the subfunctor $\mathbf{G}_m \subseteq \mathbf{A}^1$. An I-tuple in R generating the unit ideal is also called a *unimodular row* of length I.

Remark 2.58*.* A set of functions $F \subseteq \mathcal{O}(X)$ induces a map $f\colon X \to \mathbf{A}^F$. By definition, we have

$$\mathrm{V}(F) = f^{-1}(0) \quad \text{and} \quad \mathrm{D}(F) = f^{-1}(\mathbf{A}^F \setminus \{0\}).$$

Warning 2.59*.* The terminology suggests that $\mathrm{D}(F)$ should in some sense be the complement of $\mathrm{V}(F)$ in X. This is true when evaluated on fields, but it is not true in the category of algebraic functors. In fact, they are not even disjoint, since $\mathrm{V}(F)(0) = \mathrm{D}(F)(0) = X(0)$. We will see later that $\mathrm{D}(F)$ is the complement of $\mathrm{V}(F)$ (i.e., the largest disjoint suboject) in various subcategories of $\mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})$, such as the category of schemes. Even then, the converse fails: $\mathrm{V}(F)$ cannot be the complement $\mathrm{D}(F)$ in general, since it can happen that $\mathrm{V}(F) \neq \mathrm{V}(F')$ while $\mathrm{D}(F) = \mathrm{D}(F')$.

Proposition 2.60 (Formal properties of V and D)**.** *Let X be an algebraic functor.*

(a) For any family $(F_i)_{i\in I}$ of subsets of $\mathcal{O}(X)$,

$$\bigcap_{i\in I} \mathrm{V}(F_i) = \mathrm{V}\Big(\bigcup_{i\in I} F_i\Big) \quad \textit{and} \quad \bigcup_{i\in I} \mathrm{D}(F_i) \subseteq \mathrm{D}\Big(\bigcup_{i\in I} F_i\Big),$$

and the inclusion is an equality on local rings.

(b) For any finite family $F_1, \ldots, F_n$ of subsets of $\mathcal{O}(X)$, we have

$$\mathrm{D}(F_1) \cap \ldots \cap \mathrm{D}(F_n) = \mathrm{D}(F_1 \cdot \ldots \cdot F_n) \quad \textit{and} \quad \mathrm{V}(F_1) \cup \ldots \cup \mathrm{V}(F_n) \subseteq \mathrm{V}(F_1 \cdot \ldots \cdot F_n),$$

and the inclusion is an equality on integral domains.

Proof. (a) We will only show equality on local rings, since the rest are straightforward. Assume that $x \in \mathrm{D}(\bigcup_{i\in I} F_i)$, then $\sum_{i\in I}(f(x))_{f\in F_i} = 1$. Therefore, there must be $(f(x))_{f\in F_i} = 1$ for some $i \in I$. Otherwise, every $(f(x))_{f\in F_i}$ would be contained in the unique maximal ideal, hence also the sum.

(b) For the first equality, one direction is straightforward. For the other direction, let $x \in \mathrm{D}(F_i)(R)$ for all $i \in I$, then 1 can be written as $r_1 f_1(x) + \ldots + r_m f_m(x)$, where $r_i \in R$ and $f_i \in F_1$. But then we may write

$$r_1 f_1(x) + \ldots + r_m f_m(x) = (r_1 f_1(x) + \ldots + r_m f_m(x)) \cdot 1,$$

and express 1 again in terms of elements in $(f(x))_{f\in F_2}$. Iterating this process $(n-1)$ times gives us $x \in \mathrm{D}(F_1 \cdot \ldots \cdot F_n)$.

The inclusion is straightforward, hence we will only show that it is an equality on integral domains. Let $x \in \mathrm{V}(F_1 \cdot \ldots \cdot F_n)(R)$, where R is an integral domain. Assume that for each $i \in \{1, \ldots, n\}$, there exists $f_i \in F_i$ such that $f_i(x) \neq 0$. Then $f_1 \cdot \ldots \cdot f_n(x) \neq 0$, since R is an integral domain.

□

Example 2.61*.* Since 2 and 3 generate the unit ideal in $\mathbf{Z}$, we have $\mathrm{V}(2) \cap \mathrm{V}(3) = \mathrm{V}(1) = \varnothing$ as subfunctors of $\mathrm{Spec}(\mathbf{Z})$ (where $\varnothing$ is the scheme of Example 2.29). On the other hand, $\mathrm{D}(2) \cup \mathrm{D}(3) \neq \mathrm{D}(1) = \mathrm{Spec}(\mathbf{Z})$. For example, $\mathrm{id}_{\mathbf{Z}} \in \mathrm{Spec}(\mathbf{Z})(\mathbf{Z})$ belongs neither to $\mathrm{D}(2)(\mathbf{Z}) = \varnothing$ nor to $\mathrm{D}(3)(\mathbf{Z}) = \varnothing$.

Proposition 2.62 (Affineness of vanishing and nonvanishing loci)**.** *Let A be a ring.*

(a) *For any subset $F \subset A$, the quotient map $A \twoheadrightarrow A/(F)$ induces an isomorphism*

$$\operatorname{Spec}(A/(F)) \xrightarrow{\sim} \mathrm{V}(F) \subseteq \operatorname{Spec}(A).$$

In particular, $\mathrm{V}(F)$ is affine.

(b) *For any $f \in A$, the localization map $A \to A_f$ induces an isomorphism*

$$\operatorname{Spec}(A_f) \xrightarrow{\sim} \mathrm{D}(f) \subseteq \operatorname{Spec}(A).$$

In particular, $\mathrm{D}(f)$ is affine.

Proof. (a) By definition, we have

$$\mathrm{V}(F)(R) = \{\varphi \colon A \to R \mid \varphi(f) = 0 \text{ for all } f \in F\} = \operatorname{Spec}(A/(F))(R),$$

by the universal property of quotient.

(b) We have

$$\mathrm{D}(f)(R) = \{\varphi \colon A \to R \mid \varphi(f) \in R^\times\} = \operatorname{Spec}(A_f)(R),$$

by the universal property of localization.

□

If $F \subseteq A$ has more than one element, $\mathrm{D}(F) \subseteq \operatorname{Spec}(A)$ is usually not an affine scheme. For example, $\mathbf{A}^n \setminus \{0\}$ is not affine for $n \geq 2$ (see Example 2.78). This motivates the following definition:

Definition 2.63 (Quasi-affine scheme)**.** Let k be a ring. An algebraic k-functor X is a *quasi-affine k-scheme*, if there exists a k-algebra A and a *finite* subset $F \subseteq A$ such that $X \simeq \mathrm{D}(F) \subseteq \operatorname{Spec}(A)$. We denote $\mathrm{QAff}_k \subseteq \operatorname{Fun}(\mathrm{CAlg}, \mathrm{Set})$ the full subcategory of quasi-affine k-schemes. A *quasi-affine scheme* is a quasi-affine $\mathbf{Z}$-scheme.

Definition 2.64 (Closed and open subfunctors)**.** Let X be an algebraic fucntor.

(a) A subfunctor $Z \subseteq X$ is *closed*, if for every $x \colon \operatorname{Spec}(R) \to X$, we have $x^{-1}(Z) = \mathrm{V}(F)$ for some $F \subseteq R$.
(b) A subfunctor $U \subseteq X$ is *open*, if for every $x \colon \operatorname{Spec}(R) \to X$, we have $x^{-1}(U) = \mathrm{D}(F)$ for some $F \subseteq R$.

Warning 2.65. Vanishing loci are always closed subfunctors and nonvanishing loci are always open subfunctors, but the converse does not hold.

The following result generalizes the functorial Nullstellensatz (see Corollary 2.24):

Proposition 2.66 (Classification of closed and open subfunctors of affine schemes)**.** *Let A be a ring.*

(a) *The construction $F \mapsto \mathrm{V}(F)$ induces an order-reversing bijection*

$$\{\textit{ideal in } A\} \xrightarrow{\sim} \{\textit{closed subfunctors of } \operatorname{Spec}(A)\}.$$

(b) *The construction $F \mapsto \mathrm{D}(F)$ induces an order-preserving bijection*

$$\{\textit{radical ideals in } A\} \xrightarrow{\sim} \{\textit{open subfunctors of } \operatorname{Spec}(A)\}.$$

Proof. (a) We will first show that for each ideal F in A, the subfunctor $\mathrm{V}(F)$ is closed. Let $\varphi \colon A \to R$ be a map of rings[4]. We have the following pullback square:

$$\begin{array}{ccc} x^{-1}(\mathrm{V}(F)) & \longrightarrow & \mathrm{V}(F) \\ \downarrow & & \downarrow \\ \operatorname{Spec}(R) & \xrightarrow{x} & \operatorname{Spec}(A) \end{array}$$

[4]Another possible proof is to use Proposition 2.62 together with the (anti-)equivalence of categories induced by Spec and the description of pushouts. As this does not work the same way for non-vanishing sets, we give another proof

Evaluating this pullback on a commutative ring S and using the description of pullbacks in sets, we see that

$$\begin{aligned} x^{-1}(\mathrm{V}(F))(S) &\cong \{h \in \mathrm{Spec}(R)(S) \mid (h \circ \varphi)(f) = 0 \; \forall f \in F\} \\ &= \left\{h \in \mathrm{Spec}(R)(S) \mid h(\tilde{f}) = 0 \; \forall \tilde{f} \in (\varphi(F))\right\} \\ &= V(\varphi(F))(S). \end{aligned}$$

For $\psi : S \to S'$, the induced maps $x^{-1}(\mathrm{V}(F))(S) \to x^{-1}(\mathrm{V}(F))(S')$ are given by postcomposition with ψ. This shows $x^{-1}(\mathrm{V}(F)) = \mathrm{V}(\varphi(F))$.

For surjectivity, let Z be a closed subfunctor of $\mathrm{Spec}(A)$, then we have the following pullback square.

$$\begin{array}{ccc} \mathrm{V}(I) & \xrightarrow{\sim} & Z \\ \downarrow & & \downarrow \\ \mathrm{Spec}(A) & \xrightarrow{\sim} & \mathrm{Spec}(A). \end{array}$$

For injectivity, consider an ideal F of A, then we have $\mathrm{V}(F) \simeq \mathrm{Spec}(A/(F)) \subseteq \mathrm{Spec}(A)$ by Proposition 2.66. By Yoneda lemma, this is induced by the map $A \twoheadrightarrow A/(F)$. Hence, F can be recovered by the kernel of this map. It is clear that this map is order reversing.

(b) Well-definedness and surjectivity is similar to that of (a). For injectivity, we will show that $f \in F$ if and only if $\mathrm{D}(f) \subseteq \mathrm{D}(F)$, where $F \subseteq A$ is a radical ideal. If $f \in F$ it is clear that $D(f) \subseteq D(F)$. If $\mathrm{D}(f) \subseteq \mathrm{D}(F)$, then $\mathrm{D}(f)(A_f) \subseteq \mathrm{D}(F)(A_f)$, and in particular, $\mathrm{D}(F)(A_f)$ is non-empty. As $f \in F$, the localization map $\rho\colon A \to A_f$ satisfies $(\rho(g))_{g\in F} = A_f$. By equalizing denominators, this implies that there exists $a \in F$ and $n \in \mathbf{N}$ such that $\frac{a}{f^n} = 1$, i.e. $f \in \sqrt{F} = F$.

We can now conclude the injectivity as follows. For any open subfunctor $U \subseteq \mathrm{Spec}(A)$, we define the ideal

$$I(U) = \{f \in A \mid D(f) \subseteq U\}.$$

Then, by the above discussion, we have $I(\mathrm{D}(F)) = F$, implying that we may recover F from $\mathrm{D}(F)$.

□

Definition 2.67 (Closed and open immersions)**.** Let $f\colon Y \to X$ be a map of algebraic functors.

(a) f is a *closed immersion* or *closed embedding*, if it is a monomorphism, whose image ia a closed subfunctor of X.
(b) f is an *open immersion* or *open embedding*, if it is a monomorphism, whose image is an open subfunctor of X.

Definition 2.68 (Locally closed subfunctor, immersion)**.** Let X be an algebraic functor.

(a) A subfunctor $Y \subseteq X$ is *locally closed*, if there exists an open subfunctor $U \subseteq X$ containing Y as closed subfunctor.
(b) A map $f\colon Y \to X$ is an *immersion* if it is a monomorphism, whose image is a locally closed subfunctor of X.

Definition 2.69 (Closure of subfunctor)**.** Let X be an algebraic functor. The *closure* of a subfunctor $Y \subseteq X$ is the smallest closed subfunctor $\bar{Y} \subseteq X$ containing Y.

Remark 2.70. The closure of a subfunctor $Y \subseteq X$ exists, since intersections of closed subfunctors are closed. In particular, $\bar{Y} = \bigcap Z$ where $Z \subseteq X$ is a closed subfunctor containing Y.

Proposition 2.71 (Closure properties of immersions)**.** *(a) Consider a cartesian square of algebraic functors*

$$\begin{array}{ccc} Y' & \xrightarrow{f'} & X' \\ \downarrow & & \downarrow \\ Y & \xrightarrow[f]{} & X \end{array}$$

If f is a closed immersion, so is f'. The same holds for open immersion and for immersions.

(b) *Consider a commutative triangle of algebraic functors*

$$\begin{array}{ccc} Z & \xrightarrow{g} & Y \\ {\scriptstyle h}\downarrow & \swarrow{\scriptstyle f} & \\ X & & \end{array}$$

If f and g are closed immersions, so is h. If h is a closed immersion and f is a monomorphism, then g is a closed immersion. The same holds for open immersions and for immersions.

Proof. (a) Assume that f is an open immersion. Since monomorphism is stable under base change, the map f' is a monomorphism. By the pasting lemma, for each map $\mathrm{Spec}(R) \to X'$, we have the following commutative diagram

$$\begin{array}{ccccc} \mathrm{D}(F) & \longrightarrow & Y' & \longrightarrow & Y \\ \downarrow & & {\scriptstyle f'}\downarrow & & {\scriptstyle f}\downarrow \\ \mathrm{Spec}(R) & \longrightarrow & X' & \longrightarrow & X \end{array}$$

Therefore, the image of f' is an open subfunctor, hence f' is an open immersion. The proof for closed immersion is completely analogous.

Now, assume that f is an immersion, then there exists an open subfunctor U of X such that Y is closed in U. Then, by the pasting lemma, we have the following pullback square:

$$\begin{array}{ccc} Y' & \longrightarrow & Y \\ \downarrow & & \downarrow \\ U' & \longrightarrow & U \\ \downarrow & & \downarrow \\ X' & \longrightarrow & X, \end{array}$$

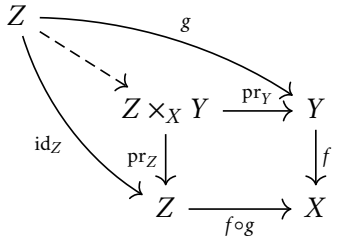
where U' is an open subfunctor of X'. Therefore, the map f' is an immersion.

(b) Let f and g be closed immersions. Since vanishing loci in affine schemes are affine, for each map $x\colon \mathrm{Spec}(R) \to X$, there exists an ideal I of R and an ideal J of R/I such that the following diagram commutes:

$$\begin{array}{ccc} \mathrm{V}(J) & \longrightarrow & Z \\ \downarrow & & \downarrow{\scriptstyle g} \\ \mathrm{Spec}(R/I) & \longrightarrow & Y \\ \downarrow & & \downarrow{\scriptstyle f} \\ \mathrm{Spec}(R) & \xrightarrow[x]{} & X \end{array}$$

But then we may consider $\mathrm{V}(J)$ as a closed subfunctor of $\mathrm{Spec}(R)$ via $\mathrm{V}(J) \simeq \mathrm{V}(\tilde{J})$, where $\tilde{J}$ is the preimage of the ideal J under the map $\pi\colon A \to A/I$. This proves that the composition of two closed immersion is again a closed immersion.

Now, assume that $f \circ g$ is a closed immersion and f is a monomorphism. Then we have the following commutative diagram:



By construction, pr_Z is a split epimorphism. Since f is a monomorphism, pr_Z is also a monomorphism, and hence an isomorphism. Therefore, the map $Z \to Z \times_X Y$ is an isomorphism, since it is a section of a monomorphism. By (a), the map pr_Y is a closed immersion, which implies that g is a closed immersion. We note that the above proof only uses the property that closed immersions are stable under base change. Therefore, the same proof holds for open immersions and immersions.

Assume that f and g are open immersions. In order to show that $f \circ g$ is an open immersion, by the following pullback diagram

$$\begin{array}{ccc} U & \longrightarrow & Z \\ \downarrow & & \downarrow g \\ \mathrm{D}(F) & \longrightarrow & Y \\ \downarrow & & \downarrow f \\ \mathrm{Spec}(A) & \longrightarrow & X, \end{array}$$

we may assume that $X = \mathrm{Spec}(A)$ for some ring A. We have to show that, for any subset $F \subseteq A$ and open subfunctor $U \subseteq \mathrm{D}(F)$, the subfunctor U is open in $\mathrm{Spec}(A)$.

For every element $f \in F$, consider the inclusion $\mathrm{D}(f) \subseteq \mathrm{D}(F)$. Since U is open in $\mathrm{D}(F)$, and $\mathrm{D}(f) \simeq \mathrm{Spec}(A_f)$ is affine (Proposition 2.62), its preimage $U \cap \mathrm{D}(f)$ in $\mathrm{Spec}(A_f)$ has the form $\mathrm{D}(\bar{H}_f)$ for some ideal $\bar{H}_f \subseteq A_f$. By clearing up the denominators of the elements in $\bar{H}_f$, we can assume that $\bar{H}_f$ is the image of a subset $H_f \subseteq A$, so that $\mathrm{D}(\bar{H}_f) \subseteq \mathrm{Spec}(A_f)$ is the preimage of $\mathrm{D}(H_f) \subseteq \mathrm{Spec}(A)$. Then by Proposition 2.60, we have

$$\mathrm{D}(fH_f) = \mathrm{D}(H_f) \cap \mathrm{D}(f) = \mathrm{D}(\bar{H}_f) = U \cap \mathrm{D}(f).$$

Let $H = \bigcup_{f \in F} fH_f \subseteq A$. We claim that $U = \mathrm{D}(H)$. We have to show that map $\varphi \colon A \to R$ is an element of $U(R)$ if and only if it is an element of $\mathrm{D}(H)(R)$.

For the inclusion $U \subseteq \mathrm{D}(H)$, suppose that the map φ is an element of $U(R)$. We then want to show that $(\varphi(h))_{h \in H} = R$. Since $U \subseteq \mathrm{D}(F)$, we have $(\varphi(f))_{f \in F} = R$. For every element $f \in F$, the following diagram commutes:

$$\begin{array}{ccc} A & \xrightarrow{\varphi} & R \\ \downarrow & & \downarrow \\ A_f & \dashrightarrow & R_{\varphi(f)} \end{array}$$

Since the functoriality of a subfunctor is given by restriction of the functoriality of the ambient functor, the map $A \to R_{\varphi(f)}$ is an element of $U(R_{\varphi(f)})$. As $\varphi \in D(f)(R_{\varphi(f)})$, the map is already an element in the intersection $(U \cap \mathrm{D}(f))\,(R_{\varphi(f)}) = \mathrm{D}(fH_f)(R_{\varphi(f)})$, by the explicit description of the pullback, i.e., the image of $\varphi(fH_f)$ in R_f generates the unit ideal. Hence, by multiplying the common denominators of the expansion of 1, the ideal $(\varphi(fH_f))$ contains some power of $\varphi(f)$, i.e. $\varphi(f) \in \sqrt{(\varphi(fH_f))}$. In particular, we have $\varphi(f) \in \sqrt{(\varphi(h))_{h \in H}}$. Taking the union gives us the inclusion $\varphi(F) \subseteq \sqrt{(\varphi(h))_{h \in H}}$. Therefore, we have

$$R = \sqrt{(\varphi(f))_{f \in F}} \subseteq \sqrt{(\varphi(h))_{h \in H}}.$$

In particular, we have $1 \in (\varphi(h))_{h\in H}$, hence $(\varphi(h))_{h\in H} = R$.

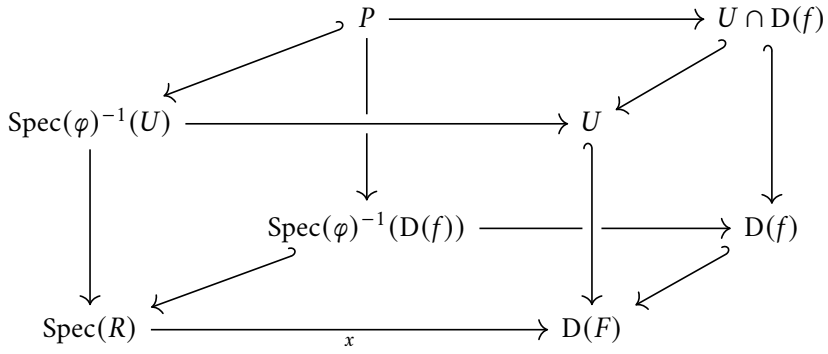
For $\mathrm{D}(H) \subseteq U$, suppose that we have a map $\varphi\colon A \to R$ such that $(\varphi(h))_{h\in H} = R$, we want to show that $\varphi \in U(R)$. By definition, we have $H \subseteq (f)_{f\in F}$, hence $\mathrm{D}(H) \subseteq \mathrm{D}(F)$. By Yoneda, we obtain a map $x\colon \operatorname{Spec}(R) \to \mathrm{D}(F)$. Since $U \subseteq \mathrm{D}(F)$ is open, its preimage in $\operatorname{Spec}(R)$ has the form $\mathrm{D}(I)$ for some radical ideal $I \subseteq R$. If we can show that $\mathrm{D}(I) \simeq \operatorname{Spec}(R)$, then every element $x\colon \operatorname{Spec}(R) \to \mathrm{D}(F)$ factors through U. Intersecting $\mathrm{D}(I)$ with the preimage $\mathrm{D}(\varphi(f))$ of $\mathrm{D}(f)$ under $\operatorname{Spec}(R) \to \operatorname{Spec}(A)$, we obtain

$$\mathrm{D}(I) \cap \mathrm{D}(\varphi(f)) \simeq \operatorname{Spec}(\varphi)^{-1}(U) \cap \operatorname{Spec}(\varphi)^{-1}(\mathrm{D}(f)),$$

since we have the following pullback diagram:

$$\begin{array}{ccccc}
\operatorname{Spec}(\varphi)^{-1}(U) = \mathrm{D}(I) & \longrightarrow & U & \xrightarrow{\sim} & U \\
\downarrow & & \downarrow & & \downarrow \\
\operatorname{Spec}(R) & \xrightarrow{x} & \mathrm{D}(F) & \hookrightarrow & \operatorname{Spec}(A).
\end{array}$$

Consider now the following commutative cube where every face is a pullback



The pasting law of pullback diagrams then gives us

$$P \cong \operatorname{Spec}(\varphi)^{-1}(U) \cap \operatorname{Spec}(\varphi)^{-1}(\mathrm{D}(f)) \simeq \operatorname{Spec}(\varphi)^{-1}(U \cap \mathrm{D}(f)) \simeq \mathrm{D}((\varphi(fH_f))).$$

Therefore, we have the following equalities:

$$\mathrm{D}(\varphi(f)I) \simeq \mathrm{D}(I) \cap \mathrm{D}(\varphi(f)) \simeq \mathrm{D}((\varphi(fH_f))).$$

By Proposition 2.66, we then have

$$\sqrt{\varphi(f)I} = \sqrt{(\varphi(fH_f))}.$$

Summing this equality over all $f \in F$ and taking the radical, we obtain

$$R = \sqrt{(\varphi(H))} = \sqrt{\varphi(F)I} = R,$$

since the radical of the sum of ideals is the radical of the sum of radicals of the ideals.

Finally, assume that f and g are immersions, we need to show that $f \circ g$ is an immersion. Since the composition of open (resp. closed) immersions is open (resp. closed), it suffices to prove the following. If $Z \subseteq X$ is closed and $U \subseteq Z$ is open, then $U \subseteq X$ is locally closed.

First assume that $X = \operatorname{Spec}(A)$. We need to find an open subfunctor $\hat{U} \subseteq X$ in which U is closed. By Proposition 2.66, we can write $Z = \operatorname{Spec}(A/I)$ and $U = \mathrm{D}(F)$ for some ideals $I \subseteq A$ and $F \subseteq A/I$. Let $\hat{F}$ be the preimage of F in A and let $\hat{U} = \mathrm{D}(\hat{F})$ be the corresponding open subfunctor of $\operatorname{Spec}(A)$. By definition, we then have $\hat{U} \cap Z = U$. Since closed immersions are stable under base change, U is closed in $\hat{U}$, as desired.

Functoriality of the pullback gives us functoriality of the construction of $\hat{U}$ is functorial in A in the following sense: given any $\varphi\colon A \to B$, we denote the corresponding pullbacks of Z and U by Z_B and U_B. Then the open subfunctor $\widehat{U_B}$ in $\operatorname{Spec}(B)$, defined as above using $U_B \subseteq Z_B \subseteq \operatorname{Spec}(B)$, is the preimage $\hat{U}_B$ of $\hat{U}$.

For X arbitrary, we can therefore define the subfunctor $\widehat{U} \subseteq X$ by

$$\widehat{U}(R) = \{x\colon \operatorname{Spec}(R) \to X \mid \widehat{x^{-1}(U)} = \widehat{U_R}\} \subseteq X(R).$$

For any $\operatorname{Spec}(A) \to X$, we then have $\hat{U}_A = \widehat{U_A} \subseteq \operatorname{Spec}(A)$, so that $\hat{U}$ is open in X and U is closed in $\hat{U}$. □

2.8. **Zariski descent.** Let $(f_i)_{i\in I}$ be a family of element in a ring R that generates the unit ideal. Then the intersection of the vanishing loci $\mathrm{V}(f_i)$ is the empty scheme, but in general, it is not true that $\operatorname{Spec}(R)$ is the union of the nonvanishing loci $\mathrm{D}(f_i)$ (see Example 2.61). In this section, we will show that this becomes true if we compute the union in the category of affine schemes. Concretely, this means that a map $\operatorname{Spec}(R) \to \operatorname{Spec}(S)$ is uniquely determined by a family of maps $\mathrm{D}(f_i) \to \operatorname{Spec}(S)$ that agrees on all the intersections $\mathrm{D}(f_i) \cap \mathrm{D}(f_j)$. This is one of the most important result in the foundations of algebraic geometry, which we will later recast as the statement that *affine schemes satisfy Zariski descent*.

Theorem 2.72 (Zariski descent for modules)**.** *Let R be a ring and $(f_i)_{i\in I}$ a family of elements of R generating the unit ideal.*

(a) (Descent for morphisms) For any R-modules M and N, the diagram

$$\operatorname{Map}(N, M) \to \prod_{i\in I} \operatorname{Map}(N_{f_i}, M_{f_i}) \rightrightarrows \prod_{i,j\in I} \operatorname{Map}(N_{f_if_j}, M_{f_if_j})$$

is an equalizer.

(b) (Descent for objects) Suppose given

- *an R_{f_i}-module M_i for each $i \in I$ and*
- *an $R_{f_if_j}$-linear isomorphism $\alpha_{ij}\colon (M_i)_{f_j} \xrightarrow{\sim} (M_j)_{f_j}$ for each $(i,j) \in I^2$,*
- *such that $\alpha_{jk} \circ \alpha_{ij} = \alpha_{ik}\colon (M_i)_{f_jf_k} \xrightarrow{\sim} (M_k)_{f_if_j}$ for each $(i,j,k) \in I^3$.*

Then there exists an R-module M with R_{f_i}-linear isomorphisms $\beta_i\colon M_{f_i} \xrightarrow{\sim} M_i$ for all $i \in I$ and such that $\alpha_{ij} \circ (\beta_i)_{f_j} = (\beta_j)_{f_i} : M_{f_if_j} \xrightarrow{\sim} (M_j)_{f_i}$ for all $(i,j) \in I^2$. Moreover, this data is unique up to unique isomorphism.

Proof. (a) For each $f \in R$, we have by the basechange adjunction along $R \to R_f$

$$\operatorname{Map}_{R_f}(N_f, M_f) \simeq \operatorname{Map}_R(N, M_f).$$

Therefore, the general case follows from the special case where $N = R$ by appyling $\operatorname{Map}(N, -)$, since $\operatorname{Map}(-,-)$ preserves limits in both variables. By unwrapping the universal property of equalizer, we need to show that the canonical map

$$M \xrightarrow{\sim} \{(m_i \in M_{f_i})_{i\in I} \mid m_i = m_j \text{ in } M_{f_if_j} \text{ for all } i, j \in I\}$$

is an isomorphism.

For injectivity, since the above map is a map between R-modules, it suffices to show that for each $m \in M$ such that $m = 0$ in M_{f_i}, we have $m = 0$. By assumption, for each $i \in I$, there exists $k_i \in \mathbf{N}$ such that $f_i^{k_i} m = 0$ in M. Since $(f_i)_{i\in I} = R$, there exists $a_i \in R$, all but finitely many zero, such that $\sum_{i\in I} a_i f_i = 1$. This implies that $1 = 1^n = (\sum_{i\in I} a_i f_i)^n$ for some large $n \in \mathbf{N}$. As almost all $a_i = 0$, we may choose n such that $(\sum_{i\in I} a_i f_i)^n \in (f_i^{k_i})_{i\in I}$. Thereby, $(f_i^{k_i})_{i\in I} = R$, and hence there exists $b_i \in R$ for $i \in I$, again almost all 0, such that $1 = \sum_{i\in I} b_i f_i^{k_i}$. But then

$$m = 1 \cdot m = \sum_{i\in I} b_i f_i^{k_i} m = 0.$$

For surjectivity, let $(m_i)_{i\in I}$ be a family of elements such that $m_i \in M_{f_i}$ and $m_i = m_j$ in $M_{f_if_j}$. We write $m_i = \frac{n_i}{f_i^{k_i}}$ for some $n_i \in M$ and $k_i \in \mathbf{N}$ for every $i \in I$. Then the condition that $m_i = m_j$ in $M_{f_if_j}$ implies

that there exists some r_{ij} such that

$$(f_if_j)^{r_{ij}}f_j^{k_j}n_i = (f_if_j)^{r_{ij}}f_i^{k_i}n_j.$$

Assume that I is finite, then we may choose $r = \max_{i,j\in I}(r_{ij})$. As before, we have $1 = \sum_{i\in I} a_i f_i^{k_i+r}$ for some $a_i \in R$. Let $m = \sum_{i\in I} a_i f_i^r n_i$, and choose $* \in I$. Then in M_{f_*}, we have

$$m = \frac{1}{f_*^{r+k_*}}\Big(\sum_{i\in I} a_i f_i^r f_*^{r+k_*} n_i\Big) = \frac{n_*}{f^{k_*}}\Big(\sum_{i\in I} a_i f_i^{r+k_i}\Big) = \frac{n_*}{f^{k_*}} = m_*.$$

Now assume that I is arbitrary. Since 1 can be expressed as a finite sum of elements from $(f_i)_{i\in I}$, there exists $J \subseteq I$ such that $(f_j)_{j\in J} = R$. Therefore, by what we have shown, there exists a unique $m \in M$ such that the image in M_{f_j} is m_j for all $j \in J$. Now, pick any $* \in I$, we need to show that $m = m_*$ in M_{f_*}. But then apply the finite case to $J \cup \{*\}$, there exist a unique $m' \in M$ such that $m' = m_i$ in M_{f_i} for all $i \in J \cup \{*\}$. By the uniqueness of m, we have $m = m'$ and $m = m_*$ in M_{f_*}.

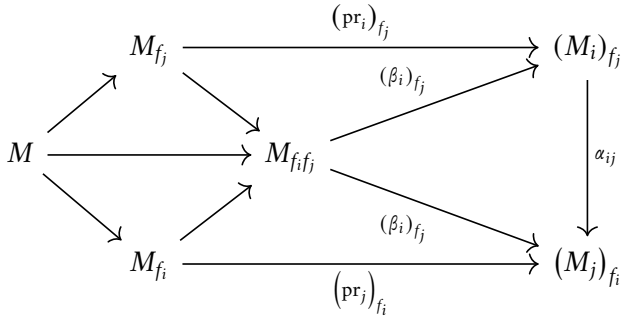
(b) We denote M as the limit of the following diagram:

$$M = \lim\Big(\prod_{i\in I} M_i \rightrightarrows \prod_{i,j\in I}(M_i)_{f_j}\Big),$$

and we denote the three morphisms as follows:

$$a_{ij}\colon \prod_{i\in I} M_i \xrightarrow{\mathrm{pr}_i} M_i \xrightarrow{\mathrm{L}_j} (M_i)_{f_j}, \quad b_{ij}\colon \prod_{i\in I} M_i \xrightarrow{\mathrm{pr}_j} M_j \xrightarrow{\mathrm{L}_i} (M_j)_{f_i} \xrightarrow{\alpha_{ji}}_{\simeq} (M_i)_{f_j}, \quad \beta_i\colon M_{f_i} \to M_i,$$

where L_i denote the map to the localization of f_i and β_i is the unique map induced from the projection $M \to \prod_{i\,\inf I} M_i \to M_i$. From the definition of β_i it follows that $\alpha_{ij}\circ(\beta_i)_{f_j} = \big(\beta_j\big)_{f_i}$. Indeed, the following diagram commutes



using the universal property of the localization, in particular the uniqueness of the induced map.

Assume that I is a finite set and fix an element $* \in I$. Since the localization functor $(-)_{f_*}$ preserves pullbacks and products, it also preserves equalizers (and hence all finite limits). Consider the solid part of the following diagram:

$$\begin{array}{ccccc}
M_{f_*} & \xrightarrow{((\beta_i)_{f_*})_{i\in I}} & \prod_{i\in I}(M_i)_{f_*} & \underset{b}{\overset{a}{\rightrightarrows}} & \prod_{i,j\in I}(M_i)_{f_*f_j} \\
{\scriptstyle\beta_*}\big\downarrow & & \big\downarrow{\scriptstyle(\alpha_{i*})_{i\in I}} & & \big\downarrow{\scriptstyle((\alpha_{i*})_{f_j})_{i,j\in I}} \\
M_* & \xrightarrow{\mathrm{can}} & \prod_{i\in I}(M_*)_{f_i} & \underset{\tilde{b}}{\overset{\tilde{a}}{\rightrightarrows}} & \prod_{i,j\in I}(M_*)_{f_if_j}
\end{array}$$

The lower row is an equalizer diagram, by (a), where

$$a_{ij}\colon \prod_{i\in I}(M_*)_{f_i} \xrightarrow{\mathrm{pr}_i} (M_*)_{f_i} \xrightarrow{\mathrm{L}_j} (M_*)_{f_if_j}, \quad b_{ij}\colon \prod_{i\in I}(M_{f_i}) \xrightarrow{\mathrm{pr}_j} (M_*)_{f_j} \xrightarrow{\mathrm{L}_i} (M_*)_{f_jf_i} \simeq (M_*)_{f_if_j}$$

The right square with a commutes as the map L_j is the counit of the basechange adjunction along $R \to R_{f_j}$ and hence natural. It commutes with b, since the following diagram commutes

$$\begin{array}{ccc} (M_j)_{f_i f_*} & \xrightarrow{\alpha_{ji}} & (M_i) f_j f_* \\ & \searrow^{\alpha_{j*}} & \downarrow{\scriptstyle \alpha_{i*}} \\ & & (M_*)_{f_i f_j}, \end{array}$$

by assumption. By the universal property of the equalizer, there exists a unique dashed map filling the diagram on the left. As the α where isomorphisms, so is the dashed map. We claim, that the dashed map is explicitly given by β_*. But this is clear, since $\alpha_{ij} \circ (\beta_i)_{f_j} = (\beta_j)_{f_i}$, thus β_* makes the whole diagram commute, and hence, the uniqueness of the map induced from the universal property of the equalizer implies the claim.

For the uniqueness, assume that there exists two data

$$((M_i)_{i \in I}, (\beta_i)_{i \in I}), \quad \text{and} \quad ((N_i)_{i \in I}, (\gamma_i)_{i \in I}).$$

Then for each $i \in I$, we have an isomorphism

$$\varphi_i \colon M_{f_i} \overset{\beta_i}{\simeq} M_i \overset{\gamma^{-1}}{\simeq} N_{f_i}.$$

Furthermore, the family $(\varphi_i)_{i \in I}$ is a compatible family of maps, in the sense that the following diagram commutes by the cocycle condition $\alpha_{jk}\alpha_{ij} = \alpha_{ik}$:

$$\begin{array}{ccccc} M_{f_i f_j} & \xrightarrow{\beta_i} & (M_i)_{f_j} & \xrightarrow{\gamma_i^{-1}} & N_{f_i f_j} \\ \simeq\downarrow & & \downarrow{\scriptstyle \alpha_{ij}} & & \downarrow\simeq \\ M_{f_j f_i} & \xrightarrow[\beta_j]{} & (M_j)_{f_i} & \xrightarrow[\gamma_j^{-1}]{} & N_{f_j f_i} \end{array}$$

Therefore, by (a), there exists a unique map $\varphi \colon M \to N$ such that $\varphi_{f_i} = \varphi_i$. Since localization is exact, we have $\ker(\varphi)_{f_i} = \ker(\varphi_{f_i})$ for each $i \in I$, hence by (a), we have $\ker(\varphi) = 0$. Similarly we can show that $\operatorname{coker}(\varphi) = 0$ and hence φ is an isomorphism.

Now assume that I is arbitrary. Then there exists a finite set $J \subseteq I$ such that $(f_j)_{j \in J} = R$ as before. Then for each finite set J' such that $J \subseteq J' \subseteq I$, we have the following diagram

$$\begin{array}{ccccc} M_{J'} & \longrightarrow & \prod_{i \in J'} M_i & \rightrightarrows & \prod_{i,j \in J'} (M_i)_{f_j} \\ \downarrow & & \downarrow & & \downarrow \\ M_J & \longrightarrow & \prod_{i \in J} M_i & \rightrightarrows & \prod_{i,j \in J} (M_i)_{f_i}, \end{array}$$

where M_J and $M_{J'}$ are the R-modules constructed as above. Since $J \subseteq J'$ the module $M_{J'}$ is also an equalizer of the lower diagram, hence we obtain an unique isomorphism $M_{J'} \to M_J$. If we take the limit[5] of the above filtered diagram indexed by finite subsets J' of I, we obtain the isomorphism $M \simeq M_J$. The uniqueness follows from the uniqueness in the finite case.

□

Corollary 2.73. *Let R be a ring and $(f_i)_{i \in I}$ a family of elements of R generating the unit ideal. For any R-module M, the diagram*

$$M \to \prod_{i \in I} M_{f_i} \rightrightarrows \prod_{i,j \in I} M_{f_i f_j}$$

is an equalizer in Mod_R.

Proof. Let $N = R$ in Theorem 2.72 gives us the result. □

[5]Remember that limits are contravariant in the diagram.

Corollary 2.74 (Zariski descent for affine schemes)**.** *Let R be a ring and let $(f_i)_{i\in I}$ be a family of elements of R generating the unit ideal. For any affine scheme X, the diagram*

$$X(R) \to \prod_{i\in I} X(R_{f_i}) \rightrightarrows \prod_{i,j\in I} X(R_{f_if_j})$$

is an equalizer.

Proof. Let $M = R$ in Corollary 2.73. Since X is affine, we have $X \simeq \mathrm{Map}(A, -)$. The result follows from that $\mathrm{Map}(-,-)$ preserves limits in both variables. □

Remark 2.75. Consider the following pushout diagram in CAlg:

$$\begin{array}{ccc} R & \longrightarrow & R_{f_j} \\ \downarrow & & \downarrow \\ R_{f_i} & \longrightarrow & R_{f_i} \otimes_R R_{f_j} \end{array}$$

Since $R_{f_i} \otimes_R R_{f_j} \simeq R_{f_if_j}$ and $\mathrm{Spec}(-)$ preserves limits, we have the following pullback diagram:

$$\begin{array}{ccc} \mathrm{Spec}(R_{f_if_j}) & \hookrightarrow & \mathrm{Spec}(R_{f_j}) \\ \downarrow & & \downarrow \\ \mathrm{Spec}(R_{f_i}) & \hookrightarrow & \mathrm{Spec}(R) \end{array}$$

Therefore, we have $\mathrm{Spec}(R_{f_if_j}) \simeq \mathrm{D}(f_i) \cap \mathrm{D}(f_j)$, since $\mathrm{Spec}(R_{f_i}) \simeq \mathrm{D}(f_i)$ and $\mathrm{Spec}(R_{f_j}) \simeq \mathrm{D}(f_j)$, by Proposition 2.62. Together with $X(R) \simeq \mathrm{Map}(\mathrm{Spec}(R), X)$ and continuity of $\mathrm{Map}(-,-)$, Corollary 2.74 says that $\mathrm{Spec}(R)$ is the union of the open subschemes $\mathrm{D}(f_i)$ in Aff. More precisely, if $\mathrm{Glue}(I)$ is the poset with objects $i \in I$ and $(i,j) \in I^2$, and morphisms $i \leftarrow (i,j) \to j$ for all $i, j \in I$, then $\mathrm{Spec}(R)$ is the colimit of the diagram $\mathrm{Glue}(I) \to \mathrm{Aff}$ sending i to $\mathrm{D}(f_i)$ and (i,j) to $\mathrm{D}(f_i) \cap \mathrm{D}(f_j)$.

The following result is a generalization of Corollary 2.74, of which it is in fact a formal consequence:

Corollary 2.76 (Functions on nonvanishing loci)**.** *Let R be a ring and $(f_i)_{i\in I}$ a family of elements of R with image $F \subseteq R$. For any affine scheme X, there is an equalizer diagram*

$$\mathrm{Map}(\mathrm{D}(F), X) \to \prod_{i\in I} X(R_{f_i}) \rightrightarrows \prod_{i,j\in I} X(R_{f_if_j}).$$

Proof. By Theorem 2.9, every presheaf can be written as the colimit of representable presheaves, i.e.

$$\mathrm{D}(F) \simeq \operatorname*{colim}_{\mathrm{El}(\mathrm{D}(F))} \mathrm{Spec}(S) \quad \text{and} \quad \mathrm{Map}(\mathrm{D}(F), X) \simeq \lim_{\mathrm{El}(\mathrm{D}(F))} X(S).$$

More explicitly, the objects of $\mathrm{El}(\mathrm{D}(F))$ are ring maps $\varphi\colon R \to S$ such that $(\varphi(f))_{f\in F} = S$, and morphisms are ring morphisms $R \to R'$ satisfying the obvious commutation condition. By Corollary 2.74, the following diagram is an equalizer diagram:

$$X(S) \to \prod_{i\in I} X(S_{\varphi(f_i)}) \rightrightarrows \prod_{i\in I} X(S_{\varphi(f_i)\varphi(f_j)}). \tag{2.77}$$

We wish to take the limit of this diagram. First observe that, for each $g \in F$, the inclusion $\mathrm{D}(g) \subseteq \mathrm{D}(F)$, induces an isomorphism

$$\lim_{\varphi\in\mathrm{El}(\mathrm{D}(F))} X(S_{\varphi(g)}) \xrightarrow{\sim} \lim_{\varphi\in\mathrm{El}(\mathrm{D}(g))} X(S_{\varphi(g)}).$$

Here the indexing category on the left runs over all rings S and morphisms $\varphi : R \to S$ such that $(\varphi(F)) = S$. Given two such objects $\varphi : R \to S$ and $\tilde{\varphi} : R \to S'$ together with a morphism $f : S \to S'$ making the obvious triangle commute, the transition morphism corresponding to f is the $X(f_{\varphi g})$ where $f_{\varphi(g)} : S_{\varphi(g)} \to S'_{\tilde{\varphi}(g)}$ is

the unique morphism induced by f. The right hand indexing category only runs over those $R \to S$, such that already $(\varphi(g)) = S$, or equivalently, such that $\varphi(g) \in S^\times$, and the transition morphisms are the same.

To see that the two limits are isomorphic, we are going to show that every leg of the left hand limit is isomorphic to a leg of the right hand limit. To do so, notice that for each object $\varphi\colon R \to S \in \mathrm{El}(\mathrm{D}(F))$, postcomposition with $L_{\varphi(g)} : S \to S_{\varphi(g)}$ gives us an object $\tilde{\varphi} : R \to S_{\varphi(g)} \in \mathrm{El}(\mathrm{D}(g)) \subseteq \mathrm{El}(\mathrm{D}(F))$. This morphism induces the following commutative diagram:

$$\begin{array}{ccc} \lim_{\mathrm{El}(\mathrm{D}(F))} X(S_{\varphi(g)}) & \longrightarrow & X(S_{\varphi(g)}) \\ & \searrow & \downarrow{\scriptstyle (L_{\varphi(g)})_{\varphi(g)}} \\ & & X((S_{\varphi(g)})_{\tilde{\varphi}(g)}) \end{array}$$

Now the right hand vertical map is an isomorphism, since the map $S \to S_{\varphi(g)}$ becomes an isomorphism after taking localization $(-)_{\varphi(g)}$. This shows the claim, and that the two limits agree.

Further, the category $\mathrm{El}(\mathrm{D}(g))$ admits an initial object $R \to R_g$. Combining these isomorphisms, we obtain the following isomorphism

$$\lim_{\mathrm{El}(\mathrm{D}(F))} X(S_g) \simeq X(R_g).$$

Thus, the proposition follows from taking the limit indexed by $\mathrm{El}(\mathrm{D}(F))$ of the equalizer diagram (2.77). □

Example 2.78. The inclusion $\mathbf{A}^I \setminus \{0\} \hookrightarrow \mathbf{A}^I$ induces an isomorphism $\mathcal{O}(\mathbf{A}^I) \xrightarrow{\sim} \mathcal{O}(\mathbf{A}^I \setminus \{0\})$, whenever $|I| \geq 2$.

Indeed, if we take $R = k[x_i \mid i \in I]$ and $(f_i)_{i\in I} = (x_i)_{i\in I}$ in Corollary 2.76 with $X = \mathbf{A}^1$, we obtain an equalizer diagram as follows:

$$\mathrm{Map}(\mathrm{D}(\{x_i\}_{i\in I}), \mathbf{A}^1) = \mathcal{O}(\mathbf{A}^I \setminus \{0\}) \longrightarrow \prod_{i\in I} R_{x_i} \rightrightarrows \prod_{i,j\in I} R_{x_i x_j}$$

We want to describe the equalizer explicitly. An element in the above equalizer is a tuple $(p_i)_{i\in I}$ with $p_i \in R_{x_i}$, such that $p_i = p_j \in R_{x_i x_j}$ for all $i, j \in I$. Using that R_{x_i} injects into $R_{x_i x_j}$ for all $i, j \in I$, using that x_j is not a zero divisor in R_{x_i}, this implies that p_i has only powers of x_i in its denominator, as well as only powers of x_j for all $j \neq i \in I$. But this is only possible, if the only elements in the denominator are units in k, hence there exists an element $p \in R$ such that $\frac{p}{1} = p_i \in R_{x_i}$ for all i. This shows that the canonical map $R \to \mathcal{O}(\mathbf{A}^1 \setminus \{0\})$ is surjective. As R injects into R_{x_i} and $R_{x_i x_j}$ for all $i, j \in I$, injective morphisms in CAlg are the same as monomorphisms, monomorphisms are limits, and limits commute, the canonical map is furthermore injective, hence an isomorphism.

Since $\mathcal{O} : \mathrm{Aff}^{\mathrm{op}} \to \mathrm{CAlg}$ is an equivalence of categories, this implies that $\mathbf{A}^I \setminus \{0\}$ is not an affine scheme. In particular, if $n \geq 2$, $\mathbf{A}^n \setminus \{0\}$ is an example of an quasi-affine scheme that is not affine.

Definition 2.79 (Zariski-local property)**.** A property P of modules (resp. of linear maps, of algebras, etc.) is *Zariski-local*, if for any ring R and family $(f_i)_{i\in I}$ generating the unit ideal in R, an R-module M (resp. an R-linear map $M \to N$, an R-algebra A, etc.) has property P if and only if, for each $i \in I$, the R_{f_i}-module M_{f_i} (resp. the R_{f_i}-linear map $M_{f_i} \to N_{f_i}$, the R_{f_i}-algebra A_{f_i}, etc.) has property P.

Proposition 2.80 (Examples of Zariski-local properties)**.** *The following properties of modules are Zariski-local:*

(a) *being zero,*
(b) *finite generation,*
(c) *finite presentation,*
(d) *projectivity,*
(e) *flatness.*

The following properties of linear maps are Zariski-local:

(f) *being zero,*
(g) *injectivity,*
(h) *surjectivity,*

(*i*) *bijectivity.*

The following properties of algebras are Zariski-local:

(*j*) *finite generation,*
(*k*) *finite presentation.*

The following properties of sequence of modules are Zariski local:

(*l*) *exactness.*

Proof. Notice that we may always choose a finite subfamily $J \subseteq I$ such that $(f_j)_{f\in J} = R$. Thus we may assume I to be finite, if necessary.

(a) Assume that the R-module M is zero. Since the localization-at-f_i functor is additive, being the same as tensoring with R_{f_i}, we have $M_{f_i} = 0$ for all $i \in I$. Conversely, assume that $M_{f_i} = 0$ for all $i \in I$. Then by Corollary 2.73, the limit M is again 0.

(b) This follows from (l). Alternatively, assume that M is finitely generated, i.e., there exists a surjection $\varphi\colon R^n \twoheadrightarrow M$ for some $n \in \mathbf{N}$. Since $(-)_{f_i} \simeq (-) \otimes_R R_{f_i}$ is a left adjoint, it preserves colimits, hence also surjections, surjections being the same as epimoprhisms in the category of R-modules. In particular, the map φ_{f_i} is a surjection for each $i \in I$.

Conversely, assume that M_{f_i} is finitely generated for all $i \in I$ and $|I| = k$. Then for each $i \in I$, there exists a map $\varphi_i\colon R^{n_i} \to M$ such that $(\varphi_i)_{f_i}$ is surjective. These maps together define a map $\sum_{i\in I} \varphi_i\colon R^{n_1+\ldots+n_k} \to M$. In order to show that this map is a surjection, it suffices by (h) to show that it is a surjection under $(-)_{f_i}$ for all $i \in I$. Observe that we have the following commutative diagram:

$$\begin{array}{ccc} R^{n_1+\ldots+n_k} & \twoheadrightarrow & M_{f_i} \\ \uparrow{\scriptstyle \iota} & \nearrow_{(\varphi_i)_{f_i}} & \\ R^{n_i} & & \end{array}$$

Therefore, the upper horizontal map must be surjective. Notice that the map ι is injective, since $(-)_{f_i}$ is exact for each $i \in I$.

(c) This follows from (l). Alternatively, assume that M is finitely presented, then there exists $n \in \mathbf{N}$, a finitely generated R-module K with an exact sequence

$$0 \to K \to R^n \to M \to 0.$$

By exactness of the localization functor, we have the following exact sequence

$$0 \to K_{f_i} \to R^n_{f_i} \to M_{f_i} \to 0.$$

Since $R^n_{f_i}$ is a finitely presented R_{f_i}-module and by (b), K_{f_i} is finitely generated, we have M_{f_i} is finitely presented.

Conversely, assume that M_{f_i} is finitely presented for each $i \in I$. Then in particular, M_{f_i} is finitely generated for each $i \in I$. Hence M is finitely generated, i.e., there exists a surjection $\pi\colon R^n \to M$ for some $n \in \mathbf{N}$. It suffices to show that the kernel K of the map π is finitely generated. Localizing at f_i gives us the following exact sequence

$$0 \to K_{f_i} \to R^n_{f_i} \to M_{f_i} \to 0.$$

Since M_{f_i} is finitely presented, $R^n_{f_i}$ is finitely generated, hence K_{f_i} is finitely generated. By (b) again, K is finitely generated.

(d) The proof of this statement is difficult, the original proof can be found in [RG71]. See also [Sta26, Tag 05A9] for an extension to the faithfully flat topology, . reference

(e) Note that M is flat if and only if $(-) \otimes_R M$ preserves injections. Assume that M is flat, and $\varphi\colon P \to Q$ be an injection in $\mathrm{Mod}_{R_{f_i}}$, then the following diagram commutes:

$$\begin{array}{ccc} P \otimes_{R_{f_i}} M_{f_i} & \xrightarrow{\varphi \otimes_{R_{f_i}} M_{f_i}} & Q \otimes_{R_{f_i}} M_{f_i} \\ \downarrow{\scriptstyle \simeq} & & \downarrow{\scriptstyle \simeq} \\ P \otimes_R M & \xrightarrow{\varphi \otimes_R M} & Q \otimes_R M \end{array}$$

Since the lower horizontal map is injective, by assumption, the upper horizontal map is injective. In other words, M_{f_i} is flat.

Conversely, assume that M_{f_i} is flat for each $i \in I$. For each injective map $\varphi\colon P \to Q$ in Mod_R. We need to show that $\varphi \otimes_R M$ is injective. By (g), we need to show that $(\varphi \otimes_R M)_{f_i}$ is injective for each $i \in I$. But unwinding the construction, we find that the map is precisely

$$(\varphi \otimes_R M)_{f_i}\colon (P \otimes_R R_{f_i}) \otimes_{R_{f_i}} M_{f_i} \to (Q \otimes_R R_{f_i}) \otimes_{R_i} M_{f_i},$$

whose injectivity follows from flatness of the R_{f_i}-module M_{f_i} and of the R-module R_{f_i}.

(f) This follows immediately from Theorem 2.72.

(g) Note that a linear map f is injective if and only if $\ker(f) = 0$. The claim follows from (a) afterwards.

(h) Note that a linear map f is surjective if and only if $\mathrm{coker}(f) = 0$. The claim then follows from (a).

(i) This follows by combining (g) and (h).

(j) Assume that an R-algebra A is finitely generated (i.e., of finite type), then there exists an isomorphism $R[x_1, \ldots, x_n]/I \simeq A$, where I is an ideal of $R[x_1, \ldots, x_n]$. Since localization preserves colimits, hence quotients, we have $R_{f_i}[x_1, \ldots, x_n]/I_{f_i} \simeq A_{f_i}$ for each $i \in I$. For the converse, the proof strategy is completely analogous to that of (b),

(k) Assume that A is a finitely presented R-algebra. It then follows from the proof of (j) and the fact that finite generation of ideals are closed under base change, that the A_{f_i} are finitely presented R_{f_i}-algebras. For the converse, by (j), there exists a exact sequence of R-modules

$$0 \to I \to R[x_1, \ldots, x_n] \to A \to 0.$$

After localizing at f_i, this gives us a exact sequence of R_{f_i}-modules

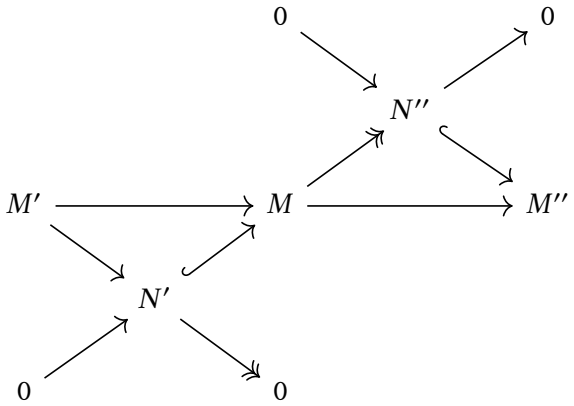
$$0 \to I_{f_i} \to R_{f_i}[x_1, \ldots, x_n] \to A_{f_i} \to 0.$$

It sufficies to show that I_{f_i} is finitely generated. We consider the following map between exact sequences:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & J & \longrightarrow & R_{f_i}[x_1, \ldots, x_m] & \longrightarrow & A_{f_i} & \longrightarrow & 0 \\ & & \vdots\downarrow & & \downarrow\!\!\downarrow & & \downarrow{\scriptstyle \simeq} & & \\ 0 & \longrightarrow & I_{f_i} & \longrightarrow & R_{f_i}[x_1, \ldots, x_n] & \longrightarrow & A_{f_i} & \longrightarrow & 0 \end{array}$$

where we have assumed that $m > n$, J finitely generated (since A is a finitely presented R_{f_i}-module), and the map $R_{f_i}[x_1, \ldots, x_m] \twoheadrightarrow R_{f_i}[x_1, \ldots, x_n]$ is given by sending x_i to 0 when $i > n$ and x_i to x_i when $1 \leq i \leq n$. Therefore, the induced map $J \to I_{f_i}$ is surjective, by four lemma. Hence, the ideal I_{f_i} is finitely generated.

(l) Let $M' \to M \to M''$ be a sequence of R-modules. Rewriting the sequence as



with N' the image of $M' \to M$ and N'' the image of $M \to M''$, it follows that it suffices to check Zariski-locality of short exact sequences. Since localization is an exact functor, it suffices to show that if $0 \to M'_{f_i} \to M_{f_i} \to M''_{f_i} \to 0$ is exact for all $i \in I$, then so is the original short sequence. But this is clear, as the exactness of $0 \to M'_{f_i} \to M_{f_i} \to M''_{f_i} \to 0$ implies that $M' \to M$ is injective by (g), that $M \to M''$ is surjective by (h), and that $\mathrm{Im}(M' \to M) \cong \ker(M \to M'')$ by (i).

□

Remark 2.81. Further Zariski-local properties of modules are: being torsion, torsion-freeness. The following properties of modules are not Zariski-local: freeness, injectivity.

Proposition 2.82 (Zariski-local nature of immersion)**.** *Let A be a ring and let $u\colon X \to \mathrm{Spec}(A)$ be a map of algebraic functors. Suppose that X satisfies Zariski descent in the sense of Corollary 2.74. Let $(f_i)_{i\in I}$ generate the unit ideal in A, and let $u_i\colon X \times_{\mathrm{Spec}(A)} \mathrm{D}(f_i) \to \mathrm{D}(f_i)$ be the base change of u to $\mathrm{D}(f_i)$. For each of the following classes of maps, if each u_i belongs to the class, so does u:*

(a) monomorphisms,
(b) closed immersions,
(c) open immersions,
(d) immersions.

Proof. We will only prove the case for monomorphism and for closed immersions, the other cases are analogous.

Assume that u is a monomorphism, then u_i is a monomorphism for each $i \in I$, since monomorphisms are closed under base change, being limits themself.

Conversely, in order to show that $u\colon X \to \mathrm{Spec}(A)$ is a monomorphism, it suffices to show that, for every ring R, the map $u_R\colon X(R) \to \mathrm{Spec}(A)(R)$ is injective. Assume that $\varphi = u_R(x) = u_R(x')$ for some $x, x' \in X(R)$. Since $(f_i)_{i\in I}$ generates the unit ideal of A, the family $(\varphi(f_i))_{i\in I}$ generates the unit ideal of R. Under the map $X(R) \to \prod_{i\in I} X(R_{\varphi(f_i)})$, assume that the element x is sent to $(x_i)_{i\in I}$ and x' is sent to $(x'_i)_{i\in I}$. By the commutativity of the diagram

$$\begin{array}{ccc} X(R) & \longrightarrow & \mathrm{Spec}(A)(R) \\ \downarrow & & \downarrow \\ X(R_{\varphi(f_i)}) & \longrightarrow & \mathrm{Spec}(A)(R_{\varphi(f_i)}) \end{array}$$

We obtain that $x \in X(R)$ is mapped to the composition $A \to A_{f_i} \xrightarrow{\varphi_i} R_{\varphi f_i} \in \mathrm{Spec}(A)(R_{\varphi(f_i)})$, which in particular implies that the elements (x_i, φ_i) and (x'_i, φ_i) actually lie in $\left(X \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A_{f_i})\right)(R_{\varphi(f_i)})$, where φ_i is the induced map $A_{f_i} \to R_{\varphi(f_i)}$. Since their image under the injection, being the pullback of a monomorphism in Set

$$(u_i)_{R_{f_i}}\colon \left(X \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A_{f_i})\right)(R_{f_i}) \to \mathrm{Spec}(A_{f_i})(R_{f_i})$$

are identical, we have $x_i = x_i'$ for all $i \in I$. Since the map $X(R) \to \prod_{i\in I} X(R_{\varphi(f_i)})$ is injective, we have $x = x'$.

Assume that u_i is a closed immersion for all $i \in I$. Then there exists an ideal $K_i \subseteq A_{f_i}$ such that

$$X \times_{\mathrm{Spec}(A_{f_i})} \mathrm{Spec}(A_{f_i}) \simeq \mathrm{Spec}(A_{f_i}/K_i),$$

over $\mathrm{Spec}(A_{f_i})$. Similar to Theorem 2.72, one can prove with no more difficulty that the functor

$$\mathrm{Idl}\colon \mathrm{CAlg} \to \mathrm{Set}, \quad R \mapsto \{\text{ideals of } R\}, \quad (f : R \to S) \mapsto (I \mapsto f(I)S)$$

satisfies Zariski descent. We start by showing that $\mathrm{Idl}(R)$ injects into the equalizer of

$$\prod_{i\in I} \mathrm{Idl}(R_{f_i}) \rightrightarrows \prod_{i,j\in I} \mathrm{Idl}(R_{f_if_j}).$$

Given ideals $J, J' \in R$, assume that $JR_{f_i} = J'R_{f_i}$ for all i. This is equivalent to the fact that for every $a \in J$ there exists an $n_i(a) \geq 1$ such that $f_i^n \in J'$ for all $n \geq n_i(a)$. As this holds for all i, as $(f_i^n)_{i\in I}$ generates the unit ideal, we obtain $a \in J'$. Indeed, choose a finite set $J \subseteq I$ such that already $\{f_i\}_{i\in J}$ generate the unit ideal, and set $n = \max\{n_i(a)\colon i \in J\}$. We can then write $1 = \sum_{i\in J} \alpha_i f_i^n$, and hence $a = \sum_{i\in J} \alpha_i f_i^n a \in J'$. As $a \in J$ was arbitrary, we have $J \subseteq J'$. By symmetry, we obtain $J = J'$.

To show surjectivity, let $(J_i)_{i\in I}$ be a tuple of ideals with $J_i \in R_{f_i}$ such that $J_iR_{f_if_j} = J_jR_{f_if_j}$. We claim that the ideal $J = \bigcap_i L_i^{-1}(J_i) \in \mathrm{Idl}(R)$ is send to J_i under the map $\mathrm{Idl}(R) \to \mathrm{Idl}(R_{f_i})$. First assume that I is finite. Fix an $i \in I$ and an element $\frac{a}{f_i^{n_i}} \in J_i$. Replacing a by $f_i^{n_i}\frac{a}{f_i^{n_i}} \in J_i$, we may take a to be in the image of R under the localization map $L_i : R \to R_{f_i}$. By the cocycle condition, there exists an $\frac{b}{f_j^{n_j}} \in J_j$ and an $r_{ij} \in \mathbf{Z}_{\geq 0}$ such that

$$(f_if_j)^{r_{ij}}(f_j^{n_j}a) = (f_if_j)^{r_{ij}}b \in R.$$

Replacing a by $f_i^{r_ij}a$, and similar for b, we obtain the relation

$$(f_j)^{r_{ij}}(f_j^{n_j}a) = (f_j)^{r_{ij}}b \in R.$$

But this is just the statement, that $a = \frac{b}{f_j^{n_j}} \in R_{f_j}$. As I was chosen to be finite, we only have to change a at most a finite number of times, hence a is still well defined, and defines an element in J, mapping to $a \in J_i$. This shows that $\mathrm{Idl}\, R$ is the equalizer for I finite, and by a similar filtered colimit argument as in Theorem 2.72, we also obtain the statement for I infinite.

Now we will show that $(K_i)_{i\in I}$ form a descent data. In other words, we need to show that $K_iA_{f_if_j} = K_jA_{f_if_j}$ for all $i, j \in I$. In fact, one can show that $K_iA_{f_if_j}$ and $K_jA_{f_if_j}$ are the ideals corresponding to the pullback of u_i and u_j along $\mathrm{Spec}(A_{f_if_j}) \to \mathrm{Spec}(A_{f_i})$ and $\mathrm{Spec}(A_{f_if_j}) \to \mathrm{Spec}(A_{f_j})$ respectively. This follows directly from the correspondence between fiber products of affine schemes and tensor products of rings. Therefore, it suffices to show that their pullbacks agree. This follows from the following pullback diagram

$$\begin{array}{ccccc} X \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A_{f_if_j}) & \longrightarrow & X \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A_{f_i}) & \longrightarrow & X \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{Spec}(A_{f_if_j}) & \longrightarrow & \mathrm{Spec}(A_{f_i}) & \longrightarrow & \mathrm{Spec}(A) \end{array}$$

and another analogous pullback square, both obtained from the pasting lemma. Therefore, there exist an ideal $K \subseteq A$ such that $KR_{f_i} = K_i$ for each $i \in I$.

We will construct a map $\mathrm{Spec}(A/K) \to X$. First observe that $(\bar{f}_i)_{i\in I}$ in A/K generates the unit ideal, where $\bar{f}_i$ is the image of f_i. Also, the localization of A/K at $\bar{f}_i$ is $A_{\bar{f}_i}/KA_{\bar{f}_i}$, since localizations commutes with quotients. By above, locally we have an isomorphism

$$\mathrm{Spec}(A_{f_i}/KA_{f_i}) \simeq X \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A_{f_i})$$

over $\mathrm{Spec}(A)$, this gives us an map $\mathrm{Spec}(A_{f_i}/KA_{f_i}) \to X$ over $\mathrm{Spec}(A)$, by postcomposition, which corresponds to an element $X(A_{f_i}/KA_{f_i})$ by the Yoneda lemma. Since X satisfies Zariski descent, this gives us a map $\mathrm{Spec}(A/K) \to X$ over $\mathrm{Spec}(A)$. In fact, this map lies above $\mathrm{Spec}(A)$, since the following diagram commutes

$$\begin{array}{ccccc} X(A/K) & \longrightarrow & \prod_{i\in I} X(A_{f_i}/KA_{f_i}) & \rightrightarrows & \prod_{i,j\in I} X(A_{f_if_j}/KA_{f_if_j}) \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{Spec}(A)(A/K) & \longrightarrow & \prod_{i\in I} \mathrm{Spec}(A)(A_{f_i}/KA_{f_i}) & \rightrightarrows & \prod_{i,j\in I} \mathrm{Spec}(A)(A_{f_if_j}/KA_{f_if_j}) \end{array}$$

and the map $\mathrm{Spec}(A_{f_i}/KA_{f_i}) \to X$ lies above $\mathrm{Spec}(A)$. To prove that u is a closed immersion, it suffices to show that this map is an isomorphism. Since u and $\mathrm{Spec}(A/K) \to \mathrm{Spec}(A)$ are monomorphisms, this map is also a monomorphism.

Unwinding the definition, to show that the map $\mathrm{Spec}(A/K) \to X$ is an isomorphism, it suffices to show that for each $x \in X(R)$, the map $u_R(x)\colon A \to R$ factors through A/K. In other words, we need to show that $K \subseteq \ker(u_R(x))$. This is clear, as one can show that inclusion of ideals is a Zariski local property, as is clear from the proof that the set of ideals of a ring is Zariski local. □

Lemma 2.83. *Let $X \simeq \mathrm{Spec}(A)$ be an affine scheme. Assume that Y is a closed subfunctor of $\mathrm{D}(F)$ for some finite set $F \subseteq A$. Then $Y = \bar{Y} \cap \mathrm{D}(F)$, where $\bar{Y}$ is the closure of Y, defined in Definition 2.69.*

Proof. As Y is closed in $\mathrm{D}(F)$, the intersection $Y \cap \mathrm{D}(f)$ is closed in $\mathrm{D}(f)$ for every $f \in F$. Let $\mathrm{J}(f) \subseteq A_f$ be the corresponding ideal, and let $I(f)$ be its image in A, so that $I(f)_f = J(f)$ and hence $\mathrm{V}(I(f)) \cap \mathrm{D}(f) = Y \cap \mathrm{D}(f)$. Let $I = \bigcap_{f\in F}(I(f))$.

We check that $Y \subseteq \mathrm{V}(I)$. Let $\varphi\colon A \to R$ be in $Y(R)$. Since $Y \subseteq \mathrm{D}(F)$, we have $(\varphi(f))_{f\in F} = R$. For each $f \in F$, the map $A \to R_f$ is an R_f-point of

$$Y \cap \mathrm{D}(f) = \mathrm{V}(\mathrm{I}(f)) \cap \mathrm{D}(f) \subseteq \mathrm{V}(I),$$

and hence the ideal $(\varphi(i))_{i\in I}$ becomes zero in each localization R_f. By Zariski descent, see Proposition 2.80 (a), we deduce that $(\varphi(i))_{i\in I} = 0$, as desired.

We now check that $\mathrm{V}(I) = \bar{Y}$, i.e., that for any ideal $K \subseteq A$ with $Y \subseteq \mathrm{V}(K)$, we have $K \subseteq I$. By the definition of I and the fact that there is a one-to-one correspondence between ideals in A_f and ideals in A not meeting $\{f^n\}_{n\geq 0}$, it suffices to check that the localization $K_f \subseteq A_f$ is contained in $J(f)$ for all $f \in F$. But $\mathrm{V}(J(f))$ is by definition $Y \cap \mathrm{D}(f)$, which is contained in $\mathrm{V}(K_f)$, so $K_f \subseteq J(f)$.

We check that $J(g) \subseteq I(f)_g$ for all $f, g \in F$. Since $(-)_g$ is exact, pullback of ideals behave well. Since $I(f)$ is the preimage of $J(f) \subseteq A_f$ in A, the ideal $I(f)_g$ is the preimage of $J(f)_g \subseteq A_{fg}$ in A_g. Since localization respects inclusion of those ideals which do not meet $\{(fg)^n\}_{n\geq 0}$, we need to show that $J(g)_f \subseteq J(f)_g$. But these are equal, since both cuts out $Y \cap \mathrm{D}(fg)$ in $\mathrm{D}(fg)$.

We will now show that $I_g = J(g)$. The inclusion $I_g \subseteq J(g)$ follows from Proposition 2.66, since

$$\mathrm{V}(J(g)) = Y \cap \mathrm{D}(g) \subseteq \bar{Y} \cap \mathrm{D}(g) = \mathrm{V}(I) \cap \mathrm{D}(g) = \mathrm{V}(I_g).$$

For the other direction, we first observe that

$$\bigcap_{f\in F} I(f)_g = \ker(A_g \to \bigoplus_{f\in F} A_g/I(f)_g) = \ker(A \to \bigoplus_{f\in F} A/I(f))_g = (\bigcap_{f\in F} I(f))_g = I_g,$$

as F is finite and localization is exact. Therefore, by what we have shown, it follows that

$$J(g) \subseteq \bigcap_{f\in F} I(f)_g = I_g.$$

Finally, we show that $\bar{Y} \cap \mathrm{D}(F) = Y$. A map $\mathrm{Spec}(R) \to \bar{Y} \cap \mathrm{D}(F)$ is a map $\varphi\colon A \to R$ such that $\varphi(I) = 0$ and $(\varphi(f))_{f\in F} = R$. Since closed immersions are stable under base change, by the following diagram

$$\begin{array}{ccccc} Y & \hookrightarrow & \mathrm{D}(F) \cap \bar{Y} & \hookrightarrow & \bar{Y} \\ \simeq\downarrow & & \downarrow & & \downarrow \\ Y & \hookrightarrow & \mathrm{D}(F) & \hookrightarrow & \mathrm{Spec}(A) \end{array}$$

Y is closed in $\bar{Y} \cap \mathrm{D}(F)$. Let $\mathrm{V}(K) \subseteq \mathrm{Spec}(R)$ be the preimage of Y. We want to show $K = 0$. For every $f \in F$, the subfunctor

$$\mathrm{V}(K_{\varphi(f)}) = \mathrm{V}(K) \cap \mathrm{D}(\varphi(f))$$

is the preimage of the subfunctor

$$Y \cap \mathrm{D}(f) = \mathrm{V}(J(f)) = \mathrm{V}(I_f) = \mathrm{V}(I) \cap \mathrm{D}(f),$$

so $K_f = (\varphi(I))_f$, which is zero by assumption. By Zariski descent, we have $K = 0$. □

Proposition 2.84. *Let X be a quasi-affine scheme and let $Y \hookrightarrow X$ be a closed immersion. Then Y is a quasi-affine scheme.*

Proof. Let $\mathrm{Spec}(A)$ be an affine scheme such that $X = \mathrm{D}(F) \subseteq \mathrm{Spec}(A)$ for some finite set $F \subseteq A$. By Lemma 2.83, we have $Y = \bar{Y} \cap \mathrm{D}(F)$, where $\bar{Y}$ is closed in $\mathrm{Spec}(A)$. Hence we may write $Y = \mathrm{Spec}(A/I) \cap \mathrm{D}(F)$ for some ideal $I \subseteq A$. But since $\mathrm{D}(F)$ is open in $\mathrm{Spec}(A)$, its restriction to $\mathrm{Spec}(A/I)$ is again open, hence we can write

$$Y = \mathrm{D}(\bar{F}) \subseteq \mathrm{Spec}(A/I) = \mathrm{V}(I),$$

where $\bar{F}$ is the image of F in A/I. This shows that Y is quasi-affine. □

2.9. **Finiteness properties.** Recall that a k-algebra is of *finite presentation*, if it is isomorphic to $k[\Sigma]$, where Σ is a system of finitely many polynomial equations in finitely many variables, and it is *of finite type*, if it is isomorphic to $k[\Sigma]$, where Σ has finitely many variables (but any number of equations). We denote the respective full subcategories of CAlg_k by $\mathrm{CAlg}_k^{\mathrm{fp}}$ and $\mathrm{CAlg}_k^{\mathrm{ft}}$. It turns outs that these finiteness conditions an naturally be expressed in terms of the algebraic k-functor $\mathrm{Spec}(A)\colon \mathrm{CAlg}_k \to \mathrm{Set}$.

Definition 2.85 (Locally of finite presentation/type)**.** Let $X\colon \mathrm{CAlg}_k \to \mathrm{Set}$ be an algebraic k-functor.

(a) X is *locally of finite presentation*, if it preserves filtered colimits.
(b) X is *locally of finite type*, it is preserves the colimits of filtered diagrams with injective transition maps.

Proposition 2.86. *Let k be a ring and A a k-algebra.*

(a) A is of finite presentation if and only if $\mathrm{Spec}(A)\colon \mathrm{CAlg}_k \to \mathrm{Set}$ is locally of finite presentation.
(b) A is of finite type if and only if $\mathrm{Spec}(A)\colon \mathrm{CAlg}_k \to \mathrm{Set}$ is locally of finite type.

write this out

Proof. This is left as an exercise to the reader, a proof for (a) can be found in [Sta26, Tag 00QL]. □

Example 2.87. (a) $\mathbf{A}_k^I$ is locally of finite type if and only if I is a finite set, in which case it is also locally of finite presentation. The same holds for the punctured affine spaces $\mathbf{A}_k^I \setminus \{0\}$.

(b) The affine k-scheme $\mathbf{A}(M)$ is locally of finite presentation (resp. of finite type) if and only if the k-module M is of finite presentation (resp. of finite type). In fact, we have $\mathbf{A}(M) \simeq \mathrm{Spec}(\mathrm{Sym}_k(M))$. The claim then follows from that the k-module M is finitely presented (resp. of finite type), if and only if the k-algebra $\mathrm{Sym}_k(M)$ is finitely presented (resp. of finite type). For example, we will show that if $\mathrm{Sym}_k(M)$ is finitely presented, then M is finitely presented. So let N be a K module, and denote by $k \oplus N$ the square zero extension of k by N. Then we have a chain of adjunctions

$$\mathrm{Map}_{(\mathrm{CAlg}_k)_{/k}}(\mathrm{Sym}_k(M), N \oplus k) \simeq \mathrm{Map}_{\mathrm{Mod}_k}(M, N),$$

where $N \oplus k$ is the square zero extension, and $\mathrm{Sym}_k(M) \to k$ is induced by the zero map $M \to k$, and $N \oplus k \to k$ is give by the projection. Furthermore, the forming square zero extensions is compatible with filtered colimits, since the forgetful functor $\mathrm{CAlg}_k \to \mathrm{Mod}_k$ preserves filtered colimits, and the underlying k-module of $N \oplus k$ is the coproduct of N and k. Therefore, if $\mathrm{Sym}_k(M)$ is finitely presented, i.e preserves filtered colimits, then $\mathrm{Map}_{\mathrm{Mod}_k}(M, -)$ also preserves filtered colimits, i.e. M is a finitely presented k-module.

(c) The algebraic k-functor $\mathbf{A}^\vee(M)$ of Remark 2.37 is locally of finite presentation for any k-module M, since the tensor product preserves colimits in each variable.

(d) The affine $\mathbf{Z}$-schemes $*$, $\varnothing$, Idem, $\mathbf{G}_m$, $\mathbf{G}_a$, Mat_n, GL_n, SL_n are all locally of finite presentation.

Remark 2.88. Since filtered colimits commute with finite limits in the category of sets, the condition of being locally of finite presentation or of finite type is stable under finite limits in $\mathrm{Fun}(\mathrm{CAlg}_k, \mathrm{Set})$.

Remark 2.89 (Compatibility with base change). Let $\varphi\colon k \to k'$ be a ring map. Since both the forgetful functor $\mathrm{CAlg}_{k'} \to \mathrm{CAlg}_k$ and its left adjoint $\varphi^*\colon \mathrm{CAlg}_k \to \mathrm{CAlg}_{k'}$ preserves filtered colimits, it follows from Corollary 2.41 that both base change along φ and Weil restriction along φ preserve the property of being locally of finite presentation or locally of finite type. On the other hand, the third functor $\varphi_\sharp$ usually does not. For example, $\mathrm{Spec}(\mathbf{C})$ is locally of finite presentation as an affine $\mathbf{R}$-scheme, but not as an affine $\mathbf{Q}$-scheme.

Recall that an algebraic k-functor can be though of as a map of algebraic functors $X \to \mathrm{Spec}(k)$ (see Corollary 2.48). Remark 2.89 implies that, for any cartesian square

$$\begin{array}{ccc} X' & \longrightarrow & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ \mathrm{Spec}(k') & \longrightarrow & \mathrm{Spec}(k) \end{array}$$

If f is locally of finite presentation or of finite type, so is f'. Consequently, we can extend Definition 2.85 to arbitrary maps of algebraic functors as follows:

Definition 2.90 (Morphism locally of finite presentation/type)**.** Let $f\colon X \to S$ be a map of algebraic functors.

(a) f is *locally of finite presentation*, if for every ring k and for every k-point $\mathrm{Spec}(k) \to S$, the algebraic k-functor $X \times_S \mathrm{Spec}(k)$ is locally of finite presentation.
(b) f is *locally of finite type*, if for every ring k and for every k-point $\mathrm{Spec}(k) \to S$, the algebraic k-functor $X \times_S \mathrm{Spec}(k)$ is locally of finite type.

Concretely, under the equivalence of Corollary 2.48, the base change of $f\colon X \to S$ along $s\colon \mathrm{Spec}(k) \to S$ is the algebraic k-functor $\mathrm{CAlg}_k \to \mathrm{Set}$ given by

$$(\varphi\colon k \to R) \mapsto \{x \in X(R) \mid f(x) = s(\varphi) \text{ in } S(R)\} = \left\{ \begin{array}{ccc} \mathrm{Spec}(R) & \overset{x}{\dashrightarrow} & X \\ {\scriptstyle \mathrm{Spec}(\varphi)}\downarrow & & \downarrow{\scriptstyle f} \\ \mathrm{Spec}(k) & \xrightarrow{s} & S \end{array} \right\}$$

Example 2.91. Any open immersion is locally of finite presentation, and any closed immersion is locally of finite type. A closed immersion $i\colon Z \hookrightarrow X$ is locally of finite presentation if and only if, for every ring R and every element $x\colon \mathrm{Spec}(R) \to X$, there exists a *finite* subset $F \subseteq R$ such that $x^{-1}(i(Z)) = \mathrm{V}(F)$.

Proposition 2.92 (Closure properties of morphisms locally of finite presentation/type)**.**

(a) Consider a commutative triangle of algebraic functors

$$\begin{array}{ccc} Z & \xrightarrow{g} & Y \\ {\scriptstyle h}\downarrow & \swarrow{\scriptstyle f} & \\ X & & \end{array}$$

If f is locally of finite presentation, then g is locally of finite presentation if and only if h is. The same holds for locally of finite type.

(b) Consider a cartesian square of algebraic functors

$$\begin{array}{ccc} Y' & \xrightarrow{f'} & X' \\ \downarrow & & \downarrow \\ Y & \xrightarrow{f} & X. \end{array}$$

If f is locally of finite presentation, so is f'. The same holds for locally of finite type.

Proof. □

Once I'm more motivated to write this

2.10. **The Nullstellensatz.** Consider a monic polynomial $f \in k[x]$ over a field k. By the functorial Nullstellensatz Corollary 2.24, we know that f is determined by its zero sets in all k-algebras. On the other hand, by the elementary theory of fields, we know that f splits into linear factors over some finite field extension of k. Hence, if we know the zero sets of f over any finite field extension of k, then we know the original polynomial f provided it is separable (i.e., does not have multiple roots). In general, the zero sets of f over finite field extensions of k determine the *radical* of f, which is the product of the prime factors of f without multiplicity. The *Nullstellensatz* of Hilbert generalizes the latter statement to systems of polynomial equations in fields. In this section, we review this result while also pointing out some shortcommings of the classical perspective.

For a ring k, define the maps

$$\{\text{subsets of } k[x_1,\dots,x_n]\} \underset{\mathrm{I}}{\overset{\mathrm{V}}{\rightleftarrows}} \{\text{subsets of } k^n\},$$

as follows:

$$\mathrm{V}(F) = \{x \in k^n \mid f(x) = 0 \text{ for all } f \in F\}, \quad \mathrm{I}(X) = \{f \in k[x_1,\dots,x_n] \mid f(x) = 0 \text{ for all } x \in X\}.$$

Note that both maps are order reversing and that $F \subseteq \mathrm{I}(\mathrm{V}(F))$ and $X \subseteq \mathrm{V}(\mathrm{I}(X))$ (in other words, this is an adjunction between posets, also known as Galois correspondence). Note also that $\mathrm{I}(X)$ is always an ideal in $k[x_1,\dots,x_n]$ and it is even a radical ideal if k is reduced (if a power of f vanishes on X, so does f). Call a subset $X \subseteq k^n$ algebraic if it lies in the image of V, or equivalently if $X = \mathrm{V}(\mathrm{I}(X))$.

Theorem 2.93 (Hilbert's Nullstellensatz)**.** *Let k be an algebraically closed field and let $n \in \mathbf{N}$. For any subset $F \subseteq k[x_1,\dots,x_n]$, we have*

$$\mathrm{I}(\mathrm{V}(F)) = \sqrt{(F)}.$$

Consequently, the maps V *and* I *define a one-to-one correspondence*

$$V : \{\textit{radical ideals in } k[x_1,\dots,x_n]\} \simeq \{\textit{algebraic subsets of } k^n\},$$

where an algebraic subset of k^n is simply a closed subset of k^n.

Proof. We assume that the readers have already seen a proof of this fact in *commutative algebra*. □

We can upgrade this result to an equivalence of categories as follows. Define the category AffSet_k of *affine algebraic set* over k as follows:

- An object of AffSet_k is a pair (n, X) with $n \geq 0$ and $X \subseteq k^n$ an algebraic subset.
- A morphism $(n, X) \to (m, Y)$ is a map $f\colon X \to Y$ such that there exists a polynomial map $k^n \to k^m$ extending f.

Recall that a ring R is reduced if 0 is the only nilpotent element of R. We denote by $\mathrm{CAlg}_k^{\mathrm{red}}$ the category of reduced k-algebras.

Corollary 2.94. *Let k be an algebraically closed field. Then there is an equivalence of categories*

$$\mathrm{AffSet}_k \xrightarrow{\sim} (\mathrm{CAlg}_k^{\mathrm{ft,red}})^{\mathrm{op}}, \quad (n, X) \mapsto k[x_1,\dots,x_n]/\mathrm{I}(X).$$

Hence AffSet_k is equivalent to the full subcategory of Aff_k spanned by the reduced affine k-schemes of finite type.

Proof. For essentially surjectivity, let A be a reduced k-algebra of finite type. We may choose a presentation $k[x_1,\dots,x_n]/I \simeq A$. Since A is reduced, I is a radical ideal. Therefore, by Theorem 2.93, we have $I = I(\mathrm{V}(\mathrm{I}))$, hence we have

$$A \simeq k[x_1,\dots,x_n]/(\mathrm{I}(\mathrm{V}(I))).$$

For fully faithfulness, observe that

$$\mathrm{Map}((n, X), (m, Y)) \simeq \{\text{polynomial maps } f\colon k^n \to k^m \mid f(X) \subseteq Y\}/\sim,$$

where $f \sim g$ if and only if $f|_X = f|_Y$. But then by definition, the right hand side just corresponds to to maps of the form

$$k[x_1,\dots,x_n]/\mathrm{I}(X) \to k[x_1,\dots,x_m]/\mathrm{I}(Y).$$

This finishes the proof. □

Remark 2.95. By Hilbert's Basissatz, *finite type* and *finite presentation* are equivalent for algebras over a field (and more generally over a noetherian ring).

We can also formulate a Nullstellensatz for an arbitrary field k as follows. Denote by $\mathrm{Field}_k^{\mathrm{fin}} \subseteq \mathrm{CAlg}_k$ the full subcategory of finite field extension of k.

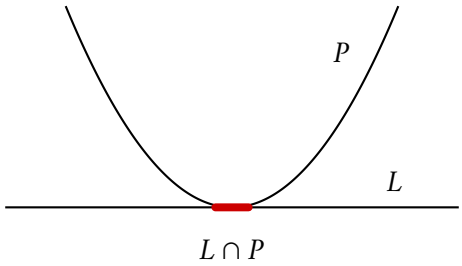
Corollary 2.96. *Let k be a field and let $n \in \mathbf{N}$. Then there is an order reversing bijection*

$$\mathrm{V}\colon \{\textit{radical ideals in } k[x_1,\dots,x_n]\} \xrightarrow{\sim} \{\textit{vanishing loci in } \mathbf{A}^n\colon \mathrm{Field}_k^{\mathrm{fin}} \to \mathrm{Set}\}.$$

Proof. □

Not sure what to put here

Example 2.97 (Non-reduced intersections). Even in the context of algebraic geometry over an algebraically closed field k, there are geometric phenomena that are not captured by only considering solutions in k. Consider for example the vanishing loci $L = \mathrm{V}(y)$ and $P = \mathrm{V}(y - x^2)$ in $\mathbf{A}^2_k = \mathrm{Spec}(k[x,y])$. Since the k-algebras $k[x,y]/(y) \simeq k[t]$ and $k[x,y]/(y-x^2) \simeq k[t]$ are reduced, these affine k-schemes are determine by the algebraic sets $L(k)$ and $P(k)$ in k^2 (by the Nullstellensatz). The intersection $L(k) \cap P(k)$ is the algebraic set $\{(0,0)\} \subseteq k^2$, which in turn corresponds to the subfunctor $\mathrm{V}(x,y) \subseteq \mathbf{A}^2_k$. However, the functorial intersection $L \cap P \subseteq \mathbf{A}^2_k$ is the subfucntor $\mathrm{V}(x^2,y)$, which is isomorphic to $\mathrm{Spec}(k[t]/(t^2))$. One can think of $\mathrm{V}(x^2,y)$ as a first-order infinitesimal neighborhood of the origin $\mathrm{V}(x,y)$ along the x-axis; this captures the fact that the line L is tangent (to first order) to the parabola P, so that they both contain the same infinitesimal horizontal segment at the origin. This residual tangency information in the intersection can only be seen by evaluating the functor $L \cap P$ on non-reduced k-algebras.



This also resolves another issue in classical algebraic geometry, which is that intersections do not vary nicely in families. Consider for example the family of horizontal line $L_a = \mathrm{V}(y-a)$ for $a \in k$. The intersection $L_a(k) \cap P(k)$ has exactly two points for any $a \neq 0$ (since k is algebraically closed), but only a single point when $a = 0$. On the other hand, the scheme-theoretic intersection $L_a \cap P$ is given by a 2-dimensional k-algebra for *all* $a \in k$, namely $k \times k$ when $a \neq 0$ and $k[t]/(t^2)$ when $a = 0$.

Example 2.98 (Geometry in mixed characteristic). Another aspect that is not captured by classical algebraic geometry over fields is algebraic geometry in *mixed characteristic*, i.e., involving rings R that do not contain any field. This is especially relevent in number theory, which studies rings of integers in finite extensions of $\mathbf{Q}$. Such rings can map to fields with different characteristics, which sometimes allows us to transport results from one characteristic to another. As a very basic example, consider the following proof that $\sqrt{2}$ is irrational (which is a reformulation of the usual argument). A positive rational number x such that $x^2 = 2$ is the same thing as an element of $X(\mathbf{Z})$, where $X \subseteq \mathbf{P}^1$ is the solution functor to the homogeneous polynomial equation $x^2 = 2y^2$. Since X is a functor, the ring map $\mathbf{Z} \to \mathbf{Z}/4$ induces a map $X(\mathbf{Z}) \to X(\mathbf{Z}/4)$. Since the squares in $\mathbf{Z}/4$ are 0 and 1, none of the six elements of $\mathbf{P}^1(\mathbf{Z}/4)$ satisfy the equation $x^2 = 2y^2$, so that $X(\mathbf{Z}/4)$ is empty. It follows that $X(\mathbf{Z})$ is also empty, i.e., that there does not exist $x \in \mathbf{Q}$ with $x^2 = 2$.

3. Projective geometry

3.1. **Projective spaces over a field.** Let k be a field. The classical projective n-space over k is the set of lines through the origin in k^{n+1}:

$$\mathbf{P}^n(k) = \{\text{1-dimensional subspaces of } k^{n+1}\}$$

Given a nonzero $(n+1)$-tuple $(a_0, \ldots, a_n) \in k^{n+1} \setminus \{0\}$, we denote by $[a_0 : \ldots : a_n]$ the 1-dimensional subspace of k^{n+1} containing $(a_0, \ldots, a_n)$. This identifies $\mathbf{P}^n(k)$ with the set of orbits of the (free)-action of $k^\times$ on $k^{n+1} \setminus \{0\}$ by scalar multiplication:

$$(k^{n+1} \setminus \{0\})/k^\times \xrightarrow{\sim} \mathbf{P}^n(k), \quad (a_0, \ldots, a_n) \mapsto [a_0, \ldots, a_n].$$

The set $\mathbf{P}^n(k)$ is the union of the $(n+1)$-subsets $U_0, \ldots, U_n$, where

$$U_i = \{[a_0, \ldots, a_n] \in \mathbf{P}^n(k) \mid a_i \text{ is a unit}\}$$

Each U_i can be identified with $\mathbf{A}^n(k) = k^n$ via

$$U_i \xrightarrow{\sim} \mathbf{A}^n(k), \quad [a_0, \ldots, a_n] \mapsto \left(\frac{a_0}{a_i}, \ldots, \frac{\hat{a}_i}{a_i}, \ldots, \frac{a_n}{a_i}\right).$$

The complement $H_i = \mathbf{P}^n(k) \setminus U_i$ is given by

$$H_i = \{[a_0, \ldots, a_n] \in \mathbf{P}^n(k) \mid a_i = 0\}$$

and can be identified with $\mathbf{P}^{n-1}$ by dropping the i-th coordinate. We often think of $\mathbf{P}^n(k) = U_0 \coprod H_0$ as the completion of $U_0 = \mathbf{A}^n(k)$ obtained by adding a point *at infinity* on every line through the origin. These points at infinity form the *hyperplane at infinity* $H_0 = \mathbf{P}^{n-1}(k)$ in $\mathbf{P}^n(k)$.

Since the k-vector space k^{n+1} is canonically self-dual, we can identify $\mathbf{P}^n(k)$ with the set of 1-dimensional *quotient spaces of* k^{n+1}:

$$\mathbf{P}^n(k) \xrightarrow{\sim} \{\text{1-dimensional quotient spaces of } k^{n+1}\}, \quad L \mapsto k^{n+1}/L^\perp,$$

where $L^\perp$ denotes the *orthogonal complement of* L. The *self-dualizability* lies in the fact that the line $[a_0, \ldots, a_n]$ corresponds to the *coimage* of the map $(a_0, \ldots, a_n) \colon k^{n+1} \twoheadrightarrow k$.

Proposition 3.1. *Let R be a ring. For an R-module V, the following conditions are equivalent:*

(a) V is finitely generated and projective;
(b) V is finitely presented and flat;
(c) V is a direct summand of R^n for some $n \in \mathbf{N}$;
(d) V is finitely presented, and for every maximal ideal $\mathfrak{m}$, the $R_\mathfrak{m}$-module $V_\mathfrak{m}$ is free;
(e) there exists some elements $f_1, \ldots, f_n \in R$ generating the unit ideal such that each V_{f_i} is a finitely generated free R_{f_i}-module;
(f) V is dualizable, i.e., there exists $V^\vee \in \mathrm{Mod}_R$ and a map $e \colon V \otimes_R V^\vee \to R$ such that $e \otimes (-)$ exhibits $V \otimes_R (-)$ as a left adjoint to $V^\vee \otimes_R (-)$. In particular, we have $V^\vee \simeq \mathrm{Hom}_R(V, -)$.

Proof. $(a) \Rightarrow (c)$: since V is finitely generated, there exists a surjective $R^n \twoheadleftarrow V$. By projectivity, this surjection splits. Therefore, V is a direct summand of R^n.

$(c) \Rightarrow (b)$: by assumption, there exists an R-module V' such that $V \oplus V' \simeq R^n$. The kernel of $R^n \twoheadrightarrow V$ is V', which is finitely generated as a direct summand of R^n. Therefore, V is finitely presented. Since the retract of every exact sequence is exact, the R-module V is flat.

$(b) \Rightarrow (d)$: a minimal generating set $R^n \twoheadrightarrow V$. After tensoring with $R_\mathfrak{m}$, we obtain an exact sequence

$$0 \to \ker(f) \to (R_\mathfrak{m})^n \xrightarrow{f} V_\mathfrak{m} \to 0.$$

Applying $(-) \otimes_{R_\mathfrak{m}} R/\mathfrak{m}$, we obtain another exact sequence

$$\mathrm{Tor}_1^{R_\mathfrak{m}}(V_\mathfrak{m}, R/\mathfrak{m}) \to \ker(f) \otimes_{R_\mathfrak{m}} R/\mathfrak{m} \to (R/\mathfrak{m})^n \to V/\mathfrak{m}V \to 0,$$

since $V_\mathfrak{m}/\mathfrak{m}V_\mathfrak{m} \simeq V/\mathfrak{m}V$. The flatness of V and of $R_\mathfrak{m}$ as R modules implies that $\operatorname{Tor}_1^{R_\mathfrak{m}}(V_\mathfrak{m}, R/\mathfrak{m}) \simeq \operatorname{Tor}_1^R(V, R/\mathfrak{m}) = 0$. By minimality of the generating set $R^n \twoheadrightarrow V$, the map $(R/\mathfrak{m})^n \to V/\mathfrak{m}V$ is an isomorphism of $R/\mathfrak{m}$-vector spaces. Therefore, we have $\ker(f) \otimes_{R_\mathfrak{m}} R/\mathfrak{m} = \ker(f)/\mathfrak{m}\ker(f) = 0$. By Nakayama's lemma, we have $\ker(f) = 0$.

$(d) \Rightarrow (e)$: recall that we can write $V_\mathfrak{m}$ as a filtered colimit $V_\mathfrak{m} = \operatorname{colim}_{f\notin\mathfrak{m}} V_f$. Let $e_1, \dots, e_n$ be a basis of the $R_\mathfrak{m}$-module $V_\mathfrak{m}$. By the explicit description of filtered colimits in Mod_R, there exists $e_i' \in V_{f_i}$ such that the image of e_i' under $V_{f_i} \to V_\mathfrak{m}$ is e_i. Taking $f = f_1 \cdot \ldots \cdot f_n$, we may assume that $e_i' \in V_f$, and that the set $\{e_i'\}$ becomes a basis after localization at $\mathfrak{m}$. Denote by $\varphi\colon (R_f)^n \to V_f$ the corresponding map. Since V is finitely presented, V_f is finitely presented, which implies that $\operatorname{coker}(\varphi)$ and $\ker(\varphi)$ are also finitely generated. As they also become 0 after localization at $\mathfrak{m}$, by unwinding the definition there exists $f_\mathfrak{m} \in R \setminus \mathfrak{m}$ such that $R_{f_\mathfrak{m}}^n \to V_{f_\mathfrak{m}}$ is an isomorphism. It is easy to verify that $R = (f_\mathfrak{m})_{\mathfrak{m}\subseteq R}$. Indeed, the ideal $(f_\mathfrak{m})_{\mathfrak{m}\subseteq R}$ is contained in no maximal ideal by construction. Since any linear combination only has finitely many non-zero coefficients, we may choose a finite subset of $\{f_\mathfrak{m}\}_{\mathfrak{m}\subseteq R}$ which still generates R.

$(e) \Rightarrow (f)$: first notice that every finite free module V is dualizable, with dual V itself. Indeed, for any R-module N we have

$$R^n \otimes_R N \cong N^n \cong \mathrm{Hom}_R(R, N)^n \cong \mathrm{Hom}(R^n, N)$$

and this chain of equivalences sends an elementary tensor $(r_i)_i \otimes n$ to the map sending the basis element e_i of R^n to $r_i n \in N$. This being natural, and $\mathrm{Hom}_(R^n, -)$ being right adjoint to $R^n \otimes_R -$ gives the claim.

Notice further that being dualizable is a Zariski local property. One direction is clear, and for the other direction we construct the dual by using Theorem 2.72 (b) on the local dual objects. The claim then follows from the fact that isomorphisms are Zariski local and the chain of isomorphisms

$$\mathrm{Hom}_R(V, N)_{f_i} \cong \mathrm{Hom}_R(V, N_{f_i}) \cong \mathrm{Hom}_{R_{f_i}}(V_{f_i}, N_{f_i}).$$

Here, the first isomorphism uses that V is finitely presented, this being again a Zariski local property, and hence categorically compact, see also Proposition 2.86. The second isomorphism is the base change adjunction.

$(f) \Rightarrow (a)$: since V is dualizable, $\mathrm{Hom}_R(V, -)$ preserves colimits. In particular, it preserves filtered colimits. Therefore, by Proposition 2.86, V is finitely presented, hence finitely generated. Recall that a map $f\colon X \to Y$ is an epimorphism if and only if the following diagram is a pushout square

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle \sim} \\ Y & \xrightarrow[\sim]{} & Y \end{array}$$

Therefore, the functor $\mathrm{Hom}_R(V, -)$ preserves epimorphism, hence V is projective. □

Remark 3.2. Since in a symmetric monoidal category, the dual of a dualizable object is again dualizable, Proposition 3.1 (f) implies that the dual of a vector space is again a vector space.

Definition 3.3 (Vector space)**.** An R-module satisfying the equivalent conditions of Proposition 3.1 is called a *vector space*[6] over R. We denote by $\mathrm{Vect}_R \subseteq \mathrm{Mod}_R$ the full subcategory spanned by the vector spaces.

Recall that any module M over a field admits a basis and that all bases have the same cardinality, which is called the dimension of M. To extend this notion to modules over a ring R, we consider R-fields, i.e., R-algebras that are also fields:

Definition 3.4 (Rank of a module)**.** Let R be a ring, M an R-module, and κ an R-field. The *rank* of M at κ, denoted by $\mathrm{rk}_\kappa(M)$, is the dimension of the κ-module $M \otimes_R \kappa$. We say that M has *constant rank* r, if $\mathrm{rk}_\kappa(M) = r$ for every R-field κ.

[6]This definition is nonstandard: when k is a field, every k-module is traditionally called a *vector space*, but only the finite dimensional ones satisfy this definition.

Remark 3.5 (Residue fields). Let R be a ring. Recall that the *residue field* of R at a prime ideal $\mathfrak{p} \subseteq R$ is the R-field

$$\kappa(\mathfrak{p}) = R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}} \simeq \mathrm{Frac}(R/\mathfrak{p}).$$

The residue fields of R form an initial family of objects of the category Field_R in the following sense: for every R-field κ, there exists a unique prime ideal $\mathfrak{p}$ such that $R \to \kappa$ factors through $\kappa(\mathfrak{p})$, and this factorization is unique. In fact, if such factorization exists, the kernel of the map $R \to \kappa$ is the kernel of $R \to \kappa(\mathfrak{p})$, which is $\mathfrak{p}$, since $\kappa(\mathfrak{p}) \to \kappa$ is injective as a field extension. This implies that, the category of R-fields decomposes as

$$\mathrm{Field}_R = \coprod_{\mathfrak{p} \subseteq R} \mathrm{Field}_{\kappa(\mathfrak{p})}$$

Extensions of scalars between fields preserves dimension, i.e., if $k \to k'$ is a field extension and V is a k-module, then $\dim_k(V) = \dim_{k'}(V \otimes_k k')$. Therefore, we have $\mathrm{rk}_\kappa(M) = \mathrm{rk}_{\kappa(p)}(M)$ for any κ in the summand indexed by $\mathfrak{p}$.

Remark 3.6. Let R be a ring. For an R-vector space V, there exists $f_1, \dots, f_n \in R$ generating the unit ideal such that $V_{f_i} \simeq R_{f_i}^{\oplus n_i}$, by Proposition 3.1. Let $\mathfrak{p} \subseteq R$ be a prime ideal. Then there exists f_i such that $f_i \notin \mathfrak{p}$ and $\mathrm{rk}_{\kappa(\mathfrak{p})}(V) = n_i$. In fact, localization commutes with coproduct, hence localization of V_{f_i} at $\mathfrak{p}$ gives us $V_{\mathfrak{p}} \simeq R_{\mathfrak{p}}^{n_i}$, and the claim follows from $V \otimes_R \kappa(\mathfrak{p}) \simeq V_{\mathfrak{p}} \otimes_{R_{\mathfrak{p}}} \kappa(\mathfrak{p})$.

Remark 3.7 (Short exact sequences of vector spaces). Since vector spaces are projective, every short exact sequence of vector spaces splits. Hence, for any ring map $\varphi\colon R \to R'$, the functor $\varphi^*\colon \mathrm{Vect}_R \to \mathrm{Vect}_{R'}$ preserves short exact sequence. In particular, if

$$0 \to U \to V \to W \to 0$$

is a short exact sequence in Vect_R and κ is any R-field, then $\mathrm{rk}_\kappa(V) = \mathrm{rk}_\kappa(U) + \mathrm{rk}_\kappa(W)$.

Remark 3.8 (Modules of constant rank 0). By Proposition 3.1 and Proposition 2.80, a vector space is zero if and only if it has constant rank 0. Note that this is not true for more general modules. For example, the $\mathbf{Z}$-module $\mathbf{Q}/\mathbf{Z}$ has constant rank 0.

Proposition 3.9. *Let R be a ring. For an R-module L, the following conditions are equivalent:*

(a) L is a vector space of constant rank 1;
(b) L is invertible, i.e., there exists $L' \in \mathrm{Mod}_R$ such that $L \otimes_R L' \simeq R$.

Proof. $(a) \Rightarrow (b)$: Consider $L^\vee = \mathrm{Hom}_R(L, R)$. Since R_f is a (R, R_f)-bimodule, the tensor-hom adjunction gives us

$$(L_f)^\vee \cong \mathrm{Hom}_{R_f}(L \otimes_R R_f, R_f) \simeq \mathrm{Hom}_R(L, \mathrm{Hom}_{R_f}(R_f, R_f)) \simeq \mathrm{Hom}_R(L, R_f),$$

for each $f \in R$. By Proposition A.1, we conclude that $(L^\vee)_f = (L_f)^\vee$. Therefore, if L is Zariski locally free of rank 1, then $L^\vee$ is also Zariski locally free of rank 1. Since localization commutes with tensor products, the R-module $L \otimes_R L^\vee$ is again Zariski locally free of rank 1. Therefore, the evaluation map $\mathrm{ev}\colon L \otimes_R L^\vee \to R$ is an isomorphism. In fact, this is true for $L = R$, and by Proposition 2.80, being an isomorphism is a Zariski local property.

$(b) \Rightarrow (a)$: Since the R-module L is invertible, the functor $L \otimes_R (-)$ is an equivalence with an inverse $L' \otimes_R (-)$ for some R-module L'. This implies that $L \otimes_R (-)$ is left adjoint to $L' \otimes_R (-)$ with counit induced by $L \otimes_R L' \simeq R$, which in particular implies that L is dualizable, hence a vector space. For each prime ideal $\mathfrak{p} \subseteq R$, the functor $\kappa(\mathfrak{p}) \otimes_R (-)$ commutes with tensor products, in the sense that $\kappa(\mathfrak{p}) \otimes_R (V \otimes_R W) \cong (\kappa(\mathfrak{p}) \otimes_R V) \otimes_{\kappa(\mathfrak{p})} (\kappa(\mathfrak{p}) \otimes_R W)$. Therefore, the rank is multiplicative. In other words, we have

$$\mathrm{rk}_{\kappa(\mathfrak{p})}(R) = \mathrm{rk}_{\kappa(\mathfrak{p})}(L \otimes_R L') \simeq \mathrm{rk}_{\kappa(\mathfrak{p})}(L) \cdot \mathrm{rk}_{\kappa(\mathfrak{p})}(L').$$

Since R is Zariski locally free of rank 1, and $\mathrm{rk}_{\kappa(\mathfrak{p})}(L) \in \mathbf{N}$, we have $\mathrm{rk}_{\kappa(\mathfrak{p})}(L) = \mathrm{rk}_{\kappa(\mathfrak{p})}(L') = 1$ for each prime ideal $\mathfrak{p} \subseteq R$. □

Definition 3.10 (Line)**.** An R-module M satisfying the equivalent conditions of Proposition 3.9 is called a *line* over R. We denote by $\mathrm{Line}_R \subseteq \mathrm{Vect}_R$ the full subcategory of lines. A line is called *trivial* if it is isomorphic to R.

Remark 3.11. The category of vector spaces over any ring is equivalent to its opposite category, by duality. However, the symmetry between injective and surjective maps does not hold anymore.

Example 3.12. Let R be a ring and let V, W be vector spaces over R. Consider a surjection $f\colon V \twoheadrightarrow W$, then $\ker(f)$ is a vector space, as we will see in Proposition 3.19. However, consider an injection $f\colon V \rightarrowtail W$, then $\operatorname{coker}(f)$ is not a vector space in general. For example, consider the exact sequence

$$0 \to \mathbf{Z} \to \mathbf{Z} \to \mathbf{Z}/2\mathbf{Z} \to 0$$

In fact, surjectivity is already a universal notion. However, injectivity is not stable under scalar extensions. Therefore, we have the following stronger notion, which is equivalent to injectivity when considering vector spaces over fields.

Definition 3.13 (Universally injective map)**.** Let R be a ring and let M, N be R-modules. An R-linear map $f\colon M \to N$ is called *universally injective* if, for every ring map $R \to S$, the map $f \otimes_R S\colon M \otimes_R S \to N \otimes_R S$ is injective.

Remark 3.14.
(a) If the R-linear map $f\colon M \to N$ is universally injective, then for every R-module P, the map $f \otimes_R P\colon M \otimes_R P \to N \otimes_R P$ is injective. This follows from applying the definition of universally injectivity to the ring map $R \to \operatorname{Sym}_R(P)$ and using that P is a direct summand of $\operatorname{Sym}_R(P)$.
(b) We need not consider the dual notion of *universally surjective map*, since surjective maps are preserved by base change, as base change preserves cokernels. In other words, every surjective map is universally surjective.

Example 3.15.
(a) By functoriality of $- \otimes_R -$, any map admitting a retraction, equivalently, any direct summand inclusion, is universally injective.
(b) The inclusion $\bigoplus_{\mathbf{N}} \mathbf{Z} \hookrightarrow \prod_{\mathbf{N}} \mathbf{Z}$ is an example of a universally injective map of abelian groups that does not admit a retraction. reference
(c) Let $f \in R$. The multiplication-by-f map $l_f : R \to R$ is injective if and only if f is a non-zero divisor, but it is universally injective if and only if f is a unit (as it becomes the zero map after base change to R/f).

Remark 3.16. By Proposition 2.80, *vector space* and *line* are Zariski-local properties of modules, and *universally injective* is a Zariski local property of linear maps. In fact, by Proposition 3.1, *vector space* and *line* are precisely the minimal weakenings of the properties *free of finite rank* and *free of rank* 1 that are Zariski-local.

Definition 3.17 (Subspace and quotient space)**.** Let V be an R-module.
(a) A *subspace* of V is a submodule $U \subseteq V$ such that U is a vector space and the inclusion map $U \hookrightarrow V$ is universally injective.
(b) A *quotient space* of V is a quotient module $V \twoheadrightarrow W$ such that W is a vector space.

The definition is mostly used when V itself as vector space. In this case, there are several useful characterizations of subspaces and of quotient spaces:

Proposition 3.18 (Characterization of subspaces)**.** *Let R be a ring and let U and V be vector spaces over R. For a map $f\colon U \to V$, the following are equivalent:*
(a) f is universally injective;
(b) for every R-field κ, the map $f \otimes_R \kappa$ is injective;
(c) f is injective and the cokernel of f is a vector space.
(d) f admits a retraction.
(e) the dual map $f^\vee\colon V^\vee \to U^\vee$ is surjective.

Proof. $(a) \Rightarrow (b)$: this is clear.
$(b) \Rightarrow (e)$: the assumption of b implies that $(f \otimes_R \kappa)^\vee$ is surjective for all fields κ. Since V is an R-vector space, we further observe that

$$\operatorname{Hom}_\kappa(V \otimes_R \kappa, \kappa) \simeq \operatorname{Hom}_R(V, \kappa) \simeq V^\vee \otimes_R \kappa.$$

Under this isomorphism, we have $(f \otimes_R \kappa)^\vee = f^\vee \otimes_R \kappa$, so that $f^\vee \otimes_R \kappa$ is also surjective for all field κ. In particular, this implies that $f^\vee \otimes_R \kappa(\mathfrak{m})$ is surjective for all maximal ideal $\mathfrak{m} \subseteq R$. By Remark 3.2 $V^\vee$ is an R-vector space, hence finitely generated. This implies that for each maximal ideal $\mathfrak{m}$, there exists some $s_\mathfrak{m} \notin \mathfrak{m}$ such that $f^\vee \otimes_R R_s$ is surjective. Since surjectivity is a Zariski local property, and $(s_\mathfrak{m})_\mathfrak{m} = R$, the map $f^\vee$ is surjective.

$(e) \Rightarrow (d)$: assume that $f^\vee \colon V^\vee \to U^\vee$ is surjective. Since tensor products commute with colimits, the map $V^\vee \otimes_R U \to U^\vee \otimes_R U$ is surjective. But $V^\vee \otimes_R U \cong \mathrm{Hom}_R(V, U)$, and similarly for $U^\vee \otimes_R U$, so that $\mathrm{Hom}_R(V, U) \to \mathrm{Hom}_R(U, U)$ is surjective. In other words, the map f has a retraction.

$(d) \Rightarrow (c)$: since f has a retraction, the map f is injective and the exact sequence

$$0 \to U \to V \to \mathrm{coker}(f) \to 0$$

splits. Therefore, $\mathrm{coker}(f)$ is a direct summand of V. Since V is an R-vector space, it is a summand of a finitely generated free module, hence $\mathrm{coker}(f)$ is also a direct summand of a finitely generated free module. By Proposition 3.1 (c), $\mathrm{coker}(f)$ is an R-vector space.

$(c) \Rightarrow (a)$: since $\mathrm{coker}(f)$ is an R-vector space and f is injective, the exact sequence

$$0 \to U \to V \to \mathrm{coker}(f) \to 0$$

splits. Therefore $f \colon U \to V$ is an inclusion of direct summand. By Example 3.15 (a), the map f is universally injective. □

Proposition 3.19 (Characterization of quotient spaces)**.** *Let R be a ring and let U and V be vector spaces over R. For a map $f \colon U \to V$, the following are equivalent:*

(a) f is surjective;
(b) for every R-field κ, $f \otimes_R \kappa$ is surjective;
(c) f is surjective and the kernel of f is a vector space;
(d) f admits a section;
(e) the dual map $f^\vee \colon W^\vee \to V^\vee$ is universally injective.

Proof. The proof is analogous to that of Proposition 3.18. □

Corollary 3.20. *Let R be a ring and let V and W be vector spaces over R. A map $f \colon V \to W$ is an isomorphism if and only if, for every R-field κ, $f \otimes_R \kappa$ is an isomorphism.*

Proof. This follows from combining Proposition 3.19 and Proposition 3.18. □

Corollary 3.21. *Let R be a ring and let V and W be vector spaces over R such that $\mathrm{rk}_\kappa(V) = \mathrm{rk}_\kappa(W)$ for all R-fields κ. For a map $f \colon V \to W$, the following are equivalent:*

(a) f is an isomorphism;
(b) f is surjective;
(c) f is universally injective.

Proof. For a field, a morphism between vector spaces in above sense is an isomorphism if and only if it is injective, if and only if it is surjective. Using that universally injective morphisms and surjective morphisms are stable under base change, this together with Corollary 3.20 implies the claim. □

Corollary 3.22. *Let R be a ring and let V be a vector space over R. There is a bijection:*

$$\{\textit{subspaces of } V\} \simeq \{\textit{quotient spaces of } V^\vee\}, \quad (U \hookrightarrow V) \mapsto (V^\vee \twoheadrightarrow U^\vee).$$

Conversely, there is also a bijection:

$$\{\textit{quotient spaces of } V\} \simeq \{\textit{subspaces of } V^\vee\}, \quad (V \twoheadrightarrow W) \mapsto (W^\vee \hookrightarrow V^\vee).$$

Proof. This follows directly from Proposition 3.19 and Proposition 3.18. □

Remark 3.23. In practice, we do not distinguish between *submodules* of M and *monomorphisms* into M, nor between *quotient modules* of M and *epimorphisms* out of M: we may always identify a monomorphism with its image and an epimorphism with its coimage. Two monomorphisms into M have the same image if and only if they are isomorphic over M (in which case the isomorphism is unique). Similarly, two epimorphisms out of M have the same coimage if and only if they are (uniquely) isomorphic under M. These identifications are happening for example in the statement of Corollary 3.22.

Remark 3.24. Let $f\colon V \to W$ be a map between vector spaces. In general, neither $\ker(f)$ nor $\operatorname{coker}(f)$ is a vector space. By Proposition 3.18 and Proposition 3.19, $\ker(f)$ is a vector space if f is surjective, and $\operatorname{coker}(f)$ is vector space if f is universally injective (but injectivity does not suffice).

3.2. **Projective spaces.**

Definition 3.25. Let k be a ring and I a set. The *projective I-space* over k is the algebraic k-functor

$$\mathbf{P}^I_k\colon \mathrm{CAlg}_k \to \mathrm{Set}, \quad R \mapsto \{\text{quotient lines of } R^{(I)}\}, \quad (R \to S) \to ((R^{(I)} \twoheadrightarrow L) \mapsto (S^{(I)} \twoheadrightarrow L \otimes_R S)).$$

We simply write $\mathbf{P}^I$, when $k = \mathbf{Z}$. For $n \geq -1$, the *projective n-space* over k is $\mathbf{P}^n_k = \mathbf{P}^{\{0,\dots,n\}}_k$. It is also called the *projective line* if $n = 1$ and the *projective plane* if $n = 2$.

Remark 3.26. (a) The projective (-1)-space $\mathbf{P}^{-1}_k = \mathbf{P}^\emptyset$ is the empty k-scheme $\emptyset$ (Example 2.29).

(b) The projective 0-space $\mathbf{P}^0_k$ is the final k-scheme $\operatorname{Spec}(k)$

(c) If I is a *finite* set, Corollary 3.22 provides a natural bijection

$$\mathbf{P}^I_k(R) \simeq \{\text{sublines of } R^I\}.$$

In fact, this follows from $(R^I)^\vee \simeq R^I$ and if W is a line, then $W^\vee$ is also a line. However, if I is infinite or more generally if we replace the R-module $R^{(I)}$ by some R-module that is not a vector space (see Example 3.48), then it is necessary to use quotient lines instead of sublines in the definition.

Notation 3.27 (Projective coordinates)**.** Let L be a line over R and let $(a_0, \dots, a_n) \in L^{n+1}$ be a family generating L as an R-module. The induced map $R^{n+1} \to L$ is then surjective and its coimage is a quotient line of R^{n+1}, which we denote by

$$[a_0 : \dots : a_n] \in \mathbf{P}^n(R)$$

Given another line M and generating family $(b_0, \dots, b_n) \in M$, we have $[a_0 : \dots : a_n] = [b_0 : \dots : b_n]$ if and only if there is a (necessarily unique) isomorphism $L \xrightarrow{\sim} M$ sending each a_i to b_i. When k is a field, we have

$$\mathbf{P}^n(k) \xrightarrow{\sim} (k^{n+1} \setminus \{0\})/k^\times, \quad (a_0, \dots, a_n) \mapsto [a_0, \dots, a_n].$$

The affine and projective spaces $\mathbf{A}^I_k$ and $\mathbf{P}^I_k$ are related via the punctured affine space $\mathbf{A}^I_k \setminus \{0\}$ of Example 2.57. Recall that $\mathbf{A}^I_k \setminus \{0\}$ is a subfunctor of $\mathbf{A}^I_k$ given by

$$(\mathbf{A}^I_k \setminus \{0\})(R) = \{a \in R^I \mid (a) = R\}.$$

Under the identification of R^I with the R-linear dual of $R^{(I)}$, I-tuples generating the unit ideal correspond to surjective maps $R^{(I)} \twoheadrightarrow R$. In particular, there is a canonical map $\mathbf{A}^I_k \setminus \{0\} \to \mathbf{P}^I_k$ sending an I-tuple to the corresponding quotient of $R^{(I)}$, which is a trivial quotient line. We therefore have a zigzag of morphisms

$$\mathbf{A}^I_k \hookleftarrow \mathbf{A}^I_k \setminus \{0\} \to \mathbf{P}^I_k$$

relating the affine space and the projective space. For $n \geq -1$, this specializes to a zigzag

$$\mathbf{A}^{n+1}_k \hookleftarrow \mathbf{A}^{n+1}_k \setminus \{0\} \to \mathbf{P}^n_k.$$

Proposition 3.28. *Let I be a set and R a ring. The map $\mathbf{A}^I \setminus \{0\} \to \mathbf{P}^I$ induces an injective map*

$$(\mathbf{A}^I \setminus \{0\})(R)/R^\times \hookrightarrow \mathbf{P}^I(R),$$

where $R^\times$ acts on $(\mathbf{A}^I \setminus \{0\})(R)$ by multiplication, whose image consists exactly of the trivial quotient lines of $R^{(I)}$. In particular, if every line over R is trivial (e.g., if R is a local ring or a principal ideal domain), then this map is bijective.

Proof. Assume that $(a_i)_{i\in I}$ and $(b_i)_{i\in I}$ define the same line in $\mathbf{P}^I(R)$. Then by definition, there exists some R-linear isomorphism such that the diagram

$$\begin{array}{ccc} R^{(I)} & \xrightarrow{(a_i)_{i\in I}} & R \\ {\scriptstyle (b_i)_{i\in I}}\downarrow & \nearrow_{\simeq} & \\ R & & \end{array}$$

commutes. Such an isomorphism is equivalent to an element $\lambda \in R^\times$ such that $\lambda a_i = b_i$ for each $i \in I$. □

Remark 3.29. The scaling action of $R^\times$ on $(\mathbf{A}^I \setminus \{0\})(R)$ is functorial in R, i.e., it is an action of the affine group scheme $\mathbf{G}_m$ on the algebraic functor $\mathbf{A}^I \setminus \{0\}$. By Proposition 3.28, the map $\mathbf{A}^I \setminus \{0\} \to \mathbf{P}^I$ induces a monomorphism of algebraic functors $(\mathbf{A}^I \setminus \{0\})/\mathbf{G}_m \hookrightarrow \mathbf{P}^I$. This map is Zariski locally surjective, in the sense that for each ring R, there exists $f_1, \ldots, f_n \in R$ such that $(f_1, \ldots, f_n) = R$ and the map is surjective on R_{f_i}.

Remark 3.30.
(a) Unlike the affine space construction $I \mapsto \mathbf{A}^I$, the constructions $I \mapsto \mathbf{P}^I$ and $I \mapsto \mathbf{A}^I \setminus \{0\}$ are not contravariantly functorial on the whole category of sets, but only on the subcategory of sets and *surjections*.
(b) A technical difference between affine and projective spaces is that $\mathbf{A}^I$ is quasi-compact for all I (this property will be defined in), whereas $\mathbf{P}^I$ and $\mathbf{A}^I \setminus \{0\}$ are quasi-compact if and only if I is finite. For this reason, many basic results in projective geometry assume a finite number of variables.

Correct reference

3.3. **Graded rings.** In the affine case, each family of polynomial equations Σ corresponds to an algebraic functor Sol_Σ, which is isomorphic to the affine scheme $\mathrm{Spec}(k[\Sigma])$. In the projective setting, we would like to construct an analogous correspondence, but this is only possible for homogeneous polynomials, i.e. polynomials lying in a single graded component of the standard graded polynomial ring.

Definition 3.31 (Homogeneous polynomial)**.** Let k be a ring, I a set and $d \in \mathbf{N}$. A polynomial in $k[x_i \mid i \in I]$ is *homogeneous of degree* d if it is a k-linear combination of monomials of the form $\prod_{i\in I} x_i^{n_i}$, where $\sum_{i\in I} n_i = d$. We denote by $k[x_i \mid i \in I]_d \subseteq k[x_i \mid i \in I]$ the k-submodule of homogeneous polynomials of degree d.

Note that there is a direct sum decomposition

$$k[x_i \mid i \in I] = \bigoplus_{d\in\mathbf{N}} k[x_i \mid i \in I]_d,$$

which is compatible with multiplication in the sense that the product of a homogeneous polynomial of degree d with one of degree e gives a homogeneous polynomial of degree $d + e$. These properties are abstracted in the notion of *graded ring*:

Definition 3.32 (Graded ring)**.** Let $(\Gamma, +, 0)$ be a commutative monoid (typically $\mathbf{N}$ or $\mathbf{Z}$). A Γ-*graded ring* is a ring A together with subgroups $A_\gamma \subseteq A$ for all $\gamma \in \Gamma$ such that:

(a) A is the direct sum of the subgroups A_γ: $\bigoplus_{\gamma\in\Gamma} A_\gamma \xrightarrow{\sim} A$;
(b) $1 \in A_0$, and for all $\gamma, \delta \in \Gamma$, we have $A_\gamma A_\delta \subseteq A_{\gamma+\delta}$.

Elements of A_γ are called *homogeneous of degree* γ. If k is a ring, a Γ-graded k-algebra is a Γ-graded ring A with a ring map $k \to A$ that lands in A_0. Let A and B be Γ-graded rings. We define a Γ-*graded map*, or simply *graded map* if the monoid defining the grading is clear from the context, $f : A \to B$ to be a map of rings $f : A \to B$, such that $f(A_\gamma) \subseteq B_\gamma$ for all $\gamma \in \Gamma$.

Definition 3.33 (Graded module)**.** Let A be a Γ-graded ring. A Γ-*graded* A-*module* is an A-module M together with subgroups $M_\gamma \subseteq M$ for all $\gamma \in \Gamma$ such that:

(a) M is the direct sum of the subgroups M_γ: $\bigoplus_{\gamma\in\Gamma} M_\gamma \xrightarrow{\sim} M$;
(b) For all $\gamma, \delta \in \Gamma$, we have $A_\gamma M_\delta \subseteq M_{\gamma+\delta}$.

An ideal $I \subseteq A$ is called *homogeneous* if it is a Γ-graded A-module with $I_\gamma = I \cap A_\gamma$.

Remark 3.34. Let A be a Γ-graded ring and M a Γ-graded A-module.

(a) A_0 is a subring of A, each A_γ is an A_0-module, and each M_γ is an A_0-module.
(b) Let $\Gamma \to \Delta$ be a map of commutative monoids. Then A is also a Δ-graded ring, where for $\delta \in \Delta$ we have $A_\delta = \bigoplus_{\gamma \mapsto \delta} A_\gamma$, the sum being indexed by all preimages of δ. A Γ-graded A-module M is similarly a Δ-graded A-module. For example, any $\mathbf{N}$-graded ring A is a $\mathbf{Z}$-graded ring with $A_n = 0$ for $n < 0$.
(c) An ideal in A is homogeneous if and only if it is generated by homogeneous elements.
(d) If $I \subseteq A$ is homogeneous ideal, the quotient ring A/I inherits a unique Γ-grading given by $(A/I)_\gamma = A_\gamma/I_\gamma$ such that $A \twoheadrightarrow A/I$ is a graded map. Moreover, the A/I-module M/IM has a unique Γ-grading such that $M \to M/IM$ is a graded map.
(e) Let $S \subseteq A$ be a multiplicative set of homogeneous elements whose degrees are invertible in Γ. Then the localized ring $A[S^{-1}]$ inherits a unique Γ-grading such that $A \to A[S^{-1}]$ is a graded map: if $a \in A_\gamma$ and $s \in S \cap A_\delta$, then $a/s \in A[S^{-1}]_{\gamma-\delta}$. Moreover, the $A[S^{-1}]$-module $M[S^{-1}]$ has a unique Γ-grading such that $M \to M[S^{-1}]$ is a graded map.

Example 3.35. Let k be a ring.

(a) For any set I, the polynomial algebra $k[x_i \mid i \in I]$ is an $\mathbf{N}$-graded k-algebra with $k[x_i \mid i \in I]_d$ the subgroup of homogeneous polynomials of degree d.
(b) The Laurent polynomial algebra $k[x^{\pm 1}]$ is a $\mathbf{Z}$-graded k-algebra with x in degree 1. More generally, inverting any set of homogeneous polynomials in $k[x_i \mid i \in I]$ yields a $\mathbf{Z}$-graded k-algebra.
(c) Let M be a k-module. The symmetric algebra $\mathrm{Sym}_k(M)$ has a canonical $\mathbf{N}$-grading

$$\mathrm{Sym}_k(M) = \bigoplus_{d \in \mathbf{N}} \mathrm{Sym}^d_k(M),$$

where $\mathrm{Sym}^d_k(M)$ is the d-th symmetric power of M. This recovers (a) for $M = k^{(I)}$.
(d) Let L be a line over k. By Proposition 3.9, L is invertible. Choose a dual by L^{-1} and an isomorphism $L \otimes_R L^{-1} \xrightarrow{\simeq} R$, and write $L^{-d} := (L-1)^d$ for $d \geq 0$. Then $\bigoplus_{d \in \mathbf{Z}} L^{\otimes d}$ is naturally a $\mathbf{Z}$-graded k-algebra.

Notation 3.36. Let M be a Γ-graded A-module, and $\gamma \in \Gamma$.

(a) We denote by $M(\gamma)$ the Γ-graded A-module whose underlying A-module is M with $M(\gamma)_\delta = M_{\gamma+\delta}$. This defines an endofunctor $M \mapsto M(\gamma)$, which is an equivalence if γ is invertible in Γ.
(b) We denote by $A^{(\gamma)}$ the $\mathbf{N}$-graded ring $\bigoplus_{n \in \mathbf{N}} A_{n\gamma}$ with $(A^{(\gamma)})_n = A_{n\gamma}$. Similarly, $M^{(\gamma)}$ is an $\mathbf{N}$-graded $A^{(\gamma)}$-module.
(c) Let $A = \bigoplus_{n \in \mathbf{Z}} A_n$ be a $\mathbf{Z}$-graded ring and f be a invertible element in degree 1. Then there exists an isomorphism

$$A_0[x^{\pm 1}] \to A, \quad x \mapsto f.$$

To see that this is an isomorphism, we only have to check this degree wise. But in this case, we have that $f^n A_0$ is the image of the map $f^n : A_0 \to A_n$, which admits an inverse given by $(f^{-1})^n : A_n \to A_0$.

Intuitively, multiplication by f shifts degrees, hence every element $x = A_n$ can be uniquely written as $x = a_0 f^n$, where $a_0 \in A_0$.
(d) Suppose γ is invertible in Γ. Given $f \in A_\gamma$, the localization A_f is a Γ-graded ring and we write $A_{(f)} = (A_f)_0$. Similarly, $M_{(f)} = (M_f)_0$ is an $A_{(f)}$-module. Note that there is an isomorphism of $\mathbf{Z}$-graded rings

$$A_{(f)}[x^{\pm 1}] \xrightarrow{\sim} A_f^{(\gamma)}, \quad x \mapsto f.$$

Indeed, explicit descriptions of $A_{(f)}$ and $A_f^{(\gamma)}$ are given by

$$A_{(f)} = \{\frac{a}{f^n} \mid a \in A_{n\gamma}\} \quad \text{and} \quad (A_f^{(\gamma)})_m = \{\frac{a}{f^n} \mid a \in A_{(n+m)\gamma}\}.$$

In particular, we find that $A_{(f)} \simeq (A_f^{(\gamma)})_0$. The isomorphism then follows from part (c).

3.4. **Homogeneous polynomial equations.**

Definition 3.37 (System of homogeneous polynomial equations)**.** Let k be a ring and let I and J be sets. A *system of J homogeneous polynomial equations in I variables* over k is a J-tuple $\Sigma = (f_j)_{j\in J}$ of homogeneous polynomials in $k[x_i \mid i \in I]$. We denote by (Σ) the homogeneous ideal in $k[x_i \mid i \in I]$ generated by $(f_j)_{j\in J}$ and by $k[\Sigma]$ the $\mathbf{N}$-graded k-algebra $k[x_i \mid i \in I]/(\Sigma)$.

The point of considering homogeneous polynomial equations is that they have well-defined solutions in projective space, as we now explain.

Remark 3.38 (Symmetric powers of lines)*.* Let L be a linebundle over a ring R.

(a) The canonical quotient map $L^{\otimes d} \to \mathrm{Sym}^d_R(L)$ is an isomorphism. Using that isomorphisms are Zariski local together with Proposition 3.9 (a) and Proposition 3.1 (e), we may assume L to be trivial. By the universal properties[7] of $R^{\otimes d}$ and $\mathrm{Sym}^d_R(R)$ it suffices to show that for any R-module M and any d-linear map $\varphi : R^{\times d} \to M$, the map φ is symmetric. But this is clear.

(b) The R-algebra $\bigoplus_{d\in\mathbf{Z}} L^{\otimes d}$ from Example 3.35 is initial among R-algebras A with an R-linear map $L \to A$ whose image generates the unit ideal. Indeed, the map $L \to \bigoplus_{d\in\mathbf{Z}} L^{\otimes d}$ is just the inclusion of the rank-one component. As $L \otimes L^{-1} \cong R$, the image generates the unit ideal. Let now A be arbitrary. Let $\varphi : L \to A$ be R linear and such, that its image generates the unit ideal in A. We obtain an A-linear surjection $L \otimes_R A \twoheadrightarrow A$. Since the ranks of L and A are constantly 1, Corollary 3.21 gives us that the above map is an isomorphism. In other words, the line bundle L becomes trivialized after base change to A. We obtain a map

$$\varphi^\vee : L^{-1} \cong \mathrm{Hom}_R(L, R) \to \mathrm{Hom}_A(L \otimes A, A) \cong \mathrm{Hom}_A(A, A) \cong A$$

where the second isomorphism is the one induced by φ. We obtain an unique R-linear map $L \oplus L^{-1} \to A$, and hence, using the universal property of the tensor algebra, a unique R-algebra homomorphism $\bigoplus_{d\in\mathbf{Z}} L^{\otimes d} \to A$, which was the claim.

Construction 3.39 (Evaluation of homogeneous polynomials)**.** Let k be a ring, I a set, $d \in \mathbf{N}$, R a k-algebra, and L a line over R. We construct an evaluation map.

$$k[x_i \mid i \in I]_d \times L^I \to L^{\otimes d}, \quad (f, a) \mapsto f(a),$$

as follows. An element $a \in L^I$ is equivalently an R-linear map $a\colon R^{(I)} \to L$. The map $f \mapsto f(a)$ is then the composite

$$k[x_i \mid i \in I]_d = \mathrm{Sym}^d_k(k^{(I)}) \to \mathrm{Sym}^d_R(R^{(I)}) \xrightarrow{\mathrm{Sym}^d_R(a)} \mathrm{Sym}^d_R(L) = L^{\otimes d}.$$

Concretely, this is the unique k-linear map sending a degree d monomial $\prod_i x_i^{n_i}$ to the tensor product $\bigotimes_i a_i^{\otimes n_i} \in L^{\otimes d}$. Note that this recovers the usual evaluation when $L = R$, in which case also $L^{\otimes d} = R$.

Definition 3.40 (Vanishing locus)**.** Let $F \subseteq k[x_i \mid i \in I]$ be a set of homogeneous polynomials. The *Vanishing locus* of F in $\mathbf{P}^I_k$ is the subfunctor $\mathrm{V}(F) \subseteq \mathbf{P}^I_k$ given by

$$\mathrm{V}(F)(R) = \{a\colon R^{(I)} \twoheadrightarrow L \mid f(a) = 0 \text{ for all } f \in F\} \subseteq \mathbf{P}^I(R).$$

This is indeed a subfunctor: for any k-algebra map $R \to S$, the induced map $\mathbf{P}^I(R) \to \mathbf{P}^I(S)$ sends $\mathrm{V}(F)(R)$ to $\mathrm{V}(F)(S)$.

It is clear that the subfunctor $\mathrm{V}(F) \subseteq \mathbf{P}^I_k$ depends only on the homogeneous ideal (F). However, unlike in the affine case, different homogeneous ideals can still have the same vanishing locus; we will discuss this later.

Definition 3.41 (Solution functor)**.** Let $\Sigma = (f_j)_{j\in J}$ be a system of J homogeneous polynomial equations in I variables over k. Its *solution functor* $\mathrm{hSol}_\Sigma\colon \mathrm{CAlg}_k \to \mathrm{Set}$ is the vanishing locus of $\{f_j \mid j \in J\}$ in $\mathbf{P}^I_k$:

$$\mathrm{hSol}_\Sigma = \mathrm{V}(\{f_j \mid j \in J\}) \subseteq \mathbf{P}^I_k.$$

[7] Using the universal properties makes it clear, that the isomorphism is induced by the comparison map as claimed.

Definition 3.42 (Projective scheme)**.** A functor $\mathrm{CAlg}_k \to \mathrm{Set}$ is a *projective k-scheme* if it is isomorphic to hSol_Σ for some system of homogeneous polynomial equations Σ in finitely many variables over k. We denote $\mathrm{Proj}_k \subseteq \mathrm{Fun}(\mathrm{CAlg}_k, \mathrm{Set})$ the full subcategory spanned by the projective k-schemes.

Remark 3.43. The finitness condition in the definition of projective k-schemes is a matter of convenience. It implies for example that any projective k-scheme is locally of finite type (see Definition 2.85). We will later see that projective k-schemes are *proper*, and for this the finitness condition is essential.

Next, we want to show that the solution functor hSol_Σ depends only on the $\mathbf{N}$-graded k-algebra $k[\Sigma]$, analogously to $\mathrm{Sol}_\Sigma \simeq \mathrm{Spec}(k[\Sigma])$ (Lemma 2.21) We first note that not every $\mathbf{N}$-graded k-algebra A is isomorphic to one of the form $k[\Sigma]$:

Proposition 3.44. *Let k ba a ring. An $\mathbf{N}$-graded k-algebra A is isomorphic to $k[\Sigma]$ if and only if A is generated by A_1, i.e. the canonical map $\mathrm{Sym}_k(A_1) \to A$ is surjective.*

Proof. Assume first that $\mathrm{Sym}_k(A_1) \twoheadrightarrow A$ is surjective. Choose a generating set $k^{(I)} \twoheadrightarrow A_1$ of A_1. Since $\mathrm{Sym}_k(-)$ is a left adjoint, it preserves colimits. In particular, the map $\mathrm{Sym}_k(k^{(I)}) \to \mathrm{Sym}_k(A_1)$ is surjective. Using $\mathrm{Sym}_k(k^{(I)}) \cong k[x_i \mid i \in I]$, we obtain a $\mathbf{N}$-graded surjection $k[x_i \mid i \in I] \twoheadrightarrow \mathrm{Sym}_k(A)$, the kernel of which is a homogeneous ideal.

Conversely, the elements $(x_i)_{i \in I}$ lies in degree 1 and generate $k[\Sigma]$. □

The following construction will be generalized to arbitrary $\mathbf{N}$-graded algebra later, but the case of algebras generated in degree 1 is by far the most important case in practice.

Construction 3.45 (Proj of $\mathbf{N}$-graded algebras generated in degree 1)**.** Let k be a ring and A an $\mathbf{N}$-graded k-algebra generated by A_1. We define the algebraic k-functor $\mathrm{Proj}(A)\colon \mathrm{CAlg}_k \to \mathrm{Set}$ by

$$\mathrm{Proj}(A)(R) = \{\text{quotient lines } A_1 \otimes_k R \twoheadrightarrow L \text{ such that } \mathrm{Sym}_k(A_1) \to \mathrm{Sym}_R(L) \text{ factors through } A\}$$

Remark 3.46. Let k be a ring and A an $\mathbf{N}$-graded k-algebra generated by A_1. We have an alternative description of $\mathrm{Proj}(A)$:

$$\mathrm{Proj}(A)(R) = \{\text{quotient } \mathbf{N}\text{-graded } R\text{-algebra of } A \otimes_k R \text{ isomorphic to symmetric algebra of a line}\}.$$

Given a quotient line $A_1 \otimes_k R \twoheadrightarrow L$, we have the following commutative diagram, since $\mathrm{Sym}_R(-)$ commutes with tensor products[8] and colimits:

$$\begin{array}{ccc} \mathrm{Sym}_R(A \otimes_k R) & \twoheadrightarrow & \mathrm{Sym}_R(L). \\ \downarrow \simeq & \nearrow & \\ \mathrm{Sym}_k(A) \otimes_k R & & \end{array}$$

Since $\mathrm{Sym}_k(A_1) \twoheadrightarrow \mathrm{Sym}_R(L)$ factors through A, we obtain from adjunction that $\mathrm{Sym}_k(A_1) \otimes_k R \twoheadrightarrow \mathrm{Sym}_R(L)$ factors through $A \otimes_k R$. This gives us a quotient map $A \otimes_k R \twoheadrightarrow \mathrm{Sym}_R(L)$.

Conversely, let $A \otimes_k R \twoheadrightarrow \mathrm{Sym}_R(L)$ be a quotient map. Then restriction to the degree 1 part gives the desired map $A_1 \otimes_k R \twoheadrightarrow L$.

Proposition 3.47. *Let Σ be a system of homogeneous polynomial equations over k. Then there exists a canonical isomorphism*

$$\mathrm{hSol}_\Sigma \simeq \mathrm{Proj}(k[\Sigma])\colon \mathrm{CAlg}_k \to \mathrm{Set}\,.$$

Hence, an algebraic k-functor X is a projective k-scheme if and only if $X \simeq \mathrm{Proj}(A)$ for some $\mathbf{N}$-graded k-algebra A that is generated by a finite subset of A_1.

Proof. We first note that there is a correspondence:

$$\mathrm{Sym}_R(-) : \{\text{quotient line } R^{(I)} \twoheadrightarrow L\} \simeq \{\text{graded maps } R[x_i \mid i \in I] \twoheadrightarrow \mathrm{Sym}_R(L)\} : (-)_1$$

[8] An alternative argument is, that $\mathrm{Sym}_R((-) \otimes_k R) \simeq \mathrm{Sym}_k(-) \otimes_k R$, since both are left adjoints of the forgetful functor.

Let $\Sigma = (f_j)_{j\in J}$ be a family of J homogeneous polynomials in I variables and $a\colon R^{(I)} \twoheadrightarrow L$ be a quotient line such that $\mathrm{t}f_j(a) = 0$ for each $j \in J$, that is a defines an element in $\mathrm{hSol}_\Sigma(R)$. The condition $f_j(a) = 0$ for all $j \in J$ is then equivalent to the condition that $\mathrm{Sym}_R(a) : R[x_i \mid i \in I] \twoheadrightarrow \mathrm{Sym}_R(L)$ factors through $k[\Sigma] \otimes_k R \cong R[\Sigma]$. This gives us a surjection $k[\Sigma] \otimes_k R \twoheadrightarrow \mathrm{Sym}(L)$. Since both $\mathrm{Sym}_R(L)$ and $k[\Sigma] \otimes_k R$ are generated by degree 1, this is equivalent to a diagram of the form

$$\begin{array}{ccc} \mathrm{Sym}_R(k[\Sigma]_1 \otimes_k R) & \twoheadrightarrow & \mathrm{Sym}_R(\mathrm{Sym}_R(L)_1) \\ \downarrow & & \downarrow{\scriptstyle \sim} \\ k[\Sigma] \otimes_k R & \twoheadrightarrow & \mathrm{Sym}_R(L) \end{array}$$

Using $\mathrm{Sym}_R((-) \otimes_k R) \simeq \mathrm{Sym}_k(-) \otimes_k R$, the above diagram is equivalent to the following diagram.

$$\begin{array}{ccc} \mathrm{Sym}_k(k[\Sigma]_1) \otimes_k R & \twoheadrightarrow & \mathrm{Sym}_R(L) \\ \downarrow & \nearrow & \\ k[\Sigma] \otimes_k R & & \end{array}$$

which is, by adjunction, furthermore equivalent to a diagram of the form

$$\begin{array}{ccc} \mathrm{Sym}_k(k[\Sigma]_1) & \longrightarrow & \mathrm{Sym}_R(L) \\ \downarrow & \nearrow & \\ k[\Sigma] & & \end{array}$$

This shows that an element of $\mathrm{hSol}_\Sigma(R)$ corresponds uniquely to a point in $\mathrm{Proj}(k[\Sigma])(R)$. Every step in the explanation is reversible. Hence, by going the above steps backwards, we see that we have an isomorphism $\mathrm{hSol}_\Sigma(R) \cong \mathrm{Proj}(k[\Sigma])(R)$ for every R. Furthermore, every step was functorial in R, hence the isomorphisms assemble into a natural isomorphism $\mathrm{hSol}_\Sigma \cong \mathrm{Proj}(k[\Sigma])$.

The second part of the statement follows from Proposition 3.44. □

Example 3.48 (The projective space of a module)*.* Let k be a ring and M a k-module. Consider the functor $\mathbf{P}(M)\colon \mathrm{CAlg}_k \to \mathrm{Set}$ defined by

$$\mathbf{P}(M)(R) = \{\text{quotient lines of } M \otimes_k R\}$$

If $\mathrm{Sym}_k(M)$ is the free k-algebra on M with its natural grading, we have $\mathbf{P}(M) \simeq \mathrm{Proj}(\mathrm{Sym}_k(M))$. In particular, if M is finitely generated, then $\mathbf{P}(M)$ is a projective k-scheme. The projective I-space is a special case of this construction: $\mathbf{P}^I_k \simeq \mathbf{P}(k^{(I)})$.

Analogously to Remark 2.37, we can also define a dual functor $\mathbf{P}^\vee(M)\colon \mathrm{CAlg}_k \to \mathrm{Set}$ by

$$\mathbf{P}^\vee(M)(R) = \{\text{sublines of } M \otimes_k R\}.$$

If M is a vector space, then $\mathbf{P}^\vee(M^\vee) \simeq \mathbf{P}(M)$. This follows from Corollary 3.22. In general, however, $\mathbf{P}^\vee(M)$ is not a scheme.

Remark 3.49*.* If A is an $\mathbf{N}$-graded k-algebra generated by A_1, then $\mathrm{Proj}(A)$ is a subfunctor of $\mathbf{P}(A_1)$, by definition.

3.5. **Loci associated with linear maps.** Consider the projective n-space $\mathbf{P}^n = \mathrm{Proj}(\mathbf{Z}[x_0, \dots, x_n])$, whose R-points are quotient lines $R^{\{0,\dots,n\}} \twoheadrightarrow L$. Given a homogeneous polynomial

$$f \in \mathbf{Z}[x_0, \dots, x_n]_d = \mathrm{Sym}^d(\mathbf{Z}^{\{0,\dots,n\}}),$$

we can consider the subfunctor $\mathrm{D}(f)$ of $\mathbf{P}^n$ consisting of the quotient lines $R^{\{0,\dots,n\}} \twoheadrightarrow L$ such that $L^{\otimes d}$ is generated by the image of f, or equivalently such that the composite

$$R \xrightarrow{f} \mathrm{Sym}^d_R(R^{\{0,\dots,n\}}) \twoheadrightarrow L^{\otimes d}$$

is an isomorphism, where the first morphism is given by multiplication with f. This is similar to the nonvanishing locus of a function, except that f is not quite a function on $\mathbf{P}^n$ (as we will see later, it is a section of the line bundle $\mathcal{O}(d)$ on $\mathbf{P}^n$). For example, $\mathrm{D}(x_i)$ can be identified with the affine n-space $\mathbf{A}^{\{0,\dots,\hat{i},\dots,n\}}$. The goal of this section is to show that $\mathrm{D}(f) \subseteq \mathbf{P}^n$ is an open subfunctor, which is moreover relatively affine, in the sense that its preimage in any affine scheme is affine. We start with some generalities. Let P be a property of modules that it preserved by base change. For any ring A and A-module M, we can then consider the locus where M has property P, which refers to the following subfunctor of $\mathrm{Spec}(A)$:

$$R \mapsto \{\varphi\colon A \to R \mid \text{the } R\text{-module } \varphi^*(M) \text{ has property } P\}$$

For example, M has a zero locus, a finite presentation locus, a flateness locus, etc. Similarly, properties of linear maps, algebras,etc., that are preserved by base change have associated loci.

Definition 3.50 (Loci associated with a linear map)**.** Let A be a ring and let $f\colon M \to N$ be a map of A-modules. We define the following subfunctors of $\mathrm{Spec}(A)$:

(a) The *vanishing locus* $\mathrm{V}(f)$ of f is the locus where f is zero;
(b) The *epimorphism locus* $\mathrm{Epi}(f)$ of f is the locus where f is surjective;
(c) The *monomorphism locus* $\mathrm{Mono}(f)$ is the locus where f is universally injective[9];
(d) The *isomorphism locus* $\mathrm{Iso}(f)$ of f is the locus where f is bijective.

If N is a line (resp. if M is a line), the epimorphism locus (resp. the monomorphism locus) of f is also called the *nonvanishing locus* of f and denoted by $\mathrm{D}(f)$.

Remark 3.51. Proposition 2.62 for $X = \mathrm{Spec}(A)$ can be viewed as a special case of Definition 3.50: if $F \subseteq A$ is a subset and $f\colon A^{(F)} \to A$ is the induced linear map, then $\mathrm{V}(F) = \mathrm{V}(f)$ and $\mathrm{D}(F) = \mathrm{D}(f)$.

Remark 3.52. Let $f\colon V \to W$ be a map of vector spaces. By Proposition 3.18, we have

$$\mathrm{Mono}(f) = \mathrm{Epi}(f^\vee) \quad \text{and} \quad \mathrm{Iso}(f) = \mathrm{Epi}(f) \cap \mathrm{Epi}(f^\vee).$$

If moreover V and W have the same rank (e.g, are both lines), Corollary 3.21 implies that

$$\mathrm{Iso}(f) = \mathrm{Epi}(f) = \mathrm{Mono}(f)$$

Proposition 3.53. *Let A be a ring and let $f\colon M \to N$ be a map of A-modules.*

(a) if N is a vector space, then $\mathrm{V}(f) \subseteq \mathrm{Spec}(A)$ is a closed subfunctor.
(b) If N is a vector space, then $\mathrm{Epi}(f) \subseteq \mathrm{Spec}(A)$ is an open subfunctor.
(c) If M and N are vector spaces, then $\mathrm{Mono}(f) \subseteq \mathrm{Spec}(A)$ is an open subfunctor.
(d) If M and N are vector spaces, then $\mathrm{Iso}(f) \subseteq \mathrm{Spec}(A)$ is an open subfunctor.

Proof. (a) Let $x \in M$ be an element considered as a map $A \to M$ of A-modules, and write $f(x)\colon A \xrightarrow{x} M \xrightarrow{f} N$. It is easy to check that $\mathrm{V}(f) = \bigcap_{x\in M} \mathrm{V}(f(x))$. Since in $\mathrm{Spec}(A)$, the intersection of closed functors is still closed, we may assume that $M = A$. Since $(-)^\vee\colon \mathrm{Vect}_A \to \mathrm{Vect}_A$ is an equivalence of categories, the map f is zero if and only if $f^\vee$ is zero. This implies that $\mathrm{V}(f) = \mathrm{V}(f^\vee)$. We now claim that $\mathrm{V}(f^\vee) = \mathrm{V}(\mathrm{im}(f^\vee))$. By definition, a ring homomorphism $\varphi\colon A \to R$ defines an R-point in $\mathrm{V}(f^\vee)$ if and only if the induced map $f^\vee \otimes \mathrm{id}_R\colon N^\vee \otimes_A R \to R$ is zero, which happens if and only if $\varphi(\mathrm{im}(f^\vee))$ is zero. In fact, we have the following commutative square:

$$\begin{array}{ccc} N^\vee & \xrightarrow{f^\vee} & A \\ \downarrow & & \downarrow{\scriptstyle \varphi} \\ N^\vee \otimes_A R & \xrightarrow[f^\vee \otimes \mathrm{id}_R]{} & R \end{array}$$

[9]We need to use universal injectivity to define the monomorphism locus, as injectivity is not preserved by base change

Since elements of the form $n \otimes 1$ generates $N^\vee \otimes_A R$, where $n \in N^\vee$, that the left-side vertical morphism is non-zero[10].

(b) Since N is a vector space, there exists some vector space N' such that $N \oplus N' \simeq A^n$. Recall that a map between A-modules $g\colon P \to P'$ is surjective if and only if $g \oplus \mathrm{id}_Q\colon P \oplus Q \to P' \oplus Q$ is surjective, using that coker(g) and coker$(g \oplus \mathrm{id}_Q)$ are isomorphic. Therefore, $\mathrm{Epi}(f) = \mathrm{Epi}(f \oplus \mathrm{id}_{N'})$ and we may assume that $N = R^n$. Furthermore, we have[11] $\mathrm{Epi}(f) = \mathrm{Epi}(\Lambda^n f)$, and hence can assume that $N = R$. In this case, for each ring map $\varphi\colon A \to R$, we have a commutative square of A-modules of the form

$$\begin{array}{ccc} M & \xrightarrow{f} & A \\ {\scriptstyle \mathrm{id}_M \otimes \varphi}\downarrow & & \downarrow{\scriptstyle \varphi} \\ M \otimes_A R & \xrightarrow[f \otimes \mathrm{id}_R]{} & R \end{array}$$

Since the image of $\mathrm{id}_M \otimes \varphi$ generates $M \otimes_A R$ as an R-module, the map $f \otimes \mathrm{id}_R$ is surjective if and only if $f \otimes \mathrm{id}_R(\mathrm{im}(\mathrm{id}_M \otimes \varphi))$ generates R, which happens, by commutativity of the square, if and only if $\varphi(\mathrm{im}(f))$ generates R. In other words, we have $\mathrm{Epi}(f) = \mathrm{D}(\mathrm{im}(f))$, which implies that $\mathrm{Epi}(f)$ is an open subfunctor of $\mathrm{Spec}(A)$.

(c) Since M and N are vector spaces, by Remark 3.52, we have $\mathrm{Mono}(f) = \mathrm{Epi}(f^\vee)$, which is an open subfunctor of $\mathrm{Spec}(A)$.

(d) Since $\mathrm{Iso}(f) = \mathrm{Mono}(f) \cap \mathrm{Epi}(f)$, and the intersection of open subfunctors of $\mathrm{Spec}(A)$ is again open, we see that $\mathrm{Iso}(f)$ is open.

□

We now study more closely the nonvanishing locus of a map of lines $s\colon L' \to L$ over R. Since L' is invertible, we can tensor s with the inverse of L' without changing its nonvanishing locus, so it suffices to consider the case $L' = R$. We can then identify the R-linear map $s\colon R \to L$ with an element $s \in L$. In the special case $L = R$, the theory of localization of rings tell us that the nonvanishing locus of any $s \in R$ is isomorphic to $\mathrm{Spec}(R_s)$ (see Proposition 2.62). This theory can be generalized to nontrivial lines as follows.

Definition 3.54 (Periodic module)**.** Let R be a ring, L a line over R, and $s \in L$. An R-module M is *s-periodic* if the map

$$\mathrm{id}_M \otimes s\colon M \cong M \otimes_R R \to M \otimes_R L$$

is an isomorphism.

Proposition 3.55. *Let R be a ring, L a line over R, and $s \in L$.*

(a) There exists an initial s-periodic R-algebra R_s.

(b) The forgetful functor $\mathrm{Mod}_{R_s} \to \mathrm{Mod}_R$ identifies Mod_{R_s} with the full subcategory of s-periodic R-modules in Mod_R. Hence, the inclusion of this subcategory has a left adjoint[12] $M \mapsto M_s$ given by $M_s = M \otimes_R R_s$.

(c) The R-module M_s can be computed as the colimit of the sequence

$$M \xrightarrow{s} M \otimes_R L \xrightarrow{s} M \otimes_R L^{\otimes 2} \xrightarrow{s} \dots.$$

Proof. We define

$$M_s = \mathrm{colim}(M \xrightarrow{s} M \otimes_R L \xrightarrow{s} M \otimes_R L^{\otimes 2} \xrightarrow{s} \dots).$$

In fact, an explicit description of M_s is the following:

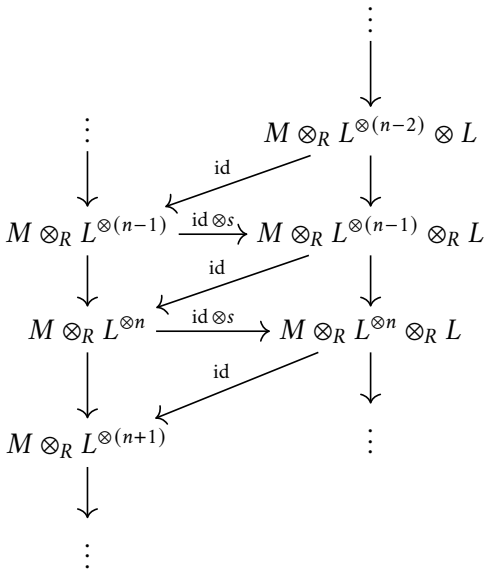
$$M_s = \{(m, n) \mid n \in \mathbf{N},\ m \in M \otimes_R L^{\otimes n}\}/\sim,$$

[10]Here we need to assume that neither N nor R are non-zero. But $R = 0$ is always contained in $V(f)$ for all maps f, while in the case $N = 0$ the map f is the zero map, hence $V(f) = \mathrm{Spec}(A)$ is trivially closed.

[11]Here Λ^n is the n-fold exterior power. We use the classical result that for a map $f : M \to R^n$, we have that $\Lambda^n(f)$ is surjective, if and only if $1 \in \mathrm{im}(\Lambda^n f)$, if and only if f itself was surjective.

[12]A localization of this form is also known as a *smashing localization*, which is equivalent to the fact that R_s will be an idempotent algebra.

where $\sim$ is the equivalence relation generated by $(m, n) \sim (m \otimes s, n+1)$. We will usually abuse notation by writing $\frac{m}{s^n}$ for (m, n). Consider now the following diagram:



Since the squares

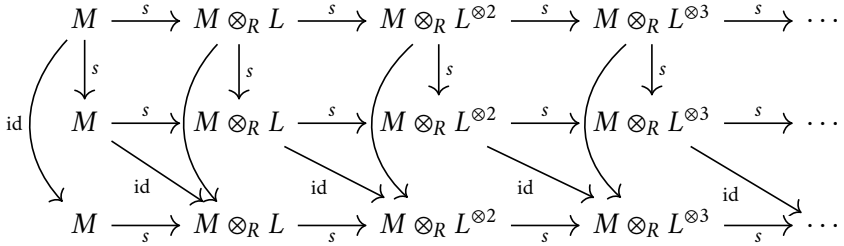
$$\begin{array}{ccc} M \otimes_R L^{\otimes(n-1)} & \xrightarrow{\mathrm{id}\otimes s} & M \otimes_R L^{\otimes(n-1)} \otimes_R L \\ \downarrow{\scriptstyle \mathrm{id}\otimes s} & & \downarrow{\scriptstyle \mathrm{id}\otimes s\otimes \mathrm{id}} \\ M \otimes_R L^{\otimes n} & \xrightarrow{\mathrm{id}\otimes s} & M \otimes_R L^{\otimes n} \otimes_R L \end{array}$$

appearing in the big diagram commute, we obtain a map between the colimits $s\colon M_s \to M_s \otimes_R L$. Consider now the other squares appearing in the big diagram above, where we explicitly spelled out the action of the maps

$$\begin{array}{ccc} M \otimes_R L^{\otimes(n-2)} \otimes_R L & \xrightarrow{\mathrm{id}} & M \otimes_R L^{\otimes(n-1)} \\ \downarrow{\scriptstyle \mathrm{id}\otimes s\otimes \mathrm{id}} & & \downarrow{\scriptstyle \mathrm{id}\otimes s} \\ M \otimes_R L^{\otimes(n-1)} \otimes_R L & \xrightarrow{\mathrm{id}} & M \otimes_R L^{\otimes n} \end{array}$$

$$\begin{array}{ccc} m \otimes l_1 \otimes \ldots \otimes l_{n-2} \otimes l & \longmapsto & m \otimes l_1 \otimes \ldots \otimes l_{n-2} \otimes l \\ \downarrow & & \downarrow \\ m \otimes l_1 \otimes \ldots \otimes l_{n-2} \otimes s \otimes l & \longmapsto & m \otimes l_1 \otimes \ldots \otimes l_{n-2} \otimes l \otimes s \end{array}$$

Recall that we have $L^d \simeq \mathrm{Sym}^d_R(L)$, i.e. Σ_d acts trivially on $L^{\otimes d}$. Take $d = 2$, we find that the above diagram commutes, hence gives us a map $t\colon M_s \otimes_R L \to M_s$. To show that $t \circ s = \mathrm{id}$, observe that the following diagram commutes:



Thus, map of diagrams defining the composition $t \circ s$ and the map of diagram inducing the identity commute, hence, by the universal property of the colimit, are the same. Similarly, $s \circ t = \mathrm{id}$, implying that M_s is s-periodic.

Notice that in the case $M = R$, we can identify the above colimit with a quotient of $T(L) \cong \mathrm{Sym}(L)$, the tensor-algebra of the line bundle L, by the (non-graded) generated ideal by $(1 - s)$. In particular, R_s acquires a canonical R-algebra structure. We will show later, that R_s is the inital s-periodic R algebra.

That $M \otimes_R R_s = M_s$ follows then from the fact that tensor product commutes with colimits. It is clear that the construction $M \mapsto M_s$ assembles into a functor $(-)_s : \mathrm{Mod}_R \to \mathrm{Mod}_{R_s}$. It now suffices to show that $(-)_s$ is the left adjoint of the inclusion $\mathrm{Mod}_{R_s} \hookrightarrow \mathrm{Mod}_R$.

There is a canonical morphism $\mathrm{can} : M \to M_s$ given by the fist leg of the colimit diagram. We want to show that this comparison morphism $\mathrm{can}^* : \mathrm{Map}(M_s, N) \to \mathrm{Map}(M, N)$ is an isomorphism natural in N and M. Using continuity of $\mathrm{Map}(-,-)$ in the first variable, this morphism is equivalent to

$$\lim_d \left(\mathrm{id}_M \otimes s^{\otimes d}\right)^* : \left(\lim_d \mathrm{Map}(M \otimes_R L^{\otimes d}, N)\right) \to \mathrm{Map}(M, N).$$

As L is invertible, so is $L^{\otimes d}$ for all d with dual $\left(L^{-1}\right)^{\otimes d}$, where L^{-1} the dual of L. In particular, $- \otimes_R L^{\otimes d}$ is left adjoint to $- \otimes_R \left(L^{-1}\right)^d$. Thus, we may identify the comparison morphism with

$$\lim_d \left(\left(s^\vee\right)^{\otimes d} \otimes \mathrm{id}_N\right)_* : \left(\lim_d \mathrm{Map}(M, \left(L^{-1}\right)^{\otimes d} \otimes_R N)\right) \to \mathrm{Map}(M, N).$$

It suffices to show that for all d the morphism $\left(\left(s^\vee\right)^{\otimes d} \otimes \mathrm{id}_N\right)_*$ is an isomorphism. Further, if N is s-periodic, so is $\left(L^{-1}\right)^{\otimes d} \otimes_R N$ for all d, so inductively, it suffices to show that

$$(s^\vee \otimes \mathrm{id}_N)_* : \mathrm{Map}(M, L^{-1} \otimes_R N) \to \mathrm{Map}(M, N)$$

is an isomorphism. To this end, we have the following commutative diagram

$$\begin{array}{ccc} L \otimes_R L^{-1} & \xrightarrow{\mathrm{id}_L \otimes s^\vee} & L \\ {\scriptstyle \mathrm{ev}}\downarrow & \nearrow_{s} & \\ R & & \end{array}$$

Here, ev is an isomorphism. After tensoring with N, the diagonal morphism becomes an isomorphism by assumption. By two-out-of-three, we find that the map $\mathrm{id}_L \otimes s^\vee \otimes \mathrm{id}_N$ is an isomorphism. As L is invertible, $s^\vee \otimes \mathrm{id}_N$ is an isomorphism implying the claim. As all these identifications were functorial, hence we obtain that $(-)_s$ is a left adjoint.

Now, for each R-module M and N, we have an isomorphism

$$(M \otimes_R N)_s \simeq M_s \otimes_R N_s,$$

since tensor product preserves colimits in both variables. In particular, if A is a R-algebra, then A_s is an R-algebra with the multiplication map given by[13] $A_s \otimes_R A_s \simeq (A \otimes_R A)_s \to A_s$. This shows again that R_s is an R-algebra. More generally, the above shows that $(-)_s$ descends to a left adjoint $\mathrm{CAlg}_R \to \mathrm{CAlg}_{R_s}$ of the inclusion $\mathrm{CAlg}_R \hookrightarrow \mathrm{CAlg}_{R_s}$, and hence that R_s is the initial s-periodic R algebra. □

Corollary 3.56. *Let R be a ring, L a line over R, and $s \in L$. Then there exists a canonical isomorphism* $\mathrm{Spec}(R_s) \xrightarrow{\sim} \mathrm{D}(s) \subseteq \mathrm{Spec}(R)$. *In particular,* $\mathrm{D}(s)$ *is an affine scheme.*

Proof. It suffices to show that there is a canonical isomorphism, natural in A:

$$\mathrm{D}(s)(A) \xrightarrow{\sim} \mathrm{Map}(R_s, A).$$

By definition, we have

$$\mathrm{D}(s)(A) = \{\varphi : R \to A \mid A \text{ is a } s\text{-periodic } R\text{-algebra via } \varphi\}.$$

[13]This property is called *symmetric monoidality*. In fact, one can show that any smashing localization induces a symmetric monoidal functor.

Let $\varphi\colon R \to A$ in $\mathrm{D}(s)(A)$. Since R_s is inital among s-periodic R-algebras, there exists an unique map $\varphi'\colon R_s \to A$ which restricts to $\varphi\colon R \to A$.

Conversely, let $\varphi'\colon R_s \to A$ be a ring map, that is, A is a R_s-module. In particular, multiplication by s induces an isomorphism. Restricting to R, we obtain a canonical R-module structure on A, for which A is s-periodic. □

Remark 3.57 (Improved Zariski descent). All the Zariski descent results from before can be generalized by replacing the family $(f_i)_{i\in I}$ generating the unit ideal in R with a family $(s_i)_{i\in I}$ generating some line L over R. For example, the generalized version of Corollary 2.74 says that, for any affine scheme X, the diagram

$$X(R) \to \prod_{i\in I} X(R_{s_i}) \rightrightarrows \prod_{i,j\in I} X(R_{s_i,s_j})$$

is an equalizer diagram (where $s_is_j := s_i \otimes s_j \in L^{\otimes 2}$). For each result, we can either repeat the proof with no essential changes, or reduce the new statement to the old one using that $\mathrm{D}(s) = \mathrm{D}(F_s)$, where $F_s \subseteq R$ is the image by $s^\vee\colon L^\vee \to R$ of a generating family of $L^\vee$, in combination with Corollary 2.76. Indeed, note that if $(s_i)_{i\in I}$ generates L, then $\mathrm{DD}((s_i)_{i\in I}) = \mathrm{Spec}(R)$, and use Corollary 3.56.

Corollary 3.58. *Let R be a ring, L a line over R and $(s_i)_{i\in I}$ be a family of elements generating L. Let M and N be R-modules, then the diagram*

$$\mathrm{Map}(M,N) \to \prod_{i\in I} \mathrm{Map}(M_{s_i}, N_{s_i}) \to \prod_{i,j\in I} \mathrm{Map}(M_{s_is_j}, N_{s_is_j})$$

is an equalizer diagram.

Proof. This follows from applying $\mathrm{Map}(N,-)$ to Remark 3.57 and use the universal property of $(-)_s$. □

Example 3.59 (Projective completion of affine spaces). Let k be a ring and M a k-module. Recall the algebraic k-functors $\mathbf{A}(M)$ and $\mathbf{P}(M)$ from Example 2.36 and Example 3.48. There is a canonical embedding

$$\mathbf{A}(M) \hookrightarrow \mathbf{P}(M \oplus k), \quad (a\colon M \otimes_k R \to R) \mapsto ((a, \mathrm{id}_R)\colon (M\otimes_k R)\oplus R \twoheadrightarrow R).$$

We claim that this is an open immersion. Indeed, given any map $\mathrm{Spec}(R) \to \mathbf{P}(M\oplus k)$ classifying a quotient line $(M\otimes_k R)\oplus R \twoheadrightarrow L$, the A-points for some commutative ring A of the preimage of $\mathbf{A}(M)$ in $\mathrm{Spec}(R)$ are given by those ring homomorphisms $R \to A$, such that $L\otimes_R A \simeq A$. That is, the preimage is the locus where the line L gets trivialized, or equivalently, it is the isomorphism locus of the composite

$$R \hookrightarrow (M\otimes_k R)\oplus R \twoheadrightarrow L,$$

which is open in $\mathrm{Spec}(R)$ by (d) of Proposition 3.53, and also affine by Corollary 3.56. Taking $M = k^{(I)}$ we obtain a canonical open immersion

$$\mathbf{A}^I \hookrightarrow \mathbf{P}^{I\sqcup\{0\}},$$

identifying $\mathbf{A}^I$ with $\mathrm{D}(x_0) \subseteq \mathbf{P}^{I\sqcup\{0\}}$, where we have define $\mathrm{D}(x_0)$ at the beginning of this section. For $I = \{1,\dots,n\}$, this is an open immersion $\mathbf{A}^n \hookrightarrow \mathbf{P}^{n+1}$.

3.6. **The functor Proj.** In this section, we define $\mathrm{Proj}(A)$ for any $\mathbf{N}$-graded ring A. If A is finitely generated as an A_0-algebra, which is usually the case in practice, we will see that $\mathrm{Proj}(A)$ reduces to Construction 3.45. Thus, the increase in generality is minimal. Instead, the main result of this section is that Construction 3.45 satisfies

$$\mathrm{Proj}(A) \simeq \mathrm{Proj}(A^{(d)}).$$

We will see that this isomorphism gives rise to the *Veronese embedding*, which is a central feature of projective geometry.

Definition 3.60 (Eventually surjective map)**.** A map of $\mathbf{N}$-graded rings $A \to B$ is called *eventually surjective* if for every $d \geq 1$ and every $b \in B_d$, there exists $n \geq 1$ such that $A_{nd}\otimes_{A_0} B_0 \to B_{nd}$ hits b^n. We denote by $\mathrm{CAlg}^{\mathbf{N},\mathrm{es}}$ the category of $\mathbf{N}$-graded rings and eventually surjective maps.

Recall from notation Notation 3.36 that $A^{(d)} = \bigoplus_{n\in\mathbf{N}} A_{nd}$ and $A_{(f)} = (A_f)_0$.

Construction 3.61. We construct a functor

$$\mathrm{Proj}\colon (\mathrm{CAlg}^{\mathbf{N},\mathrm{es}})^{\mathrm{op}} \to \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})$$

with a natural transformation $\mathrm{Proj}(-) \to \mathrm{Spec}\,((-)_0)$. If A is an $\mathbf{N}$-graded k-algebra, we can thus view $\mathrm{Proj}(A)$ as an algebraic k-functor, and we will see in Theorem (ii), that is recovers Construction 3.45, when A is generated by A_1.

correct reference

Let A be an $\mathbf{N}$-graded ring. We first define

$$\mathrm{Proj}_1(A) \in \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})_{/\mathrm{Spec}(A_0)} \simeq \mathrm{Fun}(\mathrm{CAlg}_{A_0}, \mathrm{Set})$$

as follows. Given an A_0-algebra R, $\mathrm{Proj}_1(A)(R)$ is the set of pairs (L, φ), where L is a quotient line of the R-module $A_1 \otimes_{A_0} R$ and $\varphi\colon A \otimes_{A_0} R \to \mathrm{Sym}_R(L)$ is a map of $\mathbf{N}$-graded R-algebras extending the quotient map[14] $A_1 \otimes_{A_0} R \twoheadrightarrow L$. Note that φ is automatically surjective, so that we can alternatively describe $\mathrm{Proj}_1(A)(R)$ as the set of quotient $\mathbf{N}$-graded algebras of $A \otimes_{A_0} R$ that are isomorphic to symmetric algebras of lines. Indeed, by assumption, the composition $A_1 \otimes_{A_0} R \to A \otimes_{A_0} R \to \mathrm{Sym}_R(L)$ is surjective on $(-)_1$. As $\mathrm{Sym}_R(L)$ is generated by its degree 1 elements, this implies the claim.

For $d \in \mathbf{N}$, we define $\mathrm{Proj}_d(A) = \mathrm{Proj}_1(A^{(d)})$. For any $n \in \mathbf{N}$, there is then a canonical map

$$\mathrm{Proj}_d(A) \to \mathrm{Proj}_{nd}(A), \quad (L, \varphi) \mapsto (L^{\otimes n}, \varphi^{(n)}),$$

where $L^{\otimes n}$ is a quotient of $A_{nd} \otimes_{A_0} R$ via φ_n.

In fact, let

$$f\colon A^{(d)} \otimes_{A_0} R \twoheadrightarrow \mathrm{Sym}_R(L)$$

be a quotient $\mathbf{N}$-graded R-algebra map. Then we can restrict and obtain the following map

$$f^{(n)}\colon A^{(nd)} \simeq (A^{(d)})^{(n)} \twoheadrightarrow \mathrm{Sym}_R(L)^{(n)} \simeq \mathrm{Sym}_R(L^{\otimes n}),$$

where the last isomorphism follows from Remark 3.38.

This defines a functor

$$\mathbf{N}^{\mathrm{div}} \to \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set}), \quad d \mapsto \mathrm{Proj}_d(A),$$

where $\mathbf{N}^{\mathrm{div}}$ is the poset of natural numbers under divisiblility[15]. We then set

$$\mathrm{Proj}(A) = \operatorname*{colim}_{d \in \mathbf{N}^{\mathrm{div}}_{>0}} \mathrm{Proj}_d(A).$$

Since 0 is the final object of $\mathbf{N}^{\mathrm{div}}$, this comes with a canonical map

$$\mathrm{Proj}(A) \to \mathrm{Proj}_0(A) = \mathrm{Proj}_1(A^{(0)}) = \mathrm{Proj}_1(A_0[x]) = \mathbf{P}^0_{A_0} = \mathrm{Spec}(A_0).$$

The second equality follows from that $A^{(0)} \simeq A_0[x]$, since at each degree, $A^{(0)}$ consists of a copy of A_0. The third equality follows from the following: an element of $\mathrm{Proj}_1(A_0[x])(R)$ corresponds to a quotient $\mathbf{N}$-graded R-algebra map

$$R[x] \simeq A_0[x] \otimes_{A_0} R \twoheadrightarrow \mathrm{Sym}_R(L).$$

Restricting to degree one, we obtain $R\{x\} \twoheadrightarrow L$, which is a surjective map between R-lines, hence an isomorphism. Therefore, the quotient $\mathbf{N}$-graded R-algebra maps corresponds to a map

$$R[x] \to \mathrm{Sym}_R(R) \simeq R[t],$$

extending the identity $R \to R$ and sending the degree 1 generator x to the degree 1 generator t. This map is uniquely determined, hence $\mathrm{Proj}_1(A_0[x])$ is isomorphic to the terminal projective A_0-scheme $\mathbf{P}^{(0)}_{A_0} = \mathrm{Spec}(A_0)$.

Let $\alpha\colon A \to B$ be a map of $\mathbf{N}$-graded rings and let (L, φ) be an R-point of $\mathrm{Proj}_d(B)$ for some $d \geq 1$. Then there exists $b_1, \ldots, b_r \in B_d$ whose images under $\varphi_d : (B \otimes_{B_0} R)_d \to L^{\otimes d}$ generate $L^{\otimes d}$. If for some $n \geq 1$, the

[14] Recall from Remark 3.23 that we made the convention that epimorphisms and quotient lines are considered interchangeably. In particular, we could also consider the R-points as equivalence classes of tuples (L, φ) where L is an R-line and $\varphi : A \otimes_{A_0} R \to \mathrm{Sym}_R L$ is a ring homomorphism inducing a surjection on $(-)_1$. Two tuples (L, φ) and (L', φ') are considered equivalent, if they are isomorphic with an isomorphism compatible with φ and φ'. We will use this in the proceedings without further mentioning.

[15] That is, for $m, n \in \mathbf{N}$ we have $\mathrm{Hom}_{\mathbf{N}^{\mathrm{div}}}(m, n) \cong *$ if and only if $m | d$.

powers $b_1^n, \dots, b_r^n$ are in the image of $\alpha_{nd}\colon A_{nd} \otimes_{A_0} B_0 \to B_{nd}$, then $(L^{\otimes n}, \varphi^{(n)} \circ \alpha^{(nd)})$ is an R-point of $\mathrm{Proj}_{nd}(A)$. Thus, if α is eventually surjective, precomposition with α defines a map $\mathrm{Proj}(\alpha)\colon \mathrm{Proj}(B) \to \mathrm{Proj}(A)$.

Remark 3.62. For every $d \geq 1$, the inclusion $d\mathbf{N}^{\mathrm{div}}_{>0} \subseteq \mathbf{N}^{\mathrm{div}}$ is cofinal and hence induces an isomorphism

$$\mathrm{Proj}(A) \simeq \mathrm{Proj}(A^{(d)}).$$

Indeed, since $\mathbf{N}^{\mathrm{div}}$ is a poset, it suffices to show that for every element $n \in \mathbf{N}^{\mathrm{div}}$ there exists an element $m \in d\mathbf{N}^{\mathrm{div}}_{>0}$ such that $\mathrm{Hom}_{\mathbf{N}^{\mathrm{div}}}(n, m)$ is non-empty. $dn \in d\mathbf{N}^{\mathrm{div}}_{>0}$ does the job.

Remark 3.63 (Degree d maps). We can enlarge the domain of definition of Proj as follows. If A and B are $\mathbf{N}$-graded rings and $d \in \mathbf{N}$, a *degree d map* from A to B is a graded ring map $A \to B^{(d)}$. If it is eventually surjective and $d \geq 1$, then by Remark 3.62, it induces

$$\mathrm{Proj}(B) \simeq \mathrm{Proj}(B^{(d)}) \to \mathrm{Proj}(A).$$

This also works if $d = 0$: we have the following composition

$$\mathrm{Proj}(B) \to \mathrm{Spec}(B_0) = \mathrm{Proj}(B^{(0)}) \to \mathrm{Proj}(A).$$

We can compose a degree d with a degree e map to obtain a degree $d + e$ map. Thus, there is an enlargement of the category $\mathrm{CAlg}^{\mathbf{N},\mathrm{es}}$ with the same objects and in which the set of maps from A to B is

$$\coprod_{d \in \mathbf{N}} \mathrm{Map}_{\mathrm{CAlg}^{\mathbf{N},\mathrm{es}}}(A, B^{(d)}),$$

and the functor Proj extends to this category.

Proposition 3.64 (Zariski descent for Proj)**.** *Let A be an $\mathbf{N}$-graded ring, let $d \in \mathbf{N}$, and let X be either $\mathrm{Proj}_d(A)$ or $\mathrm{Proj}(A)$. For every ring R, every line L over R, and every generating family $(s_i)_{i \in I}$ of L, the diagram*

$$X(R) \to \prod_{i \in I} X(R_{s_i}) \rightrightarrows \prod_{i,j \in I} X(R_{s_i s_j})$$

is an equalizer.

Proof. We first prove the case for $X = \mathrm{Proj}_d(A)$, where $d \in \mathbf{N}$ and A is a $\mathbf{N}$-graded ring. Since $\mathrm{Proj}_d(A) \simeq \mathrm{Proj}_1(A^{(d)})$, we may assume that $d = 1$. Let $(L_i, \varphi_i) \in X(R_{s_i})$ such that $(L_i, \varphi_i)_{s_j} \simeq (L_j, \varphi_j)_{s_i}$ in $X(R_{s_i s_j})$, for every $i, j \in I$. In particular, there exists an unique isomorphism such that the diagram commutes

$$\begin{array}{ccc} A_1 \otimes_{A_0} R_{s_i s_j} & \twoheadrightarrow & (L_i)_{s_j} \\ \downarrow & \overset{\simeq}{\swarrow}_{\exists! \alpha_{ij}} & \\ (L_j)_{s_i}. & & \end{array}$$

Due to this uniqueness, the isomorphisms of lines are compatible. In particular, since there are surjections from $A_1 \otimes_{A_0} R_{s_i s_j s_k}$ to all of objects in the following triangle, the triangle commutes

$$\begin{array}{ccccc} (L_i)_{s_j s_k} & & \xrightarrow{\alpha_{ij}} & & (L_j)_{s_i s_k} \\ & {\scriptstyle \alpha_{ik}} \searrow & & \swarrow {\scriptstyle \alpha_{jk}} & \\ & & (L_k)_{s_i s_j}. & & \end{array}$$

Therefore, by Theorem 2.72, there exists an R-module L over R such that there is an isomorphism $\beta_i\colon L_{s_i} \simeq L_i$ for each $i \in I$. This also implies that L is a line, since being a line is Zariski local. Furthermore by Corollary 3.58, using that $\mathrm{Sym}_R(L) \otimes_R R_s \simeq \mathrm{Sym}_{R_s}(L_s)$, we obtain a map

$$\varphi\colon A \otimes_{A_0} R \to \mathrm{Sym}_R(L)$$

such that $\varphi_{s_i} = \varphi_i$. In particular, the map φ is surjective, since being surjective is Zariski local. This gives us the element (L, φ) lifting (L_i, φ_i). The uniqueness also follows from Theorem 2.72.

We will now consider the case where $X = \mathrm{Proj}(A)$. If I is finite, then the result follows from that filtered colimits commutes with finite limits in the category of sets. If I is infinite, mimicking the proof of Theorem 2.72, we may reduce to the case where I is finite. □

Remark 3.65. In the proof of Proposition 3.64, we have regarded $\mathrm{Proj}(A)$ as a functor

$$\mathrm{Proj}(A)\colon \mathrm{CAlg}_{A_0} \to \mathrm{Set}\,.$$

The same statements hold, when we view the functor $\mathrm{Proj}(A)$ as a functor $\mathrm{Proj}(A)\colon \mathrm{CAlg} \to \mathrm{Set}$ under the equivalence

$$\mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})_{/\mathrm{Spec}(A_0)} \simeq \mathrm{Fun}(\mathrm{CAlg}_{A_0}, \mathrm{Set}).$$

In fact, we will see later, that the above equivalence restricts to an equivalence of categories of Zariski sheaves

$$\mathrm{Sh}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}})_{/\mathrm{Spec}(A_0)} \simeq \mathrm{Sh}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}}_{A_0}).$$

Lemma 3.66. *Let A be an $\mathbf{N}$-graded ring, let $d \geq 1$, and let $f \in A_d$. Let $\mathrm{D}(f) \subseteq \mathrm{Proj}_d(A)$ be the subfunctor of pairs (L, φ) such that $\varphi(f)$ generates L (which is open by Proposition 3.53).*

(a) There is a canonical isomorphism $\mathrm{D}(f) \simeq \mathrm{Spec}(A_{(f)})$.

(b) For every $n \geq 1$, the map $\mathrm{Proj}_d(A) \to \mathrm{Proj}_{nd}(A)$ induces an isomorphism $\mathrm{D}(f) \xrightarrow{\sim} \mathrm{D}(f^n)$. Moreover, the square

$$\begin{array}{ccc} \mathrm{D}(f) & \xrightarrow{\sim} & \mathrm{D}(f^n) \\ \downarrow & & \downarrow \\ \mathrm{Proj}_d(A) & \longrightarrow & \mathrm{Proj}_{nd}(A) \end{array}$$

is cartesian.

(c) Let $f_1, \ldots, f_k \in A$ be homogeneous elements of positive degrees. Under the isomorphism in (b), we have $\mathrm{D}(f_1, \ldots, f_k) = \mathrm{D}(f_1) \cap \ldots \cap \mathrm{D}(f_k)$ inside $\mathrm{Proj}_d(A)$ for all sufficiently divisible d.

Proof. (a) We will prove the statement by constructing inverse natural transformations. We start with the natural transformation $\mathrm{D}(f) \to \mathrm{Spec}(A_{(f)})$. An R-point of $\mathrm{D}(f)$ is a quotient map $\varphi\colon A^{(d)} \otimes_{A_0} R \twoheadrightarrow \mathrm{Sym}_R(L)$ such that $\varphi(f)$ generates L. After postcomposition with $\mathrm{Sym}_R(L) \hookrightarrow \bigoplus_{n\in\mathbf{Z}} L^{\otimes n}$, the element f maps to a unit, with inverse given by $f^\vee \in L^{-1}$. Therefore, by the universal property of localization, there exists an unique map $\tilde{\varphi}$ such that the right square in the following diagram commutes, making the whole diagram commutative:

$$\begin{array}{ccccc} A^{(d)} & \xrightarrow{\psi} & A^{(d)} \otimes_{A_0} R & \xrightarrow{\varphi} & \mathrm{Sym}_R(L) \\ \downarrow & & \downarrow & & \downarrow \\ A_f^{(d)} & \xrightarrow{\tilde{\psi}} & A_f^{(d)} \otimes_{A_0} R & \xrightarrow{\exists!\tilde{\varphi}} & \bigoplus_{n\in\mathbf{Z}} L^{\otimes n} \end{array}$$

Consider the map

$$\mathrm{ev} : \bigoplus_{\mathbf{Z}} L^{\otimes n} \cong \bigoplus_{\mathbf{N}} (L^{\otimes n} \oplus L^{\otimes(-n)}) \xrightarrow{\bigoplus_{\mathbf{N}} \mathrm{ev}_n} R$$

The composition $(\mathrm{ev} \circ \tilde{\varphi} \circ \tilde{\psi})_1$ is the element in $\mathrm{Spec}(A_{(f)})(R)$. That the obtained maps assemble into a natural transformation is clear.

Now, we will construct a map $\mathrm{Spec}(A_{(f)}) \to \mathrm{D}(f)$. Consider a map $g\colon A_{(f)} \to R$, then we have $A_f^{(d)} \simeq A_{(f)}[x^{\pm 1}]$ with x a degree d generator. As the image of the $\mathbf{Z}$-graded map $A^{(d)} \to R[x^{\pm 1}]$ lies only in the positive degree of $R[x^{\pm 1}]$, we obtain the dashed factorization making the following diagram

commute

$$\begin{array}{ccc} A^{(d)} & \overset{\exists!}{\dashrightarrow} & R[x] \\ \downarrow & & \downarrow \\ A_f^{(d)} & \xrightarrow{\cong} A_{(f)}[x^{\pm 1}] \longrightarrow & R[x^{\pm 1}]. \end{array}$$

After extension of scalars, we obtain a map $A^{(d)} \otimes_{A_0} R \twoheadrightarrow \mathrm{Sym}_R(R)$, which is surjective, since x is the image of f. It is again clear that these maps assemble into a natural transformation.

To see that the two natural transformations constructed above are inverse to each other, it suffices to notices that an R-point $(L, \varphi) \in \mathrm{D}(f)(R)$ also provides a fixed isomorphism $R \xrightarrow{\cong} L$ given by right-multiplication with $\varphi(f)$.

(b) In order to show that the diagram is cartesian, it sufficies to show that, for each map $\varphi\colon A^{(d)} \otimes_{A_0} R \twoheadrightarrow \mathrm{Sym}_R(L)$, the element $\varphi(f)$ generates L if and only if the element $\varphi(f^n)$ generates $L^{\otimes n}$.

In fact, we will prove a stronger statement. Let $(s_i)_{i\in I}$ be a finite family of elements in L. Then s_i generates L for each $i \in I$ if and only if $\otimes_{i\in I} s_i$ generates $L^{\otimes I}$. Assume that for each $i \in I$, the map $s_i\colon R \to L$ is a surjection. Then $\otimes_{i\in I} s_i\colon R \simeq R^{\otimes I} \to L^{\otimes I}$ is again a surjection, using that tensor products are right exact. Conversely, assume that $\otimes_{i\in I} s_i\colon R \to L^{\otimes I}$ is a surjection. Choose a family $(f_j)_{j\in J}$ generating R such that $L_{f_j} \simeq R_{f_j}$ for each $j \in J$, that is $(f_j)_{j\in J}$ form a trivializing cover of R for R. Then the map

$$(\otimes_{i\in I} s_i)_{f_j}\colon R_{f_j} \to \otimes_{i\in I} R_{f_j} \cong R_{f_j}, \quad 1 \mapsto \prod_{i\in I} s_i'$$

is a surjection, where s_i' is the image of s_i under the localization. Therefore, the elements s_i' are invertible in R_{f_j} for each $i \in I$ and $j \in J$. Since being an isomorphism is a Zariski local property, the maps $s_i\colon R \to L$ are isomorphisms for all $i \in I$ finishing the claim.

We will now proceed to show that $\mathrm{D}(f) \simeq \mathrm{D}(f^n)$. We have the following commutative square:

$$\begin{array}{ccc} \mathrm{D}(f) & \longrightarrow & \mathrm{D}(f^n) \\ \simeq\downarrow & & \downarrow\simeq \\ \mathrm{Spec}(A_{(f)}) & \xrightarrow{\sim} & \mathrm{Spec}(A_{(f^n)}) \end{array}$$

By (a), the two vertical maps are isomorphisms. By the universal property of the localization implies that $A_{(f)}$ and $A_{(f^n)}$ are isomorphic, hence the lower horizontal map is an isomorphism. Therefore, the upper horizontal map is an isomorphism.

(c) Taking powers of $f_1, \ldots, f_k$ and using (b), we may assume all f_i to be of the same degree. Using the identification in (a) gives the claim.

□

Theorem 3.67 (Structure of Proj)**.** *Let A be an $\mathbf{N}$-graded ring and let $d \geq 1$.*

(a) The map $\mathrm{Proj}_d(A) \to \mathrm{Proj}(A)$ is an open immersion.

(b) Suppose that A is generated as an A_0-algebra by homogeneous elements whose degrees divide d. Then the map $\mathrm{Proj}_d(A) \to \mathrm{Proj}(A)$ is an isomorphism. In particular, if A is generated by $A_{\leq 1}$, then $\mathrm{Proj}_1(A) \xrightarrow{\sim} \mathrm{Proj}(A)$.

Proof. The guiding idea for the proof is the following hope, which is, unfortunately, not true

$$\text{"}\mathrm{Proj}_d(A) \simeq \bigcup_{f\in A_d} \mathrm{D}(f)\text{"}.$$

However note that given an R-point $(L, \varphi) \in \mathrm{Proj}_d(A)(R)$ represented by a map $(x : \mathrm{Spec}(R) \to \mathrm{Proj}_d(A)$, we have a cartesian diagram

$$\begin{array}{ccc} \mathrm{Spec}(R_{\varphi(f)}) & \longrightarrow & \mathrm{D}(f) \\ \downarrow & \lrcorner & \downarrow \\ \mathrm{Spec}(R) & \xrightarrow{x} & \mathrm{Proj}_d(A) \end{array}$$

But by assumption on φ, the family $(\varphi(f))_{f \in A_d}$ generates L, so we can use Zariski descend to reduce to affines. We will make this precise in the following.

(a) We will first show that, for each $n \in \mathbf{N}$, the canonical map $\mathrm{Proj}_d \to \mathrm{Proj}_{nd}$ is injective. Let $x_1, x_2 \colon \mathrm{Spec}(R) \to \mathrm{Proj}_d(A)$ be two elements, representing

$$\varphi_1 \colon A^{(d)} \otimes_{A_0} R \twoheadrightarrow \mathrm{Sym}_R(L_1), \quad \varphi_2 \colon A^{(d)} \otimes_{A_0} R \twoheadrightarrow \mathrm{Sym}_R(L_2),$$

respectively, such that their image coincide in $\mathrm{Proj}_{nd}(A)$. Form the following two pullback squares, which we pack into one diagram for simplicity

$$\begin{array}{ccc} P_{1,2} & \overset{y_1}{\underset{y_2}{\rightrightarrows}} & \mathrm{D}(f) \\ \downarrow{\scriptstyle j_f} & & \downarrow \\ \mathrm{Spec}(R) & \overset{x_1}{\underset{x_2}{\rightrightarrows}} & \mathrm{Proj}_d(A) \end{array}$$

where $P_1 = \mathrm{Spec}(R_{\varphi_1(f)})$ and $P_2 = \mathrm{Spec}(R_2\varphi_(f))$ by the remark at the beginning of the proof.

Using Lemma 3.66 we obtain the following commutative diagram where every square is a pullback square

$$\begin{array}{ccccc} \mathrm{Spec}(R_{\varphi_1(f)}) & \overset{y_1}{\underset{y_2}{\rightrightarrows}} & \mathrm{D}(f) & \xrightarrow{\sim} & \mathrm{D}(f^n) \\ \downarrow{\scriptstyle j_f} & & \downarrow & & \downarrow \\ \mathrm{Spec}(R) & \overset{x_1}{\underset{x_2}{\rightrightarrows}} & \mathrm{Proj}_d(A) & \longrightarrow & \mathrm{Proj}_{nd}(A) \end{array}$$

By the pasting law, and since the two composites $\mathrm{Spec}(R) \xrightarrow{x_1,x_2} \mathrm{Proj}_d(A) \to \mathrm{Proj}_{nd+(A)}$ coincide, we obtain an isomorphism $P_1 \cong P_2 \cong \mathrm{Spec}(R_{\varphi_1(f)})$. Therefor we can use that the right hand square is a pullback to obtain an equality $y_1 = y_2 : \mathrm{Spec}(R_{\varphi_1(f)}) \to \mathrm{D}(f)$, and hence $x_1 \circ j_f = x_2 \circ j_f$. This holds for all f.

Since $\varphi_1 : A^{(d)} \otimes_{A_0} R \to \mathrm{Sym}_R(L)$ is surjective, the family of elements $(\varphi_1(f))_{f \in A_d}$ generates the line L_1. By Zariski descend, see Remark 3.57, there is a monomorphism

$$\mathrm{Proj}_d(A)(R) \hookrightarrow \prod_{f \in A_d} \mathrm{Proj}_d(A)(R_{\varphi_1(f)}).$$

The claim now follows from $x_1 \circ j_f = x_2 \circ j_f$ for all f as was shown above.

Now, we will show that the immersion $\mathrm{Proj}_d(A) \to \mathrm{Proj}_{nd}(A)$ is open. We claim that for each element $x \colon \mathrm{Spec}(R) \to \mathrm{Proj}_{nd}(A)$ classiyfing the element $\varphi \colon A^{(nd)} \otimes_{A_0} R \twoheadrightarrow \mathrm{Sym}_R(L)$, there exists a dotted filler making the following diagram commutative and cartesian

$$\begin{array}{ccc} \mathrm{Epi}(\psi) & \longrightarrow & \mathrm{Spec}(R) \\ \vdots & & \downarrow{\scriptstyle x} \\ \mathrm{Proj}_d(A) & \longrightarrow & \mathrm{Proj}_{nd}(A), \end{array}$$

where $\psi \colon R^{(A_d)} = \bigoplus_{f \in A_d} R \to L$ is given by the family of elements $(\varphi(f^n))_{f \in A_d}$. By Proposition 3.53, $\mathrm{Epi}(\psi)$ is an open subfunctor of $\mathrm{Spec}(R)$, implying the open subfunctor claim.

Let S be a commutative ring. An S-point of the pullback is an R-algebra structure on S and a unique tuple (L_S, φ_S), such that $(L_S^{\otimes n}, \varphi_S^{(n)}) = (L \otimes_R S, \varphi \otimes \mathrm{id})$. An S-point of $\mathrm{Epi}(\psi)$ is an R algebra structure

in S, such that $(\varphi(f^n) \otimes \mathrm{id}_S)_{f \in A_d}$ generate $L \otimes S$. Thus, to show the claim, we need to show that for each R-algebra S there exists an "n-th root" of $\varphi \otimes \mathrm{id}_S : A^{(nd)} \otimes_{A_0} S \to \mathrm{Sym}_S(L \otimes_R S)$, that is, $\mathrm{Spec}(S) \to \mathrm{Spec}(R) \to \mathrm{Proj}_{nd}(A)$ factors over $\mathrm{Proj}_d(A)$, if and only if $L \otimes_R S$ is generated by the family $(\varphi(f^n) \otimes \mathrm{id}_S)_{f \in A_d}$.

Assume that $g\colon A^{(nd)} \otimes_{A_0} S \to \mathrm{Sym}_S(L \otimes_R S)$ lands in $\mathrm{Proj}_d(A)(S)$, then there exists a map g' and a R-line M such that the following diagram of $\mathbf{N}$-graded S-algebra commutes

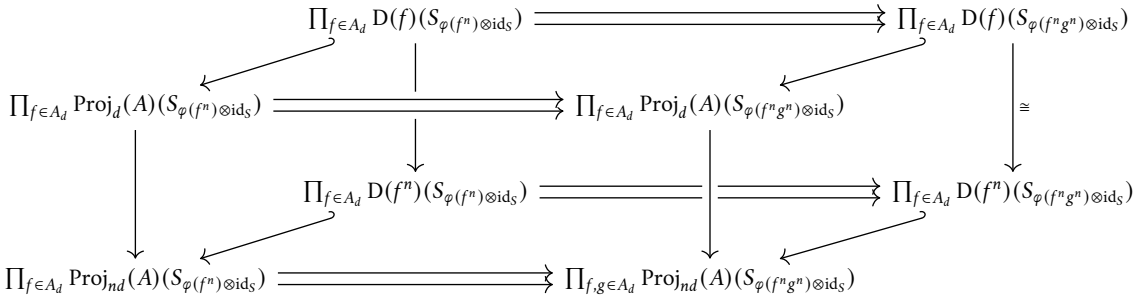
$$\begin{array}{ccc} A^{(nd)} \otimes_{A_0} S & \xrightarrow{g'^{(n)}} & \mathrm{Sym}_S(M^{\otimes n}) \\ & \searrow^{g} & \downarrow \cong \\ & & \mathrm{Sym}_S(L \otimes_R S) \end{array}$$

Taking $(-)_1$, we obtain $M^{\otimes n} \simeq L$. By definition, M is generated by the family $(\varphi(f))_{f \in A_d}$, hence $L \simeq M^{\otimes n}$ is generated by $(\varphi(f^n))_{f \in A_d}$.

For the converse, that $(\varphi(f^n) \otimes \mathrm{id}_S)_{f \in A_d}$ generates $L \otimes_R S$ gives us the existence of the following diagram by Proposition 3.64, where both rows are equalizers

$$\begin{array}{ccccc} \mathrm{Proj}_d(A)(S) & \longrightarrow & \prod_{f \in A_d} \mathrm{Proj}_d(A)(S_{\varphi(f^n)\otimes \mathrm{id}_S}) & \rightrightarrows & \prod_{f \in A_d} \mathrm{Proj}_d(A)(S_{\varphi(f^n g^n)\otimes \mathrm{id}_S}) \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{Proj}_{nd}(A)(S) & \longrightarrow & \prod_{f \in A_d} \mathrm{Proj}_{nd}(A)(S_{\varphi(f^n)\otimes \mathrm{id}_S}) & \rightrightarrows & \prod_{f,g \in A_d} \mathrm{Proj}_{nd}(A)(S_{\varphi(f^n g^n)\otimes \mathrm{id}_S}) \end{array}$$

We want to lift the element $x = (L \otimes_R S, \varphi \otimes \mathrm{id}_S) \in \mathrm{Proj}_{nd}(A)(S)$ to $\mathrm{Proj}_d(A)(S)$. Notice that, by assumption, the map $\mathrm{Proj}_{nd}(A)(S) \to \prod_{f \in A_d} \mathrm{Proj}_{nd}(A)(S_{\varphi(f^n)\otimes \mathrm{id}_S})$ send x into $\prod_{f \in A_d} \mathrm{D}(f^n)(S_{\varphi(f^n)\otimes \mathrm{id}_S})$. But $\mathrm{D}(f^n)(S_{\varphi(f^n)\otimes \mathrm{id}_S}) \cong \mathrm{D}(f)(S_{\varphi(f^n)\otimes \mathrm{id}_S})$ by Lemma 3.66, and $\mathrm{D}(f)(S_{\varphi(f^n)\otimes \mathrm{id}_S}) \subseteq \mathrm{Proj}_d(A)(S_{\varphi(f^n)\otimes \mathrm{id}_S})$. The situation is summarized in the following diagram, where we left out the $\mathrm{Proj}_d(A)(S)$ and $\mathrm{Proj}_{nd}(A)(S)$ for space reasons.



The equalizer of the upper front row being $\mathrm{Proj}_d(A)(S)$, we obtain an element $x' \in \mathrm{Proj}_d(A)(S)$ mapping to $x \in \mathrm{Proj}_{nd}(A)(S)$, finishing the claim.

(b) Assume A is generated as an A_0-algebra by homogeneous elements whose degrees divide d. It sufficies to show that the map $\mathrm{Proj}_d(A) \to \mathrm{Proj}_{nd}(A)$ is surjective. Let $x\colon \mathrm{Spec}(R) \to \mathrm{Proj}_{nd}(A)$ be an element that classifies the map $\varphi\colon A^{(nd)} \otimes_{A_0} R \twoheadrightarrow \mathrm{Sym}_R(L)$. We want to lift this element to $\mathrm{Proj}_d(A)$.

By definition, every element $f \in A_{nd}$ is a A_0 linear combination of some elements whose degrees divide d. Considering only monomials, the line L is generates by the family

$$T := \{f \in A_{nd} \mid f = f_1 \dots f_k \colon deg(f_i) | d\}.$$

Consider such an element, say $f = f_1 \cdot \ldots \cdot f_k \in T$. By assumption, there exists an $m \in \mathbf{N}$ such that $\deg(f_1)m = d$. Then we have the following commutative diagram, which is cartesian by Lemma 3.66

$$\begin{array}{ccc} \mathrm{D}(f_1^m) & \xhookrightarrow{\cong} & \mathrm{D}(f_1^{mn}) \\ \downarrow & & \downarrow \\ \mathrm{Proj}_d(A) & \hookrightarrow & \mathrm{Proj}_{nd}(A) \end{array}$$

As in the proof of (b) in Lemma 3.66, we have $\mathrm{D}(f) \subseteq \mathrm{D}(f_1^{mn})$. Therefore, the inclusion $\mathrm{D}(f) \hookrightarrow \mathrm{Proj}_{nd}(A)$ can be lifted to $\mathrm{Proj}_d(A)$. Since $f \in T$ was arbitrary, we can redo the diagram chase from part (a) of the theorem, finishing the claim.

□

Corollary 3.68. *Let A be a finitely generated $\mathbf{N}$-graded k-algebra such that $k \to A_0$ is surjective. Then $\mathrm{Proj}(A)$ is a projective k-scheme.*

Proof. In the statement, we view $\mathrm{Proj}(A)$ as an algebraic k-functor via the map $k \to A$. Since A is a finitely generated $\mathbf{N}$-graded k-algebra, there exists elements $s_1, \dots, s_n$ homogeneous of degree $d_1, \dots, d_n$ respectively that generates A over A_0. Choose any common multiple d of $d_1, \dots, d_n$. By Theorem 3.67, we have

$$\mathrm{Proj}(A) \simeq \mathrm{Proj}_d(A) \simeq \mathrm{Proj}_1(A^{(d)}).$$

Viewed as an algebraic k-functor, the functor $\mathrm{Proj}_1(A^{(d)})$ coincides with the functor defined in Construction 3.45. Therefore, it suffices to show that $A^{(d)}$ is generated over k by finitely many elements in A_d. This is clear, since $k \to A_0$ is surjective, and $A^(d)$ is generated over A_0 by elements of the from $s_1^{k_1} \dots s_n^{k_n}$ with $\sum k_i d_i = d$, and hence it is finitely generated over k. □

Remark 3.69. The functor Proj extends the functor Spec as follows: there is a commutative triangle (up to natural isomorphism)

$$\begin{array}{ccc} \mathrm{CAlg}^{\mathrm{op}} & \xrightarrow{(-)[x]} & (\mathrm{CAlg}^{\mathbf{N},\mathrm{es}})^{\mathrm{op}} \\ & \searrow^{\mathrm{Spec}} & \downarrow{\scriptstyle \mathrm{Proj}} \\ & & \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set}) \end{array}$$

where the horizontal functor sends R to the $\mathbf{N}$-graded ring $R[x]$. The follows from Theorem 3.67 as

$$\mathrm{Proj}(R[x]) \simeq \mathrm{Proj}_1(R[x]) = \mathbf{P}^0_R = \mathrm{Spec}(R).$$

Definition 3.70 (Irrelevant ideal)**.** Let A be an $\mathbf{N}$-graded ring. The irrelevant ideal of A is the homogeneous ideal $A_+ = \bigoplus_{d \geq 1} A_d$.

Remark 3.71 (Spec vs. Proj)*.* The relation between $\mathbf{A}^I$ and $\mathbf{P}^I$ from before extends to a relation between $\mathrm{Spec}(A)$ and $\mathrm{Proj}(A)$. Namely, let $\mathrm{D}(A_+) \subseteq \mathrm{Spec}(A)$ be the nonvanishing locus of the irrelevant ideal. Then there is a zigzag

$$\mathrm{Spec}(A) \hookleftarrow \mathrm{D}(A_+) \to \mathrm{Proj}(A)$$

over $\mathrm{Spec}(A_0)$, natural in $A \in \mathrm{CAlg}^{\mathbf{N},\mathrm{es}}$, where the second map is defined as follows. Since any family generating the unit ideal contains a finite generating subfamily, we have

$$\mathrm{D}(A_+) = \operatorname*{colim}_{d \in \mathbf{N}^{\mathrm{div}}_{>0}} \mathrm{D}(A_d).$$

The map $\mathrm{D}(A_+) \to \mathrm{Proj}(A)$ is then the colimit of the maps

$$\mathrm{D}(A_d) \to \mathrm{Proj}_d(A), \quad \left((\varphi \colon A \to R) \mapsto (\bigoplus_{n \in \mathbf{N}} \varphi|_{A_{nd}} \colon A^{(d)} \otimes_{A_0} R \twoheadrightarrow \mathrm{Sym}_R(R) \right).$$

There is a canonical action of the affine group scheme $\mathbf{G}_m$ on $\mathrm{Spec}(A)$, which restricts to each $\mathrm{D}(A_d)$: for any $\varphi \colon A \to R$ and $\lambda \in R^\times$, we define $\lambda\varphi \colon A \to R$ by

$$(\lambda\varphi)(a) = \lambda^d \varphi(a), \quad \text{for all } a \in A_d.$$

Each map $\mathrm{D}(A_d) \to \mathrm{Proj}_d(A)$ is then $\mathbf{G}_m$-invariant, and we have the following generalization of Proposition 3.28: for $d = 1$, the induced map $\mathrm{D}(A_1)/\mathbf{G}_m \to \mathrm{Proj}_1(A)$ is a monomorphism whose image consists exactly of the trivial quotient lines; in particular, it is a bijection on local rings and principal ideal domains. This does not holds for arbitrary $d \in \mathbf{N}$, but it will turn out that $\mathrm{Proj}_d(A)$ is still the quotient $\mathrm{D}(A_d)/\mathbf{G}_m$ in the category of *schemes*.

Definition 3.72 (Vanishing and nonvanishing loci)**.** Let A be an $\mathbf{N}$-graded ring and let $F \subset A$ be a homogeneous subset.

(a) The vanishing locus of F is the subfunctor $\mathrm{V}(F) \subseteq \mathrm{Proj}(A)$, which is the colimit over $d \in \mathbf{N}^{\mathrm{div}}_{>0}$ of the subfunctors $\mathrm{V}_d(F) \subseteq \mathrm{Proj}_d(A)$ given by

$$\mathrm{V}_d(F)(R) = \{\varphi\colon A^{(d)} \otimes_{A_0} R \twoheadrightarrow \mathrm{Sym}_R(L) \mid \varphi(f) = 0 \text{ for all } f \in (F)^{(d)}\}.$$

Here, $(F)^{(d)} := (F) \cap A^{(d)}$ is the degree d-part of the homogenous ideal generated by F.

(b) The nonvanishing locus of F is the subfunctor $\mathrm{D}(F) \subseteq \mathrm{Proj}(A)$, which is the colimit over $d \in \mathbf{N}^{\mathrm{div}}_{>0}$ of the subfunctors $\mathrm{D}_d(F) \subseteq \mathrm{Proj}_d(A)$ given by

$$\mathrm{D}_d(F)(R) = \{\varphi\colon A^{(d)} \otimes_{A_0} R \twoheadrightarrow \mathrm{Sym}_R(L) \mid \varphi((F)_{nd}) \text{ generates } L^{\otimes n} \text{ for some } n \in \mathbf{N}\}.$$

Remark 3.73. It is clear that the composition $\mathrm{V}_d(F) \to \mathrm{Proj}_d(A) \to \mathrm{Proj}_{nd}(A)$ factors over $\mathrm{V}_{nd}(F)$, so $\mathrm{V}(F)$ is well defined. To see that $\mathrm{D}(F)$ is well defined, it suffices to notice that if $\big(\varphi : A^{(d)} \otimes_{A_0} R \twoheadrightarrow \mathrm{Sym}_R(L)\big) \in \mathrm{D}(F)$, that is, $\varphi((F)_{md})$ generates $L^{\otimes m}$ for some m, then $\varphi((F)_{md})^{\otimes n} \subset \varphi((F)_{mnd})$ generates $L^{\otimes n}$.

Remark 3.74. One can easily check that V and D satisfies exactly the same formal properties as in the affine case Proposition 2.60. Note also that $\mathrm{V}(F)$ depends only on (F) and $\mathrm{D}(F)$ only on $\sqrt{(F)}$. If A is generated by $A_{\leq 1}$, we have the following simpler description as subfunctors of $\mathrm{Proj}_1(A) \simeq \mathrm{Proj}(A)$:

$$\mathrm{V}(F)(R) = \{\varphi\colon A \otimes_{A_0} R \twoheadrightarrow \mathrm{Sym}_R(L) \mid \varphi(f) = 0 \text{ for all } f \in F\}$$
$$\mathrm{D}(F)(R) = \{\varphi\colon A \otimes_{A_0} R \twoheadrightarrow \mathrm{Sym}_R(L) \mid (\varphi(F)) \text{ contains } L^{\otimes n} \text{ for some } n \in \mathbf{N}\}.$$

Remark 3.75. Let A be an $\mathbf{N}$-graded ring.

(a) For any homogeneous subset $F \subseteq A$, there is a canonical isomorphism

$$\mathrm{Proj}(A/(F)) \xrightarrow{\sim} \mathrm{V}(F).$$

(b) Let $f \in A_d$ with $d \geq 1$. Then the nonvanishing locus $\mathrm{D}(f)$ of Definition 3.72 is contained in $\mathrm{Proj}_d(A)$ and matches the one from Lemma 3.66, by part (b) of Lemma 3.66. By part (a) of the lemma, there is a canonical isomorphism $\mathrm{Spec}(A_{(f)}) \xrightarrow{\sim} \mathrm{D}(f)$.

(c) If $f \in A_0$, there is a canonical isomorphism $\mathrm{Proj}(A_f) \xrightarrow{\sim} \mathrm{D}(f)$.

Proposition 3.76. *Let A be an $\mathbf{N}$-graded ring and let $F \subseteq A$ be a homogeneous subset. then*

(a) The subfunctor $\mathrm{V}(F)$ is a closed subfunctor of $\mathrm{Proj}(A)$.
(b) The subfunctor $\mathrm{D}(F)$ is an open subfunctor of $\mathrm{Proj}(A)$.

Proof. (a) By Remark 3.74, we have $\mathrm{V}(F) = \bigcap_{f\in F} \mathrm{V}(f)$. Since arbitrary intersection of closed functors is again closed[16], it suffices to show that $\mathrm{V}(f)$ is closed in $\mathrm{Proj}(A)$. Let $x\colon \mathrm{Spec}(R) \to \mathrm{Proj}(A)$ be an element classifying the map $\varphi\colon A \otimes_{A_0} R \twoheadrightarrow \mathrm{Sym}_R(L)$. Similar to the proof of (a) in Theorem 3.67, the following commutative square is cartesian:

$$\begin{array}{ccc} \mathrm{V}(\psi) & \longrightarrow & \mathrm{V}(f) \\ \downarrow & & \downarrow \\ \mathrm{Spec}(R) & \xrightarrow{x} & \mathrm{Proj}(A). \end{array}$$

where $\psi\colon R \to L^{\otimes d}$ is given by the element $\varphi(f)$. By Proposition 3.53, $\mathrm{V}(\psi)$ is closed.

(b) Again, consider the following cartesian square:

$$\begin{array}{ccc} x^{-1}(\mathrm{D}(F)) & \longrightarrow & \mathrm{D}(F) \\ \downarrow & & \downarrow \\ \mathrm{Spec}(R) & \xrightarrow{x} & \mathrm{Proj}(A). \end{array}$$

[16]This can be deduse y reducing to affines, using that limits commute.

By the condition of x landing in $\mathrm{D}(F)$, we obtain

$$x^{-1}(\mathrm{D}(F)) = \bigcup_{n\in\mathbf{N}} \mathrm{Epi}(\psi_n),$$

where $\psi_n \colon R^{((F)_n)} \twoheadrightarrow L^{\otimes n}$. By Proposition 3.53, $\mathrm{Epi}(\psi_n)$ is an open subfunctor of $\mathrm{Spec}(R)$. Therefore, for each $n \in \mathbf{N}$, we have $\mathrm{Epi}(\psi_n) \simeq \mathrm{D}(I_n) \subseteq \mathrm{Spec}(R)$, for some radical ideal $I_n \subseteq R$. Furthermore, from the inclusion of epimorphism loci and Proposition 2.66, we have

$$\ldots \subseteq I_{n-1} \subseteq I_n \subseteq I_{n+1} \subseteq \ldots.$$

That this is a ascending chain gives us the second equality in the following chain of equalities

$$x^{-1}(\mathrm{D}(F)) = \bigcup_{n\in\mathbf{N}} \mathrm{D}(I_n) = \mathrm{D}(\bigcup_{n\in\mathbf{N}} I_n).$$

By definition $\mathrm{D}(\bigcup_{n\in\mathbf{N}} I_n)$ is open, giving the claim.

□

Example 3.77 (The Veronese embedding)*.* Let k be a ring, M a k-module, and $d \in \mathbf{N}$. The *Veronese embedding of degree* d is the map of algebraic k-functors

$$\rho_d \colon \mathbf{P}(M) \to \mathbf{P}(\mathrm{Sym}^d_k(M))$$

sending a quotient line $M \otimes_k R \twoheadrightarrow L$ to its d-th symmetric power

$$\mathrm{Sym}^d_k(M) \otimes_k R \twoheadrightarrow \mathrm{Sym}^d_R(L) \simeq L^{\otimes d}.$$

If $M = k^{(I)}$, the Veronese embedding of degree d is the map

$$\rho_d \colon \mathbf{P}^I_k \to \mathbf{P}^{\mathrm{Sym}^d(I)}_k,$$

where $\mathrm{Sym}^d(I)$ is the quotient of I^d be the action of the symmetric group Σ_d. If $M = k^{n+1}$, this becomes the map

$$\rho_d \colon \mathbf{P}^n_k \to \mathbf{P}^{\binom{n+d}{n}-1},$$

which is given in coordinates by the formula

$$[a_0 : \ldots : a_n] \mapsto [\text{all degree } d \text{ monomials in the } a_i\text{'s}].$$

We claim that ρ_d is a closed immersion if $d \geq 1$. Indeed, ρ_d is Proj of a surjective degree d map of graded rings. Explicitly, it is the composite

$$\mathrm{Proj}_1(\mathrm{Sym}_k(M)) \to \mathrm{Proj}_d(\mathrm{Sym}_k(M)) \to \mathrm{Proj}_1(\mathrm{Sym}_k(\mathrm{Sym}^d_k(M))),$$

where the first map sends (L, φ) to $(L^{\otimes d}, \varphi^{(d)})$ and the second map is Proj_1 of the surjective map

$$\mathrm{Sym}_k(\mathrm{Sym}^d_k(M)) \twoheadrightarrow \mathrm{Sym}_k(M)^{(d)}.$$

If $d \geq 1$, the first map is an isomorphism by Theorem 3.67 (b), and the second map is a closed immersion by Proposition 3.76, identifying $\mathrm{Proj}_d(\mathrm{Sym}_k(M))$ with $\mathrm{V}(\ker(\mathrm{Sym}_k(\mathrm{Sym}^d_k(M)) \twoheadrightarrow \mathrm{Sym}_k(M)^{(d)}))$.

Analogously to quasi-affine scheme, we define quasi-projective schemes as nonvanishing loci in projective scheme:

Definition 3.78 (Quasi-projective scheme)**.** Let k be a ring. An algebraic k-functor X is a *quasi-projective k-scheme* if there exists an $\mathbf{N}$-graded k-algebra A, generated by a finite subset of A_1, and a finite homogeneous subset $F \subseteq A$ such that $X \simeq \mathrm{D}(F) \subseteq \mathrm{Proj}(A)$. We denote by $\mathrm{QProj}_k \subseteq \mathrm{Fun}(\mathrm{CAlg}_k, \mathrm{Set})$ the full subcategory spanned by the quasi-projective k-schemes.

Note that we could have assumed $F \subseteq A_1$ without changing the definition: first, we can assume $F \subseteq A_+$ by multiplying F by a finite subset of A_1 that generates A. Indeed, let S be a finite set generating A_+. Then $\mathrm{D}(S) = \mathrm{D}(A_+) = \mathrm{Proj}(A)$, where the last equality follows directly from the definition. Thus

$$\mathrm{D}(F \cdot S) = \mathrm{D}(F) \cap \mathrm{D}(S) = \mathrm{D}(F).$$

Second, if $d \geq 1$ is divisible by all degree in F, we can then replace A by $A^{(d)}$ and each element of F by its power in A_d, since $\mathrm{D}(F)$ depends only on $\sqrt{(F)}$.

We will see in that every quasi-affine k-scheme of finite type is quasi-projective. Thus, under this finiteness assumption, quasi-projective k-schemes subsume all types of k-schemes discussed so far.

correct reference

3.7. **Saturation.** The goal of this section is to classify closed and open subfunctors of $\mathrm{Proj}(A)$. This is more subtle than the analogous result for $\mathrm{Spec}(A)$ (Proposition 2.66), since there can be different homogeneous ideals in A with the same vanishing locus in $\mathrm{Proj}(A)$:

Example 3.79. The homogeneous ideal (x) and (x^2, xy) in $\mathbf{Z}[x, y]$ have the same vanishing locus in $\mathbf{P}^1 = \mathrm{Proj}(\mathbf{Z}[x, y])$. Since $(x^2, xy) \subseteq (x)$, we have $\mathrm{V}(x) \subseteq \mathrm{V}(x^2, xy)$. For the converse, consider an arbitrary quotient line $(a, b) \colon R^2 \twoheadrightarrow L$ in $\mathbf{P}^1(R)$ such that $a^2 = 0$ and $ab = 0$ in $L^{\otimes 2}$. Since $\mathrm{Sym}^2_R(R^2) \to L^{\otimes 2}$ is surjective, $L^{\otimes 2}$ is generated by (a^2, ab, b^2), hence by b^2 alone. Since L is invertible, this implies that b generates L. From $ab = 0 \in L^{\otimes 2}$, we then deduce that already $a = 0$, as desired.

Definition 3.80 (Saturation)**.** Let A be an $\mathbf{N}$-graded ring and let $H \subset A$ be a homogeneous ideal. The *saturation* of H is the homogeneous ideal

$$H^{\mathrm{sat}} = \{x \in A \mid \text{for all } f \in A_+, \text{ there exists } n \in \mathbf{N} \text{ such that } f^n x \in H\}.$$

We say that H is *saturated*, if $H^{\mathrm{sat}} = H$.

Remark 3.81. The following are some basic properties of saturation.

(a) Saturated ideals are determined by their "tail": if $H \subseteq A$ is a homogeneous ideal and $D \subseteq \mathbf{N}$ is an infinite subset, then $H^{\mathrm{sat}} = (\bigoplus_{d \in D} H_d)^{\mathrm{sat}}$. Indeed, since $\bigoplus_{d \in D} H_d \subseteq H$, we have $(\bigoplus_{d \in D} H_d)^{\mathrm{sat}} \subseteq H^{\mathrm{sat}}$. Conversely, assume that $x \in H^{\mathrm{sat}}$, then for all $f \in A_+$, there exists $n \in \mathbf{N}$, such that $f^n x \in H$. Since D is infinite, we may choosing a bigger $n \in \mathbf{N}$ we may assume $f^n x \in H_d$ for some $d \in D$, that is $x \in \left(\bigoplus_{d \in D} H_d\right)^{\mathrm{sat}}$.

(b) If A is generated by $A_{\leq 1}$, we can replace A_+ by A_1 in the definition of H^{sat}.

(c) If $H \subseteq A$ is a radical homogeneous ideal, then H^{sat} is also radical: assume that H is radical and $x^n \in H^{\mathrm{sat}}$. For each $f \in A_+$, there exists $m \in \mathbf{N}$ such that $f^m x^n \in H$. But then $f^{mn} x^n = (f^m x)^n \in H$, and since H is radical, we have $f^m x \in H$, hence $x \in H^{\mathrm{sat}}$.

(d) A homogeneous prime ideal $\mathfrak{p} \subseteq A$ is saturated if and only if $\mathfrak{p}$ does not contain A_+. Assume that $\mathfrak{p}$ does not contain A_+. Then there exists $f \in A_+ \setminus \mathfrak{p}$. Let $x \in \mathfrak{p}^{\mathrm{sat}}$. Then, for some n we have $f^n x \in \mathfrak{p}$. Since $\mathfrak{p}$ is prime and $f \notin \mathfrak{p}$, we must have $x \in \mathfrak{p}$. Conversely, assume that $\mathfrak{p}$ contains A_+. Then we have $\mathfrak{p}^{\mathrm{sat}} = A$, hence $\mathfrak{p}$ cannot be saturated. More generally it is true that any ideal $I \subseteq A$, such that $A_+ \subseteq I$, we have that $I^{\mathrm{sat}} = A$.

correct as stated?

Example 3.82. The following are some examples of saturation.

(a) In $\mathbf{Z}[x, y]$, we have $(x^2, xy)^{\mathrm{sat}} = (x)$. By the above remark, the ideal (x) is saturated, since it is prime and does not contain A_+. Since $(x^2, xy) \subseteq (x)$, we have $(x^2, xy)^{\mathrm{sat}} \subseteq (x)$. Thus it suffices to show that $x \in (x^2, xy)^{\mathrm{sat}}$. By Remark 3.81 (b), we may replace A_+ by $A_{\leq 1}$. It then suffices to check for x and y, which is true by definition.

(b) In any polynomial ring $k[x_i \mid i \in I]$, the ideal (0) is saturated. Again, this follow from that (0) is prime and does not contain $k[x_i \mid i \in I]_+$.

We briefly consider a more general context for this definiton:

Definition 3.83 (I-nilpotent, I-local module)**.** Let R be a ring, $I \subseteq R$ a subset, and M an R-module. An element $x \in M$ is called *I-nilpotent* if, for every $f \in I$, there exists $n \in \mathbf{N}$ such that $f^n x = 0$. The I-nilpotent elements of M form a submodule $\Gamma_I M \subseteq M$. The module M is called:

(a) *I-nilpotent* if $\Gamma_I M = M$;

(b) *I-local* if every R-linear map $h\colon P \to Q$ with I-nilpotent kernel and cokernel induces a bijection

$$h^*\colon \operatorname{Map}(Q, M) \xrightarrow{\sim} \operatorname{Map}(P, M).$$

We denote the corresponding full subcategories of Mod_R by $\operatorname{Mod}_R^{I-\mathrm{nil}}$ and $\operatorname{Mod}_R^{I-\mathrm{loc}}$.

Note that these conditions depend only on the radical ideal generated by I. The categories $\operatorname{Mod}^{I-\mathrm{nil}}$ and $\operatorname{Mod}_R^{I-\mathrm{loc}}$ are abelian and fit in a short exact sequence of Grothendieck abelian categories

$$\operatorname{Mod}_R^{I-\mathrm{nil}} \underset{\Gamma_I}{\overset{}{\rightleftarrows}} \operatorname{Mod}_R \overset{\mathrm{L}_I}{\underset{}{\rightleftarrows}} \operatorname{Mod}_R^{I-\mathrm{loc}}$$

ref where all functors are exact[17]. In particular, an R-module M is I-nilpotent if and only if $\mathrm{L}_I M = 0$, and $\Gamma_I M$ is the kernel of the unit map $M \to \mathrm{L}_I M$.

Remark 3.84. An R-module M is I-nilpotent if and only if $M_f = 0$ for all $f \in I$. Hence, an R-linear map $M \to N$ induces an isomorphism $\mathrm{L}_I M \xrightarrow{\sim} \mathrm{L}_I N$ if and only if it induces isomorphisms $M_f \xrightarrow{\sim} N_f$ for all $f \in I$. In particular, for any $f \in R$, "$\{f\}$-local" is a synonymous with f-periodic, and $\mathrm{L}_{\{f\}} M = M_f$.

Remark 3.85. We will later give in Theorem 4.49 a geometric interpretation of $\operatorname{Mod}_R^{I-\mathrm{loc}}$ when I is finite: it can be identified with the category of quasi-coherent modules on the open subscheme $\mathrm{D}(I) \subseteq \operatorname{Spec}(R)$. In this case, $\mathrm{L}_I M$ can be computed explicitly as the equalizer

$$\mathrm{L}_I M \to \prod_{f \in I} M_f \rightrightarrows \prod_{f,g \in I} M_{fg},$$

and we have already seen in Corollary 2.76 that $\mathcal{O}(\mathrm{D}(I)) = \mathrm{L}_I(R)$.

Warning 3.86. For a subset $I \subseteq R$, the construction $\mathrm{L}_I M$ should not be confused with the construction $M[I^{-1}]$, which are both called "localization". They coincide when I has a single element but generalize this case in different directions: for any $f \in I$, there are canonical maps

$$\mathrm{L}_I M \to \mathrm{L}_f M \simeq M_f \to M[I^{-1}].$$

The geometric interpretation of these constructions is the following. If I is finite, then $\mathrm{L}_I(R)$ is the ring of functions on $\mathrm{D}(I)$, which is the union $\bigvee_{f \in I} \mathrm{D}(f)$ in the poset of open subfunctors of $\operatorname{Spec}(R)$ and is usually not affine. On the other hands, the intersection $\bigcap_{f \in I} \mathrm{D}(f)$ is always affine (but usually not open in $\operatorname{Spec}(R)$, unless I is finite) and isomorphic to $\operatorname{Spec}(R[I^{-1}])$.

Example 3.87. The following are some basic properties:

(a) Let L be a line over R and let $s\colon R \to L$. Then an R-module is s-periodic if and only if it is $\operatorname{im}(s^\vee)$-local. Indeed, if we assume M to be s-periodic, we have to show that for any $\operatorname{im}(s^\vee)$-nilpotent module I, the derived Hom-sets[18] $\mathbb{R}\operatorname{Hom}_R(I, M)$ vanish. Writing I as the filtered colimit of its finitely generated submodules, we may assume I to be finitely generated. In particular there exists a global bound n such that $\operatorname{im}(s^\vee)^n I = 0$, using that $\operatorname{im}(s^\vee)$, being the image of a map out of a finitely generated module, is itself finitely generated. Since M is s-periodic, we can write precomposition by $I \otimes L^{-1} \xrightarrow{s^\vee} I$ as the composition of isomorphisms

$$\mathbb{R}\operatorname{Hom}_R(I, M) \xrightarrow[\cong]{(s \otimes id)_*} \mathbb{R}\operatorname{Hom}_R(I, L \otimes_R M) \cong \mathbb{R}\operatorname{Hom}_R(I \otimes_R L^{-1}, M).$$

[17] This is an example of a so-called *recollment*, also known as *localizing sequence*. They will appear later on when one decomposes a scheme in to a closed subscheme and its complementary open

[18] We use the $\mathbb{R}\mathrm{Hom}$ here since working with the cokernel may introduce extension problems due to Hom being only left exact.

We obtain an isomorphism $\mathbb{R}\mathrm{Hom}_R(I,M) \cong \mathrm{colim}(\mathbb{R}\mathrm{Hom}_R(I,M) \xrightarrow{(s^\vee)^*} \mathbb{R}\mathrm{Hom}_R(I\otimes L^{-1},M) \xrightarrow{(s^\vee\otimes\mathrm{id})^*} \ldots)$, given that the transition maps are isomorphism. Since I is $\mathrm{im}(s^\vee)$ nilpotent with a global bound, this colimit vanishes. For the converse, assume that M is $\mathrm{im}(s^\vee)$-local. Since $K := \ker(s^\vee : L^{-1} \to R)$ and $C := \mathrm{coker}(s^\vee : L^{-1} \to R)$ are both $\mathrm{im}(s^\vee)$-nilpotent, we obtian an isomorphism $s \otimes \mathrm{id} : M \cong \mathrm{Hom}_R(R,M) \xrightarrow[\cong]{(s^\vee)^*} \mathrm{Hom}(L^{-1},M) \cong L \otimes M$, which was the claim.

(b) Let A be an $\mathbf{N}$-graded ring. The saturation of a homogeneous ideal $H \subseteq A$ is the kernel of the map $A \to A/H \to \mathrm{L}_{A_+}(A/H)$. In particular, $(0)^{\mathrm{sat}} = \Gamma_{A_+}A$.

Proposition 3.88 (Conservativity of Proj)**.** *Let A and B be $\mathbf{N}$-graded rings and let $\alpha : A \to B$ be an eventually surjective map. The following conditions are equivalent:*

(a) $\mathrm{Proj}(B) \to \mathrm{Proj}(A)$ *is an isomorphism.*
(b) For all $d \geq 1$, $\mathrm{L}_{A_d}A^{(d)} \to \mathrm{L}_{A_d}B^{(d)}$ *is an isomorphism.*

If $A \to B$ is surjective, these are further equivalent to:

(c) *For all* $d \geq 1$, $\ker(A \to B)^{(d)}$ *is A_d-nilpotent.*

If A is generated by $A_{\leq 1}$, it sufficies to consider $d = 1$ in (b) *and* (c).

Proof. We will only prove the case where A and B are generated in degree ≤ 1 and there exists a surjection $A_1 \otimes_{A_0} B_0 \twoheadrightarrow B_1$. Generalize (seems annoying...)

(a) $\Rightarrow$ (b): by Remark 3.84,to show that the map $L_{A_1}(\alpha) : \mathrm{L}_{A_1}A \to \mathrm{L}_{A_1}B$ is an isomorphism, we only have to show that for all $f \in A_1$, the map $\alpha_f : A_f \to B_f$ is an isomorphism. Consider the pullback square

$$\begin{array}{ccc} \mathrm{D}(\alpha(f)) & \longrightarrow & \mathrm{D}(f) \\ \downarrow & \lrcorner & \downarrow \\ \mathrm{Proj}(B) & \longrightarrow & \mathrm{Proj}(A). \end{array}$$

Since the map $\mathrm{Proj}(\alpha) : \mathrm{Proj}(B) \to \mathrm{Proj}(A)$ is an isomorphism by assumption, the map $\mathrm{D}(\alpha(f)) \to \mathrm{D}(f)$ is an isomorphism. Unwinding the definition, the ring map $A_{(f)} \to B_{(f)}$ is an isomorphism. Since $A_f \simeq A_{(f)}[x^{\pm 1}]$, we obtain an isomorphism $A_f \to B_f$.

(b) $\Rightarrow$ (a): by assumption, we have a surjective map $A_1 \otimes_{A_0} B_0 \twoheadrightarrow B_1$. Hence, for every $R \in \mathrm{CAlg}_{A_0}$, the map $A_1 \otimes_{A_0} R \to B_1 \otimes_{B_0} R$ is surjective, hence an epimorphism. It follows that $\mathrm{Proj}(B) \to \mathrm{Proj}(A)$ is a monomorphism. Indeed, given two quotient lines $\varphi : B_1 \otimes B_0 R \twoheadrightarrow L$ and $\varphi' : B_1 \otimes B_0 R \twoheadrightarrow L'$ defining points of $\mathrm{Proj}(B)(R)$, the map to the same point in $\mathrm{Proj}(A)(R)$ if and only if the two compositions $A_1 \otimes_{A_0} R \twoheadrightarrow B_1 \otimes_{B_0} R \to L^{(\prime)}$ agree, which implies that already $L = L'$ and $\varphi = \varphi'$.

It then suffices to show that, for each $R \in \mathrm{CAlg}$, the map $\mathrm{Proj}(B)(R) \to \mathrm{Proj}(A)$ is surjective. By assumption we have $\mathrm{D}(\alpha(f)) \cong \mathrm{D}(f)$, so we can copy the proof of Theorem 3.67.

(b) $\Leftrightarrow$ (c): we denote $K = \ker(A \to B)$. The goal is to show that K is A_1-nilpotent if and only if $\mathrm{L}_{A_1}A \to \mathrm{L}_{A_1}B$ is an isomorphism.

Since $A \to B$ is surjective, we have an exact sequence as follows:

$$0 \to K \to A \to B \to 0.$$

Since $\mathrm{L}_{A_1}(-)$ is exact, the following sequence is exact:

$$0 \to \mathrm{L}_{A_1}K \to \mathrm{L}_{A_1}A \to \mathrm{L}_{A_1}B \to 0.$$

Therefore, the map $\mathrm{L}_{A_1}A \to \mathrm{L}_{A_1}B$ is an isomorphism if and only if $\mathrm{L}_{A_1}K$. But this is exactly the condition that K is A_1-nilpotent. □

Corollary 3.89 (Functoriality projective Nullstellensatz)**.** *Sending a homogeneous subset $F \subseteq k[x_i \mid i \in I]$ to its vanishing locus $\mathrm{V}(F) \subseteq \mathbf{P}^I_k$ induces an order-resersing bijection*

$$\mathrm{V} : \{\textit{saturated homogeneous ideals in } k[x_i \mid i \in I]\} \xrightarrow{\sim} \{\textit{vanishing loci in } \mathbf{P}^I_k\}.$$

Proof. By definition, the map $\mathrm{V}(-)$ is order reversing. Denote $A = k[x_i \mid i \in I]$. All we have to show that, for saturated homogeneous ideals H and H' in A, we have $H = H'$ if and only if $\mathrm{V}(H) = \mathrm{V}(H')$. In fact, we may assume that $H \subseteq H'$, then the general case follows by replacing H' with $H \cup H'$ and that $\mathrm{V}(H \cup H') = \mathrm{V}(H) \cap \mathrm{V}(H')$. Since $H \subseteq H'$, we have by Remark 3.75 (a) a monomorphism $\mathrm{V}(H') \subseteq \mathrm{V}(H)$, i.e. a monomorphism $\mathrm{Proj}(A/H') \to \mathrm{Proj}(A/H)$, which is induced by the map $\pi \colon A/H \to A/H'$. By Proposition 3.88 and Example 3.87, we have $\mathrm{V}(H) = \mathrm{V}(H')$ if and only if $H'/H = \ker(\pi) \subseteq (0)^{\mathrm{sat}}$, which holds if and only if $H' \subseteq H^{\mathrm{sat}}$, i.e. $H^{\mathrm{sat}} = H'^{\mathrm{sat}}$. □

For simplicity, we state the next result for $\mathbf{N}$-graded rings that are generated in degrees ≤ 1.

Proposition 3.90 (Classification of closed and open subfunctors of projective schemes)**.** *Let A be an $\mathbf{N}$-graded ring generated by $A_{\leq 1}$.*

(a) The construction $F \mapsto \mathrm{V}(F)$ induces an order-reversing injection

$$\{\textit{saturated homogeneous ideals in } A\} \hookrightarrow \{\textit{closed subfunctors of } \mathrm{Proj}(A)\},$$

which is bijective if A is finitely generated as an A_0-algebra.

(b) The construction $F \mapsto \mathrm{D}(F)$ induces an order-preserving bijection

$$\{\textit{radical homogeneous ideals in } A_+\} \xrightarrow{\sim} \{\textit{open subfunctors of } \mathrm{Proj}(A)\}.$$

Proof. (a) Injectivity follows as in Corollary 3.89. For the second part of the statement, assume that A_1 is generated as an A_0-algebra by a finite subset $F \subseteq A_1$. We will show that every closed subfunctor $Z \subseteq \mathrm{Proj}(A)$ corresponds to some vanishing locus. Consider the following pullback square:

$$\begin{array}{ccc} \mathrm{V}(I(f)) & \hookrightarrow & Z \\ \downarrow & & \downarrow \\ \mathrm{D}(f) & \hookrightarrow & \mathrm{Proj}(A_{(f)}), \end{array}$$

where $I(f)$ is an ideal of $A_{(f)}$. Since $I(f)[x^{\pm 1}] \subseteq A_f$, we may consider the preimage $H(f)$ of $I(f)[x^{\pm 1}]$ under the localization $A \to A_f$. Let $H = \bigcap_{f \in F} H(f)$, we claim that $Z = \mathrm{V}(H)$. In fact, by a similar Zariski descent argument in Theorem 3.67, it sufficies to show that for each $f \in F$, we have

$$Z \cap \mathrm{D}(f) = \mathrm{V}(H) \cap \mathrm{D}(f),$$

which is equivalent to the statement

$$\mathrm{V}(I(f)) = \mathrm{V}(H_{(f)})$$

in $\mathrm{Spec}(A_{(f)})$. In other words, we need to show that $I(f) = H_{(f)}$. The rest of the proof is similar to that of Lemma 2.83, and we refer to this proof for details.

(b) For injectivity, let $F \subseteq A_+$ be a radical homogeneous ideal, we want to show that we can recover F from $\mathrm{D}(F)$. Effectively by definition, we have the following pullback square

$$\begin{array}{ccc} \mathrm{D}(F_{(f)}) & \hookrightarrow & \mathrm{D}(F) \\ \downarrow & & \downarrow \\ \mathrm{D}(f) & \hookrightarrow & \mathrm{Proj}(A). \end{array}$$

Therefore, from $\mathrm{D}(F)$, we may recover $\sqrt{F_{(f)}}$ for all $f \in A_d$. Since $\sqrt{F_f}^{(d)} \cong \sqrt{F_{(f)}}[x^{\pm 1}]$, we may also recover $\sqrt{F_f}^{(d)}$. We further have that $a \in A_d$ is an element in $\sqrt{F_f}^{(d)} \subseteq A_f^{(d)}$ if and only if there exists $k, n \in \mathbf{N}$ such that $a^k a^n \in F$, i.e. $a \in F$, since F is radical. Therefore, we may recover F.

For surjectivity, we will omit the proof, as the proof is similar to that of Proposition 2.71 (b) about open immersions.

□

Remark 3.91. Proposition 3.90 generalizes to all $\mathbf{N}$-graded ring A as follows: (b) remains unchanged, but in (a) we must replace "saturated" by the following stronger condition on an ideal H: H contains every homogeneous element $x \in A$ that satisfies $(x)_d \subseteq \ker(A \to \mathrm{L}_{A_d}(A/H))$ for all $d \geq 1$. This is arguably the correct definition of "saturated" in general, but it agrees with Definition 3.80 when A is generated by $A_{\leq 1}$ or when H is radical.

Remark 3.92. For any $\mathbf{N}$-graded ring A, the construction $F \mapsto F \cap A_+$ defines a bijection

$$\{\text{saturated radical homogeneous ideals in } A\} \simeq \{\text{homogeneous ideals } H \subseteq A_+,\ \sqrt{H} \cap A_+ = H\}$$

with inverse $H \mapsto H^{\mathrm{sat}}$. This gives an alternative formulation of (b) in Proposition 3.90.

It follows from (a) in Proposition 3.90 that every closed subfunctor of a projective k-scheme is again a projective k-scheme. We also have the following analogue of Proposition 2.84:

Proposition 3.93. *Let X be a quasi-projective k-scheme and let $Z \hookrightarrow X$ be a closed immersion. Then Z is a quasi-projective k-scheme.*

Proof. todo □

3.8. **Projective closure.** Recall from Example 3.59 that any affine space $\mathbf{A}^I$ admits a canonical open embedding $\mathbf{A}^I \hookrightarrow \mathbf{P}^{I \sqcup \{0\}}$, which identifies $\mathbf{A}^I$ with the nonvanishing locus of x_0 in $\mathbf{P}^{I \sqcup \{0\}}$. The goal of this section is to compare vanishing and nonvanishing loci in $\mathbf{A}^I$ and $\mathbf{P}^{I \sqcup \{0\}}$.

Proposition 3.94. *Let I be a set not containing 0, let $F \subseteq k[x_0, x_i \mid i \in I]$ be a homogeneous subset, and let $F_0 \subseteq k[x_i \mid i \in I]$ be obtained from F by setting $x_0 = 1$. Consider the vanishing loci $\mathrm{V}(F) \subseteq \mathbf{P}^{I \sqcup \{0\}}$ and $\mathrm{V}(F_0) \subseteq \mathbf{A}^I_k$ and the vanishing loci $\mathrm{D}(F) \subseteq \mathbf{P}^{I \sqcup \{0\}}$ and $\mathrm{D}(F_0) \subseteq \mathbf{A}^I_k$. Then*

$$\mathrm{V}(F) \cap \mathbf{A}^I_k = \mathrm{V}(F_0) \quad \text{and} \quad \mathrm{D}(F) \cap \mathbf{A}^I_k = \mathrm{D}(F_0).$$

Proof. This follows immediately unwinding the definitions. □

Next, we want to describe the projective closure of affine vanishing loci.

Definition 3.95 (Homogenization of polynomials)**.** Let k be a ring and I a set (not containing 0).

(a) Let $f \in k[x_i \mid i \in I]$. The *homogenization* of f is the homogeneous polynomial

$$f^{\mathrm{h}} = x_0^d f((\frac{x_i}{x_0})_{i \in I}) \in k[x_0, x_i \mid i \in I],$$

where d is the maximal degree of a monomial with nonzero coefficient in f, or equivalently the smallest integer for which the above expression is a polynomial.

(b) Let $F \subseteq k[x_i \mid i \in I]$ be an ideal. The *homogenization* of F is the homogeneous ideal

$$F^{\mathrm{h}} = (f^{\mathrm{h}} \mid f \in F) \subseteq k[x_0, x_i \mid i \in I].$$

Remark 3.96. The following are some basic examples of homogenization:

(a) We can recover any polynomial or ideal from its homogenization by setting $x_0 = 1$ is in F. It follows that homogenization preserves radical and prime ideals.

(b) We have $(f^{\mathrm{h}}) = (f)^{\mathrm{h}}$, but this does not generalize to ideals with two or more generators. For example, if F is the ideal in $k[x_1, x_2]$ generated by $f = x_1 + x_2^2$ and $g = x_2$, then $F^{\mathrm{h}} = (x_1, x_2)$, but the ideal generated by $f^{\mathrm{h}} = x_0 x_1 + x_2^2$ and $g^{\mathrm{h}} = g$ does not contain x_1.

Corollary 3.97 (Projective closure of vanishing loci)**.** *Let I be a finite set and $F \subseteq k[x_i \mid i \in I]$ an ideal with vanishing locus $\mathrm{V}(F) \subseteq \mathbf{A}^I_k$. Then $\mathrm{V}(F^{\mathrm{h}}) \subseteq \mathbf{P}^{I \sqcup \{0\}}$ is the closure of $\mathrm{V}(F)$.*

Proof. The closure is the smallest closed subfunctor of $\mathbf{P}^{I \sqcup \{0\}}$ containing $\mathrm{V}(F)$, which by Proposition 3.90 corresponds to the largest saturated homogeneous ideal such that the dehomogenization is contained in F. This is exactly the homogenization of F. □

Corollary 3.98 (Quasi-affine schemes of finite type are quasi-projective)**.** *Let k be a ring, A a k-algebra of finite type, and $F \subseteq A$ a finite subset. Then the subfunctor $\mathrm{D}(F) \subseteq \mathrm{Spec}(A)$ is a quasi-projective k-scheme.*

Proof. Consider $\mathrm{D}(F) \subseteq \mathrm{Spec}(A)$, where $A = k[x_1, \dots, x_n]/I$ for some ideal $I \subseteq A$. Consider the homogenizations I^{h} and F^{h} so that in particular $I^{\mathrm{h}}|_{x_0=1} = I$ and $F^{\mathrm{h}}|_{x_0=1} = F$. Set $B = k[x_0, \dots, x_n]/I^{\mathrm{h}}$. WE claim that in this case we have $\mathrm{D}(F) = \mathrm{D}(F^{\mathrm{h}} \cdot x_0) \subseteq \mathrm{Proj}(B) = \mathrm{V}(I^{\mathrm{h}}) \subseteq \mathbb{P}^I_k$. Indeed, we have

$$\mathrm{D}(F) = \mathrm{V}(I) \cap \mathrm{D}(F) = \mathrm{V}(I^{\mathrm{h}}) \cap \mathbf{A}^n_k \cap \mathrm{D}(F^{\mathrm{h}}) = \mathrm{V}(I^{\mathrm{h}}) \cap \mathrm{D}(F^{\mathrm{h}} \cdot x_0) = \mathrm{D}(F^{\mathrm{h}} \cdot x_0),$$

where we used $\mathrm{V}(I^{\mathrm{h}}) \cap \mathbf{A}^n_k = \mathrm{V}(I)$ and $\mathbf{A}^n_k \cap \mathrm{D}(F^{\mathrm{h}}) = \mathrm{D}(F \cdot x_0)$, identifying $\mathbf{A}^n_k = \mathrm{D}(x_0)$. This shows that $\mathrm{D}(F)$ is a quasi-projective k-scheme. □

Definition 3.99 (Finite scheme)**.** Let k be a ring. An algebraic k-functor X is a finite k-scheme if there exists a finite k-algebra A such that $X \simeq \mathrm{Spec}(A)$.

Proposition 3.100 (Finite schemes are projective)**.** *Let $X \subseteq \mathbf{A}^I_k$ be a closed subfunctor such that every function on X is integral over k, i.e. satisfies a monic polynomial equation over k. Then X is closed in $\mathbf{P}^{I \sqcup \{0\}}_k$. In particular, every finite k-scheme is projective.*

todo *Proof.* . □

Remark 3.101. Conversely, if an affine k-scheme is a projective, then it finite; we will prove this later using the notion of properness. This means that any closed subscheme of $\mathbf{A}^n_k$, if not finite, must "go to infinity" and not remain closed in $\mathbf{P}^n_k$. This is in stark contrast to the analogous situation in differential geometry, where there exists positive dimensional closed submanifold of $\mathbf{R}^n$ that are "away from infinity", i.e., remain closed in $\mathbf{RP}^n$.

3.9. Example of projective schemes.

Example 3.102 (Weighted projective spaces). Let I be a set and $w \colon I \to \mathbf{N}_{>0}$ a map. Then we can equip the polynomial k-algebra $k[x_i \mid i \in I]$ with the $\mathbf{N}$-graded structure in which x_i is homogeneous of degree $w(i)$. The resulting algebraic k-functor $\mathbf{P}^w_k = \mathrm{Proj}(k[x_i \mid i \in I])$ is called the *weighted projective space* with weights w. It is a projective k-scheme if I is a finite, by Corollary 3.68.

Example 3.103 (The Segre embedding). Let k be a ring and let M and N be k-modules. The *Segre embedding* is the map of algebraic k-functors

$$\varsigma \colon \mathbf{P}(M) \times \mathbf{P}(N) \to \mathbf{P}(M \otimes_k N)$$

that sends a pair of quotient lines $M \otimes_k R \twoheadrightarrow K$ and $N \otimes_k R \twoheadrightarrow L$ to their tensor product $(M \otimes_k N) \otimes_k R \twoheadrightarrow K \otimes_R L$. This is well-defined since the tensor product $\otimes$ preserves lines and epimorphisms. If $M = k^{(I)}$ and $N = k^{(J)}$, the Segre embedding is a map

$$\varsigma \colon \mathbf{P}^I_k \times \mathbf{P}^J_k \to \mathbf{P}^{I \times J}_k.$$

If $M = k^{m+1}$ and $N = k^{n+1}$, this becomes the map

$$\varsigma \colon \mathbf{P}^m_k \times \mathbf{P}^n_k \to \mathbf{P}^{mn+m+n}_k,$$

which is given in coordinates by the formula

$$([a_0 : \dots : a_m], [b_0 : \dots : b_n]) \mapsto [\text{all products } a_i b_j].$$

More generally, we have a Segre embedding

$$\varsigma \colon \prod_{i=1}^n \mathbf{P}(M_i) \to \mathbf{P}(\bigotimes_{i=1}^n M_i)$$

for any finite family of k-modules $M_1, \dots, M_n$. We claim that the Segre embedding is a closed immersion. Given surjective maps $M' \twoheadrightarrow M$ and $N' \twoheadrightarrow N$, we have a commutative square

$$\begin{array}{ccc} \mathbf{P}(M) \times \mathbf{P}(N) & \xrightarrow{\varsigma} & \mathbf{P}(M \otimes_k N) \\ \downarrow & & \downarrow \\ \mathbf{P}(M') \times \mathbf{P}(N') & \xrightarrow{\varsigma} & \mathbf{P}(M' \otimes_k N') \end{array}$$

where the vertical maps are closed immersions, since they can be viewed as certain vanishing loci. Hence if the bottom horizontal map is closed immersion, so is the top horizontal map by Proposition 2.71 (b). We may therefore assume that M and N are free with bases $(e_i)_{i \in I}$ and $(f_j)_{j \in J}$, so that $M \otimes_k N$ has basis $(e_i \otimes f_j)_{i,j}$. For any map $\operatorname{Spec}(R) \to \mathbf{P}(M \otimes_k N)$, the pullback $(\mathbf{P}(M) \times \mathbf{P}(N)) \times_{\mathbf{P}(M \otimes_k N)} \operatorname{Spec}(R)$ satisfies Zariksi descent, by Proposition 3.64 and Corollary 2.74. Using Proposition 2.82, we see that it is enough to show that each restriction

make it precise that these or V(kernel)

$$\varsigma_{ij} \colon \varsigma^{-1}(\mathrm{D}(e_i \otimes f_j)) \mapsto \mathrm{D}(e_i \otimes f_j)$$

of ς is a closed immersion. The open subfunctor $\mathrm{D}(e_i \otimes f_j) \subseteq \mathbf{P}(M \otimes_k N)$ is an affine space on the set $(I \times J) \setminus \{(i,j)\}$. Its preimage $\varsigma^{-1}(\mathrm{D}(e_i \otimes f_j)) \subseteq \mathbf{P}(M) \times \mathbf{P}(N)$ is the subfunctor $\mathrm{D}(e_i) \times \mathrm{D}(f_j)$, which is an affine space on the set $(I \setminus \{i\}) \sqcup (J \setminus \{j\})$. By inspection, the map ς_{ij} is then Spec of the map of polynomial rings

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$$k[z_{kl} \mid (k,l) \in (I \times J) \setminus \{i,j\}] \to k[x_k, y_l \mid k \in I, l \in J]/(x_i = 1, y_j = 1), \quad z_{kl} \mapsto x_k y_l,$$

where $x_i = y_j = 1$. Since $z_{kj} \mapsto x_k$ and $z_{il} \mapsto y_l$, this map is surjective, so that $\varsigma_{i,j}$ is a closed immersion as desired.

Corollary 3.104. *Let X and Y be projective k-schemes. Then $X \times Y$ is a projective k-scheme.*

Proof. This follows immediately from that the Segre embedding is a closed immersion and that every closed subfunctor of a projective k-scheme is again a projective k-scheme. □

Remark 3.105. The Veronese embedding of Example 3.77 is determined by the Segre embedding as follows: for any k-module M and $d \geq 1$, there is a commutative square

$$\begin{array}{ccc} \mathbf{P}(M) & \xrightarrow{\rho_d} & \mathbf{P}(\operatorname{Sym}^d_K(M)) \\ {\scriptstyle\Delta}\downarrow & & \downarrow \\ \mathbf{P}(M)^{\times d} & \xrightarrow{\varsigma} & \mathbf{P}(M^{\otimes d}) \end{array}$$

where the vertical maps are the inclusions of the Σ_d-fixed points. This gives another proof that ρ_d is a closed immersion.

Next we consider a generalization of projective spaces where we replace quotient lines with quotient spaces of arbitrary rank.

Definition 3.106 (Grassmannian)**.** Let k be a ring, M a k-module, and $n \in \mathbf{N}$. The rank n *Grassmannian* of M is the algebraic k-functor $\mathrm{Gr}_n(M)$ given by

$$\mathrm{Gr}_n(M)(R) = \{\text{quotient spaces of } M \otimes_k R \text{ of constant rank } n\}.$$

Remark 3.107. By definition, $\mathrm{Gr}_1(M) = \mathbf{P}(M)$. If M is a vector space of constant rank r, then duality induces isomorphism $\mathrm{Gr}_n(M) \simeq \mathrm{Gr}_{r-n}(M^\vee)$(Corollary 3.22).

Example 3.108 (The Plücker embedding)*.* Let M be a k-module and let $n \in \mathbf{N}$. The *Plücker embedding* is the map of algebraic k-functors

$$\wp \colon \mathrm{Gr}_n(M) \to \mathbf{P}(\Lambda^n_k(M))$$

that sends a quotient space of rank n to its n-th exterior power, which is a quotient line. We claim that the Plücker embedding is a closed immersion. For any surjective map $M' \twoheadrightarrow M$, there is a commutative diagram

$$\begin{array}{ccc} \mathrm{Gr}_n(M) & \xrightarrow{\bar{\omega}} & \mathbf{P}(\Lambda^n_k(M)) \\ \downarrow & & \downarrow \\ \mathrm{Gr}_n(M') & \xrightarrow{\bar{\omega}} & \mathbf{P}(\Lambda^n_k(M')) \end{array}$$

where the vertical maps are closed immersions. As before, we can therefore assume that M is free with a basis $(e_i)_{i\in I}$, so that $\Lambda^n_k(M)$ has an induced basis $(e_J)_J$ (defined up to signs) indexed by the set $\binom{I}{n}$ of n-element subsets of I. Zariski descent for modules implies, as in Proposition 3.64, that $\mathrm{Gr}_n(M)$ satisfies Zariski descent. By Proposition 2.82, it thus suffices to show that each map $\bar{\omega}_J \colon \bar{\omega}^{-1}(\mathrm{D}(e_J)) \to \mathrm{D}(e_J)$ is a closed immersion. Then open subfunctor $\mathrm{D}(e_J) \subseteq \mathbf{P}(\Lambda^n_k)(M)$ is an affine space on the set $\binom{I}{n} \setminus \{J\}$. The preimage $\bar{\omega}^{-1}(\mathrm{D}(e_J)) \subseteq \mathrm{Gr}_n(M)$ is the subfunctor consisting of quotient spaces $R^{(I)} \twoheadrightarrow V$ such that the composite $R^{(J)} \hookrightarrow R^{(I)} \twoheadrightarrow V$ is an isomorphism, which is an affine space on the set $(I \setminus J) \times I$. By inspection, the map $\bar{\omega}_J$ is $\mathbf{A}(-)$ of the map of free k-modules given by

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$$e_K \mapsto \begin{cases} e_{(i,j)} & \text{if } K = (J \setminus \{j\} \cup \{j\}), \\ 0 & \text{else} \end{cases}$$

As this map is evidently surjective, $\bar{\omega}_J$ is a closed immersion, as desired. If M is of finite type, it follows that the Grassmannian $\mathrm{Gr}_n(M)$ is projective k-scheme.

Definition 3.109 (Flag schemes)**.** Let M be a k-module and let $n = (n_1, \ldots, n_s)$ be an increasing sequence of natural numbers.

(a) A *flag of type* n on M is a sequence of quotient modules

$$M \twoheadrightarrow F_s \twoheadrightarrow \ldots \twoheadrightarrow F_1$$

where each F_i is a vector space of constant rank n_i.

(b) The type n *flag scheme* of M is the algebraic k-functor $\mathrm{Flag}_n(M)$ given by

$$\mathrm{Flag}_n(M)(R) = \{\text{flags of type } n \text{ on } M \otimes_k R\}.$$

Remark 3.110. By definition, $\mathrm{Flag}_{(n)}(M) = \mathrm{Gr}_n(M)$. If M is a vector space of constant rank r, then duality induces isomorphisms $\mathrm{Flag}_{(n_1,\ldots,n_s)}(M) \simeq \mathrm{Flag}_{(r-n_s,\ldots,r-n_1)}(M^\vee)$.

Example 3.111 (Projectivity of flag schemes)*.* Sending a flag to its components define a monomorphism

$$\mathrm{Flag}_n(M) \hookrightarrow \prod_{i=1}^{s} \mathrm{Gr}_{n_i}(M).$$

We claim that it is a closed immersion, and hence that $\mathrm{Flag}_n(M)$ is a projective k-scheme if M is of finite type. Indeed, for any R-point $x \colon \mathrm{Spec}(R) \to \prod_{i=1}^s \mathrm{Gr}_{n_i}(M)$ classifying quotient spaces $F_1, \ldots, F_s$ of $M \otimes_k R$, the preimage of $\mathrm{Flag}_n(M)$ by x is exactly the joint vanishing locus of the R-linear maps

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$$\ker(M \otimes_k R \twoheadrightarrow F_{i+1}) \hookrightarrow M \otimes_k R \twoheadrightarrow F_i$$

for $1 \le i \le s-1$, which is a closed subfunctor of $\mathrm{Spec}(R)$ by Proposition 3.53.

3.10. **The projective Nullstellensatz.** Hilbert's Nullstellensatz (Theorem 2.93) implies an analogous result for solutions of homogeneous polynomial equations in projective space. We record this classical result here, although it is mostly of historical interest.

For a ring k, define the maps

$$\{\text{homogeneous subsets of } k[x_0, \ldots, x_n]\} \underset{\mathrm{I}}{\overset{\mathrm{V}}{\rightleftarrows}} \{\text{subsets of } \mathbf{P}^n(k)\}$$

as follows:

$$(V)(F) = \{x \in \mathbf{P}^n(k) \mid f(x) = 0 \text{ for all } f \in F\}$$
$$\mathrm{I}(X) = \{f \in k[x_0, \ldots, x_n] \text{ homogeneous} \mid f(x) = 0 \text{ for all } x \in X\}$$

Note that both maps are order-reversing and that $F \subseteq \mathrm{I}(\mathrm{V}(F))$ and $X \subseteq \mathrm{V}(\mathrm{I}(X))$ (in other words, this is an adjunction between posets). Note also that $\mathrm{I}(X)$ is always saturated homogeneous ideal in $k[x_0, \ldots, x_n]$, which is radical if k is reduced. Call a subset $X \subseteq \mathbf{P}^n(k)$ *algebraic* if it lies in the image of V, or equivalently if $X = \mathrm{V}(\mathrm{I}(X))$.

Proposition 3.112 (Projective Nullstellensatz)**.** *Let k be an algebraically closed field and let $n \in \mathbf{N}$. For any homogeneous subset $F \subseteq k[x_0, \ldots, x_n]$, we have*

$$\mathrm{I}(\mathrm{V}(F)) = \left(\sqrt{(F)}\right)^{\mathrm{sat}}.$$

Consequently, the maps V *and* I *define a one-to-one correspondence*

$$\{\textit{saturated radical homogeneous ideals in } k[x_0, \ldots, x_n]\} \underset{\mathrm{I}}{\overset{\mathrm{V}}{\rightleftarrows}} \{\textit{algebraic subsets of } \mathbf{P}^n(k)\}$$

Proof. This is left as an exercise. □

Corollary 3.113. *Let k be a field and let $n \in \mathbf{N}$. Then there exists an order reversing bijection*

$$\{\textit{saturated radical homogeneous ideals in } k[x_0, \ldots, x_n]\} \simeq \{\textit{vanishing loci in } \mathbf{P}^n \colon \mathrm{Field}_k^{\mathrm{fin}} \to \mathrm{Set}\}.$$

Proof. This is left as an exercise. □

4. Quasi-coherent modules

4.1. Limits of categories.

Definition 4.1 (Category-valued functor)**.** Let $\mathcal{J}$ be a category. A *functor*

$$F\colon \mathcal{J} \to \mathrm{Cat}$$

consists of

(a) for every object $I \in \mathcal{J}$, a category $F(I)$;
(b) for every map $f\colon I \to J$ in $\mathcal{J}$, a functor $F(f)\colon F(I) \to F(J)$;
(c) for every object $I \in \mathcal{J}$, a natural isomorphism $\eta_I\colon \mathrm{id}_{F(I)} \simeq F(\mathrm{id}_I)$;
(d) for every pair of maps $f\colon I \to J$ and $g\colon J \to K$ in $\mathcal{J}$, a natural isomorphism $\mu_{f,g}\colon F(g) \circ F(f) \simeq F(g \circ f)$;

satisfying the following conditions:

(a) for every map $f\colon I \to J$, the following diagrams commute:

$$\begin{array}{ccc} F(f) & \xrightarrow{\eta_I} & F(f) \circ F(\mathrm{id}_I) \\ & \searrow^{\mathrm{id}} & \downarrow{\scriptstyle \mu_{\mathrm{id}_I, f}} \\ & & F(f), \end{array} \qquad \begin{array}{ccc} F(f) & \xrightarrow{\eta_J} & F(\mathrm{id}_J) \circ F(f) \\ & \searrow^{\mathrm{id}} & \downarrow{\scriptstyle \mu_{f, \mathrm{id}_J}} \\ & & F(f); \end{array}$$

(b) for every triple of maps $f\colon I \to J$, $g\colon J \to K$ and $h\colon K \to L$ in $\mathcal{J}$, the following square commutes:

$$\begin{array}{ccc} F(h) \circ F(g) \circ F(f) & \xrightarrow{\mu_{f,g}} & F(h) \circ F(g \circ f) \\ \downarrow{\scriptstyle \mu_{g,h}} & & \downarrow{\scriptstyle \mu_{g\circ f, h}} \\ F(h \circ g) \circ F(f) & \xrightarrow[\mu_{f, h\circ g}]{} & F(h \circ g \circ f) \end{array}$$

Example 4.2 (Self-indexing functor)*.* Let $\mathcal{C}$ be a category with pullbacks. There is then a functor

$$\mathcal{C}^{\mathrm{op}} \to \mathrm{Cat}, \quad X \mapsto \mathcal{C}_{/X},$$

called the *self-indexing functor* of $\mathcal{C}$, sending a map $f\colon X \to Y$ to the pullback functor

$$f^*\colon \mathcal{C}_{/X} \to \mathcal{C}_{/Y}, \quad U \mapsto U \times_X Y.$$

The natural transformation $\eta_X\colon \mathrm{id}_{\mathcal{C}_{/X}} \simeq \mathrm{id}_X^*$ and $\mu_{f,g}\colon f^* \circ g^* \simeq (g \circ f)^*$ are given by the canonical isomorphisms $U \simeq U \times_X X$ and $(U \times_X Y) \times_Y Z \cong U \times_X (Y \times_Y Z) \cong U \times_X Z$, induced by the universal property of pullbacks.

Example 4.3 (Categories of modules)*.* The assignment $R \mapsto \mathrm{Mod}_R$ is functorial in three ways:

(a) There is a functor

$$\mathrm{Mod}^*\colon \mathrm{CAlg} \to \mathrm{Cat}, \quad R \mapsto \mathrm{Mod}_R,$$

sending a ring map $f\colon R \to S$ to the extension of scalars functor

$$f^*\colon \mathrm{Mod}_R \to \mathrm{Mod}_S, \quad M \mapsto M \otimes_R S.$$

The natural isomorphisms $\eta_R\colon \mathrm{id}_{\mathrm{Mod}_R} \simeq \mathrm{id}_R^*$ and $\mu_{f,g}\colon g^* \circ f^* \simeq (g \circ f)^*$ are given by the canonical isomorphisms $M \simeq M \otimes_R R$ and $(M \otimes_R S) \otimes_S T \simeq M \otimes_R T$, induced by the universal property of scalar extension.

(b) There is a functor

$$\mathrm{Mod}_*\colon \mathrm{CAlg}^{\mathrm{op}} \to \mathrm{Cat}, \quad R \mapsto \mathrm{Mod}_R,$$

sending a ring map $f\colon R \to S$ to the restriction of scalars functor

$$f_*\colon \mathrm{Mod}_S \to \mathrm{Mod}_R, \quad M \mapsto M.$$

In this case, the natural isomorphisms μ_R and $\eta_{f,g}$ are take to be the identity.

(c) Finally there is a functor

$$\mathrm{Mod}^!\colon \mathrm{CAlg} \to \mathrm{Cat}, \quad R \mapsto \mathrm{Mod}_R,$$

sending a ring map $f\colon R \to S$ to the coextension of scalars functor

$$f^!\colon \mathrm{Mod}_R \to \mathrm{Mod}_S, \quad M \mapsto \mathrm{Hom}_R(S, M).$$

As in (a), the natural isomorphisms μ_R and $\eta_{f,g}$ are induced by the universal property of scalar coextension.

Example 4.4 (Categories of algebras)*.* The assignment $R \mapsto \mathrm{CAlg}_R$ is both covariant and contravariant in R as in (a) and (b) of Example 4.3. There is however no analogue of (c) of Example 4.3 for algebras.

Example 4.5 (Subcategories of modules)*.* Any property of modules that is preserved by extension of scalars defines a subfunctor of $\mathrm{Mod}^*\colon \mathrm{CAlg} \to \mathrm{Cat}$. This holds for the following properties, and any combination thereof: finitely generated, finitely presented, flat, projective, free, vector space, line, etc.

Example 4.6 (Posets of ideals)*.* If we view posets as categories, then a poset-valued functor $\mathcal{C} \to \mathrm{Pos}$ defines a category-valued functor (where η_I and $\mu_{f,g}$ are identities). For example, we have the following functors $\mathrm{CAlg} \to \mathrm{Pos}$:

(a) The poset Idl_R of ideals in R is functorial in R: a map $R \to S$ induces $\mathrm{Idl}_R \to \mathrm{Idl}_S, \quad I \mapsto IS$.
(b) The poset Rad_R of radical ideals in R is functorial in R: a map $R \to S$ induces $\mathrm{Rad}_R \to \mathrm{Rad}_S,\ I \mapsto \sqrt{IS}$.

Notation 4.7. Let $\mathcal{J}$ be a category. The *left cone* $\mathcal{J}^\triangleleft$ on $\mathcal{J}$ is the category obtained from $\mathcal{J}$ by adjoining a new object $-\infty$ which is strictly initial, i.e., such that $\mathrm{Map}(-\infty, I) = *$ for all $I \in \mathcal{J}^\triangleleft$ and $\mathrm{Map}(I, -\infty) = \varnothing$ for all $i \in \mathcal{J}$. Notice that we can simply define a category in this way, so the above description of $\mathcal{J}^\triangleleft$ is a formally rigorous definition. Dually, the *right cone* $J^\triangleright$ is obtained from $\mathcal{J}$ by adding a strictly final object ∞.

Definition 4.8 (Limits of categories)**.** Let $F\colon \mathcal{J} \to \mathrm{Cat}$ be a functor. The *limit* of F, denoted by $\lim F$ or $\lim_{I \in \mathcal{J}} F(I)$, is the following category:

- An object of $\lim F$ consists of objects $x_I \in F(I)$ for all $I \in \mathcal{J}$ and isomorphisms

$$\alpha_f\colon F(f)(x_I) \xrightarrow{\sim} x_J$$

for all $f\colon I \to J$ in $\mathcal{J}$. These data are required to satisfy the following *cocycle conditions*:
  (a) for every $I \in \mathcal{J}$, the following triangle commutes

$$\begin{array}{ccc} x_I & \xrightarrow{\eta_I(x_I)} & F(\mathrm{id}_I)(x_I) \\ & \searrow^{\mathrm{id}} & \downarrow{\scriptstyle \alpha_{\mathrm{id}_I}} \\ & & x_I. \end{array}$$

  (b) for every pair of morphisms $f\colon I \to J$ and $g\colon J \to K$ in $\mathcal{J}$, the following square commutes:

$$\begin{array}{ccc} F(g)(F(f)(x_I)) & \xrightarrow{F(g)(\alpha_f)} & F(g)(x_J) \\ \downarrow{\scriptstyle \mu_{f,g}(x_I)} & & \downarrow{\scriptstyle \alpha_g} \\ F(g \circ f)(x_I) & \xrightarrow[\alpha_{g\circ f}]{} & x_K. \end{array}$$

- A morphsim in $\lim F$ from (x, α) to (y, β) consists of morphisms $\varphi_I\colon x_I \to y_I$ for all $I \in \mathcal{J}$, such that for each map $f\colon I \to J$ in $\mathcal{J}$, the following square commutes:

$$\begin{array}{ccc} F(f)(x_I) & \xrightarrow{F(f)(\varphi_I)} & F(f)(y_I) \\ \downarrow{\scriptstyle \alpha_I} & & \downarrow{\scriptstyle \beta_f} \\ x_J & \xrightarrow[\varphi_J]{} & y_J. \end{array}$$

- Identities and composition are defined pointwise.

By construction, there is an extension $\bar{F}\colon \mathcal{J}^{\triangleleft} \to \mathrm{Cat}$ of F with $\bar{F}(-\infty) = \lim F$. If $\bar{F}\colon \mathcal{J}^{\triangleleft} \to \mathrm{Cat}$ is any extension of F, then $\bar{F}$ induces a functor $\bar{F}(-\infty) \to \lim F$, and we say that $\bar{F}$ is a limit diagram if that functor is an equivalence.

Remark 4.9 (Colimits of categories). Once limits have been defined, we can also define colimits: given $F\colon \mathcal{J} \to \mathrm{Cat}$, an extension $\bar{F}\colon \mathcal{J}^{\triangleleft} \to \mathrm{Cat}$ is called a *colimit diagram* if, for every category $\mathcal{E}$, the functor

$$\mathrm{Fun}(\bar{F}(-), \mathcal{E})\colon (\mathcal{J}^{\mathrm{op}})^{\triangleleft} \to \mathrm{Cat}$$

is a limit diagram. One can show that colimits of categories always exists, although they are difficult to describe explicitly in general.

Example 4.10 (Pullbacks of categories). Given functors $\mathcal{C} \xrightarrow{f} \mathcal{E} \xleftarrow{g} \mathcal{D}$, the pullback $\mathcal{C} \times_{\mathcal{E}} \mathcal{D}$ is the category whose objects are triples (x, y, α) with $x \in \mathcal{C}$, $y \in \mathcal{D}$ and $\alpha\colon f(x) \xrightarrow{\sim} g(y)$ in $\mathcal{E}$.

Example 4.11. Let $\mathcal{C}$ be a category and consider the diagram $\mathcal{C} \rightrightarrows \mathcal{C}$, where both functors are the identity. The limit of this diagram is equivalent to the category $\mathrm{Fun}(B\mathbf{Z}, \mathcal{C})$ of $\mathbf{Z}$-equivariant objects of $\mathcal{C}$, or equivalently, the category of tuples (c, α) where $c \in \mathcal{C}$ and $\alpha : c \xrightarrow{\sim} c$ is an automorphism.

Example 4.12 (Zariski descent for modules). Limits of categories allow us to reformulate Theorem 2.72 more succinctly: it is exactly the statement that, if $(f_i)_{i\in I}$ generates the unit ideal in R, then the diagram of categories

$$\mathrm{Mod}_R \to \prod_{i\in I} \mathrm{Mod}_{R_{f_i}} \rightrightarrows \prod_{i,j\in I} \mathrm{Mod}_{R_{f_i f_j}} \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow}\; \prod_{i,j,k\in I} \mathrm{Mod}_{R_{f_i f_j f_k}}.$$

obtained by restricting the functor Mod^* from Example 4.3 is a limit diagram. In the case $I = \{1, 2\}$, this is equivalent to the simpler statement that the square of categories

$$\begin{array}{ccc} \mathrm{Mod}_R & \longrightarrow & \mathrm{Mod}_{R_{f_1}} \\ \downarrow & & \downarrow \\ \mathrm{Mod}_{R_{f_2}} & \longrightarrow & \mathrm{Mod}_{R_{f_1 f_2}} \end{array}$$

is cartesian. Moreover, these results hold for any subfunctor of Mod^* defined by a Zariski-local property (as in Example 4.5), such as $R \mapsto \mathrm{Vect}_R$ and $R \mapsto \mathrm{Line}_R$.

4.2. **Quasi-coherence.**

Definition 4.13 (Quasi-coherent objects)**.** Let $F\colon \mathrm{CAlg} \to \mathrm{Cat}$ be a functor. For any algebraic functor X, we define the category $F(X)$ of quasi-coherent F-objects over X, or simply *F-objects* over X, as the limit

$$F(X) = \lim_{\mathrm{Spec}(R)\to X} F(R),$$

where the indexing category is the category of elements of X. For example:

(a) The category of *quasi-coherent modules* over X is $\mathrm{Mod}_X = \lim_{\mathrm{Spec}(R)\to X} \mathrm{Mod}_R$.
(b) The category of quasi-coherent vector spaces over X is $\mathrm{Vect}_X = \lim_{\mathrm{Spec}(R)\to X} \mathrm{Vect}_R$. These are called *vector bundles* over X.
(c) The category of *quasi-coherent lines over X* is $\mathrm{Line}_X = \lim_{\mathrm{Spec}(R)\to X} \mathrm{Line}_R$. These are called *line bundles over X*.
(d) The category of *quasi-coherent algebras* over X is $\mathrm{CAlg}_X = \lim_{\mathrm{Spec}(R)\to X} \mathrm{CAlg}_R$.
(e) The poset of *quasi-coherent ideals* over X is $\mathrm{Idl}_X = \lim_{\mathrm{Spec}(R)\to X} \mathrm{Idl}_R$.
(f) The poset of *quas-coherent radical ideals* over X is $\mathrm{Rad}_X = \lim_{\mathrm{Spec}(R)\to X} \mathrm{Rad}_R$.

Remark 4.14 (Functoriality of quasi-coherent objects). (a) The assignment $X \mapsto F(X)$ has a structure of functor $\mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})^{\mathrm{op}} \to \mathrm{Cat}$. In particular, for every map of algebraic functors $f\colon Y \to X$, we have a *base change* or *pullback* functor

$$f^*\colon F(X) \to F(Y),$$

given by precomposition with the induced functor $\mathrm{El}(Y) \to \mathrm{El}(X)$. The right adjoint to f^*, if it exists, is called the *pushforward* functor and denoted by f_*.

(b) For $X = \mathrm{Spec}(R)$, evaluation at id_R induces an equivalence of categories $F(X) \xrightarrow{\sim} F(R)$. In other words, the construction of Definition 4.2 is an extension of F along the Yoneda embedding $\mathrm{CAlg} \hookrightarrow \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})^{\mathrm{op}}$ (in fact, it is the *right Kan extension*). For example, a quasi-coherent module (algebra, ideal, etc.) over $\mathrm{Spec}(R)$ is the same as an R-module (algebra, ideal, etc.).

(c) There is an obvious notion of a *natural transformation* between category valued functors. Any natural transformation $\alpha\colon F \to G$ between functors $\mathrm{CAlg} \to \mathrm{Cat}$ extends automatically to a natural transformation between their quasi-coherent extensions $\mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})^{\mathrm{op}} \to \mathrm{Cat}$. In particular, for any algebraic functor X, there is an induced functor $\alpha_X\colon F(X) \to G(X)$.

Example 4.15 (Quasi-coherent modules). Let us spell out explicitly the definition of a quasi-coherent module. A quasi-coherent module M over an algebraic functor X consists of the following data:

(a) for every $x\colon \mathrm{Spec}(R) \to X$, an R-module $M(x)$;

(b) for every commutative triangle

$$\begin{array}{ccc} \mathrm{Spec}(S) & \xrightarrow{y} & X \\ {\scriptstyle\varphi}\downarrow & \nearrow_{x} & \\ \mathrm{Spec}(R) & & \end{array}$$

an S-linear isomorphism $\varphi^*(M(x)) \cong M(x) \otimes_R S \simeq M(y)$;

such that the isomorphism in (b) satisfy the cocyle conditions of Definition 4.8. Under the equivalence $\mathrm{Mod}_{\mathrm{Spec}(R)} \simeq \mathrm{Mod}_R$ of Remark 4.14, the R-module $M(x)$ in (a) is identified with the pullback $x^*(M)$, and the isomorphism in (b) is an instance of the natural isomorphism $\mu_{\mathrm{Spec}(\varphi),x}\colon \mathrm{Spec}(\varphi)^* \circ x^* \simeq y^*$, which is part of the functor $\mathrm{Mod}\colon \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})^{\mathrm{op}} \to \mathrm{Cat}$ of Remark 4.14.

Remark 4.16 (Pushforward of quasi-coherent modules). Let $f\colon Y \to X$ be a map of algebraic functors. For quasi-coherent modules and algebras, the pushforward functor f_* always exists, moduluo "size issues" as in Remark 2.53. That is, if M is a quasi-coherent module over Y, then $f_*(M)$ is potentially a large quasi-coherent module over X. This does not happen if X and Y are accessible, hence does not happen in practice.

Example 4.17 (Set-valued functors). As a special case, we can apply Definition 4.2 to any set-valued functor $F\colon \mathrm{CAlg} \to \mathrm{Set}$. By Corollary 2.11, we then have $F(X) = \mathrm{Map}(X, F)$ for any algebraic functor X. In other words, the quasi-coherent extension of F is $\mathrm{Map}(-, F)$.

Remark 4.18. Let $F\colon \mathrm{CAlg} \to \mathrm{Cat}$ be a functor. Any properties or constructions within the categories $F(R)$ that are compatible with base change automatically extend to the categories of quasi-coherent F-objects. For example:

(a) (Properties of modules) A quasi-coherent module can be of *finite type*, of *finite presentation*, *flat*, *projective*, *free*, etc.

(b) (Properties of module maps) A morphism of quasi-coherent modules can be *zero*, *surjective*, *universally injective*, etc. On the other hand, injectivity does not a priori make sense, as this property is not preserved by base change.

(c) (Sums of module maps) Two morphisms $f, g\colon M \to N$ in Mod_X have a sum $f + g\colon M \to N$. This defines an abelian group structure on $\mathrm{Map}_{\mathrm{Mod}_X}(M, N)$.

(d) (Colimits of modules) The category of quasi-coherent modules over X has a symmetric monoidal structure with $(M \otimes N)(x) = M(x) \otimes N(x)$. For any $d \in \mathbf{N}$, there are symmetric, exterior, and divided power functors $\mathrm{Sym}^d, \Lambda^d, \Gamma^d\colon \mathrm{Mod}_X \to \mathrm{Mod}_X$.

(e) (Operations with ideals) Quasi-coherent ideals $I, J \in \mathrm{Idl}_X$ have a sum $I + J$ and a product IJ, defined by $(I + J)(x) = I(x) + J(x)$ and $(IJ)(x) = I(x)J(x)$.
(f) (Properties of algebras) A quasi-coherent algebra can be of *finite type*, of *finite presentation*, *finite*, *free*, etc. A quasi-coherent $\mathbf{N}$-graded algebra can be *generated* by A_1, *finitely generated as an* A_0*-algebra*, etc.
(g) (Properties of algebra maps) A morphism of quasi-coherent algebras can be *surjective*, *a localization*, have a *nilpotent kernel*, etc. A morphism of quasi-coherent $\mathbf{N}$-graded algebras can also be *essentially surjective*.
(h) (Underlying module and symmetric algebra) There is a forgetful functor $\mathrm{CAlg}_X \to \mathrm{Mod}_X$ with a left adjoint Sym, such that the Underlying quasi-coherent module of $\mathrm{Sym}(M)$ is $\bigoplus_{d\in\mathbf{N}} \mathrm{Sym}^d(M)$.
(i) (Kernel of a quasi-coherent algebra) There is a *kernel* functor $\mathrm{CAlg}_X \to \mathrm{Idl}_X$, which extends the functors $\mathrm{CAlg}_R \to \mathrm{Idl}_R$ sending a ring map $R \to A$ to its kernel.
(j) etc.

Warning 4.19. Beware that the forgetful functors $\mathrm{Rad}_R \hookrightarrow \mathrm{Idl}_R \to \mathrm{Mod}_R$ are *not* compatible with base change, and so do not extend to arbitrary algebraic functors X (although they will when X is a scheme): a quasi-coherent radical ideal over X need not have an underlying quasi-coherent ideal, and a quasi-coherent ideal over X need not have an underlying quasi-coherent module.

Notation 4.20 (The quasi-coherent algebra of functions)**.** Let X be an algebraic functor. We denote $\mathcal{O}(X) \in \mathrm{CAlg}_X$ the quasi-coherent algebra given by $\mathcal{O}_X(x\colon \mathrm{Spec}(R) \to X) = \mathcal{O}(\mathrm{Spec}(R)) = R$. It is the initial object in CAlg_X, and its underlying quasi-coherent module is the unit of the tensor product in Mod_X. For any map $f\colon Y \to X$, there is a canonical isomorphism $f^*(\mathcal{O}_X) \simeq \mathcal{O}_Y$ in CAlg_Y.

Definition 4.21 (Global sections of a quasi-coherent module)**.** Let X be an algebraic functor and M a quasi-coherent module over X. A *global section* of M is a map $\mathcal{O}_X \to M$ in Mod_X. We denote by $M(X)$ or $\Gamma(X, M)$ the abelian group of global sections of M. By definition of Mod_X, it is the limit

$$M(X) = \Gamma(X, M) = \lim_{(R,x)\in \mathrm{El}(X)^{\mathrm{op}}} \mathrm{Map}_{\mathrm{Mod}_R}(R, M(x)) = \lim_{(R,x)\in \mathrm{El}(X)^{\mathrm{op}}} M(x).$$

Remark 4.22 (Functions as global sections)*.* A global section of $\mathcal{O}_X$ is exactly a function on X in the sense of Definition 2.51: $\Gamma(X, \mathcal{O}_X) = \mathcal{O}(X)$. Hence, for any $M \in \mathrm{Mod}_X$, the abelian group $M(X)$ has a canonical structure of a module over the ring of functions $\mathcal{O}(X)$. We can also see this module structure as follows: if $a\colon X \to \mathrm{Spec}(\mathcal{O}(X))$ is the canonical map (see Remark 2.54), then $M(X) = a_*(M)$ an an $\mathcal{O}(X)$-module.

4.3. **Classification of closed and open subfunctors.**

Proposition 4.23 (Classification of closed and open subfunctors)**.** *Let X be an algebraic functor.*

(a) There is an isomorphism of posets

$$\mathrm{V}\colon \{\textit{quasi-coherent ideal over } X\}^{\mathrm{op}} \xrightarrow{\sim} \{\textit{closed subfunctors of } X\}.$$

(b) There is an isomorphism of posets

$$\mathrm{D}\colon \{\textit{quasi-coherent radical ideals over } X\} \xrightarrow{\sim} \{\textit{open subfunctors of } X\}$$

Proof. We denote $\mathrm{Sub}(X)$ the poset of subfunctors of X. Then the canonical map

$$\mathrm{Sub}(X) \to \lim_{(R,x)\in \mathrm{El}(X)^{\mathrm{op}}} \mathrm{Sub}(\mathrm{Spec}(R))$$

is an equivalence. One can show that the functor sending $(Y_x \subseteq \mathrm{Spec}(R))_{(R,x)\in\mathrm{El}(X)}$ to the subfunctor Y with sections defined by

$$Y(R) = \{x\colon \mathrm{Spec}(R) \to X \mid Y_x = \mathrm{Spec}(R)\}$$

is the inverse. Alternatively, since te category of algebraic functors is a topos, one could also argue that the functor $\mathrm{Sub} : \mathrm{Fun}(\mathrm{CAlg}_k, \mathrm{Set}) \to \mathrm{Set}$ is representable by a so-called subobject classifier.

(a) The above equivalence restricts to

$$\begin{array}{ccc} \mathrm{Sub}(X) & \xrightarrow{\sim} & \lim_{(R,x)\in \mathrm{El}(X)^{\mathrm{op}}} \mathrm{Sub}(\mathrm{Spec}(R)) \\ \uparrow & & \uparrow \\ \mathrm{Closed}(X) & \xrightarrow{\sim} & \lim_{(R,x)\in \mathrm{El}(X)^{\mathrm{op}}} \mathrm{Closed}(\mathrm{Spec}(R)), \end{array}$$

by definition. There exists a canonical natural isomorphism

$$\mathrm{V}\colon \mathrm{Closed}(\mathrm{Spec}(R)) \xrightarrow{\sim} \mathrm{Idl}_R^{\mathrm{op}},$$

which induces an isomorphism

$$\mathrm{V}\colon \lim_{(R,x)\in \mathrm{El}(X)^{\mathrm{op}}} (\mathrm{Closed}(\mathrm{Spec}(R))) \simeq \lim_{(R,x)\in \mathrm{El}(X)^{\mathrm{op}}} \mathrm{Idl}_R^{\mathrm{op}}.$$

Combined with the above diagram, we have proved the statement.

(b) The proof is similar to that of (a).

□

Definition 4.24 (Open complement)**.** Let $Z \subseteq X$ be closed subfunctor defined by the quasi-coherent ideal I. The *open complement* $X \setminus Z$ of Z in X is the open subfunctor defined by the quasi-coherent radical ideal $\sqrt{I}$.

Example 4.25. Let I be a set.

(a) If A is a ring and $F \subseteq A$ is a subset, then $\mathrm{D}(F)$ is the open complement of $\mathrm{V}(F)$ in $\mathrm{Spec}(A)$. For example, the punctured affine space $\mathbf{A}^I \setminus \{0\}$ (Example 2.57) is the open complement of 0 in $\mathbf{A}^I$.

(b) If A is an $\mathbf{N}$-graded ring and $F \subseteq A$ is a homogeneous subset, then $\mathrm{D}(F)$ is the open complement of $\mathrm{V}(F)$ in $\mathrm{Proj}(A)$. For example, in $\mathbf{P}^{I \sqcup \{0\}}$, the affine space $\mathrm{D}(x_0) \simeq \mathbf{A}^I$ is the open complement of the hyperplane at infinity $\mathrm{V}(x_0) \simeq \mathbf{P}^I$.

Warning 4.26. Different closed subfunctors can have the same open complement, since different ideals can have the same radical (e.g., (x) and (x^2) in $\mathbf{Z}[x]$). For this reason, there is no notion of "closed complement" of an open subfunctor.

Example 4.27 (Closed image). Any map of ring $\varphi\colon A \to B$ admits a canonical factroization $A \twoheadrightarrow A/\ker(\varphi) \hookrightarrow B$, so that $\mathrm{V}(\ker(\varphi)) \subseteq \mathrm{Spec}(A)$ is the smallest closed subfunctor through which $\mathrm{Spec}(\varphi)$ factors. This factorization generalizes to algebraic functors as follows. Let $f\colon X \to Y$ be a map of algebraic functors. The *closed image* of f, also called the *scheme-theoretical image*, is the smallest closed subfunctor of X through which f factors. By Proposition 4.23, the closed image of f is $\mathrm{V}(I) \subseteq X$, where I is the largest quasi-coherent ideal such that $\mathcal{O}_X \to f_*(\mathcal{O}_Y)$ factors through $\mathcal{O}_X/I$ in CAlg_X.

Remark 4.28 (Loci associated with maps of quasi-coherent modules). The loci from Definition 3.50 have a direct generalization to any algebraic functor X: a map $f\colon M \to N$ in Mod_X defines subfunctors $\mathrm{V}(f)$, $\mathrm{Epi}(f)$, $\mathrm{Mono}(f)$, and $\mathrm{Iso}(f)$ on X. These are characterized by the property that their preimage along any R-point $x\colon \mathrm{Spec}(R) \to X$ is the corresponding locus for the R-linear map $f(x)\colon M(x) \to N(x)$. Proposition 3.53 then immediately generalizes to this setting: if N is a vector bundle, then $\mathrm{V}(f)$ is a closed subfunctor and $\mathrm{Epi}(f)$ is an open subfunctor of X, and if both M and N are vector bundles, then also $\mathrm{Mono}(f)$ and $\mathrm{Iso}(f)$ are open subfunctors of X.

Call a subfunctor *clopen*, if it is both closed and open. Recall that $\mathrm{Idem}(R)$ is the set of idempotent elements in a ring R.

Proposition 4.29 (Classification of clopen subfunctors)**.** *For any algebraic functor X, there is a bijection*

$$\mathrm{Idem}(\mathcal{O}(X)) \xrightarrow{\sim} \{\textit{clopen subfunctors of } X\}, \quad e \mapsto \mathrm{V}(e) = \mathrm{D}(1-e).$$

Proof. Todo □

Remark 4.30. By Example 2.30 and Remark 2.54, there is a bijection

$$\mathrm{Map}(X, \mathrm{Spec}(\mathbf{Z} \times \mathbf{Z})) \xrightarrow{\sim} \mathrm{Idem}(\mathcal{O}(X)), \quad f \mapsto f^*(1,0).$$

Thus, clopen subfunctors are also in bijection with maps to $\mathrm{Spec}(\mathbf{Z} \times \mathbf{Z})$, which is the coproduct $* \sqcup *$ in the category of affine schemes. This is analogous to the following (much more obvious) topological statement: if T is a topological space, there is a bijection between clopen subsets of T and continuous maps $T \to * \sqcup *$.

4.4. **Relative Spec and Proj.** The constructions Spec and Proj are compatible with base change in the following sense. If A is an R-algebra (resp. an $\mathbf{N}$-graded R-algebra) and $R \to S$ is a ring map, then the following squares are cartesian:

$$\begin{array}{ccc} \mathrm{Spec}(A \otimes_R S) & \longrightarrow & \mathrm{Spec}(A) \\ \downarrow & & \downarrow \\ \mathrm{Spec}(S) & \longrightarrow & \mathrm{Spec}(R), \end{array} \qquad \begin{array}{ccc} \mathrm{Proj}(A \otimes_R S) & \longrightarrow & \mathrm{Proj}(A) \\ \downarrow & & \downarrow \\ \mathrm{Spec}(S) & \longrightarrow & \mathrm{Spec}(R). \end{array}$$

This allows us to extend Spec and Proj to quasi-coherent algebras:

Definition 4.31 (Relative Spec and Proj)**.** Let X be an algebraic functor.

(a) Let A be a quasi-coherent algebra over X. The algebraic functor $\mathrm{Spec}(A)$ over X is defined by

$$\mathrm{Spec}(A)(R) = \{(x, a) \mid x \in X(R) \text{ and } a \text{ is a section of } \mathrm{Spec}(A(x)) \to \mathrm{Spec}(R)\}.$$

This defines a functor $\mathrm{Spec}\colon \mathrm{CAlg}_X^{\mathrm{op}} \to \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})_{/X}$.

(b) Let A be a quasi-coherent $\mathbf{N}$-graded algebra over X. The algebraic functor $\mathrm{Proj}(A)$ over X is defined by

$$\mathrm{Proj}(A)(R) = \{(x, a) \mid x \in X(R) \text{ and } a \text{ is a section of } \mathrm{Proj}(A(x)) \to \mathrm{Spec}(R)\}.$$

This defines a functor $\mathrm{Proj}\colon (\mathrm{CAlg}_X^{\mathbf{N},\mathrm{es}})^{\mathrm{op}} \to \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})_{/X}$.

Definition 4.32 (Relative affine and projective spaces)**.** Let X be an algebraic functor and let M be a quasi-coherent module over X.

(a) The *affine space* $\mathbf{A}(M)$ over X is $\mathrm{Spec}(\mathrm{Sym}(M))$. Explicitly:

$$\mathbf{A}(M)(R) = \{(x, a) \mid x \in X(R) \text{ and } a\colon M(x) \to R \text{ is an } R\text{-linear map}\}.$$

(b) The *punctured affine space* $\mathbf{A}(M) \setminus \{0\}$ over X is $\mathrm{D}(M) \subseteq \mathrm{Spec}(\mathrm{Sym}(M))$. Explicitly:

$$(\mathbf{A}(M) \setminus \{0\})(R) = \{(x, a) \mid x \in X(R) \text{ and } a\colon M(x) \twoheadrightarrow R \text{ is a surjective } R\text{-linear map}\}.$$

(c) The *projective space* $\mathbf{P}(M)$ over X is $\mathrm{Proj}(\mathrm{Sym}(M))$. Explicitly:

$$\mathbf{P}(M)(R) = \{(x, a) \mid x \in X(R) \text{ and } a\colon M(x) \twoheadrightarrow L \text{ is a quotient } R\text{-line}\}.$$

Remark 4.33 (Morphisms into Spec and Proj)*.* Using Corollary 2.11, we obtain the following descriptions of maps into $\mathrm{Spec}(A)$ or $\mathrm{Proj}(A)$. Let X and Y be algebraic functors.

(a) For any $A \in \mathrm{CAlg}_X$, a map $Y \to \mathrm{Spec}(A)$ consists of a map $f\colon Y \to X$ and a map $f^*(A) \to \mathcal{O}(Y)$ in CAlg_Y (cf. Remark 2.54).

(b) Let A be a quasi-coherent $\mathbf{N}$-graded algebra over X generated by A_1. Then a map $Y \to \mathrm{Proj}(A)$ consists of a map $f\colon Y \to X$ and a quotient line bundle $f^*(A_1) \twoheadrightarrow L$ in Mod_Y such that the induced map $\mathrm{Sym}(f^*(A_1)) \twoheadrightarrow \mathrm{Sym}(L)$ factors through $f^*(A)$.

The relative Spec and Proj functors allows us to relativize the notions of affine, quasi-affine, projective, and quasi-projective schemes:

Definition 4.34 (Affine and quasi-affine morphisms)**.** Let $f\colon Y \to X$ ve a map of algebraic functors.

(a) We say that f is *affine*, or that Y is *affine over* X, if there exists $A \in \mathrm{CAlg}_X$ and an isomorphism $Y \simeq \mathrm{Spec}(A)$ over X. We denote by Aff_X the category of algebraic functors affine over X.

(b) We say that f is *quasi-affine*, or that Y is *quasi-affine over* X, if there exists a affine algebraic X-functor $V \in \mathrm{Aff}_X$, a quasi-coherent, finitely generated radical ideal $I \in \mathrm{Rad}_V^{\mathrm{fg}}$, and an isomorphism $Y \simeq \mathrm{D}(I)$ over X. We denote by QAff_X the category of algebraic functors quasi-affine over X.

Definition 4.35 (Projective and quasi-projective morphism)**.** Let $f\colon Y \to X$ be a map of algebraic functors.

(a) We say that f is *projective*, or that Y is projective over X, if there exists $A \in \mathrm{CAlg}_X^{\mathbf{N},\mathrm{es}}$, generated by A_1 an finitely generated, and an isomorphism $Y \simeq \mathrm{Proj}(A)$ over X. We denote by Proj_X the category of algebraic functors projective over X.

(b) We say that f is *quasi-projective*, or that Y is *quasi-projective over* X, if there exists a projective algebraic X-functor $V \in \mathrm{Proj}_X$, a quasi-coherent, finitely generated radical ideal $I \in \mathrm{Rad}_V^{\mathrm{ft}}$, and an isomorphism $Y \simeq \mathrm{D}(I)$ over X. We denote by QProj_X the category of algebraic functors quasi-projective over X.

Affineness turns out to be a *quasi-coherent property* in the sense that the functor $X \mapsto \mathrm{Aff}_X$ is right Kan extended from affine schemes:

Proposition 4.36 (Characterization of affine morphisms)**.** *Let X be an algebraic functor.*

(a) A map $Y \to X$ is affine if and only if, for every ring R and every map $\mathrm{Spec}(R) \to X$, the pullback $Y \times_X \mathrm{Spec}(R)$ is affine.

(b) The functor Spec *defines an equivalence of categories*

$$\mathrm{Spec}\colon \mathrm{CAlg}_X^{\mathrm{op}} \xrightarrow{\sim} \mathrm{Aff}_X,$$

whose inverse sends $f\colon Y \to X$ to $f_*(\mathcal{O}_Y)$.

Proof. (a) It sufficies to prove that the canonical map

$$\mathrm{Aff}_X \to \lim_{(R,x)\in \mathrm{El}(X)^{\mathrm{op}}} \mathrm{Aff}_R$$

is an equivalence of categories. Essentially surjectivity follows from the following diagram

$$\begin{array}{ccc} \mathrm{CAlg}_X^{\mathrm{op}} & \longrightarrow & \lim_{(R,x)\in \mathrm{El}(X)^{\mathrm{op}}} \mathrm{CAlg}_R^{\mathrm{op}} \\ \downarrow & & \downarrow{\scriptstyle \mathrm{Spec}} \\ \mathrm{Aff}_X & \longrightarrow & \lim_{(R,x)\in \mathrm{El}(X)^{\mathrm{op}}} \mathrm{Aff}_R, \end{array}$$

where the right vertical map is an isomorphism, so that the upper composition is essentially surjective, and hence so is the lower horizontal arrow.

For fully faithfulness, note that we have the following pullback square:

$$\begin{array}{ccc} \mathrm{Aff}_X & \longrightarrow & \lim_{(R,x)\in \mathrm{El}(X)^{\mathrm{op}}} \mathrm{Aff}_R \\ \downarrow & & \downarrow \\ \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})_{/X} & \longrightarrow & \lim_{(R,x)\in \mathrm{El}(X)^{\mathrm{op}}} \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})_{/\mathrm{Spec}} \end{array}$$

It then sufficies to show that the lower horizontal map is fully faithful.

Let $\mathcal{C}$ be a category, and $X = \mathrm{colim}_{i\in I} X_i$ in $\mathrm{P}(\mathcal{C}) = \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathrm{Set})$. We will show that the canonical functor

$$\mathrm{P}(\mathcal{C})_{/X} \to \lim_{i\in I} \mathrm{P}(\mathcal{C})_{/X_i},$$

sending $Y \to X$ to the collection $\{Y \times_X X_i\}_{i\in I}$, is fully faithful. But this functor admits a left adjoint by sending $(Y_i \to X_i)_{i\in I}$ to $\mathrm{colim}_{i\in I} Y_i \to X$. Therefore, it suffices to show that the counit $\mathrm{colim}_{i\in I}(X_i \times_X Y) \to Y$ is an isomorphism. This follows from the fact the functor

$$\mathrm{P}(\mathcal{C})_{/X} \to \mathrm{P}(\mathcal{C})_{/Y}, \quad Z \mapsto Z \times_X Y$$

preserves colimits. Indeed, colimits and limits of functors are calculated sectionwise, hence it suffices to check that for sets X, Y the functor

$$\mathrm{Set}_{/X} \to \mathrm{Set}_{/Y}, \quad Z \mapsto Z \times_X Y$$

preserves colimits. But this is clear, since this functor admits a right adjoint

$$\mathrm{Set}_{/Y} \to \mathrm{Set}_{/X}, \quad T \mapsto \amalg_{x \in X} \mathrm{Hom}_{/Y}(Y_x, T),$$

where Y_x is the fiber of the map $Y \to X$ over $x \in X$, and the structure morphism is the canonical map

$$\amalg_{x \in X} \mathrm{Hom}_{/Y}(Y_x, T) \to \amalg_{x \in X} * \cong X$$

To see that this really defines an adjunction, let $\alpha : Z \to X \in \mathrm{Set}_{/X}$ and $\beta : T \to Y \in \mathrm{Set}_{/Y}$. We define morphisms

$$\varphi : \mathrm{Hom}_{\mathrm{Set}_{/Y}}(Z \times_X Y, T) \rightleftarrows \mathrm{Hom}_{\mathrm{Set}_{/X}}(Z, \amalg_{x \in X} \mathrm{Hom}_{\mathrm{Set}_{/Y}}(Y_x, T)) : \psi,$$

where φ sends a function $(z, y) \mapsto f(z, y)$ to the function $\varphi(f) : (z \mapsto f|_{Y_{\alpha(z)}}(z, -))$, while ψ sends a function[19] $\big(z \mapsto g_{\alpha(z)} : (Y_{\alpha(z)} \ni z \mapsto g_{\alpha(z)}(y))\big)$ to the function $((z, y) \mapsto g_{\alpha(z)}(y))$, i.e. ψ 'glues the function defined on fibers together'. That φ and ψ and mutually inverses is left as an exercise to the reader.

(b) Consider again the diagram

$$\begin{array}{ccc} \mathrm{CAlg}_X^{\mathrm{op}} & \longrightarrow & \lim_{(R,x) \in \mathrm{El}(X)^{\mathrm{op}}} \mathrm{CAlg}_R^{\mathrm{op}} \\ \downarrow & & \downarrow \mathrm{Spec} \\ \mathrm{Aff}_X & \longrightarrow & \lim_{(R,x) \in \mathrm{El}(X)^{\mathrm{op}}} \mathrm{Aff}_R, \end{array}$$

The upper horizontal map is an equivalence by definition of CAlg_X, the right vertical map is an equivialence by the definition fo Aff_R for a ring R. In (a) we showed that the lower horizontal morphism is an equivalence. Hence the left vertical morphism is an equivalence, using the two-out-of-three property. This shows the first part of the claim. The explicit description follows that this is trivially true for algebras, using the definition of $\mathcal{O}$.

□

Warning 4.37. The analogue of (a) in Proposition 4.36 does not hold for quasi-affine, projective, and quasi-projective morphisms.

Example 4.38 (Closed immersions as affine morphisms)*.* By Proposition 4.36, any closed immersion $Z \hookrightarrow X$ is affine, and $Z \simeq \mathrm{Spec}(i_*(\mathcal{O}_Z))$. Moreover, an affine map $f\colon Y \to X$ is a closed immersion if and only if the induced map $\mathcal{O}_X \to f_*(\mathcal{O}_Y)$ is surjective. Indeed, if $Y \to X$ is a closed immersion, this is true by definition and the description just given. If $\mathcal{O}_X \to f_*(\mathcal{O}_Y)$ on the other hand is surjective, denote it's kernel by $\mathcal{I}$. Then $\mathrm{Spec}(f_*\mathcal{O}_Y) \cong \mathrm{Spec}(\mathcal{O}_X/\mathcal{I})$, so that for any map $\mathrm{Spec}(R) \to X$ we have $Y \times_X \mathrm{Spec}(R) \cong \mathrm{Spec}(R/\mathcal{I}(R)) = \mathrm{V}(\mathcal{I}(R))$ proving the claim.

4.5. **Modules over quasi-projective schemes.**

Definition 4.39 (Zariski-local epimorphism)**.** A map of algebraic functors $Y \to X$ is called *Zariski-local epimorphism*, if for every $\mathrm{Spec}(R) \to X$, there exists $f_1, \ldots, f_n \in R$ generating the unit ideal such that each composite $\mathrm{Spec}(R_{f_i}) \hookrightarrow \mathrm{Spec}(R) \to X$ lifts to Y.

Example 4.40*.* The following are some examples of Zariski-local epimorphism:

(a) Any epimorphism is a Zariski-local epimorphism. In partiuclar, any map with a section, such as the projection $\mathbf{A}^n \times X \to X$, is a Zariski-local epimorphism.

[19]Notice that we are only considering functions living over X, hence the image function of $z \in Z$ under any function $g \in \mathrm{Hom}_{\mathrm{Set}_{/X}}(Z, \amalg_{x \in X} \mathrm{Hom}_{\mathrm{Set}_{/Y}}(Y_x, T))$ has to live in the component indexed by $\alpha(z)$.

(b) Let $(F_i)_{i\in I}$ be a family of subsets of $\mathcal{O}(X)$ whose union generates the unit ideal. Then the map $\coprod_{i\in I} \mathrm{D}(F_i) \to X$ is a Zariski-local epimorphism.
(c) Let L be a line bundle over X. A family $(s_i)_{i\in I}$ of global sections of L is called *generating*, if for every $x \in \mathrm{El}(X)$, the family $(s_i(x))_{i\in I}$ generates the line $L(x)$. In this case, the map $\coprod_{i\in I} \mathrm{D}(s_i) \to X$ is a Zariski-local epimorphism.
(d) The map $\mathbf{A}^I \setminus \{0\} \to \mathbf{P}^I$ is a Zariski-local epimorphism, by Proposition 3.28. More generally, for any $\mathbf{N}$-graded ring A, the map $\mathrm{D}(A_1) \to \mathrm{Proj}_1(A)$ is a Zariski-local epimorphism (see Remark 3.71).

Proposition 4.41 (Zariski descent for quasi-coherent modules)**.** *Let $(Y_i \to X)_{i\in I}$ be family of maps of algebraic functors such that $\coprod_{i\in I} Y_i \to X$ is a Zariski-local epimorphism. Then the following diagram of categories is a limit diagram:*

$$\mathrm{Mod}_X \to \prod_{i\in I} \mathrm{Mod}_{Y_i} \rightrightarrows \prod_{i,j\in I} \mathrm{Mod}_{Y_i\times_X Y_j} \;\substack{\rightarrow\\ \rightarrow}\; \prod_{i,j,k\in I} \mathrm{Mod}_{Y_i\times_X Y_j\times_X Y_k},$$

where the maps are given by pulling back.

In particular, the functor $\mathrm{Mod}_X \to \prod_{i\in I} \mathrm{Mod}_{Y_i}$ is conservative.

Remark 4.42. Proposition 4.41 is a formal consequence of the Zariski descent property of Mod^* from Example 4.12: for any functor $F\colon \mathrm{CAlg} \to \mathrm{Cat}$ that satisfies the latter property, its quasi-coherent extension $X \mapsto F(X)$ also satisfies Proposition 4.41. Indeed, the quasi-coherent extension is given by the right Kan-extension, in particular commuting with limits. For example, there are analogous statements for quasi-coherent algebras, ideals, and radical ideals.

Corollary 4.43 (Zariski descent for functions)**.** *Let $(Y_i \to X)_{i\in I}$ be a family of maps of algebraic functors such that $\coprod_{i\in I} Y_i \to X$ is a Zariski-local epimorphism. Then the following diagram of rings is an equalizer:*

$$\mathcal{O}(X) \to \prod_{i\in I} \mathcal{O}(Y_i) \rightrightarrows \prod_{i,j\in I} \mathcal{O}(Y_i \times_X Y_j).$$

Proof. This follows from Proposition 4.41, using $\mathcal{O} : \mathrm{P}(\mathrm{CAlg}^{\mathrm{op}}) \to \mathrm{CAlg}$ is a functor with values in an algebraic category over Set, so the only the first two stages of the limit are necessary. □

For the following examples, recall the notion of I-local module and the associated localization functor $\mathrm{L}_I\colon \mathrm{Mod}_A \to \mathrm{Mod}_A^{I-\mathrm{loc}}$ (see Definition 3.83).

Example 4.44 (Modules over nonvanishing loci)*.* Let A be a ring and $I \subseteq A$ a subset. Then the map $\coprod_{f\in I} \mathrm{Spec}(A_f) \to \mathrm{D}(I)$ is a Zariski-local epimorphism. By Proposition 4.41, there is a limit diagram

$$\mathrm{Mod}_{\mathrm{D}(I)} \to \prod_{f\in I} \mathrm{Mod}_{A_f} \rightrightarrows \prod_{f,g\in I} \mathrm{Mod}_{A_{fg}} \;\substack{\rightarrow\\ \rightarrow}\; \prod_{f,g,h\in I} \mathrm{Mod}_{A_{fgh}}.$$

In particular, the pullback functor $\mathrm{Mod}_A \to \mathrm{Mod}_{\mathrm{D}(I)}$ factors through the localization functor L_I,using Remark 3.85.

Example 4.45 (Modules over Proj)*.* Let A be an $\mathbf{N}$-graded ring.

(a) Let $I \subseteq A_+$ be a homogeneous subset such that $A_+ \subseteq \sqrt{(I)}$. Then the map

$$\coprod_{f\in I} \mathrm{Spec}(A_{(f)}) \to \mathrm{Proj}(A)$$

is a Zariski-local epimorphism. Proposition 4.41 gives the limit diagram

$$\mathrm{Mod}_{\mathrm{Proj}(A)} \to \prod_{f\in I} \mathrm{Mod}_{A_{(f)}} \rightrightarrows \prod_{f,g\in I} \mathrm{Mod}_{A_{(fg)}} \;\substack{\rightarrow\\ \rightarrow}\; \prod_{f,g,h\in I} \mathrm{Mod}_{A_{(fgh)}}.$$

(b) Consider $\mathrm{D}(A_1) \subseteq \mathrm{Spec}(A)$. By Remark 3.71, the canonical map $\mathrm{D}(A_1) \to \mathrm{Proj}_1(A)$ is a Zariski-local epimorphism, and it induces a monomorphism $\mathrm{D}(A_1)/\mathbf{G}_m \hookrightarrow \mathrm{Proj}_1(A)$. Hence, the n-fold fiber product

of $\mathrm{D}(A_1)$ over $\mathrm{Proj}_1(A)$ is the same as n-fold fiber product over $\mathrm{D}(A_1)/\mathbf{G}_m$, which is $\mathrm{D}(A_1) \times \mathbf{G}_m^{n-1}$, since the action of $\mathbf{G}_m$ on $\mathrm{D}(A_1)$ is free. By Proposition 4.41, we get a limit diagram of the form

$$\mathrm{Mod}_{\mathrm{Proj}_1(A)} \to \mathrm{Mod}_{\mathrm{D}(A_1)} \rightrightarrows \mathrm{Mod}_{\mathrm{D}(A_1)\times\mathbf{G}_m} \mathrel{\substack{\rightarrow\\[-1ex]\rightarrow\\[-1ex]\rightarrow}} \mathrm{Mod}_{\mathrm{D}(A_1)\times\mathbf{G}_m\times\mathbf{G}_m}.$$

Example 4.46 (Modules over $\mathbf{P}^1$). In case $\mathbf{P}^1_k = \mathrm{Proj}(k[x,y])$, Example 4.45 (a) applies with the two element set $I = \{x, y\}$. We have $k[x,y]_{(x)} = k[u]$ with $u = y/x$, $k[x,y]_{(y)} = k[v]$ with $v = x/y$, and $k[x,y]_{(xy)} = k[t^{\pm 1}]$ with $t = y/x$. Hence, the limit diagram may be rewritten as a cartesian square

$$\begin{array}{ccc} \mathrm{Mod}_{\mathbf{P}^1_k} & \longrightarrow & \mathrm{Mod}_{k[u]} \\ \downarrow & & \downarrow{\scriptstyle u\mapsto t} \\ \mathrm{Mod}_{k[v]} & \xrightarrow{v\mapsto t^{-1}} & \mathrm{Mod}_{k[t^{\pm 1}]}. \end{array}$$

A quasi-coherent module over $\mathbf{P}^1_k$ is therefore a triple (M, N, α), where M is a $k[u]$-module, N is a $k[v]$-module, and $\varphi\colon M[u^{-1}] \xrightarrow{\sim} N[v^{-1}]$ is an isomorphism to $k[t^{\pm 1}]$-modules (i.e., an isomorphism of k-module such that the action of u on $M[u^{-1}]$ is *inverse* to the action of v on $N[v^{-1}]$).

Example 4.47 (Modules over the affine line with doubled origin). Let X be the affine line with doubled origin over k, which is the algebraic k-functor defined by

$$X(R) = \{(f, e) \mid f \in R,\ e \in R/(f),\ \text{and } e^2 = e\}.$$

Let $X_0 \subseteq X$ and $X_1 \subseteq X$ be the loci where $e = 0$ and $e = 1$, respectively. One can check that:

- X_0 and X_1 are open subfunctors of X such that $X_0 \sqcup X_1 \to X$ is a Zariski-local epimorphism;
- The map $X \to \mathbf{A}^1_k$, $(f, e) \mapsto f$, restricts to isomorphism

$$X_0 \xrightarrow{\sim} \mathbf{A}^1_k, \quad X_1 \xrightarrow{\sim} \mathbf{A}^1_k, \quad \text{and} \quad X_0 \cap X_1 \xrightarrow{\sim} \mathbf{G}_{m,k}.$$

Indeed, for $\mathrm{Spec}(R) \to X$ representing (f, e), we have $X_0 \times_X \mathrm{Spec}(R) = \mathrm{D}(1-e)$, while $X_1 \times_X \mathrm{Spec}(R) = \mathrm{D}(e)$. That X_0 and X_1 define a Zariski-local epimorphism follows then directly from $e + (1 - e) = 1$. That the intersection is $\mathbf{G}_{m,k}$ follows from the fact, that if $\mathrm{Spec}(R) \to X$ represents (f, e), then $0 = e = 1 \in R/(f)$, so that $R/(f) = 0$, that is $f \in R^\times$.

By Proposition 4.41, we obtain a cartesian square

$$\begin{array}{ccc} \mathrm{Mod}_X & \longrightarrow & \mathrm{Mod}_{k[x]} \\ \downarrow & & \downarrow{\scriptstyle x\mapsto x} \\ \mathrm{Mod}_{k[x]} & \xrightarrow{x\mapsto x} & \mathrm{Mod}_{k[x^{\pm 1}]}. \end{array}$$

A quasi-coherent module over X is therefore a triple (M, N, α), where M is a $k[x]$-module, N is a $k[x]$-module, and $\alpha\colon M[x^{-1}] \xrightarrow{\sim} N[x^{-1}]$ is an isomorphism of $k[x^{\pm 1}]$-modules (or equivalently of $k[x]$-modules).

Example 4.48 (Modules over Grassmannian). Let M be a k-module and let $n \in \mathbf{N}$. Let $\mathrm{St}_n(M)$ be the algebraic k-functor given by

$$\mathrm{St}_n(M)(R) = \{\text{surjective } R\text{-linear maps } M \otimes_k R \to R^n\},$$

called the *Stiefel scheme* of M. By Proposition 3.53, this is an open subfunctor of the affine space $\mathbf{A}(M^n)$. There is an obvious map $\mathrm{St}_n(M) \to \mathrm{Gr}_n(M)$, given section wise by sending a map $M \otimes_k R \twoheadrightarrow R^n$ to its coimage. This map is a Zariski-local epimorphism, since vector spaces are Zariski-locally free. Moreover, there is a free action of the affine group scheme GL_n on St_n given by postcomposition with a R-linear isomorphism $R^n \to R^n$, such that $\mathrm{St}_n(M) \to \mathrm{Gr}_n(M)$ is GL_n-invariant and induces a monomorphism $\mathrm{St}_n(M)/\mathrm{GL}_n \hookrightarrow \mathrm{Gr}_n(M)$; in the case $n = 1$, this recovers the known monomorphism $(\mathbf{A}(M) \setminus \{0\})/\mathbf{G}_m \hookrightarrow \mathbf{P}(M)$. As in Example 4.45, we then obtain a limit diagram of the form

$$\mathrm{Mod}_{\mathrm{Gr}_n(M)} \to \mathrm{Mod}_{\mathrm{St}_n(M)} \rightrightarrows \mathrm{Mod}_{\mathrm{St}_n(M)\times\mathrm{GL}_n} \mathrel{\substack{\rightarrow\\[-1ex]\rightarrow\\[-1ex]\rightarrow}} \mathrm{Mod}_{\mathrm{St}_n(M)\times\mathrm{GL}_n\times\mathrm{GL}_n},$$

where each term can be further expanded using Example 4.44.

Using Example 4.44, we can identify quasi-coherent modules over quasi-affine schemes:

Theorem 4.49 (Quasi-coherent modules over quasi-affine schemes)**.** *Let A be a ring and let $I \subseteq A$ be a finite subset with nonvanishing locus $\mathrm{D}(I) \subseteq \mathrm{Spec}(A)$. Then the pullback functor $\mathrm{Mod}_A \to \mathrm{Mod}_{\mathrm{D}(I)}$ induces an equivalence of categories*

$$\mathrm{Mod}_A^{I\text{-loc}} \xrightarrow{\sim} \mathrm{Mod}_{\mathrm{D}(I)}\,.$$

Proof. By Example 4.44, there exists a unique conservative functor such that the following diagram commutes:

$$\begin{array}{ccc} \mathrm{Mod}_A & \longrightarrow & \mathrm{Mod}_{\mathrm{D}(I)} \\ {\scriptstyle \mathrm{L}_I}\downarrow & \nearrow & \\ \mathrm{Mod}_A^{I\text{-loc}} & & \end{array}$$

Furthermore, this functor admits a right adjoint, see [HM24, Lemma D.4.7]. It suffices to show that the right adjoint is fully faithful, i.e. the canonical map

$$\lim\left(\prod_{f\in I} M(f) \rightrightarrows \prod_{f,g\in I} M(f)_g\right) \otimes_A A_h \to M(h)$$

is an isomorphism, for every $(M(f), \alpha_{fg})_{f,g\in I}$ and $h \in I$. This follows from $(-)_h$ being left exact and Zariski-descent applied to the ring A_h. □

this seems like an overkill, maybe prove it directly for the 1-categorical case

Corollary 4.50 (Affine completion of quasi-affine schemes)**.** *Let X be a quasi-affine scheme. Then the canonical map $X \to \mathrm{Spec}(\mathcal{O}(X))$ is an open immersion defined by a finitely generated radical ideal.*

Construction 4.51 (Quasi-coherent modules associated with a $\mathbf{Z}$-graded module)**.** Let A be an $\mathbf{N}$-graded ring. We define a functor

$$\{\mathbf{Z}\text{-graded } A\text{-module}\} \to \mathrm{Mod}_{\mathrm{Proj}(A)}, \quad M \mapsto \tilde{M},$$

as follows. Let M be a $\mathbf{Z}$-graded A-module and let $x\colon \mathrm{Spec}(R) \to \mathrm{Proj}(A)$ classify a quotient $\mathbf{N}$-graded algebra $\varphi\colon A^{(d)} \otimes_{A_0} R \twoheadrightarrow \mathrm{Sym}_R(L)$ for some $d \geq 1$ (if A is generated by $A_{\leq 1}$, we can take $d = 1$). Let $\tilde{\varphi}$ be the induced map of $\mathbf{Z}$-graded rings $A^{(d)} \to \bigoplus_{n\in\mathbf{Z}} L^{\otimes n}$. We set

$$\tilde{M}(x) = \left(\tilde{\varphi}^*(M^{(d)})\right)_0 = \left(M^{(d)} \otimes_{A^{(d)}} \left(\bigoplus_{n\in\mathbf{Z}} L^{\otimes n}\right)\right)_0 \in \mathrm{Mod}_R\,.$$

Note that for every $d \geq 1$ and $f \in A_d$, the pullback functor $\mathrm{Mod}_{\mathrm{Proj}(A)} \to \mathrm{Mod}_{A(f)}$ sends $\tilde{M}$ to $M_{(f)}$. In particular, by Example 4.45 (a), the functor $M \mapsto \tilde{M}$ factors through the localization functor L_{A_+}.

Using either (a) or (b) of Example 4.45, we can make the following computation:

Theorem 4.52 (Quasi-coherent modules over quasi-projective schemes)**.** *Let A be an $\mathbf{N}$-graded ring generated by $A_{\leq 1}$ and $I \subseteq A_1$ a finite subset with nonvanishing locus $\mathrm{D}(I) \subseteq \mathrm{Proj}(A)$. Then the functor $M \mapsto \tilde{M}$ induces an equivalence*

$$\{I\text{-local } \mathbf{Z}\text{-graded } A\text{-modules}\} \xrightarrow{\sim} \mathrm{Mod}_{\mathrm{D}(I)}\,.$$

In particular, if A is a finitely generated A_0-algebra, then

$$\{A_1\text{-local } \mathbf{Z}\text{-graded } A\text{-modules}\} \xrightarrow{\sim} \mathrm{Mod}_{\mathrm{Proj}(A)}$$

Proof. The proof is similar to that of Theorem 4.49. □

4.6. **Serre twists.** Let M be a $\mathbf{Z}$-graded module over a $\mathbf{Z}$-graded ring A and let $d \in \mathbf{Z}$. Recall from Notation 3.36 that $M(d)$ denotes the $\mathbf{Z}$-graded A-module with $M(d)_n = M_{n+d}$.

Definition 4.53 (Serre twists)**.** Let A be an $\mathbf{N}$-graded ring and let $d \in \mathbf{Z}$. We denote by $\mathcal{O}(d)$ the quasi-coherent module over $\mathrm{Proj}(A)$ associated with the $\mathbf{Z}$-graded A-module $A(d)$ (see Construction 4.51). Given $M \in \mathrm{Mod}_{\mathrm{Proj}(A)}$, the d-th *Serre twists* of M is the quasi-coherent module over $\mathrm{Proj}(A)$ given by

$$M(d) := M \otimes \mathcal{O}(d).$$

Remark 4.54. If A is generated by $A_{\leq 1}$, then an R-point $x \in \mathrm{Proj}(A)(R)$ is a quotient line $A_1 \otimes_{A_0} R \twoheadrightarrow L$ satisfying some condition. Unraveling the definition, we see that $\mathcal{O}(d)(x) = L^{\otimes d}$.

Example 4.55 (Serre twists on $\mathbf{P}^1$)*.* Over $\mathbf{P}^1$, there is a short exact sequence of quasi-coherent modules

$$0 \to \mathcal{O}(-1) \to \mathcal{O}^{\otimes 2} \to \mathcal{O}(1) \to 0.$$

Indeed, for any R-point $\mathrm{Spec}(R) \to \mathbf{P}^1$, classifying a quotient line $(a, b)\colon R^2 \twoheadrightarrow L$, the kernel of (a, b) is identified with $L^\vee = L^{-1}$ via the map $(b^\vee, -a^\vee)\colon L^{-1} \to R^2$.

Proposition 4.56 (Properties of Serre twists)**.** *Let A be an $\mathbf{N}$-graded ring generated by $A_{\leq 1}$.*

(a) For any $d \in \mathbf{Z}$, $\mathcal{O}(d)$ is a line bundle over $\mathrm{Proj}(A)$.

(b) For any $\mathbf{Z}$-graded A-module M and $d \in \mathbf{Z}$, there is an isomorphism $\tilde{M}(d) \xrightarrow{\sim} \widetilde{M(d)}$. In particular, for all $d, e \in \mathbf{Z}$, there are isomorphisms

$$\mathcal{O}(d) \otimes \mathcal{O}(e) \xrightarrow{\sim} \mathcal{O}(d+e) \quad \text{and} \quad \mathcal{O}(1)^{\otimes d} \xrightarrow{\sim} \mathcal{O}(d),$$

so that $\bigoplus_{d \in \mathbf{Z}} \mathcal{O}(d)$ has a structure of quasi-coherent $\mathbf{Z}$-graded algebra over $\mathrm{Proj}(A)$.

(c) There is a canonical map of $\mathbf{Z}$-graded rings

$$\mathrm{L}_{A_1} A \to \bigoplus_{d \in \mathbf{Z}} \Gamma(\mathrm{Proj}(A), \mathcal{O}(d)),$$

and, for any $\mathbf{Z}$-graded A-module M, a canonical map of $\mathbf{Z}$-graded A-modules

$$\mathrm{L}_{A_1} M \to \bigoplus_{d \in \mathbf{Z}} \Gamma(\mathrm{Proj}(A), \tilde{M}(d)),$$

which are isomorphisms if A is finitely generated as an A_0-algebra.

Proof. Let $x\colon \mathrm{Spec}(R) \to \mathrm{Proj}(A)$ classify a quotient $\mathbf{N}$-graded algebra $\varphi\colon A \otimes_{A_0} R \twoheadrightarrow \mathrm{Sym}_R(L)$. This induces a map of $\mathbf{Z}$-graded rings $\tilde{\varphi}\colon A \to \bigoplus_{n \in \mathbf{Z}} L^{\otimes n}$.

(a) This is exactly the statement of Remark 4.54. which is a R-line.

(b) Unwinding the constructions, we have

$$\tilde{M}(d)(x) \simeq \tilde{M}(x) \otimes_R \mathcal{O}(d)(x) \simeq \tilde{\varphi}^*(M)_0 \otimes_R L^{\otimes d} \simeq \tilde{\varphi}^*(M(d))_0 \simeq \widetilde{M(d)}(x),$$

since tensor product commutes with grade-shift and $(-)_0$.

(c) By Construction 4.51, we have an map induced by $\widetilde{(-)}$

$$\mathrm{Map}_{\mathrm{Mod}_A^{A_1\text{-loc},\mathbf{Z}}}(\mathrm{L}_{A_1} A, \mathrm{L}_{A_1} A(d)) \to \mathrm{Map}_{\mathrm{Proj}(A)}(\mathcal{O}_{\mathrm{Proj}(A)}, \mathcal{O}(d)).$$

The right hand side is $\Gamma(\mathrm{Proj}(A), \mathcal{O}(d))$, by definition. The left hand side can be computed as

$$\mathrm{Map}_{\mathrm{Mod}_A^{A_1\text{-loc},\mathbf{Z}}}(\mathrm{L}_{A_1} A, \mathrm{L}_{A_1} A(d)) \sim \mathrm{Map}_{\mathrm{Mod}_A^{\mathbf{Z}}}(A, \mathrm{L}_{A_1} A(d)) \simeq \mathrm{Map}_{\mathrm{Mod}_A^{\mathbf{Z}}}(A(-d), \mathrm{L}_{A_1} A) \simeq (\mathrm{L}_{A_1} A)_d.$$

That is, we obtain for all d a map

$$(\mathrm{L}_{A_1} A)_d \to \Gamma(\mathrm{Proj}(A), \mathcal{O}(d)).$$

Taking the sum over all d we obtain our desired map. Since the functor $\widetilde{(-)}$ is *symmetric monoidal*, this map descends to a map between $\mathbf{Z}$-graded ring. If A is finitely generated as an A_0-algeba, this map is furthermore an isomorphism by Theorem 4.52.

The statement for $\mathbf{Z}$-graded A-modules is analogous.

□

Remark 4.57. If A is an arbitrary $\mathbf{N}$-graded ring and M a $\mathbf{Z}$-graded A-module, there are still canonical maps

$$\tilde{M}(d) \to \widetilde{M(d)} \quad \text{and} \quad (\mathrm{L}_{A_+}M)_0 \to \Gamma(\mathrm{Proj}(A), \tilde{M})$$

but they need not be isomorphisms. If A is generated over A_0 by homogeneous elements whose degrees divide d, then $\mathcal{O}$ is a line bundle over $\mathrm{Proj}(A)$ and the first map is an isomorphism.

Definition 4.58 (Tautological line bundle)**.** Let A be an $\mathbf{N}$-graded ring generated by $A_{\leq 1}$. The line bundle $\mathcal{O}(1)$ over $\mathrm{Proj}(A)$ is called the *tautological line bundle*.

Example 4.59. If $A = k[x_0, \ldots, x_n]$, then

$$\mathrm{L}_{A_1}A = \begin{cases} k[x_0^{\pm 1}] & n = 0 \\ A & n \geq 1 \end{cases}$$

(this is exactly the computation of $\mathcal{O}(\mathbf{A}^{n+1}_k \setminus \{0\})$, see Example 2.78). Hence, $\Gamma(\mathbf{P}^n_k, \mathcal{O}(d)) \simeq k[x_0, \ldots, x_n]_d$ for all $d \geq 0$ and $\Gamma(\mathbf{P}^n_k, \mathcal{O}(d)) = 0$ for all $d < 0$ and $n \geq 1$. For $d = 0$, this says that $\mathcal{O}(\mathbf{P}^n_k) = k$, i.e., that every function on projective n-space is constant.

Example 4.60. Under the description of $\mathrm{Mod}_{\mathbf{P}^1_k}$ from Example 4.46, the quasi-coherent module $\mathcal{O}(d)$ over $\mathbf{P}^1_k$ is the triple $(k[u], k[v], \alpha)$, where $\alpha\colon k[x^{\pm 1}] \xrightarrow{\sim} k[v^{\pm 1}]$ is the $k[t^{\pm 1}]$-linear isomorphism sending 1 to v^d (hence u^n to v^{d-n}).

Example 4.61 (Tautological vector bundles over Grassmannian)*.* Let k be a ring, M a k-module, and $n \in \mathbf{N}$. The *Tautological vector bundle* $\mathcal{T}$ on $\mathrm{Gr}_n(M)$ is the rank n vector bundle such that, for any R-point $x\colon \mathrm{Spec}(R) \to \mathrm{Gr}_n(M)$ classifying a quotient space $M \otimes_k R \twoheadrightarrow V$, $\mathcal{T}(x) = V$. For $n = 1$, this recovers the tautological line bundle $\mathcal{O}(1)$ on $\mathbf{P}(M) = \mathrm{Proj}(\mathrm{Sym}_k(M))$. If $\mathfrak{o}\colon \mathrm{Gr}_n(M) \hookrightarrow \mathbf{P}(\Lambda^n_k(M))$ is the Plücker embedding (Example 3.108), then $\mathfrak{o}^*(\mathcal{O}(1)) \simeq \Lambda^n(\mathcal{T})$ in $\mathrm{Vect}_{\mathrm{Gr}_n(M)}$.

Remark 4.62. Let A be an $\mathbf{N}$-graded ring generated by $A_{\leq 1}$. If $f \in A$ is homogeneous of degree d, it defines by Proposition 4.56 a global section of $\mathcal{O}(d)$, or equivalently a map $\mathcal{O}(-d) \to \mathcal{O}$ in $\mathrm{Mod}_{\mathrm{Proj}(A)}$. Hence, any homogeneous subset $F \subseteq A$ induces a map $\bigoplus_{f \in F} \mathcal{O}(-d_f) \to \mathcal{O}$, where f has degree d_f. One can check that the loci $\mathrm{V}(F)$ and $\mathrm{D}(F)$ in Proj defined in Remark 3.74 coincide with the vanishing and nonvanishing loci of this map, in the sense of Remark 4.28. This is analoguous to Remark 3.51 in the affine case. Thus, the loci associated with maps of quasi-coherent modules unify the vanishing and nonvanishing loci in affine and projective geometry.

Remark 4.63 (Proj and line bundles)*.* Note that the tautological line bundle $\mathcal{O}(1)$ on $\mathrm{Proj}(A)$ depends on the $\mathbf{N}$-graded ring A and not just on the algebraic functor $\mathrm{Proj}(A)$. For example, under the isomorphism $\mathrm{Proj}(A) \simeq \mathrm{Proj}(A^{(d)})$ for $d \geq 1$, the line bundle $\mathcal{O}(1)$ on $\mathrm{Proj}(A^{(d)})$ corresponds to the line bundle $\mathcal{O}(d)$ on $\mathrm{Proj}(A)$. In fact, by (c) of Proposition 4.56, the data of this line bundle is exactly what allows us to recover (the A_1-localization of) A from the algebraic functor $\mathrm{Proj}(A)$.

We can make this more precise as follows. Let $\mathrm{CAlg}^{\mathbf{N},1} \subseteq \mathrm{CAlg}^{\mathbf{N},\mathrm{es}}$ be the full subcategory spanned by the $\mathbf{N}$-graded rings generated in degrees ≤ 1. Let LineBdl be the category of pairs (X, L), where X is an algebraic functor and $L \in \mathrm{Line}_X$, whose morphisms $(X', L') \to (X, L)$ are pairs (f, λ) with $f\colon X' \to X$ and $\lambda\colon L' \xrightarrow{\sim} f^*(L)$. Then there is a functor

$$(\mathrm{Proj}(-), \mathcal{O}(1))\colon (\mathrm{CAlg}^{\mathbf{N},1})^{\mathrm{op}} \to \mathrm{LineBdl},$$

which is fully faithful on the subcategory of $\mathbf{N}$-graded rings of the form $(\mathrm{L}_{A_1}A)_{\geq 0}$, where A is finitely generated over A_0.

5. Locales and topological spaces

5.1. **Pointless topology.** The starting point of *pointless topology* is the observation that most topological spaces (including for example all Hausdorff spaces) are determined by their posets of open subsets. Since posets are ubiquitous in mathematics, this leads to the appearance of topology in unexpected places. In this chapter, we will see that the poset of open subfunctors of any algebraic functor Z is isomorphic to the poset of open subsets of an associated topoogical space $|X|$, which we will describe explicitly when $X = \mathrm{Spec}(A)$ or $\mathrm{Proj}(A)$.

Notation 5.1. Let P be a poset. Given a family $(x_i)_{i\in I}$ in P, we denote by $\bigvee_{i\in I} x_i$ its supremum (least upper bound) and by $\bigwedge_{i\in I} x_i$ its infimum (greatest lower bound), if they exist.

Remark 5.2. Infima and suprema are related to limits and colimits as follows:

(a) If we view a poset as a category, suprema are colimits and infima are limits.
(b) The supremum of the empty family (i.e., the colimit of the empty diagram) is the smallest element (i.e., the initial object). Dually, the infimum of the empty family is the largest element.
(c) A poset admits all suprema if and only if it admits all infima. Indeed, the infimum of a family is the supremum of all elements below the family.

Definition 5.3 (Locale)**.** A *locale* is a poset $\mathcal{O}$ satisfying the following condition

(a) (Completeness) $\mathcal{O}$ admits all suprema (hence all infima).
(b) (Distributivity) For any $u \in \mathcal{O}$ and any family $(v_i)_{i\in I}$,
$$u \wedge (\bigvee_{i\in I} v_i) = \bigvee_{i\in I} (u \wedge v_i).$$

A morphism of locales $f\colon \mathcal{O}' \to \mathcal{O}$ is a morphism of posets $f^*\colon \mathcal{O} \to \mathcal{O}'$ (in the other direction!) that preserves colimits and finite limits. We denote by Loc the category of locales.

Example 5.4 (Totally ordered sets)*.* Any totally ordered set with suprema is a locale. For example, the posets $\mathbf{N} \cup \{+\infty\}$ and $\mathbf{R} \cup \{\pm\infty\}$ are locales.

Example 5.5 (The locale of a topological space)*.* Let T be a topological space. The poset $\mathrm{Open}(T)$ of open subsets of T is a locale:

- Given a family of open subsets $(U_i)_{i\in I}$, its supremum is the union $\bigcup_{i\in I} U_i$ and its infimum is the interior of the intersection $\bigcap_{i\in I} U_i$.
- The distributivity law follows from the set-theoretic distributive law and the fact that *finite* intersections of open subsets are open.

If $f\colon T \to S$ is a map of topological spaces, then $f^{-1}\colon \mathrm{Open}(S) \to \mathrm{Open}(T)$ is a morphism of posets that preserves colimits and finite limits. Hence, it is a morphism of locales $\mathrm{Open}(T) \to \mathrm{Open}(S)$. This defines a functor
$$\mathrm{Open}\colon \mathrm{Top} \to \mathrm{Loc}\,.$$

Remark 5.6. Locales are also called *frames* or *complete Heyting algebras*. However, these other names come with different notions of morphisms, hence describe different categories.

Remark 5.7 (Limits and colimits of locales)*.* We denote $\mathrm{Frm} \simeq \mathrm{Loc}^{\mathrm{op}}$ the category of *frames*.

(a) Let $(P_i)_{i\in I}$ be a family in Frm and $P = \lim_{i\in I} P_i$. Then colimits and finite limits are computed componentwise in P.
(b) The category Loc admits limits and colimits, and the inclusion $\mathrm{Loc} \hookrightarrow \mathrm{Pos}^{\mathrm{op}}$ preserves colimits.
(c) The initial object of Loc is $0 = \mathrm{Open}(\varnothing) = \{\varnothing\}$.
(d) The final object of Loc is $1 = \mathrm{Open}(*) = \{\varnothing \to *\}$.

Definition 5.8 (Points of a locale)**.** Let $\mathcal{O}$ be a locale. We define a topological space $\mathrm{Pt}(\mathcal{O})$ as follows:

- The elements of $\mathrm{Pt}(\mathcal{O})$ are the *points* of $\mathcal{O}$, i.e. the morphisms of locales $1 \to \mathcal{O}$.
- The open sets of $\mathrm{Pt}(\mathcal{O})$ are the subsets $\mathrm{Pt}(u) = \{p\colon 1 \to \mathcal{O} \mid p^*(u) = *\}$ for all $u \in \mathcal{O}$.

Given a morphism of locales $f\colon \mathcal{O}' \to \mathcal{O}$, the induced map $\mathrm{Pt}(\mathcal{O}') \to \mathrm{Pt}(\mathcal{O})$ is continous, since the preimage of $\mathrm{Pt}(u)$ is $\mathrm{Pt}(f^*(u))$. Hence, we have a functor

$$\mathrm{Pt}\colon \mathrm{Loc} \to \mathrm{Top}.$$

Remark 5.9 (Explicit description of points). Let P be a poset. A map of *sets* $f\colon P \to \{\varnothing \to *\}$ is determined by the subset $F = f^{-1}(*) \subseteq P$. It is a map of posets if and only if:

(a) F is upward-closed.

If this is the case and P has finite limits, then f preserves finite limits if and only if

(b) F is nonempty and $x \wedge y \in F$ whenver $x, y \in F$.

If moreover P is complete, then f preserves colimits if and only if:

(c) whenever $\bigvee_{i\in I} x_i \in F$, there existss $i \in I$ with $x_i \in F$.

Subsets $F \subset P$ satisfying (a) and (b) are called *filters* on P, and they are called *completely prime* if they also satisfy (c). Thus, we can identify the points of a locale $\mathcal{O}$ with the completely prime filters on $\mathcal{O}$.

Proposition 5.10 (The adjunction between topological spaces and locales)**.** *The functor*

$$\mathrm{Open}\colon \mathrm{Top} \to \mathrm{Loc}$$

is left adjoint to the functor $\mathrm{Pt}\colon \mathrm{Loc} \to \mathrm{Top}$*. This adjunction is idempotent, i.e., it restricts to an equivalence between the essential images of both functors.*

Proof. Let T be a topological space and L a locale. We will show that there exists an equivalence:

$$\mathrm{Map}_{\mathrm{Top}}(T, \mathrm{Pt}(L)) \simeq \mathrm{Map}_{\mathrm{Loc}}(\mathrm{Open}(T), L).$$

Let $f\colon T \to \mathrm{Pt}(L)$ be a continuous map. We obtain a map $\mathrm{Open}(T) \to L$ between locales, given by $u \mapsto f^{-1}(\mathrm{Pt}(u))$. Conversely, let $g\colon \mathrm{Open}(T) \to L$ be a map between locales. We obtain a map $T \to \mathrm{Pt}(L)$, given by $t \mapsto g \circ \mathrm{Open}(t)$. It is left to the readers to verify that they are inverses to each other.

By a general fact in category theory, in order to show idempotence, we only need to show that the counit map

$$\mathrm{Open}(\mathrm{Pt}(\mathrm{Open}(T))) \to \mathrm{Open}(T)$$

is an equivalence for every topological space T. In fact, the unit map

$$T \to \mathrm{Pt}(\mathrm{Open}(T)), \quad t \mapsto (t^{-1}\colon \mathrm{Open}(T) \to \mathrm{Open}(*))$$

induces an inverse to the counit, by the definition of the topology on $\mathrm{Pt}(\mathrm{Open}(T))$. □

Definition 5.11 (Sober spaces and spatial locales)**.** We have the following definitions:

(a) A topological space T is *sober*, if it lies in the essential image of $\mathrm{Pt}\colon \mathrm{Loc} \to \mathrm{Top}$ or equivalently if the unit map $T \to \mathrm{Pt}(\mathrm{Open}(T))$ is an isomorphism.

(b) A locale $\mathcal{O}$ is *spatial* if it lies in the essential image of $\mathrm{Open}\colon \mathrm{Top} \to \mathrm{Loc}$, or equivalently if the counit map $\mathrm{Open}(\mathrm{Pt}(\mathcal{O})) \to \mathcal{O}$ is an isomorphism.

Remark 5.12. By Proposition 5.10, the adjunction $\mathrm{Open} \dashv \mathrm{Pt}$ restricts to an equivalence between the category $\mathrm{Top}^{\mathrm{sob}}$ of sober topological spaces and the category of $\mathrm{Loc}^{\mathrm{spa}}$ of spatial locales. Moreover, the inclusion $\mathrm{Top}^{\mathrm{sob}} \hookrightarrow \mathrm{Top}$ has a left adjoint given by $\mathrm{Pt} \circ \mathrm{Open}$, called *soberification*, and the inclusion $\mathrm{Loc}^{\mathrm{spa}} \hookrightarrow \mathrm{Loc}$ has a right adjoint given by $\mathrm{Open} \circ \mathrm{Pt}$ called *spatialization*. In partiuclar, limits of sober spaces are sober and colimits of spatial locales are spatial.

Proposition 5.13 (Characterization of sober spaces)**.** *Let T be a topological space. Then the points of the locale $\mathrm{Open}(T)$ can be identified with the irreducible closed subsets of T, and the unit map $T \to \mathrm{Pt}(\mathrm{Open}(T))$ sends t to the closure of $\{t\}$. Hence, a topological space T is sober if and only if the map*

$$\{\textit{points of } T\} \to \{\textit{irreducible closed subsets of } T\}, \quad t \mapsto \overline{\{t\}},$$

is a bijection, i.e., if and only if every irreducible closed subsets of T has a unique generic point.

Proof. We will construct a bijection

$$\{\text{closed irreducible subsets of } T\} \xrightarrow{\sim} \mathrm{Pt}(\mathrm{Open}(T)).$$

Let Z be a closed irreducible subset of T. Then the subset $\{U \mid U \cap Z \neq \varnothing\}$ lies in $\mathrm{Pt}(\mathrm{Open})$. Conversely, Let $F \in \mathrm{Pt}(\mathrm{Open}(T))$. We may define a closed irreducible subset of T by $Z = T \subseteq W$, where $W = \bigcup_{U \in \mathrm{Open}(T) \setminus F} U$. It is left to the readers to verify that this defines a bijection. □

Remark 5.14. We have the following properties:

(a) Every Hausdorff space is sober, as its irreducible subsets are singletons.
(b) Being sober is a local property: if T admits an open covering by sober spaces, then T is sober.

5.2. **Locales of radical ideals.** Let R be a ring. Recall that the poset of open subfunctors of $\mathrm{Spec}(R)$ is isomorphic to the poset Rad_R of radical ideals in R (Proposition 2.66). We claim that is is a locale. The supremum of a family of radical ideals $(K_i)_{i \in I}$ is the radical of $\sum_{i \in I} K_i$. The distributivity law reads

$$K \cap \sqrt{\sum_{i \in I} L_i} = \sqrt{\sum_{i \in I} (K \cap L_I)},$$

and it follows from the following facts: we have $K \cap L = \sqrt{KL}$ for any radical ideals K and L, and the *product* of ideals distributes over sums. By contrast, the poset Idl_R of all ideals is usually not a locale, since the intersection of ideals does not distribute over sums.

For any ring map $R \to S$, the base change map $\mathrm{Idl}_R \to \mathrm{Idl}_S$, $I \mapsto IS$, preserves products and sums of ideals. It follows that the base change map $\mathrm{Rad}_R \to \mathrm{Rad}_S$, $I \mapsto \sqrt{IS}$, preserves finite intersections and suprema, hence is a morphism of locales $\mathrm{Rad}_S \to \mathrm{Rad}_R$.

We now computes the points of the locale Rad_R.

Definition 5.15 (Prime spectrum)**.** Let R be a ring. The *prime spectrum* of R is the topological space $\mathrm{Prim}(R)$ defined as follows:

- The elements of $\mathrm{Prim}(R)$ are the prime ideals of R.
- The open sets of $\mathrm{Prim}(R)$ are the subsets $\mathrm{Prim}(I) = \{\mathfrak{p} \mid I \nsubseteq \mathfrak{p}\}$ for all $I \subseteq R$.

Since prime ideals are radical, $\mathrm{Prim}(I)$ depends only on the radical ideal $\sqrt{(I)}$ generated by I.

Proposition 5.16 (The locale of a ring)**.** *Let R be a ring.*

(a) There is a homeomorphism

$$\mathrm{Prim}(R) \xrightarrow{\sim} \mathrm{Pt}(\mathrm{Rad}_R), \quad \mathfrak{p} \mapsto \{I \in \mathrm{Rad}_R \mid I \nsubseteq \mathfrak{p}\},$$

under which the open subset $\mathrm{Pt}(I)$ corresponds to the open subset $\mathrm{Prim}(I)$.

(b) The locale Rad_R is spatial, i.e., the map from (a) induces an isomorphism $\mathrm{Open}(\mathrm{Prim}(R)) \xrightarrow{\sim} \mathrm{Rad}_R$.

(c) The topological space $\mathrm{Prim}(R)$ is spectral, i.e., it is sober and its quasi-compact open subsets form a basis of the topology that is closed under finite intersections.

Proof. We denote $F_{\mathfrak{p}} = \{I \in \mathrm{Rad}_R \mid I \nsubseteq \mathfrak{p}\}$.

(a) Let $\mathfrak{p}$ be a prime ideal. It is clear that $F_{\mathfrak{p}}$ is a completely prime filter on $\mathrm{Rad}_{\mathfrak{p}}$. We will now show that this map is bijective. In fact, we can define a inverse map by

$$\mathrm{Pt}(\mathrm{Rad}_R) \to \mathrm{Prim}(R), \quad F \mapsto \beta_F,$$

where $\beta_F = \max(\mathrm{Rad}_R \setminus F) = \bigcup_{I \in \mathrm{Rad}_R \setminus F}$. It is now left to the readers to show that this defines an iverse.

(b) There exists a map

$$\mathrm{Rad}_R \to \mathrm{Open}(\mathrm{Prim}(R)), \quad I \mapsto \mathrm{Prim}(I).$$

This defines an inverse.

(c) The open subsets $\{\mathrm{Prim}(f)\}_{f\in R}$ consists of a basis of the topological space $\mathrm{Prim}(R)$. In fact, we have $\mathrm{Prim}(I) = \bigcup_{f\in I}\mathrm{Prim}(f)$. Since $\mathrm{Prim}(I) = \mathrm{Prim}(R)$ implies that $\sqrt{(I)} = R$, $\mathrm{Prim}(R)$ is quasi-compact. Furthermore, $\mathrm{Prim}(f) = \mathrm{Prim}(R_f)$ is also quasi-compact. Finally, we have $\mathrm{Prim}(f)\cap\mathrm{Prim}(g) = \mathrm{Prim}(fg)$ and $\mathrm{Prim}(1) = \mathrm{Prim}(R)$.

□

Remark 5.17. One can show that, conversely, every spectral space is homeomorphic to $\mathrm{Prim}(R)$ for some ring R (which is however far from unique).

Let now A be an $\mathbf{N}$-graded ring, and let hRad_A be the poset of saturated radical homogeneous ideals in A. By Proposition 3.90 (and Remark 3.91), the poset hRad_A is isomorphic to the poset of open subfunctors of $\mathrm{Proj}(A)$. This poset is also a locale: the distributivity law follows from the one in Rad_A and the fact that saturation preserves finite intersections. We now compute the points of hRad_A.

Definition 5.18 (Homogeneous prime spectrum)**.** Let A be an $\mathbf{N}$-graded ring. The *homogeneous prime spectrum* of A is the topological space $\mathrm{hPrim}(A)$ defined as follows:

- The elements of $\mathrm{hPrim}(A)$ are the saturated homogeneous prime ideals of A, or equivalently the homogeneous prime ideals of A that do not conatain A_+ (see (d) of Remark 3.81).
- The open sets of $\mathrm{hPrim}(R)$ are the subsets $\mathrm{hPrim}(I) = \{\mathfrak{p} \mid I \not\subseteq \mathfrak{p}\}$ for all homogeneous subsets $I \subseteq A$.

Note that $\mathrm{hPrim}(I)$ depends only on the saturated radical ideal $\sqrt{(I)}^{\mathrm{sat}}$ generated by I.

Proposition 5.19 (The locale of an $\mathbf{N}$-graded ring)**.** *Let A be an $\mathbf{N}$-graded ring.*

(a) There is a homeomorphism

$$\mathrm{hPrim}(A) \xrightarrow{\sim} \mathrm{Pt}(\mathrm{hRad}_A), \quad \mathfrak{p} \mapsto \{I \in \mathrm{hRad}_A \mid I \not\subseteq \mathfrak{p}\},$$

under which the open subset $\mathrm{Pt}(I)$ corresponds to the open subset $\mathrm{hPrim}(I)$.

(b) The locale hRad_A is spatial, i.e., the map from (a) induces an isomorphism

$$\mathrm{Open}(\mathrm{hPrim}(A)) \xrightarrow{\sim} \mathrm{hRad}_A\,.$$

(c) The topological space $\mathrm{hPrim}(A)$ is sober and its quasi-compact open subsets form a basis of the topology that is closed under binary intersections. If A is finitely generated as an A_0-algebra, then $\mathrm{hPrim}(A)$ is spectral.

Proof. The proof of (a) and (b) are similar to the case of $\mathrm{Spec}(R)$. We will only prove (c).

Let $f \in A_+$. We have

$$\mathrm{hPrim}(f) = \{\mathfrak{p} \mid f \notin \mathfrak{p}\} \simeq \mathrm{Prim}(A_{(f)}).$$

This implies that $\mathrm{hPrim}(f)\cap\mathrm{hPrim}(g) = \mathrm{hPrim}(fg)$. Furthermore, if $A_+ \subseteq \sqrt{(f_1,\dots,f_n)}$. Then we have $\mathrm{hPrim}(A) = \bigcup_{i=1}^n \mathrm{hPrim}(f_i)$, hence $\mathrm{hPrim}(A)$ is quasi-compact. □

5.3. **The topological space of an algebraic functor.** LKet $\mathrm{Open}(X)$ denote the poset of open subfunctors of an algebraic functor X. Since the preimage of an open subfunctor is open, we have a functor

$$\mathrm{Open}\colon \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set}) \to \mathrm{Pos}^{\mathrm{op}}\,.$$

By Proposition 4.23, $\mathrm{Open}(X)$ is isomorphic to the poset Rad_X of quasi-coherent radical ideals over X, so that there is an isomorphism of posets

$$\mathrm{Open}(X) \xrightarrow{\sim} \lim_{\mathrm{Spec}(R)\to X} \mathrm{Open}(\mathrm{Spec}(R)).$$

In Section 5.2, we saw that the restriction of $\mathrm{Open}(X)$ to the subcategory $\mathrm{Aff} \subseteq \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})$ of affine schemes lands in the subcategory of locales. Combining these facts with Remark 5.7, we immediately deduce the following result:

Proposition 5.20. *The functor* $\mathrm{Open}\colon \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set}) \to \mathrm{Pos}^{\mathrm{op}}$ *lands in the subcategory* $\mathrm{Loc} \subseteq \mathrm{Pos}^{\mathrm{op}}$*, and the induced functor*

$$\mathrm{Open}\colon \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set}) \to \mathrm{Loc}$$

is colimit preserving.

Since colimits of spatial locales are spatial (Remark 5.12), we deduce:

Corollary 5.21. *For any algebraic functor* X*, the locale* $\mathrm{Open}(X)$ *is spatial.*

Remark 5.22. The final object in $\mathrm{Open}(X)$ is $\mathrm{D}(1) = X$ and the initial object is $\mathrm{D}(0) = \varnothing_X$, where $\varnothing_X(0) = X(0)$ and $\varnothing_X(R) = \varnothing$ for all $R \neq 0$. For any $U, V \in \mathrm{Open}(X)$, we have $U \wedge V = U \cap V$.

We now define an explicit topological space $|X|$ whose locale of open sets is $\mathrm{Open}(X)$. If $X = \mathrm{Spec}(R)$, the prime spectrum $\mathrm{Prim}(R)$ is the unique sober space with this property, by Proposition 5.16. On the other hand, by Remark 3.5, there is a bijection

$$\{\text{connected components of } \mathrm{Field}_R\} \xrightarrow{\sim} \mathrm{Prim}(R), \quad (\varphi\colon R \to k) \mapsto \ker(\varphi),$$

with inverse given by $\mathfrak{p} \mapsto \varkappa(\mathfrak{p})$.

Construction 5.23 (Underlying space of an algebraic functor)**.** Let X be an algebraic functor and let $\mathrm{Field}_X \subseteq \mathrm{El}(X)^{\mathrm{op}}$ be the full subcategory spanned by the field-valued points of X. We define a topological space $|X|$ as follows:

- The points of $|X|$ are the connected components of the category Field_X.
- A subset $U \subseteq |X|$ is open if and only if, for every ring R and every R-point $x\colon \mathrm{Spec}(R) \to X$, the preimage $|x|^{-1}(U) \subseteq \mathrm{Prim}(R)$ is open.

Proposition 5.24. *The functor*

$$\mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set}) \to \mathrm{Top}, \quad X \mapsto |X|,$$

is colimit-preserving. Hence, for any algebraic functor X*, there is a natural isomorphism of posets*

$$\mathrm{Open}(X) \xrightarrow{\sim} \mathrm{Open}(|X|), \quad U \mapsto |U|.$$

Proof. It suffices to show that for an algebraic functor X, the canonical map

$$\left(\operatorname*{colim}_{\mathrm{Spec}(R) \to X} \mathrm{Prim}(R)\right) \to |X|$$

is an homeomorphism. The functor $\mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set}) \to \mathrm{Top}$ is then left Kan extended from $\mathrm{CAlg}^{\mathrm{op}}$. Surjectivity is clear. Injectivity follows from following fact: let $k_0 \hookrightarrow k_1$ and $k_0 \hookrightarrow k_2$ be field homomorphisms. Consider the pushout

$$\begin{array}{ccc} k_0 & \longrightarrow & k_1 \\ \downarrow & & \downarrow \\ k_2 & \longrightarrow & k_1 \otimes_{k_0} k_2 \end{array}$$

in CAlg. Since $k_1 \otimes_{k_0} k_2 \neq 0$, it contains a maximal ideal. Therefore, there exists a field k_3 and a homomorphism $k_1 \otimes_{k_0} k_2 \to k_3$. □

Remark 5.25. We have the following observations:

(a) Since every field admits an algebraic closure field; such a k-point is sometimes called a *geometry point* of X, although the exact menaning of "geometric point" varies from sources to sources.
(b) If X is affine, then $|X|$ is sober. However, as colimits of sober spaces need not to be sober, $|X|$ is not sober in general. Its soberification is the space $\mathrm{Pt}(\mathrm{Open}(X))$.
(c) If $X = \mathrm{Proj}(A)$, then $|X|$ is sober as it is covered by the open subspaces $|\mathrm{Spec}(A_{(f)})|$, which are sober. Hence, the canonical map $|X| \to \mathrm{Pt}(\mathrm{Open}(X)) \simeq \mathrm{hPrim}(A)$ is homeomorphism.

Remark 5.26 (Points in algebraic geometry). We now have introduced two distinct notions of "points" of an algebraic functor X:

(a) If R is any ring, an *R-point* or *R-valued point* of X is an element of $X(R)$.
(b) A *point* of X usually refers to a point of the topological space $|X|$, which is an equivalence class of field-valued points of X.

For fixed R, the R-points of X only form a set $X(R)$, but for varying R they form a category, namely the category of elements $\mathrm{El}(X)$. On the other hand, the point of X in the sense of (b) form a topological space $|X|$. These two notions of points are related by the zigzag of functors

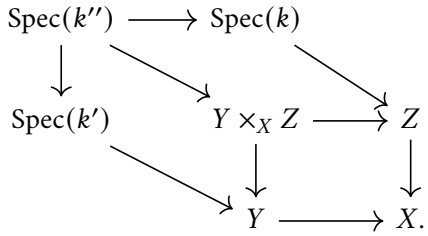
$$\mathrm{El}(X)^{\mathrm{op}} \hookleftarrow \mathrm{Field}_X \twoheadrightarrow |X|.$$

Lemma 5.27 (Underlying space of pullbacks). *Let $Y \to X \leftarrow Z$ be maps of algebraic functors. Then the canonical map*

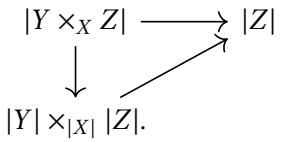
$$|Y \times_X Z| \to |Y| \times_{|X|} |Z|$$

is surjective. It is bijective if $Y \to X$ is a monomorphism.

Proof. Let $\mathrm{Spec}(k) \to Y$ and $\mathrm{Spec}(k') \to Z$ be points of Y and Z respectively, such that their image in X coincide. Similar to the proof of Proposition 5.24, there exists field homomorphisms $k'' \to k$ and $k \to k'$ such that the following diagram commutes



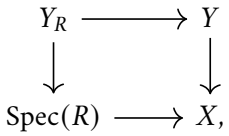
Assume that $Y \to X$ is a monomorphism. Then the diagonal map induces an isomorphism $Y \simeq Y \times_X Y$. Therefore, the diagonal map $|Y| \to |Y| \times_{|X|} |Y|$ is an surjective, by the first part of the statement. Hence, the map $|Y| \to |X|$ is a monomorphism. Furthermore, since monomorphisms are closed under base change, the map $Y \times_X Z \to Z$ is an monomorphism, hence $|Y \times_X Z \to Z$ is also injective. Observe that we have the following commutative square:



Therefore, the map $|Y \times_X Z| \to |Y| \times_{|X|} |Z|$ is injective, hence bijective. □

Proposition 5.28 (Underlying space of immersions). *Let X be an algebraic functor and let $Y \hookrightarrow X$ be an open immersion, a closed immersion, or an immersion. Then the induced map $|Y| \to |X|$ is an open embedding, a closed embedding, or a locally closed embedding, respectively.*

Proof. We will only prove the case for closed immersion, the other cases are left to the readers to verify. Let $Y \hookrightarrow X$ be a closed immersion. Then for every map $x\colon \mathrm{Spec}(R) \to X$, we have the following pullback square:



where $Y_R \simeq \operatorname{Spec}(R/I)$ for some ideal $I \subseteq R$. By Lemma 5.27, the following commutative square is a pullback square of sets:

$$\begin{array}{ccc} |Y_R| & \longrightarrow & |Y| \\ \downarrow & & \downarrow \\ |\operatorname{Spec}(R)| & \longrightarrow & |X|. \end{array}$$

It suffices to show that $Z \subseteq |Y|$ is closed in $|Y|$ if and only if it is closed in $|X|$. By the map $Y \to X$ being a monomorphism, and the definition of colimit topology, a subset $Z \subseteq |Y|$ is closed if and only if, for every map $x\colon \operatorname{Spec}(R) \to X$, its restriction Z_R is closed in $|Y_R|$. This is the case if and only if Z_R is closed in $|\operatorname{Spec}(R)|$, since the map $|Y_R| \to |\operatorname{Spec}(R)|$ is a closed immersion. Finally, this is equivalent to Z being closed in X, as a consequence of the definition of colimit topology. □

Remark 5.29 (Closed subfunctors vs. closed subspaces)*.* By Proposition 5.28, there is a map of posets

$$\operatorname{Closed}(X) \to \operatorname{Closed}(|X|), \quad Z \mapsto |Z|.$$

In contrast to Proposition 5.24, this map to almost never bijective, but it is surjective if $X = \operatorname{Spec}(A)$ or if $X = \operatorname{Proj}(A)$: a preimage of a closed subset of $|X|$ is given by the vanishing locus of the radical ideal in A corresponding to the open complement. In general, if $Z \subseteq X$ is a closed subfunctor with open complement U (Definition 4.24), then $|Z| = |X| \setminus |U|$.

Definition 5.30 (Topological properties of algebraic functors)**.** Let P be a property of topological spaces, such as *connected*, *locally connected*, *irreducible*, or *discrete*. We say that an algebraic functor X has property P if $|X|$ has property P[20].

Example 5.31. If $X = \operatorname{Spec}(R)$, so that $|X|$ is the prime spectrum $\operatorname{Prim}(R)$, we have the following standard results from commutative algebra:

(a) X is connected if and only if R has exactly two idempotent element (see Remark 5.43 below for a generalization of this statement to arbitrary algebraic functors).
(b) X is irreducible if and only if the nilradical $\sqrt{0} \subseteq R$ is prime.
(c) X is discrete if and only if R is artinian.

Definition 5.32 (Topological properties of maps of algebraic functors)**.** Let P be a property of maps of topological spaces, such as *surjective*, *injective*, *bijective*, *dominant*, *submersive*, *open*, *closed*, or *homeomorphism*[21] A map of algebraic functors $f\colon Y \to X$ is said to have property P if the map $|f|\colon |Y| \to |X|$ has property P. We say that f is *universally* P if for every $X' \to X$, the base change $Y \times_X X' \to X'$ has property P.

Proposition 5.33. *Let $f\colon Y \to X$ be a map of algebraic functors.*

(a) Let P be any of the following properties: surjective, injective, bijective, dominant, submersive, open, closed, or homeomorphism. Then f is universally P if and only if, for every ring R and every $\operatorname{Spec}(R) \to X$, the base change $Y \times_X \operatorname{Spec}(R) \to \operatorname{Spec}(R)$ has property P.
(b) f is universally surjective if and only if it is surjective.
(c) If f is a monomorphism, then it is universally injective.
(d) If f is a Zariski-local epimorphism, then it is universally submersive and in particular universally surjective.

Proof. (a) We will proof the case where P is submersive. The rest is left to the readers. For every map $x\colon \operatorname{Spec}(R) \to X$, we have the following pullback square:

$$\begin{array}{ccc} Y_R & \longrightarrow & Y \\ \downarrow{\scriptstyle f_R} & & \downarrow{\scriptstyle f} \\ \operatorname{Spec}(R) & \xrightarrow[x]{} & X. \end{array}$$

[20]Provided this does not conflict with any other definitions. For example, an affine scheme X is called *noetherian* if the ring $\mathcal{O}(X)$ is noetherian, which is strictly stronger than $|X|$ being noetherian; the latter property is then called *topologically noetherian*

[21]A continuous map $f\colon T \to S$ is *dominant* if $f(T)$ is dense in S, and it is *submersive*, if it is surjective and S had the quotient topology.

Then the map $Y \to X$ can be identified with the colimit

$$\operatorname*{colim}_{\operatorname{Spec}(R)\to X} Y_R \to \operatorname*{colim}_{\operatorname{Spec}(R)\to X} \operatorname{Spec}(R).$$

We will first show that f is surjective if and only if every f_R is surjective. One direction is clear, since surjectivity is closed under base change in a presheaf category. Assume that f_R is surjective for each R. Since the functor $|-|$ preserves colimits, we have

$$\operatorname*{colim}_{\operatorname{Spec}(R)\to X} |\operatorname{Spec}(R)| = X \quad \text{and} \quad \operatorname*{colim}_{\operatorname{Spec}(R)\to X} |Y_R| = Y.$$

Therefore, every $x \in |X|$ comes from some $|\operatorname{Spec}(R)|$, hence from $|Y_R|$, i.e. from Y.

Being submersive if closed under base change, hence we only need to show that f is submersive if every f_R is submersive. By the colimit topology on $|X|$, a subset $U \subseteq |X|$ is open if and only if the preimage of U is open in every $|\operatorname{Spec}(R)|$. This is the case if and only if the preimage of U is open in every $|Y_R|$, by assumption. But this is furthermore equivalent to the preimage of U being open in $|Y|$.

(b) This follows from Lemma 5.27 combined with surjectivity being closed under base change.

(c) This follows from the fact that pullback is closed under base change and Lemma 5.27.

(d) By (a), we may assume without losses of generality that $X = \operatorname{Spec}(R)$. Since f is a Zariski-local epimorphism, there exists elements $f_1, \dots, f_n \in R$ such that $(f_1, \dots, f_n) = R$ and $\operatorname{Prim}(R) = \bigcup_{i=1}^n \operatorname{Prim}_{R_{f_i}}$. In other words, for every $\operatorname{Spec}(R_{f_i})$, there exists a unique map such that the following diagram commutes:

$$\begin{array}{ccc} & & Y \\ & \overset{\exists}{\nearrow} & \downarrow f \\ \operatorname{Spec}(R_{f_i}) & \hookrightarrow & \operatorname{Spec}(R). \end{array}$$

Therefore, the map $|f|$ is surjectivity. Now we will show that the map is submersive. Let $U \subseteq \operatorname{Prim}(R)$ be a subset such that its preimage in $|Y|$ is open. This implies that its preimage $U \cap \operatorname{Prim}(R_{f_i})$ is open in $\operatorname{Prim}(R_{f_i})$ for every $i \in I$. Since $\operatorname{Prim}(R_{f_i})$ is open cover of $\operatorname{Prim}(R)$. This implies that U is open in $\operatorname{Prim}(R)$.

□

Example 5.34. With the exception of surjectivity, the properties listed in Equation (5.33) are not automatically universal, even for maps between affine schemes:

(a) The map $\operatorname{Spec}(\mathbf{C}) \to \operatorname{Spec}(\mathbf{R})$ is a homeomorphism, but it is not universally injective: its pullback along itself is $\operatorname{Spec}(\mathbf{C} \times \mathbf{C}) \to \operatorname{Spec}(\mathbf{C})$, which induces the map of topological spaces $* \sqcup * \to *$.

(b) If k is a field, then $|\operatorname{Spec}(k)| = *$ and hence any map $X \to \operatorname{Spec}(k)$ is closed. The map $\mathbf{A}^1_k \to \operatorname{Spec}(k)$ is however not universally closed: its pullback along itself is the projection $\mathbf{A}^2_k \to \mathbf{A}^1_k$, which is not closed as it restricts to an isomorphism $\mathrm{V}(xy - 1) \xrightarrow{\sim} \mathbf{A}^1_k \setminus \{0\}$.

(c) An example of an open morphism that is not universally open is

$$f\colon \operatorname{Spec}(k[t]) \to \operatorname{Spec}(k[x,y])/(y^2 - x^3 - x^2), \quad x \mapsto t^2 - 1, \quad y \mapsto t(t^2 - 1),$$

where k is a field of characteristic not 2, which is the resolution of singularities of the affine nodal cubic. The map $|f|$ is the quotient map $|\mathbf{A}^1_k| \twoheadrightarrow |\mathbf{A}^1_k|/(-1 \sim 1)$, which is open but whose pullback along itself is not open. By Lemma 5.27, this implies that the pullback of f along itself is not open.

Corollary 5.35 (Zariski codescent for the underlying space)**.** *Let $(Y_i \to X)_{i\in I}$ be a family of maps of algebraic functors such that $\coprod_{i\in I} Y_i \to X$ is Zariski-local epimorphism. Then the diagram*

$$\coprod_{i,j\in I} |Y_i \times_X Y_j| \rightrightarrows \coprod_{i\in I} |Y_i| \to |X|$$

is a coequalizer diagram of topological spaces.

Proof. By (d) of Equation (5.33), the map $\coprod_{i\in I} |Y_i| \to |X|$ is universally submersive. Therefore, the following is a coequalizer diagram:

$$\coprod_{i,j\in I} |Y_i| \times_X |Y_j| \rightrightarrows \coprod_{i\in I} |Y_i| \to |X|.$$

By Lemma 5.27, there exists an epimorphism $\coprod_{i,j\in I} |Y_i \times_X Y_j| \to \coprod_{i,j\in I} |Y_i| \times_X |Y_j|$. By checking the relations in the category of sets, we obtain the claim. □

Definition 5.36 (Residue field)**.** Leet X be an algebraic functor and let $x \in |X|$. Let $\mathrm{Field}_{(X,x)}$ be the full subcategory of Field_X spanned by the field-valued points hitting x. If the category $\mathrm{Field}_{(X,x)}$ has an initial object, we call it the *residue field* of X at x and denote it by $\kappa(x)$.

Let $f\colon Y \to X$ be a map of algebraic functors, let $y \in |Y|$ and let $x = f(y) \in |X|$. If both residue fields $\kappa(y)$ and $\kappa(x)$ exist, then the universal property of $\kappa(x)$ gives a map of fields $\kappa(x) \to \kappa(y)$, called the *residual field extension* of f at y.

Proposition 5.37 (Existence of residue fields)**.** *Let X be an algebraic functor, let $Y \hookrightarrow X$ be an immersion, and let $y \in |Y|$ be a point with image $x \in |X|$. Then the functor $\mathrm{Field}_{(Y,y)} \to \mathrm{Field}_{(X,x)}$ is an equivalence. In particular, the residue field of X at x exists if and only if the residue field of Y at y exists, in which case the residual field extension $\kappa(x) \to \kappa(y)$ is an isomorphism.*

Proof. Let $x\colon \mathrm{Spec}(k) \to X$ be a map. It suffices to solve the lifting problem:

$$\begin{array}{ccc} & & Y \\ & \swarrow & \downarrow \\ \mathrm{Spec}(k) & \xrightarrow[x]{} & X \end{array}$$

Since $Y \to X$ is an immersion, the map $Y \times_X \mathrm{Spec}(k) \to \mathrm{Spec}(k)$ is again an immersion. By Lemma 5.27, the map $|Y \times_X \mathrm{Spec}(k)| \to |Y| \times_{|X|} * \simeq *$ is surjective. Therefore, we have $Y \times_X \mathrm{Spec}(k) \neq \mathrm{Spec}(0)$ and hence we have an isomorphism $Y \times_X \mathrm{Spec}(k) \simeq \mathrm{Spec}(k)$, i.e., the lifting problem can be solved. □

Example 5.38. We have the following examples of residue fields:

(a) Let $X = \mathrm{Spec}(R)$ and let $x \in |X|$ be given by the prime ideal $\mathfrak{p} \subseteq R$. Then the residue field $\kappa(x)$ exists and is the usual residue field $\kappa(\mathfrak{p})$, by Remark 3.5.
(b) Let $X = \mathrm{Proj}(A)$ and let $x \in |X|$ be given by the saturated homogeneous prime ideal $\mathfrak{p} \subseteq A$. By Remark 3.81, there exists a homogeneous element $f \in A_+$ with $f \notin \mathfrak{p}$. Then $\mathfrak{p}_{(f)}$ is a prime ideal in $A_{(f)}$ and the open immersion $\mathrm{Spec}(A_{(f)}) \hookrightarrow \mathrm{Proj}(A)$ sends $\mathfrak{p}_{(f)}$ to $\mathfrak{p}$. By Proposition 5.37 and (a), the residue field $\kappa(x)$ exists and is given by $\kappa(\mathfrak{p}_{(f)}) \simeq \kappa(\mathfrak{p})_0$.

5.4. **Open coverings.** The notion of *open covering* of a topological space T depends only on the locale $\mathrm{Open}(T)$: a family of open subsets $(U_i \subseteq T)_{i\in I}$ is an open covering if and only if its supremum in $\mathrm{Open}(T)$ equals T. In the locale of an algebraic functor, there are several useful characterization of this property:

Proposition 5.39. *Let X be an algebraic functor, let $(U_i \subseteq X)_{i\in I}$ be a family of open subfunctors, and let $K_i \in \mathrm{Rad}_X$ be the quasi-coherent radical ideal of U_i. The following conditions are equivalent:*

(a) $X = \bigvee_{i\in I} U_i$ in the poset $\mathrm{Open}(X)$.
(b) $\mathcal{O}(X) = \bigvee_{i\in I} K_i$ in the poset Rad_X.
(c) For every ring R and every $x\colon \mathrm{Spec}(R) \to X$, $\bigcup_{i\in I} K_i(x)$ generates the unit ideal in R.
(d) For every local ring R, $X(R) = \bigcup_{i\in I} U_i(R)$.
(e) For every field k, $X(k) = \bigcup_{i\in I} U_i(k)$.
(f) The induced map $\coprod_{i\in I} U_i \to X$ is a Zariski-local epimorphism (Definition 4.39).
(g) The induced map $\coprod_{i\in I} U_i \to X$ is surjective (Proposition 5.30).

Proof. (a) ⇔ (b): this follows from the equivalence

$$\mathrm{Open}(X) \xrightarrow{\sim} \mathrm{Rad}_X\,.$$

(b) $\Leftrightarrow$ (c): this follows from

$$\mathrm{Rad}_X = \lim_{\mathrm{Spec}(R)\to X} \mathrm{Rad}_R$$

and suprema in Rad_X are computed pointwise.

(a) $\Leftrightarrow$ (f): this follows from the equivalence

$$\mathrm{Open}(X) \xrightarrow{\sim} \mathrm{Open}(|X|),$$

given by sending U to $|U|$.

(c) $\Rightarrow$ (d): let $x\colon \mathrm{Spec}(R) \to X$, where R is local. From $\sum_{i\in I} K_i(x) = R$, we can conclude that there exist $i \in I$ such that $1 \in K_i(x)$, i.e. $x \in U_i(x)$.

(d) $\Rightarrow$ (e), e $\Rightarrow$ (g) are trivial.

(f) $\Rightarrow$ (g): this follows from Equation (5.33).

(c) $\Rightarrow$ (f): let $x\colon \mathrm{Spec}(R) \to X$ be a map. Since $f_1, \dots, f_n$ generate the unit ideal, we have $\sum_{i\in I} K_i(x) = R$. If $f \in K_i(x)$, then the restriction of x to $\mathrm{Spec}(R_{f_i})$ lies in $U_i(R_f)$. Therefore, the map $\coprod_{i\in I} U_i \to X$ is an Zariski-local epimorphism. □

Definition 5.40 (Open covering)**.** A family of open subfunctors $(U_i \subseteq X)_{i\in I}$ is called an *open covering* of X if it satisfies the equivalent conditions of Proposition 5.39

Remark 5.41. By Proposition 5.39, open coverings of an algebraic functor X are equivalent to open coverings of its underlying space $|X|$.

Example 5.42 (Open coverings of affine and projective schemes)*.* The following are some examples of open coverings:

(a) Let R be a ring and $(F_i)_{i\in I}$ a family of subsets of R. Then $(\mathrm{D}(F_i) \subseteq \mathrm{Spec}(R))_{i\in I}$ is an open covering of $\mathrm{Spec}(R)$ if and only if $\bigcup_{i\in I} F_i$ generates the unit ideal of R, by Proposition 2.66. In general, the subfunctor $\mathrm{D}(F_i)$ form an open covering of $\mathrm{D}(\bigcup_{i\in I} F_i)$.

(b) Let R be a ring, L a line over R, and $(s_i)_{i\in I}$ a family of elements of L. Then $(D(s_i) \subseteq \mathrm{Spec}(R))_{i\in I}$ is an open covering of $\mathrm{Spec}(R)$ if and only if $(s_i)_{i\in I}$ generates L.

(c) Let A be an $\mathbf{N}$-graded ring and $(F_i)_{i\in I}$ a family of homogeneous subsets of R. Then $(\mathrm{D}(F_i) \subseteq \mathrm{Proj}(A))_{i\in I}$ is an open covering of $\mathrm{Proj}(A)$ if and only if the saturated radical ideal generated by $\bigcup_{i\in I} F_i$ is the unit ideal of A, by Proposition 3.90. For example, the affine open subschemes $\mathrm{D}(f) \simeq \mathrm{Spec}(A_{(f)})$ with $f \in A_+$ homogeneous form an open covering of $\mathrm{Proj}(A)$. In general, the subfunctors $\mathrm{D}(F_i)$ form an open covering of $\mathrm{D}(\bigcup_{i\in I} F_i)$.

(d) Let k be a ring, M a k-module, and $n \in \mathbf{N}$. For every $\alpha \in M^n$, let $U(\alpha) \subseteq \mathrm{Gr}_n(M)$ be the subfunctor consisting of the quotient spaces $\varphi\colon M \otimes_k R \twoheadrightarrow V$ such that $\varphi \circ \alpha\colon R^n \to V$ is an isomorphism, which is open by Proposition 3.53. If $(\alpha_i)_{i\in I}$ is a family in M^n, then $(U(\alpha_i) \subseteq \mathrm{Gr}_n(M))_{i\in I}$ is an open covering of $\mathrm{Gr}_n(M)$ if and only if the image of $(\alpha_i)_{i\in I}$ in $\Lambda^n_k M$ is a generating family (Example 3.108).

Remark 5.43 (Clopen subfunctors vs. clopen subspaces)*.* For any algebraic functor X, the map

$$\mathrm{Clopen}(X) \to \mathrm{Clopen}(|X|), \quad U \mapsto |U|,$$

is bijective. Indeed, let $U \subseteq X$ be an open subfucntor such that $|U|$ is clopen in $|X|$, and let $V \subseteq X$ be the open subfunctor such that $|X| = |U| \sqcup |V|$. Then U and V form an open covering of X such that $U \cap V = \varnothing_X$, so that the map $\mathcal{O}(X) \to \mathcal{O}(U) \times \mathcal{O}(V)$ is an isomorphism by Corollary 4.43. We therefore find an idempotent function $e \in \mathcal{O}(X)$ such that $U \subseteq \mathrm{D}(e)$ and $\mathrm{V} \subseteq \mathrm{D}(1-e)$. Since $U \vee V = X$ and $\mathrm{D}(e) \wedge \mathrm{D}(1-e) = \varnothing_X$ in $\mathrm{Open}(X)$, the distributivity law implies that $U = \mathrm{D}(e)$. As e is idempotent, we have $\mathrm{D}(e) = \mathrm{V}(1-e)$, so that U is closed, as desired. In particular, by Proposition 4.29, X is connected (in the sense of Proposition 5.30) if and only if the ring $\mathcal{O}(X)$ has exactly two idempotent elements.

Proposition 5.44 (Closure properties of open coverings)**.** *We have the following properties of open coverings:*

(a) (Refinement) A family of open subfunctors $(U_i \subseteq X)_{i\in I}$ is an open covering if it is refined by an open covering $(V_j \subseteq X)_{j\in J}$, i.e., if there exists $\alpha\colon J \to I$ such that $V_j \subseteq U_{\alpha(j)}$ for all $j \in J$.

(b) *(Composition) If* $(U_i \subseteq X)_{i\in I}$ *and* $(V_{ij} \subseteq U_i)_{j\in J_i}$ *are open coverings, then* $(V_{ij} \subseteq X)_{(i,j)\in\coprod_{i\in I} J_i}$ *is an open covering.*

(c) *(Intersection) If* $(U_i \subseteq X)_{i\in I}$ *and* $(V_j \subseteq X)_{j\in J}$ *are open coverings.*

(d) *(Base change) If* $(U_i \subseteq X)_{i\in I}$ *is an open covering and* $f\colon Y \to X$ *is a map of algebraic functors, then* $(f^{-1}(U_i) \subseteq Y)_{i\in I}$ *is an open covering.*

Proof. The proofs are left to the readers. □

6. Sheaves

In this chapter, we introduce the formalism of Grothendieck topologies and sheaves. A Grothendieck topology τ on a category $\mathcal{C}$ determines (and is determined by) a subcategory $\mathrm{Sh}_\tau(\mathcal{C})$ of the presheaf category $\mathrm{P}(\mathcal{C})$, whose objects are called τ-sheaves on $\mathcal{C}$. These are presheaves on $\mathcal{C}$ satisfying a certain local-to-global condition with respect to the topology τ, called τ-descent.

In algebraic geometry, we typically use this formalism in the following situations:

(a) $\mathcal{C}$ is the category of open subsets of a topological space T. This was the original notion of sheaf discovered by Leray in the 1940s, before Grothendieck's categorical generalization in the 1950s.
(b) $\mathcal{C} = \mathrm{CAlg}_k^{\mathrm{op}}$ is the opposite of the category of k-algebras, or equivalently the category Aff_k of affine k-schemes, on which there is a plethora of commonly used Grothendieck's topologies. The main ones are the Zariski, Nisnevich, étale, fppf ("fidèlement plat de présentation finie"), and fpqc ("fidèlement plat quasi-compac") topologies.
(c) The topologies in (b) extend to various enlargements of Aff_k, such as the category Sch_k of k-schemes, which we will introduce later, and even the whole category $\mathrm{Fun}(\mathrm{CAlg}_k, \mathrm{Set})$ of algebraic k-functors.

All the "Zariski descent" statements established so far are examples of descent statements for the so-called Zariski topology on $\mathrm{CAlg}^{\mathrm{op}}$ (e.g., Corollary 2.74 and Proposition 3.64) or on $\mathrm{Fun}(\mathrm{CAlg}_k, \mathrm{Set})$ (e.g., Corollary 4.43).

6.1. Sieves and descent.

Definition 6.1 (Sieve)**.** Let $\mathcal{C}$ be a category and let $X \in \mathcal{C}$. A *sieve* on X is a subpresheaf of the representable presheaf $\text{ﾒ}(X) \colon \mathcal{C}^{\mathrm{op}} \to \mathrm{Set}$.

Remark 6.2 (Sieves as subcategories)*.* A *left closed subcategory* of a category $\mathcal{C}$ is a full subcategory $\mathcal{D} \subseteq \mathcal{C}$ such that, if $X \in \mathcal{D}$ and $Y \to X$ is any morphism in $\mathcal{C}$, then $Y \in \mathcal{D}$. If R is a sieve on an object $X \in \mathcal{C}$, its category of element $\mathrm{El}(R)$ is a left closed subcategory of $\mathrm{El}(\text{ﾒ}(X)) = \mathcal{C}_{/X}$. This defines an order-preserving bijection

$$\mathrm{El} \colon \{\text{sieves on } X\} \xrightarrow{\sim} \{\text{left closed subcategories of } \mathcal{C}_{/X}\}.$$

In practice, we think of sieves on X either as subfunctors of $\text{ﾒ}(X)$ or as left closed subcategories of $\mathcal{C}_{/X}$, depending on the situation.

Notation 6.3 (Pullback of sieves)**.** Let R be a sieve on X and let $f \colon Y \to X$ be a map. We denote by $f^*(R)$ the pullback $R \times_{\text{ﾒ}(X)} \text{ﾒ}(Y)$, which is a sieve on Y. Concretely, $f^*(R)$ consists of all maps to Y whose composition with f belongs to R.

Definition 6.4 (Generated sieve)**.** Let $\mathcal{C}$ be a category and let $X \in \mathcal{C}$. The *sieve on X generated by a family of maps* $(Y_i \to X)_{i \in I}$ is the image of the map $\coprod_{i \in I} \text{ﾒ}(Y_i) \to \text{ﾒ}(X)$, i.e., the sieve consisting of all maps to X that factor through Y_i for some i.

Definition 6.5 (Descent)**.** Let $\mathcal{C}$ be a category, $F \in \mathrm{P}(\mathcal{C})$ a presheaf on $\mathcal{C}$ and R a sieve on $X \in \mathcal{C}$. We say that F *satisfies descent* along R if the inclusion $R \subseteq \text{ﾒ}(X)$ induces an isomorphism

$$F(X) = \mathrm{Map}(\text{ﾒ}(X), F) \xrightarrow{\sim} \mathrm{Map}(R, F).$$

Example 6.6 (The empty sieve)*.* For any $X \in \mathcal{C}$, the empty subpresheaf of $\text{ﾒ}(X)$ is a sieve on X, called the *empty sieve* on X. Since the empty presheaf is an initial object in $\mathrm{P}(\mathcal{C})$, a presheaf F satifies descent along this sieve if and only if $F(X)$ is a one-point set.

Example 6.7 (Sieve generated by two subobjects)*.* Let $X \in \mathcal{C}$, let $U, V \subseteq X$ be a pair of subobjects of X, and let R be the sieve on X generated by U and V. If the intersection $U \cap V$ exists, then R is the pushout $\text{ﾒ}(U) \sqcup_{\text{ﾒ}(U \cap V)} \text{ﾒ}(V)$ in $\mathrm{P}(\mathcal{C})$. Hence, a presheaf F on $\mathcal{C}$ satisfies descent along R if and only if the square

$$\begin{array}{ccc} F(X) & \longrightarrow & F(U) \\ \downarrow & & \downarrow \\ F(V) & \longrightarrow & F(U \cap V) \end{array}$$

is cartesian.

Lemma 6.8 (Presentation of sieves)**.** *Let R be a sieve on X and let $(Y_i \to X)_{i\in I}$ be a family of maps generating R. The the maps $\text{ɬ}(Y_i) \to R$ induce an isomorphism of presheaves*

$$\operatorname{colim}(\coprod_{i,j\in I} \text{ɬ}(Y_i) \times_{\text{ɬ}(X)} \text{ɬ}(Y_j) \rightrightarrows \coprod_{i\in I} \text{ɬ}(Y_i)) \xrightarrow{\sim} R.$$

Proof. By definition, we have

$$R = \operatorname{im}(\coprod_{i\in I} \text{ɬ}(Y_i) \to \text{ɬ}(X)).$$

In general, if $f\colon F \to G$ is a map in $\mathrm{P}(\mathcal{C})$, then

$$\operatorname{im}(f) = \operatorname{colim}(F \times_G F \rightrightarrows F),$$

since this is true section wise, this being the formula for the image of a morphism in the category of sets, and thus it is true globally. □

Proposition 6.9 (Characterization of descent)**.** *Let R be a sieve on $X \in \mathcal{C}$ and let $(Y_i \to X)_{i\in I}$ be a family of maps generating R. For a presheaf $F \in \mathrm{P}(\mathcal{C})$, the following conditions are equivalent:*

(a) F satisfies descent along R.

(b) The canonical map

$$F(X) \to \lim_{Y\in \mathrm{El}(R)} F(Y)$$

is an isomorphism.

(c) The diagram

$$F(X) \to \prod_{i\in I} F(Y_i) \rightrightarrows \prod_{i,j\in I} \operatorname{Map}(\text{ɬ}(Y)_i \times_{\text{ɬ}(X)} \text{ɬ}(Y_j), F)$$

is an equalizer

Proof. $(a) \Leftrightarrow (b)$ is just the equivalence $R \cong \operatorname{colim}_{Y\in\mathrm{El}(R)} Y$, while $(a) \Leftrightarrow (c)$ is just Lemma 6.8. □

Remark 6.10. If in Proposition 6.9 the pullbacks $Y_i \times_X Y_j$ exists in $\mathcal{C}$, we can rewrite condition (c) as follows:

(c') The diagram

$$F(X) \to \prod_{i\in I} F(Y_i) \rightrightarrows \prod_{i,j\in I} F(Y_i \times_X Y_j)$$

is an equalizer.

Example 6.11 (Monogenic sieve)*.* Let R be the sieve on $X \in \mathcal{C}$ generated by a single map $Y \to X$. If the pullback $Y \times_X Y$ exists in $\mathcal{C}$, then a presheaf $F \in \mathrm{P}(\mathcal{C})$ satisfies descent along R if and only if the following diagram is an equalizer

$$F(X) \to F(Y) \rightrightarrows F(Y \times_X Y).$$

Remark 6.12 (Descent for categories)*.* We can define descent for a presheaf of categories $F\colon \mathcal{C}^{\mathrm{op}} \to \mathrm{Cat}$ (in the sense of Definition 4.1) via Condition (b) of Proposition 6.9. Namely, if R is a sieve on $X \in \mathcal{C}$, we say that F *satisfies descent along* R if the canonical functor

$$F(X) \to \lim_{Y\in\mathrm{El}(R)} F(Y)$$

is an equivalence of categories. It turns out that this is *not* equivalent to Condition (c) of Proposition 6.9. Instead, assuming that the relevant fiber products exist in $\mathcal{C}$ for simplicity, it is equivalent to the following condition.

(c'') The diagram of categories

$$F(X) \to \prod_{i\in I} F(Y_i) \rightrightarrows \prod_{i,j\in I} F(Y_i \times_X Y_j) \mathrel{\substack{\rightarrow\\\rightarrow\\\rightarrow}} \prod_{i,j,k} \prod_{i,j,k\in I} F(Y_i \times_X Y_j \times_X Y_k)$$

is a limit diagram.

If we would have stopped after the first product, we could only guarantee that the objects agree. However, there are non-trivial morphisms between objects, so we need one more "coherence". This explains why the descent condition takes this more complicated form for presheaves of categories. Note that (c'') is equivalent to (c') if F is a presheaf of posets, as in this case, there is at most one morphism between objects.

6.2. **Grothendieck topologies.**

Definition 6.13 (Grothendieck topologies, site)**.** A *Grothendieck topology* τ on a category $\mathcal{C}$, or *topology* for short, assigns to each object $X \in \mathcal{C}$ a collection of sieves on X, called *τ-covering sieves*, subject to the following conditions:

(a) If R is a τ-covering sieve on X and $f\colon Y \to X$ is any map, then the pullback $f^*(R)$ is a τ-covering sieve on Y.
(b) Let S be a sieve on X and let R be a τ-covering sieve on X. If for all $f\colon Y \to X$ in R, that is $f \in R(Y)$, the pull back sieve $f^*(S)$ is τ-covering, then S is itself τ-covering.
(c) The maximal sieve $\text{ɬ}(X)$ on any X is τ-covering.

A family of maps $(Y_i \to X)_{i\in I}$ is called a *τ-covering family* or a *τ-cover*, if it generates a τ-sieve. A category equipped with a Grothendieck topology is called a *site*

Remark 6.14. The following further properties of a Grothendieck topology τ are consequences of the axioms $(a) - (c)$:

(d) If R is a τ-covering sieve on X and S is any sieve on X containing R, then S is a τ-covering.
(e) Any finite intersection of τ-covering sieves on X is τ-covering.

Indeed, (d) follows at once from (a) and (b), using that if $f \in R(Y)$, then for any $g : Z \to Y$ we have $f \circ g \in R(Z)$, so for any $f \in R(Y)$, $f^*(S) = f^*(R)$. (e) follows from (d) and (b) by rewriting $R \cap S = R \times_{\text{ɬ}(X)} S$ as $\operatorname{colim}_{(Y,f)\in \mathrm{El}(R)}(\text{ɬ}(Y) \times_{\text{ɬ}(X)} S) \cong \operatorname{colim}_{(Y,f)\in \mathrm{El}(R)}(f^*(S)) = \bigcup_{(Y,f)\in \mathrm{El}(R)} f^*S$.

Definition 6.15 (Sheaf)**.** Let $\mathcal{C}$ be a category and τ a topology on $\mathcal{C}$. A presheaf $F \in \mathrm{P}(C)$ is called a *τ-sheaf* if it satisfies descent along every τ-covering sieve. We denote by $\mathrm{Sh}_\tau(\mathcal{C}) \subseteq \mathrm{P}(\mathcal{C})$ the full subcategory of τ-sheaves.

Remark 6.16 (Posets of topologies)*.* Let τ and ρ be topologies on $\mathcal{C}$, we say that τ is *coarser* than ρ and ρ is *finer* than τ, and we write $\tau \leq \rho$, if every τ-covering sieve is a ρ-covering sieve. The collection of topologies on $\mathcal{C}$ is a poset under the relation $\leq$. By definition, any intersection of topologies on $\mathcal{C}$ is again a topology. The poset of topologies on $\mathcal{C}$ therefore admits all infima, and hence also all suprema. In particular, any collection σ of sieves on objects of $\mathcal{C}$ generates a topology $\bar{\sigma}$, which is the coarsest topology on $\mathcal{C}$ containing σ.

Example 6.17. On any category $\mathcal{C}$, we can consider the following topologies:

(a) The *discrete topology*, where the only covering sieves are the trivial sieves and the empty sieves. This is the finest topology: all sieves are covering. The final presheaf $*$ is the only sheaf in the discret topology.
(b) The *indiscrete topology* if the coarsest topology: only maximal sieves are covering. All presheaves are sheaves in the indiscrete topology

Example 6.18 (Locales and topological spaces)*.* Let $\mathcal{O}$ be a locale, viewed as a category. If we define a sieve on $u \in \mathcal{O}$ to be a covering if its supremum is u, we obtain a topology on $\mathcal{O}$, called the *canonical topology*. Indeed, Conditions (a) and (b) follow from the distributivity law, an Condition (c) is clear. In particular, if T is a topological space, this defines the canonical topology on the category $\mathrm{Open}(T)$.

Example 6.19 (Weiss topologies)*.* Let T be a topological space and κ cardinal. The *κ-Weiss topology* on $\mathrm{Open}(T)$ is defined as follows: a sieve R on U is covering if any subset of U of size $< \kappa$ contained in some element of R. This recovers the canonical topology for $\kappa = 2$ (Example 6.18), the discrete topology for $\kappa = 0$ and the indiscrete topology for $\kappa > |T|$ (Example 6.17). For $\kappa = 1$, the covering sieves are exactly the nonempty sieves, and the corresponding sheaves are the constant presheaves.

Warning 6.20. The notion of Grothendieck topology is *not* a direct generalization of the notion of topology on a set. Example 6.18 explains the relation between the two notions. Historically, Grothendieck named the new concepty "topology" because it replaced topological spaces in the context of sheaf theory.

Proposition 6.21 (The adjunction between topologies and subcategories)**.** *Let $\mathcal{C}$ be a category. The map of posets*

$$\{\textit{topologies on } \mathcal{C}\}^{\mathrm{op}} \to \{\textit{subcategories of } \mathrm{P}(\mathcal{C})\}, \quad \tau \mapsto \mathrm{Sh}_\tau(\mathcal{C}),$$

has a left adjoint $\mathcal{E} \to \tau_\mathcal{E}$, i.e., there exists a finest topology $\tau_\mathcal{E}$ such that $\mathcal{E} \subseteq \mathrm{Sh}_{\tau_\mathcal{E}}(\mathcal{C})$. A sieve R on X is $\tau_\mathcal{E}$ covering if and only if, for every map $f\colon Y \to X$, every $F \in \mathcal{E}$ satisfies descent along $f^*(R)$.

Proof. We will show that the collection τ of sieves we defined above is a topology. It is then clearly the finest topology with $\mathcal{E} \subseteq \mathrm{Sh}_\tau(\mathcal{C})$.

The condition (a) and (c) in Definition 6.13 is clear. Therefore it suffices to check condition (b). Let $R, S \subseteq \text{ɬ}(X)$ such that R is a τ-covering and for every $f\colon Y \to X$ in R, $f^*(S) \subseteq \text{ɬ}(Y)$ is a τ-covering. We need to show that S is a τ-covering. In other words, we have to show that for every $F \in \mathcal{E}$ satifies descent along S, i.e we need to show that the upper horizontal map of the following diagram is bijective:

$$\begin{array}{ccc} F(X) & \longrightarrow & \mathrm{Map}(S, F) \\ \downarrow & & \downarrow \\ \mathrm{Map}(R, F) & \longrightarrow & \mathrm{Map}(R \cap S, F). \end{array}$$

This amounts to show that all the other maps in the diagram are bijective.

The left vertical map is bijective, since R is a τ-covering sieve. Since $R = \mathrm{colim}_{(Y,f)\in \mathrm{El}(R)}\, \text{ɬ}(Y)$ and $R \cap S = \mathrm{colim}_{(Y,f)\in\mathrm{El}(R)}(\text{ɬ}(Y) \times_{\text{ɬ}(X)} S) \cong \mathrm{colim}_{(Y,f)\in\mathrm{El}(R)}(f^*(S))$ the lower horizontal map can be identified with

$$\lim_{(Y,f)\in\mathrm{El}(R)} (F(Y) \to \mathrm{Map}(f^*(S), F)),$$

which is bijective by the assumption that f^*S is a τ sheaf for all $f \in R$. Similarly $S = \mathrm{colim}_{Y\in\mathrm{El}(S)}\, \text{ɬ}(Y)$, so the right vertical map can be identified with

$$\lim_{(Y,f)\in\mathrm{El}(S)} (F(Y) \xrightarrow{\sim} \mathrm{Map}(f^*(R), F)),$$

which is bijective, by definition of R as a τ-covering sieve. □

Corollary 6.22. *Let $(\tau_i)_{i\in I}$ be a family of topologies on $\mathcal{C}$ and let $\tau = \bigvee_{i\in I} \tau_i$ be their supremum. Then*

$$\mathrm{Sh}_\tau(\mathcal{C}) = \bigcap_{i\in I} \mathrm{Sh}_{\tau_i}(\mathcal{C}).$$

Proof. Let F be a τ-sheaf. As $\tau_i \subseteq \tau$ for all i, the presheaf F is a τ_i-sheaf for all i, hence defines an element in $\bigcap_{i\in I} \mathrm{Sh}_{\tau_i}(\mathcal{C})$. Let us now assume that $F \in \bigcap_{i\in I} \mathrm{Sh}_{\tau_i}(\mathcal{C})$, and let R be a τ-covering sieve on some object X. We have to show that F satisfies descent with respect to R. As τ is the supremum of all τ_i, it is the coarsest topology containing all τ_i, that is, it contains all the sieves of all the τ_i, and additionally those sieves such that there exists some $i \in I$ and some sieve $S \in \tau_i$, such that for every $f : Y \to X$, the sieve $f^*(R)$ is a τ_j-covering sieve on Y, where j may depend on f. In particular we have $F(Y) \simeq \mathrm{Map}(f^*(R), F)$. Just as before, we obtain a commutative square

$$\begin{array}{ccc} F(X) & \longrightarrow & \mathrm{Map}(R, F) \\ \downarrow & & \downarrow \\ \mathrm{Map}(S, F) & \longrightarrow & \mathrm{Map}(R \cap S, F) \end{array}$$

Here the left vertical map is an isomorphism as F is τ_i sheaf, the lower horizontal map is an isomorphism by rewriting $R \cap S \cong \mathrm{colim}_{(Y,f)\in\mathrm{El}(S)} f^*(R)$ and the discussion above, and the right vertical map is an isomorphism by writing $R = \mathrm{colim}_{(Y,f\in\mathrm{El}(R))}\, \text{ɬ}(Y)$ as well as $S \cap R \cong \mathrm{colim}_{(Y,f)\in\mathrm{El}(R)} f^*(S)$ and identifying the right vertical map with

$$\lim_{(Y,f\in\mathrm{El}(R))} \mathrm{Map}(\text{ɬ}(Y), F) \cong \lim_{(Y,f\in\mathrm{El}(R))} F(Y) \cong \lim_{(Y,f\in\mathrm{El}(R))} \mathrm{Map}(f^*(S), F) \cong \mathrm{Map}(S \cap R, F).$$

By two-out-of-three, the top horizontal morphism is an isomorphism, hence we are done. □

Corollary 6.23. *Let $\mathcal{C}$ be a category and let σ be a collection of sieves on $\mathcal{C}$ that is closed under pullbacks (i.e, satisfies Condition (a) of Definition 6.13). Let $\bar{\sigma}$ be the topology generated by σ. Then a presheaf F on $\mathcal{C}$ is a $\bar{\sigma}$-sheaf if and only if it satifies descent along all sieves in σ.*

Proof. We apply Proposition 6.21 to

$$\mathcal{E} = \{F \in \mathrm{P}(\mathcal{C}) \mid F \text{ satifies descent along } \sigma\}.$$

Then there exists a finest topology τ such that $\mathcal{E} \subseteq \mathrm{Sh}_\tau(\mathcal{C})$. Since σ is stable under base change, $\bar{\sigma}$ is finer than τ, which implies that $\mathrm{Sh}_\tau(\mathcal{C}) \subseteq \mathrm{Sh}_{\bar{\sigma}}(\mathcal{C}) \subseteq \mathcal{E}$. Since we also have $\mathcal{E} \subseteq \mathrm{Sh}_\tau(\mathcal{C})$, we conclude that $\mathrm{Sh}_\tau(\mathcal{C}) = \mathrm{Sh}_{\bar{\sigma}}(\mathcal{C})$. □

In practice, topologies are often defined by means of covering families rather than covering sieves:

Definition 6.24 (Pretopology)**.** A *pretopology* π on a category $\mathcal{C}$ assigns to each object $X \in \mathcal{C}$ a collection of families of $\mathcal{C}_{/X}$, called *π-covering families* or *π-covers*, subject to the following conditions:

(a) If $(U_i \to X)_{i\in I}$ is a π-covering family and $f\colon Y \to X$ is any map, then the pullbacks $U_i \times_X Y$ exist in $\mathcal{C}$ and the family $(U_i \times_X Y \to Y)_{i\in I}$ is π-covering.
(b) If $(U_i \to X)_{i\in I}$ and for all i the collections $(V_{ij} \to U_i)_{j\in J}$ are π-covering families, then $(V_{ij} \to X)_{i\in I, j\in J_i}$ is π-covering.
(c) The singleton family $(\mathrm{id}_X\colon X \to X)$ is π-covering.

The *topology associated with* a pretopology π is the coarsest topology for which π-coverings families generate covering sieves (which exists by Remark 6.16).

Example 6.25. Let τ be a topology on $\mathcal{C}$. If (and only if) $\mathcal{C}$ has pullbacks, then the τ-covering families form a pretopology on $\mathcal{C}$, whose associated topology is τ. For example, if X is a topological space or an algebraic functor, covering families for the canonical topology on $\mathrm{Open}(X)$ are precisely open coverings, which in particular form a pretopology on $\mathrm{Open}(X)$.

Proposition 6.26 (Pretopologies and descent)**.** *Let $\mathcal{C}$ be a category, let π be a pretopology on $\mathcal{C}$ and let τ be the topology associated with π.*

(a) A sieve is τ-covering if and only if it contains a π-covering family.
(b) A presheaf F is a τ-sheaf if and only if, for every π-covering family $(U_i \to X)_{i\in I}$, the following diagram is an equalizer:

$$F(X) \to \prod_{i\in I} F(U_i) \rightrightarrows \prod_{i,j\in I} F(U_i \times_X U_j).$$

Proof. Let τ be the collection of all sieves that contains a π-covering family, we will show that it is a topology on $\mathcal{C}$. We first show condition (a) of Definition 6.13. Assume that $R \subseteq \text{ɟ}(X)$ is a sieve that contains a π-covering family $(U_i \to X)_{i\in I}$ and $f\colon Y \to X$. Then $f^*(R)$ contains $(U_i \times_X Y \to Y)_{i\in I}$, which is a π-covering family by condition (a) of Definition 6.24. For condition (b), let $R, S \subseteq \text{ɟ}(X)$ such that R contains a π-covering family $f_i : U_i \to X_{i\in i}$ and such that for every $f : Y \to X$ the pullback sieve $f^*(S)$ contains a π-covering family. In particular, $f_i^*(S)$ contains a π-covering family $\{V_{ij} \to U_i\}_{j\in J}$ for all $i \in I$. But then S contains the composites $\{V_{ij} \to U_i \to X\}_{ij}$, which is a π-covering family by condition (b) of Definition 6.24. Condition (c) is clearly satisfied, due to condition (c) of Definition 6.24. Thus, τ defines a topology. As this topology contains exactly those sieves which contain a sieve generated by a π-covering family, and every sieve containing a covering sieve is itself covering, τ is the topology generated by the family of sieves generated by π-covering families, and hence the topology associated with the pretopology π.

Part (b) of the proposition is Proposition 6.9 together with Corollary 6.23, as τ is generated by the family of sieves generated by π-covering families, □

Example 6.27 (The standard topology on Top)*.* Open coverings form a pretopology on the category Top of all topological spaces (and many similar categories, like that of smooth manifolds). If τ is the assoicated topology, Proposition 6.26 implies the following:

(a) A sieve on X is τ-covering if and only if it contains an open covering of X. Hence, a family $(Y_i \to X)_{i\in I}$ is τ-covering if and only if it locally has sections, i.e., if and only if every point of X has a neighborhood that maps to some Y_i over X.
(b) A presheaf $F\colon \mathrm{Top}^{\mathrm{op}} \to \mathrm{Set}$ is a τ-sheaf if and only if $F_{|\,\mathrm{Open}(T)}$ is a sheaf on T for each $T \in \mathrm{Top}$.

For example, for every $X \in \mathrm{Top}$, the presheaf $\underline{\mathbb{R}}(X)$ of X-valued functions is a τ-sheaf (by Example 6.18).

Example 6.28 (The Zariski topology)*.* There is a pretopology on $\mathrm{CAlg}^{\mathrm{op}}$ whose covers are the families $(R \to R_{f_i})_{i\in I}$ suchthat $(f_i)_{i\in I}$ generates the unit ideal in R. The assoicated topology is called the *Zariski topology*. By Proposition 6.26, we have the following:

(a) A sieve $S \subseteq \mathrm{Spec}(R)$ is Zariski-covering if and only if the set of $f \in R$ such that $R \to R_f$ belongs to S generates the unit ideal. Consequently, an arbitrary family $(R \to R_i)_{i\in I}$ is Zariski-covering if and only if $\coprod_{i\in I} \mathrm{Spec}(R_i) \to \mathrm{Spec}(R)$ is a Zariski-local epimorphism (Definition 4.39).
(b) An algebraic functor $X\colon \mathrm{CAlg} \to \mathrm{Set}$ is a Zariski sheaf if and only if, for any ring R and any family $(f_i)_{i\in I}$ generating the unit ideal in R, the following diagram is an equalizer:

$$X(R) \to \prod_{i\in I} X(R_{f_i}) \rightrightarrows \prod_{i,j\in I} X(R_{f_i f_j}).$$

For example, any affine scheme is a Zariski sheaf by Corollary 2.74, as is $\mathrm{Proj}(A)$ for any $\mathbf{N}$-graded ring A by Proposition 3.64. Note that if we only consider families $(f_i)_{i\in I}$ where I is finite, we obtain another pretopology on $\mathrm{CAlg}^{\mathrm{op}}$, whose associated topology is *also* the Zariski topology. This uses that if $(f_i)_{i\in I} = R$, there exists some finite subset $J \subseteq I$ such that $(f_j)_{f\in J=R}$. Hence, the condition in (b) is *equivalent* to the seemingly weaker condition where we require that I be finite. On the other hand, by Proposition 6.9, it is also equivalent to the following seemingly stronger condition: for every Zariski-covering family $(R \to R_i)_{i\in I}$, the diagram

$$X(R) \to \prod_{i\in I} X(R_i) \rightrightarrows \prod_{i,j\in I} X(R_i \otimes_R R_j)$$

is an equalizer.

Example 6.29 (The fpqc and fppf topologies)*.* The definition of the Zariski topology on $\mathrm{CAlg}^{\mathrm{op}}$ is a special case of a more general construction. Let E be a class of ring maps, which contains isomorphism and is closed under composition and cobase change. Then there is a pretopology on $\mathrm{CAlg}^{\mathrm{op}}$ whose covers are the families $(R \to R_i)_{i\in I}$ such that:

- I is finite;
- $\coprod_{i\in I} \mathrm{Spec}(R_i) \to \mathrm{Spec}(R)$ is surjective;
- each map $R \to R_i$ belongs to E.

For example:

(a) If E is the class of localizations $R \to R_f$, this recovers the Zariski topology of Example 6.28.
(b) If E is the class of flat maps, the associated topology on $\mathrm{CAlg}^{\mathrm{op}}$ is called the *fpqc topology*.
(c) If E is the class of flat maps of finite presentation, the associated topology on $\mathrm{CAlg}^{\mathrm{op}}$ is called the *fppf topology*.

Since the localization maps $R \to R_f$ are flat and of finite presentation, we have the following relations between these topologies: Zar $\subseteq$ fppf $\subseteq$ fpqc.

Definition 6.30 (Dense subcategory)**.** Let $(\mathcal{C}, \tau)$ be a site. A *dense subcategory* of $(\mathcal{C}, \tau)$ is a (small) full subcategory $\mathcal{D} \subseteq \mathcal{C}$ such that every $X \in \mathcal{C}$ admits a τ-covering family $(U_i \to X)_{i\in I}$ with $U_i \in \mathcal{D}$.

Remark 6.31*.* The smallness assumption on $\mathcal{D}$ in the previous definition has the property to ensure the existence of the Kan extensions $u_\sharp, u_* : \mathrm{P}(\mathcal{D}) \to \mathrm{P}(\mathcal{C})$.

Proposition 6.32 (The "comparison lemma")**.** *Let $(\mathcal{C}, \tau)$ be a site and $\mathcal{D} \subseteq \mathcal{C}$ a dense subcategory. Let $\tau|_{\mathcal{D}}$ be the collection of sieves $R|_{\mathcal{D}} \subseteq \underline{\mathbb{R}}_{\mathcal{D}}(X)$ for all $X \in \mathcal{D}$ and all τ-covering sieves $R \subseteq \underline{\mathbb{R}}_{\mathcal{C}}(X)$. Then $\tau|_{\mathcal{D}}$ is a topology on $\mathcal{D}$ and the functor $\mathrm{P}(\mathcal{C}) \to \mathrm{P}(\mathcal{D})$, $F \mapsto F|_{\mathcal{D}}$, restricts to an equivalence of categories $\mathrm{Sh}_\tau(\mathcal{C}) \xrightarrow{\sim} \mathrm{Sh}_{\tau|_{\mathcal{D}}}(\mathcal{D})$.*

Proof. We start by showing that $\tau|_{\mathcal{D}}$ is a topology. We will do so by showing that for $X \in \mathcal{D}$ a sieve $S \in \mathrm{P}(\mathcal{D})$ is $\tau|_{\mathcal{D}}$ covering if and only if the sieve $\tilde{S} \in \mathrm{P}(\mathcal{C})$ generated by the morphisms in S is τ-covering. Condition (c) will then follow at once. For condition (a), we note that for any $f : Y \to X \in \mathcal{D}$, we have $\widetilde{f^*(S)} \subseteq f^*(\tilde{S})$, and that any sieve containing a covering sieve is itself a sieve. For condition (b), let S be a sieve on $S \in \mathrm{P}(\mathcal{D})$ be a sieve on $X \in \mathcal{D}$ and let $R|_{\mathcal{D}}$ be a $\tau|_{\mathcal{D}}$ covering sieve, with $R \in \mathrm{P}(\mathcal{C})$ being τ-covering, such that for any $U \in \mathcal{D}$ and any $f \in R(D)$ we have that f^*S is $\tau|_{\mathcal{D}}$-covering. By the claim, it suffices to show that the sieve $\tilde{S} \in \mathcal{C}$ is τ covering. Let Y be an object in $\mathcal{C}$ and $f \in R(Y)$. We will show that $f^*(\tilde{S})$ is a τ-covering sieve on Y. By assumption on $\mathcal{D}$ there exists a τ-covering family $\{f_i : U_i \to Y\}_{i\in I}$ with $U_i \in \mathcal{D}$. Let R_I be the τ-covering sieve generated by $\{f_i : U_i \to Y\}_{i\in I}$. Using condition (b) for the topology τ with $f^*(\tilde{S})$ and R_I, and that R_I is generated by the f_i, it suffices to show that $f_i^*\left(f^*(\tilde{S})\right) = (f \circ f_i)^*(\tilde{S})$ is a τ-covering sieve on U_i. But $f \circ f_i \in R(U_i)$, so that we know $\widetilde{(f \circ f_i)^*(S)}$, the τ-sieve generated by the members of $(f \circ f_i)^*(S)$, is a τ-sieve by assumption on S and R. But by construction $\widetilde{(f \circ f_i)^*(S)} \subseteq (f \circ f_i)^*\tilde{S}$, and every sieve containing a covering sieve is itself covering.

So for the first part of the proposition, we only have to prove the claim. Let $S \in \mathcal{D}$ be a sieve on X. We have that $\tilde{S}|_{\mathcal{D}} = S$ by construction, so we only have to show that if S is $\tau|_{\mathcal{D}}$-covering, then $\tilde{S}$ is τ-covering. By definition of $\tau|_{\mathcal{D}}$, there exists some sieve R such that $R|_{\mathcal{D}} = S$, in particular, $\tilde{S} \subseteq R$ and $\tilde{S}|_{\mathcal{D}} = R|_{\mathcal{D}}$. Let $Y \in \mathcal{C}$, and take any $f \in R(Y)$. Consider $f^*(\tilde{S})$. Choose a τ-covering family $\{f_i : U_i \to Y\}_{i\in I}$ and let R_I be the sieve generated by the f_i. Since $f \in R(Y)$, we have that $f \circ f_i \in R(U_i) = \tilde{S}(U_i)$, which implies that[22] $(f \circ f_i)^*(\tilde{S}) = \text{ɬ}_{\mathcal{C}}(U_i)$. Since $\text{ɬ}_{\mathcal{C}}(U_i)$ is a τ-covering sieve for any i, the sieve $(f \circ f_i)^*(\tilde{S})$ is τ-covering. Since R_I is a τ-covering sieve generated by the f_i, condition (b) for the topology τ implies that $f^*(\tilde{S})$ is a covering sieve. As $(Y, f) \in \mathrm{El}(R)$ was arbitrary, condition (b) for the topology τ implies that $\tilde{S}$ is a τ-covering sieve, finishing the proof of the claim.

All that is left to show that the restriction $u^* : \mathrm{P}(\mathcal{C}) \to \mathrm{P}(\mathcal{D})$ induces an equivalence $\mathrm{Shv}_\tau(\mathcal{C}) \to \mathrm{Shv}_{\tau|_{\mathcal{D}}}(\mathcal{D})$. Let F be a τ-sheave, we start by showing that u^*F is indeed a $\tau|_{\mathcal{D}}$-covering sheaf. Let S be a $\tau|_{\mathcal{D}}$-sieve on $X \in \mathcal{D}$ and denote by $\tilde{S} \in \mathrm{P}(\mathcal{C})$ the extended sieve. Since F is a sheaf, we know that $F(X) \cong \lim_{(Y,f)\in\mathrm{El}(\tilde{S})} F(Y)$. We want to show that $F(X)$ is also a limit of the diagram $\{F(Y)\}_{(Y,f)\in\mathrm{El}(S)}$. For this, let $\{T \to F(Y)\}_{(Y,f)\in\mathrm{El}(S)}$ be a cone over the diagram. Our goal is to extend this cone uniquely to a cone $\{T \to F(Y)\}_{(Y,f)\in\mathrm{El}(\tilde{S})}$. As $\tilde{S}$ is generated by S, for any $(Z, g) \in \mathrm{El}(\tilde{S})$ there exists an $(U, f) \in \mathrm{El}(S)$ and a map $\varphi : (Z, g) \to (U, f)$. We now define $T \to F(Z)$ as the composition $T \to F(U) \xrightarrow{F(\varphi)} F(Z)$. We need to show that this morphism is actually well defined, that is, independent of the choice of (U, f). For this, assume we have two morphisms $(U, \varphi), (U', \varphi') \to (Z, g)$ and hence to legs, which we, by abuse of notation, also denote by $\varphi, \varphi' : T \to F(Z)$. Choose a τ-covering family $\{f_i : U_i \to Z\}$. As F is a sheaf, we have that $F(Z) \hookrightarrow \prod_{i\in I} F(U_i)$ is a monomorphism. But, using that $\{T \to F(Y)\}_{(Y,f)\in\mathrm{El}(S)}$ is a diagram, the maps $T \to F(U) \to F(Z) \to \prod_{i\in I} F(U_i)$ and $T \to F(U') \to F(Z) \to \prod_{i\in I} F(U_i)$ are the same. Using the defining property of monomorphisms, we obtain that $\varphi = \varphi' : T \to F(Z)$. Hence the map $T \to F(Z)$ was well defined, an the diagram $\{T \to F(Y)\}_{(Y,f)\in\mathrm{El}(S)}$ extends to a diagram $\{T \to F(Y)\}_{(Y,f)\in\mathrm{El}(\tilde{S})}$, hence factors uniquely over $F(X)$. This shows that $F(X) = \lim_{(Y,f)\in\mathrm{El}(S)}(u^*F)(Y)$, finishing the claim that u^* preserves sheaves. Thus $u^* : \mathrm{Shv}_\tau(\mathcal{C}) \to \mathrm{Shv}_{\tau|_{\mathcal{D}}}(\mathcal{D})$. Notice further, that the right adjoint u_* of u^* also preserves sheaves. Indeed, let $F \in \mathrm{Shv}_{\tau|_{\mathcal{D}}}(\mathcal{D})$ be a sheaf and let R be a τ-covering sieve on $X \in \mathcal{C}$. We then have

$$\begin{aligned}
u_*F(X) &\cong \mathrm{Map}_{\mathrm{P}(\mathcal{C})}(\text{ɬ}_{\mathcal{C}}(X), u_*F) \\
&\cong \mathrm{Map}_{\mathrm{P}(\mathcal{D})}(u^*\text{ɬ}_{\mathcal{C}}(X), F) \\
&\cong \lim_{(Y,f)\in\mathrm{El}(u^*X)} \mathrm{Map}_{\mathrm{P}(\mathcal{D})}(\text{ɬ}_{\mathcal{D}}(Y), F) \\
&\cong \lim_{(Y,f)\in\mathrm{El}(u^*X)} \mathrm{Map}_{\mathrm{P}(\mathcal{D})}(f^*(u^*R), F) \\
&\cong \mathrm{Map}_{\mathrm{P}(\mathcal{D})}(\operatorname*{colim}_{(Y,f)\in\mathrm{El}(u^*X)} f^*(u^*R), F)
\end{aligned}$$

[22] Indeed, if S is a sieve, and $f \in S(Y)$, then for every $Z \in \mathcal{C}$ and any $g : Z \to Y$, we have that $g^* : S(Y) \to S(Z)$ sends f to $f \circ g \in S(Z)$. But this just means that for any Z and any $g \in \text{ɬ}_{\mathcal{C}}(Y)(Z)$, we have that $g \in f^*(S)(Z)$.

$$\begin{aligned}
&\cong \mathrm{Map}_{\mathrm{P}(\mathcal{D})}\big(\big(u^* \yoneda_{\mathcal{C}}(X)\big) \times_{\left(u^* \yoneda_{\mathcal{C}}(X)\right)} \left(u^* R\right), F\big) \\
&\cong \mathrm{Map}_{\mathrm{P}(\mathcal{D})}(u^* R, F) \\
&\cong \mathrm{Map}_{\mathrm{P}(\mathcal{C})}(R, u_* F).
\end{aligned}$$

The isomorphisms are, in order, the Yoneda lemma, the (u^*, u_*)-adjunction, the Yoneda density theorem, using that F is a $\tau|_{\mathcal{D}}$-sheaf and that for $f : Y \to X$ with $Y \in \mathcal{D}$, the sieve $f^*(u^* R)$ is a $\tau|_{\mathcal{D}}$-covering sieve of Y by definition, continuity of Map in the first component, universality of colimits and that for $f : Y \to X$ we have $f^* R \cong \yoneda(Y) \times_{\yoneda(X)} R$, that $u^* R$ is a subpresheaf[23] of $u^* \yoneda_{\mathcal{C}}(X)$, and lastly the (u^*, u_*)-adjunction again.

Thus, the adjunction $(u^*, u_*) : \mathrm{P}(\mathcal{C}) \leftrightarrows \mathrm{P}(\mathcal{D})$ descends to an adjunction $(u^*, u_*) : \mathrm{Shv}_\tau(\mathcal{C}) \leftrightarrows \mathrm{Shv}_{\tau|_{\mathcal{D}}}(\mathcal{D})$, where the right adjoint is still fully faithful, as can be seen from the commutative diagram

$$\begin{array}{ccc}
\mathrm{P}(\mathcal{C}) & \xleftarrow{u_*} & \mathrm{P}(\mathcal{D}) \\
\uparrow & & \uparrow \\
\mathrm{Shv}_\tau(\mathcal{C}) & \xleftarrow[u_*]{} & \mathrm{Shv}_{\tau|_{\mathcal{D}}}(\mathcal{D}).
\end{array}$$

Thus, we only have to show that u^* is conservative[24]. So let $F \to G \in \mathrm{Shv}_\tau(\mathcal{C})$ be a map of sheaves, such that $u^* F \to u^* G$ is an isomorphism. It suffices to check that $F \to G$ is section wise an isomorphism, so let $X \in \mathcal{C}$ be any object. Choose a covering family $\{f_i : U_i \to X\}_{i \in I}$ and denote by S the covering sieve generated by the U_i. We obtain a map of diagrams $\{F(Y)\}_{(Y,f) \in \mathrm{El}(S)} \to \{G(Y)\}_{(Y,f) \in \mathrm{El}(S)}$ This map is an isomorphism on all $(U, g) \in \mathrm{El}(S)$ with $U \in \mathcal{D}$ by the assumption that $u^* F \to u^* G$ is an isomorphism. In particular, we obtain an isomorphism of the subdiagrams. We can then extend this isomorphism over the whole of the diagram $\mathrm{El}(S)$, first extending the map $\{G(Y)\}_{(Y,f) \in \mathrm{El}(S) \,:\, Y \in \mathcal{D}} \to \{F(Y)\}_{(Y,f) \in \mathrm{El}(S) \,:\, Y \in \mathcal{D}}$, which we can do as the jointly initial objects of the diagram are exactly the $F(U_i)$, respectively the $G(U_i)$, and then checking that this really defines inverses. The well-definedness of the construction and that the are indeed isomorphisms follows the same principle as how we showed that u^* preserves sheaves, that is, by checking after postcomposition with the monomorphism to the product over another covering family. We leave the details to the reader. □

Example 6.33 (Comparison lemma for presheaves). Let $\mathcal{D} \subseteq \mathcal{C}$ be a full subcategory. Then $\mathcal{D}$ is dense with respect to the indiscrete topology ((*b*) of Example 6.17) if and only if every object of $\mathcal{C}$ is a retract of an object of $\mathcal{D}$. In this case, the comparison lemma says that $\mathrm{P}(\mathcal{C}) \xrightarrow{\sim} \mathrm{P}(\mathcal{D})$.

Example 6.34 (Basis of a topological space). Let T be a topological space and let $\mathcal{B} \subseteq \mathrm{Open}(T)$ be a basis of its topology. By definition, $\mathcal{B}$ is dense with respect to the canonical topology on $\mathrm{Open}(T)$, and the restriction of the canonical topology to $\mathcal{B}$ consists of sieves generated by open coverings in $\mathcal{B}$. Hence, the comparison lemma provides an equivalence $\mathrm{Sh}(T) \xrightarrow{\sim} \mathrm{Sh}(\mathcal{B})$.

Example 6.35 (Sheaves on manifolds). Let Man_n be the category of n-dimensional topological manifolds, equipped with the standard topology (induced by the pretopology of open coverings). Since by definition every n-manifold admits an open covering by manifolds homeomorphic to $\mathbf{R}^n$, the full subcategory $\{\mathbf{R}^n\} \subseteq \mathrm{Man}_n$ spanned by $\mathbf{R}^n$ is dense. By comparison lemma, $\mathrm{Sh}(\mathrm{Man}_n) \simeq \mathrm{Sh}(\{\mathbf{R}^n\})$.

Example 6.36 (Sheaves on $\mathrm{Spec}(R)$). Let R be a ring. Consider the following two full subcategories of the locale $\mathrm{Open}(\mathrm{Spec}(R)) \simeq \mathrm{Open}(\mathrm{Prim}(R)) \simeq \mathrm{Rad}_R$:

(a) $\mathrm{Open}^{\mathrm{pr}}(\mathrm{Spec}(R))$ consists of open subfunctors of the form $\mathrm{D}(f)$ with $f \in R$ (called the *principal open subschemes* of $\mathrm{Spec}(R)$);
(b) $\mathrm{Open}^{\mathrm{aff}}(\mathrm{Spec}(R))$ consists of the *affine* open subfunctors of $\mathrm{Spec}(R)$.

[23] This follows from R being a subpresheaf of $\yoneda_{\mathcal{C}}(X)$ and u^* commutes with limits, hence with injections for presheaves of sets.

[24] If we have an adjunction $(L, R) : \mathcal{C} \leftrightarrow \mathcal{D}$ such that one of the functors is fully faithful. Then this is an adjoint equivalence, if and only if the other one is conservative. Indeed, one direction is clear, for the other, let us assume that the left adjoint L is fully faithful, and that R is conservative. It suffices to check that L is essentially surjective. Consider the counit map $LRX \to X$. Applying R again, we obtain $RLRX \to RX$. By the triangle identity, this is a section of the unit map $RX \to RLRX$, which is an isomorphism since L is fully faithful. Hence the R of the counit map is an isomorphism, and, by conservativity, so is the counit map itself.

We have $\mathrm{Open}^{\mathrm{pr}}(\mathrm{Spec}(R)) \subseteq \mathrm{Open}^{\mathrm{aff}}(\mathrm{Spec}(R))$, and both subcategories are dense in $\mathrm{Open}(\mathrm{Spec}(R))$, since $\mathrm{D}(I) = \bigvee_{f\in I} \mathrm{D}(f)$. By the comparison lemma, the three categories of sheaves are equivalent. In particular, every sheaf on $\mathrm{Open}^{\mathrm{pr}}(\mathrm{Spec}(R))$ *extends uniquely* to a sheaf on $\mathrm{Open}(\mathrm{Spec}(R))$.

Example 6.37 (Sheaves on presheaves)*.* Let $\mathcal{C}$ be a small category, which we view as a full subcategory of $\mathrm{P}(\mathcal{C})$ via the Yoneda embedding. Any topology τ on $\mathcal{C}$ extends canonically to a topology $\hat{\tau}$ on $\mathrm{P}(\mathcal{C})$ as follows: a sieve R on a presheaf F is $\hat{\tau}$-covering if, for any $\text{ɣ}(X) \to F$, the sieve of all $Y \to X$, such that $\text{ɣ}(Y) \to \text{ɣ}(X) \to F$ is in $R(\text{ɣ}(Y))$, is τ-covering. Since every object of $\mathrm{P}(\mathcal{C})$ is a quotient of a coproduct of representable presheaves, the comparison lemma applies to the Yoneda embedding $\mathcal{C} \hookrightarrow \mathrm{P}(\mathcal{C})$, so that

$$\mathrm{Sh}_\tau(\mathcal{C}) \simeq \mathrm{Sh}_{\hat{\tau}}(\mathrm{P}(\mathcal{C})).$$

(The assumption that $\mathcal{C}$ is small can be removed either by considering $\widehat{\mathrm{Set}}$-valued sheaves or by replacing $\mathrm{P}(\mathcal{C})$ by the full subcategory of presheaves that are *small* colimits of representables, cf. Remark 2.53)

Remark 6.38*.* The comparison lemma remains true for sheaves of categories (as in Remark 6.12). Indeed, in the proof, we only used the sieve description of descend, thus the proof generalizes without change to presheaves of categories. Applied to $\mathcal{C} = \mathrm{CAlg}^{\mathrm{op}}$ and τ the Zariski topology, Example 6.37 shows that a Zariski sheaf of categories on $\mathrm{CAlg}^{\mathrm{op}}$ extends uniquely to a sheaf of categories on $\mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})$. By the proof of the comparison lemma, this extension is given by right Kan extension, i.e. by forming categories of quasi-coherent objects (Definition 4.13). Granted this, Example 6.37 justifies Remark 4.42.

6.3. **Sheafification.**

Theorem 6.39. *Let $\mathcal{C}$ be a small category and τ a topology on $\mathcal{C}$. Then the inclusion $\mathrm{Sh}_\tau(\mathcal{C}) \subseteq \mathrm{P}(\mathcal{C})$ admits a left adjoint*

$$\alpha_\tau \colon \mathrm{P}(\mathcal{C}) \to \mathrm{Sh}_\tau(\mathcal{C}),$$

called τ-sheafification. Moreover:

(a) α_τ is left exact, i.e., preserves finite limits.
(b) A sieve $R \subseteq \text{ɣ}(X)$ is τ-covering if and only if $\mathrm{a}_\tau(R) \xrightarrow{\sim} \mathrm{a}_\tau(\text{ɣ}(X))$.

Proof. For $X \in \mathcal{C}$, define the partially ordered set of τ-covering sieve $\mathrm{Cov}_\tau(X) \subset \{\text{sieves on } X\}$ on X. For $Y \to X \in \mathcal{C}$ we obtain a canonical morphism $f^* : \mathrm{Cov}_\tau(X) \to \mathrm{Cov}_\tau(Y)$. We define a functor

$$L : \mathrm{P}(\mathcal{C}) \to \mathrm{P}(\mathcal{C})$$

by sending a presheaf $F \in \mathrm{P}(\mathcal{C})$ to the presheaf LF defined section wise on $X \in \mathcal{C}$ as

$$LF(X) := \operatorname*{colim}_{R \in \mathrm{Cov}_\tau(X)^{\mathrm{op}}} \mathrm{Map}(R, F),$$

with functoriality of LF coming from functoriality of the colimit in the indexing category. Functoriality of L in F comes from postcomposition. As the trivial sieve $\text{ɣ}(X) \to \text{ɣ}(X)$ is a covering sieve, we obtain a canonical morphism $(\eta_F)_X : F(X) \to LF(X)$, and hence a natural transformation $\eta_F : F \to LF$. Since all constructions where natural and functorial in F, we obtain a natural transformation of functors $\eta : \mathrm{id} \Rightarrow L$.

We now claim the following two things.

- For every sheaf $G \in \mathrm{Shv}_\tau(\mathcal{C})$ with a morphism $F \to G$ there exists a unique morphism $LF \to G$ making the following diagram commute

$$\begin{array}{ccc} F & \longrightarrow & G \\ {\scriptstyle \eta_F}\downarrow & \nearrow_{\exists!} & \\ LF & & \end{array}$$

- $L(LF)$ is a sheaf.

In particular, for every sheaf G with a morphism of presheaves $F \to G$, there exists a unique morphism of sheaves $L^2F \to G$. This shows that $\alpha_\tau := L^2$ is the desired left adjoint. We now proof the claims.

yea, todo

Our next goal is to show that α_τ is left exact. It suffices to show that L is left exact. For this, notice that $\mathrm{Cov}_\tau(X)^{\mathrm{op}}$ is a filtered partially orderd set, since the finite intersection of covering sieves is again a covering sieve. But in Set, filtered colimtis commute with finite limits, and colimits and limits of presheaves are calculated sectionwise. This shows the claim.

All that is left is to show that a sieve $R \hookrightarrow \text{ю}(X)$ is a τ-covering if and only if the induced morphism $\alpha_\tau R \xrightarrow{\cong} \alpha_\tau \text{ю}(X)$ is an isomorphism. Assume first that R is a τ-covering sieve. Then we have for every sheaf $F \in \mathrm{Shv}_\tau(\mathcal{C})$ a the commutative diagram

$$\begin{array}{ccc} \mathrm{Map}_{\mathrm{Shv}_\tau(\mathcal{C})}(\alpha_\tau \text{ю}(X), F) & \longrightarrow & \mathrm{Map}_{\mathrm{Shv}_\tau(\mathcal{C})}(\alpha_\tau R, F) \\ \cong\uparrow & & \uparrow\cong \\ \mathrm{Map}_{\mathrm{P}(\mathcal{C})}(\text{ю}(X), F) & \xrightarrow[\cong]{} & \mathrm{Map}_{\mathrm{P}(\mathcal{C})}(R, F) \end{array}$$

Here the vertical morphisms are isomorphisms by the adjunction, and the lower morphism is an isomorphism as R is a τ covering sieve. Hence the top morphism is an isomorphism. As this holds functorially for all sheaves F, we obtain that the induced morphism $\alpha_\tau R \xrightarrow{\cong} \alpha_\tau \text{ю}(X)$ induces a natural isomorphism $\mathrm{Map}_{\mathrm{Shv}_\tau(\mathcal{C})}(\alpha_\tau \text{ю}(X), -) \xrightarrow{\cong} \mathrm{Map}_{\mathrm{Shv}_\tau(\mathcal{C})}(\alpha_\tau R, -)$. By the Yoneda lemma, the claim follows. Conversely assume that $\alpha_\tau(R) \to \alpha_\tau(\text{ю}(X))$ is an isomorphism. Consider the following diagram

$$\begin{array}{ccc} R & \hookrightarrow & \text{ю}(X) \\ \downarrow & & \downarrow \\ LR & \longrightarrow & L\text{ю}(X) \\ \downarrow & & \downarrow \\ L^2R & \xrightarrow{\cong} & L^2\text{ю}(X). \end{array}$$

By adjunction, there exists further a commutative diagram

$$\begin{array}{ccc} \text{ю}(X) & \overset{\exists!}{\dashrightarrow} & L^2R \\ {\scriptstyle \eta_{\text{ю}(X)}}\downarrow & \nearrow_{\cong} & \\ L^2\text{ю}(X) & & \end{array}$$

By the explicit definition of L, the existence of a morphism $\text{ю}(X) \to L^2R$ means that there exists a τ-covering sieve S of X such that

$$\begin{array}{ccc} S & \hookrightarrow & \text{ю}(X) \\ \vdots & & \downarrow \\ LR & \longrightarrow & L\text{ю}(X) \end{array}$$

Indeed, we have

$$\mathrm{Map}(X, L^2R) \cong L^2R(X) = \operatorname*{colim}_{S \in \mathrm{Cov}_\tau(X)^{\mathrm{op}}} \mathrm{Map}(S, LR)$$

and the left hand side is non-empty.

By condition (b) of Definition 6.13, to show that R is covering, it suffices to show that for any $f \in S(Y)$ the pullback $f^*(R)$ is a covering sieve of Y. Using that L is left exact, we hence may replace S and Y by X. But thus

gives us a commutative diagram

$$\begin{array}{ccc} ⅄(X) & \xrightarrow{\mathrm{id}} & ⅄(X) \\ \downarrow{\scriptstyle \mathrm{id}} & & \downarrow \\ LR & \longrightarrow & L⅄(X) \end{array}$$

Again, using the explicit description of L, this implies that there exists a $T \in \mathrm{Cov}_\tau(X)$ such that

$$\begin{array}{ccc} T & \hookrightarrow & ⅄(X) \\ \vdots & & \downarrow{\scriptstyle \mathrm{id}} \\ R & \hookrightarrow & ⅄(X) \end{array}$$

In particular $T \hookrightarrow R$, so R contains a τ-covering sieve, and is hence itself covering. □

Remark 6.40. The assumption that $\mathcal{C}$ is small can be replaced with the following weaker assumption: for every $X \in \mathcal{C}$, there exists a small collection J of τ-covering sieves on X such that every τ-covering sieve on X contains a sieve in J. This assumption holds for example for the Zariski and fppf topologies on $\mathrm{CAlg}^{\mathrm{op}}$, so that Zariski sheafification and fppf sheafification of algebraic functors make sense. However, it does not hold for the fpqc topology on $\mathrm{CAlg}^{\mathrm{op}}$, and indeed fpqc sheafification does *not* exists.

Corollary 6.41. *Let $(\mathcal{C}, \tau)$ be a small site. Then the category $\mathrm{Sh}_\tau(\mathcal{C})$ admits limits and colimits: limits are computed objectwise, while colimits are computed by sheafifying objectwise colimits.*

Proof. This is simply the usual reflexive subcategory yoga, see for example. □ reference

Corollary 6.42. *Let $\mathcal{C}$ be a small category. Then there is an isomorphism of posets*

$$\{\textit{topologies on } \mathcal{C}\} \xrightarrow{\sim} \{\textit{full subcategories of } \mathrm{P}(\mathcal{C}) \textit{ with left exact left adjoint}\},$$

given by $\tau \mapsto \mathrm{Sh}_\tau(\mathcal{C})$.

Proof. This is a combination of Theorem 6.39 and Proposition 6.21. □

Definition 6.43 (Local epimorphism, monomorphism, isomorphism)**.** Let $(\mathcal{C}, \tau)$ be a site, let $f\colon F \to G$ be a map in $\mathrm{P}(\mathcal{C})$, and let $\Delta_f\colon F \to F \times_G F$ be its diagonal.

(a) f is a *τ-local epimorphism* if, for every $X \in \mathcal{C}$ and every $x \in G(X)$, the sieve on X consisting of all $u\colon Y \to X$ such that $u^*(x) \in G(Y)$ is in the image of $f_Y : F(Y) \to G(Y)$ is a τ-covering sieve.
(b) f is a *τ-local monomorphism*, if its diagonal Δ_f is a τ-local epimorphism.
(c) f is a *τ-local isomorphism* if it is both a τ-local epimorphism and a τ-local monomorphism.

Proposition 6.44. *Let $(\mathcal{C}, \tau)$ be a small site and let f be a monomorphism in $\mathrm{P}(\mathcal{C})$. Then $\mathrm{a}_\tau(f)$ is an epimorphism in $\mathrm{Sh}_\tau(\mathcal{C})$ if and only if f is a τ-local epimorphism. The same statement holds for monomorphisms and for isomorphisms.*

Proof. We define *image of f* as the coequalizer

$$F \times_G F \underset{\pi_2}{\overset{\pi_1}{\rightrightarrows}} F \longrightarrow \mathrm{im}(f), \qquad f\colon F \to G \text{ (with } \mathrm{im}(f) \hookrightarrow G\text{)}$$

Notice that the induced morphism $\mathrm{im}(f) \to G$ is injective, as it is so section wise. Applying α_τ and using that α_τ preserves colimits as a right adjoint, hence epimorphisms, we obtain that the map $\alpha_\tau(F) \to \alpha_\tau(\mathrm{im}(f))$ is an epimorphism.

We start with a general remark: $f : F \to G$ is a τ-local epimorphism if and only if $\alpha_\tau \operatorname{im}(f) \to \alpha_\tau G$ is an isomorphism. Indeed, by definition that f is a τ-local epimorphism isequivalent to the statement that for all $\text{ɬ}(X) \to G$, we have that in the pullback diagram

$$\begin{array}{ccc} \operatorname{im}(f) \times_G \text{ɬ}(X) & \hookrightarrow & \text{ɬ}(X) \\ \downarrow & \lrcorner & \downarrow \\ \operatorname{im}(f) & \hookrightarrow & G \end{array}$$

$\operatorname{im}(f) \times_G \text{ɬ}(X)$ is a τ-covering sieve on X. Using Theorem 6.39 (b), after applying α_τ, the top map is an isomorphism. Writing $G = \operatorname{colim}_{(X,f)\in \mathrm{El}(G)} \text{ɬ}(X)$ and $\operatorname{im}(f) = \operatorname{colim}_{(X,f)\in \mathrm{El}(G)} \big(\operatorname{im}(f) \times_G \text{ɬ}(X)\big)$, where we used universality of colimits in the second case, we see that $\alpha_\tau \operatorname{im}(f) \to \alpha_\tau G$ is an isomorphism. On the other hand, if $\alpha_\tau \operatorname{im}(f) \to \alpha_\tau G$ is an isomorphism, so is $\alpha_\tau \big(\operatorname{im}(f) \times_G \text{ɬ}(X)\big) \to \alpha_\tau \text{ɬ}(X)$, using that by Theorem 6.39 (a), sheafification preserves finite limits. But then, by Theorem 6.39 (b), $\operatorname{im}(f) \times_G \text{ɬ}(X)$ is a τ-covering sieve on X.

This now implies the first part of the proposition immediately. Indeed, if f is a τ-local epi, by the above $\alpha_\tau \operatorname{im}(f) \to \alpha_\tau G$ is an isomorphism. but $\alpha_\tau f$ factors as $\alpha_\tau F \twoheadrightarrow \alpha_\tau \operatorname{im}(f) \xrightarrow{\cong} \alpha_\tau G$, hence $\alpha_\tau f$ is an epimorphism.

Conversly, assume that $\alpha_\tau f$ is an epimorphism. Consider the following pullback

$$\begin{array}{ccc} F & & \\ & \searrow & \\ & \operatorname{coim}(f) & \longrightarrow G \\ & \downarrow & \lrcorner \quad \downarrow \\ & G & \longrightarrow G \amalg_F G \end{array}$$

and recall that in the category of sets, the coimage is isomorphic to the image. Applying α_τ, the outer square stays a pushout by cocontinuity of α_τ. We obtain that $\alpha_\tau G \to \alpha_\tau(G \amalg_F G)$ is an isomorphism, as $\alpha_\tau(f)$ was assumed to be an epimorphism. Using that $\operatorname{im}(G \to G \amalg_F G) \cong G$, we obtain from the general remark above, that $G \to G \amalg_F G$ is a τ-local epimorphism. Since τ-local epimorphisms are closed under pullback, basically by definition, we obtain that $\operatorname{im}(f) \cong \operatorname{coim}(f) \to G$ is a τ-local epimorphism. As $F \twoheadrightarrow \operatorname{im}(f)$ was an epimorphism, and epimorphisms are in particular τ-local epimorphisms, f is a composition of two τ-local epimorphisms, hence itself a τ-local epimorphism.

If now f is assumed to be a τ-local monomorphism, this is equivalent by definition to Δ_f to be a τ-local epimorphism. Using that in the category of sets, the image of the diagonal $\Delta_f : F \to F \times_G F$ is isomorphic to F, we obtain from the general remark that $\alpha_\tau(\Delta_f)$ is an isomorphism. But $\alpha_\tau(\Delta_f) = \Delta_{\alpha_\tau f}$, using that α_τ preserves finite limits, which is equivalent to α_τ being an monomorphism.

At last, assume that f is a τ-local isomorphism. This is equivalent to f being a τ-local epimorphism, and a τ-local monomorphism, which in turn, by what we have already proven, is equivalent to $\alpha_\tau(\operatorname{im}(f)) \xrightarrow{\cong} \alpha_\tau G$ being an isomorphism and $\alpha_\tau(\Delta_f)$ being an isomorphism. The second condition is equivalent to $\alpha_\tau(\pi_1) = \alpha_\tau(\pi_2)$ being an isomorphisms. But this implies that $\alpha_\tau F \xrightarrow{\cong} \alpha_\tau(\operatorname{im}(f))$. This finishes the proof. □

Corollary 6.45 (Epi-mono factorization)**.** *Let $(\mathcal{C}, \tau)$ be a small site. Every map $f\colon F \to G$ in $\mathrm{Sh}_\tau(\mathcal{C})$ factors uniquely (up to unique isomorphism) as*

$$F \xrightarrow{f_{\mathrm{epi}}} \operatorname{im}_\tau(f) \xrightarrow{f_{\mathrm{mono}}} G,$$

where f_{epi} is an epimorphism and f_{mono} is a monomorphism. This factorization is obtained by applying a_τ to the factorization $F \to \operatorname{im}(f) \to G$ in $\mathrm{P}(\mathcal{C})$. In particular, f_{epi} is the coequalizer in $\mathrm{Sh}_\tau(\mathcal{C})$ of the two projections $F \times_G F \rightrightarrows F$.

Warning 6.46. A map in $\mathrm{Sh}_\tau(\mathcal{C})$ is a monomorphism in $\mathrm{Sh}_\tau(\mathcal{C})$ if and only if it is a monomorphism in $\mathrm{P}(\mathcal{C})$ (since the inclusion $\mathrm{Sh}_\tau(\mathcal{C}) \hookrightarrow \mathrm{P}(\mathcal{C})$ preserves limits and hence monomrophisms). However, the corresponding

statements for epimorphisms does not hold: epimorphisms in $\mathrm{Sh}_\tau(\mathcal{C})$ are usually not epimorphisms in $\mathrm{P}(\mathcal{C})$, i.e., they are not objectwise surjective. For example, if R is a ring and $(f, g) = R$, then $\mathrm{Spec}(R_f \times R_g) \to \mathrm{Spec}(R)$ is an epimorphism in $\mathrm{Sh}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}})$, since it is a Zariski-local epimorphism, but it is not objectwise surjective.

In the case of sheaves on a topological space, we can rephrase the criterion of Proposition 6.44 in terms of *stalks*:

Definition 6.47 (Stalk)**.** Let T be a topological space, let $x \in T$ and let $F \in \mathrm{P}(\mathrm{Open}(T))$. The *stalk* of F at x is the set

$$F_x = \operatorname*{colim}_{x \in U} F(U),$$

where the colimit is indexed by the poset of open neighborhoods of x.

Remark 6.48. Since the colimit in the definition of the stalk F_x is a filtered colimit, the functor

$$\mathrm{P}(\mathrm{Open}(T)) \to \mathrm{Set}, \quad F \mapsto F_x,$$

preserves colimits and finite limits.

Proposition 6.49 (Stalkwise characterization of epimorphisms, monomorphisms, and isomorphisms)**.** *Let T be a topological space, let $f\colon F \to G$ be a map of $\mathrm{P}(\mathrm{Open}(T))$, and let $\mathrm{a}\colon \mathrm{P}(\mathrm{Open}(T) \to \mathrm{Sh}(T))$ be the sheafification functor. Then $\mathrm{a}(f)$ is an epimorphism (resp. a monomorphism; an isomorphism) in $\mathrm{Sh}(T)$ if and only if, for every $x \in T$, the induced map of stalks $f_x\colon F_x \to G_x$ is surjective (resp. injective, bijective).*

Proof. □ see Wedhorn

Corollary 6.50. *Let T be a topological space and let $F \in \mathrm{P}(\mathrm{Open}(T))$. Then the unit map $F \to \mathrm{a}(F)$ induced a bijection on all stalks.*

Remark 6.51 (Underlying space and Zariski sheafification)*.* Let X be an algebraic functor. By Corollary 5.35 and Proposition 6.44, the unit map $X \mapsto \mathrm{a}_{\mathrm{Zar}}(X)$ induces an isomorphism of topological spaces $|X| \xrightarrow{\sim} |\mathrm{a}_{\mathrm{Zar}}(X)|$. This implies that there is a factorization

$$\begin{array}{ccc} \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set}) & \xrightarrow{|\cdot|} & \mathrm{Top}, \\ {\scriptstyle \mathrm{a}_{\mathrm{Zar}}}\downarrow & \nearrow & \\ \mathrm{Sh}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}}) & & \end{array}$$

where the dashed arrow is colimit-preserving.

7. Schemes

Definition 7.1 (Scheme)**.** A *scheme* is a functor $X\colon \mathrm{CAlg} \to \mathrm{Set}$ satisfying the following two conditions:

(a) X is a sheaf for the Zariski topology on $\mathrm{CAlg}^{\mathrm{op}}$ (Example 6.28);
(b) X admits an open covering by affine schemes (Definition 5.40).

We denote by $\mathrm{Sch} \subseteq \mathrm{Sh}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}})$ the full subcategory spanned by *schemes*. For any scheme S, we denote by Sch_S the slice category $\mathrm{Sch}_{/S}$, whose objects are called *S-schemes*. When $S = \mathrm{Spec}(k)$ for some ring k, an S-scheme is also called a *k-scheme*.

Our goal in this section is to investigate the closure properties of the category of schemes inside the larger category of Zariski sheaves.

Proposition 7.2. *If X is a scheme and $Y \subseteq X$ is a locally closed subfunctor, then Y is a scheme.*

Example 7.3 (Examples of schemes)*.* We have the following examples of schemes:

(a) For any ring A, the affine scheme $\mathrm{Spec}(A)$ is a scheme, since it is a Zariski sheaf by Corollary 2.74. By Proposition 7.2, every quasi-affine scheme is also a scheme.
(b) For any $\mathbf{N}$-graded ring A, $\mathrm{Proj}(A)$ is a scheme: it is a Zariski sheaf by Proposition 3.64, and it admits an open covering by the affine schemes $\mathrm{D}(f) \simeq \mathrm{Spec}(A_{(f)})$ for $f \in A_+$ homogeneous (Example 5.42). In particular, for any ring k, every projective k-scheme is a scheme. By Proposition 7.2, every quasi-projective k-scheme is also a scheme.
(c) The affine line with doubled origin (Example 4.47) is a scheme: it admits an open covering by two copies of $\mathbf{A}^1$, and it follows from Zariski descent for modules that it is a Zariski sheaf.
(d) We have the following relative generalization of (a) and (b): if X is a scheme and $Y \to X$ is either a quasi-affine morphism (Definition 4.34) or quasi-projective morphism (Definition 4.34), then Y is also a scheme.
(e) Let k be a ring, M a k-module, and $n \in \mathbf{N}$. Proposition 7.2 and the Plücker embedding from Example 3.108 show that the Grassmannian $\mathrm{Gr}_n(M)$ is a scheme. More generally, for any increasing sequence of natural numbers $n = (n_1, \dots, n_s)$, the flag scheme $\mathrm{Flag}_n(M)$ is a scheme, being a closed subfunctor of the projective space $\mathbf{P}(\bigoplus_{i=1}^s \bigwedge_k^{n_i} M)$ by Example 3.111, Example 3.108, and Example 3.103.

Remark 7.4 (Residue fields of schemes)*.* Let X be a scheme and let $x \in |X|$. By Proposition 5.37, the residue field $\kappa(x)$ of X at x exists, and it coincides with the residue field at x of any open subscheme $U \subseteq X$ containing x. Moreover, by taking U to be affine, we see that the canonical map $\mathrm{Spec}(\kappa(x)) \to X$ is a monomorphism.

Definition 7.5 (Equivalence relation)**.** Let $\mathcal{C}$ be a category. An *equivalence relation* in $\mathcal{C}$ is a diagram of the form

$$R \rightrightarrows X$$

such that, for every object $Y \in \mathcal{C}$, the induced map $\mathrm{Map}(Y, R) \to \mathrm{Map}(Y, X) \times \mathrm{Map}(Y, X)$ is injective and defines an equivalence relation on $\mathrm{Map}(Y, X)$. The coequalizer of the diagram, if it exists, is called the *quotient* of X by R and denoted by X/R.

Remark 7.6 (Effective epimorphism)*.* In a category with pullbacks, any map $f\colon X \to Y$ defines an equivalence relation $X \times_Y X \rightrightarrows X$. The map f is called an *effective epimorphism* if it is recovered by taking the quotient of this relation. In the category Set, and hence in any category of sheaves $\mathrm{Sh}_\tau(\mathcal{C})$, all epimorphism are effective (Corollary 6.45), but this is not true in general. In Top, epimorphisms are surjective maps, but there are non-surjective epimorphisms, such as $\mathbf{Z} \hookrightarrow \mathbf{Q}$.

Definition 7.7 (Locally trivial relations)**.** An equivalence relation $s, t\colon R \rightrightarrows X$ in $\mathrm{Sh}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}})$ is called *locally trivial* if there is an open covering $(U_i \subseteq X)_{i \in i}$ such that the maps

$$s^{-1}(U_i) \hookrightarrow R \xrightarrow{t} X$$

are open immersions.

Lemma 7.8. *Let $(X_i)_{i\in I}$ be a family in $\mathrm{Sh}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}})$ with coproduct X. Then the canonical maps $X_i \hookrightarrow X$ are clopen immersions. In particular, $(X_i \hookrightarrow X)_{i\in I}$ is an open covering of X.*

Example 7.9 (Gluing datum). A *gluing datum* in $\mathrm{Sh}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}})$ consists of families of objects $(U_i)_{i\in I}$ and $(U_{ij})_{i,j\in I}$ and maps $U_{ij} \to U_i \times U_j$ such that:

(a) for all $i, j \in I$, the maps $U_{ij} \to U_i$ and $U_{ij} \to U_j$ are open immersions.
(b) the induced diagram
$$\coprod_{i,j\in I} U_{ij} \rightrightarrows \coprod_{i\in I} U_i$$
in $\mathrm{Sh}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}})$ is an equivalence relation.

By Lemma 7.8, the relation in (b) is the locally trivial. Condition (b) means explicitly the following:

- (Reflexivity) The diagonal map $U_i \to U_i \times U_i$ lands in U_{ii}.
- (Symmetry) The map $U_{ij} \to U_i \times U_j \xrightarrow{\sim} U_j \times U_i$ lands in U_{ji}.
- (Transitivity) The map $U_{ij} \times_{U_j} U_{jk} \to U_i \times U_k$ lands in U_{ik}.

If X is the quotient of the relation in (b), then the maps $U_i \to X$ are open immersion and form an open covering of X such that $U_{ij} \xleftarrow{\sim} U_i \times_X U_j$. Conversely, given a Zariski sheaf X and an open covering $(U_i \subseteq X)_{i\in I}$, we obtain a gluing datum by setting $U_{ij} = U_i \cap U_j$, such that the quotient of the associated relation is X.

Theorem 7.10 (Limits and colimits of schemes)**.** *The full subcategory $\mathrm{Sch} \subseteq \mathrm{Sh}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}})$ is closed under the following constructions:*

(a) finite limits;
(b) limits of cofiltered diagrams with affine transition maps;
(c) arbitrary coproducts;
(d) quotients of locally trivial equivalence relations.

Example 7.11 (The initial scheme). Since the empty sieve on the zero ring is Zariski-covering, every Zariski sheaf X satisfies $X(0) = *$ (Example 6.6). Hence, the empty scheme $\varphi = \mathrm{Spec}(0)$ (Example 2.29) is the initial object in $\mathrm{Sh}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}})$ and in Sch.

Example 7.12 (Coproducts of affine schemes). Let R and S be rings. In CAlg, the following square is a pushout square:
$$\begin{array}{ccc} R \times S & \longrightarrow & R \\ \downarrow & & \downarrow \\ S & \longrightarrow & 0 \end{array}$$
By Example 6.7, every Zariski sheaf X satisfies $X(R \times S) \xrightarrow{\sim} X(R) \times X(S)$. This means that
$$\mathrm{Spec}(R) \sqcup \mathrm{Spec}(S) \xrightarrow{\sim} \mathrm{Spec}(R \times S)$$
in $\mathrm{Sh}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}})$ and hence in Sch.

Warning 7.13. By Example 7.11 and Example 7.12, the inclusion $\mathrm{Aff} \subseteq \mathrm{Sch}$ preserves finite coproducts. Note however that is does not preserve infinite coproducts: given an infinite family of nonzero rings $(R_i)_{i\in I}$, the scheme $X = \coprod_{i\in I} \mathrm{Spec}(R_i)$ is not affine. Indeed, X admits an open covering by the subschemes $\mathrm{D}(e_i) = \mathrm{Spec}(R_i)$, but the family of functions $(e_i)_{i\in I}$ does not generate the unit ideal in $\mathcal{O}(X) = \prod_{i\in I} R_i$.

Example 7.14 (Coproducts of projective spaces). Let k be a ring and let M and N be k-modules. The two projections $M \oplus N \twoheadrightarrow M$ and $M \oplus N \twoheadrightarrow N$ induce a canonical map of k-schemes
$$\vartheta\colon \mathbf{P}(M) \sqcup \mathbf{P}(N) \to \mathbf{P}(M \oplus M),$$
which is an "additive" analogue of the Segre embedding (Example 3.103). If $M = k^{(I)}$ and $N = k^{(J)}$, this is a map
$$\vartheta\colon \mathbf{P}^I_k \sqcup \mathbf{P}^J_k \to \mathbf{P}^{I\sqcup J}_k.$$

If $M = k^{m+1}$ and $N = k^{n+1}$, this becomes the map

$$\vartheta\colon \mathbf{P}^m_k \sqcup \mathbf{P}^n_k \to \mathbf{P}^{m+n+1}_k,$$

which is given in coordinates by the formulas

$$[x_0, : \ldots : x_m] \mapsto [x_0 : \ldots : x_m : 0 : \ldots : 0]$$
$$[y_0 : \ldots : y_n] \mapsto [0 : \ldots : 0 : y_0 : \ldots : y_n].$$

We claim that ϑ is a closed immersion. If $(e_i)_{i\in I}$ and $(f_j)_{j\in J}$ are generating families of M and N, then $\mathbf{P}(M \oplus N)$ has an open covering by the subschemes $\mathrm{D}(e_i, 0)$ and $\mathrm{D}(0, f_j)$. As ϑ is a map of Zariski sheaves, it suffices by Proposition 2.82 to show that the pullback of ϑ to these open subschemes is a closed immersion. Since the preimage of $\mathrm{D}(e_i, 0)$ in $\mathbf{P}(N)$ is the empty scheme $\mathrm{D}(0)$, the pullback of ϑ to $\mathrm{D}(e_i, 0)$ is the same as the pullback of $\vartheta|_{\mathbf{P}(M)}$, which is a closed immersion. Similarly, the pullback of ϑ to $\mathrm{D}(0, f_j)$ is also the pullback of the closed immersion of $\vartheta|_{\mathbf{P}(M)}$, which proves the claim. More generally, there is a canonical monomorphism of k-schemes $\coprod_{i\in I} \mathbf{P}(M_i) \hookrightarrow \mathbf{P}(\bigoplus_{i\in I} M_i)$ for any family of k-modules $(M_i)_{i\in I}$, which is a closed immersion when I is finite.

Proposition 7.15. *Let S be a scheme. Then the subcategories*

$$\mathrm{Aff}_S \subseteq \mathrm{QAff}_S \subseteq \mathrm{Sch}_S \supset \mathrm{QProj}_S \supset \mathrm{Proj}_S$$

of affine, quasi-affine, projective, and quasi-projective S-schemes are closed under finite limits and finite coproducts.

Proposition 7.16 (Schemes as colimits of affine schemes)**.** *Let X be a scheme and let $(U_i \subseteq X)_{i\in I}$ and $(U_{ijk} \subseteq U_i \cap U_j)_{k\in K(i,j)}$ be open coverings by affine schemes. Then the diagram*

$$\coprod_{i,j,k} U_{ijk} \rightrightarrows \coprod_i U_i \to X$$

is a coequalizer in $\mathrm{Sh}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}})$ and hence in Sch.

Example 7.17 (Proj via gluing)*.* Let A be an $\mathbf{N}$-graded ring and let $I \subseteq A_+$ be a homogeneous subset such that $A_+ \subseteq \sqrt{(I)}$, so that $(\mathrm{D}(f) \subseteq \mathrm{Proj}(A))_{f\in I}$ is an open covering. Since $\mathrm{D}(f) \simeq \mathrm{Spec}(A_{(f)})$ and $\mathrm{D}(f) \cap \mathrm{D}(g) = \mathrm{D}(fg)$, we obtain a coequalizer diagram

$$\coprod_{f,g\in I} \mathrm{Spec}(A_{(fg)}) \rightrightarrows \coprod_{f\in I} \mathrm{Spec}(A_{(f)}) \to \mathrm{Proj}(A)$$

in $\mathrm{Sh}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}})$, which presents $\mathrm{Proj}(A)$ as gluing of the affine schemes $\mathrm{Spec}(A_{(f)})$.

Example 7.18 (Projective spaces via gluing)*.* Let $n \geq -1$ and let us specialize Example 7.17 to the case $A = \mathbf{Z}[x_0, \ldots, x_n]$, so that $\mathrm{Proj}(A) = \mathbf{P}^n$. Recall that $\mathbf{P}^n(R)$ is the set of quotient lines of $R^{\{0,\ldots,n\}} \twoheadrightarrow L$ such that a_i is an isomorphism, which is isomorphism $\mathbf{A}^n$. The intersection $\mathrm{D}(x_i) \cap \mathrm{D}(x_j) = \mathrm{D}(x_i x_j)$ is the subfunctor where both a_i and a_j are isomorphisms, which is isomorphic $\mathbf{A}^{n-1} \times \mathbf{G}_m$ if $i \neq j$. We therefore have a coequalizer in $\mathrm{Sh}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}})$ of the form

$$\coprod_{i<j} (\mathbf{A}^{n-1} \times \mathbf{G}_m) \rightrightarrows \coprod_{i=0}^{n} \mathbf{A}^n \to \mathbf{P}^n.$$

Example 7.19*.* Let X be the affine line with doubled origin (see Example 4.47). Similar to $\mathbf{P}^1$, X is obtained by gluing two copies of $\mathbf{A}^1$ along a copy of $\mathbf{G}_m$. However, the gluing data for X and for $\mathbf{P}^1$ are different: if $u\colon \mathbf{G}_m \hookrightarrow \mathbf{A}^1$ is the open immersion given by $t \mapsto t$ and $v\colon \mathbf{G}_m \hookrightarrow \mathbf{A}^1$ is the open immersion given by $t \mapsto t^{-1}$, then we have the following pushout squares of open immersions in Sch:

$$\begin{array}{ccc} \mathbf{G}_m & \xrightarrow{u} & \mathbf{A}^1 \\ {\scriptstyle v}\downarrow & & \downarrow \\ \mathbf{A}^1 & \longrightarrow & \mathbf{P}^1, \end{array} \qquad \begin{array}{ccc} \mathbf{G}_m & \xrightarrow{u} & \mathbf{A}^1 \\ {\scriptstyle u}\downarrow & & \downarrow \\ \mathbf{A}^1 & \longrightarrow & X. \end{array}$$

Example 7.20 (Projective space via group actions). Let $n \geq 0$. The affine group scheme $\mathbf{G}_m$ acts on the punctured affine space $\mathbf{A}^n \setminus \{0\}$ by scaling. By Proposition 3.28, the canonical map

$$(\mathbf{A}^{\setminus}\{0\})/\mathbf{G}_m \to \mathbf{P}^{n-1}$$

is a monomorphism as well as a Zariski-local epimorphism. By Proposition 6.44, it is thus an *isomorphism* if we compute the quotient in $\mathrm{Sh}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}})$. In other words, the diagram

$$\mathbf{G}_m \times (\mathbf{A}^n \setminus \{0\}) \underset{\text{proj}}{\overset{\text{act}}{\rightrightarrows}} \mathbf{A}^n \setminus \{0\} \longrightarrow \mathbf{P}^{n-1}$$

is a coequalizer in $\mathrm{Sh}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}})$. Since it is a diagram of schemes, it is also a coequalizer in the category of schemes, i.e., $\mathbf{P}^{n-1}$ is the quotient of $\mathbf{A}^n \setminus \{0\}$ by scaling action of $\mathbf{G}_m$ in the category of schemes. More generally, if A is an $\mathbf{N}$-graded ring generated by $A_{\leq 1}$, then $\mathrm{Proj}(A) \simeq \mathrm{D}(A_+)/\mathbf{G}_m$ as Zariski sheaves and hence as schemes, where $\mathrm{D}(A_+) \subseteq \mathrm{Spec}(A)$ (see Remark 3.71). For A an arbitrary $\mathbf{N}$-graded ring, one can show that $\mathrm{Proj}(A)$ is still the quotient $\mathrm{D}(A_+)/\mathbf{G}_m$ in the category of schemes, even though this is not true anymore in $\mathrm{Sh}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}})$.

Remark 7.21 (Zariski topology on the category of schemes). By Proposition 7.2, any open subfunctor of a scheme is a scheme. Hence, open covering form a pretopology on Sch, whose associated topology is called the *Zariski topology*. The inclusion $\mathrm{CAlg}^{\mathrm{op}} \simeq \mathrm{Aff} \subseteq \mathrm{Sch}$ satisfies the assumption of the comparison lemma (Lemma 6.32), so that the restriction functor $\mathrm{P}(\mathrm{Sch}) \to \mathrm{P}(\mathrm{CAlg}^{\mathrm{op}})$ induces an equivalence of categories

$$\mathrm{Sh}_{\mathrm{Zar}}(\mathrm{Sch}) \xrightarrow{\sim} \mathrm{Sh}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}}).$$

7.1. Separatedness.

Definition 7.22 (Locally separated, separated)**.** A morphism of algebraic functors $f\colon Y \to X$ is *locally separated* if its diagonal $\Delta_f\colon Y \to Y \times_X Y$ is an immersion, and it is *separated* if Δ_f is a closed immersion. An algebraic functor X is *locally separated* (resp. *separated*) if the unique map $X \to \mathrm{Spec}(\mathbf{Z})$ is, i.e., if the diagonal $\Delta_X\colon X \to X \times X$ is an immersion (resp. a closed immersion).

For schemes, "locally separated" turns out to be vacuous, so this notion is rarely used:

Proposition 7.23 (Diagonal of schemes)**.** *Any morphism of schemes $f\colon Y \to X$ is locally separated, i.e., its diagonal $\Delta_f\colon Y \to Y \times_X Y$ is an immersion.*

Proposition 7.24 (Closure properties of separated morphisms)**.** *The following are some closure properties of separated morphisms.*

(a) Consider a commutative triangle of algebraic functors

$$\begin{array}{ccc} Z & \xrightarrow{g} & Y \\ & \searrow^{h} & \downarrow{\scriptstyle f} \\ & & X. \end{array}$$

If f and g are separated, so is h. If h is separated, then so is g. If h is a closed immersion and f is separated. The same results holds with "locally separated" and "immersion".

(b) Consider a cartesian square of algebraic functors

$$\begin{array}{ccc} Y' & \xrightarrow{f'} & X' \\ \downarrow & & \downarrow{\scriptstyle g} \\ Y & \xrightarrow{f} & X \end{array}$$

If f is (locally) separated, so is f'. The converse holds if X and Y are Zariski sheaves and g is a Zariski local epimorphism.

Corollary 7.25 (Sections of morphisms of schemes)**.** *Let $f\colon X \to S$ be a morphism of schemes and $s\colon S \to X$ a section of f. Then s is an immersion. If f is separated, then s is a closed immersion.*

Example 7.26. (a) Any monomorphism is separated, since its diagonal is an isomorphism.
(b) Any affine scheme is separated: the diagonal of $\mathrm{Spec}(R)$ can be identified with Spec of the multiplication map $R \otimes R \to R$ (which is the diagonal of R in $\mathrm{CAlg}^{\mathrm{op}}$), which is surjective. More generally, any affine morphism and hence by (a) any quasi-affine morphism is separated.
(c) If A is an $\mathbf{N}$-graded ring, then $\mathrm{Proj}(A)$ is separated: using the standard affine open covering of $\mathrm{Proj}(A)$ and (b) of Proposition 7.24, this follows from the observation that the multiplication maps $A_{(f)} \otimes A_{(g)} \to A_{(fg)}$ are surjective. More generally, if A is a quasi-coherent $\mathbf{N}$-graded algebra over X, then $\mathrm{Proj}(A) \to X$ is separated. In particular, any projective morphism and hence by (a) any quasi-projective morphism is separated.
(d) The affine line with doubled origin (Example 4.47) is not separated.

7.2. **Quasi-compactness and quasi-separatedness.** By Proposition 7.16, any scheme can be written as the colimit of a diagram of affine schemes and open immersions. A scheme is called *quasi-compact* and *quasi-separated*, or *qcqs* for short, if a *finite* such diagram exists. In this section, we discuss in more details this finiteness condition, which makes sense more generally for algebraic functors.

Definition 7.27 (Quasi-compact and quasi-separated)**.** (a) Any algebraic functor X is called *quasi-compact* if, for every Zariski-local epimorphism $\coprod_{i\in I} Y_i \to X$, there exists a finite subset $J \subseteq I$ such that $\coprod_{i\in J} Y_i \to X$ is a Zariski-local epimorphism.
(b) A morphism of algebraic functors $f\colon Y \to X$ is called *quasi-compact* if, for every morphism $Z \to X$ with Z quasi-compact, the pullback $Y \times_X Z$ is quasi-compact.
(c) An algebraic functor X is called *quasi-separated* if, for every diagram $Y \to X \leftarrow Z$ with Y and Z quasi-compact, the pullback $Y \times_X Z$ is quasi-compact.
(d) A morphism of algebraic functors $f\colon Y \to X$ is called *quasi-separated* if its diagonal $\Delta_f\colon Y \to Y \times_X Y$ is quasi-compact.

Proposition 7.28 (Characterization of quasi-compactness)**.** *For an algebraic functor X, the following conditions are equivalent:*

(a) X is quasi-compact;
(b) the unique map $X \to \mathrm{Spec}(\mathbf{Z})$ is quasi-compact;
(c) there exists a Zariski-local epimorphism $\coprod_{i\in I} Y_i \to X$ where I is finite and each Y_i affine.

Proposition 7.29 (Characterization of quasi-separatedness)**.** *For an algebraic functor X, the following conditions are equivalent:*

(a) X is quasi-separated;
(b) the unique map $X \to \mathrm{Spec}(\mathbf{Z})$ is quasi-separated;
(c) for all maps $Y \to X$ and $Z \to X$, where Y and Z are affine, $Y \times_X Z$ is quasi-compact.

Proposition 7.30 (Composition of qc/qs morphisms)**.** *Consider a commutative triangle of algebraic functors*

$$\begin{array}{ccc} Z & \xrightarrow{g} & Y \\ & \searrow^{f} & \downarrow{\scriptstyle f} \\ & & X. \end{array}$$

(a) If f and g are quasi-compact, then h is quasi-compact.
(b) If f and g are quasi-separated, then h is quasi-separated.
(c) If h is quasi-compact and f is quasi-separated, then g is quasi-compact.
(d) If h is quasi-separated, then g is quasi-separated.
(e) If h is quasi-compact and g is a Zariski-local epimorphism, then f is quasi-compact.
(f) If h is quasi-separated and g is quasi-compact and a Zariski local epimorphism, then f is quasi-separated.

Proposition 7.31 (Base change of qc/qs morphisms)**.** *Consider a cartesian square of algebraic functors*

$$\begin{array}{ccc} Y' & \xrightarrow{f'} & X' \\ \downarrow & & \downarrow g \\ Y & \xrightarrow{f} & X \end{array}$$

If f is quasi-compact, then f' is quasi-compact. The converse holds if g is a Zariski-local epimorphism. The same statements holds with "quasi-separated".

Example 7.32 (Quasi-compact immersions)*.*
(a) By Proposition 7.28, a closed subfunctor of a quasi-compact algebraic functor is again quasi-compact. It follows that any closed immersion is quasi-compact. In particular, as the terminology suggests any separated morphism is quasi-separated.
(b) Let R be a ring and let $I \subseteq R$. Consider the open covering $(\mathrm{D}(f) \subseteq \mathrm{D}(I))_{f\in I}$ of $\mathrm{D}(I)$, we see that the schemes $\mathrm{D}(I)$ is quasi-compact if and only if the radical ideal $\sqrt{(I)}$ is finitely generated (i.e., equals $\sqrt{(J)}$ for some finite set J). Hence, an open immersion is quasi-compact if and only if the correpsonding quasi-coherent radical ideal is finitely generated.
(c) Any monomorphism is quasi-separated, since its diagonal is an isomorphism.

Example 7.33 (Quasi-affine and quasi-projective schemes)*.* The following are some examples of quasi-affine and quasi-projective schemes.

(a) By (b) of Example 7.32, any quasi-affine scheme is quasi-compact. It is also quasi-separated, since it is separated by (b) of Example 7.26.
(b) Let A be an $\mathbf{N}$-graded ring and let $I \subseteq A$ be a homogenization subset. As in (b) of Example 7.32, $\mathrm{D}(I)$ is quasi-compact if and only if the saturated radical ideal generated by I is finitely generated. In particular, for any ring k, any quasi-projective k-scheme is quasi-compact. It is also quasi-separated, since it is separated by (c) of Example 7.26.
(c) It follows from (a) and (b) and Proposition 7.31 that any quasi-affine or quasi-projective morphism of algebraic functors is quasi-compact and quasi-separated.

For schemes, these finiteness conditions depend only on the underlying topological space, and they can be checked using any affine open covering:

Proposition 7.34 (Quasi-compactness for schemes)**.** *Let X be a scheme and $(U_i \subseteq X)_{i\in I}$ an open covering by affine schemes. The following are equivalent:*

(a) X is quasi-compact;
(b) the topological space $|X|$ is quasi-compact;
(c) there exists a finite subset $J \subseteq I$ such that $(U_i \subseteq X)_{i\in J}$ is an open covering of X.

Proposition 7.35 (Quasi-separatedness for schemes)**.** *Let X be a scheme and $(U_i \subseteq X)_{i\in I}$ an open covering by affine schemes. The following are equivalent:*

(a) X if quasi-separated;
(b) the intersection of any two quasi-compact open subsets of $|X|$ is quasi-compact;
(c) for every $i, j \in I$, the intersection $U_i \cap U_j$ is quasi-compact.

Remark 7.36*.* By Proposition 7.34 and Proposition 7.35, a scheme is quasi-compact and quasi-separated (often abbreviated as *qcqs*) if and only if it can be written as a *finite* colimit of affine schemes as in Proposition 7.16. The finiteness property propagates to many invariants: for example, the underlying space of a qcqs scheme is spectral, and its category of quasi-coherent modules is compactly generated.

As Example 7.33 shows, most schemes of interest are qcqs. Examples of schemes that are not quasi-compact are infinite coproducts of nonempty schemes and projective spaces $\mathbf{P}^I_k$, where I is an infinite set and k a nonzero ring. An infinite-dimensional affine space with doubled origin is an example of a quasi-compact scheme that is not quasi-separated.

The following standard definition combines the local "algebraic" finiteness property of Section 2.9 with the global "topological" finiteness property of this section:

Definition 7.37 (Morphism of finite presentation/type)**.** Let $f\colon X \to S$ be a map of algebraic fucntors.

(a) f is of *finite presentation* if it is locally of finite presentation, quasi-compact, and quasi-separated.
(b) f is of *finite type* if it is locally of finite type and quasi-compact.

For example, any quasi-projective morphism is of finite type.

7.3. **Locally ringed spaces.** If $f\colon X \to Y$ is a continuous map of topological spaces, the induced map of posets $f^{-1}\colon \mathrm{Open}(Y) \to \mathrm{Open}(X)$ preserves open coverings. It follows that the pushforward functor

$$f_* = (f^{-1})^*\colon \mathrm{P}(\mathrm{Open}(X)) \to \mathrm{P}(\mathrm{Open}(Y)), \quad f_*(F)(V) = F(f^{-1}(U)),$$

preserves sheaves:

$$\begin{array}{ccc} \mathrm{P}(\mathrm{Open}(X)) & \xrightarrow{f_*} & \mathrm{P}(\mathrm{Open}(Y)) \\ \uparrow & & \uparrow \\ \mathrm{Sh}(X) & \xrightarrow[f_*]{} & \mathrm{Sh}(Y). \end{array}$$

Proposition 7.38 (Pullbacks of sheaves)**.** *Let $f\colon X \to Y$ be a continuous map. Then the pushforward functor $f_*\colon \mathrm{Sh}(X) \to \mathrm{Sh}(Y)$ admits a left exact adjoint $f^*\colon \mathrm{Sh}(Y) \to \mathrm{Sh}(X)$. Moreover, for any $F \in \mathrm{Sh}(Y)$ and any $x \in X$, there is a canonical isomorphism of stalks $f^*(F)_x \simeq F_{f(x)}$.*

Remark 7.39. Let $f\colon X \to Y$ be a continuous map. Since both functors f_* and f^* preserve finite products, they preserve algebraic structures (see Remark 2.13). Hnece, they induce adjunctions between categories of sheaves of monoids, groups, rings, etc.

Definition 7.40 (Ringed space)**.** Let k be a ring. A *k-ringed space* $(X, \mathcal{O}_X)$ is a topological space X together with a sheaf of k-algebras $\mathcal{O}_X$ on X. A morphism of k-ringed spaces $(f, \varphi)\colon (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a continuous map $f\colon X \to Y$ together with a map of sheaves of k-algebras $\varphi\colon \mathcal{O}_Y \to f_*(\mathcal{O}_X)$. We denote by RS_k the category of k-ringed spaces. When $k = \mathbf{Z}$, we write $\mathrm{RS} = \mathrm{RS}_{\mathbf{Z}}$ for the category of *ringed spaces*.

Remark 7.41. By Proposition 7.38, a map of sheaves of k-algebras $\varphi\colon \mathcal{O}_Y \to f_*(\mathcal{O}_Y)$ is equivalently a map of sheaves of k-algebras $f^*(\mathcal{O}_Y) \to \mathcal{O}_X$. Moreover, for any point $x \in X$, the induced map on stalks has the form

$$\varphi_x\colon \mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}.$$

Definition 7.42 (Local map)**.** A ring map $\varphi\colon A \to B$ is *local* if it detects units, i.e., if $A^\times = \varphi^{-1}(B^\times)$.

Remark 7.43 (Local maps between local rings)*.* Let A and B by local rings with maximal ideals $\mathfrak{m}_A$ and $\mathfrak{m}_B$. For a ring map $\varphi\colon A \to B$, the following are equivalent:

(a) φ is local;
(b) $\mathfrak{m}_A = \varphi^{-1}(\mathfrak{m}_B)$;
(c) $\varphi(\mathfrak{m}_A) \subseteq \mathfrak{m}_B$.

Definition 7.44 (Locally ringed space)**.** Let k be a ring. A *locally k-ringed space* is a k-ringed space $(X, \mathcal{O}_X)$ such that, for each $x \in X$, the stalk $\mathcal{O}_{X,x}$ is a local ring. If $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ are locally k-ringed spaces, a morphism of locally k-ringed spaces $(f, \varphi)\colon (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a morphism of k-ringed spaces such that, for each $x \in X$, the map $\varphi_x\colon \mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}$ is a local ring map.

We denote by LRS_k the category of locally k-ringed spaces, which is a (non-full!) subcategory of RS_k. When $k = \mathbf{Z}$, we write $\mathrm{LRS} = \mathrm{LRS}_{\mathbf{Z}}$ for the category of *locally ringed spaces*.

Remark 7.45 (Alternative description of locally ringed spaces)*.* Let $(X, \mathcal{O}_X)$ be a ringed space. Given an open $U \subseteq X$ and a section $s \in \mathcal{O}_X(U)$, let

$$U_s = \{x \in U \mid s_x \in \mathcal{O}_{X,x} \text{ is a unit}\}.$$

By definition of the stalk, U_s is an open subset of U. Moreover, since $\mathcal{O}_X$ is a sheaf, the restriction of s to U_s is a unit, so that U_s is the largest open subset of U with this property. Recall that a ring R is loal if and only if the non-units form a subgroup of $(R,+)$, and a map of rings is local if and only if it sends non-units to non-units. We deduce the following description of the subcategory LRS $\subseteq$ RS:

(a) A ringed space $(X,\mathcal{O}_X)$ is a locally ringed space if and only if $X_0 = X$ and for all open subsets $U \subseteq X$ and sections $s,t \in \mathcal{O}_X(U)$, $U_{s+t} \subseteq U_s \cup U_t$.
(b) If $(X,\mathcal{O}_X)$ and $(Y,\mathcal{O}_Y)$ are locally ringed spaces and $(f,\varphi)\colon (X,\mathcal{O}_X) \to (Y,\mathcal{O}_Y)$ is a map in RS, then (f,φ) is a map in LRS if and only if, for every open $V \subseteq Y$ and every section $t \in \mathcal{O}_Y(V)$, we have $f^{-1}(V_s) = f^{-1}(V)_{\varphi(s)}$.

This alternative description makes no reference to the points of the underlying spaces and can be used to define the category of *locally ringed locales*, which enlarges the category of locally ringed sober spaces.

It turns out that the concept of locally ringed spaces unifies many kinds of geometric objects in mathematics. To exemplify this, we introduce some notation:

Notation 7.46. Let k be a ring and let $\mathcal{M}$ be a collection of locally k-ringed spaces. We denote by $\mathrm{LRS}_k(\mathcal{M}) \subseteq \mathrm{LRS}_k$ the full subcategory of locally k-ringed spaces $(X,\mathcal{O}_X)$ with the following property: every $x \in X$ admits an open neighborhood U such that $(U,\mathcal{O}_{X|U})$ is isomorphic to an object of $\mathcal{M}$.

Example 7.47 (Manifolds as locally ringed spaces). (a) Let $r \in \mathbf{N} \cup \{\infty,\omega\}$. For any open subset $U \subseteq \mathbf{R}^n$, denote by $\mathcal{C}^r_U$ the sheaf of ($\mathbf{R}$-valued) functions of class C^r on U. Let $\mathcal{D}\mathrm{isk}_{C^r}$ be the collection of locally $\mathbf{R}$-ringed spaces $(U,\mathcal{C}^r_U)$, where U is the open unit disk in $\mathbf{R}^n$ for some $n \in \mathbf{N}$. Then there are equivalences of categories

$$\mathrm{LRS}(\mathcal{D}\mathrm{isk}_{C^r}) \simeq \mathrm{LRS}(\mathcal{D}\mathrm{isk}_{C^r}) \simeq \{\mathrm{C}^r\text{-manifolds}\}.$$

In particular, we get *topological manifolds* for $r = 0$, *smooth manifolds* for $r = \infty$, and *analytic manifolds* for $r = \omega$.[25]

(b) For an open subset $U \subseteq \mathbf{C}^n$, denote by $\mathcal{H}_U$ the sheaf of ($\mathbf{C}$-valued) holomorphic functions on U. Let $\mathcal{D}\mathrm{isk}_{\mathbf{C}}$ be the collection of locally $\mathbf{C}$-ringed spaces $(U,\mathcal{H}_U)$, where U is the open unit disk in $\mathbf{C}^n$ for some $n \in \mathbf{N}$. Then there is an equivalence of categories

$$\mathrm{LRS}_{\mathbf{C}}(\mathcal{D}\mathrm{isk}_{\mathbf{C}}) \simeq \{\text{complex manifolds}\}.$$

(c) More generally, many categories of manifolds with boundary, with corners, and with various types of singularities are equivalent to categories of the form $\mathrm{LRS}_{\mathbf{R}/\mathbf{C}}(\mathcal{M})$.

Remark 7.48 (Colimits an limits of ringed spaces). The category RS_k admits colimits and limits. If $(X_i,\mathcal{O}_X)_{i\in\mathcal{J}}$ is a diagram of k-ringed spaces, and if $\iota_i\colon X_i \to \mathrm{colim}_{i\in\mathcal{J}} X_i$ and $\pi\colon \lim_{i\in\mathcal{J}} X_i \to X_i$ are the canonical maps in Top, then

$$\begin{aligned}\operatorname*{colim}_{i\in\mathcal{J}}(X_i,\mathcal{O}_{X_i}) &= (\operatorname*{colim}_{i\in\mathcal{J}} X_i, \lim_{i\in\mathcal{J}}(\iota_i)_*(\mathcal{O}_{X_i})),\\ \lim_{i\in\mathcal{J}}(X_i,\mathcal{O}_{X_i}) &= (\lim_{j\in\mathcal{J}} X_i, \operatorname*{colim}_{i\in\mathcal{J}}(\pi_i)^*(\mathcal{O}_{X_i})).\end{aligned}$$

Proposition 7.49 (Colimits of locally ringed spaces)**.** *The category* LRS_k *admits colimits and the inclusion functor* $\mathrm{LRS}_k \hookrightarrow \mathrm{RS}_k$ *preserves colimits.*

Remark 7.50. The category LRS_k also admits limits, but this is more subtle as the inclusion $\mathrm{LRS}_k \hookrightarrow \mathrm{RS}_k$ does not preserve them. For example, it follows from Theorem 7.54 below that the final object of LRS_k is the underlying space of $\mathrm{Spec}(k)$ (i.e., the prime spectrum of k) equipped with the sheaf of functions.

[25] It is common to further require C^r-manifolds to be Hausdorff, second-countable, and of bounded dimension; these conditions are equivalent to being embeddable in $(\mathbf{R}^n,\mathcal{C}^r)$ for some n.

7.4. **Schemes as locally ringed spaces.** Let X be an algebraic functor. Denote by $\mathcal{O}_X$ the presheaf of functions on $\mathrm{Open}(X) \simeq \mathrm{Open}(|X|)$, given By

$$\mathcal{O}_X(U) = \mathcal{O}(U) = \mathrm{Map}(U, \mathbf{A}^1).$$

By Corollary 4.43, $\mathcal{O}_X$ is then a sheaf of rings on the underlying topological space $|X|$. This construction defines a functor

$$\mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set}) \to \mathrm{RS}, \quad X \mapsto (|X|, \mathcal{O}_X).$$

By Zariski descent for the underlying space (Remark 6.51) and for functions (Corollary 4.43), this functor factor through the Zariski sheafification functor $\mathrm{a}_{\mathrm{Zar}} \colon \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set}) \to \mathrm{Sh}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}})$.

Warning 7.51. The notation $\mathcal{O}_X$ is used for two different objects associated with the algebraic functor X: it is the unit objecty in the category Mod_X of quasi-coherent modules over X (Notation 4.20), and it is the sheaf of rings $\mathcal{O}_{\mathrm{Open}(X)}$ on the underlying space $|X|$. These are two different ways of restricting the functor $\mathcal{O} \colon \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})^{\mathrm{op}} \to \mathrm{CAlg}$ to X. We will see in Section 7.5 in which sense these two objects are equivalent when X is a scheme.

Example 7.52 (Prime spectra as locally ringed spaces)*.* Let R be a ring and let $X = \mathrm{Spec}(R)$. Then the ringed space $(|X|, \mathcal{O}_X)$ has the following explicit description:

- The topological space $|X|$ is the prime spectrum $\mathrm{Prim}(R)$.
- Under the equivalence of categories $\mathrm{Sh}(\mathrm{Prim}(R)) \simeq \mathrm{Sh}(\mathrm{Open}^{\mathrm{pr}}(\mathrm{Spec}(R)))$ of Example 6.36, the sheaf $\mathcal{O}_X$ is given by $\mathcal{O}_X(\mathrm{D}(f)) = R_f$.

Given a prime ideal $\mathfrak{p} \in \mathrm{Prim}(R)$, the stalk of $\mathcal{O}_X$ at $\mathfrak{p}$ is the local ring $R_{\mathfrak{p}}$:

$$\mathcal{O}_{X,\mathfrak{p}} = \operatorname*{colim}_{\mathfrak{p} \in |\mathrm{D}(f)|} \mathcal{O}_X(\mathrm{D}(f)) = \operatorname*{colim}_{f \notin \mathfrak{p}} R_f = R_{\mathfrak{p}}.$$

Hence, the ringed space $(\mathrm{Prim}(R), \mathcal{O}_{\mathrm{Spec}(R)})$ is in fact a locally ringed space. Moreover, for any ring map $\varphi \colon R \to S$, the induced map $(\mathrm{Prim}(S), \mathcal{O}_{\mathrm{Spec}(S)}) \to (\mathrm{Prim}(R), \mathcal{O}_{\mathrm{Spec}(R)})$ is a map of locally ringed spaces: for any prime ideal $\mathfrak{q} \in \mathrm{Prim}(S)$, the induced map on stalks is the local map $\varphi_{\mathfrak{q}} \colon R_{\varphi^{-1}} \to S_{\mathfrak{q}}$.

Recall that both functors

$$|-| \colon \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set}) \to \mathrm{Top} \quad \text{and} \quad \mathcal{O} \colon \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set}) \to \mathrm{CAlg}^{\mathrm{op}}$$

preserves colimits (Proposition 5.24 and (d) of Remark 2.52). Using the explicit description of colimits in RS from Remark 7.48, we deduce that the functor at the beginning of the this section is colimit-preserving. By the universal property of presheaves (Theorem 2.9), it is therefore the unique colimit-preserving extension of the functor

$$\mathrm{CAlg}^{\mathrm{op}} \to \mathrm{RS}, \quad R \mapsto (\mathrm{Prim}(R), \mathcal{O}_{\mathrm{Spec}(R)}).$$

This functor lands in the (non-full) subcategory $\mathrm{LRS} \subseteq \mathrm{RS}$ by Example 7.52. By Proposition 7.49, we deduce that its colimit preserving extension also lands in the subcategory $\mathrm{LRS} \subseteq \mathrm{RS}$. The following proposition summarizes these observation:

Proposition 7.53. *There is a factorization*

$$\begin{array}{ccc} \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set}) & \xrightarrow{(|-|,\mathcal{O})} & \mathrm{RS} \\ {\scriptstyle \mathrm{a}_{\mathrm{Zar}}}\downarrow & & \uparrow \\ \mathrm{Sh}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}}) & \dashrightarrow & \mathrm{LRS}. \end{array}$$

Theorem 7.54 (The adjunction between rings and locally ringed spaces)**.** *The global section functor*

$$\mathrm{LRS} \to \mathrm{CAlg}^{\mathrm{op}}, \quad (X, \mathcal{O}_X) \mapsto \mathcal{O}_X(X)$$

has a right adjoint given By

$$\mathrm{CAlg}^{\mathrm{op}} \to \mathrm{LRS}, \quad R \mapsto (\mathrm{Prim}(R), \mathcal{O}_{\mathrm{Spec}(R)}).$$

Since $\mathcal{O}(\mathrm{Spec}(R)) \simeq R$, the counit of this adjunction is and isomorphism, which implies the following:

Corollary 7.55. *The prime spectrum functor*

$$\mathrm{CAlg}^{\mathrm{op}} \to \mathrm{LRS}, \quad R \mapsto (\mathrm{Prim}(R), \mathcal{O}_{\mathrm{Spec}(R)})$$

is fully faithful.

The following result follows from Corollary 7.55 by "standard descent arguments":

Theorem 7.56 (Schemes as locally ringed spaces)**.** *Let* $\mathcal{P}\mathrm{rim}$ *be the collection of locally ringed spaces* $(\mathrm{Prim}(R), \mathcal{O}_{\mathrm{Spec}(R)})$*, where* R *is any ring. Then the functor* $(|-|, \mathcal{O})\colon \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set}) \to \mathrm{LRS}$ *restricts to an equivalence of categories*

$$\mathrm{Sch} \xrightarrow{\sim} \mathrm{LRS}(\mathcal{P}\mathrm{rim}).$$

Remark 7.57. Both the algebraic functor $\mathrm{Spec}(R)$ and the locally ringed space $\mathrm{Prim}(R)$ are universal objects with ring of functions R in their respective categories:

(a) For any algebraic functor X and ring map $\varphi\colon R \to \mathcal{O}(X)$, there is a unique map $X \to \mathrm{Spec}(R)$ inducing φ on rings of functions.
(b) For any locally ringed space X and ring map $\varphi\colon R \to \mathcal{O}(X)$, there is a unique map $X \to \mathrm{Prim}(R)$ inducing φ on ring of functions.

For $\mathrm{Spec}(R)$, this holds by "abstract nonsense" (Remark 2.54), whereas for $\mathrm{Prim}(R)$ it is the content of Theorem 7.54.

7.5. **Quasi-coherent modules on schemes.** If X is an algebraic functor, the category Mod_X of quasi-coherent modules over X is by definition a limit of a diagram of categories of modules over rings, with transition functors given by extension of scalars (Definition 4.13). For any ring R, the category Mod_R is an abelian category, and it is easy to show that the limit of a diagram of abelian category and exact functors is again an abelian category. However, extension of scalars is only right exact and not exact in general, so that the category Mod_X is not necessarily abelian.

If X is a scheme, the situation is much better. Indeed, in the category of Zariski sheaves, we can write X as the colimit of a diagram of affine schemes and *open immersions*. If $j\colon \mathrm{Spec}(B) \hookrightarrow \mathrm{Spec}(A)$ is an open immersion of affine schemes, the ring map $A \to B$ is *flat*, so that the functor $j^*\colon \mathrm{Mod}_A$ *to* Mod_B is exact. Hence, using Zariski descent, for the category of quasi-coherent modules (Proposition 4.41), we can write Mod_X as the limit of a diagram of abelian categories and exact functors. This implies that Mod_X is an abelian category, and also that for any open immersions $j\colon U \hookrightarrow X$, the functor $j^*\colon \mathrm{Mod}_X \to \mathrm{Mod}_U$ is exact. Thus, for any open covering $(U_i \subseteq X)_{i\in I}$, the functor $\mathrm{Mod}_X \to \prod_{i\in I} \mathrm{Mod}_{U_i}$ is both exact and conservative, so that it preserves and detects monomorphisms, epimorphisms, and isomorphisms.

Furthermore, while the forgetful functors $\mathrm{Rad}_R \hookrightarrow \mathrm{Idl}_R \hookrightarrow (\mathrm{Mod}_R)_{/R}$ are not compatible with arbitrary base change and hence do not extends to algebraic functors, they are compatible with *flat* base change, so that they do extend to schemes. These considerations lead to the following result:

Theorem 7.58 (Quasi-coherent modules and ideals on schemes)**.** *Let* X *be a scheme, let* $(U_i \subseteq X)_{i\in I}$ *be an open covering by affine schemes, and let* $u_i\colon \mathrm{Spec}(R_i) \simeq U_i \hookrightarrow X$ *be the corresponding points.*

(a) The category Mod_X *is abelian.*
(b) For any open immersion $j\colon U \hookrightarrow X$*, the pullback functor* $j^*\colon \mathrm{Mod}_X \to \mathrm{Mod}_U$ *is exact.*
(c) A map $f\colon M \to N$ *in* Mod_X *is a monomorphism if and only if the maps* $f(u_i)\colon M(u_i) \to N(u_i)$ *are monomorphisms in* Mod_{R_i} *for all* $i \in I$*. The same statement holds for epimorphisms and isomorphisms.*
(d) A quasi-coherent module $M \in \mathrm{Mod}_X$ *is a vector bundle (resp. a line bundle) if and only if each* $M(u_i)$ *is a vector space (resp. a line) over* R_i*.*
(e) There is an isomorphism of posets

$$\mathrm{Idl}_X \simeq \{\textit{subobjects of } \mathcal{O}_X \textit{ in } \mathrm{Mod}_X\}.$$

(f) There is an isomorphism of posets

$$\mathrm{Rad}_X \simeq \{I \in \mathrm{Idl}_X \mid \textit{for all } i \in I\textit{, the ideal } I(u_i) \subseteq R_i \textit{ is radical}\}.$$

Finally, we explain how to identify quasi-coherent modules over a scheme X with certain sheaves on the underlying topological space $|X|$.

Definition 7.59 (Quasi-coherent sheaf)**.** Let $(X, \mathcal{O}_X)$ be a ringed space and M an $\mathcal{O}_X$-module in $\mathrm{Sh}(X)$. We say that M is a *quasi-coherent sheaf* on $(X, \mathcal{O}_X)$ if it admits a presentation locally on X, i.e., if for every $x \in X$ there is an open neighborhood U of x and an exact sequence of the form

$$\mathcal{O}_U^{(J)} \to \mathcal{O}_U^{(I)} \to M|_U \to 0,$$

where I and J are sets.

Proposition 7.60 (Quasi-coherent modules as sheaves)**.** *Let X be scheme. Then the functor*

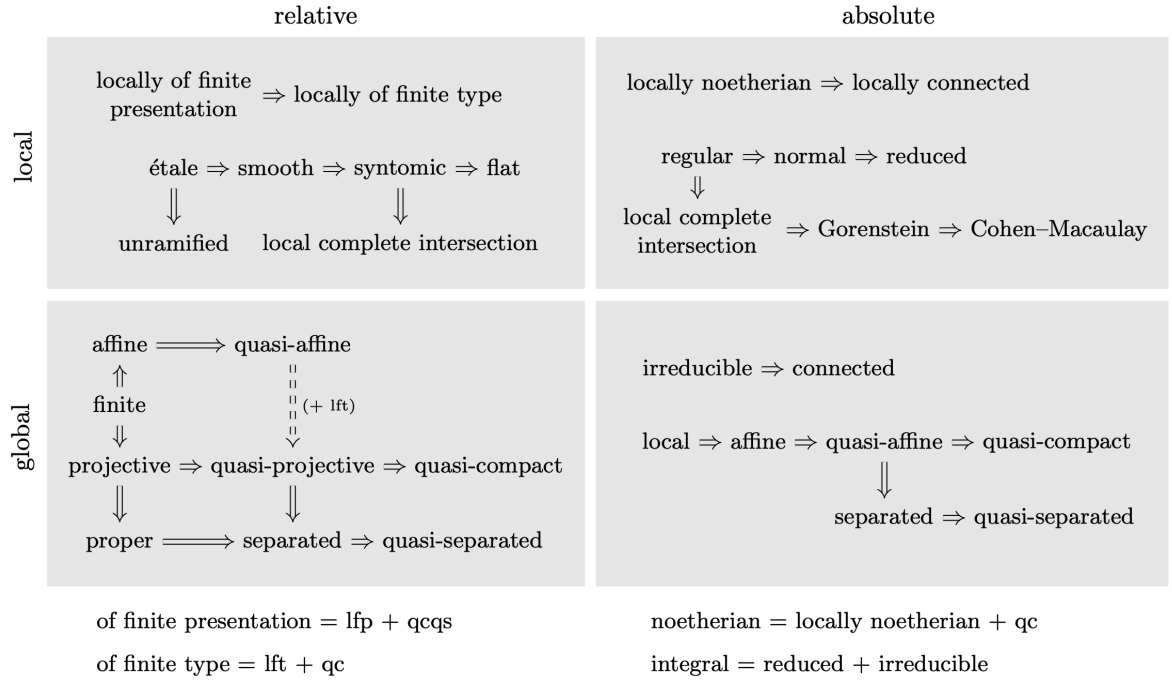
$$\mathrm{Mod}_X \to \mathrm{Mod}_{\mathcal{O}_X}(\mathrm{Sh}(|X|)), \quad M \mapsto (U \mapsto \Gamma(U, M)),$$

is fully faithful and identifies Mod_X *with the category of quasi-coherent sheaves on* $(|X|, \mathcal{O}_X)$.

8. Properties of schemes

Broadly speaking, there are four different types of properties in geometry. A property is called *local* if it can be checked in a neighborhood of every point, and it is *global* otherwise. In algebraic geometry, local properties are determined by their values on affine schemes and hence come from commutative algebra. For example, as the name implies, "locally of finite presentation" is a local property, which in the affine setting corresponds to the notion of finitely presented algebra (Proposition 2.84). We will see many more examples of local properties of schemes in this chapter. On the other hand, "affine" and "connected" are global properties: every scheme is locally affine, so affineness cannot be checked locally, and connectedness is unrelated to local connectedness. In a different direction, there are *absolute* properties, which are properties of the geometry objects themselves, and *relative properties*, which are properties of "objects over a base", i.e. properties of morphism. For example, projectivity is a property of S-schemes which heavily depends on the base scheme S, and is thus a relative property: a nonempty projective $\mathbf{C}$-scheme will never be projective as a $\mathbf{Q}$-scheme.

The following table lists a few important examples of all four types of properties in algebraic geometry, as well as some implications between them:



of finite presentation = lfp + qcqs
of finite type = lft + qc
noetherian = locally noetherian + qc
integral = reduced + irreducible

Remark 8.1. Every relative property has an associated absolute property, since every scheme can be regarded as a scheme over $\mathrm{Spec}(\mathbf{Z})$. For example, the absolute versions of "affine", "quasi-affine", "quasi-compact", "quasi-separated", and "separated" are obtained from the relative versions in this way. But for many relative properties, like smoothness and properness, the associated absolute property is not used, since it is very restrictive (e.g., a nonempty $\mathbf{C}$-scheme can never be smooth or proper over $\mathrm{Spec}(\mathbf{Z})$).

8.1. Local properties.

Given a property P of rings, we can consider schemes that admit an open covering by spectra of rings with property P; this property of schemes is usually called "locally P" or simply "P". This is only well-behaved when the property P is *local* in the following sense:

Definition 8.2 (Local property of rings)**.** A property P of rings is *local*, if the following holds:

(a) If R has property P and $f \in R$, then R_f has property P.
(b) If $(f_1, \dots, f_n) = R$ and each R_{f_i} has property P, then R has property P.

Definition 8.3 (Local property of schemes)**.** A property P of schemes is *local* if the following holds:

(a) If X has property P and $U \subseteq X$ is open, then U has property P.

(b) If $(U_i \subseteq X)_{i\in I}$ is an open covering of X and each U_i has property P, then X has property P.

Remark 8.4. Condition (a) of Definition 8.3 says that P defines (for every scheme X) a presheaf of truth values on $\mathrm{Open}(X)$, and condition (b) says that this presheaf is a sheaf for the canonical topology.

Proposition 8.5 (Extending absolute local properties)**.** *Every local property of rings extends uniquely to a local property of schemes.*

Proof. We define the corresponding property as follows: a scheme X has property P if and only if there exists an affine open covering $(U_i \subseteq X)_{i\in I}$ such that for any $i \in I$, the ring of global functions $\mathcal{O}(U_i)$ has property P.

We will first show that this extends the property P of rings, i.e., the affine scheme $\mathrm{Spec}(R)$ had property P if and only if R has property P. If R has property P, take the affine open covering consisting of the single subfunctor $\mathrm{Spec}(R)$ and we find that $\mathrm{Spec}(R)$ has property P. Conversely, assume that $(\mathrm{Spec}(R_i) \subseteq \mathrm{Spec}(R))_{i\in I}$ is an affine open covering such that every R_i has property P. By Example 6.36 and Proposition 5.44, the open subfunctors $\mathrm{D}(f)$ such that $\mathrm{D}(f) \subseteq \mathrm{Spec}(R_i)$ form an open covering of $\mathrm{Spec}(R)$. It suffices to show that R_f has property P for every f, where $\mathrm{D}(f) \subseteq \mathrm{Spec}(R_i)$ for some $i \in I$. By $\mathrm{D}(f) \cap \mathrm{Spec}(R_i) = \mathrm{D}(f)$ and pullback of affine schemes being computed as pushout of rings, the ring $R_f \simeq (R_i)_f$ has property P.

Furthermore, we show that P is a local property of schemes. Suppose that $(U_i \subseteq X)_{i\in I}$ is an open covering such that U_i has property P. By construction, there exists affine open coverings $(V_{ij} \subseteq U_i)_{J_i}$ such that $\mathcal{O}(V_{ij})$ has property P. By Proposition 5.44, $(V_{ij} \subseteq X)_{(i,j)\in I\times J}$ is an open covering such that $\mathcal{O}(V_{ij})$ has property P, hence X has property P.

Let $U \subseteq X$ be an open subscheme and assume that X has property P. We will show that U also has property P. If X is affine, then U admits an open covering $(\mathrm{D}(f_i) \subseteq U)_{i\in I}$, where $f_i \in \mathcal{O}(X)$. Since X is affine and has property P, any $\mathrm{D}(f_i)$ also has property P. Therefore, by construction U also has property P.

Let X be arbitrary and has property P. By construction, there is an affine open covering $(V_i \subseteq X)_{i\in I}$ such that V_i has property P. Let $U_i = U \cap V_i \subseteq V_i$. By the affine case, every open subscheme of V_i has property P, hence U_i also has property P, i.e., there exists an affine open covering of U_i with property P. This again gives us an affine covering of U with property P, i.e., U has property P. □

For properties of morphisms, locality has two aspects: locality on the source and locality on the base:

Definition 8.6. Let P be a property of morphisms of schemes.

(a) P is *local on the source* if, for any morphism $f\colon Y \to X$:
- If f has property P and $V \subseteq Y$ is an open subscheme, then $f|V\colon V \to X$ has property P.
- If $(V_i \subseteq V)_{i\in I}$ is an open covering and each $f|V_i\colon V_i \to X$ has property P, then f has property P.

(b) P is *local on the base* if, for any morphism $Y \to X$:
- If f has property P and $U \subseteq X$ is open, then $f_U\colon Y \times_X U \to U$ has property P.
- If $(U_i \subseteq X)_{i\in I}$ is an open covering and each $f_{U_i}\colon Y \times_X U_i \to U_i$ has property P, then f has property P.

(c) P is *local* if it is both local on the source and local on the base.

(d) P is *stable under base change* if, any cartesian square of schemes

$$\begin{array}{ccc} Y' & \longrightarrow & Y \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ X' & \longrightarrow & X, \end{array}$$

if f has property P, then f' has property P.

(e) P is *stable under composition* if it contains isomorphism and if $g \circ f$ has property P whenever both f and g has property P.

Definition 8.6 has an obvious analogue for morphisms of rings, and we have the following relative version of Proposition 8.5 (recall from Proposition 4.36(a) that a morphism $Y \to X$ is *affine* if and only if, for every $\mathrm{Spec}(R) \to X$, the pullback $Y \times_X \mathrm{Spec}(R)$ is an affine scheme):

Proposition 8.7 (Extending relative local properties)**.** *Let P be a property of morphisms of rings.*

(a) *If P is local on the base, then it extends uniquely to a property $\hat{P}$ of affine morphisms of schemes that is local on the base.*
(b) *If P is local, then it extends uniquely to a local property $\hat{P}$ of morphisms of schemes.*

In both cases, if P is stable under base change or stable under composition, then so is $\hat{P}$.

Proof. (a) We define the corresponding property as follows: a morphism of schemes $Y \to X$ has property $\hat{P}$ if and only if it is affine and there exists an affine open covering $(U_i \subseteq X)_{i\in I}$ such that for any $i \in I$, the pullback $Y \times_X U_i \to U_i$ has property P.
(b) We define the corresponding property as follows: a morphism of schemes $Y \to X$ has property $\hat{P}$ if and only if, for every $y \in Y$, there exists an affine open neighborhoods V of y and U of $f(y)$ such that $f(V) \subseteq U$ and $\mathcal{O}(U) \to \mathcal{O}(V)$ has property P. Indeed, this is an equivalent characterization of local property of morphisms of schemes.

The verification that $\hat{P}$ uniquely extends P in both cases is similar to the proof of Proposition 8.5. The statements about stability under base change and composition follows immediately from pasting law for cartesian squares. □

Remark 8.8. A property of algebras (i.e. of morphisms of rings) is Zariski-local in the sense of Definition 2.79 if and only if it is local on the *base* in the sense of Definition 8.6.

Remark 8.9. Most relevant properties of morphism are stable under base change and composition. For such properties, locality on the source implies locality on the base.

Example 8.10 (Local finiteness properties). For morphism of schemes, the properties "locally of finite presentation" and "locally of finite type" from Definition 2.90 are local. Hence, a morphism of schemes $f\colon Y \to X$ is locally of finite presentation (resp. locally of finite type) if and only if, for every $y \in |Y|$, there exist affine open neighborhoods V of y and U of $f(y)$ such that $f(V) \subseteq U$ and $\mathcal{O}(V)$ is an $\mathcal{O}(U)$-algebra of finite presentation (resp. of finite type). In the following, we will demonstrate the locality of "locally of finite presentation".

Let P be the property "finitely presented" of morphisms of rings. By Proposition 2.80, it is a local on the base. We will now show that P is also local on the base. Let $R \to S$ be a morphism of rings and $(f_1, \dots, f_r) = S$, where $f_i \in S$. It suffices to show that $R \to S$ is finitely presented if and only if, for any $i \in I$, the morphism $R \to S_{f_i}$ is finitely presented.

One implication follows from the fact that $S_{f_i} = S[x]/(xf_i - 1)$ is a finitely presented S-algebra and finitely presented morphisms of rings are stable under composition. The converse is [Sta26, Tag 00EP].

YW: Marc's proof here is slippery. I wasn't satisfied here.

Therefore, by Proposition 8.7, we may uniquely extend P to a local property $\hat{P}$ of morphisms of schemes. It suffices to show that $\hat{P}$ is the property "locally of finite presentation". Without loss of generality, we may assume that $X = \operatorname{Spec}(R)$ for $R \in \mathrm{CAlg}$. We will first show that "locally of finite presentation" implies $\hat{P}$. Let $(V_i \subseteq Y)_{i\in I}$ be an affine open covering of Y. By Example 2.91, open immersions are locally of finite presentation. Furthermore, by Proposition 2.92, morphisms locally of finite presentation are stable under composition, hence $V_i \to \operatorname{Spec}(R)$ is locally of finite presentation. Since both sides are affine schemes, $V_i \to \operatorname{Spec}(R)$ satisfies $\hat{P}$, hence $Y \to \operatorname{Spec}(R)$ satisfies $\hat{P}$.

Conversely, we will show that $\hat{P}$ implies "locally of finite presentation". Since we are working over $\operatorname{Spec}(R)$, it suffices to show that the algebraic R-functor $Y\colon \mathrm{CAlg}_R \to \mathrm{Set}$ preserves filtered colimits. Let $(V_i \subseteq Y)_{i\in I}$ be an affine open covering of Y and $(V_{ijk} \subseteq V_i \cap V_j)_{k\in K_{ij}}$ be an affine open covering of $V_i \cap V_j$. By Proposition 7.16, Y can be written as a coequalizer

$$\coprod_{ijk} V_{ijk} \rightrightarrows \coprod_{i\in I} V_i \to Y$$

in $\mathrm{Sh}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}})$. By assumption, the algebraic R-functors $V_i, V_{ijk}\colon \mathrm{CAlg}_R \to \mathrm{Set}$ preserve filtered colimits. Therefore, suffices to show that if $F \in \mathrm{Fun}(\mathrm{CAlg}_R, \mathrm{Set})$ preserves filtered colimit, then its Zariski sheafification $a_{\mathrm{Zar}}(F)$ also preserves filtered colimits.

Indeed, the category CAlg_R is *compactly generated*, i.e., $\mathrm{CAlg}_R \simeq \mathrm{Ind}(\mathrm{CAlg}_R^{\mathrm{fp}})$. By the universal property of Ind-completion, the restriction functor induces an equivalence of categories

$$\mathrm{res}\colon\ \mathrm{Fun}^{\mathrm{filt}}(\mathrm{CAlg}_R, \mathrm{Set}) \to \mathrm{Fun}(\mathrm{CAlg}_R^{\mathrm{fp}}, \mathrm{Set}),$$

where the left hand side denotes the filtered colimits-preserving functors. The inverse is given by the left Kan extension along the restriction. Indeed, if $F\colon\ \mathrm{CAlg}_R^{\mathrm{fp}} \to \mathrm{Set}$ is an arbitrary functor and the R-algebra can be written as the filtered colimit $S = \mathrm{colim}_{i\in I} S_i$, where $S_i \in \mathrm{CAlg}_R^{\mathrm{fp}}$, then $L(F)(S) = \mathrm{colim}_{i\in I} F(S_i)$.

We will show that L preserves Zariski-sheaves. In other words, if $S \in \mathrm{CAlg}_R$ and $(f_1, \dots, f_n) = S$, we need to show that the following diagram is an equalizer diagram:

$$L(F)(S) \to \prod_{i\in I} L(F)(S_{f_i}) \rightrightarrows \prod_{i,j\in I} L(F)(S_{f_i f_j}).$$

Write $S = \mathrm{colim}_{k\in K} S_k$, where S_k are finitely presented R-algebras. At some stage $k \in K$, there exist $f_{i,k} \in S_k$ whose image in S is f_i such that $(f_{1,k}, \dots, f_{n,k}) = S_k$. Since localization commutes with colimits, the following diagram is an equalizer diagram:

$$F(S_k) \to \prod_{i\in I} F\left((S_k)_{f_{i,k}}\right) \rightrightarrows \prod_{i,j\in I} F\left((S_k)_{f_{i,k}f_{j,k}}\right).$$

Since filtered colimits commute with finite limits in Set, taking the filtered colimit over K of the above diagram yields the desired claim, by cofinality.

By the universal property of L and $\mathrm{a}_{\mathrm{Zar}}$, the following diagram commutes:

$$\begin{array}{ccc} \mathrm{Fun}(\mathrm{CAlg}_R^{\mathrm{fp}}, \mathrm{Set}) & \xrightarrow{L} & \mathrm{Fun}(\mathrm{CAlg}_R, \mathrm{Set}) \\ \downarrow{\scriptstyle \mathrm{a}_{\mathrm{Zar}}} & & \downarrow{\scriptstyle \mathrm{a}_{\mathrm{Zar}}} \\ \mathrm{Sh}_{\mathrm{Zar}}(\mathrm{CAlg}_R^{\mathrm{op}}) & \dashrightarrow^{L} & \mathrm{Sh}_{\mathrm{Zar}}(\mathrm{CAlg}_R^{\mathrm{fp,op}}). \end{array}$$

Since F preserves filtered colimits, it is the Ind-completion of itself, i.e., $F = L(\mathrm{res}(F))$. Therefore, $\mathrm{a}_{\mathrm{Zar}}(F)$ is the Ind-completion of $\mathrm{a}_{\mathrm{Zar}}(\mathrm{res}(F))$, hence preserves filtered colimits.

Example 8.11 (Flatness). Recall that a map of rings $\varphi\colon\ R \to S$ is *flat* if the functor $\varphi^*\colon\ \mathrm{Mod}_R \to \mathrm{Mod}_S$ is exact. This property of morphisms of rings is clearly stable under base change and under composition. It is moreover a local property: in fact, $\varphi\colon\ R \to S$ is flat if and only if, for every prime ideal $\mathfrak{q} \subseteq S$, the induced map of local rings $R_{\varphi^{-1}(\mathfrak{q})} \to S_{\mathfrak{q}}$ is flat (this follows immediately from the fact that, for any ring R, the functor $\mathrm{Mod}_R \to \prod_{\mathfrak{p}} \mathrm{Mod}_{R_{\mathfrak{p}}}$ is exact and conservative). Hence, by Proposition 8.7, flatness extends uniquely to a local property of morphisms of schemes, which has the following characterization: $f\colon\ Y \to X$ is flat if and only if, for every $y \in |Y|$, the map of local rings $f^*\colon\ \mathcal{O}_{X,f(y)} \to \mathcal{O}_{Y,y}$ is flat.

Example 8.12 (Properties local on the base). The following properties of morphisms of schemes are local on the base, but not on the source (see Proposition 2.82, Proposition 7.24(b), Proposition 7.31):

(a) affine;
(b) quasi-affine;
(c) quasi-compact;
(d) quasi-separated;
(e) separated;
(f) of finite presentation;
(g) of finite type.

All these properties are also stable under base change and composition. The first four are properties of affine morphisms and hence arise from Proposition 8.7(a). Properties that are not local on the base are somewhat pathological; these include "projective" and "quasi-projective". Properness is a weaknening of projectivity that is

local on the base, which we will study in Section 11. With the exception of "immersion", one can show that all of the above properties are even *fpqc-local* on the base.

8.2. **Reduced schemes.** Recall that a ring R is *reduced* if its nilradical $\mathrm{Nil}(R) = \sqrt{0}$ is the zero ideal. The *reduction* of a ring R is $R_{\mathrm{red}} = R/\mathrm{Nil}(R)$.

Definition 8.13 (Reduced scheme)**.** A scheme X is *reduced* if there is an affine open covering $(U_i \subseteq X)_{i\in I}$ such that each ring $\mathcal{O}(U_i)$ is reduced.

Lemma 8.14. *"Reduced" is a local property of rings.*

Proof. Let R be a ring and $f \in R$. Unwinding the definitions, if R is reduced, then R_f is also reduced. Assume that $(f_1, \dots, f_n) = R$ and R_{f_i} is reduced. By Zariski descent, the canonical map of rings $R \hookrightarrow \prod_{i\in I} R_{f_i}$ is injective. If $x \in R$ is nilpotent, then its image in R_{f_i} is nilpotent, hence zero for every $i \in I$. Therefore, x itself must be 0. □

By Proposition 8.5, we deduce:

Proposition 8.15. *"Reduced" is a local property of schemes. Moreover,* $\mathrm{Spec}(R)$ *is reduced if and only if* R *is reduced.*

Remark 8.16.

(a) A scheme is reduced if and only if, for every point $x \in |X|$, the local ring $\mathcal{O}_{X,x}$ is reduced. The forward implication follows from the fact that localizations of reduced rings are reduced, and the converse from the fact that any ring is a subring of $\prod_{\mathfrak{p}} R_{\mathfrak{p}}$.

(b) If X is a reduced scheme, then the ring $\mathcal{O}(X)$ is reduced. Indeed if $(U_i \subseteq X)_{i\in I}$ is an open covering of X by reduced affine schemes, then $\mathcal{O}(X)$ is a subring of the reduced ring $\prod_{i\in I} \mathcal{O}(U_i)$.

Example 8.17 (Reduced Proj)*.* Let A be an $\mathbf{N}$-graded ring and let $I \subseteq A_+$ be a homogeneous subset such that $A_+ \subseteq \sqrt{(I)}$. Recall that $\mathrm{Proj}(A)$ has an open covering by the affine schemes $\mathrm{Spec}(A_{(f)})$ for $f \in I$. Hence, $\mathrm{Proj}(A)$ is reduced if and only if each ring $A_{(f)}$ with $f \in I$ is reduced. A sufficient condition is that A itself is reduced. For example, if R is reduced and $n \geq -1$, then $\mathbf{P}^n_R$ is reduced.

If X is a scheme, recall that there is an embedding

$$\mathrm{Rad}_X \hookrightarrow \mathrm{Idl}_X$$

(Theorem 7.58(f)), so that "radical" is a meaningful property of quasi-coherent ideals over X. Concretely, if $K \in \mathrm{Idl}_X$ and $(x_i\colon \mathrm{Spec}(R_i) \hookrightarrow X)_{i\in I}$ is an affine open covering of X, then K is radical if and only if each ideal $K(x_i) \in \mathrm{Idl}_{R_i}$ is radical.

YW: During the lecture, Marc proved some facts in chapter 7. Wrote them down

Proposition 8.18 (Classification of reduced closed subschemes)**.** *Let X be a scheme. A closed subscheme $Z \subseteq X$ is reduced if and only if the quasi-coherent ideal I over X such that $\mathrm{V}(I) = Z$ is radical.*

Proof. Let $(\mathrm{Spec}(R_i) \subseteq X)_{i\in I}$ be an affine open covering of X, where $R_i \in \mathrm{CAlg}$. Since closed immersions are closed under base change, the following commutative square is cartesian, for every $i \in I$:

$$\begin{array}{ccc} \mathrm{Spec}(R_i/I(u_i)) & \hookrightarrow & Z \\ \downarrow & & \downarrow \\ \mathrm{Spec}(R_i) & \xrightarrow{u_i} & X. \end{array}$$

Since $(\mathrm{Spec}(R_i) \subseteq X)_{i\in I}$ is an open covering of X, the collection of subfunctors $(\mathrm{Spec}(R_i/I(u)) \subseteq Z)_{i\in I}$ is again an open covering of Z (see Proposition 5.44). Therefore, the closed subscheme Z is reduced if and only if, for any $i \in I$, the open subscheme $\mathrm{Spec}(R_i/I(u))$ is reduced, i.e., $R_i/I(u_i)$ is reduced. Equivalently, this is the same as the ideal $I(u_i)$ being radical. □

Corollary 8.19. *Let X be a scheme. Them the composite*

$$\mathrm{Closed}^{\mathrm{red}}(X) \hookrightarrow \mathrm{Closed}(X) \xrightarrow{|-|} \mathrm{Closed}(|X|)$$

is bijective. In particular, the second map is surjective.

The following commutative diagram of posets summarizes the classification of closed and open subschemes of a scheme X, where "c" means taking the complement (see Proposition 4.23, Proposition 5.24, Remark 5.29, Proposition 8.18):

$$\begin{array}{ccccc}
 & & \mathrm{Closed}(|X|) & \xrightarrow[\simeq]{\mathrm{c}} & \mathrm{Open}(|X|) \\
 & \overset{\simeq}{\nearrow} & \big|-\big| \uparrow & & \big|-\big| \uparrow \simeq \\
\mathrm{Closed}^{\mathrm{red}}(X)^{\mathrm{op}} & \hookrightarrow & \mathrm{Closed}(X)^{\mathrm{op}} & \twoheadrightarrow^{\mathrm{c}} & \mathrm{Open}(X) \\
\simeq \uparrow & & \mathrm{V}(-) \uparrow \simeq & & \mathrm{D} \uparrow \simeq \\
\mathrm{Rad}_X & \hookrightarrow & \mathrm{Idl}_X & \xrightarrow[\sqrt{-}]{} & \mathrm{Rad}_X\,.
\end{array}$$

Example 8.20 (Reduced closed subschemes of Proj)*.* Let A be an $\mathbf{N}$-graded ring. Recall that the closed subsets of $|\operatorname{Proj}(A)|$ are in bijection with the saturated radical homogeneous ideals $I \subseteq A$ (Remark 3.91). By Corollary 8.19 and Example 8.17, we obtain a bijection

$$\{\text{saturated radical homogeneous ideals in } A\} \xrightarrow{\sim} \mathrm{Closed}^{\mathrm{red}}(X), \quad I \mapsto \mathrm{V}(I).$$

Definition 8.21 (Reduction)**.** Let X be a scheme. The *reduction* of X is the closed subscheme

$$X_{\mathrm{red}} = \mathrm{V}(\sqrt{0}) \subseteq X.$$

Proposition 8.22. *Let* $\mathrm{Sch}^{\mathrm{red}} \subseteq \mathrm{Sch}$ *be the full subcategory of reduced schemes. Then the inclusion* $\mathrm{Sch}^{\mathrm{red}} \hookrightarrow \mathrm{Sch}$ *has a right adjoint given by* $X \mapsto X_{\mathrm{red}}$*.*

Proof. Let Y be a reduced scheme and X be a scheme. We need to show that the induced map

$$\mathrm{Map}(Y, X_{\mathrm{red}}) \to \mathrm{Map}(Y, X)$$

is an isomorphism, i.e. any map $f\colon Y \to X$ factors through X_{red}. Consider the following pullback square:

$$\begin{array}{ccc}
\mathrm{V}(f^*\sqrt{0}) & \longrightarrow & \mathrm{V}(\sqrt{0}) \\
\downarrow & & \downarrow \\
Y & \xrightarrow{f} & X.
\end{array}$$

Since $X_{\mathrm{red}} \to X$ is a monomorphism, it suffices to show that the canonical map $\mathrm{V}(f^*\sqrt{0}) \hookrightarrow Y$ is an isomorphism. By Proposition 4.23, this is the case if and only if the quasi-coherent ideal $f^*\sqrt{0}$ is zero. Let $(\operatorname{Spec}(R_i) \subseteq X)_{i\in I}$ be an affine open covering of X. By restricting to Y, we obtain an open covering $(\operatorname{Spec}(S_i) \subseteq Y)_{i\in I}$. By Theorem 7.58, it suffices to show that after base change, the ideal $\mathrm{Nil}(R_i)S_j \simeq 0$ for any $i, j \in I$. By the explicit description of $\mathrm{Nil}(R_i)S_j$, we find that $\mathrm{Nil}(R_i)S_j \subseteq \mathrm{Nil}(S_j) \subseteq S_j$. Since Y is reduced, S_i is radical for every $j \in I$, hence $\mathrm{Nil}(R_i)S_j = 0$. □

Remark 8.23*.* Let X be a scheme. Then the closed immersion $X_{\mathrm{red}} \hookrightarrow X$ induces an isomorphism on underlying spaces. Indeed, the open complement of $X_{\mathrm{red}} = \mathrm{V}(\sqrt{0})$ in X is the empty scheme $\mathrm{D}(\sqrt{0}) = \varnothing$. Hence, as a locally ringed space, X_{red} has the form $(|X|, \mathcal{O}_X^{\mathrm{red}})$, where $\mathcal{O}_X^{\mathrm{red}}$ is given by $U \mapsto \mathcal{O}(U_{\mathrm{red}})$. Note that the sheaf $\mathcal{O}_X^{\mathrm{red}}$ is the sheafification of the presheaf of rings $U \mapsto \mathcal{O}_X(U)_{\mathrm{red}}$, since the canonical map $\mathcal{O}_X(-)_{\mathrm{red}} \to \mathcal{O}_X^{\mathrm{red}}$ is an isomorphism on affine opens.

8.3. **Noetherian schemes.** Recall that a ring R is *noetherian* if every ascending chain of ideals $I_0 \subseteq I_1 \subseteq I_2 \subseteq \ldots$ in R stablizes. Equivalently, R is noetherian if every descending sequence of closed subschemes $Z_0 \supseteq Z_1 \supseteq Z_2 \supseteq \ldots$ in $\operatorname{Spec}(R)$ stablizes.

Proposition 8.24 (Characterization of noetherian rings)**.** *For a ring* R*, the following conditions are equivalent:*

(a) *R is noetherian.*
(b) *Every submodule of a finitely generated R-module is finitely generated.*
(c) $\mathrm{Mod}_R^{\mathrm{ft}}$ *is an abelian category such that the inclusion* $\mathrm{Mod}_R^{\mathrm{ft}} \hookrightarrow \mathrm{Mod}_R$ *is exact.*
(d) *Every R-module of finite type is of finite presentation:* $\mathrm{Mod}_R^{\mathrm{ft}} = \mathrm{Mod}_R^{\mathrm{fp}}$*.*

(e) *Every R-algebra of finite type is of finite presentation:* $\mathrm{CAlg}_R^{\mathrm{ft}} = \mathrm{CAlg}_R^{\mathrm{fp}}$.

Proof. (a) $\Rightarrow$ (e): let S be a finitely generated R-algebra. Then $S \simeq R[x_1, \dots, x_n]/I$. By the Hilbert basis theorem, the ring $R[x_1, \dots, x_n]$ is noetherian, hence the ideal I is finitely generated, i.e, the R-algebra S is finitely presented.

(e) $\Rightarrow$ (d): let M be a finitely generated R-module. Then there exists a short exact sequence

$$0 \to \ker(f) \to R^n \xrightarrow{f} M \to 0,$$

Applying $\mathrm{Sym}_R(-)$, we obtain a short exact sequence

$$0 \to \ker(\mathrm{Sym}_R(f)) \to R[x_1, \dots, x_n] \xrightarrow{\mathrm{Sym}_R(f)} \mathrm{Sym}_R(M) \to 0,$$

since $\mathrm{Sym}_R(-)$ is right exact. By (e), the ideal $\ker(\mathrm{Sym}_R(f))$ is finitely generated. Therefore, there exists a graded map $\bigoplus_{i \in I} R[x_1, \dots, x_n](-d_i) \twoheadrightarrow \ker(\mathrm{Sym}_R(f))$ for some $d_i \in \mathbf{N}$ and I finite. By taking the degree-1 part, we obtain a surjection $R^m \twoheadrightarrow \ker(f)$ for some $m \in \mathbf{N}$.

(d) $\Rightarrow$ (b): let M be a finitely generated R-module and $N \subseteq M$ a submodule. Since M is finitely generated, the quotient M/N is again finitely generated, hence by (d) finitely presented. Consider the following short exact sequence:

$$0 \to N \to M \to M/N \to 0.$$

Since M and M/N are finitely presented, by a standard result in commutative algebra, the R-module N is finitely generated.

(b) $\Rightarrow$ (c): the category $\mathrm{Mod}_R^{\mathrm{ft}} \subseteq \mathrm{Mod}_R$ is closed under quotient and extensions. By (b), it is also closed under subobjects, which implies (c).

(c) $\Rightarrow$ (a): let $I \subseteq R$ be an ideal. Since $\mathrm{Mod}_R^{\mathrm{ft}}$ is closed under quotient, the map $\pi\colon R \twoheadrightarrow R/I$ lies in $\mathrm{Mod}_R^{\mathrm{ft}}$. Since $\mathrm{Mod}_R^{\mathrm{ft}}$ is abelian and the inclusion $\mathrm{Mod}_R^{\mathrm{ft}} \subseteq \mathrm{Mod}_R$ is exact, $\ker(\pi)$ is finitely generated. □

Remark 8.25. Condition (e) in Proposition 8.24 implies that, if R is noetherian, then any finitely generated R-algebra is again noetherian. The fact that $R[x_1, \dots, x_n]$ is noetherian is Hilbert's basis theorem.

Definition 8.26 (Locally noetherian, noetherian scheme)**.** A scheme X is *locally noetherian* if there is an affine open covering $(U_i \subseteq X)_{i \in I}$ such that each ring $\mathcal{O}(U_i)$ is noetherian. A scheme is *noetherian* if it is locally noetherian and quasi-compact.

Lemma 8.27. *"Noetherian" is a local property of rings.*

Proof. Let R be a noetherian ring and $\mathrm{L}_f\colon R \to R_f$ be the localization map. Consider an ascending sequence of ideals $I_0 \subseteq I_1 \subseteq \dots R_f$. Since L_f^{-1} is order preserving and R is noetherian, the ascending sequence

$$\mathrm{L}_f^{-1}(I_0) \subseteq \mathrm{L}_f^{-1}(I_1) \subseteq \dots \subseteq R$$

stablizes. It follows from $I_i = (\mathrm{L}_f^{-1}(I_i))_f$ that R_f is noetherian.

Assume that there exists $f_1, \dots, f_n \in R$ such that $(f_1, \dots, f_n) = R$ and R_{f_i} is noetherian. Let M be a finitely generated R-module. The R_{f_i}-module M_{f_i} is finitely generated, hence finitely presented, by Proposition 8.24. Since being finitely presented is a Zariski local property (see Proposition 2.80), the R-module M is again finitely presented. □

By Proposition 8.5, we deduce:

Proposition 8.28. *"Locally noetherian" is a local property of schemes. Moreover,* $\mathrm{Spec}(R)$ *is locally noetherian if and only if R is noetherian.*

Example 8.29 (Noetherian Proj)*.* Let A be an $\mathbf{N}$-graded ring and let $I \subseteq A_+$ be a homogeneous subset such that $A_+ \subseteq \sqrt{(I)}$. Then $\mathrm{Proj}(A)$ is locally noetherian if and only if each ring $A_{(f)}$ with $f \in I$ is noetherian. It is moreover quasi-compact if and only if a finite such set I exists. In particular, if A itself is noetherian, then A_+ is a finitely generated ideal and hence $\mathrm{Proj}(A)$ is noetherian. For example, if R is noetherian and $n \geq -1$, then $\mathbf{P}_R^n$ is noetherian.

Remark 8.30 (Coherent rings). A ring R is called *coherent* if $\mathrm{Mod}_R^{\mathrm{fp}}$ is an abelian category such that the inclusion $\mathrm{Mod}_R^{\mathrm{fp}} \hookrightarrow \mathrm{Mod}_R$ is exact. By Proposition 8.24, noetherian rings are coherent, but the converse fails. For example, any valuation ring is coherent, and if R is noetherian and I is any set, then the polynomial ring $R[x_i \mid i \in I]$ is coherent. One can show that "coherent" is also a local property of rings, so that it extends uniquely to a local property of schemes.

Proposition 8.31. *A scheme X is noetherian if and only if every descending sequence of closed subschemes $Z_0 \supseteq Z_1 \supseteq Z_2 \supseteq \ldots$ in X stablizes.*

YW: Prove this.

Proof. This is left as an exercise. □

Definition 8.32 (Noetherian space)**.** A topological space T is noetherian if every descending sequence of closed subsets $Z_0 \supseteq Z_1 \supseteq Z_2 \supseteq \ldots$ in T stablizes.

Remark 8.33. If T is a noetherian space, then T is quasi-compact. Moreover, any subspace of T is again noetherian, and hence quasi-compact.

Remark 8.34. By Proposition 8.31 and Corollary 8.19, if X is a noetherian scheme, then the underlying space $|X|$ is a noetherian space. However, the converse is not true. For example, if k is a field, then $R = k[x_1, x_2, \ldots]/(x_1^2, x_2^2, \ldots)$ is a non noetherian ring such that $|\operatorname{Spec}(R)|$ is a single point (since $R_{\mathrm{red}} = k$).

Remark 8.35 (Noetherian induction). To show that all closed subsets of a noetherian space have a certain property P, it suffices to show that Z has property P whenever all $W \subsetneq Z$ have property P. The same holds for closed subschemes of a noetherian scheme (by Proposition 8.31). The methode of proof is called *noetherian induction*.

Just like for noetherian rings, there is no difference between "finite presentation" and "finite type" for noetherian schemes:

Proposition 8.36 (Finiteness properties for noetherian schemes)**.**

(a) *A locally noetherian scheme is quasi-separated.*
(b) *If X is locally noetherian and $f\colon Y \to X$ is locally of finite type, then Y is locally noetherian and f is locally of finite presentation.*
(c) *If X is noetherian and $f\colon Y \to X$ is of finite type, then Y is noetherian and f is of finite presentation.*

Proof. (a) Let X be locally noetherian and $(U_i \subseteq X)_{i\in I}$ an affine open covering such that U_i is noetherian. By Proposition 7.35, it suffices to show that $U_i \cap U_j$ is quasi-compact for any $i, j \in I$. By Remark 8.33 and Remark 8.34, $|U_i \cap U_j|$ is quasi-compact, hence by Proposition 7.34, $U_i \cap U_j$ is quasi-compact.

(b) Since X is locally noetherian, there exists an affine open covering $(U_i \subseteq X)_{i\in I}$ such that each U_i is noetherian. Consider V_i defined by the following cartesian square:

$$\begin{array}{ccc} V_i & \hookrightarrow & Y \\ \downarrow & & \downarrow \\ U_i & \hookrightarrow & X. \end{array}$$

As before, $(V_i \subseteq Y)_{i\in I}$ is an open covering of Y and $Y_i \to U_i$ is locally of finite type. If $y \in |Y|$, then $y \in |Y_i|$ for some $i \in I$. By Example 8.10, we may reduce to showing that, for a noetherian ring R and a R-algebra of finite type S, the ring S is again noetherian. But this follows from the Hilbert basis theorem immediately.

(c) As $f\colon Y \to X$ is of finite type, in particular, it is quasi-compact. Since quasi-compact maps are closed under composition, Y is again quasi-compact. By (b), Y is noetherian and f is of finite presentation.

□

8.4. **Irreducible components.** Recall that a topological space T is *irreducible* if it is nonempty and for every nonempty open subsets $U, V \subseteq T$, the intersection $U \cap V$ is nonempty. If T is sober (e.g., the underlying space of a scheme), this is equivalent to the existence of a *generic point*, i.e., a point $\eta \in T$ such that $\overline{\{\eta\}} = T$ (Proposition 5.13). According to Proposition 5.30, a scheme X is *irreducible* if its underlying space $|X|$ is irreducible.

Definition 8.37 (Irreducible component)**.**

(a) The *irreducible components* of a topological space T are the maximal irreducible subsets of T.
(b) The *irreducible components* of a scheme X are the irreducible components of the underlying space $|X|$.

Remark 8.38. Let T be a topological space.

(a) By Zorn's lemma, every irreducible subset of T is contained in an irreducible component of T. In particular, since points are irreducible, T is the union of its irreducible components.
(b) A subset $A \subseteq T$ is irreducible if and only if its closure $\overline{A} \subseteq$ is. It follows that the irreducible components of T are closed.

Remark 8.39 (Irreducible components of affine schemes). Let R be a ring. Under the bijection

$$\mathrm{Rad}_R \xrightarrow{\sim} \{\text{closed subsets of } \mathrm{Prim}(R)\}, \quad I \mapsto \{\mathfrak{p} \mid I \subseteq \mathfrak{p}\},$$

the irreducible closed subsets of $\mathrm{Prim}(R)$ correspond to the prime ideals of R. Hence, the irreducible components of $\mathrm{Spec}(R)$ correspond to the minimal prime ideals of R. In particular, $\mathrm{Spec}(R)$ is irreducible if and only if R has a unique minimal prime ideal. Since the intersection of all prime ideals is always the nilradical, this is also equivalent to the nilradical $\mathrm{Nil}(R)$ being prime.

Example 8.40 (Irreducible components of Proj). Let A be an $\mathbf{N}$-graded ring. Recall that the closed subsets of $|\mathrm{Proj}(A)|$ are in bijection with the saturated radical homogeneous ideals $I \subseteq A$ (Remark 3.91). Under this bijection, the irreducible components correspond to the minimal such ideals. In particular, $\mathrm{Proj}(A)$ is irreducible if and only if $\mathrm{Nil}(A)^{\mathrm{sat}}$ is prime.

Example 8.41. Let k be a field.

(a) The affine k-scheme $\mathrm{Spec}(k[x,y]/(xy))$ has two irreducible components, corresponding to the minimal prime ideals (x) and (y) in $k[x,y]/(xy)$. They meet in a single point, correpsonding to the maximal ideal (x,y).
(b) The projective k-scheme $\mathrm{Proj}(k[x,y,z]/(xyz))$ has three irreducible components, correpsonding to the minimal saturated prime ideals (x), (y) and (z) in $k[x,y,z]/(xyz)$. These components are in a triangle configuration, each pair of components meeting in a single point away from the third component.

Proposition 8.42 (Irreducible components of noetherian spaces)**.** *A noetherian topological space has finitely many irreducible components. In particular, a noetherian scheme has finitely many irreducible components.*

Proof. By Remark 8.35, assume the result holds for any closed subset $Z \subsetneq T$. Assume that T is not irreducible, then there exists closed subsets $A, B \subseteq T$ such that $T = A \cup B$. Then A and B has finitely many irreducible components, hence T has finitely many irreducible components. □

Definition 8.43 (Integral scheme, function field)**.** A scheme X is *integral* if it is irreducible and reduced. The *function field* of X (or field of rational functions on X) is the residue field $\kappa(\eta)$ of the generic point $\eta \in |X|$.

Example 8.44 (Integral affine schemes). Let R be a ring. Then $\mathrm{Spec}(R)$ is integral if and only if R is an integral domain (in which case the function field of $\mathrm{Spec}(R)$ is the fraction field $\mathrm{Frac}(R)$). This follows from Proposition 8.15 and Remark 8.39.

Proposition 8.45. *Let X be an integral scheme with generic point η.*

(a) For every nonempty open subscheme $U \subseteq X$, the map $\mathcal{O}(U) \to \kappa(\eta)$ is injective.
(b) For every nonempty affine open subscheme $U \subseteq X$, the map $\mathcal{O}(U) \to \kappa(\eta)$ exhibits $\kappa(\eta)$ as the fraction field of $\mathcal{O}(U)$.
(c) For every point $x \in |X|$, the map $\mathcal{O}_{X,x} \to \kappa(\eta)$ exhibits $\kappa(\eta)$ as the fraction field of $\mathcal{O}_{X,x}$.

Proof. This is left as an exercise. □ YW: Prove this.

Example 8.46. (Integral Proj) If A is an integral $\mathbf{N}$-graded ring such that $A_+ \neq 0$, then $\operatorname{Proj}(A)$ is an integral scheme: this follows from Example 8.17 and Example 8.40 (the assumption that $A_+ \neq 0$ ensures that $(0)^{\mathrm{sat}} = (0)$). By Proposition 8.45(b), the function field of $\operatorname{Proj}(A)$ is then the fraction field of $A_{(f)}$ for any nonzero homogeneous element $f \in A_+$, i.e., it is $\operatorname{Frac}(A)_0$. For example, if R is an integral domain with fraction field K and $n \geq 0$, then $\mathbf{P}^n_R$ is an integral scheme with function field $K(x_1, \ldots, x_n)$.

Remark 8.47 (Rational function). The notion of rational function can be defined for any scheme X as follows. The *sheaf of rational functions* $\mathcal{K}_X$ on $|X|$ is the sheafification of the presheaf of rings $U \mapsto \mathcal{O}(U)[S_U^{-1}]$, where $S_U \subseteq \mathcal{O}(U)$ is the multiplicative subset of all functions f such that $f|V$ is not a zero divisor in $\mathcal{O}(V)$ for all opens $V \subseteq U$. Since sheafification is left exact, there is a canonical monomorphism of sheaves $\mathcal{O}_X \hookrightarrow \mathcal{K}_X$. If X is integral with generic point η, then $\mathcal{K}_X$ is the constant sheaf with $\mathcal{K}_X(X) = \kappa(\eta)$. If X is locally noetherian, one can show that

$$\mathcal{K}_X(X) = \operatorname*{colim}_{U \subseteq X} \mathcal{O}(U),$$

where the colimit ranges over the open subschemes $U \subseteq X$ whose closure is X (Definition 2.69).

8.5. **Dimension theory.** We review the basics of dimension theory. Define the *length* of a poset P to be the supremum of the lengths of chains in P:

$$\operatorname{length}(P) = \sup\{n \in \mathbf{N} \mid \text{there is a chain } x_0 \lneq \ldots \lneq x_n \text{ in } P\} \in \mathbf{N} \cup \{\pm\infty\}.$$

Definition 8.48 (Krull dimension, codimension)**.** Let T be a topological space.

(a) The *Krull dimension* of T, denoted by $\dim(T)$, is the length of the poset of closed irreducible subsets of T.
(b) Given an irreducible subset $K \subseteq T$, the *codimension* of K in T, denoted by $\operatorname{codim}(K, T)$, is the length of the poset of closed irreducible subsets of T containing K.

If X is a scheme, we also apply these definitions to the underlying space of X.

Remark 8.49 (Specialization). If T is a topological space and $x, y \in T$, we say that x *specializes to* y, and write $x \rightsquigarrow y$, if $y \in \overline{\{x\}}$. This defines a partial order on the underlying set of T. If T is sober, there is an isomorphism of posets

$$(T, \rightsquigarrow) \xrightarrow{\sim} (\{\text{irreducible closed subsets of } T\}, \supseteq), \quad x \mapsto \overline{\{x\}},$$

by Proposition 5.13. We can therefore rephrase Definition 8.48 in terms of the poset of points of T. For example,

$$\dim(T) = \operatorname{length}(T, \rightsquigarrow).$$

Remark 8.50. Let T be a topological space.

(a) By definition, we have $\dim(T) = \sup_K \operatorname{codim}(K, T)$, where K ranges over the irreducible closed subsets of T. If T is sober, we equivalently have

$$\dim(T) = \sup_{x \in T} \operatorname{codim}(x, T).$$

(b) If $(T_i \subseteq T)_{i \in I}$ is either an open covering of T or a finite closed covering of T,[26] then

$$\dim(T) = \sup_{i \in I} \dim(T_i).$$

(c) Given $x \in T$, we define the *Krull dimension of T at the point x* to be $\dim_x(T) = \min_U \dim(U)$, where U ranges over the open neighborhoods of x. By (b), we then have

$$\dim(T) = \sup_{x \in T} \dim_x(T).$$

Definition 8.51 (Krull dimension of schemes and rings)**.**

(a) If X is a scheme, the *Krull dimension* of X is the Krull dimension of its underlying space:

$$\dim(X) = \dim(|X|).$$

We similarly define $\operatorname{codim}(K, X)$ for K an irreducible subset of $|X|$ or subscheme of X.

[26] If T is sober, then this holds for any closed covering.

(b) If R is ring, the *Krull dimension* of R is

$$\dim(R) = \dim(\mathrm{Spec}(R)) = \dim(\mathrm{Prim}(R)) = \mathrm{length}(\mathrm{Prim}(R), \subseteq).$$

Remark 8.52. Let X be a scheme. For any $x \in |X|$ and any open neighborhood U of x, we have

$$\mathrm{codim}(x, X) = \mathrm{codim}(x, U) = \dim(\mathcal{O}_{X,x}).$$

Indeed, every point in X specializing to x lies in U, so the first two numbers are the lengths of the same poset, and taking U to be affine, we see that this poset is isomorphic to the poset of prime ideals in $\mathcal{O}_{X,x}$ (since the prime ideals of $R_\mathfrak{p}$ correspond to the prime ideals of R contained in $\mathfrak{p}$). By Remark 8.50(a), we deduce that

$$\dim(X) = \sup_{x \in X} \dim(\mathcal{O}_{X,x}).$$

Because of this, local rings play a central role in dimension theory.

Example 8.53. If R is principal ideal domain, then

$$\dim(R) = \begin{cases} 0 & \text{if } R \text{ is a field,} \\ 1 & \text{otherwise.} \end{cases}$$

Hence, if X is a scheme all of whose local rings are principal ideal domains, then $\dim(X) \leq 1$.

Theorem 8.54. *Let R be a noetherian ring and let $X = \mathrm{Spec}(R)$.*

(a) (Krull's principal ideal theorem) Let $f \in R$ and let K be an irreducible component of $\mathrm{V}(f) \subseteq X$. Then $\mathrm{codim}(K, X) \leq 1$. The equality holds if f is not a zero divisor in R_{red}.

(b) (Krull's height theorem) Let $f_1, \ldots, f_n \in R$ and let $Y = \mathrm{V}(f_1, \ldots, f_n) \subseteq X$. For every irreducible subset $K \subseteq |Y|$,

$$\mathrm{codim}(K, X) \leq \mathrm{codim}(K, Y) + n.$$

(c) Let $K \subseteq |X|$ be an irreducible closed subset with $\mathrm{codim}(K, X) = n$. Then there there exist $f_1, \ldots, f_n \in R$ such that K is an irreducible component of $\mathrm{V}(f_1, \ldots, f_n) \subseteq X$.

Proof. See [Sta26, Tag 00KV], [Sta26, Tag 0BBZ] and [Sta26, Tag 00KQ]. □

Corollary 8.55. *Let R be a noetherian ring and let $\mathfrak{p} \subseteq R$ be a prime ideal. Then*

$$\mathrm{codim}(V(\mathfrak{p}), \mathrm{Spec}(R)) \leq \min\{n \in \mathbf{N} \mid \textit{there exists } f_1, \ldots, f_n \in \mathfrak{p} \textit{ with } \mathfrak{p} = \sqrt{(f_1, \ldots, f_n)}\} < \infty.$$

Proof. This follows immediately from Theorem 8.54(b). □

We now investigate the behavior of Krull dimension under morphisms.

Definition 8.56 (Going-up and going-down properties)**.** Let $f \colon Y \to X$ be a morphism of topological spaces.

(a) f has the *going-up property* if, for every $y \in Y$ and every specialization $f(y) \rightsquigarrow x'$, there exists a specialization $y \rightsquigarrow y'$ with $f(y') = x'$.

(b) f has the *going-down property* if, for every $y \in Y$ and every specialization $x' \rightsquigarrow f(y)$, there exists a specialization $y' \rightsquigarrow y$ with $f(y') = x'$.

Note that any closed map has the going-up property, while any open map of schemes has the going-down property.

The following assertions follow immediately from the definitions:

Proposition 8.57. *Let $f \colon Y \to X$ be a map of sober topological spaces, let $y \in Y$ and let $x = f(y)$.*

(a) If f has the going-up property, then $\dim(Y) \geq \dim(f(Y))$.

(b) If f has the going-down property, then $\mathrm{codim}(y, Y) \geq \mathrm{codim}(x, X) + \mathrm{codim}(y, Y_x)$.

(c) If all fibers of f have Krull dimension $\leq d$, then $\mathrm{codim}(y, Y) \leq (d+1)\mathrm{codim}(x, X) + d$.

We recall the Cohen-Seidenberg theorems about integral ring maps:

Theorem 8.58 (Cohen-Seidenberg)**.** *Let $\varphi\colon R \to S$ be an integral ring map and let $f = \operatorname{Spec}(\varphi)$.*

(a) (Going-up) *The map $|f|$ is closed and in particular has going-up property.*
(b) (Incomparability) *All fibers of $|f|$ have Krull dimension ≤ 0.*
(c) (Lying-over) *If φ is injective, then $|f|$ is surjective.*
(d) (Going-down) *If φ is injective, S is a domain, and R is integrally closed, then $|f|$ has the going-down property.*

Proof. See [Sta26, Tag 00GU], [Sta26, Tag 00GT], [Sta26, Tag 00GQ] and [Sta26, Tag 00H8]. □

Corollary 8.59 (Integral maps and dimension)**.** *If $f\colon Y \to X$ is a surjective integral map of schemes, then* $\dim(Y) = \dim(X)$.

Proof. By Remark 8.52, we may assume that $f\colon Y \to X$ is an integral map of affine schemes. By Theorem 8.58(a) and Proposition 8.57(a), $\dim(Y) \geq \dim(X)$. Conversely, by Proposition 8.57(c) with $d = 0$,

$$\dim(Y) = \sup_{y \in Y} \operatorname{codim}(y, Y) \leq \sup_{y \in Y} \operatorname{codim}(f(y), X) \leq \dim(X),$$

where the first equality is Remark 8.50(a). □

Proposition 8.60 (Going-down for flat maps)**.** *Let $\varphi\colon R \to S$ be a flat ring map and let $f = \operatorname{Spec}(\varphi)$. Then $|f|$ has the going-down property.*

Proof. See [Sta26, Tag 00HS]. □

Corollary 8.61 (Flat maps and dimension)**.** *Let $f\colon Y \to X$ be a morphism of schemes, let $y \in |Y|$, and let $x = f(y)$. If $f^*\colon \mathcal{O}_{X,x} \to \mathcal{O}_{Y,y}$ is flat, then*

$$\operatorname{codim}(y, Y) \geq \operatorname{codim}(x, X) + \operatorname{codim}(y, Y_x).$$

Proof. This follows immediately from Proposition 8.60, Proposition 8.57(b) and Remark 8.52. □

The following converse inequality is a consequence of Theorem 8.54(b, c):

Proposition 8.62 (Dimension formula)**.** *Let $f\colon Y \to X$ be a morphism of schemes, let $y \in |Y|$ and let $x = f(y)$. If $\mathcal{O}_{X,x}$ and $\mathcal{O}_{Y,y}$ are noetherian, then*

$$\operatorname{codim}(y, Y) \leq \operatorname{codim}(x, X) + \operatorname{codim}(y, Y_x).$$

Since $\mathbf{A}^1_k$ has Krull dimension 1 when k is a field, Corollary 8.61 and Proposition 8.57(c) imply that

$$\dim(X) + 1 \leq \dim(X \times \mathbf{A}^1) \leq 2\dim(X) + 1.$$

In the locally noetherian case, we have a much better result thanks to Proposition 8.62:

Corollary 8.63. *If X is a locally noetherian scheme, then* $\dim(X \times \mathbf{A}^1) = \dim(X) + 1$.

Proof. Without loss of generality, we may assume that $X = \operatorname{Spec}(R)$. In this case, the projection map $X \times \mathbf{A}^1 \to X$ corresponds to the inclusion $R \to R[x]$, which is flat, as $R[x]$ is a free R-module. By Corollary 8.61 and Proposition 8.62, for every $y \in Y$ and $x = f(y) \in X$,

$$\operatorname{codim}(y, Y) = \operatorname{codim}(x, X) + \operatorname{codim}(y, Y_x).$$

By $Y_x \simeq \operatorname{Spec}(\kappa(x)) \times \mathbf{A}^1 \simeq \operatorname{Spec}(\kappa(x)[t])$, we obtain that $\operatorname{codim}(y, Y_x) \leq \dim(Y_x) = 1$, hence $\operatorname{codim}(y, Y) \leq \dim(X) + 1$. By taking the supremum, we conclude that $\dim(Y) \leq \dim(X) + 1$.

Conversely, for every $x \in X$, we may choose $y_x \in Y_x$ corresponding to the prime ideal $(t) \subseteq \kappa(x)[t]$ and

$$\dim(Y) \geq \sup_{x \in X} (\operatorname{codim}(x, X) + 1) = \dim(X) + 1,$$

since $\operatorname{codim}(y_x, Y_x) = 1$ and $\operatorname{codim}(y_x, Y) = \operatorname{codim}(x, X) + 1$. □

Example 8.64 (Affine and projective spaces). Let X be a locally noetherian scheme and let V be a vector bundle of constant rank $n \geq 1$ over X. Then

$$\dim(\mathbf{A}(V)) = \dim(X) + n \quad \text{and} \quad \dim(\mathbf{P}(V)) = \dim(X) + n - 1.$$

Indeed, as both statements are local on X by Remark 8.50(b), we can assume that V is free, so that $\mathbf{A}(V) \simeq X \times \mathbf{A}^n$ and $\mathbf{P}(V) \simeq X \times \mathbf{P}^{n-1}$. Since $\mathbf{P}^{n-1}$ has an open covering by copies of $\mathbf{A}^{n-1}$, the claims now follow from Corollary 8.63.

Theorem 8.65 (Dimension of schemes of finite type over a field)**.** *Let k be a field and let X be an integral k-scheme locally of finite type with generic point η.*

(a) The Krull dimension *of X equals the transcendence degree of the function field $\kappa(\eta)$ over k.*
(b) For any irreducible closed subscheme $K \subseteq X$, we have $\dim(X) = \dim(K) + \operatorname{codim}(K, X)$.

Proof. See [Sta26, Tag 0A21]. □

Warning 8.66. Theorem 8.65(b) does not hold more generally for locally noetherian integral schemes. For example, let $X = \operatorname{Spec}(\mathbf{Z}_{(p)}[x])$ and $K = \mathrm{V}(px - 1)$. Then X has dimension 2 (by Example 8.53 and Corollary 8.63), K has codimension 1 in X (by Theorem 8.54(a)), but $K \simeq \operatorname{Spec}(\mathbf{Q})$ has dimension zero.

8.6. **Regular schemes.** Combining Theorem 8.54(b) and Corollary 8.55, we obtain the following characterization of the dimension of noetherian local rings:

Proposition 8.67 (Dimension of noetherian local rings)**.** *Let R be a noetherian local ring with maximal ideal $\mathfrak{m}$. Then*

$$\dim(R) = \min\{n \in \mathbf{N} \mid \textit{there exists } f_1, \dots, f_n \in \mathfrak{m} \textit{ with } \mathfrak{m} = \sqrt{(f_1, \dots, f_n)}\} < \infty.$$

Definition 8.68 (System of parameters)**.** Let R be a noetherian local ring of Krull dimension d with maximal ideal $\mathfrak{m}$. A d-tuple $(f_1, \dots, f_d)$ such that $\mathfrak{m} = \sqrt{(f_1, \dots, f_d)}$ is called a *system of parameters* of R. By Proposition 8.67, every noetherian local ring admits a system of parameters.

The folowing result is a direct consequence of Proposition 8.67:

Corollary 8.69 (Quotients of noetherian local rings)**.** *Let R be a noetherian local ring of Krull dimension d and let $f_1, \dots, f_n \in \mathfrak{m}$. The following assertions are equivalent:*

(a) The local ring $R/(f_1, \dots, f_n)$ has Krull dimension $d - n$.
(b) The sequence $(f_1, \dots, f_n)$ can be completed to a system of parameters of R.

Using Proposition 8.67 and Theorem 8.54(a) inductively, we deduce:

Corollary 8.70. *Let R be a noetherian local ring and let $(f_1, \dots, f_n)$ be regular sequence in $\mathfrak{m}$, i.e., such that f_i is not a zero divisor in $R/(f_1, \dots, f_{i-1})$ for each i. Then*

$$\dim(R/(f_1, \dots, f_n)) = \dim(R) - n.$$

Proof. By induction, it suffices to show that if $f \in \mathfrak{m}$ is not a zero divisor in R_{red}, then $\dim(R/f) = \dim(R) - 1$. Indeed, let K be the irreducible component of $\mathrm{V}(f)$. By definition,

$$\operatorname{codim}(K, \operatorname{Spec}(R)) + \dim(K) \leq \operatorname{Spec}(R).$$

By Theorem 8.54(a), $\operatorname{codim}(K, \operatorname{Spec}(R)) = 1$. Since the irreducible components of $\mathrm{V}(f)$ form a closed covering of $\mathrm{V}(f)$, Remark 8.50(b) implies that $\dim(\mathrm{V}(f)) = \sup_K \dim(K)$. Therefore, $\dim(R/f) \leq \dim(R) - 1$.

Conversely, R/f is again a noetherian local ring, hence there exists $f_1, \dots, f_n \in \mathfrak{m}$ such that $\overline{f_1}, \dots, \overline{f_n}$ is a system of parameters of R/f, where $n = \dim(R/f)$. Therefore, $\dim(R/(f_1, \dots, f_n, f)) = \mathfrak{m}$, hence $\dim(R) \leq n + 1$. □

Definition 8.71 (Depth)**.** Let R be a noetherian local ring. The *depth* of R is the maximal length of a regular sequence in $\mathfrak{m}$. By Corollary 8.70, we have $\operatorname{depth}(R) \leq \dim(R)$.

Definition 8.72 (Cohen-Macauley, regular local ring)**.** Let R be a noetherian local ring.

(a) R is called *Cohen-Macauley*, if $\operatorname{depth}(R) = \dim(R)$, i.e., if $\mathfrak{m}$ contains a regular sequence of length $\dim(R)$.
(b) R is called *regular* if it admits a system of parameters generating $\mathfrak{m}$, i.e., if $\mathfrak{m}$ can be generated by $\dim(R)$ elements.

Proposition 8.73 (Regular ⇒ Cohen-Macauley)**.** *Let R be a regular local ring of Krull dimension d. If $\mathfrak{m} = (f_1, \dots, f_d)$, then $(f_1, \dots, f_d)$ is a regular sequence. In particular, R is Cohen-Macauley.*

Proof. See [Sta26, Tag 00NQ]. □

Example 8.74 (The coordinate axes in the plane)*.* Let k be a field and let R be the local ring of

$$\operatorname{Spec}(k[x,y]/(xy)) \subseteq \mathbf{A}^2_k$$

at the origin, with maximal ideal $\mathfrak{m} = (x, y)$. Then

$$\sqrt{0} \neq \mathfrak{m} = \sqrt{(x+y)},$$

so that $x + y$ is a system of parameters of R and $\dim(R) = 1$. As $x + y$ is not a zero divisor in R, we also have $\operatorname{depth}(R) = 1$, so that R is Cohen-Macauley. However, $\mathfrak{m}$ cannot be generated by fewer than two elements, so that R is not regular.

Example 8.75 (Regular local rings of dimension ≤ 1)*.* Let R be a noetherian local ring. Then R is regular of Krull dimension 0 if and only if it is a field, as $\mathfrak{m}$ must be the zero ideal, and it is regular of Krull dimension 1 if and only if it is a discrete valuation ring, i.e, a noetherian domain in which divisibility is a total order (or equivalently a local principal ideal domain) that is not a field. Indeed, if R is principal and not a field, then it is regular of dimension 1 by definition. Conversely, if R is regular of dimension 1, then $\mathfrak{m} = (f)$ for some $f \neq 0$. Since $\bigcap_{n \in \mathbf{N}} \mathfrak{m}^n = 0$ by Krull's intersection theorem, every element of R has the form uf^n with $n \in \mathbf{N}$ and $u \in R^\times$, so that R is principal.

Definition 8.76 (Zariski (co)tangent space)**.** Let $(X, \mathcal{O}_X)$ be a locally ringed space and let $x \in X$.

(a) The *Zariski cotangent space* at x is the $\kappa(x)$-module $\mathrm{T}^*_x(X) = \mathfrak{m}_x/\mathfrak{m}_x^2$.
(b) The *Zariski tangent space* at x is the $\kappa(x)$-module $\mathrm{T}_x(X) = (\mathfrak{m}_x/\mathfrak{m}_x^2)^\vee$.

Example 8.77 (Tangent spaces in differential geometry)*.* If the locally ringed space $(X, \mathcal{O}_X)$ is a C^∞-manifold (see Example 7.47), then there is a surjective map

$$\{\mathrm{C}^\infty - \text{maps } \gamma\colon (-1,1) \to X \text{ such that } \gamma(0) = x\} \twoheadrightarrow \mathrm{T}_x(X), \quad \gamma \mapsto ([f] \mapsto (f \circ \gamma)'(0)),$$

which allows us to identify tangent vectors at x with equivalence classes of curves through x.

Remark 8.78 (Regularity via the cotangent space)*.* Let R be a noetherian local ring. The Zariski cotangent space of $\operatorname{Spec}(R)$ at the closed point is the $\kappa(\mathfrak{m})$-module $\mathfrak{m}/\mathfrak{m}^2 = \mathfrak{m} \otimes_R \kappa(\mathfrak{m})$. By Nakayama's lemma, we have

$$\dim_{\kappa(\mathfrak{m})}(\mathfrak{m}/\mathfrak{m}^2) = \min\{n \in \mathbf{N} \mid \text{there exists } f_1, \dots, f_n \in \mathfrak{m} \text{ with } \mathfrak{m} = (f_1, \dots, f_n)\}.$$

By Proposition 8.67, we therefore have $\dim(R) \leq \dim_{\kappa(\mathfrak{m})}(\mathfrak{m}/\mathfrak{m}^2)$. Moreover, R is regular if and only if $\dim(R) = \dim_{\kappa(\mathfrak{m})}(\mathfrak{m}/\mathfrak{m}^2)$.

From Remark 8.78 and Nakayama's lemma, we get the following useful criterion:

Corollary 8.79 (Quotients of regular local rings)**.** *Let R be a regular local ring of Krull dimension d and let $f_1, \dots, f_n \in \mathfrak{m}$. The following assertions are equivalent:*

(a) The local ring $R/(f_1, \dots, f_n)$ is regular of Krull dimension $d - n$.
(b) The images of $f_1, \dots, f_N$ in the Zariski cotangent space $\mathfrak{m}/\mathfrak{m}^2$ are $\kappa(\mathfrak{m})$-linearly independent.

Example 8.80 (The cuspidal cubic)*.* Let k be a field and let $X = \operatorname{Spec}(k[x,y]/(y^2 - x^3))$. We have $\dim(X) = 1$ (for example by Theorem 8.65(a)). The Zariski cotangent space at a point $(a, b) \in X(k)$ is then

$$\mathrm{T}^*_{(a,b)}(X) = (x-a, y-b)/(x-a, y-b)^2 \in \mathrm{Mod}_k,$$

where the ideals are in the ring $k[x,y]/(y^2-x^3)$. A straightforward computation shows that

$$\mathrm{T}^*_{(a,b)}(X) \simeq (k \oplus k)/(2b, -3a^2).$$

This has dimension 2 if $(a,b) = (0,0)$ and dimension 1 otherwise, so that the origin is the only singular k-point of X.

Theorem 8.81 (Properties of regular local rings)**.** *Let R be regular local rings. Then:*

(a) The canonical map of $\mathbf{N}$-graded $\kappa(\mathfrak{m})$-algebras

$$\mathrm{Sym}_{\kappa(\mathfrak{m})}(\mathfrak{m}/\mathfrak{m}^2) \to \mathrm{gr}_{\mathfrak{m}(R)} = \bigoplus_{i\in\mathbf{N}} \mathfrak{m}^i/\mathfrak{m}^{i+1}$$

is an isomorphism.

(b) (Auslander Buchsbaum theorem) *R is a unique factorization domain.*

(c) For every prime ideal $\mathfrak{p} \subseteq R$, $R_\mathfrak{p}$ is a regular local ring.

Proof. See [Sta26, Tag 00NO], [Sta26, Tag 0AGo] and [Sta26, Tag 0AFS]. □

Definition 8.82 (Regular ring, regular scheme)**.**

(a) A ring R is called *regular* if it noetherian and, for every prime ideal $\mathfrak{p} \subseteq R$, $R_\mathfrak{p}$ is a regular local ring.

(b) A scheme X is called *regular* if is locally noetherian and, for every point $x \in |X|$, the local ring $\mathcal{O}_{X,x}$ is a regular local ring.

From these definitions and Proposition 8.28, we immediately deduce:

Proposition 8.83. *"Regular" is a local property of schemes. Moreover,* $\mathrm{Spec}(R)$ *is regular if and only if R is regular.*

For completeness, let us mention the following module-theoretic characterization of regular rings, which is the key ingredient in the proof of Theorem 8.81(c):

Theorem 8.84 (Serre's homological characterization of regularity)**.** *Let R be a noetherian ring and let $n \in \mathbf{N} \cup \{\infty\}$. The following conditions are equivalent:*

(a) R is regular of Krull dimension $\leq n$.

(b) Every R-module admits a finite projective resolution of length $\leq n$.

(c) For every ideal $I \subseteq R$, the R-module R/I admits a finite projective resolution of length $\leq n$.

Proof. See [Sta26, Tag 065T], [Sta26, Tag 00OC] and [Sta26, Tag 00O9]. □

8.7. **Normal schemes.** Let R be a ring and A an R-algebra. Recall that an element $a \in A$ is *integral* over R if it satisfies a monic polynomial equation with coefficients in R, or equivalently if the R-subalgebra of A generated by a is finite over R. The set of all such elements is an R-subalgebra of A, called the *integral closure* of R in A. An integral domain R is *integrally closed* if it equals its integral closure in its fraction field $\mathrm{Frac}(R)$.

Example 8.85. By Gauss's lemma on primitive polynomials, unique factorization domains are integrally closed. In particular, by the Auslander-Buchsbaum theorem (Theorem 8.81), regular local rings are integrally closed domains.

Proposition 8.86. *Let R be an integral domain.*

(a) If R is integrally closed, then $R[S^{-1}]$ is integrally closed for any subset $S \subseteq R \setminus \{0\}$.

(b) If $R_\mathfrak{m}$ is integrally closed for every maximal ideal $\mathfrak{m} \subseteq R$, then R is integrally closed.

Proof. See [Sta26, Tag 00GY] and [Sta26, Tag 030B]. □

Definition 8.87 (Normal ring, normal scheme)**.**

(a) A ring R is called *normal* if, for every prime ideal $\mathfrak{p} \subseteq R$, $R_\mathfrak{p}$ is an integrally closed domain.

(b) A scheme X is called *normal* if, for every point $x \in |X|$, $\mathcal{O}_{X,x}$ is an integrally closed domain.

The following result follows immediately from the definitions:

Proposition 8.88. *"Normal" is a local property of schemes. Moreover,* $\mathrm{Spec}(R)$ *is normal if and only if* R *is normal.*

Remark 8.89. By Proposition 8.86, an integral domain is normal if and only if it is integrally closed.

Remark 8.90 (Schemes whose local rings are domains). For a scheme X, the following conditions are equivalent (cf. Remark 8.16):

(a) For every $x \in |X|$, $\mathcal{O}_{X,x}$ is an integral domain.
(b) X is reduced and its irreducible components are pairwise disjoint.

In general, this does not imply that X is a coproduct of integral schemes, but this is true if X is locally noetherian, by Proposition 8.42 (if a topological space has finitely many irreducible components that are pairwise disjoint, then they are clopen). In particular, every locally noetherian normal scheme is a coproduct of integral normal schemes.

Construction 8.91 (Normalization)**.** Let X be an integral scheme. The *normalization* of X is a morphism $X^\nu \to X$, which is uniquely determined by the following properties:

(a) X^ν is a normal integral scheme;
(b) $X^\nu \to X$ is integral;
(c) $X^\nu \to X$ induces an isomorphism of function fields.

By (b), X^ν must be $\mathrm{Spec}(A)$ for some integral quasi-coherent algebra A over X. To construct it, recall from Section 7.5 that

$$\mathrm{CAlg}_X \xrightarrow{\sim} \lim_{\substack{\mathrm{Spec}(R) \hookrightarrow X \\ \text{open immersion}}} \mathrm{CAlg}_R\,.$$

Let K be the function field of X. For any open immersion $u\colon \mathrm{Spec}(R) \hookrightarrow X$ with $R \neq 0$, R is a domain with fraction field K (Proposition 8.45(b)). Define $A(u)$ to be 0 if $R = 0$ and the integral closure of R in K otherwise. Using Proposition 8.86(a), we see that for any open immersion $j\colon \mathrm{Spec}(S) \hookrightarrow \mathrm{Spec}(R)$, the canonical map $A(u) \otimes_R S \to A(u \circ j)$ is an isomorphism. The algebras $A(u)$ and these canonical isomorphisms form a quasi-coherent algebra A over X with the desire properties.

Example 8.92 (Normalization of the nodal cubic). Let $X = \mathrm{Spec}(R)$, where $R = k[x,y]/(y^2 - x^3 - x^2)$ and k is a field. There is an isomorphism of k-algebras

$$R \xrightarrow{\sim} k[t^2, t^3 - t] \subseteq k[t], \quad x \mapsto t^2 - 1, \quad y \mapsto t(t^2 - 1).$$

Since t^2 comes from R, t is integral over R, so that $k[t]$ is an integral (even finite) R-algebra. Moreover, the map $R \hookrightarrow k[t]$ becomes an isomorphism after inverting x, so that R and $k[t]$ have the same field of fractions $k(t)$. Hence, the normalization of X is $\mathbf{A}^1_k = \mathrm{Spec}(k[t])$.

For a non-affine example, consider the projective closure $\bar{X} = \mathrm{Proj}(k[x,y,z]/(y^2z - x^3 - x^2z))$ of X in $\mathbf{P}^2_k$. The complement of the node (the locus where x is a unit) is isomorphic to $\mathrm{D}(y^2 - 1) \subseteq \mathrm{Spec}(k[y])$, which is already normal. Hence, the normalization of $\bar{X}$ is $\mathbf{P}^1_k$.

Proposition 8.93 (Characterization of integrally closed noetherian domains)**.** *A noetherian integral domain* R *is integrally closed if and only if the following conditions holds:*

(a) For every prime ideal $\mathfrak{p} \subseteq R$ *of codimension* 1, $R_\mathfrak{p}$ *is a discrete valuation ring (or equivalently, by Example 8.75, a regular local ring).*
(b) R *is the intersection of its localization* $R_\mathfrak{p} \subseteq \mathrm{Frac}(R)$ *at the prime ideals* $\mathfrak{p}$ *of codimension* 1.

Proof. See [Sta26, Tag 031T], [Sta26, Tag 00PD] and [Sta26, Tag 031S]. □

Corollary 8.94. *Let* X *be a locally noetherian normal scheme.*

(a) For every point $x \in |X|$ *of codimension* ≤ 1, *the local ring* $\mathcal{O}_{X,x}$ *is regular.*
(b) (Hartog's theorem) *Let* $U \subseteq X$ *be an open subscheme containing all points of codimension* ≤ 1. *Then the map* $\mathcal{O}(X) \to \mathcal{O}(U)$ *is bijective, i.e., every function on* U *extends uniquely to* X.

Proof. (a) Suppose that $x \in |X|$ has codimension 1. By assumption, $\mathcal{O}_{X,x}$ is an integrally closed noetherian domain. The unique maximal ideal has codimension 1, hence by Proposition 8.93(a), $\mathcal{O}_{X,x}$ is regular. Suppose that $x \in |X|$ has codimension 0. In particular, $\mathcal{O}_{X,x}$ is a reduced local ring of Krull dimension 0, hence it is a field. Indeed, the nilradical (0) is the unique maximal ideal of $\mathcal{O}_{X,x}$. In particular, $\mathcal{O}_{X,x}$ is regular.

(b) Without loss of generality, we may assume that X is noetherian, hence has finitely many irreducible components. By Remark 8.90, X is a coproduct of integral normal schemes. Therefore, we may further assume that X is integral.

Let $U \subseteq X$ be an open subscheme containing all points of codimension ≤ 1 and $(U_i \subseteq U)_{i \in I}$ an affine open covering such that $U_i \neq \emptyset$. By Zariski descent, we obtain the following equalizer diagram:

$$\mathcal{O}(U) \to \prod_{i \in I} \mathcal{O}(U_i) \rightrightarrows \prod_{(i,j) \in I^2} \mathcal{O}(U_i \cap U_j).$$

By Proposition 8.45, $\mathcal{O}(U)$ and $\mathcal{O}(U_i)$ are subrings of the function field $\kappa(\eta)$ of X, where $\eta \in |X|$ is the generic point. Therefore, $\mathcal{O}(U) = \bigcap_{i \in I} \mathcal{O}(U_i)$ in $\kappa(\eta)$. By Proposition 8.93(b), $\mathcal{O}(U_i) = \bigcap_{x \in U_i} \mathcal{O}_{X,x}$, where x ranges over all the points of U_i with codimension 1. Thus, $\mathcal{O}(U) = \bigcap_{x \in U} \mathcal{O}_{X,x}$, where x ranges over all the points of U with codimension 1. The same story holds for $\mathcal{O}(X)$. Since U and X have the same codimension points of codimension 1, this proves the claim.

□

Example 8.95 (Dedekind schemes). Let X be a locally noetherian scheme of Krull dimension ≤ 1. By Proposition 8.93(a), X is normal if and only if it is regular. Such a scheme X is called a *Dedekind scheme*. An affine scheme $\mathrm{Spec}(R)$ is a Dedekind scheme if and only if R is a finite product of Dedekind domains (by Remark 8.90).

If X is any locally noetherian integral scheme of Krull dimension ≤ 1, the Krull-Akizuki theorem implies that the normalization X^ν is again locally noetherian, and is therefore a Dedekind scheme.

Remark 8.96 (Serre's condition). Let X be a locally noetherian scheme. We can consider the following conditions for all $k \in \mathbf{N}$:

(R_k) : For every point $x \in |X|$ of codimension $\leq k$, $\mathcal{O}_{X,x}$ is regular.
(S_k) : For every point $x \in |X|$, $\mathrm{depth}(\mathcal{O}_{X,x}) \geq \min\{k, \mathrm{codim}(x, X)\}$.

Note that these are local properties by definition, and that $(\mathrm{R}_{k+1}) \Rightarrow (\mathrm{R}_k)$ and $(\mathrm{S}_{k+1}) \Rightarrow (\mathrm{S}_k)$. These conditions can be used to give a uniform treatments of various properties of locally noetherian schemes:

(a) X is reduced if and only if it satisfies (R_0) and (S_1).
(b) X is normal if and only if it satisfies (R_1) and (S_2).
(c) X is regular if and only if it satisfies (R_k) for all k.
(d) X is Cohen-Macauley if and only if it satisfies (S_k) for all k.

9. Smoothness

9.1. **Smooth and étale morphisms.** Smooth morphisms in algebraic geometry are the analogue of *submersions* in differential geometry, i.e., morphisms along which tangent vectors (and more general tangential data) can be lifted. In particular, there is no relation to the usage of "smooth" to mean "C^∞ in analysis (all maps in algebraic geometry are "smooth" in that sense, since they are polynomial maps). Étale morphisms are smooth morphisms of relative dimension zero, and they are analogous to *local diffeomorphisms* in differential geometry (indeed, submersions of relative dimension zero are local diffeomorphisms by the inverse function theorem).

We begin with some recollections from differential geometry. If $f_1, \dots, f_m$ are C^∞-functions on $\mathbf{R}^n$, the implicit function theorem gives a sufficient condition under which the vanishing locus

$$X = V(f_1, \dots, f_m) \subseteq \mathbf{R}^n$$

acquires a structure of C^∞-manifold: it suffices that, for every point $p \in X$, the Jacobian matrix

$$\left(\frac{\partial f_i}{x_j}(p)\right) \in \mathrm{Mat}_{m\times n}(\mathbf{R})$$

have rank m. In that case, the m rows of this matrix form a basis of the normal space to X at p, whose orthogonal complement in $\mathbf{R}^n$ is the tangent space to X at p.

Remark 9.1. Like in algebraic geometry, the vanishing locus $X = V(f_1, \dots, f_m) \subseteq \mathbf{R}^n$ is a priori a (possibly singular) geometric object called a "C^∞-scheme". The Jacobian criterion above is then more precisely a sufficient condition for the map of C^∞-schemes $X \to *$ to be a submersion.

Remark 9.2. A peculiar feature of real fields k is that the standard scalar product on k^n remains non-degenerate on any subspace of k^n (this is not the case over other fields, not even over $\mathbf{C}$!). This implies that every subspace of k^n is canonically isomorphism to its dual space, which can cause some confusion. Generalizing the above discussion away from the real numbers reveals that the row space of the Jacobian matrix is more canonically the *conormal* space, i.e., the dual of the normal space, whose elements are the linear forms that vanish on tangent vectors.

We can copy this Jacobian criterion to define smooth morphisms in algebraic geometry:

Definition 9.3 (Smooth, unramified, and étale morphisms)**.** Let $f\colon Y \to X$ be a morphism of schemes. We say that f is *smooth* (resp. *unramified*; *étale*) if, for every point $y \in |Y|$, there are affine open subschemes $V \subseteq Y$ and $U \subseteq X$ with $y \in |V|$ and $f(V) \subseteq U$ such that $f|_V$ factors as

$$V \xrightarrow{\sim} V(H) \subseteq \mathbf{A}^n_U \to U,$$

for some $n \in \mathbf{N}$ and some finite subset $H \subset \mathcal{O}(\mathbf{A}^n_U) = \mathcal{O}(U)[x_1, \dots, x_n]$ with $|H| \le n$ (resp. $|H| \ge n$; $|H| = n$) such that, for every $v \in |V|$, the Jacobian matrix

$$\left(\frac{\partial h}{\partial x_j}(v)\right)_{h,j} \in \mathrm{Mat}_{H\times n}(\kappa(v))$$

has maximal rank. Note that smooth and étale morphisms are locally of finite presentation (as H is finite), while unramified morphisms are locally of finite type, by Example 8.10.

Definition 9.3, while fairly concrete, is difficult to work with. For example, it is not at all obvious that smoothness is a local property, and it is not easy to show from this definition that a morphism is *not* smooth. In this chapter, we will establish several characterizations of smoothness, which address these issues. We record and sloganize the most important ones in the following theorems (flatness was defined in Example 8.11; the cotangent and conormal modules are defined in Section 9.2 and formal smoothness is defined in Definition 9.79):

Theorem 9.4 (Characterizations of smoothness)**.** *Let $f\colon Y \to X$ be a morphism of schemes locally of finite presentation. The following assertions are equivalent:*

(a) (Jacobian criterion) *f is smooth.*

(b) (Infinitesimal lifting property) *f is formally smooth.*
(c) (Cotangent complex is a vector bundle) *The cotangent module $\Omega_f \in \mathrm{Mod}_Y$ is a vector bundle and the conormal module $\mathcal{N}_f \in \mathrm{Mod}_Y$ is zero.*
(d) (Flat and cotangent module is a vector bundle) *f is flat and the cotangent module $\Omega_f \in \mathrm{Mod}_Y$ is a vector bundle of rank $y \mapsto \dim_y(Y_{f(y)})$.*
(e) (Flat and geometric fibers are regular) *f is flat and for every algebraically closed field k and every map $x\colon \mathrm{Spec}(k) \to X$, the scheme Y_x is regular.*

Theorem 9.5 (Characterizations of étaleness)**.** *Let $f\colon Y \to X$ be a morphism of schemes locally of finite presentation. The following assertions are equivalent:*

(a) (Jacobian criterion) *f is étale.*
(b) (Unique Infinitesimal lifting property) *f is formally étale.*
(c) (Cotangent complex vanishes) *The cotangent module $\Omega_f \in \mathrm{Mod}_Y$ and the conormal module $\mathcal{N}_f \in \mathrm{Mod}_Y$ are both zero.*
(d) (Flat and cotangent module vanishes) *f is flat and the cotangent module $\Omega_f \in \mathrm{Mod}_Y$ is zero.*
(e) (Flat with discrete geometric fibers) *f is flat and, for every algebraically closed field k and every map $x\colon \mathrm{Spec}(k) \to X$, the k-scheme Y_x is a coproduct of copies of $\mathrm{Spec}(k)$.*

Theorem 9.6 (Characterizations of unramifiedness)**.** *Let $f\colon Y \to X$ be a morphism of schemes locally of finite type. The following assertions are equivalent:*

(a) (Jacobian criterion) *f is unramified.*
(b) (Infinitesimal lifting property) *f is formally unramified.*
(c) (Cotangent module vanishes) *The cotangent module $\Omega_f \in \mathrm{Mod}_Y$ is zero.*
(d) (Geometric fibers are sets) *For every algebraically closed field k and every map $x\colon \mathrm{Spec}(k) \to X$, the k-scheme Y_x is a coproduct of copies of $\mathrm{Spec}(k)$.*

Corollary 9.7. *"Smooth", "unramified", and "étale" are local properties of morphisms of schemes that are stable under base change and composition.*

Remark 9.8. One can show that these properties are even fpqc-local on the base.

Corollary 9.9. *A morphism of schemes is étale if and only if it is smooth and unramified.*

Example 9.10. Immersions are unramified: we can take $n = 0$ in Definition 9.3. Open immersions are even étale, as we can also take $H = \varnothing$. We will see later that, conversely, a smooth immersion is an open immersion (Remark 9.55(b)). In fact, one can show that smooth monomorphisms are open immersions, but this is more difficult.

Example 9.11. Let k be a ring and consider the affine curve $C = \mathrm{Spec}(k[x,y]/(y^2 - x^3 + x))$ from Section 1.1. By definition, we have $C \simeq \mathrm{V}(y^2 - x^3 + x) \subseteq \mathbf{A}^2_k$. The corresponding Jacobian matrix is the 1×2-matrix $(-3x^2 + 1, 2y)$ over $k[x,y]$. A point $v \in |C|$ corresponds to a pair $(a,b) \in \kappa(v)^2$ with $b^2 = a^3 - a$, and evaluating the matrix at v yields the matrix $(-3a^2 + 1, 2b)$ over $\kappa(v)$. This matrix only fails to have rank 1 if $\mathrm{char}(\kappa(v)) = 2$ and $(a,b) = (1,0)$. Thus, if k is a $\mathbf{Z}[\frac{1}{2}]$-algebra, then the map $C \to \mathrm{Spec}(k)$ is smooth by Definition 9.3.

If on the other hand 2 is not a unit in k, then $C \to \mathrm{Spec}(k)$ is *not* smooth: one can see this via (e) of Theorem 9.4, by computing that the Zariski cotangent space at the k-point $(1,0)$ has dimension 2 when k is a field of characteristic 2, so that the base change of C to any such field is not regular.

YW: Return to this example.

Example 9.12 (Finite separable field extensions are étale)*.* Let k be a field and let $k \subseteq k'$ be a finite separable field extension. By the primitive element theorem, there is an $a \in k'$ such that $k' = k[a]$. The choice of a is equivalent to a choice of closed immersion $\mathrm{Spec}(k') \hookrightarrow \mathbf{A}^1_k$ over $\mathrm{Spec}(k)$. Let $m \in k[t]$ be the minimal polynomial of a. As a is separable over k, a is not a multiple root of m, so that $\frac{\partial m}{\partial t}(a) \neq 0$ in k'. By Definition 9.3, this shows that $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$ is étale.

Conversely, if $k \subseteq k'$ is a finite field extension such that $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$ is smooth, then k' is a separable extension of k. We can see this via Theorem 9.4(e) and the fact that if k' is not separable over k and $\overline{k}$ is an algebraically closed extension of k, then $k' \otimes_k \overline{k}$ is not reduced and hence not regular.

YW: Return to this example.

YW: Return to this remark.

Remark 9.13 (Local structure of smooth and unramified maps). An advantage of the above definitions in terms of the Jacobian criterion is that they immediately imply the following local structure theorems for smooth and unramified maps, which are very useful:

(a) (Local structure of smooth maps) A map $f\colon Y \to X$ is smooth if and only if, for every $y \in |Y|$, there exist affine open subschemes $V \subseteq Y$ and $U \subseteq X$ with $y \in |V|$ and $f(V) \subseteq U$ such that $f|_V$ factors as
$$V \to \mathbf{A}^n_U \to U,$$
where $V \to \mathbf{A}^n_U$ is étale and $\mathbf{A}^n_U \to U$ is the canonical map.

(b) (Local structure of unramified maps) A map $f\colon Y \to X$ is unramified if and only if, for every $y \in |Y|$, there exist affine open subschemes $V \subseteq Y$ and $U \subseteq X$ with $y \in |V|$ and $f(V) \subseteq U$ such that $f|_V$ factors as
$$V \hookrightarrow V' \to U,$$
where $V \hookrightarrow V'$ is a closed immersion and $V' \to U$ is étale.

9.2. Cotangent and conormal modules.

Definition 9.14 (Derivation)**.** Let $R \to S$ be a ring map and let M be an S-module. An *R-derivation* of S into M is a map (of sets) $\delta\colon S \to M$ such that:

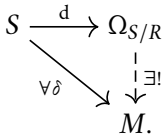
(a) δ is R-linear;
(b) δ satisfies the *Leibniz rule*: for every $a, b \in S$, $\delta(ab) = a\delta(b) + b\delta(a)$.

We denote by $\mathrm{Der}_R(S, M)$ the set of R-derivations of S into M, which is an S-module under pointwise addition and scalar multiplication.

Remark 9.15. An R-derivation $\delta\colon S \to M$ is equivalently a $\mathbf{Z}$-derivation such that $\delta|_R = 0$.

Remark 9.16 (Split square-zero extensions). The following alternative description of derivations is often useful. If M is an S-module, the *split square-zero extension* of S by M is the ring whose underlying abelian group is $S \oplus M$ with multiplication given by
$$(a, m)(b, n) = (ab, bm + an).$$
Note that both canonical maps $S \hookrightarrow S \oplus M \twoheadrightarrow S$ are ring maps, and that $M \hookrightarrow S \oplus M$ is a square-zero ideal. There is then a bijection
$$\mathrm{Der}_R(S, M) \xrightarrow{\sim} \{R\text{-algebra sections of } S \oplus M \twoheadrightarrow S\}, \quad \delta \mapsto (\mathrm{id}_S, \delta).$$

Definition 9.17 (Cotangent module, differential forms)**.** Let $R \to S$ be a ring map. We denote by $\mathrm{d}\colon S \to \Omega_{S/R}$ the *universal* R-derivation of S, i.e., the R-derivation such that, for every R-derivation $\delta\colon S \to M$, there is a unique S-linear map $f\colon \Omega_{S/R} \to M$ with $f \circ \mathrm{d} = \delta$:



The S-module $\Omega_{S/R}$ is called *cotangent module* of S over R, and its elements are called the *differential forms* or simply the *differentials* of S over R.

Remark 9.18. The functor $\mathrm{Der}_R(S, -)\colon \mathrm{Mod}_S \to \mathrm{Mod}_R$ has a left adjoint given by $M \mapsto \Omega_{S/R} \otimes_R M$.

Example 9.19. Let $R \to S$ be an epimorphism in CAlg, for example a surjection or a localization. Then it follows from Remark 9.16 that $\mathrm{Der}_R(S, M) = 0$ for any S-module M, so that $\Omega_{S/R} = 0$.

Proposition 9.20 (Presentation of the cotangent module)**.** *Let R be a ring and let S be an R-algebra with a presentation $S = R[x_i \mid i \in I]/(f_j \mid j \in J)$. Then the S-module $\Omega_{S/R}$ is generated by the differentials $\mathrm{d}x_i$ for all $i \in I$ with relations given by $\sum_{i \in I} \frac{\partial f_j}{x_i} \mathrm{d}x_i$ for all $j \in J$.*

YW: Solve this exercise.

Proof. This is left as an exercise. □

Example 9.21 (Cotangent modules of polynomials and quotient algebras). The following computations are important special cases of Proposition 9.20:

(a) If $P = R[x_i \mid i \in I]$ is polynomial R-algebra, then $\Omega_{P/R}$ is a free P-module with basis $(\mathrm{d}x_i)_{i\in I}$.
(b) If $R \twoheadrightarrow S$ is a surjection, then $\Omega_{S/R} = 0$.

Corollary 9.22 (Finiteness of the cotangent module). *If $R \to S$ is a morphism of finite presentation (resp. of finite type), then the S-module $\Omega_{S/R}$ is of finite presentation (resp. of finite type).*

Proof. The follows immediately from Proposition 9.20. □

Remark 9.23 (The de Rham complex). As in differential geometry, we define *differential n-forms* for $n \in \mathbf{N}$ by taking exterior powers of the cotangent module $\Omega^n_{S/R} = \bigwedge^n_S \Omega_{S/R}$. The universal derivation d then extends to a cochain complex of S-modules

$$\Omega^0_{S/R} = S \xrightarrow{\mathrm{d}} \Omega^1_{S/R} = \Omega_{S/R} \xrightarrow{\mathrm{d}} \Omega^2_{S/R} \xrightarrow{\mathrm{d}} \Omega^3_{S/R} \xrightarrow{\mathrm{d}} \dots, \quad \mathrm{d}(g\mathrm{d}f_1 \wedge \dots \wedge \mathrm{d}f_n) = \mathrm{d}g \wedge \mathrm{d}f_1 \wedge \dots \wedge \mathrm{d}f_n,$$

which is a commutative differential graded S-algebra under the exterior product. This complex is called the *de Rham complex* of S over R, and its cohomology is called the *de Rham cohomology* of S over R.

We now discuss the functoriality of the cotangent module. Consider a commutative square of rings

$$\begin{array}{ccc} R & \longrightarrow & S \\ \downarrow & & \downarrow \\ R' & \longrightarrow & S'. \end{array}$$

The map $S \to S' \xrightarrow{\mathrm{d}} \Omega_{S'/R'}$ is then an R-derivation of S, which induces an S-linear map $\Omega_{S/R} \to \Omega_{S'/R'}$, or equivalently an S'-linear map $\Omega_{S/R} \otimes_S S' \to \Omega_{S'/R'}$. This construction is easily seen to be compatible with vertical composition of squares. To make this precise, consider the following category CAlgMod:

- The objects of CAlgMod are pairs (R, M) where R is a ring and M is an R-module.
- A morphism $(R, M) \to (S, N)$ in CAlgMod is a pair (φ, f), where $\varphi\colon R \to S$ is a ring map and f is an R-linear map $M \to N$ (or equivalently an S-linear map $M \otimes_R S \to N$).
- Composition is defined componentwise.

There is a forgetful functor CAlgMod $\to$ CAlg sending (R, M) to R (also known as the *cocartesian fiberation* classified by the functor $\mathrm{Mod}^*\colon$ CAlg $\to$ Cat (a) of Example 4.3). We can then view the assignment $(R \to S) \mapsto (S, \Omega_{S/R})$ as a functor making the following triangle commute:

$$\begin{array}{ccc} & & \mathrm{CAlgMod} \\ & \overset{\Omega}{\nearrow} & \downarrow \\ \mathrm{Ar(CAlg)} & \xrightarrow[\text{target}]{} & \mathrm{CAlg}. \end{array}$$

Proposition 9.24 (Properties of the cotangent module). *Let $R \to S$ be a ring map.*

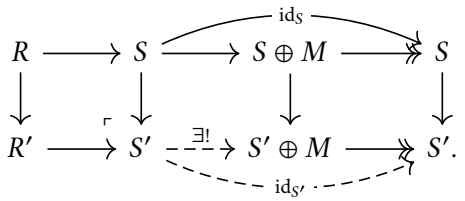
(a) (Base change) *For any base change $R' \to S'$ of $R \to S$, the S'-linear map $\Omega_{S/R} \otimes_S S' \to \Omega_{S'/R'}$ is an isomorphism.*
(b) (Localization) *For any localization $S \to S'$, the S'-linear map $\Omega_{S/R} \otimes_S S' \to \Omega_{S'/R}$ is an isomorphism.*
(c) (Quotients) *For any surjection $S \twoheadrightarrow S'$ with kernel I, the following sequence of S'-modules is exact:*

$$I/I^2 \xrightarrow{\mathrm{d}} \Omega_{S/R} \otimes_S S' \to \Omega_{S'/R} \to 0.$$

Proof. (a) Let M be a S'-module. By the universal property of cotangent modules, it suffices to show that

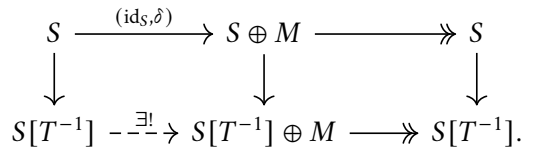
$$\mathrm{Der}_R(S, M) \simeq \mathrm{Der}_{R'}(S', M).$$

Let $S \to S \oplus M$ be a R-linear section of $S \oplus M \twoheadrightarrow S$. Consider the following diagram:



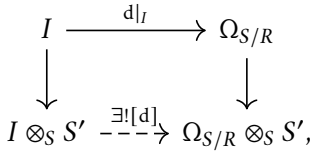
By the universal property of cocartesian squares, there exist unique dashed arrows such that the above diagram commutes. By uniqueness, the map $S' \to S' \oplus M$ is a section of $S' \oplus M \twoheadrightarrow S'$. Conversely, if $(\mathrm{id}_{S'}, \delta') \colon S' \to S' \oplus M$ is a section of $S' \oplus M \twoheadrightarrow S'$, then $(\mathrm{id}_S, \delta) \colon S \to S \oplus M$ is a section of $S \oplus M \twoheadrightarrow S$, where δ is the precomposition of δ' with the map $S \to S'$.

(b) Write $S' = S[T^{-1}]$ and let M be a S'-module. Consider the following diagram



There exists a unique dashed arrow such that the above diagram commutes. Indeed, it suffices to show that $S \to S \oplus M \to S[T^{-1}] \oplus M$ sends T to units. Let $t \in T$. The inverse of $(t, \delta(t)) \in S[T^{-1}] \oplus M$ is given by $(t^{-1}, -t^{-2}\delta(t))$.

(c) Let $\mathrm{d} \colon S \to \Omega_{S/R}$ be the universal R-derivation of S. Consider the following diagram:



where $I \otimes_S S \simeq I/I^2$. For every $a, b \in I$, $\mathrm{d}(ab) = a\mathrm{d}(b) + \mathrm{d}(a)b \equiv 0 \mod I$. Therefore, there exists a unique map $[\mathrm{d}]$ such that the above diagram commutes. Moreover, it is S'-linear. Indeed, let $[a] \in I/I^2$ and $[s] \in S'$ be elements represented by $a \in I$ and $s \in S$ respectively. Clearly, $[s][a] = [sa]$. By

$$[\mathrm{d}]([s][a]) = [\mathrm{d}]([sa]) = \mathrm{d}(sa) \equiv s\mathrm{d}(a) \mod I,$$

the map $[\mathrm{d}]$ is S'-linear.

Finally, it suffices to show that $\mathrm{coker}([\mathrm{d}]) \simeq \Omega_{S'/R}$. Let M be an S'-module and $\alpha \colon \Omega_{S/R} \otimes_S S' \to M$ be an S'-linear map, corresponding to an R-derivation $\delta \colon S \to M$. Unwinding the definitions, α factors through $\mathrm{coker}([\mathrm{d}])$ if and only if $\delta(i) = 0$ for every $i \in I$, i.e., if and only if the map δ factors through S'. □

Example 9.25. Let k be a ring and consider the ring maps

$$k \xrightarrow{f} k[x] \xrightarrow{g} k[x]/(x^2) = R.$$

By Proposition 9.20, we have

$$g^*(\Omega_f) = R\mathrm{d}x \quad \text{and} \quad \Omega_{gf} = R\mathrm{d}x/(2x\mathrm{d}x).$$

We thus have the following exact sequence, as predicted by Proposition 9.24(c):

$$(x^2)/(x^4) \xrightarrow{\mathrm{d}} g^*(\Omega_f) \to \Omega_{gf} \to 0,$$

Note that the first map is not injective (if $k \neq 0$), since $\mathrm{d}(x^3) = 3x^2\mathrm{d}x = 0$.

Proposition 9.26 (The fundamental exact sequence of cotangent modules)**.** *Let $R \to S \to T$ be ring maps. Then the induced sequence of T-modules*

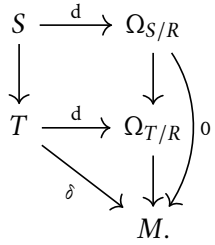
$$\Omega_{S/R} \otimes_S T \xrightarrow{\alpha} \Omega_{T/R} \to \Omega_{T/S} \to 0$$

is exact.

Proof. Let M be a T-module. By universal property of $\Omega_{T/S}$, it suffices to show that

$$\mathrm{Map}_T(\mathrm{coker}(\alpha), M) \simeq \mathrm{Der}_S(T, M).$$

The left hand side can be identified with T-linear maps $\Omega_{T/R} \to M$ such that the precomposition with $\Omega_{S/R} \to \Omega_{T/R}$ is zero. Consider the following commutative diagram:



Therefore, $\mathrm{Map}_T(\mathrm{coker}(\alpha), M)$ can be further identified with R-derivations $\delta\colon T \to M$ such that $\delta|_S$. By definition, this is $\mathrm{Der}_S(T, M)$. □

Example 9.27. Let k be a ring and consider the k-algebra maps

$$k \xrightarrow{f} k[x,y]/(y^2 - x^3) \xhookrightarrow{g} k[t], \quad g(x) = t^2, \quad g(y) = t^3.$$

By Proposition 9.20, we have

$$\Omega_{gf} = k[t]\mathrm{d}t \quad \text{and} \quad g^*(\Omega_f) = (k[t]\mathrm{d}x \oplus k[t]\mathrm{d}y)/(2t^3\mathrm{d}y - 3t^4\mathrm{d}x),$$

and the $k[t]$-linear map $g^*(\Omega_f) \to \Omega_{gf}$ sends $\mathrm{d}x$ to $2t\mathrm{d}t$ and $\mathrm{d}y$ to $3t^2\mathrm{d}t$. By Proposition 9.26, we obtain the following presentation of Ω_g:

$$\Omega_g = \mathrm{coker}(g^*(\Omega_f) \to \Omega_{gf}) = k[t]\mathrm{d}t/(2t\mathrm{d}t, 3t^2\mathrm{d}t).$$

Note that the map $g^*(\Omega_f) \to \Omega_{gf}$ also has a nontrivial kernel (if $k \neq 0$), as it identifies $3t\mathrm{d}x$ and $2\mathrm{d}y$.

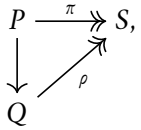
As Example 9.27 shows, the fundamental exact sequence of Proposition 9.26 is not always a short exact sequence. Our next goal is to extend this sequence to the left, generalizing Proposition 9.24(c).

Construction 9.28. Let $R \to S$ be a ring map and let $\pi\colon P \twoheadrightarrow S$ be a surjection where P is a polynomial R-algebra. Let $I \subseteq P$ be the kernel of π. By the Leibniz rule, the universal derivation $\mathrm{d}\colon P \to \Omega_{P/R}$ induces an S-linear map

$$\mathrm{d}_\pi\colon I/I^2 \to \Omega_{P/R} \otimes_P S,$$

whose cokernel is $\Omega_{S/R}$ by Proposition 9.24(c). We denote the kernel of d_π by $\mathcal{N}_{S/R}(\pi)$.

Lemma 9.29. *Given two polynomial R-algebras P and Q and a commuting triangle*



the induced map $\mathcal{N}_{S/R}(\pi) \to \mathcal{N}_{S/R}(\rho)$ is an isomorphism.

YW: From here until the end of the section, I feel like some of the proofs are not very well written as I've hoped...

Proof. Since Q is an polynomial R-algebra and $\pi\colon P \to S$ is surjective, we may lift the image of the generators of Q to construct a map of R-algebras $Q \to P$ such that the following diagram commutes:

$$\begin{array}{ccc} Q & \xrightarrow{\rho} & S. \\ \downarrow & \nearrow_{\pi} & \\ P & & \end{array}$$

To show that these maps induce mutual inverses on conormal modules, it suffices to show that, any maps $\alpha, \beta\colon P \to Q$ satisyfing $\pi \circ \alpha = \rho$ and $\pi \circ \beta = \rho$ induce the same map on conormal modules. Consider the following commutative diagram:

$$\begin{array}{ccccc} \mathcal{N}_{S/R}(\pi) & \hookrightarrow & K/K^2 & \xrightarrow{\mathrm{d}_\pi} & \Omega_{P/R} \otimes_P S \\ \downarrow & & \downarrow & \swarrow_{h} & \downarrow \\ \mathcal{N}_{S/R}(\rho) & \hookrightarrow & L/L^2 & \xrightarrow{\mathrm{d}_\rho} & \Omega_{Q/P} \otimes_Q S, \end{array}$$

where $K = \ker(\pi)$ and $L = \ker(\rho)$. It suffices to construct a map $h\colon \Omega_{P/R} \otimes_P S \to L/L^2$ such that $h \circ \mathrm{d}_\pi = \alpha - \beta$. Consider the following commutative diagram:

$$\begin{array}{ccc} P & \xrightarrow{\alpha-\beta} & L \\ & \searrow_{\overline{\alpha}-\overline{\beta}} & \downarrow \\ & & L/L^2. \end{array}$$

Unwinding the constructions, it suffices to show that the map $\overline{\alpha} - \overline{\beta}$ is an R-derivation. For every $p, q \in P$,

$$\begin{aligned} (\alpha - \beta)(pq) &= \alpha(p)(\alpha - \beta)(q) + \beta(q)(\alpha - \beta)(p) \\ &= \pi(p)(\alpha - \beta)(q) + \pi(q)(\alpha - \beta)(p) \\ &= p(\alpha - \beta)(q) + q(\alpha - \beta)(p) \end{aligned}$$

in L/L^2, where the second equality follows from the definition of the S-module structure of L/L^2. □

Thus, the S-module $\mathcal{N}_{S/R}(\pi)$ is canonically independent of π. For definiteness, we can define the conormal module using the canonical choice of π:

Definition 9.30 (Conormal module)**.** Let $R \to S$ be a ring map and let $\pi\colon R[x_s \mid s \in S] \twoheadrightarrow S$ be the R-algebra map with $\pi(x_s) = s$. The S-module $\mathcal{N}_{S/R}(\pi)$ is called the *conormal module* of S over R and is denoted by $\mathcal{N}_{S/R}$.

Example 9.31 (Conormal modules of polynomial and quotient algebras)*.* We can compute the conormal module $\mathcal{N}_{S/R}$ using any choice of $\pi\colon P \twoheadrightarrow S$, by Lemma 9.29. For example:

(a) If P is any polynomial R-algebra, then $\mathcal{N}_{P/R} = 0$ (choose $\pi = \mathrm{id}_P$).
(b) If $R \twoheadrightarrow S$ is a surjection with kernel I, then $\mathcal{N}_{S/R} = I/I^2$ (choose $P = R$).

The conormal module has the same functoriality in $R \to S$ as the cotangent module. We begin with a commutative square

$$\begin{array}{ccc} R & \longrightarrow & S \\ \downarrow & & \downarrow \\ R' & \longrightarrow & S'. \end{array}$$

This gives us the following commutative diagram:

$$\begin{array}{ccccc} R & \longrightarrow & P & \xrightarrow{\pi} & S \\ \downarrow & & \downarrow & & \downarrow \\ R' & \longrightarrow & P' & \xrightarrow{\pi'} & S'. \end{array}$$

Hence, we obtain the the following map, by the commutative diagram

$$\begin{array}{ccccc} \mathcal{N}_{S/R} & \longrightarrow & I/I^2 & \xrightarrow{\mathrm{d}_\pi} & \Omega_{P/R} \otimes_P S \\ \downarrow & & \downarrow & & \downarrow \\ \mathcal{N}_{S'/R'} & \longrightarrow & I'/I'^2 & \xrightarrow{\mathrm{d}_{\pi'}} & \Omega_{P'/R'} \otimes_{P'} S', \end{array}$$

where $P = R[x_s \mid s \in S]$, $P' = R'[x_t \mid t \in S']$, $I = \ker(\pi)$, $I' = \ker(\pi')$, and $P \to P'$ is induced by $S \to S'$.

Proposition 9.32 (Properties of the conormal module)**.** *Let $R \to S$ be a ring map.*

(a) (Base change) *For any base change $R' \to S'$ of $R \to S$ such that $\mathrm{Tor}_1^R(S, R') = 0$, the S'-linear map $\mathcal{N}_{S/R} \otimes_S S' \to \mathcal{N}_{S'/R'}$ fits in an exact sequence*

$$\mathrm{Tor}_2^S(\Omega_{S/R}, S') \to \mathcal{N}_{S/R} \otimes_S S' \to \mathcal{N}_{S'/R'} \to \mathrm{Tor}_1^S(\Omega_{S/R}, S') \to 0.$$

In particular, if $R \to R'$ is flat or if $R \to S$ and $\Omega_{S/R}$ are flat, then $\mathcal{N}_{S/R} \otimes_S S' \xrightarrow{\sim} \mathcal{N}_{S'/R'}$.

(b) (Localization) *For any localization $S \to S'$, the S'-linear map $\mathcal{N}_{S/R} \otimes_S S' \to \mathcal{N}_{S'/R}$ is an isomorphism.*

Proof. (a) Let P be a polynomial R-algebra and let $P \twoheadrightarrow S$ be a surjection. Since $\mathrm{Tor}_1^R(S, R') = 0$, the following sequence is exact:

$$0 \to I \otimes_R R' \to P \otimes_R R' \to S' \to 0,$$

where $P \otimes_R R'$ is a polynomial R'-algebra. By Proposition 9.26, there is the following exact sequence:

$$0 \to \mathcal{N}_{S/R} \to I/I^2 \xrightarrow{\mathrm{d}} \Omega_{P/R} \otimes_P S \to \Omega_{S/R} \to 0.$$

We may split this into the following two short exact sequences:

$$0 \to \mathcal{N}_{S/R} \to I/I^2 \xrightarrow{\sim} \mathrm{im}(\mathrm{d}) \to 0 \quad \text{and} \quad 0 \to \mathrm{im}(\mathrm{d}) \to \Omega_{P/R} \otimes_P S \to \Omega_{S/R} \to 0.$$

Since $\Omega_{P/R} \otimes_P S$ is a free S-module, the long exact Tor-sequence for the second short exact sequence is

$$0 \to \mathrm{Tor}_2^S(\Omega_{S/R}, S') \xrightarrow{\sim} \mathrm{Tor}_1^S(\mathrm{im}(\mathrm{d}), S') \to 0 \to \mathrm{Tor}_1^S(\Omega_{S/R}, S') \to \mathrm{im}(\mathrm{d}) \otimes_S S' \to \Omega_{P/R} \otimes_P S' \to 0.$$

Since $- \otimes_S S'$ is left exact, the first map in the following composite is surjective:

$$I/I^2 \otimes_S S' \xrightarrow{f} \mathrm{im}(\mathrm{d}) \otimes_S S' \xrightarrow{g} \Omega_{P/R} \otimes_S S'.$$

By the kernel-cokernel sequence for a composite, the following sequence is exact:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \ker(f) & \longrightarrow & \ker(g \circ f) & \longrightarrow & \ker(g) & \longrightarrow & 0 \\ & & \downarrow \simeq & & \downarrow \simeq & & \downarrow \simeq & & \\ 0 & \longrightarrow & \mathrm{im}(\mathrm{d} \otimes_S S') & \longrightarrow & \mathcal{N}_{S'/R'} & \longrightarrow & \mathrm{Tor}_1^S(\Omega_{S/R}, S') & \longrightarrow & 0. \end{array}$$

Finally, consider the following long exact Tor-sequence:

$$\mathrm{Tor}_1^S(\mathrm{im}(\mathrm{d}), S') \to \mathcal{N}_{S/R} \otimes_S S' \to I/I^2 \otimes_S S' \xrightarrow{\mathrm{d} \otimes_S S'} \mathrm{im}(\mathrm{d}) \otimes_S S' \to 0.$$

By $\mathrm{Tor}_2^S(\Omega_{S/R}, S') \simeq \mathrm{Tor}_1^S(\mathrm{im}(\mathrm{d}), S')$ and the kernel-cokernel sequence, we deduce the claim.

(b) Write $S' = S[T^{-1}]$. Let P be a polynomial R-algebra, $P \twoheadrightarrow S$ a surjection with kernel I,

$$P[x_t | t \in T] \twoheadrightarrow S', \quad x_t \mapsto t^{-1}$$

a surjection with kernel J and $S[x_t] \twoheadrightarrow S'$ a surjection with kernel K. By the kernel-cokernel exact sequence for $P[x_t] \twoheadrightarrow S[x_t] \to S'$, the following sequence is exact:

$$0 \to I[x_t | t \in T] \to J \to K \to 0.$$

By explicit computations, this implies that the following sequence is exact:

$$I/I^2 \otimes_S S' \simeq I[x_t]/I[x_t]^2 \otimes_{S[x_t]} S' \to K/K^2 \to J/J^2 \to 0.$$

By Proposition 9.24(c), we may consider the following commutative diagram:

$$\begin{array}{ccccccccc} 0 & \dashrightarrow & I/I^2 \otimes_S S' & \longrightarrow & J/J^2 & \longrightarrow & K/K^2 & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \mathrm{d}} & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \Omega_{P/R} \otimes_P S' & \longrightarrow & \Omega_{P[x_t]/R} \otimes_{P[x_t]} S' & \longrightarrow & \Omega_{S[x_t]/R} \otimes_{S[x_t]} S' & \longrightarrow & 0. \end{array}$$

The kernel of the second vertical map is $\mathcal{N}_{S'/R}$. Since localization is flat, the kernel of the first column is $\mathcal{N}_{S/R} \otimes_S S'$. It suffices to show that the first row is left exact and that the third vertical map is an isomorphism. Then the claim follows from the snake lemma.

We will first show that the map $I/I^2 \otimes_S S' \to J/J^2$ admits a retraction. Let $\tilde{T} \subseteq P$ be a set of lifts of $T \subseteq S$. We may factor the map $P[x_t] \to S[T^{-1}]$ as

$$P[x_t] \to P[\tilde{T}^{-1}] \to S[T^{-1}],$$

where the kernel of the second map is $I[T^{-1}]$. Since J is the kernel of the composite, there exists a map $J \to I[T^{-1}]$. The induced map $J/J^2 \to I/I^2 \otimes_S S'$ is the desired retraction.

Finally, we will show that the third vertical map is an isomorphism. By definition, K is generated by elements of the form $tx_t - 1$, where $t \in T$. Since $S[x_t]/K \simeq S'$, K/K^2 is a S'-module, i.e., every $t \in T$ acts invertibly on K/K^2. Therefore, $K[T^{-1}]/K[T^{-1}]^2 \simeq K/K^2$. Furthermore, the map

$$\varphi\colon S'[x_t] \to S'[x_t], \quad x_t \mapsto tx_t - 1,$$

defines an isomorphism $(x_t | t \in T)/(x_t | t \in T)^2 \simeq K[T^{-1}]/K[T^{-1}]^2$. Indeed, its inverse is given by $\varphi^{-1}(x_t) = t^{-1}(x_t + 1)$. This implies that K/K^2 is a free S'-module generated by the basis $tx_t - 1$. Since $\mathrm{d}(tx_t - 1) = t\mathrm{d}x_t$ and t is invertible, the map $\mathrm{d}\colon K/K^2 \to \bigoplus_{t\in T} S'\mathrm{d}t$ is an isomorphism.

□

Corollary 9.33. *If $R \to S$ is a flat epimorphism in* CAlg*, for example a localization, then $\mathcal{N}_{S/R} = 0$.*

Theorem 9.34 (The fundamental exact sequence of cotangent and conormal modules)**.** *Let $R \to S \to T$ be ring maps. Then there is an exact sequence of T-modules*

$$? \to \mathcal{N}_{T/R} \to \mathcal{N}_{T/S} \to \Omega_{S/R} \otimes_S T \to \Omega_{T/R} \to \Omega_{T/S} \to 0,$$

where $?$ fits in an exact sequence

$$\mathrm{Tor}_2^S(\Omega_{S/R}, T) \to \mathcal{N}_{S/R} \otimes_S T \to ? \to \mathrm{Tor}_1^S(\Omega_{S/R}, T) \to 0.$$

In particular, if $\mathrm{Tor}_1^S(\Omega_{S/R}, T) = 0$ (e.g. if either $\Omega_{S/R}$ or T is flat over S), then there is an exact sequence

$$\mathcal{N}_{S/R} \otimes_S T \to \mathcal{N}_{T/R} \to \mathcal{N}_{T/S} \to \Omega_{S/R} \otimes_S T \to \Omega_{T/R} \to \Omega_{T/S} \to 0.$$

Proof. Let $R[x_i | i \in I] \twoheadrightarrow S$ be a surjection with kernel I. By Proposition 9.24, the sequence

$$I/I^2 \to \bigoplus_{i\in I} S\mathrm{d}x_i \to \Omega_{S/R} \to 0$$

is exact. Since $- \otimes_S T$ is right exact, the following sequence is again exact:

$$0 \to ? \to I/I^2 \otimes_S T \to \bigoplus_{i\in I} T\mathrm{d}x_i \to \Omega_{S/R} \otimes_S T \to 0.$$

Let $S[y_j | j \in J] \twoheadrightarrow T$ be a surjection with kernel J and $R[x_i, y_j \mid i \in I, j \in J] \twoheadrightarrow T$ a surjection with kernel K. By Proposition 9.24, we may consider the following commutative diagram:

$$\begin{array}{ccccccccc}
0 & \longrightarrow & ? & \longrightarrow & I/I^2 \otimes_S T & \longrightarrow & \bigoplus_{i\in I} T\mathrm{d}x_i & \longrightarrow & \Omega_{S/R} \otimes_S T \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & \mathcal{N}_{T/R} & \longrightarrow & K/K^2 & \xrightarrow{\mathrm{d}} & \bigoplus_{k\in I\sqcup J} T\mathrm{d}x_k & \longrightarrow & \Omega_{T/R} \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & \mathcal{N}_{T/S} & \longrightarrow & J/J^2 & \xrightarrow{\mathrm{d}} & \bigoplus_{j\in J} T\mathrm{d}x_j & \longrightarrow & \Omega_{T/S} \longrightarrow 0.
\end{array}$$

where the third column is exact. We will now demonstrate how to construct the second column and show that it is right exact. Since Mod_R is abelian, it follows from kernel-cokernel sequence that

$$0 \to I[y_j | j \in J] \to K \to J \to 0$$

is exact. This implies that the following sequence is exact:

$$I/I^2 \otimes_S T \simeq I[y_j]/I[y_j]^2 \otimes_{S[y_j]} T \to K/K^2 \to J/J^2 \to 0.$$

For the second claim, recall that ? is the kernel of the map $I/I^2 \otimes_S T \to \bigoplus_{i\in I} T\mathrm{d}x_i$, which factors as

$$I/I^2 \otimes_S T \twoheadrightarrow \mathrm{im}(d) \otimes_S T \to \bigoplus_{i\in I} T\mathrm{d}x_i.$$

Therefore, ? can be identified with elements whose image in $\mathrm{im}(\mathrm{d}) \otimes_S T$ lies in the kernel of $\mathrm{im}(\mathrm{d}) \otimes_S T \to \bigoplus_{i\in I} T\mathrm{d}x_i$. Since $\bigoplus_{i\in I} T\mathrm{d}x_i$ is a free S-module, we obtain the following exact sequence:

$$0 \to \mathrm{Tor}_2^S(\Omega_{S/R}, T) \xrightarrow{\sim} \mathrm{Tor}_1^S(\mathrm{im}(\mathrm{d}), T) \to 0 \to \mathrm{Tor}_1^S(\Omega_{S/R}, T) \to \mathrm{im}(\mathrm{d}) \otimes_S T \to \bigoplus_{i\in I} T\mathrm{d}x_i.$$

In particular, there is a surjection $? \twoheadrightarrow \mathrm{Tor}_1^S(\Omega_{S/R}, T)$ whose kernel is given by the kernel of the map $I/I^2 \otimes_S T \to \mathrm{im}(\mathrm{d}) \otimes_S T$. We may patch this surjection with the following exact sequence:

$$\mathrm{Tor}_1^S(\mathrm{im}(\mathrm{d}), T) \to \mathcal{N}_{S/R} \otimes_S T \to I/I^2 \otimes_S T \to \mathrm{im}(d) \otimes_S T.$$

By $\mathrm{Tor}_2^S(\Omega_{S/R}, T) \simeq \mathrm{Tor}_1^S(\mathrm{im}(\mathrm{d}), T)$, we deduce the second claim. The third claim follows from Proposition 9.32. □

Remark 9.35 (The cotangent complex). The exact sequence of Theorem 9.34 can be extended further to a long exact sequence, which (like all long exact sequences) comes from a *fiber sequence* in a higher category. Namely, if we repeat the definition of the cotangent module $\Omega_{S/R}$ but replace the category Mod_S with its derived category $\mathrm{D}(\mathrm{Mod}_S)$, we obtain the definition of cotangent complex $\mathrm{L}_{S/R} \in \mathrm{D}(\mathrm{Mod}_S)$, such that

$$\mathrm{H}_0(\mathrm{L}_{S/R}) = \Omega_{S/R}, \quad \mathrm{H}_1(\mathrm{L}_{S/R}) = \mathcal{N}_{S/R}, \quad \mathrm{H}_{<0}(\mathrm{L}_{S/R}) = 0.$$

The proof of Proposition 9.26 carries over to this setting: for every ring maps $R \to S \to T$, we obtain a fiber sequence

$$\mathrm{L}_{S/R} \otimes_S T \to \mathrm{L}_{T/R} \to \mathrm{L}_{T/S}$$

in $\mathrm{D}(\mathrm{Mod}_T)$, hence a long exact sequence of homology modules. This expains the mysterious module in Theorem 9.34: $? = \mathrm{H}_1(\mathrm{L}_{S/R} \otimes_S T)$. It also explains the different nature of the definitions of $\Omega_{S/R}$ and $\mathcal{N}_{S/R}$: the universal property of the cotangent complex $\mathrm{L}_{S/R}$ induces a universal property for $\Omega_{S/R}$ because $\mathrm{H}_0 \colon \mathrm{D}(\mathrm{Mod}_S)_{\geq 0} \to \mathrm{Mod}_S$ is a left adjoint, but $\mathrm{H}_1 \colon \mathrm{D}(\mathrm{Mod}_S)_{\geq 0} \to \mathrm{Mod}_S$ is not left adjoint, so $\mathcal{N}_{S/R}$ does not inherit any universal property.

9.3. **The cotangent module of an algebraic functor.** In this section, we globalize the constructions of the cotangent and conormal modules from rings to schemes. In fact, the universal property of $\Omega_{S/R}$ can be formulated for any morphism of algebraic functors:

Definition 9.36 (Cotangent module of a morphism of algebraic functors)**.** Let $f\colon Y \to X$ be map of algebraic functors. We say that f *admits a cotangent module* if the following conditions hold:

(a) For every ring R and every R-point $y\colon \operatorname{Spec}(R) \to Y$, the functor $\operatorname{Mod}_R \to \operatorname{Set}$ sending M to the set of lifts

$$\begin{array}{ccccc} \operatorname{Spec}(R) & & \xrightarrow{\quad y \quad} & & Y \\ \downarrow & \nearrow & & & \downarrow f \\ \operatorname{Spec}(R \oplus M) & \longrightarrow & \operatorname{Spec}(R) & \xrightarrow[f(y)]{} & X. \end{array}$$

is representable by an R-module $\Omega_f(y)$.

(b) For every ring map $\varphi\colon R \to S$ and every R-point $y\colon \operatorname{Spec}(R) \to Y$, the canonical S-linear map $\Omega_f(y) \otimes_R S \to \Omega_f(\varphi^*(y))$ is an isomorphism.

When these conditions hold, the module from (a) and isomorphisms from (b) define a quasi-coherent module $\Omega_f = \Omega_{Y/X}$ over Y, called the *cotangent module* of f

Example 9.37. Any monomorphism of algebraic functors admits a cotangent module, which is zero.

Example 9.38. If $\varphi\colon R \to S$ is a ring map and $f = \operatorname{Spec}(\varphi)$, then f admits a cotangent module and $\Omega_f = \Omega_{S/R}$. To see this, we must show that, for any map $y\colon \operatorname{Spec}(T) \to \operatorname{Spec}(S)$, $y^*(\Omega_{S/R})$ represents the functor $\operatorname{Mod}_T \to \operatorname{Set}$ from Definition 9.36(a). For any T-module M, the square

$$\begin{array}{ccc} \operatorname{Spec}(T) & \xrightarrow{\quad y \quad} & \operatorname{Spec}(S) \\ \downarrow & & \downarrow \\ \operatorname{Spec}(T \oplus M) & \longrightarrow & \operatorname{Spec}(S \oplus y^*(M)) \end{array}$$

is cocartesian in the category of affine schemes, since the corresponding square of rings is cartesian. Together with Remark 9.16, this show that the functor is isomorphic to $\operatorname{Der}_R(S, y_*(-))$. But we have natural isomorphisms $\operatorname{Der}_R(S, y_*(-)) \simeq \operatorname{Map}(\Omega_{S/R}, y_*(-)) \simeq \operatorname{Map}(y^*(\Omega_{S/R}), -)$, as desired.

Example 9.39 (Cotangent modules of affine and projective spaces)*.* Let X be an algebraic functor and let $M \in \operatorname{Mod}_X$ be a quasi-coherent module over X.

(a) The map $f\colon \mathbf{A}(M) \to X$ admits a cotangent module, given by $\Omega_f = f^*(M) \in \operatorname{Mod}_{\mathbf{A}(M)}$.

(b) Consider the map $f\colon \mathbf{P}(M) \to X$. Let $x\colon \operatorname{Spec}(R) \to \mathbf{P}(M)$ be an R-point classifying a quotient line $q\colon M(f(x)) \twoheadrightarrow L$. Using the fact that the functor $\operatorname{Line}_R \to \operatorname{Line}_{R\oplus N}$ is essentially surjective, we see that the functor $\operatorname{Mod}_R \to \operatorname{Set}$ of (b) of Definition 9.36(a) is given by

$$N \mapsto \operatorname{coker}\left(q^*\colon \operatorname{Hom}_R(L, N \otimes_R L) \hookrightarrow \operatorname{Hom}_R(M(f(x)), N \otimes_R L)\right).$$

Since the short exact sequence $\ker(q) \hookrightarrow M(f(x)) \twoheadrightarrow L$ is split, this functor is represented by the R-module $\ker(q) \otimes_R L^{-1}$, which is preserved by any scalar extension $R \to S$. Hence, f admits a cotangent module and we have a short exact sequence

$$0 \to \Omega_f \to f^*(M)(-1) \to \mathcal{O}_{\mathbf{P}(M)} \to 0$$

in $\operatorname{Mod}_{\mathbf{P}(M)}$, called the *Euler sequence*. When $M = \mathcal{O}^2$, we obtain an isomorphism $\Omega_{\mathbf{P}^1} \simeq \mathcal{O}(-2)$ using Example 4.55[27].

[27]This isomorphism is not completely canonical, as there are two possible choices of signs in Example 4.55

The cotangent module of Definition 9.36 has a similar functoriality for ring maps. Namely, for any commutative square of algebraic functors

$$\begin{array}{ccc} Y' & \xrightarrow{h} & Y \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ X' & \longrightarrow & X, \end{array}$$

if both f and f' admit cotangent modules, then there is a canonical map $h^*(\Omega_f) \to \Omega_{f'}$ in $\mathrm{Mod}_{Y'}$. This makes $f \mapsto \Omega_f$ into a functor on the full subcategory of $\mathrm{Ar}(\mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set}))$ consisting of maps that admit a cotangent module.

The following two propositions are generalizations of Proposition 9.24 and Proposition 9.26; as in the affine case, they are formal consequences of the universal property defining the cotangent module.

Proposition 9.40 (Properties of the cotangent module)**.** *Let $f\colon Y \to X$ be a map of algebraic functors admitting a cotagnent module.*

(a) (Base change) *For any cartesian square of algebraic functors*

$$\begin{array}{ccc} Y' & \xrightarrow{h} & Y \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ X' & \longrightarrow & X, \end{array}$$

f' admits a cotangent module and the map $h^*(\Omega_f) \to \Omega_{f'}$ is an isomorphism.

(b) (Open immersion) *If $j\colon U \hookrightarrow Y$ is an open immersion, then $f \circ j$ admits a cotangent module and the map $j^*(\Omega_f) \to \Omega_{f\circ j}$ is an isomorphism.*

(c) (Closed immersion) *If $i\colon Z \hookrightarrow Y$ is a closed immersion, then $f \circ i$ admits a cotangent module and the map $i^*(\Omega_f) \to \Omega_{f\circ i}$ is an epimorphism. If $I \in \mathrm{Idl}_Y$ is the quasi-coherent ideal corresponding to i and if $z\colon \mathrm{Spec}(R) \to Z$ is any point, the kernel of $\Omega_f(i \circ z) \twoheadrightarrow \Omega_{f\circ i}(z)$ is exactly the pullback of I by the canonical map $\mathrm{Spec}(R \oplus \Omega_f(i \circ z)) \to Y$.*

Proof. □

YW: Maybe write down a proof.

Proposition 9.41 (The fundamental exact sequence of cotangent modules)**.** *Consider a commutative triangle of algebraic functors*

$$\begin{array}{ccc} Z & \xrightarrow{g} & Y \\ & {\scriptstyle h}\searrow & \downarrow{\scriptstyle f} \\ & & X. \end{array}$$

If f and h admit cotangent modules, then g admits a cotangent module, and the induced square

$$\begin{array}{ccc} g^*(\Omega_f) & \longrightarrow & \Omega_h \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & \Omega_g \end{array}$$

in Mod_Z is cocartesian.

Proof. Maybe write down a proof. □

Proposition 9.42 (Existence of cotangent modules for schemes)**.** *Let $f\colon Y \to X$ be a schematic morphism of algebraic functors.*

(a) f admits a cotangent module.

(b) For every $\mathrm{Spec}(R) \to X$ and every open immersion $\mathrm{Spec}(S) \hookrightarrow Y \times_X \mathrm{Spec}(R)$, the pullback of Ω_f to $\mathrm{Spec}(S)$ is $\Omega_{S/R}$.

Proof. (a) Let $y\colon \operatorname{Spec}(R) \to Y$ be an R-point of Y and $F_y\colon \operatorname{Mod}_R \to \operatorname{Set}$ be the functor sending M to the set of lifts in Definition 9.36(a). Let $f_1, \dots, f_n \in R$ such that $(f_1, \dots, f_n) = 1$. Since f is schematic, the target of the liftings can be reduced to the scheme $Y \times_X$. As localization commutes with square zero extensions, we obtain the following equalizer diagram:

$$F_y(M) \to \prod_{i \in I} F_{y_{f_j}}(M_{f_j}) \rightrightarrows \prod_{i,j \in I} F_{y_{f_i f_j}}(M_{f_i f_j}).$$

(b) This follows from (a) and Proposition 9.40(a, b).

□

We now discuss the connection between the cotangent module and tangent vectors, which we already considered in Section 1.1.

Notation 9.43 (Dual numbers)**.** If R is a ring, we denote by $R[\varepsilon]$ the ring of dual numbers $R[x]/(x^2)$, which is the split square-zero extension of R by the R-module R. If X is an algebraic functor, we write $X[\varepsilon]$ for $X \times \operatorname{Spec}(\mathbf{Z}[\varepsilon])$.

Definition 9.44 (Tangent functor, tangent vector)**.** Let X be an algebraic functor. The *tangent functor* $\mathrm{T}(X)$ of X is the algebraic functor defined by

$$\mathrm{T}(X)(R) = X(R[\varepsilon]).$$

In other words, $\mathrm{T}(X)$ is the Weil restriction $\mathrm{R}_{\mathbf{Z}[\varepsilon]/\mathbf{Z}](X[\varepsilon])}$ (see Definition 2.42). The ring maps $\mathbf{Z}[\varepsilon] \twoheadrightarrow \mathbf{Z}$ and $\mathbf{Z} \hookrightarrow \mathbf{Z}[\varepsilon]$ induce a canonical map $\mathrm{T}(X) \to X$ with a section $X \hookleftarrow \mathrm{T}(X)$, called the *zero section* of the tangent functor. Given an R-point $x \in X(R)$, a preimage of x by the map $\mathrm{T}(X) \to X$ is called a *tangent vector* to X at x.

If $f\colon Y \to X$ is a map of algebraic functors, the *tangent functor* $\mathrm{T}(f) = \mathrm{T}(Y/X)$ of f is defined by the cartesian square

$$\begin{array}{ccc} \mathrm{T}(f) & \longrightarrow & X \\ \downarrow & & \downarrow \\ \mathrm{T}(Y) & \longrightarrow & \mathrm{T}(Y). \end{array}$$

In other words, the elements of $\mathrm{T}(f)$ are the tangent vectors to Y that becomes zero in $\mathrm{T}(X)$. Note that $\mathrm{T}(X) = \mathrm{T}(X/\operatorname{Spec}(\mathbf{Z}))$.

Example 9.45. Let k be a ring, I a set $f \in k[x_i \mid i \in i]$ a polynomial in I variables, and R a k-algebra. For any family of dual numbers $a + u\varepsilon \in R[\varepsilon]^I$, we have

$$f(a + u\varepsilon) = f(a) + \nabla f(a)(u)\varepsilon \in R[\varepsilon],$$

where the gradient $\nabla f(a) = ((\partial f/\partial x_i)(a))_{i \in I} \in R^{(I)}$ is viewed as an R-linear map $R^I \to R$.

Given an affine k-scheme $X = \operatorname{Spec}(k[x_i] \mid i \in I)/(f_j \mid j \in J)$, it follows that an R-point of $\mathrm{T}(X/k)$ is a pair (a, u), where a is an R-point of X and $u \in R^I$ is a vector satisfying $\nabla f_j(a)(u) = 0$ for all $j \in J$.

Let $R \to S$ be a ring map. By Remark 9.16, the set of R-algebra sections of $S[\varepsilon] \twoheadrightarrow S$ can be identified with the tangent module $\Omega^\vee_{S/R} = \operatorname{Hom}_S(\Omega_{S/R}, S)$. The following result generalizes this observation to maps of algebraic functors. Recall that any quasi-coherent module $M \in \operatorname{Mod}_X$ has an associated affine space $\mathbf{A}(M) = \operatorname{Spec}(\operatorname{Sym}(M))$ over X (Definition 4.32). By Proposition 4.36(b), sections of $\mathbf{A}(M) \to X$ are in bijection with maps $M \to \mathcal{O}_X$ in Mod_X.

Proposition 9.46 (Relation between cotangent module and the tangent functor)**.** *Let $f\colon Y \to X$ be a map of algebraic functors admitting a cotangent module. Then there is a canonical isomorphism $\mathbf{A}(\Omega_f) \simeq \mathrm{T}(f)$ over Y.*

We now turn our attention to the conormal module. Since it cannot be characterized by a universal property like the cotangent module, we construct the conormal module of a map of schemes "by hand", by gluing conormal modules of maps of rings.

Remark 9.47. Let $f\colon Y \to X$ be a morphism of schemes . Then Y is covered by its affine open subschemes that the map to affine open subschemes of X. By Zariski descent, to define a quasi-coherent module over Y, it suffices to specify the cocyle conditions. More precisely, we can write

$$\mathrm{Mod}_Y \xrightarrow{\sim} \lim_{(x,y)} \mathrm{Mod}_S,$$

where the limit is indexed by the full subcategory of $\mathrm{Ar}(\mathrm{Sch})_{/f}$ spanned bt the squares

$$\begin{array}{ccc} \mathrm{Spec}(S) & \xrightarrow{y} & Y \\ \downarrow & & \downarrow f \\ \mathrm{Spec}(R) & \xrightarrow{x} & X, \end{array}$$

where x and y are open immersions.

Construction 9.48 (Conormal module of a morphism of schemes)**.** Let $f\colon Y \to X$ be a morphism of schemes. We will construct a quasi-coherent module $\mathcal{N}_f = \mathcal{N}_{Y/X}$ over Y, called the *conormal module* of f, which is uniquely determined by the following properties:

(a) $f \mapsto \mathcal{N}_f$ is a functor on $\mathrm{Ar}(\mathrm{Sch})^{\mathrm{op}}$, which extends the conormal module functor for ring maps.
(b) For any square (x, y) as in Remark 9.47, the map $y^*(\mathcal{N}_f) \to \mathcal{N}_{S/R}$ is an isomorphism.

For any such (x, y), set $\mathcal{N}(x, y) = \mathcal{N}_{S/R} \in \mathrm{Mod}_S$. Given a commutative square

$$\begin{array}{ccc} \mathrm{Spec}(S') & \xrightarrow{k} & \mathrm{Spec}(S) \\ \downarrow & & \downarrow \\ \mathrm{Spec}(R') & \xrightarrow{j} & \mathrm{Spec}(R), \end{array}$$

where j and k are open immersions, we get and induced S'-linear map $\varphi_{j,k}\colon \mathcal{N}(x, y) \otimes_S S' \to \mathcal{N}(xj, yk)$ by the functoriality of conormal modules, and these maps automatically satisfies the cocyle conditions. By the compatibility of $\mathcal{N}$ with flat base change and localization (Proposition 9.32), $\varphi_{j,k}$ is an isomorphism. Thus, the modules $\mathcal{N}(x, y)$ and the isomorphisms $\varphi_{j,k}$ define an object $\mathcal{N}_f$ of $\mathrm{Mod}_Y \simeq \lim_{(x,y)} \mathrm{Mod}_S$.

The following proposition and theorem globalize Proposition 9.32 and Theorem 9.34, from which they directly follow by Zariski descent:

Proposition 9.49 (Properties of the conormal module)**.** *Let $f\colon Y \to X$ be a morphism of schemes.*

(a) (Base change) *For any cartesian square of schemes*

$$\begin{array}{ccc} Y' & \xrightarrow{h} & Y \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ X' & \xrightarrow{g} & X, \end{array}$$

if g is flat or if f and Ω_f are flat, then the map $h^*(\mathcal{N}_f) \to \mathcal{N}_{f'}$ is an isomorphism.

(b) (Open immersion) *If $j\colon U \hookrightarrow Y$ is an open immersion, then the map $j^*(\mathcal{N}_f) \to \mathcal{N}_{f\circ j}$ is an isomorphism.*

Theorem 9.50 (Fundamental exact sequence of cotangent and conormal modules)**.** *Consider a commutative triangle of schemes*

$$\begin{array}{ccc} Z & \xrightarrow{g} & Y \\ & {\scriptstyle h}\searrow & \downarrow{\scriptstyle f} \\ & & X. \end{array}$$

Then there is an exact sequence in Mod_Z

$$? \to \mathcal{N}_h \to \mathcal{N}_g \to g^*(\Omega_f) \to \Omega_h \to \Omega_g \to 0,$$

where ? fits an exact sequence

$$\mathrm{L}_2 g^*(\Omega_f) \to g^*(\mathcal{N}_f) \to ? \to \mathrm{L}_1 g^*(\Omega_f) \to 0.$$

In particular, if either Ω_f or g is flat, then $? = g^*(\mathcal{N}_f)$.

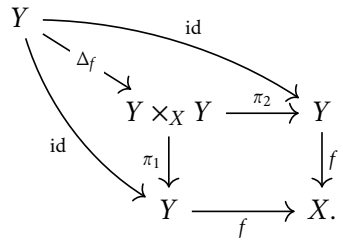
Remark 9.51 (Relation with the Zariski cotangent space). Let $(X, \mathcal{O}_X)$ be a locally ringed space. By Example 9.21(b), the Zariski cotangent space of X at x is the conormal module of $\mathcal{O}_{X,x} \twoheadrightarrow \kappa(x)$.

If X is a scheme, Proposition 9.49(a) with the flat map $\operatorname{Spec}(\mathcal{O}_{X,x}) \hookrightarrow X$ yields a canonical isomorphism

$$\mathcal{N}_{\operatorname{Spec}(\kappa(x))/X} \simeq \mathrm{T}^*_x(X).$$

In words, the Zariski cotangent space at a point is the conormal module of the inclusion of the point.

Remark 9.52 (Conormal module of the diagonal). Let $f\colon Y \to X$ be a map of schemes. Consider the diagram



By the fundamental exact sequence and the compatibility of Ω with base change, we obtain two canonical isomorphisms

$$\alpha_1\colon \mathcal{N}_{\Delta_f} \simeq \Delta_f^*(\Omega_{\pi_1}) \simeq \Delta_f^*\pi_2^*(\Omega_f) \simeq \Omega_f$$

and

$$\alpha_2\colon \mathcal{N}_{\Delta_f} \simeq \Delta_f^*(\Omega_{\pi_2}) \simeq \Delta_f^*\pi_1^*(\Omega_f) \simeq \Omega_f,$$

which differ by a sign: $\alpha_1 = -\alpha_2$. Indeed, they differ by the automorphism $\mathcal{N}_{\Delta_f} \xrightarrow{\sim} \mathcal{N}_{\Delta_f}$ induced by the cartesian square

$$\begin{array}{ccc} Y & \xrightarrow{\mathrm{id}} & Y \\ {\scriptstyle \Delta_f}\downarrow & & \downarrow{\scriptstyle \Delta_f} \\ Y \times_X Y & \xrightarrow[\sigma]{\sim} & Y \times_X Y, \end{array}$$

where σ is the swap map, which is seen to be $-\mathrm{id}$ by considering the affine case.

The following theorem establishes the equivalence (a) $\Leftrightarrow$ (b) of Theorem 9.4:

Theorem 9.53 (Smoothness via the cotangent complex)**.** *Let $f\colon Y \to X$ be a morphism of schemes.*

(i) f is smooth if and only if it is locally of finite presentation, Ω_f is a vector bundle, and $\mathcal{N}_f = 0$.
(ii) f is unramified if and only if it is locally of finite type and $\Omega_f = 0$.
(iii) f is étale if and only if it is locally of finite presentation, $\Omega_f = 0$, and $\mathcal{N}_f = 0$.

Remark 9.54. Theorem 9.53 already establishes Corollary 9.7. Indeed, smoothness is clearly stable under base change, while the property "Ω_f is a vector bundle and $\mathcal{N}_f$ is zero" is local by Proposition 9.40(a, b) and Proposition 9.49(a, b) and stable under base change by Theorem 9.50.

Remark 9.55. In addition to Corollary 9.9, Theorem 9.53 has the following immediate consequences:

(a) Any monomorphism locally of finite type is unramified, by Example 9.37. Hence, a monomorphism is smooth if and only if it is étale.
(b) An immersion is smooth if and only if it is an open immersion. Indeed, if $I \subset R$ is a finitely generated ideal with $I/I^2 = 0$, then there exists $f \in 1 + I$ with $fI = 0$ by Nakayama, so that $\mathrm{V}(I) = \mathrm{D}(f)$.
(c) A map f is unramified if and only if it is locally of finite type and Δ_f is an open immersion. Indeed, as $\Omega_f \simeq \mathcal{N}_{\Delta_f}$ by Remark 9.52 and Δ_f is an immersion by Proposition 7.23, the claim follows from (ii).

Remark 9.56 (Smoothness of closed subschemes). Let $R \to S$ be a smooth ring map, let $I \subset S$ be a finitely generated ideal, and let $T = S/I$. Set $X = \operatorname{Spec}(R)$, $Y = \operatorname{Spec}(S)$, and $Z = \operatorname{Spec}(T)$. Consider the commutative triangle

$$\begin{array}{ccc} Z & \xhookrightarrow{i} & Y \\ & \searrow^{g} & \downarrow{\scriptstyle f} \\ & & X \end{array}$$

and the associated fundamental exact sequence

$$0 \to \mathcal{N}_g \to I/I^2 \xrightarrow{\mathrm{d}} i^*(\Omega_f) \to \Omega_g \to 0.$$

By Theorem 9.53 and Proposition 3.18, an open subscheme $U \subset Z$ is smooth over X if and only if $(I/I^2)|_U$ is a vector bundle and $\mathrm{d}|_U$ is universally injective. Given $z \in |Z|$, we deduce from Nakayama's lemma that g is smooth in a neighborhood of z if and only if there exist $f_1, \dots, f_m \in I$ with $m = \operatorname{rk}_z(\Omega_f) - \operatorname{rk}_z(\Omega_g)$ whose images generate the $\kappa(z)$-module $I/I^2 \otimes_T \kappa(z)$.

Proposition 9.57 (Local structure of smooth closed subschemes). *Let $f\colon Y \to X$ be a smooth morphism of schemes and let $Z \subseteq Y$ be a closed subscheme such that $f|_Z\colon Z \to X$ is smooth. For every $z \in |Z|$, there exists an open subscheme $V \subseteq Y$ containing z and an étale map $g\colon V \to \mathbf{A}^n_X$ over X such that $Z \cap V = g^{-1}(\mathbf{A}^{n-m}_X)$, where $n = \operatorname{rk}_z(\Omega_f)$ and $m = \operatorname{rk}_z(\mathcal{N}_{Z/Y})$.*

9.4. **Smoothness over a field.** We now relate smoothness over a field to regularity.

Remark 9.58 (Separable field extensions). Recall that a finitely generated field extension $k \to k'$ is *separable* if it admits a separating transcendence basis, i.e., if it factors as

$$k \to k(x_1, \dots, x_r) \to k',$$

where each x_i is transcendent over k and where k' is a finite separable extension of $k(x_1, \dots, x_r)$. By Example 9.12, k is then étale over $k(x_1, \dots, x_r)$, and by the compatibility of Ω and $\mathcal{N}$ with localization, the cotangent module of $k \to k(x_1, \dots, x_r)$ is free of rank r and its conormal module vanishes. By the fundamental exact sequence, we deduce that

$$\Omega_{k'/k} = \bigoplus_{i=1}^{r} k' \mathrm{d}x_i \quad \text{and} \quad \mathcal{N}_{k'/k} = 0.$$

Remark 9.59 (Fundamental exact sequence of the inclusion of a point). Let k be a field, X a k-scheme locally of finite presentation, and $x \in |X|$. The residue field $\kappa(x)$ is then a finitely generated field extension of k. Consider the commutative triangle

$$\begin{array}{ccc} \operatorname{Spec}(\kappa(x)) & \xrightarrow{x} & X \\ & \searrow & \downarrow \\ & & \operatorname{Spec}(k). \end{array}$$

The associated fundamental exact sequence of $\kappa(x)$-modules is

$$? \to \mathcal{N}_{\kappa(x)/k} \to \mathfrak{m}/\mathfrak{m}^2 \to \Omega_{X/k}(x) \to \Omega_{\kappa(x)/k} \to 0,$$

where $\mathfrak{m}$ is the maximal ideal of $\mathcal{O}_{X,x}$. If $\kappa(x)$ is separable over k, it reduces by Remark 9.58 to a short exact sequence

$$0 \to \mathfrak{m}/\mathfrak{m}^2 \to \Omega_{X/k}(x) \to \kappa(x)^r \to 0,$$

where $r = \operatorname{trdeg}_k(\kappa(x))$.

Remark 9.60 (Dimension at a point). Let X be a scheme locally of finite presentation over a field k and let $x \in |X|$. If X is irreducible, then we have $\dim(X) = \dim(U)$ for any nonempty open $U \subset X$ by Theorem 8.65(a), and hence $\dim_x(X) = \dim(X)$ (see Remark 8.50(c) for the definition of $\dim_x$). In general, $\dim_x(X)$ is therefore the

maximum of the dimensions of the irreducible components of X passing through x. Using Theorem 8.65(b, a), we deduce that

$$\dim_x(X) = \operatorname{codim}(x, X) + \dim(\overline{\{x\}}) = \dim(\mathcal{O}_{X,x}) + \operatorname{trdeg}_k(\kappa(x)).$$

Combining Remark 9.59 and Remark 9.60, we obtain the following result:

Lemma 9.61. *Let k be a field, X a k-scheme locally of finite presentation, and $x \in |X|$ a point such that $\kappa(x)$ is separable over k. Then $\mathcal{O}_{X,x}$ is regular if and only if $\operatorname{rk}_x(\Omega_{X/k}) = \dim_x(X)$.*

Lemma 9.62. *Let $\varphi\colon R \twoheadrightarrow S$ be a surjective map of rings. If R is a domain and $\dim(R) \leq \dim(S) < \infty$, then φ is an isomorphism.*

Theorem 9.63 (Smoothness over a field)**.** *Let k be a field, let X be a k-scheme locally of finite presentation, and let $x \in |X|$. Consider the following conditions:*

(i) There exists an open neighborhood of x that is smooth over k.
(ii) $\operatorname{rk}_x(\Omega_{X/k}) = \dim_x(X)$.
(iii) $\mathcal{O}_{X,x}$ is a regular local ring.

Then (i) $\Leftrightarrow$ (ii) $\Rightarrow$ (iii)*. If $\kappa(x)$ is a separable extension of k, then also* (iii) $\Rightarrow$ (i)*.*

Corollary 9.64 (Smoothness over a perfect field)**.** *Let k be a perfect field and X a k-scheme locally of finite presentation. Then X is smooth over k if and only if X is regular.*

Applying (iii) $\Rightarrow$ (i) to the generic points of the irreducible components of X, we get the following useful corollary:

Corollary 9.65. *Let k be a perfect field and let X be a reduced k-scheme locally of finite presentation. Then there exists a dense open subscheme $U \subset X$ that is smooth over k.*

Remark 9.66.

(a) The implication (iii) $\Rightarrow$ (i) in Theorem 9.63 does not always hold if $\kappa(x)$ is not separable over k. For example, if k' is an inseparable finite field extension of k, then $k \to k'$ is not smooth as $\Omega_{k'/k}$ has rank ≥ 1: if $k' = k(a)$, it has rank exactly 1 by Example 9.12, and in general we deduce that it has a rank 1 quotient by the fundamental exact sequence.
(b) If X is smooth over k and $x \in |X|$, then $\kappa(x)$ need not be separable over k. Indeed, any finitely generated field extension of k is the residue field of a point in $\mathbf{A}^n_k$ for some $n \in \mathbf{N}$.

Corollary 9.67 (Étaleness over a field)**.** *Let k be a field and X a k-scheme. The following conditions are equivalent:*

(i) X is étale over k.
(ii) X is unramified over k.
(iii) $X \simeq \coprod_{i\in I} \operatorname{Spec}(k_i)$ for some family $(k_i)_{i\in I}$ of finite separable field extensions of k.

Note that the implication (ii) $\Rightarrow$ (iii) in Theorem 9.63 proves the implication (iv) $\Rightarrow$ (v) in Theorem 9.4. Given Corollary 9.64, the following result establishes the implication (v) $\Rightarrow$ (i) in Theorem 9.4:

Proposition 9.68 (Fiberwise smoothness)**.** *Let $f\colon Y \to X$ be a flat morphism locally of finite presentation such that, for every algebraically closed field k and every map $x\colon \operatorname{Spec}(k) \to X$, the scheme Y_x is smooth over k. Then f is smooth.*

9.5. **Formal smoothness.**

Definition 9.69 (Thickening, square-zero extension)**.** Let R be a ring. A *thickening* of R is a surjective ring map $R' \twoheadrightarrow R$ whose kernel I is a nilideal (i.e., an ideal whose elements are nilpotent). It is called an *nth-order thickening* if $I^{n+1} = 0$. A first-order thickening is also called a *square-zero extension*.

Example 9.70.

(a) A split square-zero extension $R \oplus M \twoheadrightarrow R$ is a first-order thickening. In fact, the construction $M \mapsto R \oplus M$ defines an equivalence between the category of R-modules and that of first-order thickenings of R with a section.
(b) For $n \in \mathbf{N}$, the *standard nth-order thickening* of a ring R is $\tau_n \colon R[x]/(x^{n+1}) \twoheadrightarrow R$, $x \mapsto 0$.
(c) Let p be a prime number and $n \in \mathbf{N}$. The map $\mathbf{Z}/p^{n+1} \twoheadrightarrow \mathbf{F}_p$ is an nth-order thickening, which is easily seen to be the initial object in the category of nth-order thickenings of $\mathbf{F}_p$. More generally, any perfect $\mathbf{F}_p$-algebra k has an initial nth-order thickening $W_n(k) = W(k)/p^{n+1}$, where $W(k)$ is the ring of p-typical Witt vectors of k.

Remark 9.71. The notion of thickening has a natural extension to algebraic functors: a *thickening* of an algebraic functor X is a closed immersion $X \hookrightarrow X'$ defined by a quasi-coherent nilideal. Equivalently, by Equation (5.33), it is a surjective immersion.

Definition 9.72 (Infinitesimal neighborhoods)**.** Let X be an algebraic functor, let $Z \subset X$ be a closed subfunctor defined by a quasi-coherent ideal $I \in \mathrm{Idl}_X$, and let $n \in \mathbf{N}$. The *nth infinitesimal neighborhood* $Z^{(n)}$ of Z in X is the closed subfunctor of X defined by I^{n+1}(Remark 4.18(g)). Note that the inclusion $Z \subset Z^{(n)}$ is an nth-order thickening.

Definition 9.73 (Completion)**.** Let X be an algebraic functor and let $K \subset |X|$ be a closed subset. The *completion* $X^\wedge_K$ of X along K is the largest subfunctor of X whose underlying space maps to K. In other words, if $I \in \mathrm{Rad}_X$ is the quasi-coherent radical ideal of the open complement of K, then

$$X^\wedge_K(R) = \{x \in X(R) \mid I(x) = \sqrt{0}\}.$$

If $Z \subset X$ is a closed subfunctor, we also write $X^\wedge_Z$ for $X^\wedge_{|Z|}$.

Remark 9.74 (Completion as union of infinitesimal neighborhoods)*.* Let X be an algebraic functor and $Z \subset X$ a closed subfunctor. There is an inclusion $\mathrm{colim}_{n\to\infty} Z^{(n)} \subset X^\wedge_Z$, which is an equality if $Z \hookrightarrow X$ is of finite presentation. Indeed, if $I \subset R$ is a finitely generated nilideal, then $I^{n+1} = 0$ for some n.

Example 9.75*.*

(a) The standard nth-order thickening $\tau_n \colon R[x]/(x^{n+1}) \twoheadrightarrow R$ is the nth infinitesimal neighborhood of the origin in $\mathbf{A}^1_R$, and the thickening $\mathbf{Z}/p^{n+1} \twoheadrightarrow \mathbf{F}_p$ is the nth infinitesimal neighborhood of $\mathrm{V}(p)$ in $\mathrm{Spec}(\mathbf{Z})$.
(b) The algebraic functor $R \mapsto \mathrm{Nil}(R)$ is the completion of $\mathbf{A}^1$ at the origin. The property "p is nilpotent" is the completion of $\mathrm{Spec}(\mathbf{Z})$ at $\mathrm{V}(p)$.

Warning 9.76*.* As the previous example shows, the completion of a scheme along a closed subset is not usually a scheme; it belongs to a larger class of algebraic functors called *formal schemes*.

Definition 9.77 (Tangential datum)**.** Let X be an algebraic functor, $x \colon \mathrm{Spec}(R) \to X$ an R-point, and $\tau \colon R' \twoheadrightarrow R$ a thickening of R. A *τ-tangential datum* at x in X is a lift of x along τ:

$$\begin{array}{ccc} \mathrm{Spec}(R) & \xrightarrow{x} & X \\ {\scriptstyle \mathrm{Spec}(\tau)}\uparrow & \nearrow & \\ \mathrm{Spec}(R') & & \end{array}$$

Example 9.78 (Jets of curves)*.* Let $n \in \mathbf{N}$. For the standard nth-order thickening $\tau_n \colon R[x]/(x^{n+1}) \twoheadrightarrow R$, a τ_n-tangential datum is also called an *n-jet*. For example, a 1-jet is a tangent vector (Definition 9.44). As in *loc. cit*, we can define the n-jet functor $\mathrm{J}^n(f)$ of a map of algebraic functors $f \colon Y \to X$ by

$$\mathrm{J}^n(f)(R) = Y(R[x]/(x^{n+1})) \times_{X(R[x]/(x^{n+1}))} X(R).$$

There is then a tower

$$\mathrm{J}^\infty(f) = \lim_{n\to\infty} \mathrm{J}^n(f) \to \cdots \to \mathrm{J}^2(f) \to \mathrm{J}^1(f) = \mathrm{T}(f) \to \mathrm{J}^0(f) = Y,$$

where the limit $\mathrm{J}^\infty(f)$ is known as the *arc space* of f.

If $X = \operatorname{Spec}(k[x_i \mid i \in I]/(f_j \mid j \in J))$ is an affine k-scheme, its n-jet functor $\mathrm{J}^n(X/k)$ is also an affine k-scheme, which is the solution functor of a system of $(n+1)J$ polynomial equations in $(n+1)I$ variables: an R-point of $\mathrm{J}^n(X/k)$ is a tuple $(a_0, \dots, a_n) \in (R^I)^{n+1}$ such that $f_j(a_0 + \cdots + a_n x^n) = 0$ in $R[x]/(x^{n+1})$ for all $j \in J$.

Definition 9.79 (Formally smooth, formally unramified, formally étale)**.** A map of algebraic functors $f\colon Y \to X$ is called *formally smooth* (resp. *formally unramified*; *formally étale*) if, for every ring R, every first-order thickening τ of R, and every R-point $y \in Y(R)$, the induced map

$$\{\tau\text{-tangential data at } y \text{ in } Y\} \to \{\tau\text{-tangential data at } f(y) \text{ in } X\}$$

is surjective (resp. injective; bijective).

Example 9.80.

(a) If $Y \subset X$ is any intersection of open subfunctors of X, then $Y \hookrightarrow X$ is formally étale. This follows directly from the fact that thickenings induce surjective maps of underlying spaces.
(b) For any projective quasi-coherent module M over X, the maps $\mathbf{A}(M) \to X$ and $\mathbf{P}(M) \to X$ are formally smooth. In fact, the induced map of τ-tangential data is surjective for any thickening $\tau\colon R' \twoheadrightarrow R$. This is clear for the former, and for the latter it follows from the fact that the functor $\tau^*\colon \mathrm{Line}_{R'} \to \mathrm{Line}_R$ is essentially surjective and from the openness of the epimorphism locus (Proposition 3.53(b)).
(c) A separable algebraic field extension is formally étale, and a separable finitely generated field extension is formally smooth. Indeed, purely transcendental extensions are formally smooth by (a) and (b), so it suffices to prove the first claim, which is an instance of Hensel's lemma: if $R' \twoheadrightarrow R$ is a square-zero extension and $m \in R'[x]$, then any root $\bar{a}$ of m in R such that $m'(\bar{a}) \in R^\times$ lifts uniquely to a root of m in R'. This follows immediately from the fact that $R' \to R$ detects units and the Taylor expansion $m(a+h) = m(a) + m'(a)h$ when $h^2 = 0$.

Example 9.81 (Nilinvariant functors)*.* An algebraic functor X is *nilinvariant* if it sends thickenings to isomorphisms. Any map between nilinvariant functors is in particular formally étale. The inclusion of the full subcategory of nilinvariant algebraic functors has a left adjoint $X \mapsto X^{\mathrm{inf}}$ given by

$$X^{\mathrm{inf}}(R) = X(R_{\mathrm{red}}),$$

which therefore sends all maps to formally étale maps. The algebraic functor X^{inf} is called the *infinitesimal stack* of X (or its *de Rham stack* if X is an algebraic $\mathbf{Q}$-functor).

To investigate the formal properties of Example 9.81, it is useful to consider a more abstract situation (which will also be relevant in the study of properness):

Definition 9.82 (Lifting properties)**.** Let $\mathcal{C}$ be a category and T a class of maps in $\mathcal{C}$. A map $f\colon Y \to X$ in $\mathcal{C}$ is called a *T-epimorphism* (resp. *T-monomorphism*; *T-isomorphism*) if, for every commutative square

$$\begin{array}{ccc} A & \longrightarrow & Y \\ {\scriptstyle t}\downarrow & \nearrow & \downarrow{\scriptstyle f} \\ B & \longrightarrow & X \end{array}$$

with $t \in T$, there exists at least one (resp. at most one; exactly one) map $B \to Y$ making both triangles commute. We also say that f has the *right lifting property* with respect to T if it is a T-epimorphism, and that it has the *unique right lifting property* with respect to T if it is a T-isomorphism.

Remark 9.83. Let T be a class of maps in $\mathcal{C}$.

(a) Every monomorphism is a T-monomorphism and every isomorphism is a T-isomorphism.
(b) A map is a T-isomorphism if and only if it is both a T-epimorphism and a T-monomorphism.
(c) Suppose that the diagonal Δ_f exists in $\mathcal{C}$. Then f is a T-monomorphism if and only if Δ_f is a T-epimorphism.

Example 9.84 (Lifting properties in Set). Let $\mathcal{C}$ be the category of sets.

(a) If $T = \{\varnothing \to *\}$, then a map is a T-epimorphism (resp. T-monomorphism; T-isomorphism) if and only if it is surjective (resp. injective; bijective). The same holds whenever T is a collection of injective maps containing at least one non-surjective map.
(b) If $T = \{* \amalg * \to *\}$, then all maps are T-monomorphisms, and a map is a T-epimorphism if and only if it is injective. The same holds whenever T is a collection of surjective maps containing at least one non-injective map.

Example 9.85 (Projective and injective modules). Let R be a ring. An R-module M is projective if and only if every surjective map has the right lifting property with respect to $\{0 \to M\}$. Dually, M is injective if and only if the map $M \to 0$ has the right lifting property with respect to all injective maps.

Proposition 9.86. *Let $\mathcal{C}$ be a category and T a class of maps in $\mathcal{C}$.*

(a) The classes of T-epimorphisms, T-monomorphisms, and T-isomorphisms are closed under base change and composition.
(b) The classes of T-monomorphisms and T-isomorphisms are closed under limits in $\mathrm{Ar}(\mathcal{C})$.
(c) If $g \circ f$ is a T-epimorphism and g is a T-monomorphism, then f is a T-epimorphism. If $g \circ f$ is a T-monomorphism, then f is a T-monomorphism. If $g \circ f$ is a T-isomorphism and g is a T-monomorphism, then f is a T-isomorphism.
(d) Let T' be the closure of T under cobase change and composition. Then T-epimorphisms and T'-epimorphisms coincide, and the same holds for monomorphisms and isomorphisms.
(e) Let T' be the closure of T under colimits in $\mathrm{Ar}(\mathcal{C})$. Then T-monomorphisms and T'-monomorphisms coincide, and the same holds for isomorphisms.

Example 9.87 (Lifting properties with respect to thickenings). Definition 9.82 recovers Definition 9.79 when $\mathcal{C} = \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})$ and

$$T = \{\mathrm{Spec}(\tau) \mid \tau \colon R' \twoheadrightarrow R \text{ is a first-order thickening}\}.$$

We list some consequences of Proposition 9.86 in this case:

(a) Proposition 9.86(a, b, c) gives various closure properties of the classes of formally smooth, formally unramified, and formally étale maps.
(b) Any nth-order thickening is a composition of n first-order thickenings. Therefore, by Proposition 9.86(d), Definition 9.79 if we replace "first-order thickening" by "finite-order thickening".
(c) Any nth-order thickening of algebraic functors is a colimit of nth-order thickenings of affine schemes. Hence, by Proposition 9.86(e), the lifting properties of formally unramified and formally étale maps also apply to finite-order thickenings of algebraic functors.
(d) Any thickening $R' \twoheadrightarrow R$ is a filtered colimit of finite-order thickenings $R' \twoheadrightarrow R_\alpha$. If f (resp. Δ_f) is locally of finite presentation (Definition 2.85) and f is formally smooth or formally étale (resp. formally unramified), it follows from (b) that f has the corresponding lifting property with respect to arbitrary thickenings.

Definition 9.88 (Cohesive, infinitesimally cohesive)**.** Let $f \colon Y \to X$ be a map of algebraic functors. We say that f is *cohesive* (resp. *infinitesimally cohesive*) if, for every cartesian square of rings

$$\begin{array}{ccc} A \times_C B & \longrightarrow & B \\ \downarrow & & \downarrow \\ A & \longrightarrow & C \end{array}$$

where the maps $A \to C \leftarrow B$ are surjective (resp. first-order thickenings), the induced square

$$\begin{array}{ccc} Y(A \times_C B) & \longrightarrow & Y(A) \times_{Y(C)} Y(B) \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f} \\ X(A \times_C B) & \longrightarrow & X(A) \times_{X(C)} X(B) \end{array}$$

is cartesian.

Proposition 9.89 (Schemes are cohesive)**.** *Any schematic morphism is cohesive.*

Example 9.90 (Addition of tangent vectors)*.* Let $f\colon Y \to X$ be a map of algebraic functors. The ring map

$$\mathbf{Z}[\varepsilon] \to \mathbf{Z}[\varepsilon][t], \quad a + u\varepsilon \mapsto a + u\varepsilon t,$$

induces a "scalar multiplication" map

$$\mathbf{A}^1 \times \mathrm{T}(f) \to \mathrm{T}(f),$$

which is an action of the multiplicative monoid $(\mathbf{A}^1, \cdot)$ on $\mathrm{T}(f)$. If f is infinitesimally cohesive, then the ring map

$$\mathbf{Z}[\varepsilon] \times_{\mathbf{Z}} \mathbf{Z}[\varepsilon] \to \mathbf{Z}[\varepsilon], \quad (a + u\varepsilon, a + v\varepsilon) \mapsto a + (u + v)\varepsilon,$$

induces an "addition" map

$$\mathrm{T}(f) \times_Y \mathrm{T}(f) \to \mathrm{T}(f).$$

For any $y \in Y(R)$, these maps equip the set of tangent vectors $\mathrm{T}(f)(R) \times_{Y(R)} \{y\}$ with an R-module structure. If f admits a cotangent module, one can check that this R-module structure coincides with the one induced by the isomorphism $\mathrm{T}(f) \simeq \mathbf{A}(\Omega_f)$ of Proposition 9.46.

More generally, if $f\colon Y \to X$ is infinitesimally cohesive, then for any $y \in Y(R)$ and any R-module M, the set of lifts

$$\begin{array}{ccccc} \mathrm{Spec}(R) & & \xrightarrow{\quad y \quad} & & Y \\ \downarrow & \nearrow & & & \downarrow f \\ \mathrm{Spec}(R \oplus M) & \longrightarrow & \mathrm{Spec}(R) & \xrightarrow[f(y)]{} & X \end{array}$$

has a canonical structure of R-module, which is isomorphic to $\mathrm{Hom}_R(\Omega_f(y), M)$ if f admits a cotangent module.

Proposition 9.91. *Let $f\colon Y \to X$ be a map of algebraic functors. If f is formally unramified, then it admits a cotangent module and $\Omega_f = 0$. The converse holds if f is either infinitesimally cohesive or locally separated (e.g., schematic).*

By definition, the cotangent module of a map $f\colon Y \to X$ tells us how to lift τ-tangential data along f, as long as τ is a *split* square-zero extension. Our next goal is to investigate the more general case where τ is an arbitrary square-zero extension. The key idea, which we carry out in Construction 9.93, is that a general square-zero extension $\tau\colon R' \twoheadrightarrow R$ with kernel I can be realized as a *deformation* of the split square-zero extension $\tau_0\colon R \oplus I \twoheadrightarrow R$, which implies that the set of lifts of any τ-tangential datum is either empty or a *torsor* under the R-module of lifts of the trivial τ_0-tangential datum.

Definition 9.92 (Deformation ring)**.** Let R be a ring, $I \subset R$ an ideal, and $\overline{R} = R/I$. The *deformation ring* of I in R (also called the *deformation to the normal cone* or the *extended Rees algebra*) is the $R[t]$-subalgebra of $R[t^{\pm 1}]$ given by

$$\mathrm{Def}_I(R) = \bigoplus_{n \in \mathbf{Z}} I^n t^{-n} \subset R[t^{\pm 1}],$$

where $I^n = R$ for $n \leq 0$. It comes with ring maps

$$R[t] \hookrightarrow \mathrm{Def}_I(R) \twoheadrightarrow \overline{R}[t].$$

Setting $t = 1$ in the deformation ring recovers the ring R, while setting $t = 0$ yields the $\mathbf{N}$-graded $\overline{R}$-algebra $\mathrm{gr}_I(R) = \bigoplus_{n \in \mathbf{N}} I^n/I^{n+1}$ (whose Spec is the *normal cone* of $\mathrm{V}(I)$ in $\mathrm{Spec}(R)$).

Construction 9.93 (The deformation ring of a square-zero extension)**.** Let $\tau\colon R' \twoheadrightarrow R$ be a square-zero extension with kernel I. The deformation ring of I in R' is then a square-zero extension of $R'[t]$-algebras

$$R'[t] \oplus It^{-1} \twoheadrightarrow R[t].$$

For any $a \in R$, we define the square-zero extension $\tau_a\colon R'(a) \twoheadrightarrow R$ by

$$R'(a) = (R'[t] \oplus It^{-1})/(t - a') \twoheadrightarrow R[t]/(t - a) \simeq R,$$

where $a' \in R'$ is any lift of a. As an R'-module, $R'(a)$ is the quotient of $R' \oplus It^{-1}$ by the relations $i \sim a'it^{-1}$ for all $i \in I$. Since the R'-module structure of I descends to an R-module structure, we see that $R'(a)$ does not depend on the choice of a'. The R-indexed family of square-zero extensions τ_a has the following properties:

(a) We have $\tau_1 = \tau$ and τ_0 is the split square-zero extension $R \oplus I \twoheadrightarrow R$.

(b) For any $a \in R$ and $u \in R$, there is a "scalar multiplication by u" map

$$R'(a) \to R'(ua), \quad r + it^{-1} \mapsto r + uit^{-1}.$$

(c) For any $a, b \in R$, there is an "addition" map

$$R'(a) \times_R R'(b) \to R'(a+b), \quad (r + it^{-1}, r + jt^{-1}) \mapsto r + (i+j)t^{-1}.$$

(d) The maps from (c) and (d) satisfy the axioms for an R-module structure.

Let now $f\colon Y \to X$ be infinitesimally cohesive and fix a lifting problem

$$\begin{array}{ccc} \operatorname{Spec}(R) & \longrightarrow & Y \\ {\scriptstyle \operatorname{Spec}(\tau)}\downarrow & & \downarrow{\scriptstyle f} \\ \operatorname{Spec}(R') & \longrightarrow & X. \end{array}$$

For any $a \in R$, the map $R' = R'(1) \to R'(a)$ from (b) gives an induced lifting problem with τ_a instead of τ. Consider the map

$$\coprod_{a \in R} \left\{ \begin{array}{ccc} \operatorname{Spec}(R) & \longrightarrow & Y \\ {\scriptstyle \operatorname{Spec}(\tau_a)}\downarrow & \nearrow & \downarrow{\scriptstyle f} \\ \operatorname{Spec}(R'(a)) & \longrightarrow & X \end{array} \right\} \to R.$$

By (d) above, the left-hand side has a structure of R-module such that this map is R-linear. Its kernel is the R-module of lifts of the trivial τ_0-tangential datum (see Example 9.90), while the preimage of $1 \in R$ is the set of solutions to the original lifting problem for τ. This shows that the latter, if nonempty, is canonically a torsor under the former.

Lemma 9.94. *Let X and Y be Zariski sheaves on $\mathrm{CAlg}^{\mathrm{op}}$, let $f\colon Y \to X$ be an infinitesimally cohesive map, and let $\tau\colon R' \twoheadrightarrow R$ be a square-zero extension. Consider a lifting problem*

$$\begin{array}{ccc} \operatorname{Spec}(R) & \longrightarrow & Y \\ {\scriptstyle \operatorname{Spec}(\tau)}\downarrow & & \downarrow{\scriptstyle f} \\ \operatorname{Spec}(R') & \longrightarrow & X. \end{array}$$

If Ω_f is projective, then the set of all $h \in R'$ for which the induced lifting problem for $\tau_h\colon R'_h \twoheadrightarrow R_h$ has a solution is a radical ideal in R'.

Theorem 9.95 (Formal smoothness via the cotangent complex)**.** *Let $f\colon Y \to X$ be a map of schemes.*

(i) f is formally smooth if and only if Ω_f is projective and $\mathcal{N}_f = 0$.

(ii) f is formally unramified if and only if $\Omega_f = 0$.

(iii) f is formally étale if and only if $\Omega_f = 0$ and $\mathcal{N}_f = 0$.

Corollary 9.96. *"Formally smooth", "formally unramified", and "formally étale" are local properties of morphisms of schemes.*

Combining Theorems 9.53 and 9.95, we deduce the equivalence $(a) \Leftrightarrow (b)$ of Theorem 9.4:

Corollary 9.97. *Let $f\colon Y \to X$ be a map of schemes.*

(i) f is smooth if and only if it is formally smooth and locally of finite presentation.

(ii) f is unramified if and only if it is formally unramified and locally of finite type.

(iii) f is étale if and only if it is formally étale and locally of finite presentation.

Remark 9.98. By Example 9.87(d), smooth, unramified, and étale morphisms of schemes satisfy their respective lifting property with respect to *arbitrary* thickenings of rings, and the latter two even with respect to arbitrary thickenings of algebraic functors.

Remark 9.99. The proof of the local nature of formal smoothness (Corollary 9.96) relies on the highly nontrivial fact that projectivity of modules is a Zariski-local property (Proposition 2.80(d)). However, this fact is not used in Theorem 9.95.

9.6. **Local structure of étale maps.** Recall that a scheme X is *discrete* if its underlying space $|X|$ is discrete, and that the *fibers* of a morphism of schemes $Y \to X$ are the schemes $Y_x = Y \times_X \operatorname{Spec}(\kappa(x))$ for $x \in |X|$.

Definition 9.100 (Locally quasi-finite, quasi-finite morphism)**.** A morphism of schemes is *locally quasi-finite* if it is locally of finite type and its fibers are discrete. It is *quasi-finite* if it is locally quasi-finite and quasi-compact.

Remark 9.101. "Locally quasi-finite" is a local property of morphisms of schemes.

Remark 9.102 (Fiberwise finiteness)*.* A map is quasi-finite if and only if it is of finite type and has finite fibers. Indeed, the following conditions are equivalent for a scheme X of finite type over a field k:

(a) X is discrete;
(b) X has Krull dimension ≤ 0;
(c) X has finitely many points;
(d) X is finite over k (Definition 3.99).

The implication (a) $\Rightarrow$ (c) holds whenever X is quasi-compact, and the implication (b) $\Rightarrow$ (a) holds whenever X is locally noetherian. The implication (d) $\Rightarrow$ (b) follows from Theorem 8.58(b). The implication (c) $\Rightarrow$ (b) follows from Theorem 8.65: the generic points of the irreducible components are open, so that their residue fields are of finite type over k and hence of transcendence degree 0. Finally, (b) and (c) imply that $X \simeq \operatorname{Spec}(R)$ where R is a noetherian k-algebra of Krull dimension ≤ 0, hence artinian. Thus, R admits a finite filtration by ideals whose successive quotients are isomorphic to residue fields of R. Since these fields are themselves finite extensions of k, this proves (d).

Example 9.103*.*

(a) By Corollary 9.67, unramified morphisms of schemes are locally quasi-finite.
(b) By Remark 9.102, finite morphisms of schemes are quasi-finite.

Theorem 9.104 (Zariski's main theorem, local version)**.** *Let $R \to S$ be a ring map of finite type, let $\mathfrak{p} \subseteq R$ be a prime ideal, and let $\mathfrak{q} \subseteq S$ be a prime ideal above $\mathfrak{p}$ that is isolated in its fiber. Then there exists a finite R-algebra $S' \subseteq S$ and $g \in S' \setminus \mathfrak{q}$ such that $S'_g \xrightarrow{\sim} S_g$.*

Remark 9.105 (Zariski's main theorem)*.* The global version of Zariski's main theorem is the following statement: if $f\colon Y \to X$ is a separated quasi-finite morphism of schemes, then f admits a factorization

$$Y \overset{j}{\hookrightarrow} \overline{Y} \overset{p}{\to} X$$

where j is an open immersion and p is integral; in particular, f is quasi-affine. Moreover, we can choose p finite if X is qcqs. The scheme $\overline{Y}$ is $\operatorname{Spec}(A)$ where $A \in \operatorname{CAlg}_X$ is the integral closure of $\mathcal{O}_X$ in $f_*(\mathcal{O}_Y)$ (cf. Construction 8.91), and the main point is then that j is an open immersion. This requires some nontrivial new arguments, as Theorem 9.104 only implies that j is an open immersion locally on Y.

Definition 9.106 (Standard étale algebra)**.** An R-algebra S is *standard étale* if $S \simeq (R[x]/(f))_g$ where $f \in R[x]$ is a monic polynomial and $g \in R[x]$ is such that the derivative f' becomes a unit in $(R[x]/(f))_g$. Note that $R \to S$ is indeed étale as the Jacobian matrix of the presentation $S \simeq R[x,y]/(f, yg-1)$ has determinant $f'g$, which is invertible in S.

Theorem 9.107 (Local structure of étale maps)**.** *Let $f\colon Y \to X$ be an étale morphism of schemes. For every $y \in |Y|$, there exist affine open subschemes $V \subset Y$ and $U \subset X$ with $y \in |V|$ and $f(V) \subset U$ such that $\mathcal{O}(V)$ is a standard étale $\mathcal{O}(U)$-algebra.*

Since standard étale algebras are clearly flat and since a smooth map locally factors as an étale map to an affine space (Remark 9.13(a)), we deduce:

Corollary 9.108. *Smooth morphisms are flat.*

Corollary 9.108 combined with Theorem 9.63 yields the implications (a) $\Rightarrow$ (d) in Theorem 9.4, which concludes the proof.

Remark 9.109 (The étale topology). By Corollary 9.7, the general recipe of Example 6.29 applies to the classes of smooth, unramified, and étale maps of rings. In particular, there is a pretopology on $\mathrm{CAlg}^{\mathrm{op}}$ whose covers are finite families of étale maps $(R \to R_i)_{i\in I}$ such that $\coprod_{i\in I} \mathrm{Spec}(R_i) \to \mathrm{Spec}(R)$ is surjective. The associated Grothendieck topology on $\mathrm{CAlg}^{\mathrm{op}}$ is called the *étale topology*. Like the Zariski topology, the étale topology extends to the category of schemes and even to the whole category of algebraic functors (see Example 6.37). The topology defined by the larger class of smooth maps turns out to coincide with the étale topology. Since open immersions are étale, the étale topology is finer than the Zariski topology, and since étale maps are flat, it is coarser than the fppf topology:

$$\mathrm{Zar} \leq \text{ét} \leq \mathrm{fppf} \leq \mathrm{fpqc}.$$

An *étale sheaf* on a scheme X is a sheaf on the site $(\mathrm{Ét}_X, \mathrm{et})$, where $\mathrm{Ét}_X \subseteq \mathrm{Sch}_{/X}$ is the full subcategory of étale X-schemes. This notion was the original motivation for the formalism of Grothendieck topologies and sheaves, as the category of étale sheaves on a scheme is usually not equivalent to any category of sheaves on a topological space.

The étale topology plays a very important role in algebraic geometry, as the Zariski topology is too coarse for some purposes, such as having a useful cohomology theory with constant coefficients. Indeed, for any irreducible scheme, constant sheaves of abelian groups have trivial higher Zariski cohomology, but their étale cohomology is interesting. A famous theorem of Grothendieck states that the étale cohomology of any quasi-projective $\mathbf{C}$-scheme $X \subset \mathbf{P}^n_{\mathbf{C}}$ with coefficients in any finite abelian group is isomorphic to the singular cohomology of $X(\mathbf{C}) \subset \mathbf{P}^n(\mathbf{C})$. But the former makes sense for any scheme and so can be used as a replacement for singular cohomology. For example, from the perspective of the étale topology, projective spaces and elliptic curves over any separably closed field look like complex projective spaces and tori. Fields that are not separably closed themselves have a nontrivial étale site: for example, it turns out that $\mathrm{Spec}(\mathbf{F}_q)$ has the same étale cohomology as a topological circle.

Another advantage of the étale topology is that it allows us to compute certain quotients. Consider for example a finite group G acting freely on a scheme X. Then the quotients $\mathrm{a}_{\mathrm{Zar}}(X/G)$ and $\mathrm{a}_{\text{ét}}(X/G)$ are usually different. Under mild technical assumptions (which hold if X is quasi-projective over some ring), $\mathrm{a}_{\text{ét}}(X/G)$ is a scheme and is in fact the quotient of X by G in the category of schemes.

9.7. **Bertini's theorem.** Bertini's theorem states that a generic hyperplane section of a smooth quasi-projective scheme over a field is again smooth. The proof uses a fundamental ideal in geometry, which is to promote a set of interest to a geometric object. In this case, the set of hyperplanes in a projective space over k can be identified with the set of k-points of the dual projective space. We then estimate the Krull dimension of the closure of set of "bad" hyperplanes (those for which the conclusion of the theorem fails) in this dual projective space to deduce that the open complement is nonempty.

YW: I haven't go through this section yet.

Definition 9.110 (Hyperplane, hyperplane section)**.** Let k be a ring and let V be a k-module.

(a) A *hyperplane* in $\mathbf{P}(V)$ is a closed subscheme of the form $\mathbf{P}(W)$, where W is a quotient module of V such that $\ker(V \twoheadrightarrow W)$ is a line.
(b) Given a subscheme $X \subseteq \mathbf{P}(V)$, a *hyperplane section* of X is a closed subscheme of X of the form $X \cap H$, where $H \subseteq \mathbf{P}(V)$ is a hyperplane.

Definition 9.111 (Dual projective space)**.** Let k be a ring and V a vector space over k. The projective space $\mathbf{P}(V^\vee)$ is called the *dual projective space* of $\mathbf{P}(V)$. As an algebraic k-functor, it is given by

$$\mathbf{P}(V^\vee)(R) = \{\text{hyperplanes in } \mathbf{P}(V)_R\}.$$

Remark 9.112. If V is an arbitrary k-module, the functor $R \mapsto \{\text{hyperplanes in } \mathbf{P}(V)_R\}$ is exactly the algebraic k-functor $\mathbf{P}^\vee(V)$ of Example 3.48. As mentioned there, it is not a scheme if V is not a vector space.

Construction 9.113 (Incidence hyperplane)**.** Let k be a ring and let V be a vector space over k. Let $\mathcal{O}(1,1)$ be the line bundle $\pi_1^*(\mathcal{O}(1)) \otimes \pi_2^*(\mathcal{O}(1))$ over the product $\mathbf{P}(V) \times \mathbf{P}(V^\vee)$. It has a canonical global section $s \in \Gamma(\mathbf{P}(V)\times\mathbf{P}(V^\vee), \mathcal{O}(1,1))$ defined as follows: for any k-algebra R and any R-point $x = (x_1, x_2)$ of $\mathbf{P}(V)\times\mathbf{P}(V^\vee)$ given by quotient lines $x_1 \colon V \otimes_k R \twoheadrightarrow L_1$ and $x_2 \colon V^\vee \otimes_k R \twoheadrightarrow L_2$, $s(x) \in L_1 \otimes L_2$ is the image of 1 by the composite

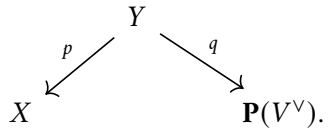
$$R \xrightarrow{\text{coev}} (V \otimes_k R) \otimes_R (V^\vee \otimes_k R) \xrightarrow{x_1 \otimes x_2} L_1 \otimes_R L_2.$$

The vanishing locus $\mathrm{V}(s) \subseteq \mathbf{P}(V) \times \mathbf{P}(V^\vee)$ is called the *incidence hyperplane* of *universal hyperplane* of $\mathbf{P}(V)$. For any k-algebra R, $\mathrm{V}(s)(R)$ consists of pairs (x, H), where $x \in \mathbf{P}(V)(R)$ and $H \subseteq \mathbf{P}(V)_R$ is a hyperplane with $x \in H$. It is universal in the following sense: for any hyperplane $H \subseteq \mathbf{P}(V)_R$, there is a unique map of k-schemes $h \colon \operatorname{Spec}(R) \to \mathbf{P}(V^\vee)$ such that H is the preimage of $\mathrm{V}(s)$ by the induced map $\mathrm{id}_{\mathbf{P}(V)} \times h \colon \mathbf{P}(V)_R \to \mathbf{P}(V) \times \mathbf{P}(V^\vee)$.

The following basic flatness criterion is a consequence of Krull's intersection theorem; it does not hold as stated without the noetherian assumption.

Proposition 9.114 (Flatness of fiberwise effective divisors)**.** *Let $R \to S$ be a flat map of noetherian rings and let $f \in S$. If for each maximal ideal $\mathfrak{m} \subseteq R$, f is not a zero divisor in $S/\mathfrak{m}S$, then f is not a zero divisor in S and S/f is flat over R.*

Remark 9.115 (Krull dimension of the locus of singular hyperplane sections)*.* Let k be a field, V a vector space of rank r over k, and $X \subseteq \mathbf{P}(V)$ a locally closed subscheme that is smooth over k. Let $Y \subseteq X \times \mathbf{P}(V^\vee)$ be the restriction of the incidence hyperplane in $\mathbf{P}(V) \times \mathbf{P}(V^\vee)$ with the two projections



and let $\Sigma \subseteq |Y|$ be the closed complement of the smooth locus of q. We seek to bound the Krull dimension of the closure of $q(\Sigma)$, to deduce that the open complement is nonempty. It follows from Theorem 8.65 that

$$\dim(\overline{q(\Sigma)}) \le \dim(\Sigma) \le \sup_{x \in |X|} (\dim_x(X)) + \dim(p^{-1}(x) \cap \Sigma).$$

Fix a field extension $k \to \kappa$ and a κ-point x of X, and let $d_x = \dim_x(X) = \dim(\mathcal{O}_{X,x})$. A preimage of x by p is the data of a hyperplane $H \subseteq \mathbf{P}(V_\kappa)$ containing x, and the fiber of q over H is the hyperplane section $X_\kappa \cap H$. Since X is smooth over k, the local ring $\mathcal{O}_{X_\kappa,x}$ is regular (Theorem 9.63) and in particular a domain (Theorem 8.81(b)). It follows from Proposition 9.114 and Corollary 8.61 that q is flat at (x, H) if and only if H does not contain the irreducible component of x in X_κ. By fiberwise smoothness (Proposition 9.68), we deduce that the κ-points of $p^{-1}(x) \cap \Sigma$ are exactly the hyperplanes $H \subseteq \mathbf{P}(V_\kappa)$ containing x such that the local ring $\mathcal{O}_{X_\kappa \cap H,x}$ is *not* regular of Krull dimension $d_x - 1$.

To understand this set of hyperplanes, consider the κ-linear maps

$$\varphi_x \colon V_\kappa \simeq \Gamma(\mathbf{P}(V_\kappa), \mathcal{O}(1)) \to \Gamma(X_\kappa, \mathcal{O}_{X_\kappa}(1)) \to \mathcal{O}_{X_\kappa}(1)_x \to \mathcal{O}_{X_\kappa}(1)_x/\mathfrak{m}_x^2\mathcal{O}_{X_\kappa}(1)_x,$$

where the isomorphism is due to Proposition 4.56(c) and $\mathcal{O}_{X_\kappa}(1)$ is the restriction of $\mathcal{O}(1)$ to X_κ. For any $e \in V_\kappa \setminus 0$ such that $x \in \mathrm{V}(e)$, the map of local rings at x induced by the inclusion $X_\kappa \cap \mathrm{V}(e) \subseteq X_\kappa$ is identified, upon trivializing $\mathcal{O}(1)$ in a neighborhood of x, with the quotient map

$$\mathcal{O}_{X_\kappa,x} \twoheadrightarrow \mathcal{O}_{X_\kappa,x}/\mathfrak{m}_x^2 \to \mathcal{O}_{X_\kappa,x}/\mathfrak{m}_x = \kappa \to 0$$

and the assumption that X is locally closed in $\mathbf{P}(V)$, we deduce that φ_x is surjective with target of rank $d_x + 1$, so that $\ker(\varphi_x)$ has rank $r - d_x - 1$ and hence $\dim(p^{-1}(x) \cap \Sigma) = r - d_x - 2$. Altogether, we get

$$\dim(\overline{q(\Sigma)}) \le \sup_{x \in |X|} (d_x + (r - d_x - 2)) = r - 2 < r - 1 = \dim(\mathbf{P}(V^\vee)).$$

Theorem 9.116 (Bertini's theorem)**.** *Let k be a field, V a vector space over k, and $X \subseteq \mathbf{P}(V)$ a locally closed subscheme that is smooth over k. Then there exists a nonempty open subscheme $U \subseteq \mathbf{P}(V^\vee)$ such that, for every point $h \in |U|$ classifying a hyperplane $H \subseteq \mathbf{P}(V)_{\kappa(h)}$, the hyperplane section $X_{\kappa(h)} \cap H$ is smooth over $\kappa(h)$ and does not contain any irreducible component of $X_{\kappa(h)}$.*

This result does not imply that there exists a hyperplane $H \subseteq \mathbf{P}(V)$ such that $X \cap H$ is smooth over k. Indeed, if k is a finite field, then $\mathbf{P}(V^\vee)$ only has finitely many k-points, so there exist nonempty open subschemes of $\mathbf{P}(V^\vee)$ that do not have any k-point. However, this cannot happen for infinite fields:

Lemma 9.117. *If k is an infinite field, then $\mathbf{A}^n(k)$ is a dense subset of $|\mathbf{A}^n_k|$.*

Corollary 9.118. *Let k be an infinite field, V a vector space over k, and $X \subseteq \mathbf{P}(V)$ a locally closed subscheme that is smooth over k. Then there exists a hyperplane $H \subseteq \mathbf{P}(V)$ such that $X \cap HY$ is smooth over k and does not contain any irreducible component of X.*

10. Divisors and blowups

Blowing up is a ubiquitous construction in algebraic geometry, which serves many purposes, such as eliminating indeterminacies of rational maps and resolving singularities. We illustrate this with two examples.

Consider the "slope" rational function y/x on $\mathbf{A}^2 = \operatorname{Spec}(\mathbf{Z}[x,y])$. Its domain of definition is the open subscheme $\mathrm{D}(x) = \mathbf{A}^2 - (0 \times \mathbf{A}^1)$, so it is a map $y/x \colon \mathbf{A}^2 - (0 \times \mathbf{A}^1) \to \mathbf{A}^1$. If we enlarge its codomain from $\mathbf{A}^1$ to $\mathbf{P}^1$, it extends further to $\mathbf{A}^2 - 0$ by setting $y/x = \infty$ if $x = 0$. More precisely, the map

$$y/x \colon \mathbf{A}^2 - 0 \to \mathbf{P}^1$$

classifies the quotient line $(x,y) \colon \mathcal{O}^2 \twoheadrightarrow \mathcal{O}$ over $\mathbf{A}^2 - 0$. However, it is not possible to assign a sensible value to y/x at the origin, as y/x is constant on each line through the origin but with different values on different lines. To extend y/x across the origin, we are forced to add to $\mathbf{A}^2 - 0$ different limit points for each line through the origin, i.e., to glue in a copy of $\mathbf{P}^1$ at the origin. This results in a scheme $\mathrm{Bl}_0(\mathbf{A}^2)$ called the *blowup* of $\mathbf{A}^2$ at the origin, which is formally defined by

$$\mathrm{Bl}_0(\mathbf{A}^2)(R) = \left\{ \begin{array}{ccc} R^2 & \xrightarrow{\quad a \quad} & R \\ & {\scriptstyle q}\searrow \quad \nearrow{\scriptstyle s} & \\ & L & \end{array} \in \mathrm{Mod}_R \;\middle|\; q \text{ is a quotient line} \right\}.$$

We have cartesian squares

$$\begin{array}{ccccc} \mathbf{P}^1 & \xhookrightarrow{e} & \mathrm{Bl}_0(\mathbf{A}^2) & \hookleftarrow & \mathbf{A}^2 \setminus 0 \\ \downarrow & & \downarrow{\scriptstyle b} & & \downarrow{\scriptstyle \simeq} \\ 0 & \longrightarrow & \mathbf{A}^2 & \longleftarrow & \mathbf{A}^2 \setminus 0 \end{array}$$

where $b(a,q) = a$ and $e(q) = (0,q)$. The scheme $\mathrm{Bl}_0(\mathbf{A}^2)$ is in fact a smooth quasi-projective $\mathbf{Z}$-scheme, which can be described in two other ways:

$$\mathbf{A}_{\mathbf{P}^1}(\mathcal{O}(1)) \simeq \mathrm{Bl}_0(\mathbf{A}^2) \simeq \mathbf{P}_{\mathbf{A}^2}((x,y)).$$

The first isomorphism is simply the observation that a is determined by q and s. The second isomorphism states that a factors through q if and only if $q(a_2, -a_1) = 0$, which one can check using Zariski descent. The map $\mathrm{Bl}_0(\mathbf{A}^2) \simeq \mathbf{A}_{\mathbf{P}^1}(\mathcal{O}(1)) \to \mathbf{P}^1$ is then the desired extension of y/x from $\mathbf{A}^2 \setminus 0$.

As another example, consider the affine nodal cubic $N = \mathrm{V}(y^2 - x^3 - x^2) \subset \mathbf{A}^2$. At the singular point $(0,0)$, there are two tangent directions to N. Blowing up N at the singular point replaces that point with two points, which has the effect of separating the two branches through $(0,0)$. The result is in fact the normalization of N, which is the map $\mathbf{A}^1 \to N$ given by $x \mapsto t^2 - 1$ and $y \mapsto t(t^2 - 1)$ (see Example 8.92). Thus, blowing up the nodal singularity "resolves" the singularity and produces a regular scheme.

In both of these examples, the key feature of the blowup construction is that the preimage of the blown-up point is an *effective Cartier divisor*, i.e., a closed subscheme locally given by the vanishing locus of a single function that is not a zero divisor. In general, the blowup of a scheme X at a closed subscheme $Z \subset X$ will be defined as the universal construction with this property: the map $b \colon \mathrm{Bl}_Z(X) \to X$ is the final object in the full subcategory of $\mathrm{Sch}_{/X}$ consisting of the maps $f \colon Y \to X$ such that $f^{-1}(Z) \subset Y$ is an effective Cartier divisor. It turns out that b is an isomorphism over $X - Z$ and that the preimage $b^{-1}(Z) \subset \mathrm{Bl}_Z(X)$ is the *projectivized normal cone* of Z in X, i.e., the scheme of normal directions to Z in X, which is a closed subscheme of $\mathbf{P}(\mathcal{N}_{Z/X})$.

Let us mention two theorems about blowups that are general forms of the above two examples:

(a) (Elimination of indeterminacies) Let S be a scheme, let X and P be S-schemes with X qcqs and P proper over S, and let $U \subset X$ be a dense quasi-compact open subscheme. For every map $f \colon U \to P$ over S, there exists a finitely presented closed subscheme $Z \subset X$ with $Z \cap U = \varnothing$ and a map $\hat{f} \colon \mathrm{Bl}_Z(X) \to P$ over S extending f.

(b) (Resolution of singularities) Let X be a scheme of finite type over a complete noetherian local $\mathbf{Q}$-algebra (e.g., a field of characteristic zero) and let $U \subset X$ be a regular open subscheme. Then there exists a closed subscheme $Z \subset X$ with $Z \cap U = \varnothing$ such that $\mathrm{Bl}_Z(X)$ is regular.

10.1. **Effective divisors.**

Definition 10.1 (Effective virtual divisor, effective Cartier divisor)**.** Let X be an algebraic functor. An *effective virtual divisor* (also called a *generalized Cartier divisor*) on X is a pair (L, s) where $L \in \mathrm{Line}_X$ is a line bundle over X and $s\colon L \to \mathcal{O}_X$ is a map in Line_X. We denote by $\mathrm{vDiv}^+(X) = ((\mathrm{Line}_X)_{/\mathcal{O}_X})^{\mathrm{op}}$ the category of effective virtual divisors on X.

If X is a scheme, an *effective Cartier divisor* or simply an *effective divisor* on X is an effective virtual divisor (L, s) where s is a monomorphism. We denote by $\mathrm{Div}^+(X) \subset \mathrm{vDiv}^+(X)$ the full subcategory of effective Cartier divisors.

Remark 10.2 (Effective divisors as closed subschemes)*.* For any scheme X, the category $\mathrm{Div}^+(X)$ is a poset, which by Theorem 7.58 is equivalent to the opposite of the poset of *invertible* quasi-coherent ideals over X:

$$\mathrm{Div}^+(X) \simeq (\mathrm{Idl}_X \times_{\mathrm{Mod}_X} \mathrm{Line}_X)^{\mathrm{op}} \subset \mathrm{Idl}_X^{\mathrm{op}}.$$

We can further identify effective divisors on X with closed subschemes of X whose quasi-coherent ideals are line bundles. Such closed subschemes are also called *effective (Cartier) divisors*. Note that we have $(L, s) \leq (L', s')$ in the poset $\mathrm{Div}^+(X)$ if and only if there is a map $(L', s') \to (L, s)$ in $(\mathrm{Line}_X)_{/\mathcal{O}_X}$. Thus, the relation $\leq$ corresponds to the inclusion relation $\subset$ between closed subschemes.

Proposition 10.3 (Characterization of effective divisors)**.** *Let X be a scheme and $Z \subset X$ a closed subscheme with quasi-coherent ideal I. The following conditions are equivalent:*

(i) Z is an effective divisor on X, i.e., I is a line bundle over X.
(ii) There exists an affine open covering $(U_i \subset X)_i$ and functions $f_i \in \mathcal{O}(U_i)$ such that $Z \cap U_i = \mathrm{V}(f_i)$ and f_i is not a zero divisor in $\mathcal{O}(U_i)$.

Warning 10.4*.* If X is affine, effective divisors on X are not necessarily of the form $\mathrm{V}(f)$ for a non-zero divisor $f \in \mathcal{O}(X)$: there can exist monomorphisms $L \hookrightarrow \mathcal{O}_X$ where L is a nontrivial line bundle over X.

Remark 10.5 (Commutative monoid structure)*.* For any algebraic functor X, the category $\mathrm{vDiv}^+(X)$ has a symmetric monoidal structure given by the tensor product. It is customary to use additive notation for this symmetric monoidal structure:

$$(L, s) + (L', s') = (L \otimes L', s \otimes s') \quad \text{and} \quad 0 = (\mathcal{O}_X, \mathrm{id}).$$

Note that 0 is also the initial object in $\mathrm{vDiv}^+(X)$.

If X is a scheme, this symmetric monoidal structure restricts to the poset $\mathrm{Div}^+(X)$ of effective divisors, since tensoring with a line bundle preserves monomorphisms. In other words, $\mathrm{Div}^+(X)$ is a partially ordered commutative monoid (all of whose elements are ≥ 0).

Remark 10.6 (Functoriality)*.* The categories of effective virtual divisors assemble into a functor

$$\mathrm{vDiv}^+\colon \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})^{\mathrm{op}} \to \mathrm{Cat},$$

which is in fact the quasi-coherent extension of its restriction to CAlg in the sense of §4.2. It is even a functor valued in symmetric monoidal categories: for any map of algebraic functors $f\colon Y \to X$, the pullback functor

$$f^*\colon \mathrm{vDiv}^+(X) \to \mathrm{vDiv}^+(Y)$$

has a canonical symmetric monoidal structure.

If $f\colon Y \to X$ is a morphism of schemes, however, then $f^*\colon \mathrm{vDiv}^+(X) \to \mathrm{vDiv}^+(Y)$ does not necessarily preserve the subcategories of effective divisors. Given an effective divisor $I \hookrightarrow \mathcal{O}_X$, its pullback to Y is non-virtual if and only if $\mathrm{L}_1 f^*(\mathcal{O}_X/I) = 0$. In particular, if f is flat, then f^* restricts to a map $\mathrm{Div}^+(X) \to \mathrm{Div}^+(Y)$. This partial functoriality of Div^+ is one of the reasons for introducing the better-behaved notion of effective virtual divisor.

Example 10.7. (i) Let X be a scheme. Every function $f \in \mathcal{O}(X)$ defines an effective virtual divisor $(\mathcal{O}_X, f)$. It is an effective divisor if and only if, for every open subscheme $U \subset X$, $f|_U \in \mathcal{O}(U)$ is not a zero divisor. In fact, it suffices to check the latter on some affine open covering of X.

(ii) Let A be an $\mathbf{N}$-graded ring generated by $A_{\leq 1}$, so that $\mathcal{O}(d)$ is a line bundle on $\mathrm{Proj}(A)$ for all $d \in \mathbf{Z}$. Every homogeneous element $f \in A_d$ defines an effective virtual divisor $(\mathcal{O}(-d), f)$. It is an effective divisor if and only if, for every $h \in A_1$, f/h^d is not a zero divisor in $A_{(h)}$. This is the case if f is not a zero divisor in A.

(iii) If R is a unique factorization domain, then any invertible ideal $I \subset R$ is principal. Moreover, two elements of R generate the same principal ideal if and only if they differ by a unit. Hence, there is an isomorphism of partially ordered commutative monoids

$$(\mathrm{Div}^+(\mathrm{Spec}(R)), +, \leq) \simeq ((R - \{0\})/R^\times, \cdot, |).$$

(iv) If R is a Dedekind domain, then any nonzero ideal $I \subset R$ is a line, so that $\mathrm{Div}^+(\mathrm{Spec}(R))$ is the poset of all closed subschemes of $\mathrm{Spec}(R)$ other than $\mathrm{Spec}(R)$. By the unique prime factorization of ideals, this is also the free commutative monoid on the set of maximal ideals of R. Consequently, if X is a Dedekind scheme, then $\mathrm{Div}^+(X)$ is the poset of closed subschemes of X that do not contain any point of codimension 0. If X is moreover quasi-compact, $\mathrm{Div}^+(X)$ is also the free commutative monoid on the set of closed points of $|X|$.

Remark 10.8 (Zariski descent for effective divisors). The Zariski descent property of $X \mapsto \mathrm{Mod}_X$ implies the same property for $X \mapsto \mathrm{vDiv}^+(X)$, i.e.,

$$\mathrm{vDiv}^+ \colon \mathrm{Fun}(\mathrm{CAlg}^{\mathrm{op}}, \mathrm{Set})^{\mathrm{op}} \to \mathrm{Cat}$$

is a Zariski sheaf of categories. If X is a scheme, then Div^+ restricts to a presheaf on $\mathrm{Open}(X)$, since open immersions are flat. Since being a monomorphism is a Zariski-local property, we deduce that

$$\mathrm{Div}^+ \colon \mathrm{Open}(X)^{\mathrm{op}} \to \mathrm{Pos}$$

is a sheaf of posets on $|X|$.

Remark 10.9 (vDiv^+ as a quotient stack). Given a ring R, the *divisibility category* $R//R$ is the category whose objects are the elements of R and with $\mathrm{Map}_{R//R}(r, s) = \{t \in R \mid tr = s\}$; composition is given by the product in R. There is a functor $R//R \to \mathrm{vDiv}^+(R)$ sending r to (R, r). This construction is natural in R and defines a functor $\mathcal{O}//\mathcal{O} \colon \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})^{\mathrm{op}} \to \mathrm{Cat}$ together with a map $\mathcal{O}//\mathcal{O} \to \mathrm{vDiv}^+$. Since line bundles are Zariski-locally trivial, this map is in fact a Zariski-local epimorphism, and one can show that it exhibits vDiv^+ as the Zariski sheafification of $\mathcal{O}//\mathcal{O}$. In particular, the maximal subgroupoid of vDiv^+ can be described as the quotient $\mathcal{O}/\mathcal{O}^\times$ computed as a Zariski sheaf of groupoids.

Proposition 10.10. *Let X be a scheme and let (L, s) be an effective virtual divisor on X. The category of X-schemes Y such that the pullback of (L, s) to Y is an effective divisor admits a final object, which is the closure of the open subscheme $\mathrm{D}(s) \subset X$, or equivalently the closed subscheme of X whose quasi-coherent ideal is the kernel of the localization map $\mathcal{O}_X \to \mathcal{O}_X[s^{-1}]$.*

Definition 10.11 (Cartier locus)**.** Let X be a scheme and let (L, s) be an effective virtual divisor on X. The closed subscheme of X defined by the previous proposition is called the *Cartier locus* of (L, s).

Remark 10.12 (Functoriality of the Cartier locus). Let $f \colon Y \to X$ be a map of schemes and let (L, s) be an effective virtual divisor on X. Let $Z \subset X$ be the Cartier locus of (L, s) and $W \subset Y$ that of $f^*(L, s)$. Since the pullback of (L, s) to W is an effective divisor, we have $f(W) \subset Z$ by definition of Z. The closed immersion $W \hookrightarrow f^{-1}(Z)$ is an isomorphism if and only if the epi-mono factorization of $\mathcal{O}_X \to \mathcal{O}_X[s^{-1}]$ is preserved by f^*, for example if f is flat.

10.2. **Blowing up.**

Definition 10.13 (Virtual blowup, exceptional divisor)**.** Let X be an algebraic functor and let $z\colon M \to \mathcal{O}_X$ be a map of quasi-coherent modules over X. The *virtual blowup* $\mathrm{vBl}_z(X)$ of X at z is the algebraic functor over X classifying factorizations of z through a quotient line of M. In other words, $\mathrm{vBl}_z(X)$ is the subfunctor of $\mathbf{P}(M)$ consisting of those quotient lines $q\colon M(x) \twoheadrightarrow L$ such that $z(x)$ factors through q.

If $b\colon \mathrm{vBl}_z(X) \to X$ is the structure map, we therefore have a canonical factorization

$$\begin{array}{ccc} b^*(M) & \xrightarrow{\quad b^*(z) \quad} & \mathcal{O} \\ & \searrow \quad \nearrow_{s} & \\ & \mathcal{O}(1) & \end{array}$$

in $\mathrm{Mod}_{\mathrm{vBl}_z(X)}$, where $\mathcal{O}(1)$ is the restriction of the tautological line bundle $\mathcal{O}(1) \in \mathrm{Mod}_{\mathbf{P}(M)}$. The effective virtual divisor $(\mathcal{O}(1), s)$ on $\mathrm{vBl}_z(X)$ is called the *exceptional divisor* of the virtual blowup.

If X is a scheme and $Z \subset X$ is a closed subscheme with quasi-coherent ideal I, the *virtual blowup* $\mathrm{vBl}_Z(X)$ of X at Z is the virtual blowup of X at $I \hookrightarrow \mathcal{O}_X$.

Remark 10.14. By the characterization of closed subfunctors of projective spaces, $\mathrm{vBl}_z(X)$ is a closed subfunctor of $\mathbf{P}(M)$. In particular, the map $b\colon \mathrm{vBl}_z(X) \to X$ is schematic, and it is projective if M is of finite type.

Definition 10.15 (Blowup, exceptional divisor)**.** Let X be a scheme and let $z\colon M \to \mathcal{O}_X$ be a map of quasi-coherent modules over X. The *blowup* $\mathrm{Bl}_z(X)$ of X at z is the Cartier locus of the exceptional divisor on $\mathrm{vBl}_z(X)$. The restriction of that divisor to $\mathrm{Bl}_z(X)$ is then an effective Cartier divisor by definition, called the *exceptional divisor* of the blowup.

If $Z \subset X$ is a closed subscheme with quasi-coherent ideal I, the *blowup* $\mathrm{Bl}_Z(X)$ of X at Z is the blowup of X at $I \hookrightarrow \mathcal{O}_X$.

Example 10.16. Let X be a scheme.

(i) If $z\colon M \to \mathcal{O}_X$ is the zero map, then $\mathrm{vBl}_z(X) = \mathbf{P}(M)$ with exceptional divisor $(\mathcal{O}(1), 0)$, and hence $\mathrm{Bl}_z(X) = \varnothing$. In particular, for $M = 0$, we get $\mathrm{vBl}_X(X) = \mathrm{Bl}_X(X) = \varnothing$.

(ii) If $z\colon L \to \mathcal{O}_X$ is an effective virtual divisor, then $\mathrm{vBl}_z(X) = X$ and $\mathrm{Bl}_z(X)$ is the Cartier locus of (L, z). Hence, if $Z \subset X$ is an effective divisor, then $\mathrm{vBl}_Z(X) = \mathrm{Bl}_Z(X) = X$. This applies in particular when $Z = \varnothing$.

(iii) Let $f \in \mathcal{O}(X)$ and $Z = \mathrm{V}(f)$. Then $\mathrm{vBl}_Z(X)$ and $\mathrm{Bl}_Z(X)$ are the closed subschemes of X defined by the quasi-coherent ideals of f-torsion and f-power torsion elements, respectively. For example, the blowup of $\mathrm{Spec}(k[x, y]/(xy))$ at $\mathrm{V}(x)$ is $\mathrm{V}(y)$.

Remark 10.17 (Geometric description of blowups)*.* Let X be an algebraic functor, $z\colon M \to \mathcal{O}_X$ a map in Mod_X, and $b\colon \mathrm{vBl}_z(X) \to X$ the virtual blowup of X at z. By the characterization of epimorphism loci, the vanishing locus $\mathrm{V}(z)$ is a closed subfunctor of X, whose open complement is the epimorphism locus $\mathrm{D}(z)$. We claim that there are cartesian squares

$$\begin{array}{ccccc} \mathbf{P}(M|_{\mathrm{V}(z)}) & \longrightarrow & \mathrm{vBl}_z(X) & \longleftarrow & \mathrm{D}(z) \\ \downarrow & & \downarrow{\scriptstyle b} & & \downarrow \\ \mathrm{V}(z) & \longrightarrow & X & \longleftarrow & \mathrm{D}(z). \end{array}$$

Indeed, if z is the zero map, then it factors uniquely through any quotient line of M, which explains the left-hand cartesian square. On the other hand, if z is an epimorphism, then it admits a unique factorization through a quotient line, which explains the right-hand cartesian square.

If X is a scheme, the description of the Cartier locus shows that $\mathrm{Bl}_z(X)$ is exactly the *closure* of $\mathrm{D}(z)$ in $\mathrm{vBl}_z(X)$ and that the exceptional divisor in $\mathrm{Bl}_z(X)$ is the "boundary" of $\mathrm{D}(z)$, i.e., the intersection $\mathbf{P}(M|_{\mathrm{V}(z)}) \cap \mathrm{Bl}_z(X)$.

Remark 10.18 (Functoriality of blowups)*.* The virtual blowup construction is a functor on the following category:

- An object is a pair $(X, z\colon M \to \mathcal{O}_X)$, where X is an algebraic functor and z is a map in Mod_X.

- A map $(Y, w\colon N \to \mathcal{O}_Y) \to (X, z\colon M \to \mathcal{O}_X)$ consists of a map $f\colon Y \to X$ and an epimorphism $\varphi\colon f^*(M) \twoheadrightarrow N$ in $(\mathrm{Mod}_Y)_{/\mathcal{O}_Y}$.

Such a map induces a commutative square

$$\begin{array}{ccc} \mathrm{vBl}_w(Y) & \xrightarrow{f'} & \mathrm{vBl}_z(X) \\ \downarrow & & \downarrow \\ Y & \xrightarrow[f]{} & X, \end{array}$$

such that the exceptional divisor pulls back to the exceptional divisor. The induced map $\mathrm{vBl}_w(Y) \to \mathrm{vBl}_z(X) \times_X Y$ is always a closed immersion, and it is an isomorphism if φ is an isomorphism. If X and Y are schemes, the map f' restricts to a map $\mathrm{Bl}_w(Y) \to \mathrm{Bl}_z(X)$ by the functoriality of the Cartier locus. Moreover, the closed immersion $\mathrm{Bl}_w(Y) \hookrightarrow \mathrm{Bl}_z(X) \times_X Y$ is an isomorphism if φ is an isomorphism and f is flat (as then f' is also flat).

Specializing to the case of closed subschemes, we see that both vBl and Bl restrict to functors on the following category:

- An object is a pair (X, Z) where X is a scheme and $Z \subset X$ is a closed subscheme.
- A map $(Y, W) \to (X, Z)$ is a map $f\colon Y \to X$ such that $f^{-1}(Z) = W$.

Such a map induces a commutative diagram

$$\begin{array}{ccc} \mathrm{Bl}_W(Y) & \xrightarrow{f'} & \mathrm{Bl}_Z(X) \\ \downarrow & & \downarrow \\ \mathrm{vBl}_W(Y) & \longrightarrow & \mathrm{vBl}_Z(X) \\ \downarrow & & \downarrow \\ Y & \xrightarrow[f]{} & X. \end{array}$$

The bottom square is cartesian if $\mathrm{L}_1 f^*(\mathcal{O}_X/I) = 0$, for example if f is flat (and more generally if Y and Z are Tor-independent over X, see Definition 10.40). If f is flat, then f' is flat and hence the top square is also cartesian.

Remark 10.19 (Blowups only depend on the vanishing locus). For any scheme X and any $z\colon M \to \mathcal{O}_X$, there is a closed immersion $\mathrm{vBl}_{\mathrm{V}(z)}(X) \hookrightarrow \mathrm{vBl}_z(X)$. Using the description of $\mathrm{Bl}_z(X)$ as the closure of $\mathrm{D}(z)$ in $\mathrm{vBl}_z(X)$, we immediately deduce that the closed immersion $\mathrm{Bl}_{\mathrm{V}(z)}(X) \hookrightarrow \mathrm{Bl}_z(X)$ is in fact an isomorphism (note that the analogous statement is not true for virtual blowups, as the zero-map example shows). Consequently, if the exceptional divisor on $\mathrm{vBl}_z(X)$ happens to be non-virtual, then $\mathrm{Bl}_{\mathrm{V}(z)}(X) \xrightarrow{\sim} \mathrm{vBl}_z(X)$.

Example 10.20 (Blowing up a point in affine or projective n-space). Let $n \geq 0$.

(i) Let $z = (x_1, \ldots, x_n)\colon \mathcal{O}_{\mathbf{A}^n}^n \to \mathcal{O}_{\mathbf{A}^n}$ be the tautological map over $\mathbf{A}^n = \mathrm{Spec}(\mathbf{Z}[x_1, \ldots, x_n])$. There is an isomorphism

$$\mathrm{vBl}_z(\mathbf{A}^n) \simeq \mathbf{A}_{\mathbf{P}^{n-1}}(\mathcal{O}(1)),$$

since a linear map $R^n \to R$ together with a factorization through a quotient line $R^n \twoheadrightarrow L$ is the same thing as quotient line $R^n \twoheadrightarrow L$ together with a linear map $L \to R$. For any line bundle L, the canonical map $L \otimes \mathrm{Sym}(L) \to \mathrm{Sym}(L)$ is a monomorphism, since locally it is the multiplication by x on $R[x]$. Hence, the exceptional divisor is non-virtual, so that $\mathrm{Bl}_0(\mathbf{A}^n) \simeq \mathbf{A}_{\mathbf{P}^{n-1}}(\mathcal{O}(1))$ by the previous remark.

(ii) Let $z = (x_1, \ldots, x_n)\colon \mathcal{O}_{\mathbf{P}^n}(-1)^n \to \mathcal{O}_{\mathbf{P}^n}$ over $\mathbf{P}^n = \mathrm{Proj}(\mathbf{Z}[x_0, \ldots, x_n])$. There is an isomorphism

$$\mathrm{vBl}_z(\mathbf{P}^n) \simeq \mathbf{P}_{\mathbf{P}^{n-1}}(\mathcal{O}(1) \oplus \mathcal{O}),$$

given on R-points by

$$\begin{pmatrix} R^{n+1} & \twoheadrightarrow & L' \\ \uparrow & & \uparrow \\ R^n & \twoheadrightarrow & L \end{pmatrix} \iff (R^n \twoheadrightarrow L \hookrightarrow L \oplus R \twoheadrightarrow L').$$

We deduce as in (i) that $\mathrm{Bl}_0(\mathbf{P}^n) \simeq \mathbf{P}_{\mathbf{P}^{n-1}}(\mathcal{O}(1) \oplus \mathcal{O})$.

(iii) Let X be a scheme and let $M, N \in \mathrm{Mod}_X$. The arguments in (i) and (ii) imply more generally the following:

$$\mathrm{Bl}_{\mathbf{A}(N)}(\mathbf{A}(M \oplus N)) = \mathbf{A}_{\mathbf{P}(M)}(\mathcal{O}(1) \oplus N)$$

and

$$\mathrm{Bl}_{\mathbf{P}(N)}(\mathbf{P}(M \oplus N)) = \mathbf{P}_{\mathbf{P}(M)}(\mathcal{O}(1) \oplus N).$$

Definition 10.21 (Strict transform, total transform)**.** Let X be a scheme and $Z \subset X$ a closed subscheme. The *strict transform* at Z is the functor

$$\mathrm{Sch}_{/X} \to \mathrm{Sch}_{/\mathrm{Bl}_Z(X)}, \quad Y \mapsto \mathrm{Bl}_{Z \times_X Y}(Y).$$

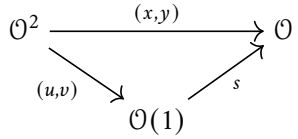
The *total transform* at Z is simply the pullback functor

$$\mathrm{Sch}_{/X} \to \mathrm{Sch}_{/\mathrm{Bl}_Z(X)}, \quad Y \mapsto \mathrm{Bl}_Z(X) \times_X Y.$$

By functoriality, there is a natural closed immersion of the strict transform into the total transform, which is an isomorphism when Y is flat over X. In general, by the geometric description of blowups, it identifies the strict transform of Y at Z with the closure of $(X - Z) \times_X Y$ in $\mathrm{Bl}_Z(X) \times_X Y$.

Remark 10.22 (Blowing up a point in a quasi-projective scheme)*.* Let k be a ring, let X be either $\mathbf{A}^n_k$ or $\mathbf{P}^n_k$, and let $b\colon \mathrm{Bl}_0(X) \to X$ be the blowup at the origin, which was computed in the previous example. Let $Y \subset X$ be a subscheme containing 0. The blowup $\mathrm{Bl}_0(Y)$ is then the strict transform of Y at 0, which is the closure of $Y - 0$ in $b^{-1}(Y)$. If Y is closed in X, this is simply the closure of $Y - 0$ in $\mathrm{Bl}_0(X)$.

Example 10.23 (Blowing up the cusp)*.* We use the previous remark to compute the blowup of the affine cuspidal curve $C = \mathrm{Spec}(k[x,y]/(y^2 - x^3)) \subset \mathbf{A}^2_k$ at the origin. The blown-up plane $\mathrm{Bl}_0(\mathbf{A}^2_k) = \mathbf{A}_{\mathbf{P}^1_k}(\mathcal{O}(1))$ is covered by the two affine open subschemes $\mathrm{D}(u)$ and $\mathrm{D}(v)$, where $\mathbf{P}^1_k = \mathrm{Proj}(k[u,v])$. Let $(\mathcal{O}(1), s)$ be the exceptional divisor:



We thus have factorizations $x = su$ and $y = sv$ in $\mathcal{O}(\mathrm{Bl}_0(\mathbf{A}^2_k))$, and $\mathrm{D}(u)$ (resp. $\mathrm{D}(v)$) is an affine plane with coordinates x and $\tilde{y} = u^{-1}v$ (resp. $\tilde{x} = v^{-1}u$ and y). The preimage of $C - 0 = \mathrm{D}(x)$ in the blowup is contained in $\mathrm{D}(u)$ and given by $\mathrm{Spec}(k[x^{\pm 1}, \tilde{y}]/(x^2\tilde{y}^2 - x^3)) = \mathrm{Spec}(k[x^{\pm 1}, \tilde{y}]/(\tilde{y}^2 - x))$; its closure in $\mathrm{D}(u)$ is therefore the parabola $P = \mathrm{Spec}(k[x, \tilde{y}]/(\tilde{y}^2 - x))$. The latter is in fact already closed in $\mathrm{Bl}_0(\mathbf{A}^2_k)$, since $P \cap \mathrm{D}(v) = \mathrm{Spec}(k[\tilde{x}, y]/(1 - \tilde{x}^3 y))$ is closed in $\mathrm{D}(v)$. Hence, P is the strict transform of C at 0, and in particular $\mathrm{Bl}_0(C) \simeq P \simeq \mathbf{A}^1_k$.

Proposition 10.24 (Universal property of blowups)**.** *Let X be a scheme and $Z \subset X$ a closed subscheme.*

(i) The exceptional divisor on $\mathrm{Bl}_Z(X)$, viewed as a closed subscheme, is the preimage of Z.

(ii) If $f\colon Y \to X$ is a map such that $f^{-1}(Z) \subset Y$ is an effective Cartier divisor, then f factors uniquely through the blowup $\mathrm{Bl}_Z(X) \to X$.

Proposition 10.25 (Smoothness of blowups)**.** *Let S be a scheme, let X be a smooth S-scheme, and let $Z \subset X$ be a closed subscheme that is also smooth over S. Then the blowup $\mathrm{Bl}_Z(X)$ is smooth over S.*

By definition, the blowup $\mathrm{Bl}_Z(X)$ is a closed subscheme of $\mathbf{P}(I)$, where I is the quasi-coherent ideal of Z. Our next goal is to compute $\mathrm{Bl}_Z(X)$ as Proj of an explicit quasi-coherent $\mathbf{N}$-graded algebra over X. Part (i) of the following definition is a globalization of the deformation ring.

Definition 10.26 (Deformation algebra, Rees algebra, associated graded algebra)**.** Let X be a scheme and let $I \in \mathrm{Idl}_X$ be a quasi-coherent ideal over X.

(i) The *deformation algebra* or *extended Rees algebra* of I is the quasi-coherent $\mathbf{Z}$-graded algebra

$$\mathrm{Def}_I(\mathcal{O}_X) = \bigoplus_{n\in\mathbf{Z}} I^n \in \mathrm{CAlg}_X^{\mathbf{Z}},$$

where $I^n = \mathcal{O}_X$ for all $n < 0$. In particular, $\mathrm{Def}_I(\mathcal{O}_X)_{\leq 0} = \mathrm{Sym}(\mathcal{O}_X)$.

(ii) The *Rees algebra* of I is the quasi-coherent $\mathbf{N}$-graded algebra

$$\mathrm{Rees}_I(\mathcal{O}_X) = \mathrm{Def}_I(\mathcal{O}_X)_{\geq 0} = \bigoplus_{n\in\mathbf{N}} I^n \in \mathrm{CAlg}_X^{\mathbf{N}}.$$

(iii) Let $Z = \mathrm{V}(I)$ and let $i\colon Z \hookrightarrow X$ be the inclusion. The *associated graded algebra* of I is the quasi-coherent $\mathbf{N}$-graded algebra

$$\mathrm{gr}_I(\mathcal{O}_X) = i^*(\mathrm{Rees}_I(\mathcal{O}_X)) \in \mathrm{CAlg}_Z^{\mathbf{N}}.$$

Note that under the equivalence of categories $i^*\colon \mathrm{CAlg}_Z \xrightarrow{\sim} \mathrm{CAlg}_{\mathcal{O}_X/I}$, $\mathrm{gr}_I(\mathcal{O}_X)$ is the associated graded algebra of the I-adic filtration of $\mathcal{O}_X$, i.e., $\bigoplus_{n\in\mathbf{N}} I^n/I^{n+1}$.

Remark 10.27 (Gradings via monoid actions)*.* Let X be an algebraic functor. Recall that there is an equivalence of categories

$$\mathrm{Spec}\colon \mathrm{CAlg}_X^{\mathrm{op}} \xrightarrow{\sim} \mathrm{Aff}_X \subset \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})_{/X}$$

between quasi-coherent algebras over X and algebraic functors affine over X. Under this equivalence, an $\mathbf{N}$-grading on a quasi-coherent algebra A corresponds to an action of $\mathbf{A}^1$ on $\mathrm{Spec}(A)$, where $\mathbf{A}^1$ is viewed as a monoid under multiplication (not to be confused with $\mathbf{G}_a$, which is $\mathbf{A}^1$ as a monoid under addition). Similarly, a $\mathbf{Z}$-grading on A corresponds to an action of the multiplicative group $\mathbf{G}_m$ on $\mathrm{Spec}(A)$. Indeed, by definition of CAlg_X, this follows from the corresponding statements for $X = \mathrm{Spec}(R)$.[28] In other words, the functor Spec induces equivalences of categories

$$(\mathrm{CAlg}_X^{\mathbf{Z}})^{\mathrm{op}} \simeq \mathbf{G}_m\,\mathrm{Aff}_X \quad \text{and} \quad (\mathrm{CAlg}_X^{\mathbf{N}})^{\mathrm{op}} \simeq \mathbf{A}^1\,\mathrm{Aff}_X,$$

where $\mathbf{G}_m\,\mathrm{Aff}_X$ (resp. $\mathbf{A}^1\,\mathrm{Aff}_X$) is the category of algebraic functors affine over X with an action of $\mathbf{G}_m$ (resp. of $\mathbf{A}^1$).

Definition 10.28 (Deformation space)**.** Let X be a scheme and let $Z \subset X$ be a closed subscheme with quasi-coherent ideal I. The *deformation space* of Z in X is the affine X-scheme with $\mathbf{G}_m$-action

$$\mathrm{Def}_Z(X) = \mathrm{Spec}(\mathrm{Def}_I(\mathcal{O}_X)).$$

The inclusion $\mathrm{Sym}(\mathcal{O}_X) = \mathrm{Def}_I(\mathcal{O}_X)_{\leq 0} \subset \mathrm{Def}_I(\mathcal{O}_X)$ induces a $\mathbf{G}_m$-equivariant map $\mathrm{Def}_Z(X) \to \mathbf{A}^1_X$.

Definition 10.29 (Normal cone)**.** Let X be a scheme and let $Z \subset X$ be a closed subscheme with quasi-coherent ideal I. The *normal cone* of Z in X is the affine Z-scheme with $\mathbf{A}^1$-action

$$C_Z(X) = \mathrm{Spec}(\mathrm{gr}_I(\mathcal{O}_X)).$$

Since $\mathrm{gr}_I(\mathcal{O}_X)$ is generated in degree 1 and since $\mathrm{gr}_I(\mathcal{O}_X)_1 = \mathcal{N}_{Z/X}$, $C_Z(X)$ is an $\mathbf{A}^1$-equivariant closed subscheme of $\mathbf{A}(\mathcal{N}_{Z/X})$.

[28] More generally, for any commutative monoid Γ, the affine scheme $\mathrm{Spec}(\mathbf{Z}[\Gamma])$ has a structure of commutative monoid induced by the diagonal $\Gamma \to \Gamma \times \Gamma$, and a Γ-grading on a quasi-coherent algebra $A \in \mathrm{CAlg}_X$ is equivalent to an action of $\mathrm{Spec}(\mathbf{Z}[\Gamma])$ on $\mathrm{Spec}(A)$.

Example 10.30 (Normal cone of the origin in the nodal cubic). Let $R = k[x,y]/(y^2 - x^3 - x^2)$ and let $I = (x,y) \subset R$. Then $I^n = (x^n, x^{n-1}y)$ and I^n/I^{n+1} is free of rank 2 over k with basis $(x^n, x^{n-1}y)$. It follows that the map $k[x,y]/(y^2 - x^2) \to \mathrm{gr}_I(R)$ is an isomorphism of $\mathbf{N}$-graded k-algebras. Thus,

$$C_0(\mathrm{Spec}(R)) = \mathrm{V}(y^2 - x^2)$$

as a subscheme of $\mathbf{A}(\mathcal{N}_{0/\mathrm{Spec}(R)}) = \mathbf{A}^2_k$.

Definition 10.31 (Cone, projectivization)**.** Let X be an algebraic functor. A *cone* over X (or "with vertex X") is an object of $\mathbf{A}^1\,\mathrm{Aff}_X$ whose associated quasi-coherent $\mathbf{N}$-graded algebra over X is generated in degree 1. The *projectivization* of a cone $C = \mathrm{Spec}(A)$ over X is the algebraic functor $\mathbf{P}(C) = \mathrm{Proj}(A)$.

Remark 10.32 (Cones vs. affine spaces). Let X be an algebraic functor.

(i) Given a quasi-coherent module $M \in \mathrm{Mod}_X$, the symmetric algebra $\mathrm{Sym}(M)$ has a canonical $\mathbf{N}$-grading and is generated in degree 1. Hence, the affine space $\mathbf{A}(M)$ has a canonical structure of cone over X. Its projectivization is the algebraic functor $\mathbf{P}(M)$ over X.

(ii) Let C be an algebraic functor with $\mathbf{A}^1$-action over X. Then C is a cone over X if and only if there exists an $\mathbf{A}^1$-equivariant closed immersion $C \hookrightarrow \mathbf{A}(M)$ for some $M \in \mathrm{Mod}_X$. Indeed, the latter is equivalent to an isomorphism $C \simeq \mathrm{Spec}(A)$, where A is a quotient of the quasi-coherent $\mathbf{N}$-graded algebra $\mathrm{Sym}(M)$. Such an A is generated in degree 1, and conversely any A generated in degree 1 is a quotient of $\mathrm{Sym}(A_1)$.

Proposition 10.33 (Deformation to the normal cone)**.** *Let X be a scheme and $Z \subset X$ a closed subscheme. Then there are $\mathbf{G}_m$-equivariant cartesian squares*

$$\begin{array}{ccccc} C_Z(X) & \longrightarrow & \mathrm{Def}_Z(X) & \longleftarrow & X \times \mathbf{G}_m \\ \downarrow & & & & \downarrow \\ 0 & \longrightarrow & \mathbf{A}^1 & \longleftarrow & \mathbf{G}_m. \end{array}$$

Lemma 10.34. *Let $R \to S$ be a ring map and $I \subset R$ an ideal. Then the S-linear map $I \otimes_R S \to S$ factors through all quotient lines of $I \otimes_R S$.*

Theorem 10.35 (Blowups via Rees algebras)**.** *Let X be a scheme and $Z \subset X$ a closed subscheme with quasi-coherent ideal I. Then the closed subschemes $\mathrm{Bl}_Z(X) \subset \mathrm{vBl}_Z(X) \subset \mathbf{P}(I)$ are given by*

$$\mathrm{vBl}_Z(X) = \mathbf{P}(I) \quad \textit{and} \quad \mathrm{Bl}_Z(X) = \mathrm{Proj}(\mathrm{Rees}_I(\mathcal{O}_X)).$$

In particular, an R-point $I(x) \twoheadrightarrow L$ of the virtual blowup belongs to the actual blowup if and only if, for every $n \geq 2$, the R-linear map $I(x)^{\otimes n} \twoheadrightarrow L^{\otimes n}$ factors through $I^n(x)$.

Corollary 10.36. *Let X be a scheme and $Z \subset X$ a closed subscheme. Then the exceptional divisor in $\mathrm{Bl}_Z(X)$ is the projectivization of the normal cone of Z in X. Hence, there are cartesian squares*

$$\begin{array}{ccccc} \mathbf{P}(C_Z(X)) & \longrightarrow & \mathrm{Bl}_Z(X) & \longleftarrow & X - Z \\ \downarrow & & \downarrow & & \downarrow \\ Z & \longrightarrow & X & \longleftarrow & X - Z. \end{array}$$

Remark 10.37 (Affine blowup algebras). Let R be a ring, $I \subset R$ an ideal, and $a \in I$. The *affine blowup algebra* $R[\frac{I}{a}]$ is the R-algebra $\mathrm{Rees}_I(R)_{(a)}$, where a denotes the copy of a in degree 1. By the previous theorem, the blowup $\mathrm{Bl}_{\mathrm{V}(I)}(\mathrm{Spec}(R))$ has an open covering by the affine schemes $\mathrm{Spec}(R[\frac{I}{a}])$, where a runs over any generating set of I.

Proposition 10.38 (Blowups via deformation spaces)**.** *Let X be a scheme and let $Z \subset X$ be a closed subscheme. Then the action of $\mathbf{G}_m$ on $\mathrm{Def}_Z(X) - \mathrm{Def}_Z(Z)$ is free and there is a canonical isomorphism*

$$\mathrm{Bl}_Z(X) \simeq (\mathrm{Def}_Z(X) - \mathrm{Def}_Z(Z))/\mathbf{G}_m$$

in $\mathrm{Sch}_{/X}$, where the quotient is computed in $\mathrm{Sh}_{\mathrm{Zar}}(\mathrm{CAlg}^{\mathrm{op}})$.

Proposition 10.39 (Deformation spaces via blowups)**.** *Let X be a scheme and let $Z \subset X$ be a closed subscheme. There is a canonical isomorphism*

$$\mathrm{Def}_Z(X) \simeq \mathrm{Bl}_{Z\times 0}(X \times \mathbf{A}^1) - \mathrm{Bl}_{Z\times 0}(X \times 0)$$

in $\mathrm{Sch}_{/X\times\mathbf{A}^1}$.

10.3. **Regular closed immersions.** Regular closed immersions are a generalization of effective divisors to higher codimension.

Definition 10.40 (Tor-independence)**.**

(i) Let R be a ring and let M and N be R-modules. We say that M and N are *Tor-independent* over R if $\mathrm{Tor}_i^R(M, N) = 0$ for all $i \geq 1$.

(ii) Let X be a scheme, let Y and Z be schemes over X, and let $M \in \mathrm{Mod}_Y$ and $N \in \mathrm{Mod}_Z$ be quasi-coherent modules. We say that M and N are *Tor-independent* over X if, for every affine open subschemes $V \subset Y$ and $W \subset Z$ mapping to an affine open subscheme $U \subset X$, $M(V)$ and $N(W)$ are Tor-independent over $\mathcal{O}(U)$. We say that Y and Z themselves are *Tor-independent* over X if $\mathcal{O}_Y$ and $\mathcal{O}_Z$ are Tor-independent over X.

(iii) A square of rings (resp. of schemes)

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ C & \longrightarrow & D \end{array} \qquad \begin{array}{ccc} W & \longrightarrow & Y \\ \downarrow & & \downarrow \\ Z & \longrightarrow & X \end{array}$$

is called *Tor-independent* if it is cocartesian (resp. cartesian) and B and C are Tor-independent over A (resp. Y and Z are Tor-independent over X).

Proposition 10.41 (Closure properties of Tor-independence)**.** *Let R be a ring and M an R-module.*

(i) Let N be an R-module and $N' \subset N$ a submodule such that M and N/N' are Tor-independent. Then M and N are Tor-independent if and only if M and N' are Tor-independent.

(ii) Let $R \to S$ be a ring map and N an S-module. If M and S are Tor-independent over R, then M and N are Tor-independent over R if and only if $M \otimes_R S$ and N are Tor-independent over S.

Corollary 10.42 (Pasting law for Tor-independent squares)**.** *Consider a commutative diagram of rings*

$$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow & & \downarrow & & \downarrow \\ A' & \longrightarrow & B' & \longrightarrow & C'. \end{array}$$

If the left square is Tor-independent, then the right square is Tor-independent if and only if the rectangle is Tor-independent.

The above proposition and its corollary have obvious generalizations to schemes.

Definition 10.43 (Regular map)**.** Let X be a scheme and $E \in \mathrm{Mod}_X$. A map $s\colon E \to \mathcal{O}_X$ is called *regular* if the induced square

$$\begin{array}{ccc} \mathrm{V}(s) & \longrightarrow & X \\ \downarrow & & \downarrow{\scriptstyle s} \\ X & \xrightarrow[0]{} & \mathbf{A}(E) \end{array}$$

is Tor-independent.

Remark 10.44 (Relation to effective divisors)*.* An effective virtual divisor $s\colon L \to \mathcal{O}_X$ is a regular map if and only if it is an effective Cartier divisor. To see this, we can assume $X = \mathrm{Spec}(R)$ and we compute $\mathrm{Tor}^*_{\mathrm{Sym}(L)}((R, 0), -)$ using the projective resolution

$$0 \to L \otimes_R \mathrm{Sym}_R(L) \to \mathrm{Sym}_R(L) \to R \to 0,$$

which yields

$$\operatorname{Tor}_i^{\operatorname{Sym}(L)}((R,0),(R,s)) = \begin{cases} \operatorname{coker}(s), & i = 0, \\ \ker(s), & i = 1, \\ 0, & i \geq 2. \end{cases}$$

Remark 10.45 (The Koszul complex). Let X be a scheme and let $E \in \operatorname{Mod}_X$ be flat. The $\operatorname{Sym}(E)$-module $\mathcal{O}_X$ (where E acts trivially) has a canonical flat resolution by the exterior powers of E, generalizing the previous remark:

$$\cdots \to \Lambda^n(E) \otimes \operatorname{Sym}(E) \to \Lambda^{n-1}(E) \otimes \operatorname{Sym}(E) \to \cdots \to E \otimes \operatorname{Sym}(E) \to \operatorname{Sym}(E) \to \mathcal{O}_X \to 0.$$

Given $s\colon E \to \mathcal{O}_X$, the extension of scalars of this complex along $s\colon \operatorname{Sym}(E) \to \mathcal{O}_X$ is called the *Koszul complex* of s. Thus, s is regular if and only if its Koszul complex is exact.

Example 10.46 (Regular sequence). Let $f_1, \ldots, f_n \in R$ be such that, for each i, multiplication by f_i on $R/(f_1, \ldots, f_{i-1})$ is injective. It then follows from the pasting law for Tor-independent squares that $(f_1, \ldots, f_n)\colon R^n \to R$ is regular.

Example 10.47.
(i) Let $\mathbf{A}^n = \operatorname{Spec}(k[x_1, \ldots, x_n])$ and let $x_0 = 1$. For any subset $I \subset \{0, \ldots, n\}$, the map $(x_i)_{i \in I}\colon \mathcal{O}_{\mathbf{A}^n}^I \to \mathcal{O}_{\mathbf{A}^n}$ is regular. This follows from the previous example.
(ii) Let $\mathbf{P}^n = \operatorname{Proj}(k[x_0, \ldots, x_n])$. For any subset $I \subset \{0, \ldots, n\}$, the map $(x_i)_{i \in I}\colon \mathcal{O}_{\mathbf{P}^n}(-1)^I \to \mathcal{O}_{\mathbf{P}^n}$ is regular. This can be checked on the standard open covering of $\mathbf{P}^n$, where it follows from (i).

Definition 10.48 (Regular closed immersion, regular quasi-coherent ideal)**.** Let X be a scheme.
(i) A closed immersion $Z \hookrightarrow X$ is *regular* if X admits an open covering by schemes U such that $Z \cap U = \mathrm{V}(s)$ for some regular maps $s\colon E \to \mathcal{O}_U$ in Vect_U.
(ii) A quasi-coherent ideal $I \in \operatorname{Idl}_X$ is *regular* if $\mathrm{V}(I) \hookrightarrow X$ is a regular closed immersion.

Remark 10.49. Since vector bundles are locally free, a closed immersion $Z \hookrightarrow X$ is regular if and only if X is covered by open subschemes U for which there exists a Tor-independent square of the form

$$\begin{array}{ccc} Z \cap U & \longrightarrow & U \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathbf{A}^n. \end{array}$$

Remark 10.50 (Noetherian case). One can show that the converse of the regular-sequence example holds when R is a noetherian local ring and $f_1, \ldots, f_n \in \mathfrak{m}$: if $(f_1, \ldots, f_n)\colon R^n \to R$ is regular, then multiplication by f_i on $R/(f_1, \ldots, f_{i-1})$ is injective for each i. In other words, the notion of regular map recovers the notion of regular sequence used in the study of noetherian local rings. This further implies that, if X is locally noetherian and $E \in \operatorname{Vect}_X$, a map $s\colon E \to \mathcal{O}_X$ is regular if and only if X admits an open covering by affine schemes U such that, under some isomorphism $E|_U \simeq \mathcal{O}_U^n$, $s|_U$ is a regular sequence. Geometrically, this means that a closed immersion $Z \hookrightarrow X$ is regular if and only if X admits an open covering by schemes U such that $Z \cap U \hookrightarrow U$ is a composition of effective Cartier divisors. If X is not locally noetherian, however, the latter condition is sufficient but not necessary.

Warning 10.51 (Relation to regular schemes). By the previous remark, a noetherian local ring R is regular if and only if its maximal ideal $\mathfrak{m} \subset R$ is regular. Suppose that X is a regular scheme and that $i\colon Z \hookrightarrow X$ is a closed immersion. If Z is regular, it follows from the local dimension theory of regular rings that i is regular. However, the converse is far from true!

Proposition 10.52 (Closure properties of regular closed immersions)**.**
(i) Compositions of regular closed immersions are regular.
(ii) Consider a Tor-independent square of schemes

$$\begin{array}{ccc} Z' & \longrightarrow & X' \\ {\scriptstyle i'}\downarrow & & \downarrow \\ Z & \xrightarrow[i]{} & X. \end{array}$$

If i is a regular closed immersion, so is i'.

Lemma 10.53. *Let $R = \mathbf{Z}[t_\alpha \mid \alpha \in A]$, let $I = (t_\alpha \mid \alpha \in A) \subset R$, and let M be an R-module such that M and R/I are Tor-independent over R. Then, for all $n \geq 0$, M and R/I^n are Tor-independent over R. In particular, the map $I^n \otimes_R M \to I^n M$ is an isomorphism.*

Proposition 10.54 (Normal cone of a regular closed immersion)**.** *Let X be a scheme and let $Z \hookrightarrow X$ be a regular closed immersion.*

(i) The conormal module $\mathcal{N}_{Z/X}$ is a vector bundle over Z.
(ii) The normal cone $C_Z(X)$ is $\mathbf{A}(\mathcal{N}_{Z/X})$.

Definition 10.55 (Codimension of a regular immersion)**.** Let $Z \hookrightarrow X$ be a regular closed immersion. The *codimension* of Z in X is the rank of the vector bundle $\mathcal{N}_{Z/X}$, which is a locally constant function $|Z| \to \mathbf{N}$. Note that effective divisors are exactly regular closed immersions of codimension 1.

The following result generalizes the computation of blowing up affine space:

Proposition 10.56 (Blowing up a regular closed immersion)**.** *Let X be a scheme, let $E \in \mathrm{Mod}_X$ be flat, and let $s \colon E \to \mathcal{O}_X$ be a regular map. Then $\mathrm{vBl}_s(X) = \mathrm{Bl}_{V(s)}(X)$. Hence, if $Z \hookrightarrow X$ is a regular closed immersion with quasi-coherent ideal I, then $\mathrm{Bl}_Z(X) = \mathbf{P}(I)$.*

Proposition 10.57 (Regular closed immersions vs. smoothness)**.** *Let S be a scheme and let $i \colon Z \hookrightarrow X$ be a closed immersion of S-schemes, where X is locally of finite presentation over S and Z is smooth over S. Then i is regular if and only if some open neighborhood of Z in X is smooth over S.*

Definition 10.58 (Local complete intersection, syntomic morphism)**.** Let $f \colon Y \to X$ be a morphism of schemes.

(i) f is a *local complete intersection* if, for every point $y \in |Y|$, there are affine open subschemes $V \subset Y$ and $U \subset X$ with $y \in |V|$ and $f(V) \subset U$ such that $f|_V$ factors as

$$V \xhookrightarrow{i} \mathbf{A}^n_U \to U,$$

where i is a regular closed immersion.
(ii) f is *syntomic* if it is flat and a local complete intersection.

Proposition 10.59. *(i) "Local complete intersection" is a local property of morphisms of schemes that is stable under composition and Tor-independent base change.*
(ii) "Syntomic" is a local property of morphisms of schemes that is stable under composition and base change.

Proposition 10.60. *For closed immersions, "local complete intersection" is equivalent to "regular".*

By the local structure of smooth maps (Definition 9.3) and of étale maps (Theorem 9.107), any smooth morphism is locally of the form

$$R \to R[x_1, \ldots, x_n]_h[x]/(f)_g = R[x_1, \ldots, x_n, x, y]/(f, ygh - 1),$$

where f is monic in x and does not contain y. By the regular-sequence example, $(f, ygh - 1)$ is a regular ideal. Hence, smooth morphisms are local complete intersections. Since they are also flat (Corollary 9.108), we deduce:

Proposition 10.61. *Smooth morphisms are syntomic.*

Remark 10.62 (Syntomicity via dimension of fibers)*.* There is an alternative characterization of syntomicity that does not refer to flatness. A ring map $R \to S$ is called a *global complete intersection* if there is an isomorphism $S \simeq R[x_1, \ldots, x_n]/(f_1, \ldots, f_m)$ and for every prime ideal $\mathfrak{p} \subset R$, $\dim(S \otimes_R \kappa(\mathfrak{p})) \in \{-\infty, n - m\}$. A map of schemes $f \colon Y \to X$ is then syntomic if and only if, for every point $y \in |Y|$, there are affine open subschemes $V \subset Y$ and $U \subset X$ with $y \in |V|$ and $f(V) \subset U$ such that $\mathcal{O}(U) \to \mathcal{O}(V)$ is a global complete intersection. Note that smooth morphisms satisfy this condition by the Jacobian criterion and Theorem 9.63. The fact that global complete intersections are flat is nontrivial, but it gives another proof that smooth morphisms are flat, which does not use the local structure of étale maps.

Proposition 10.63. *Consider a commutative triangle of schemes*

$$\begin{array}{ccc} Z & \xrightarrow{g} & Y \\ & \searrow^{h} & \downarrow f \\ & & X. \end{array}$$

If f is smooth and h is a local complete intersection, then g is a local complete intersection. In particular, any map of smooth X-schemes is a local complete intersection.

Remark 10.64. If X and Y are regular schemes and $f\colon Y \to X$ is locally of finite type, then f is a local complete intersection.

Remark 10.65 (Local complete intersections via the cotangent complex). Local complete intersection morphisms have a simple characterization using the cotangent complex: a morphism $f\colon Y \to X$ is a local complete intersection if and only if it is locally of finite presentation and, for any ring R and any $y \in Y(R)$, $\mathrm{L}_f(y) \in \mathrm{D}(\mathrm{Mod}_R)$ is perfect and concentrated in degrees $[0, 1]$. Using this characterization, the previous proposition is a direct consequence of the fundamental fiber sequence of cotangent complexes $g^*(\mathrm{L}_f) \to \mathrm{L}_h \to \mathrm{L}_g$.

10.4. **Cartier divisors.** Let X be a scheme. The sheaf of rational functions $\mathcal{K}_X$ on $|X|$ was defined earlier: it is a sheaf of rings containing $\mathcal{O}_X$ as a subsheaf, and it is the maximal localization of $\mathcal{O}_X$ with this property. In particular $\mathcal{K}_X$ is an $\mathcal{O}_X$-algebra in $\mathrm{Sh}(|X|)$. In what follows we implicitly use the equivalence between quasi-coherent modules over X and suitable $\mathcal{O}_X$-modules in $\mathrm{Sh}(|X|)$.

Notation 10.66. If X is a scheme, we write $K(X) = \mathcal{K}_X(X)$ for the ring of rational functions on X. Note that for any open subscheme $U \subset X$, we have $K(U) = \mathcal{K}_X(U)$.

Warning 10.67. Let X be a scheme.

(i) By definition, $\mathcal{K}_X$ is the sheafification of a presheaf whose value on any affine open $U \subset X$ is the total ring of fractions $\mathrm{Frac}(\mathcal{O}(U))$. Hence, if X is affine, $K(X)$ contains $\mathrm{Frac}(\mathcal{O}(X))$ as a subring, but it *can* be strictly bigger, in which case it is not a localization of $\mathcal{O}(X)$. This does not happen if X is noetherian.

(ii) If $V \subset U$ are affine open subschemes of X, the canonical map $K(U) \otimes_{\mathcal{O}(U)} \mathcal{O}(V) \to K(V)$ is injective but not necessarily surjective. In other words, $\mathcal{K}_X$ need not be a quasi-coherent sheaf. This injectivity implies however that the kernel and the image of any map $M \to \mathcal{K}_X$ in $\mathrm{Mod}_{\mathcal{O}_X}(\mathrm{Sh}(|X|))$ with M quasi-coherent are quasi-coherent.

Definition 10.68 (Virtual divisor, Cartier divisor)**.** Let X be a scheme. A *virtual divisor* on X is a pair (L, s) where $L \in \mathrm{Line}_X$ is a line bundle over X and $s\colon L \to \mathcal{K}_X$ is an $\mathcal{O}_X$-linear map. We denote by $\mathrm{vDiv}(X) \subset (\mathrm{Mod}_{\mathcal{O}_X}(\mathrm{Sh}(|X|))_{/\mathcal{K}_X})^{\mathrm{op}}$ the full subcategory of virtual divisors on X.

A *Cartier divisor* or simply a *divisor* on X is a virtual divisor (L, s) where s is a monomorphism. We denote by $\mathrm{Div}(X) \subset \mathrm{vDiv}(X)$ the full subcategory of Cartier divisors, which is a poset.

Remark 10.69 (Symmetric monoidal structure). Since $\mathcal{K}_X$ is a localization of the ring $\mathcal{O}_X$ in $\mathrm{Sh}(|X|)$, it is an $\mathcal{O}_X$-algebra (in fact an idempotent one: $\mathcal{K}_X \otimes_{\mathcal{O}_X} \mathcal{K}_X \xrightarrow{\sim} \mathcal{K}_X$). The tensor product and the multiplication on $\mathcal{K}_X$ define a symmetric monoidal structure on $\mathrm{Mod}_{\mathcal{O}_X}(\mathrm{Sh}(|X|))_{/\mathcal{K}_X}$, which restricts to a symmetric monoidal structure on $\mathrm{vDiv}(X)$. This symmetric monoidal structure extends the one on $\mathrm{vDiv}^+(X)$, and as above it restricts to the full subcategory $\mathrm{Div}(X)$. We again write $(L, s) + (L', s')$ for the tensor product $(L \otimes L', s \otimes s')$.

The following proposition shows that $\mathrm{Div}(X)$ is in fact a *partially ordered abelian group*:

Proposition 10.70 (Characterization of Cartier divisors)**.** *Let X be a scheme and let (L, s) be a virtual divisor on X. The following conditions are equivalent:*

(i) (L, s) is a Cartier divisor, i.e., s is a monomorphism.

(ii) (L, s) is an invertible object of the symmetric monoidal category $\mathrm{vDiv}(X)$.

(iii) s induces an isomorphism $\mathcal{K}_X \otimes_{\mathcal{O}_X} L \xrightarrow{\sim} \mathcal{K}_X$.

When this holds, the inverse $-(L,s)$ of (L,s) is the divisor

$$L^{-1} \hookrightarrow \mathcal{K}_X \otimes_{\mathcal{O}_X} L^{-1} \xrightarrow{s^{-1}} \mathcal{K}_X.$$

In summary, for any scheme X, we have a square of fully faithful symmetric monoidal functors

$$\begin{array}{ccc} \mathrm{Div}^+(X) & \longrightarrow & \mathrm{Div}(X) \\ \downarrow & & \downarrow \\ \mathrm{vDiv}^+(X) & \longrightarrow & \mathrm{vDiv}(X), \end{array}$$

where:

- $\mathrm{Div}(X) \subset \mathrm{vDiv}(X)$ is the full subcategory of invertible objects;
- $\mathrm{vDiv}^+(X) \subset \mathrm{vDiv}(X)$ is the full subcategory $\mathrm{vDiv}(X)_{0/}$;
- $\mathrm{Div}^+(X) = \mathrm{Div}(X) \cap \mathrm{vDiv}(X)^+$, i.e., the square is cartesian;
- $\mathrm{Div}(X)$ and $\mathrm{Div}^+(X)$ are posets.

Remark 10.71 (Functoriality)*.* The functoriality of $\mathrm{vDiv}(X)$ in X is more subtle than for vDiv^+. Given a map of schemes $f\colon Y \to X$, both $\mathcal{O}_Y$-modules $\mathcal{K}_Y$ and $f^*(\mathcal{K}_X)$ are localizations of $\mathcal{O}_Y$, and we can only define the pullback functor $f^*\colon \mathrm{vDiv}(X) \to \mathrm{vDiv}(Y)$ if $\mathcal{K}_Y$ is a further localization of $f^*(\mathcal{K}_X)$, or equivalently if $\mathcal{O}_Y \to f^*(\mathcal{K}_X)$ is a monomorphism. This holds if f is flat, in which case f^* also restricts to $f^*\colon \mathrm{Div}(X) \to \mathrm{Div}(Y)$.

Remark 10.72 (Div as the group completion of Div^+)*.* Let X be a scheme. Since open immersions are flat, there is a well-defined presheaf of abelian groups

$$\mathrm{Div}\colon \mathrm{Open}(X)^{\mathrm{op}} \to \mathrm{Ab},$$

which is in fact a sheaf. By definition of the sheaf $\mathcal{K}_X$ and the fact that line bundles are locally trivial, the inclusion $\mathrm{Div}^+ \hookrightarrow \mathrm{Div}$ has the following universal property: for any sheaf of abelian groups A on $|X|$, any monoid map $\mathrm{Div}^+ \to A$ extends uniquely to Div. In other words, Div is precisely the group completion of the monoid Div^+ in $\mathrm{Sh}(|X|)$.

Example 10.73*.* (i) If R is a unique factorization domain with fraction field K, then

$$(\mathrm{Div}(\mathrm{Spec}(R)), +, \leq) \simeq (K^\times/R^\times, \cdot, |),$$

where $a|b$ means $a^{-1}b \in R$.

(ii) If R is a Dedekind domain, then $\mathrm{Div}(\mathrm{Spec}(R))$ is the free abelian group on the set of maximal ideals of R. Consequently, if X is a quasi-compact Dedekind scheme (e.g., a smooth curve of finite type over a field), then $\mathrm{Div}(X)$ is the free abelian group on the set of closed points of $|X|$.

Definition 10.74 (Divisor of a rational function, principal divisor)**.** Let X be a scheme.

(i) The *divisor* $\div(f)$ of a rational function $f \in K(X)$ is the virtual divisor $\mathcal{O}_X \to \mathcal{K}_X$, $1 \mapsto f$.
(ii) A virtual divisor on X is called *principal* if it is isomorphic to $\div(f)$ for some $f \in K(X)$.

Proposition 10.75. *Let X be a scheme.*

(i) The map $\div\colon K(X) \to \mathrm{vDiv}(X)$ has a symmetric monoidal structure. In particular, there are canonical isomorphisms $\div(1) \simeq 0$ and $\div(fg) \simeq \div(f) + \div(g)$ for all $f, g \in K(X)$.
(ii) Let $f \in K(X)$. Then $\div(f)$ is effective if and only if $f \in \mathcal{O}(X)$.
(iii) Let $f \in K(X)$. Then $\div(f)$ is a Cartier divisor if and only if $f \in K(X)^\times$.
(iv) The map $\div$ induces an isomorphism of abelian groups

$$\div\colon \Gamma(|X|, \mathcal{K}_X^\times/\mathcal{O}_X^\times) \xrightarrow{\sim} \mathrm{Div}(X).$$

Example 10.76*.* The divisor of a rational function f records the zeros and poles of f with multiplicity. For example:

(i) Consider the rational function $f = t^2$ in $K(\mathbf{P}^1) = \mathbf{Q}(t)$, where $\mathbf{P}^1 = \mathrm{Proj}(\mathbf{Z}[x, y])$ and $t = x/y$. It has a zero of order two at $0 = \mathrm{V}(x)$ and a pole of order two at $\infty = \mathrm{V}(y)$. Hence, $\div(f)$ is the Cartier divisor $2 \cdot 0 - 2 \cdot \infty$ on $\mathbf{P}^1$.

(ii) Consider the rational function $f = s/(t - s^2)$ in $K(\mathbf{P}^2) = \mathbf{Q}(s, t)$, where $\mathbf{P}^2 = \mathrm{Proj}(\mathbf{Z}[x, y, z])$, $s = x/z$, $t = y/z$. It vanishes to order one on the y-axis $\mathrm{V}(x)$ and on the line at infinity $\mathrm{V}(z)$, and has a pole of order one on the ellipse $\mathrm{V}(yz - x^2)$. Hence, we have

$$\div(f) = \mathrm{V}(x) + \mathrm{V}(z) - \mathrm{V}(yz - x^2)$$

in $\mathrm{Div}(\mathbf{P}^2)$. We can also view f as a rational function on $\mathbf{P}^1 \times \mathbf{P}^1 = \mathrm{Proj}(\mathbf{Z}[x, u]) \times \mathrm{Proj}(\mathbf{Z}[y, v])$ with $t = x/u$ and $s = y/v$. Then

$$\div(f) = \mathrm{V}(x) + \mathrm{V}(u) + \mathrm{V}(v) - \mathrm{V}(yu^2 - x^2v)$$

in $\mathrm{Div}(\mathbf{P}^1 \times \mathbf{P}^1)$.

Remark 10.77 (Blowing up at fractional ideals)*.* A *fractional ideal* over a scheme X is an $\mathcal{O}_X$-submodule $I \subset \mathcal{K}_X$ in $\mathrm{Sh}(|X|)$ which is a quasi-coherent sheaf. The image I^n of $I^{\otimes n} \to \mathcal{K}_X^{\otimes n} = \mathcal{K}_X$ is then again quasi-coherent, so that the Rees algebra $\bigoplus_{n \in \mathbf{N}} I^n$ is a quasi-coherent $\mathbf{N}$-graded algebra over X. We can then define the *blowup* of X at I to be $\mathrm{Proj}(\bigoplus_{n \in \mathbf{N}} I^n)$. This blowup has the following universal property: it is a final object in the full subcategory of $\mathrm{Sch}_{/X}$ consisting of maps $f \colon Y \to X$ such that $\mathcal{K}_Y$ is a further localization of $f^*(\mathcal{K}_X)$ and the image of $f^*(I)$ in $\mathcal{K}_Y$ is a Cartier divisor.

10.5. **Weil divisors.** Weil divisors are an alternative approach to divisors that is in some sense much simpler, as it only depends on the underlying topological space of a scheme. Although Weil divisors are not well-behaved in general, we will show that they are equivalent to Cartier divisors on regular schemes.

Definition 10.78 (Cycle in a topological space)**.** Let T be a topological space and A a commutative monoid. A *cycle in T with coefficients in A* is a formal sum

$$\sum_V a_V \cdot V, \qquad a_V \in A,$$

where V runs over all irreducible closed subsets of T, which is *locally finite*: every point has a neighborhood that intersects only finitely many V's with $a_V \neq 0$. We denote by $\mathrm{Z}^*(T, A)$ the commutative monoid of cycles in T with coefficients in A.

If $n \in \mathbf{N} \cup \{\infty\}$, a cycle $\sum_V a_V \cdot V$ has *codimension n* if $a_V = 0$ whenever $\mathrm{codim}(V, T) \neq n$. We denote by $\mathrm{Z}^n(T, A) \subset \mathrm{Z}^*(T, A)$ the submonoid of codimension n cycles.

Remark 10.79.

(i) If T is sober, cycles in T are equivalently locally finite sums indexed by the set of points of T.

(ii) If A is an abelian group, then $\mathrm{Z}^*(T, A)$ and $\mathrm{Z}^n(T, A)$ are also abelian groups. If A is a partially ordered commutative monoid, then $\mathrm{Z}^*(T, A)$ inherits a partial order, defined termwise.

(iii) If T is quasi-compact, cycles in T are finite sums. Hence, $\mathrm{Z}^n(T, A) = \mathrm{Z}^n(T, \mathbf{N}) \otimes A$, and $\mathrm{Z}^n(T, \mathbf{N})$ (resp. $\mathrm{Z}^n(T, \mathbf{Z})$) is the free commutative monoid (resp. the free abelian group) on the set of irreducible closed subsets of T of codimension n. Moreover,

$$\mathrm{Z}^*(T, A) = \bigoplus_{n \in \mathbf{N} \cup \{\infty\}} \mathrm{Z}^n(T, A).$$

(iv) We can restrict cycles in T to any open subspace $U \subset T$ by intersecting with U and setting empty intersections to zero. This makes $\mathrm{Z}^*(-, A)$ into a presheaf of commutative monoids on $\mathrm{Open}(T)$, which is in fact a sheaf.

Definition 10.80 (Weil divisor, effective Weil divisor)**.** Let X be a scheme. A *Weil divisor* on X is a cycle of codimension 1 in $|X|$ with coefficients in $\mathbf{Z}$, i.e., an element of the partially ordered abelian group $\mathrm{Z}^1(|X|, \mathbf{Z})$. Elements of the submonoid $\mathrm{Z}^1(|X|, \mathbf{N})$ are called *effective Weil divisors*.

Recall that the length of an R-module M is the length of the poset of submodules of M:

$$\operatorname{length}_R(M) \in \mathbf{N} \cup \{\infty\}.$$

For every short exact sequence $0 \to M \to N \to P \to 0$, it satisfies

$$\operatorname{length}_R(N) = \operatorname{length}_R(M) + \operatorname{length}_R(P).$$

The length of a ring is its length as a module over itself. A ring is called *artinian* if it has finite length, equivalently if it is noetherian of Krull dimension at most 0.

Example 10.81 (Fundamental cycle)*.* Let X be a locally noetherian scheme. The set of points of codimension 0 in $|X|$ is locally finite. Moreover, if x is such a point, then $\mathcal{O}_{X,x}$ is noetherian of Krull dimension 0 and hence has finite length. The *fundamental cycle* of X is

$$[X] = \sum_x \operatorname{length}(\mathcal{O}_{X,x}) \cdot x \in \mathrm{Z}^0(|X|, \mathbf{N}),$$

where x ranges over the points of codimension 0. More generally, for any closed subscheme $Z \subset X$, its fundamental cycle $[Z] \in \mathrm{Z}^*(|X|, \mathbf{N})$ is defined through the inclusions

$$[Z] \in \mathrm{Z}^0(|Z|, \mathbf{N}) \subset \mathrm{Z}^*(|Z|, \mathbf{N}) \subset \mathrm{Z}^*(|X|, \mathbf{N}).$$

Construction 10.82 (Multiplicity of a divisor)**.** Let X be a locally noetherian scheme and let $x \in |X|$ have codimension 1. We define a map of partially ordered abelian groups

$$\operatorname{mult}_x \colon \operatorname{Div}(X) \to \mathbf{Z}.$$

First let R be a noetherian local ring of Krull dimension at most 1, and let $S \subset R$ be the commutative monoid of non-zero divisors, partially ordered by divisibility. For $f \in S$, the ring $R/(f)$ has finite length. The map

$$S \to \mathbf{N}, \qquad f \mapsto \operatorname{length}(R/(f)),$$

is a map of partially ordered commutative monoids that vanishes on $R^\times \subset S$. Indeed, for $f, g \in S$ there is a short exact sequence

$$0 \to R/(f) \xrightarrow{g} R/(fg) \to R/(g) \to 0,$$

and $R/(u) = 0$ for every unit u. Passing to group completions gives

$$\operatorname{mult} \colon \operatorname{Div}(\operatorname{Spec}(R)) \simeq \operatorname{Frac}(R)^\times / R^\times \to \mathbf{Z}.$$

Since the canonical map $i_x \colon \operatorname{Spec}(\mathcal{O}_{X,x}) \to X$ is flat, pullback defines $i_x^* \colon \operatorname{Div}(X) \to \operatorname{Div}(\operatorname{Spec}(\mathcal{O}_{X,x}))$. We define mult_x as the composite

$$\operatorname{Div}(X) \xrightarrow{i_x^*} \operatorname{Div}(\operatorname{Spec}(\mathcal{O}_{X,x})) \xrightarrow{\operatorname{mult}} \mathbf{Z}.$$

Lemma 10.83. *Let X be a locally noetherian scheme and let $D \in \operatorname{Div}(X)$. The set of codimension 1 points $x \in |X|$ such that $\operatorname{mult}_x(D) \neq 0$ is locally finite in $|X|$.*

Construction 10.84 (Cycle associated with a divisor)**.** Let X be a locally noetherian scheme. Define

$$\operatorname{cyc} \colon \operatorname{Div}(X) \to \mathrm{Z}^1(|X|, \mathbf{Z}), \qquad D \mapsto \sum_x \operatorname{mult}_x(D) \cdot x,$$

where x ranges over the codimension 1 points of $|X|$. This is a map of partially ordered abelian groups, and it restricts to a map $\operatorname{cyc} \colon \operatorname{Div}^+(X) \to \mathrm{Z}^1(|X|, \mathbf{N})$.

Example 10.85*.* For $(L, s) \in \operatorname{Div}^+(X)$, the cycle $\operatorname{cyc}(L, s)$ is the fundamental cycle of the closed subscheme $\mathrm{V}(s) \subset X$. By the universal property of the inclusion $\operatorname{Div}^+ \hookrightarrow \operatorname{Div}$, this uniquely determines cyc as a map of sheaves of abelian groups $\operatorname{Div} \to \mathrm{Z}^1(-, \mathbf{Z})$ on $\operatorname{Open}(X)$.

Remark 10.86 (Flat pullback of cycles). Let $f\colon Y \to X$ be a flat map of locally noetherian schemes. Define

$$f^*\colon \mathrm{Z}^*(|X|, A) \to \mathrm{Z}^*(|Y|, A), \qquad \sum_V a_V \cdot V \mapsto \sum_V a_V [f^{-1}(V)],$$

where an irreducible closed subset $V \subset |X|$ is viewed as a reduced closed subscheme and $[-]$ is the fundamental cycle. Flat pullback preserves codimension n cycles and satisfies $f^*([Z]) = [f^{-1}(Z)]$ for every closed subscheme $Z \subset X$. Consequently, the square

$$\begin{array}{ccc} \mathrm{Div}(X) & \xrightarrow{\mathrm{cyc}} & \mathrm{Z}^1(|X|, \mathbf{Z}) \\ {\scriptstyle f^*}\downarrow & & \downarrow{\scriptstyle f^*} \\ \mathrm{Div}(Y) & \xrightarrow[\mathrm{cyc}]{} & \mathrm{Z}^1(|Y|, \mathbf{Z}) \end{array}$$

commutes.

Lemma 10.87. *Let X be a locally noetherian normal scheme and let $D \in \mathrm{Div}(X)$. If $\mathrm{cyc}(D)$ is effective, then D is effective.*

Theorem 10.88 (Comparison between Cartier divisors and Weil divisors)**.** *Let X be a locally noetherian normal scheme. Then*

$$\mathrm{cyc}\colon \mathrm{Div}(X) \to \mathrm{Z}^1(|X|, \mathbf{Z})$$

is injective and order-reflecting. It is surjective, and hence an isomorphism of partially ordered abelian groups, if and only if all the local rings of X are unique factorization domains; this holds, for example, if X is regular.

10.6. **The Picard group.**

Definition 10.89 (Picard group)**.** Let X be an algebraic functor. The *Picard group* $\mathrm{Pic}(X)$ of X is the set of isomorphism classes of line bundles over X, with the abelian group structure induced by the tensor product:

$$[L] + [L'] = [L \otimes L'].$$

Remark 10.90 (Cohomological interpretation of Pic). If X is a scheme, the groupoid of line bundles over X is equivalent to that of $\mathcal{O}_X^\times$-torsors in $\mathrm{Sh}(|X|)$. Indeed, any line bundle is locally isomorphic to $\mathcal{O}_X$ and $\mathrm{Aut}(\mathcal{O}_U) \simeq \mathcal{O}(U)^\times$. Consequently, there is an isomorphism

$$\mathrm{Pic}(X) \simeq \mathrm{H}^1(|X|, \mathcal{O}_X^\times)$$

between the Picard group and the group of isomorphism classes of $\mathcal{O}_X^\times$-torsors.

Proposition 10.91 (The unit–Picard exact sequence)**.** *For any scheme X, there is an exact sequence of abelian groups*

$$0 \to \mathcal{O}(X)^\times \to K(X)^\times \xrightarrow{\div} \mathrm{Div}(X) \to \mathrm{Pic}(X).$$

The last map is surjective if X is reduced and its set of irreducible components is locally finite.

Example 10.92. If R is an integral domain, the unit–Picard exact sequence identifies $\mathrm{Pic}(R)$ with the quotient of the group of invertible fractional ideals, i.e., sublines of $\mathrm{Frac}(R)$, by the subgroup of principal fractional ideals. In particular, if R is a Dedekind domain, then $\mathrm{Pic}(R) = 0$ if and only if R is a principal ideal domain, equivalently a unique factorization domain.

Remark 10.93. The unit–Picard exact sequence is the beginning of the long exact sequence of cohomology groups induced by the short exact sequence of sheaves

$$0 \to \mathcal{O}_X^\times \to \mathcal{K}_X^\times \to \mathcal{K}_X^\times/\mathcal{O}_X^\times \to 0.$$

Indeed, we have $\mathrm{H}^0(|X|, \mathcal{K}_X^\times/\mathcal{O}_X^\times) \simeq \mathrm{Div}(X)$ by the characterization of principal divisors and $\mathrm{H}^1(|X|, \mathcal{O}_X^\times) \simeq \mathrm{Pic}(X)$ by the previous remark. If X is reduced with a locally finite set of irreducible components, then $\mathcal{K}_X^\times$ is a flasque sheaf and hence $\mathrm{H}^1(|X|, \mathcal{K}_X^\times) = 0$, which explains the surjectivity assertion in the previous proposition.

The following theorem has an elementary proof using Zariski descent for line bundles when X is locally a unique factorization domain. The key ingredient in the general case is Horrocks' theorem on vector bundles: for any ring R and $n \geq 0$, any vector bundle on $\mathbf{A}^n_R$ that extends to $\mathbf{P}^n_R$ comes from $\operatorname{Spec}(R)$.

Theorem 10.94 (Picard group of projective spaces)**.** *Let X be a scheme, let V be a vector bundle over X of rank ≥ 2, and let $p\colon \mathbf{P}(V) \to X$ be the associated projective space. Then there is an isomorphism*

$$\operatorname{Pic}(X) \oplus \operatorname{Map}(|X|, \mathbf{Z}) \xrightarrow{\sim} \operatorname{Pic}(\mathbf{P}(V)), \quad ([L], n) \mapsto [p^*(L)(n)].$$

In particular, if X is connected, then $\operatorname{Pic}(\mathbf{P}(V)) \simeq \operatorname{Pic}(X) \oplus \mathbf{Z}$.

The next theorem can be proved by reducing to the case $M = \mathcal{O}_X$ and using the unit–Picard exact sequence together with the comparison between Cartier and Weil divisors. The latter is the source of the normality assumption; the statement fails, for example, for the cuspidal cubic over any field.

Theorem 10.95 (Picard group of affine spaces)**.** *Let X be a scheme, let M be a quasi-coherent module over X, and let $p\colon \mathbf{A}(M) \to X$ be the associated affine space. If X is normal and M is flat, then there is an isomorphism*

$$\operatorname{Pic}(X) \xrightarrow{\sim} \operatorname{Pic}(\mathbf{A}(M)), \quad [L] \mapsto [p^*(L)].$$

11. Properness

11.1. **Prim and proper morphisms.** Properness is the most fundamental global property in geometry: it is essentially about the existence and uniqueness of limits, in the sense of analysis. In classical algebraic geometry, it is called "completeness", although that name has a different, but closely related, meaning in differential geometry. We start by reviewing properness in topology.

Let X and Y be topological spaces. A continuous map $f\colon Y \to X$ is called *universally closed* if, for every continuous map $Z \to X$, the base change $Y \times_X Z \to Z$ is a closed map. By the tube lemma, f is universally closed if and only if it is closed and all its fibers are quasi-compact. Thus, the unique map $X \to *$ is universally closed if and only if X is quasi-compact. The map f is called *separated* if its diagonal Δ_f is closed, or equivalently universally closed since it is injective. The unique map $X \to *$ is separated if and only if X is Hausdorff. Finally, f is called *proper* if it is both universally closed and separated.

Universal closedness has a standard characterization in terms of lifting limit points. Namely, $f\colon Y \to X$ is universally closed if and only if, for every generalized sequence $s\colon I \to Y$ and every limit point x of $f \circ s$, there exists a subsequence of s with a limit point y such that $f(y) = x$. This is almost a right lifting property with respect to $I \hookrightarrow I \cup \{\infty\}$, except that we may need to pass to a subsequence because ∞ can be approached from I in more than one way. Universal closedness can instead be characterized by a right lifting property with respect to inclusion maps $V' \hookrightarrow V$ such that every point in V can be approached from V' in one and only one way. The simplest example is the inclusion of the generic point of the Sierpiński space, $\{*\} \hookrightarrow \{* \rightsquigarrow \infty\}$; another is $[0,1) \hookrightarrow [0,1]$.

Once universal closedness is characterized by a right lifting property, separatedness and properness are characterized by the "at most one" and "exactly one" versions of the same lifting property. In algebraic geometry, the actual definition of properness also involves a finiteness condition: a morphism of schemes is called *prim* if it is universally closed and locally of finite type, and proper if it is both prim and separated. Ignoring finiteness conditions, the analogy with smooth, unramified, and étale morphisms is summarized by

class of maps T	thickenings	valuations
T-epimorphism	smooth	prim
T-monomorphism	unramified	separated
T-isomorphism	étale	proper.

Definition 11.1 (Prim, separated, and proper morphisms)**.** Let $f\colon Y \to X$ be a morphism of schemes.

(i) f is *prim* if it is locally of finite type and universally closed.
(ii) f is *separated* if its diagonal $\Delta_f\colon Y \to Y \times_X Y$ is prim.
(iii) f is *proper* if it is prim and separated.

Remark 11.2. (i) The diagonal of a morphism of schemes is always an immersion, immersions are locally of finite type, and closed immersions are universally closed. Thus, the definition of separatedness above agrees with the earlier one.
(ii) Proper morphisms are often defined with the additional condition of being quasi-compact, but quasi-compactness is a consequence of universal closedness. Hence, prim and proper morphisms are of finite type.

Proposition 11.3. *The properties "prim", "separated", and "proper" are local on the base and stable under composition and base change.*

Remark 11.4. These properties are even fpqc-local on the base, but none of them is local on the source: an open immersion that is not closed is not prim, and any morphism whose source is affine is separated.

Example 11.5 (Closed immersions)*.* Closed immersions are prim and separated, hence proper. Conversely, suppose that $i\colon Y \hookrightarrow X$ is a prim immersion and factor it as a closed immersion $Y \hookrightarrow U$ followed by an open immersion $U \hookrightarrow X$. The open subschemes U and the complement of the image of $|i|$ cover X, and i is a closed immersion over each of them. Hence, i is a closed immersion. In fact, every prim monomorphism is a closed immersion.

Example 11.6 (Integral and finite morphisms)*.* By the going-up theorem, integral maps of algebraic functors are universally closed. They are also separated, since they are affine. Since "finite" is equivalent to "integral and locally of finite type", finite maps of schemes are proper.

Remark 11.7 (Nagata compactification)*.* Zariski's main theorem is an ingredient in the broader Nagata compactification theorem: if X is qcqs and $f\colon Y \to X$ is a separated morphism of finite type, then f admits a non-canonical factorization

$$Y \overset{j}{\hookrightarrow} \overline{Y} \overset{p}{\longrightarrow} X,$$

where j is an open immersion and p is proper. As in algebraic topology, such a compactification can be used to define the cohomology with compact support of Y relative to X.

11.2. **Valuative criteria.** To express the existence and uniqueness of limits in algebraic geometry, we need a class of inclusions $X' \hookrightarrow X$ such that every point of X may be approached from X' in one and only one way. We want X' to be dense in X, in the sense that functions on X are determined by their restrictions to X'. Given a closed subscheme $Z \subset X$ disjoint from X', the directions of approach to Z in X are classified by the projectivized normal cone $\mathbf{P}(C_Z(X))$, the preimage of Z in $\mathrm{Bl}_Z(X)$. Requiring $\mathrm{Bl}_Z(X) \to X$ to be an isomorphism for every such Z is, by the universal property of blowups, equivalent to requiring every such Z to be an effective divisor. This motivates the following definition.

Definition 11.8 (Prüfer pair, Prüfer domain, Prüfer scheme)**.**
(i) Let $j\colon X' \hookrightarrow X$ be a monomorphism of schemes. The pair (X, X') is a *Prüfer pair* if j is quasi-compact, X' is the intersection of its open neighborhoods in X, the map $\mathcal{O}_X \to j_*(\mathcal{O}_{X'})$ in Mod_X is a monomorphism, and, for every open subscheme $U \subset X$, every finitely presented closed subscheme $Z \subset U$ with $Z \cap X' = \varnothing$ is an effective divisor on U.
(ii) A *Prüfer domain* is an integral domain R such that $(\mathrm{Spec}(R), \mathrm{Spec}(\mathrm{Frac}(R)))$ is a Prüfer pair.
(iii) A *Prüfer scheme* is a scheme admitting an affine open covering by Prüfer domains.

Remark 11.9 (Affine Prüfer pairs)*.* Let $R \to R'$ be a localization. Then $(\mathrm{Spec}(R), \mathrm{Spec}(R'))$ is a Prüfer pair if and only if $R \to R'$ is injective and every finitely generated ideal $I \subset R$ with $IR' = R'$ is a line. In particular, a domain is Prüfer if and only if every nonzero finitely generated ideal is a line. Thus, the locally noetherian Prüfer schemes are exactly the Dedekind schemes.

Remark 11.10 (Prüfer implies normal)*.* Prüfer domains are integrally closed, and hence Prüfer schemes are normal. Indeed, consider a fraction r/s over a Prüfer domain R. The ideal (r, s) is a line, so there are elements $f_1, \dots, f_n \in R$ generating the unit ideal such that r/s simplifies over each R_{f_i} to r_i/s_i with $(r_i, s_i) = R_{f_i}$. If r/s satisfies an integral equation of degree d over R, then $r_i^d/s_i \in R_{f_i}$ and hence $r_i/s_i \in R_{f_i}$. Zariski descent now implies $r/s \in R$.

Definition 11.11 (Valuation)**.** Let K be a field and Γ a totally ordered abelian group. A map $v\colon K \to \Gamma \cup \{\infty\}$ is a *valuation* if

(i) $v(1) = 0$ and $v(xy) = v(x) + v(y)$ for all $x, y \in K$;
(ii) $v(0) = \infty$ and $v(x + y) \geq \min\{v(x), v(y)\}$ for all $x, y \in K$.

Then

$$\mathcal{O}_v = \{x \in K \mid v(x) \geq 0\}$$

is a local subring of K with fraction field K, maximal ideal $\mathfrak{m}_v = \{x \in K \mid v(x) > 0\}$, unit group $\mathcal{O}_v^\times = \{x \in K \mid v(x) = 0\}$, and residue field $\kappa(v) = \mathcal{O}_v/\mathfrak{m}_v$.

Proposition 11.12. *Let R be a domain with fraction field K. The following conditions are equivalent:*

(i) R is a local Prüfer domain.
(ii) The divisibility relation on R is total.
(iii) The poset of ideals in R is totally ordered.
(iv) The poset of divisors $\mathrm{Div}(\mathrm{Spec}(R))$ *is totally ordered.*

(v) *For every* $x \in K^\times$, *either* x *or* x^{-1} *belongs to* R.
(vi) *There exists a valuation* $v\colon K \to \Gamma \cup \{\infty\}$ *such that* $R = \mathcal{O}_v$.

Definition 11.13 (Valuation ring)**.** A *valuation ring* is an integral domain satisfying the equivalent conditions of Proposition 11.12.

Remark 11.14 (Value group)*.* If R is a valuation ring with fraction field K, there is an initial choice of valuation $v\colon K \to \Gamma \cup \{\infty\}$ with $R = \mathcal{O}_v$, namely

$$\div\colon K^\times \twoheadrightarrow K^\times/R^\times = \mathrm{Div}(\mathrm{Spec}(R)).$$

The totally ordered abelian group $\mathrm{Div}(\mathrm{Spec}(R))$ is therefore also called the *value group* of R.

Example 11.15. (i) Fields are precisely the valuation rings with trivial value group.
(ii) A valuation ring is *discrete* if its value group is isomorphic to $\mathbf{Z}$. Since $\mathbf{N} \subset \mathbf{Z}$ is well-ordered, such a ring is a principal ideal domain. Conversely, a local principal ideal domain that is not a field is a discrete valuation ring.
(iii) Let Γ be a totally ordered cancellative commutative monoid and let k be a ring. The ring of *Hahn series* $k[[t^\Gamma]]$ consists of formal sums

$$\sum_{\gamma\in\Gamma} a_\gamma t^\gamma$$

with $a_\gamma \in k$ and well-ordered support. Taking the infimum of the support defines a surjection

$$v\colon k[[t^\Gamma]] \twoheadrightarrow \Gamma \cup \{\infty\}.$$

If k is a field and Γ a group, then $k[[t^\Gamma]]$ is a field and v a valuation. Hence, $k[[t^{\Gamma_{\geq 0}}]] \subset k[[t^\Gamma]]$ is a valuation ring with value group Γ. When $\Gamma = \mathbf{Q}$, Hahn series are also called *Puiseux series*.

Remark 11.16 (Noetherian valuation rings)*.* Since finitely generated ideals in a valuation ring are principal, noetherian valuation rings are principal ideal domains. Hence, they are either fields or discrete valuation rings. Equivalently, they are precisely the regular local rings of Krull dimension at most 1.

Proposition 11.17 (Valuation rings as maximal local subrings)**.** *Let* K *be a field and* $R \subset K$ *a subring. The following conditions are equivalent:*

(i) R *is a valuation ring and* $\mathrm{Frac}(R) = K$.
(ii) R *is a maximal local subring of* K *with respect to local inclusions.*

Definition 11.18 (Formally prim, formally separated, formally proper)**.** Let T be the collection of maps $\mathrm{Spec}(K) \hookrightarrow \mathrm{Spec}(V)$, where V is a valuation ring with fraction field K. A morphism of algebraic functors is called

(i) *formally prim* if it is a T-epimorphism;
(ii) *formally separated* if it is a T-monomorphism;
(iii) *formally proper* if it is a T-isomorphism.

Proposition 11.19 (Specializations via valuations)**.** *Let* X *be a scheme, let* $y \rightsquigarrow x$ *be a specialization in* $|X|$*, and let* $\kappa(y) \hookrightarrow K$ *be a field extension. Then there exist a valuation* v *on* K *and an extension*

$$\begin{array}{ccccc} \mathrm{Spec}(K) & \longrightarrow & \mathrm{Spec}(\kappa(y)) & \xrightarrow{y} & X \\ \uparrow & & & \nearrow & \\ \mathrm{Spec}(\mathcal{O}_v) & & & & \end{array}$$

sending the closed point to x.

Lemma 11.20. *Let* $f\colon Y \to X$ *be a quasi-compact map of schemes. If the image of* $|f|$ *is closed under specialization in* $|X|$*, then it is a closed subset of* $|X|$.

Theorem 11.21 (Valuative criterion for universal closedness)**.** *Let $f\colon Y \to X$ be a map of schemes. The following conditions are equivalent:*

(i) f is universally closed.
(ii) f is formally prim and quasi-compact.

Corollary 11.22 (Valuative criteria for primness, separatedness, and properness)**.** *Let $f\colon Y \to X$ be a map of schemes.*

(i) f is prim if and only if it is formally prim and of finite type.
(ii) f is separated if and only if it is formally separated and quasi-separated.
(iii) f is proper if and only if it is formally proper, of finite type, and quasi-separated.

Example 11.23. Let X be the affine line with doubled origin and let $p\colon X \to \mathbf{A}^1$ be the finite-presentation map $(f, e) \mapsto f$. Then p is formally prim but not formally separated: given a valuation ring V with fraction field K and $f \in V$, every idempotent in $K/(f)$, which is either 0 or 1, lifts to an idempotent in $V/(f)$, but not uniquely if $f \notin V^\times \cup \{0\}$. Hence, by Corollary 11.22, p is prim but not proper.

As for smooth morphisms, the finiteness property of prim morphisms allows us to enlarge the class of maps with respect to which they have the right lifting property.

Proposition 11.24 (Completing curves in prim schemes)**.** *Let C be a Prüfer scheme and let $C' \subset C$ be a subscheme containing every point of codimension 0. Suppose that C' is a cofiltered intersection of quasi-compact open immersions $U_\alpha \hookrightarrow C$ with locally finite complement in $|C|$. Then every prim morphism of schemes has the right lifting property with respect to $C' \hookrightarrow C$.*

Example 11.25. For a Dedekind scheme C, the following are basic examples of Proposition 11.24:

(i) $C - x \hookrightarrow C$ for any codimension 1 point $x \in |C|$. For example, $\operatorname{Spec}(\mathbf{Z}[1/p]) \hookrightarrow \operatorname{Spec}(\mathbf{Z})$ for a prime p, or $\mathbf{A}^1_k \hookrightarrow \mathbf{P}^1_k$ for a field k.
(ii) $\coprod_\eta \operatorname{Spec}(\kappa(\eta)) \hookrightarrow C$, where η runs over the codimension 0 points of C. For example, $\operatorname{Spec}(\mathbf{Q}) \hookrightarrow \operatorname{Spec}(\mathbf{Z})$, or $\operatorname{Spec}(k(t)) \hookrightarrow \mathbf{P}^1_k$.

Remark 11.26. The conclusion of Proposition 11.24 propagates to separated and proper morphisms by applying it to the diagonal. The inclusion $C' \subset C$ in the statement is a Prüfer pair. For separated and proper morphisms, uniqueness allows a further generalization: they satisfy their respective lifting properties with respect to arbitrary Prüfer pairs.

11.3. **Relation to projectivity and finiteness.** Projective morphisms are of finite type and separated. The valuative criterion for universal closedness now implies the following.

Proposition 11.27. *Projective morphisms of schemes are proper.*

Corollary 11.28. *Let X be a scheme and let $Z \subset X$ be a closed subscheme of finite presentation. Then $\operatorname{Bl}_Z(X) \to X$ is proper.*

Corollary 11.29. *Let X be a scheme and let $Z \subset X$ be a closed subscheme of finite presentation. Then $\operatorname{Bl}_Z(X) \to X$ is surjective if and only if $|X| - |Z|$ is dense in $|X|$.*

Since prim immersions are closed immersions, proper quasi-projective morphisms are projective; Proposition 11.27 gives the converse.

Corollary 11.30. *A quasi-projective morphism of schemes is projective if and only if it is proper.*

Remark 11.31. The finiteness condition in projectivity serves three independent purposes. It implies that projective morphisms are locally of finite type and quasi-compact, and it ensures that they are formally prim. Indeed:

(i) If I is infinite, $\mathbf{P}^I$ is not formally prim over $\operatorname{Spec}(\mathbf{Z})$. If $\varphi\colon I \twoheadrightarrow \mathbf{N}$ is a surjection, the quotient line

$$(1/p^{\varphi(i)})_{i\in I}\colon \mathbf{Q}^{(I)} \twoheadrightarrow \mathbf{Q}$$

does not lift to a quotient line of $\mathbf{Z}^{(I)}_{(p)}$.

(ii) If $Z \subset X$ is not of finite presentation, then $\mathrm{Bl}_Z(X) \to X$ need not be formally prim. For example, let I be infinite and let $z\colon \mathcal{O}^{(I)} \to \mathcal{O}$ be the tautological map over $\mathbf{A}^I$. Then $\mathrm{Bl}_0(\mathbf{A}^I) = \mathrm{vBl}_z(\mathbf{A}^I)$, and this map to $\mathbf{A}^I$ is not formally prim since it admits $\mathbf{P}^I \to \mathrm{Spec}(\mathbf{Z})$ as a base change.

Remark 11.32 (Elimination theory). The fact that $\mathbf{P}^n_R \to \mathrm{Spec}(R)$ is closed is the classical main theorem of elimination theory. For every system Σ of homogeneous polynomial equations in finitely many variables over R, there exists a unique radical ideal $I \subset R$ such that, for every R-field k, the system Σ has a solution in some algebraic closure of k if and only if I maps to zero in k. In other words, all variables can be eliminated, leaving a system of equations in zero variables. The radical ideal I is called the *resultant* of Σ.

The converse of Proposition 11.27 does not hold: over every field there are proper schemes that are not projective. Proper schemes are nevertheless close to projective schemes and can always be obtained from projective schemes by "blowing down".

Theorem 11.33 (Chow's lemma)**.** *Let S be a qcqs scheme, let X be a separated S-scheme of finite type, and let $U \subset X$ be an open subscheme that is quasi-projective over S, for example any affine open subscheme. Then there exists a finitely presented closed subscheme $Z \subset X$ with $Z \cap U = \varnothing$ such that $\mathrm{Bl}_Z(X)$ is quasi-projective over S. Hence, if X is proper over S, then $\mathrm{Bl}_Z(X)$ is projective over S.*

Lemma 11.34. *Let φ be a ring map. Then $\mathrm{Spec}(\varphi)$ is dominant if and only if $\ker(\varphi)$ is a nilideal.*

Theorem 11.35 (Valuative criterion for integrality)**.** *Let $f\colon Y \to X$ be a map of schemes. The following conditions are equivalent:*

(i) f is integral.
(ii) f is affine and formally prim.

Combining the valuative criteria for properness and integrality gives the following characterization.

Corollary 11.36 (Characterizations of finiteness)**.** *Let $f\colon Y \to X$ be a map of schemes. The following conditions are equivalent:*

(i) f is finite.
(ii) f is affine and proper.
(iii) f is affine and projective.

Remark 11.37. By Zariski's main theorem, the conditions in Corollary 11.36 are also equivalent to:

(iv) f is quasi-finite and proper.
(v) f is locally quasi-finite, universally closed, and separated.

Appendix A. Facts about commutative algebra

Proposition A.1. *Let R be a ring and V be an R-vector space. For any R-module N and P, there is a canonical isomorphism*

$$\mathrm{Hom}_R(V, N \otimes_R P) \simeq \mathrm{Hom}_R(V, N) \otimes_R P$$

Proof. By Proposition 3.1, there exists V' such that the functor $V' \otimes_R (-)$ is right adjoint to $V \otimes_R (-)$. Since right adjoint is essentially unique, there exists a canonical isomorphism $V' \otimes_R (-) \simeq \mathrm{Hom}_R(V, -)$. The claim then follows from

$$\mathrm{Hom}_R(V, N \otimes_R P) \simeq V' \otimes_R (N \otimes_R P) \simeq (V' \otimes_R N) \otimes_R P \simeq \mathrm{Hom}_R(V, N) \otimes_R P.$$

□

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